CONSOLIDATED RESEARCH MANUSCRIPT

OEIS Conjecture Hunt

Collected Proofs, Counterexamples, and Corrections

Rounds I–X

KEIO

Consolidated edition: 21 October 2026

Source manuscripts: 20 October 2026

Abstract

This single English manuscript consolidates the complete mathematical text of ten OEIS conjecture-hunt dossiers. Their historical selection ledger contains 1,008 assertion packages: 107 in Round I, 101 in Round II, and 100 in each of Rounds III–X. The subjects include triangular and Hankel matrices, character sums, divisor identities, congruences, formal power series, digital sequences, floor identities, asymptotics, differential recurrences, and finite-state enumeration. The constituent proofs, exact counterexamples, corrected statements, coefficient lists, printed certificates, source attributions, and qualifications are retained. One title, one contents list, continuous pagination, round-qualified numbering, internal links, and an OEIS identifier locator provide a common structure. The selection total is a project ledger, not a count of worldwide-first discoveries or OEIS editorial closures. In particular, the retrospective priority correction recorded in Round V for five Round IV entries remains explicit. This is a consolidation of the supplied manuscripts, not a new mathematical validation or priority audit.

A standalone pdfLaTeX source with all manuscript text embedded.
Original companion verification archives remain separate supporting records.

Editorial guide to the consolidated edition

The ten supplied manuscripts are reproduced in their original round order. Their mathematical bodies have not been shortened: all written proofs, recurrence signatures, tables, counterexamples, boundary qualifications, and source discussions remain. The original separate cover pages and contents lists have been replaced by the common front matter. Cover-page outcome counts are collected below. Local definitions and reference lists are retained with the arguments they support, even when they repeat material from an earlier round.

Round	Selected	Outcomes reported by the source manuscript
I	107	104 proof-only; 1 mixed proof/refutation; 2 refutation-only.
II	101	100 proof-only; 1 mixed proof/refutation.
III	100	95 proof-only; 4 refutations; 1 mixed package.
IV	100	97 proof-only; 2 refutations; 1 boundary correction with a proved tail.
V	100	90 original claims proved; 2 refutations; 8 corrected statements proved.
VI	100	94 original claims proved; 5 recurrence-boundary corrections; 1 period-index correction.
VII	100	91 selected claims proved; 4 recurrence-boundary corrections; 4 other corrections; 1 asymptotic refutation.
VIII	100	98 selected claims proved; 1 support correction; 1 classification refutation.
IX	100	96 selected assertions proved; 4 source formulas corrected and proved.
X	100	97 selected assertions or questions proved; 3 source formulas corrected and proved.
Total	**1,008**	Historical selection ledger; the categories retain each source's conventions.

Priority correction carried forward. Round V explicitly withdraws the earlier new-resolution assessment for A176605, A176607, A176610, A176611, and A176612: Georg Fischer's 26 January 2020 family derivation already implies the selected recurrences. The Round IV arguments are retained, together with the later correction. The 1,008 figure must therefore be read as a selection total. Uncounted repairs, supplements, exclusions, and previously known inputs retain the status assigned in the originals.

Numbering and navigation. Part r is Round r. Numbered sections and automatic equation numbers carry the round prefix; theorem numbers inherit the section prefix. For example, Section 4.3 is original Round IV, Section 3. Descriptive manual equation tags are kept as printed in the source and are local to their round. All reference links point to the corresponding location in this combined document, and all page references use continuous pagination. The contents lists the main sections; PDF bookmarks retain the finer hierarchy. The final identifier locator lists every mentioned OEIS identifier, including uncounted and excluded entries, with links to the rounds in which it occurs.

Printed mathematics and companion files. This `.tex` file compiles independently: it does not read any external proof fragments, images, bibliography databases, or data files. It includes all material printed in the ten supplied manuscripts. Some original arguments additionally refer to expanded machine-readable matrices, coefficient vectors, audit records, and verification programs in the corresponding verification ZIP archives. Those payloads are not newly transcribed into this manuscript. References to a round's original filename, builder, README, or archive describe those preserved supporting records. Standalone compilation should not be confused with embedding every file from the original verification archives.

Editorial scope and provenance. Source dates, attributions, and statements about earlier audits remain historical statements from the original dossiers. Consolidation does not update OEIS

status, certify novelty, supply missing external certificates, or convert an internal check into external peer review. The edition was prepared with AI assistance at KEIO's request. The source TeX files were checked against the SHA-256 index in the supplied Rounds 01–10 archive before assembly. The consolidation check compares the retained bodies, mathematical environments, and references against those sources.

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Part I

Structural identities, arithmetic, floors, and generating functions

How to read and use this dossier

The original goal was to settle about one hundred accessible OEIS problems. The resulting packet contains 107 selected conjecture or empirical assertion groups, each tied to a named entry and a specific mathematical statement. The proofs resolve the specified statements for their full stated ranges. Explicit starting indices and finite exceptions matter: none are silently discarded.

The catalogue uses **P** for a proof, **R** for a refutation, and **P/R** where one conjecture is proved and another is refuted in the same entry. A394578 is the latter case: its recurrence is valid, while its proposed ratio limit is the reciprocal of the actual value. Four standalone index or definition repairs are presented separately and are excluded from the headline total.

Counting convention. We use an entry-level assertion group, not an arbitrary number of corollaries. All three A329649 conjectures count as one selection. Fixed-base or fixed-block conjectures on different entries receive separate selections, even when one parameterized theorem proves them together. The universal square assertion copied across A113249–A113256 counts once; the A127751 copy of A127752, A334110 copy of A334109, and A224511 consequence of A224515 add no count. A284005 and A329369 share a transform identity, but A329369 also has a separately proved Eulerian-sum assertion.

What the source audit establishes. The targeted statements were still conjectural or empirical in the inspected OEIS text, subject to the source qualifications explicitly given below. Known earlier solutions discovered in the entries or their linked documents were excluded. An additional audit compared all 100 claims in Appendix A of Tom Adamczewski's *OEIS Open* with the selected entries. That appendix contains the 100 costliest of 153 combined-model resolutions. Its only overlapping entry was A224515, where the existing result concerns existence; our selection is the separately stated uniqueness claim. The repository metadata could not be inspected completely, so the audit does not establish worldwide priority. See https://arxiv.org/abs/2608.11941 and the audit JSON accompanying this dossier.

What makes these proofs, rather than term fitting. The structural sections derive identities from definitions, with explicit inductions or counting arguments. The floor sections bound remainders for every index beyond a finite threshold using rational inequalities and positive polynomial coefficients. The recurrence section applies exact differential operators to known generating functions and reduces the result to zero or to a finite boundary polynomial. The companion programs check these identities with integer or rational arithmetic. Finite enumeration is supplementary verification of indexing and construction; it is never the reason an infinite assertion is accepted.

Scope of the supporting formulas. Where an established generating function or a previously proved recurrence is used as a premise, the text identifies it. For example, A054120 uses Hendel's already recorded proof of its first conjecture to prove its second. The recurrence of A174193 is addressed from its defining generating function; a separate tentative combinatorial interpretation is outside that selection. The harmonic-block work covers the six specified entries, not every polynomial in the broader A227012 discussion.

Companion files. The standalone `oeis_hunt_dossier.tex` contains the complete report. The verification bundle contains the detailed ledger, per-entry mathematical certificates, proof fragments, reproducible Python programs, and recorded verification results. The JSON ledger preserves the exact selected statement and its qualifications. OEIS page edits and submissions were not performed.

1.1 Catalogue of the selected assertions

Within each group, entry identifiers link directly to the corresponding OEIS source. The subsequent proof sections supply the complete arguments.

Structural and matrix (14)

No.	Entry	Result	Selected assertion
1	A329649	P	Nonvanishing, divisibility, and the exact quadratic L-value
2	A355401	P	The q-power triangle, first column, and inverse
3	A355282	P	The q-integer triangle and first column
4	A350557	P	Signed inverse identification
5	A136097	P	Gaussian central-entry divisibility for general q
6	A146955	P	Closed formula for the row sum
7	A127752	P	Inverse row sums have Feigenbaum parity
8	A081199	P	Collinear 4-subsets of a discrete cube
9	A081200	P	Collinear 5-subsets of a discrete cube
10	A081201	P	Collinear 6-subsets of a discrete cube
11	A054120	P	The second McGarvey identity and bivariate series
12	A078121	P	Binary-partition formula for matrix entries
13	A078122	P	Ternary-partition formulas for entries and row sums
14	A113249	P	Universal odd-index perfect-square assertion

Arithmetic and digital (19)

No.	Entry	Result	Selected assertion
15	A049803	P	Base-3 remainder recurrence and series
16	A049804	P	Base-4 remainder recurrence and series
17	A330358	P	Base-5 remainder recurrence and series
18	A323689	P	Integer coefficients; odd exactly at squares
19	A324300	P	Integer coefficients; odd exactly at k(k+2)
20	A362560	P	Divisibility is invariant under partition reversal
21	A085765	P	Growth exceeds every fixed power
22	A100311	P	The binary XOR recurrence
23	A156834	P	Prime-detecting inequality
24	A156833	P	Prime-detecting Mobius-sum inequality
25	A008578	P	The divisor-sum characterization
26	A006068	P	Inverse Gray-code recurrence
27	A016789	P	Negabinary XOR identity
28	A284005	P	Three binary-mask identities
29	A329369	P	Inverse mask transform and Eulerian sums
30	A334109	P	Exact first-occurrence minimization
31	A216971	P	Every entry in row n is divisible by n
32	A334592	P	Strict zero-count records in XOR triangles
33	A224515	P	Uniqueness for the XOR of consecutive squares

Floors and symmetric functions (23)

No.	Entry	Result	Selected assertion
34	A203150	P	Empirical periodic symmetric-function series
35	A203164	P	Periodic symmetric-function series and recurrence
36	A203165	P	Periodic symmetric-function series and recurrence
37	A203166	P	Periodic symmetric-function series and recurrence
38	A203167	P	Periodic symmetric-function series and recurrence
39	A203231	P	Periodic symmetric-function series and recurrence
40	A203234	P	Periodic symmetric-function series and recurrence
41	A203235	P	Periodic symmetric-function series and recurrence
42	A203157	P	Triangular-number symmetric-function recurrence
43	A203159	P	Even-integer symmetric-function recurrence
44	A024220	P	Arithmetic-progression floor series and recurrence
45	A024385	P	Arithmetic-progression floor series and recurrence
46	A024399	P	Arithmetic-progression floor series and recurrence
47	A024389	R	Empirical generating function fails at n=39
48	A024390	P	Empirical floor series with finite corrections
49	A024403	P	Empirical floor series with finite corrections
50	A173693	P	Rounded recurrence via a parity period
51	A227012	P	Exact floor formula and harmonic-block series
52	A227013	P	Exact floor formula and harmonic-block series
53	A227014	P	Exact floor formula and harmonic-block series
54	A227015	P	Exact floor formula and harmonic-block series
55	A227016	P	Exact floor formula and harmonic-block series
56	A227017	P	Exact floor formula and harmonic-block series

Generating functions and recurrences (51)

No.	Entry	Result	Selected assertion
57	A174193	P	Specified polynomial recurrence
58	A346074	P	Specified polynomial recurrence
59	A247173	P	Specified polynomial recurrence
60	A247102	P	Specified polynomial recurrence
61	A234269	P	Specified polynomial recurrence
62	A268600	P	Specified polynomial recurrence
63	A268601	P	Specified polynomial recurrence
64	A108296	P	Specified polynomial recurrence
65	A244884	P	Specified polynomial recurrence
66	A244886	P	Specified polynomial recurrence
67	A257104	P	Specified polynomial recurrence
68	A099038	P	Specified polynomial recurrence
69	A337500	P	Specified polynomial recurrence
70	A127905	P	Specified polynomial recurrence
71	A392976	P	Specified polynomial recurrence
72	A186185	P	Specified polynomial recurrence
73	A220433	P	Specified polynomial recurrence
74	A071264	P	Specified polynomial recurrence
75	A392211	P	Specified polynomial recurrence

No.	Entry	Result	Selected assertion
76	A091539	P	Specified polynomial recurrence
77	A027175	P	Specified polynomial recurrence
78	A027182	P	Specified polynomial recurrence
79	A177258	P	Specified polynomial recurrence
80	A177259	P	Specified polynomial recurrence
81	A177260	P	Specified polynomial recurrence
82	A351933	P	Specified polynomial recurrence
83	A001722	P	Specified polynomial recurrence
84	A345651	P	Specified polynomial recurrence
85	A111942	P	Specified polynomial recurrence
86	A394578	P/R	Recurrence true; ratio limit is 4, not 1/4
87	A396724	P	Specified polynomial recurrence
88	A121580	P	Specified polynomial recurrence
89	A180195	P	Specified polynomial recurrence
90	A121639	P	Specified polynomial recurrence
91	A121751	P	Specified polynomial recurrence
92	A121747	P	Specified polynomial recurrence
93	A121584	P	Specified polynomial recurrence
94	A121696	P	Specified polynomial recurrence
95	A121553	P	Specified polynomial recurrence
96	A187252	P	Specified polynomial recurrence
97	A001277	P	Specified polynomial recurrence
98	A001278	P	Specified polynomial recurrence
99	A001279	P	Specified polynomial recurrence
100	A001280	P	Specified polynomial recurrence
101	A045742	P	Specified polynomial recurrence
102	A121752	P	Specified polynomial recurrence
103	A027050	P	Specified polynomial recurrence
104	A027058	P	Specified polynomial recurrence
105	A027059	P	Specified polynomial recurrence
106	A027088	R	Printed recurrence fails, including eventually
107	A027089	P	Specified polynomial recurrence

Four standalone repairs excluded from the count

Entry	Qualification
A059837	The printed recurrence uses a shifted sequence index; a corrected recurrence is proved.
A142243	The row-sum recurrence is shifted by one; the literal version is false and the corrected version is proved.
A024388	The title's block length disagrees with the data and cross-reference formula.
A117436	The diagonal formula requires a one-based column variable; a later formula already implies it.

1.2 Quadratic character sums: all three assertions in A329649

The entry A329649, by Jianing Song (2019), defines

$$B(D)=\sum_{a=1}^{D}\chi_D(a)a^2,\qquad \chi_D(a)=\left(\frac{D}{a}\right),\quad D>0,\quad D\equiv 0,1 \pmod 4.$$

The symbol is the Kronecker symbol. The inspected entry still labels three assertions as conjectures: nonvanishing; a divisibility statement with square, 5 times square, and odd powers of 2 exceptions; and an exact L-value formula for fundamental discriminants. We prove these assertions together.

Theorem 1.2.1. *Suppose $D=df^2$ is not a square, where $d>1$ is the positive fundamental discriminant of its quadratic field. Then*

$$B(D)=f^2B(d)\prod_{p\mid f}(1-p\chi_d(p)),\qquad B(d)=\frac{d^{5/2}}{\pi^2}L(2,\chi_d)>0. \tag{1.1}$$

In particular $B(D)\neq 0$ for every allowed D, including squares. If D is neither a square nor 5 times a square nor 2^t with odd $t>1$, then $4D\mid B(D)$. If $D=5f^2$, then $4f^2\mid B(D)$. For a fundamental discriminant $d>8$, the quotient $B(d)/(4d)$ is a positive integer.

Proof. First let $\chi=\chi_d$ be nonprincipal, primitive, real and even. Character orthogonality and the substitution $a\mapsto d-a$ give

$$\sum_{a=1}^{d}\chi(a)=0,\qquad \sum_{a=1}^{d}a\chi(a)=0.$$

Consequently periodic lifting gives

$$\sum_{a=1}^{Md}\chi(a)a^2=MB(d)\quad (M\geq 1). \tag{1.2}$$

Since $\chi_D(a)=\chi_d(a)\mathbf{1}_{\gcd(a,f)=1}$, inclusion–exclusion over primes dividing f but not d gives

$$\begin{aligned}B(D)&=\sum_{t\mid \mathrm{rad}(f),\ (t,d)=1}\mu(t)\chi_d(t)t^2\sum_{b=1}^{D/t}\chi_d(b)b^2\\&=f^2B(d)\sum_{t\mid\mathrm{rad}(f)}\mu(t)\chi_d(t)t,\end{aligned}$$

which is the product in (1.1).

For completeness, the analytic step uses the classical quadratic Gauss-sum identity

$$\sum_{a=1}^{d}\chi_d(a)e^{2\pi ika/d}=\sqrt{d}\,\chi_d(k)$$

and the absolutely convergent Fourier expansion

$$B_2(x)=x^2-x+\tfrac16=\frac{1}{\pi^2}\sum_{k\geq 1}\frac{\cos(2\pi kx)}{k^2}\quad (0\leq x\leq 1).$$

Summing the latter against $\chi_d(a)$ at $x=a/d$ yields $B(d)/d^2=\sqrt{d}\,L(2,\chi_d)/\pi^2$. The Euler product for $L(2,\chi_d)$ is strictly positive. Every factor $1-p\chi_d(p)$ is nonzero. If D is a square, χ_D is a principal character and the defining sum is positive. This proves nonvanishing.

We give the divisibility argument explicitly. For odd fundamental d, pairing a with $d-a$ gives

$$B(d) = 2 \sum_{1 \le a < d/2} \chi_d(a) a(a-d),$$

because the half-sum of the character is zero. Each $a(a-d)$ is even, so $4 \mid B(d)$. If d has at least two prime factors, the Chinese remainder theorem shows that $B(d) \equiv 0 \pmod{p}$ for every $p \mid d$: the sum over the other prime factors contains a nonprincipal character sum and vanishes. Since d is squarefree, $d \mid B(d)$. If $d = p$ is prime, then $p \equiv 1 \pmod 4$ and Euler's criterion gives

$$B(p) \equiv \sum_{a=1}^{p-1} a^{(p-1)/2+2} \pmod{p}.$$

This vanishes unless $p = 5$, by the elementary power-sum identity over $\mathbb{F}_p$. Thus $4d \mid B(d)$ for odd $d > 5$, whereas $B(5) = 4$.

Next let $d = 4m$, where m is odd, squarefree and $m \equiv 3 \pmod 4$. The primitive character satisfies $\chi_d(a+2m) = -\chi_d(a)$ for odd a, and is even. Grouping the four residues $a, 2m-a, 2m+a, 4m-a$, for odd $0 < a < m$ coprime to m, gives contribution

$$8m\chi_d(a)(m-a).$$

It is divisible by $16m = 4d$ because $m - a$ is even. These groups partition all nonzero character terms.

Finally let $d = 8m$, where m is odd and squarefree. The corresponding shift relation is $\chi_d(a+4m) = -\chi_d(a)$. The groups $a, 4m-a, 4m+a, 8m-a$, for odd $0 < a < 2m$ coprime to m, give

$$B(8m) = 16m \sum_{\substack{1 \le a < 2m \\ a \bmod 2,\ (a,m)=1}} \chi_{8m}(a)(2m-a).$$

For $m > 1$ the sum has $\varphi(m)$ odd summands, an even number, so $32m = 4d$ divides $B(d)$. At $m = 1$ direct evaluation gives $B(8) = 16$.

Now apply (1.1). Fundamental $d > 8$ immediately gives $4D \mid B(D)$. At $d = 5$ the product gives the asserted $4f^2$ divisibility. At $d = 8$, any odd prime dividing f supplies the even factor $1 - p\chi_8(p)$; if no such prime exists then $D = 2^{3+2r}$. These are precisely the exceptional nonsquare powers of 2 in the entry. This completes all three assertions. □

Corollary 1.2.2. *For nonsquare $D = df^2$, the sign of $B(D)$ is $(-1)^s$, where s counts the distinct primes $p \mid f$ with $\chi_d(p) = 1$. In particular the negative terms are explained exactly by the primes splitting in the primitive quadratic character.*

Source correction. The entry's displayed $D = 5$ numerical example omits a factor 4 in one expression: the correct value is $L(2, \chi_5) = 4\pi^2/(25\sqrt{5})$. The general formula in its conjecture (c) has the correct normalization. The classical Gauss-sum and Bernoulli Fourier formulas are imported standard identities; the argument does not claim their novelty.

1.3 A general cancellation theorem for triangular inverses

Let $f(0) = 0$ and let $f(1), f(2), \ldots$ be arbitrary elements of a commutative ring. Define a unit lower triangular array M by $M(0,0) = 1$, $M(n,0) = 0$ for $n > 0$, and

$$M(n,k) = M(n-1,k-1) + f(k)M(n-1,k) \quad (1 \le k \le n).$$

Write $I = M^{-1}$. Here h_j and e_j denote complete and elementary symmetric polynomials, respectively, with $h_0 = e_0 = 1$.

Theorem 1.3.1. *For $1 \le k \le n$, put $m = n - k$. Then*

$$\sum_{i=0}^{m} I(m,i)M(n-1+i,n-1) = h_m(f(m), f(m+1), \dots, f(n-1)). \tag{1.3}$$

At $m = 0$ both sides equal 1. In particular the first-column values at row $n+1$ are $f(n)^n$, with the usual convention $0^0 = 1$.

Proof. The defining recurrence gives $M(r+i,r) = h_i(f(1), \dots, f(r))$. The coefficients of $x\prod_{j=1}^{m-1}(x - f(j))$ form row m of I, since these monic polynomials and the monomials are inverse bases under the recurrence. Thus $I(m, m-j) = (-1)^j e_j(f(1), \dots, f(m-1))$. The left side of (1.3) is the coefficient of z^m in

$$\frac{\prod_{j=1}^{m-1}(1 - f(j)z)}{\prod_{j=1}^{n-1}(1 - f(j)z)} = \prod_{j=m}^{n-1}(1 - f(j)z)^{-1}.$$

This is the asserted complete symmetric polynomial. The $m = 0$ case follows directly without using a product with a negative upper bound. □

1.3.1 A355401: all its displayed q-triangle conjectures

The construction in A355401 takes $f(j) = q^{j-1}$ for $j \ge 1$. Theorem 1.3.1 gives, for $1 \le k \le n$,

$$T_q(n,k) = q^{(n-k)(n-k-1)} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q. \tag{1.4}$$

Indeed, with $m = n-k$, factor q^{m-1} from the k variables in h_m and use $h_m(1, q, \dots, q^{k-1}) = \begin{bmatrix} m+k-1 \\ m \end{bmatrix}_q$. This proves the full array formula, and hence the first-column formula $T_q(n+1,1) = q^{n(n-1)}$. The identity is polynomial in q and extends to all specializations where the displayed definitions are interpreted polynomially.

The inverse conjecture also follows. Set $c_m = q^{m(m-1)}$ and define $g_0 = 1$ and

$$g_m = -\sum_{i=0}^{m-1} \begin{bmatrix} m \\ i \end{bmatrix}_q g_i c_{m-i}.$$

Then the inverse has entries $R_q(n,k) = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q g_{n-k}$ for $k > 0$, with $R_q(n,0) = \mathbf{1}_{n=0}$. To check this, use

$$\begin{bmatrix} n-1 \\ j-1 \end{bmatrix}_q \begin{bmatrix} j-1 \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q \begin{bmatrix} n-k \\ j-k \end{bmatrix}_q.$$

The matrix product reduces to the displayed convolution defining g, which is 0 off the diagonal and 1 on it. This also proves $g_n = R_q(n+1,1)$. These formulas settle the special case $q = 2$ and the general version recorded in the comments of the same entry.

1.3.2 A355282: the q-Stirling counterpart

Here $f(j) = [j]_q = (q^j - 1)/(q-1)$ and the construction is exactly that of Theorem 1.3.1. It follows at once that $T_q(n+1,1) = [n]_q^n$. For $m = n - k$ its full conjectured formula is

$$T_q(n,k) = \frac{1}{(q-1)^m} \sum_{i=0}^{m} (-1)^i \binom{n-1}{i} \begin{bmatrix} n-1-i \\ k-1 \end{bmatrix}_q q^{m(m-i)}. \tag{1.5}$$

To prove it, use the translation identity in k variables

$$h_m(x_1 - 1, \dots, x_k - 1) = \sum_{i=0}^{m} (-1)^i \binom{k+m-1}{i} h_{m-i}(x_1, \dots, x_k).$$

It follows by writing $\prod_j (1-(x_j-1)z)^{-1} = (1+z)^{-k}\prod_j (1-x_j z/(1+z))^{-1}$ and extracting coefficients. Apply it with $(x_1,\dots,x_k) = (q^m,\dots,q^{n-1})$ and use $h_{m-i}(q^m,\dots,q^{n-1}) = q^{m(m-i)}\begin{bmatrix} n-1-i \\ k-1 \end{bmatrix}_q$. The resulting expression is (1.5). Although written as a quotient, it has a removable specialization at $q=1$ by the complete-symmetric-polynomial form. Both the first-column assertion and the full formula in A355282 are therefore proved.

1.3.3 A350557: an explicit inverse

Let T be the array in A350557, and put $u_n = (2n-1)!!$, including $u_0 = 1$. Its definition is

$$T(n,0) = u_n, \qquad T(n,k) = u_n \sum_{i=k}^{n} \binom{i-1}{k-1}\frac{1}{u_i} \quad (k \geq 1).$$

Let L be lower bidiagonal with $L(n,n) = 1$ and $L(n,n-1) = -(2n-1)$, and let $P = 1 \oplus \text{Pascal}$, so that $P(n,k) = \binom{n-1}{k-1}$ for $n,k \geq 1$ and its column 0 is the zeroth unit vector. The defining sum telescopes to $LT = P$. Thus $T^{-1} = P^{-1}L$. Direct multiplication gives

$$(T^{-1})(n,0) = (-1)^n,$$

and, for $1 \leq k \leq n$,

$$(T^{-1})(n,k) = (-1)^{n-k}\left\{\binom{n-1}{k-1} + (2k+1)\binom{n-1}{k}\right\}.$$

The bracket equals $\binom{n-1}{k-1}(2k(n-k)+n)/k$, the defining expression for A350512. If $J = \mathrm{diag}((-1)^n)$, this says $T^{-1} = JA350512J$ and hence $(A350512)^{-1} = JTJ$. This is exactly the signed-inverse assertion of A350557.

1.3.4 A136097: divisibility of a Gaussian central entry

The inverse of the Gaussian array has entries

$$G^{-1}(n,k) = (-1)^{n-k}q^{\binom{n-k}{2}}\begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Multiplying by G proves the formula using the finite q-binomial identity $\sum_{j=0}^{r}(-1)^j q^{\binom{j}{2}}\begin{bmatrix} r \\ j \end{bmatrix}_q = \prod_{j=0}^{r-1}(1-q^j)$, which is zero for $r > 0$. Moreover,

$$\frac{\begin{bmatrix} 2n \\ n \end{bmatrix}_q}{[n+1]_q} = \begin{bmatrix} 2n \\ n \end{bmatrix}_q - q\begin{bmatrix} 2n \\ n-1 \end{bmatrix}_q \in \mathbb{Z}[q].$$

At $n = 0$, interpret the second Gaussian coefficient as 0. Therefore the central inverse entry is divisible, at every integer $q > 1$, by $[n+1]_q q^{n(n-1)/2}$. This proves the general conjecture in A136097. Its $q = 2$ connection to the q-Catalan numbers is already recorded in the entry; the proof uses classical identities and is not a claim of a new q-Catalan theorem.

1.4 A direct row sum and a diagonal indexing correction

1.4.1 A146955

The row polynomial of A146955, for $n \geq 1$, is

$$p_n(x) = (1+x)^n + 2^{n-4}\sum_{m=1}^{n-1}(2^m+2m)(x^m+x^{n-m}).$$

Its row sum is $p_n(1)$, and evaluation gives

$$p_n(1) = 2^n + 2^{n-3}((2^n - 2) + n(n-1)) = \frac{2^n(2^n + 6 - n + n^2)}{8}.$$

The empty sum at $n = 1$ obeys the same calculation. This proves the explicit row-sum conjecture without fitting any terms.

1.4.2 A117436: state the column convention

Write $c_d = d![z^d]\sec(2z)$. The established column exponential generating function of A117436 gives

$$T(k+d, k) = c_d \binom{k+d}{d}.$$

With a *one-based* column variable $x = k + 1$, this is $(-1)^d c_d \binom{-x}{d}$. Odd d have $c_d = 0$; the first nonzero constants are $c_0 = 1, c_2 = 4, c_4 = 80, c_6 = 3904$. Thus the proportionality conjecture has an exact formula under the one-based convention. If its x is instead the zero-based column index used by $T(n, k)$, the literal statement is false: at $d = 2, k = 0$ it compares 4 with a multiple of $\binom{0}{2} = 0$. We record this as an indexing clarification, separately from unambiguous conjecture proofs. The later explicit formula already in the entry implies the corrected statement.

1.5 A127752: the inverse row sums and their Feigenbaum parity

For an integer $c \geq 1$, let

$$A_c(n,k) = \begin{cases} (cn+1)^{-1}, & \lceil n/2 \rceil \leq k \leq n, \\ 0, & \text{otherwise}, \end{cases} \qquad n, k \geq 0,$$

and let $s_c(n)$ be the sum of row n of A_c^{-1}. Then

$$s_c(n) = c(\nu_2(n+1) + 1) + (1-c)\mathbf{1}_{n+1 \text{ is a power of } 2}. \tag{1.6}$$

Indeed $A_c s_c = \mathbf{1}$ gives

$$\sum_{k=\lceil n/2 \rceil}^{n} s_c(k) = cn + 1.$$

Subtract the preceding equation. At even $n = 2m \geq 2$, the lower endpoint is unchanged, so $s_c(2m) = c$. At odd $n = 2m + 1$, the lower endpoint increases by one, so $s_c(2m+1) = c + s_c(m)$. Together with $s_c(0) = 1$, these relations prove (1.6) by induction: iterate through the trailing binary 1 digits, stopping at a positive even number or at 0. At $c = 3$, the correction term in (1.6) is even, and hence

$$s_3(n) \equiv \nu_2(n+1) + 1 \pmod 2.$$

The right side is exactly the one-based sequence A035263$(n+1)$, proving the conjecture in A127752. The same assertion is repeated in A127751; that copy receives no additional count. Related older inverse-matrix assertions with $c = 2$ are omitted from the claimed closures because a proof for their underlying matrix is already linked in A111967.

1.6 Full-length collinear subsets in a discrete cube

The entries A081199, A081200, and A081201 conjecture interpretations as collinear q-subsets of $\{0, 1, \ldots, q-1\}^d$, with $q = 4, 5, 6$, respectively. The following count proves all three.

Theorem 1.6.1. *For all integers $q \geq 2$ and $d \geq 0$, the number of unordered q-element collinear subsets of $\{0, \ldots, q-1\}^d$ is*

$$\frac{(q+2)^d - q^d}{2}.$$

Proof. For $d = 0$ the count is zero. Otherwise a line containing two integer points has a primitive integer direction vector v, and its integer points have integer parameters in that direction. To see the latter claim, choose integers c_j with $\sum_j c_j v_j = 1$; the parameter of an integer displacement tv is then $\sum_j c_j(tv_j)$. Write the parameters of our q points in increasing order as $t_1 < \cdots < t_q$. Their span is at least $q-1$. For any nonzero coordinate v_j, containment in the cube gives

$$q - 1 \geq |v_j|(t_q - t_1) \geq q - 1.$$

Thus every nonzero v_j equals 1 or -1, the span is exactly $q-1$, and the parameters are consecutive. Each subset is therefore an entire q-point line. If the support of v has size k, choose the support in $\binom{d}{k}$ ways, its signs in 2^{k-1} ways after identifying the two orientations, and the constant coordinates in q^{d-k} ways. Each varying coordinate is forced to run from one boundary value to the other. The count is

$$\sum_{k=1}^{d} \binom{d}{k} 2^{k-1} q^{d-k} = \frac{(q+2)^d - q^d}{2}.$$

□

Only the three explicitly conjectured instances are entered in the closure ledger; the general theorem is not used to manufacture more sequence entries.

1.7 A054120: the second McGarvey assertion

Let $R(n,k)$ denote A081577 in triangular coordinates, with out-of-range entries zero. Its bivariate generating function, in coordinates $(u,v) = (k, n-k)$, is

$$\sum_{n \geq 0} \sum_{k=0}^{n} R(n,k) u^k v^{n-k} = \frac{1}{1 - u - v - 2uv}.$$

Consequently R obeys $R(n,k) = R(n-1,k-1) + R(n-1,k) + 2R(n-2,k-1)$ away from the origin. Set $S(n,k) = R(n,k) - R(n-2,k-1)$. Then

$$\sum_{n,k} S(n,k) u^k v^{n-k} = \frac{1 - uv}{1 - u - v - 2uv}.$$

Its first three rows are (1), $(1,1)$, and $(1,3,1)$; thereafter it obeys the same recurrence. Hendel's proof already recorded in A054120 establishes that recurrence for its geometrically defined array T, for $n \geq 3$, with exactly these initial rows. Induction therefore proves $S = T$. This proves the second McGarvey assertion and the later tentative bivariate generating function. The first assertion, explicitly proved in the entry in 2015, is used as a cited premise and is not counted as a new resolution.

1.8 Matrix self-similarity and partitions into powers of a base

Fix an integer $b \geq 2$. Let M be the lower unit triangular matrix with first column 1 and with

$$M^b(n,k) = M(n+1, k+1).$$

These conditions recursively determine M row by row. They are the definitions of A078121 and A078122 at $b = 2, 3$. The prose names of both entries reverse the indices in the first-column condition; we use the row–column convention specified by their examples and programs. Let

$$p_r(N) = [x^N] \prod_{j=0}^{r} (1 - x^{b^j})^{-1}, \qquad p_r(N) = 0 \ (N < 0).$$

Theorem 1.8.1. *For $0 \le k \le n$,*

$$M(n,k) = p_{n-k}(b^n - b^{n-k}).$$

Moreover $\sum_{k=0}^{n} M(n,k)$ counts partitions of b^n into powers of b.

Proof. For nonnegative integer t define $P_0(t) = 1$, and for $r \ge 1$ define $P_r(0) = 0$ and $P_r(t) = p_r(b^r(t-1))$ when $t \ge 1$. Classifying a partition according to the number of parts b^r gives, for $t \ge 1$,

$$P_r(t) = \sum_{u=0}^{t-1} p_{r-1}(b^r u).$$

It follows, also at $t = 0$, that

$$P_r(t+1) - P_r(t) = P_{r-1}(bt+1). \tag{1.7}$$

Induction on r in this identity gives

$$P_r(t+1) = \sum_{j=0}^{r} P_{r-j}(b^j t). \tag{1.8}$$

For $r = 0$ this just states $1 = 1$.

Deleting the first row and first column is an algebra homomorphism S on lower triangular matrices, because their entrywise matrix products involve finite sums. Therefore $S^k(M^t) = M^{b^k t}$. Put $F_r(t) = (M^t)(r, 0)$. This gives

$$(M^t)(n,k) = F_{n-k}(b^k t).$$

Since the first column of M is 1, multiplication by M gives

$$F_r(t+1) = \sum_{j=0}^{r} F_{r-j}(b^j t), \qquad F_r(0) = \mathbf{1}_{r=0}.$$

These are exactly (1.8) and its initial values. Induct first on r, proving the assertion for every t at all smaller r, and then on t for that fixed r. The term $j = 0$ uses the preceding t at the same r, whereas every other term uses a smaller r. This proves $F_r(t) = P_r(t)$ without assuming a matrix formula in advance. At $t = 1$ the desired formula is $F_{n-k}(b^k) = p_{n-k}(b^n - b^{n-k})$.

Finally, classify partitions of b^n by their largest part b^r, $0 \le r \le n$, and remove one such part. The resulting number for that class is $p_r(b^n - b^r)$. Summing over $r = n - k$ gives the row sum. □

This proves the coefficient conjectures in both entries, and the row-sum conjecture in A078122. The 2025 MathOverflow link from A078121 discusses a different proposed closed form along fixed diagonals; it is not used as a premise here.

1.9 A113249: the universal perfect-square assertion

For any integer m, define the sequence by

$$a_0 = -1, \quad a_1 = 4, \quad a_2 = 3m^2 - 16, \quad a_3 = (m^2 - 8)^2,$$

$$a_n = -4a_{n-1} + 4m^2 a_{n-3} + m^4 a_{n-4} \quad (n \geq 4).$$

The conjecture in A113249, repeated in A113250–A113256, asks whether every odd-indexed term is a square for every m. Define integers

$$c_0 = 1, \quad c_1 = 2, \qquad c_{r+2} = 4c_{r+1} - m^2 c_r.$$

Then

$$a_{2r+1} = c_{r+1}^2 \quad (r \geq 0). \tag{1.9}$$

Here is an algebraic proof that includes degenerate parameters. Temporarily take m indeterminate and let α, β be the roots of $z^2 - 4z + m^2 = 0$, so $c_j = (\alpha^j + \beta^j)/2$. Set

$$A_n = \frac{m^{n+1} + (-m)^{n+1} + (-\alpha)^{n+1} + (-\beta)^{n+1}}{4}.$$

Each of the four geometric terms is annihilated by

$$(z^2 - m^2)(z^2 + 4z + m^2) = z^4 + 4z^3 - 4m^2 z - m^4.$$

Direct evaluation of A_0, A_1, A_2, A_3 gives the four initial values above, so $A_n = a_n$. At odd index $2r + 1$ one obtains

$$A_{2r+1} = \frac{2m^{2r+2} + \alpha^{2r+2} + \beta^{2r+2}}{4} = \left(\frac{\alpha^{r+1} + \beta^{r+1}}{2}\right)^2,$$

using $\alpha\beta = m^2$. Both sides of (1.9) are polynomials in m, so the identity specializes to every integer m, including $m = 0$ and $m = \pm 2$. The c_j recurrence establishes their integrality directly.

The entry already attributes the $m = 3$ case to Robert Israel (2017), and subsequent formulas cover some other individual parameters. We claim only a proof here of the remaining universal statement. Its eight copied appearances receive one ledger item, not eight; no priority claim is made for the elementary generalization.

1.10 Arithmetic, digital, and parity conjectures

The results in this section address precisely the assertions stated below. Some of the cited entries contain further conjectures which are not addressed. The first-occurrence assertion in A334109 and A334110 is the same conjecture recorded twice. Likewise the transform assertions in A284005 and A329369 are equivalent formulations. These overlaps should be retained in any count of independent problems. All source pages were inspected on October 20, 2026.

1.10.1 Remainder sums: A049803, A049804, and A330358

For an integer $b \geq 2$ and $n \geq 1$, define

$$S_b(n) = \sum_{j=1}^{\lfloor \log_b n \rfloor} (n \bmod b^j), \qquad L_b(n) = \lfloor \log_b n \rfloor + 1.$$

The three entries concern $b = 3, 4, 5$, respectively. Each records a recurrence conjecture of Petros Hadjicostas from December 2019. Sources: https://oeis.org/A049803, https://oeis.org/A049804, https://oeis.org/A330358.

Theorem. For $n \geq 1$ and $0 \leq r < b$,

$$S_b(bn + r) = bS_b(n) + rL_b(n).$$

Writing $A_b(x) = \sum_{n \geq 1} S_b(n)x^n$, one also has the formal identity

$$A_b(x) = b(1 + x + \cdots + x^{b-1})A_b(x^b) + \frac{x + 2x^2 + \cdots + (b-1)x^{b-1}}{1 - x^b} \sum_{j \geq 1} x^{b^j}.$$

Proof. Put $L = L_b(n)$. The number $bn + r$ has $L + 1$ base-b digits, and

$$(bn + r) \bmod b^j = b(n \bmod b^{j-1}) + r \qquad (1 \leq j \leq L).$$

Sum this equality, using $n \bmod 1 = 0$. For the generating function, sum the recurrence over residue classes and use

$$\sum_{n \geq 1} L_b(n)z^n = \frac{\sum_{j \geq 0} z^{b^j}}{1 - z}.$$

The omitted integers $1 \leq n < b$ have $S_b(n) = 0$, so there is no additional initial polynomial. This proves both assertions. □

1.10.2 Delannoy divisor pairing: A323689 and A324300

Sources: https://oeis.org/A323689 and https://oeis.org/A324300. These entries conjecture restrictions on the positions of odd coefficients in two series involving $i = \sqrt{-1}$.

Lemma. For $u, v \geq 0$, put

$$D(u, v) = \sum_{j \geq 0} 2^j \binom{u}{j} \binom{v}{j}.$$

Then $D(u, v) = D(v, u)$, every $D(u, v)$ is odd, and

$$[y^v] \frac{(y + i)^u}{(1 + iy)^{u+1}} = i^{u-v} D(u, v).$$

Proof. Symmetry and oddness follow from the displayed sum. Summing first over v and using $\sum_{v\ge 0}\binom{v}{j}z^v = z^j/(1-z)^{j+1}$ gives

$$\sum_{v\ge 0} D(u,v)z^v = \frac{(1+z)^u}{(1-z)^{u+1}}.$$

Substitute $z=-iy$ and multiply by i^u. □

Theorem. For A323689, $a(N)$ is odd if and only if N is a square. For A324300, $a(N)$ is odd if and only if $N=k(k+2)$ for some $k\ge 0$. Both series have integer coefficients under their specified constant-term conventions.

Proof. For A323689 and $N\ge 1$, coefficient extraction gives the finite sum

$$a(N) = \sum_{\substack{uv=N\\ u,v\ge 1}} i^{u-v}D(u,v).$$

Pair the terms (u,v) and (v,u) when $u\neq v$. Their sum is

$$\bigl(i^{u-v}+i^{v-u}\bigr)D(u,v),$$

which is either zero or an even integer. There is a diagonal term precisely when N is a square, and that term is odd. The prescribed $a(0)=1$ is also consistent. This reasoning extracts only positive coefficients, so it does not assume convergence of the original constant terms.

For A324300, the summand is multiplied by x^u and y is replaced by x^{u+1}. Consequently the relevant exponent is $N=u+(u+1)v=(u+1)(v+1)-1$, and

$$a(N) = \sum_{\substack{(u+1)(v+1)=N+1\\ u,v\ge 0}} i^{u-v}D(u,v).$$

The same pairing proves that the coefficient is odd exactly when $N+1$ is a square. □

1.10.3 Reversing weighted partitions: A362560

Source: https://oeis.org/A362560. For a finite integer sequence $(y_1,\dots,y_k)$ of positive total n, let $W=\sum_{j=1}^{k} jy_j$. The weighted sum of its reversal is

$$W_{\mathrm{rev}} = \sum_{j=1}^{k}(k+1-j)y_j = (k+1)n - W.$$

Thus $n\mid W$ if and only if $n\mid W_{\mathrm{rev}}$, proving the stated partition conjecture. In fact $\gcd(n,W)=\gcd(n,W_{\mathrm{rev}})$, and no ordering assumption on the sequence is needed. □

1.10.4 Superpolynomial growth: A085765

Sources: https://oeis.org/A085765 and https://oeis.org/A086450. The parity conjecture on these pages was already proved in 2007 and is not counted here. We address the remaining growth conjecture.

Let $b(0)=1$, $b(2n+1)=b(n)$, and $b(2n)=\sum_{j=0}^{n} b(j)$. Then A085765 is $A(N)=\sum_{j=0}^{N} b(j)$.

Theorem. One has

$$\lim_{N\to\infty}\frac{\log A(N)}{\log N} = +\infty.$$

Proof. The function A is positive and increasing. For $N \geq 4$,

$$\begin{aligned} A(N) &\geq \sum_{j=\lceil N/4 \rceil}^{\lfloor N/2 \rfloor} b(2j) = \sum_{j=\lceil N/4 \rceil}^{\lfloor N/2 \rfloor} A(j) \\ &\geq \lfloor N/4 \rfloor A(\lfloor N/4 \rfloor). \end{aligned}$$

In particular, iteration at powers of 4 gives $A(4^r) \geq 4^{r(r-1)/2}$. If $4^r \leq N < 4^{r+1}$, then

$$\frac{\log A(N)}{\log N} \geq \frac{r(r-1)}{2(r+1)} \longrightarrow +\infty.$$

The argument proves a limit, not just an unbounded subsequence. □

1.10.5 A Pascal-parity XOR identity: A100311

Source: https://oeis.org/A100311. The selected 2012 conjecture is an immediate consequence of Pascal's identity. The page also cites Vladimir Shevelev's established polynomial framework, so this proof does not support a claim of worldwide priority.

By definition,

$$a(n) = \sum_{k=0}^{n} \left(\binom{n}{k} \bmod 2 \right) 8^k.$$

The binary digits can be nonzero only at positions $3k$. Multiplying by 8 shifts them by three positions, while bitwise XOR adds their coefficients modulo two. Therefore Pascal's identity yields

$$(8a(n)) \text{ XOR } a(n) = \sum_{k} \left(\binom{n}{k-1} + \binom{n}{k} \bmod 2 \right) 8^k = a(n+1).$$

Together with $a(0) = 1$, this proves the conjecture. The same proof applies with 8 replaced by any 2^s, $s \geq 1$. □

1.10.6 A totient-weighted prime characterization: A156834

Source: https://oeis.org/A156834. The displayed formula is

$$a(n) = \sum_{d|n} \varphi(d) \binom{d + n/d - 2}{d - 1}.$$

Theorem. For $n > 1$, $a(n) = n$ if and only if n is prime; otherwise $a(n) > n$.

Proof. Every binomial coefficient in the sum is at least one, hence $a(n) \geq \sum_{d|n} \varphi(d) = n$. For $d = 1$ and $d = n$ the coefficient equals one. Any proper divisor $1 < d < n$ has $d \geq 2$ and $n/d \geq 2$, making its binomial coefficient strictly greater than one. Since $\varphi(d) > 0$, the inequality is strict for every composite n, and is an equality for every prime. □

1.10.7 Primitive compositions: A156833

Source: https://oeis.org/A156833. Its matrix-product definition gives

$$a(n) = \sum_{r|n} \mu(n/r) \sum_{d|r} d \binom{d + r/d - 2}{d - 1}.$$

Theorem. For $n > 1$, $a(n) = n$ if and only if n is prime; otherwise $a(n) > n$.

Proof. Reverse the finite divisor sums to obtain

$$a(n) = \sum_{d|n} dP_d(n/d), \qquad P_d(m) = \sum_{q|m} \mu(m/q) \binom{d+q-2}{d-1}.$$

The binomial coefficient $\binom{d+m-2}{d-1}$ counts integer tuples $(x_1, \ldots, x_d)$ with $x_1 \geq 1$, all other coordinates nonnegative, and sum m. Sorting these tuples by the gcd of their coordinates shows, by Möbius inversion, that $P_d(m)$ counts the tuples with gcd one. Thus all $P_d(m)$ are nonnegative, $P_d(1) = 1$, and $P_1(m) = 0$ for $m > 1$. The summand $d = n$ is exactly n. If n is prime it is the only nonzero summand. For composite n choose $1 < d < n$ and put $m = n/d \geq 2$; the tuple $(m-1, 1, 0, \ldots, 0)$ has gcd one, so $P_d(m) \geq 1$. This gives the strict inequality for composites. □

1.10.8 Abundancy and prime-factor multiplicity: A008578

Source: https://oeis.org/A008578. We prove Irina Gerasimova's 2013 characterization, recorded there together with partial observations.

Theorem. For a positive integer n,

$$\sigma(n) > n\Omega(n) \quad \Longleftrightarrow \quad n = 1 \text{ or } n \text{ is prime.}$$

Among composites, equality $\sigma(n) = n\Omega(n)$ occurs only at $n = 6$.

Proof. For a composite n, let $r = \omega(n)$ and $k = \Omega(n)$, and order its distinct prime factors as $p_1 < \cdots < p_r$. Since $p_j \geq j + 1$, if $k \geq r + 1$ then

$$\frac{\sigma(n)}{n} < \prod_{j=1}^{r} \frac{p_j}{p_j - 1} \leq \prod_{j=1}^{r} \frac{j+1}{j} = r + 1 \leq k.$$

If $k = r$, the integer is squarefree and $k \geq 2$, so

$$\frac{\sigma(n)}{n} = \prod_{j=1}^{k} (1 + p_j^{-1}) \leq \prod_{j=1}^{k} \frac{j+2}{j+1} = \frac{k+2}{2} \leq k.$$

Equality in the latter case requires $k = 2$, $p_1 = 2$, $p_2 = 3$, giving $n = 6$. The values $n = 1$ and prime n satisfy the asserted strict inequality directly. □

1.10.9 An inverse Gray-code recurrence: A006068

Sources: https://oeis.org/A006068, https://oeis.org/A063946, https://oeis.org/A053645, and https://oeis.org/A053644. Let $G(n) = n \,\mathrm{XOR}\, \lfloor n/2 \rfloor \,\mathrm{XOR}\, \lfloor n/4 \rfloor \,\mathrm{XOR} \cdots$. This is the inverse Gray-code permutation A006068.

Theorem. With the conventions of the cited entries,

$$G(n) = G(\mathrm{A053645}(\mathrm{A063946}(n))) + \mathrm{A053644}(n), \qquad n > 0.$$

Proof. Write $n = 2^L + m$, $0 \le m < 2^L$. The map G is linear for XOR and $G(2^L) = 2^{L+1} - 1$, whence

$$G(n) = 2^L + (2^L - 1 - G(m)).$$

For $L \ge 1$, flipping the second most significant bit and deleting the first bit produces $h = m \mathrm{XOR} 2^{L-1}$. Therefore $G(h) = G(m)$ XOR $(2^L - 1) = 2^L - 1 - G(m)$, which proves the identity. The case $n = 1$ follows directly. □

1.10.10 A negabinary XOR identity: A016789

Sources: https://oeis.org/A016789, https://oeis.org/A005351, https://oeis.org/A005352. Let $B(t)$ be the integer obtained by reading the negabinary digits of a signed integer t as ordinary binary digits.

Theorem. For every $n \ge 0$,

$$n \,\mathrm{XOR}\, B(n+1) \,\mathrm{XOR}\, B(-n-1) = 3n + 2.$$

Proof. Set $X(t) = B(t)$ XOR $B(-t)$ for $t \ge 0$. Negabinary digit extraction gives, for every signed u, $B(2u) = 2B(-u)$ and $B(2u+1) = 2B(-u) + 1$. Consequently

$$X(2u) = 2X(u), \quad X(4u+1) = 4X(u) + 2, \quad X(4u+3) = 4X(u+1) + 2.$$

Together with $X(0) = 0$, these determine X by induction. The function $F(t) = t$ XOR $3t$ obeys the same recurrences. For the last recurrence use

$$u \,\mathrm{XOR}\, (3u+2) = (u+1) \,\mathrm{XOR}\, 3(u+1) :$$

decrementing positive integers t and $3t$ toggles identical bit masks, because $v_2(t) = v_2(3t)$. Hence $X(t) = t$ XOR $3t$. Applying the same decrement observation with $t = n + 1$ yields

$$(t-1) \,\mathrm{XOR}\, X(t) = 3t - 1 = 3n + 2,$$

as required. □

1.10.11 Digital products and excedance sets: A284005 and A329369

Sources: https://oeis.org/A284005 and https://oeis.org/A329369. Write $\mathrm{wt}(n)$ for binary popcount and set

$$A(0) = 1, \qquad A(n) = (1 + \mathrm{wt}(n))A(\lfloor n/2 \rfloor).$$

This is A284005. Equivalently, $A(n) = \prod_{j \ge 0}(1 + \mathrm{wt}(\lfloor n/2^j \rfloor))$, with all sufficiently late factors equal to one.

Theorem (trailing-zero identity). For $m, n \ge 0$,

$$A(2^m(2n+1)) = \sum_{k=0}^{m+1} \binom{m+1}{k} A(2^k n).$$

Proof. One has $A(2^k n) = (\mathrm{wt}(n) + 1)^k A(n)$ and $A(2^m(2n+1)) = (\mathrm{wt}(n) + 2)^{m+1} A(n)$. Both statements also hold at $n = 0$. The ordinary binomial theorem finishes the proof. □

Theorem (leading-bit identity). Let $n = 2^d + m$, $0 \le m < 2^d$, $d \ge 1$, and let ϵ_k be bit k of m. Then

$$A(n) = 2A(m) + \sum_{k=0}^{d-1} A(m + 2^k(1 - \epsilon_k)).$$

Proof. Put $c_k = 1 + \mathrm{wt}(\lfloor m/2^k \rfloor)$ for $0 \le k \le d$. Then $c_d = 1$, $c_k - c_{k+1} = \epsilon_k$. Define

$$P_{-1} = \prod_{j=0}^{d-1} c_j = A(m), \qquad P_k = \prod_{j=0}^{k} (c_j + 1) \prod_{j=k+1}^{d-1} c_j.$$

Now $A(n) = 2P_{d-1}$, and $P_k = A(m + 2^k)$ whenever $\epsilon_k = 0$. The identity

$$(c_{k+1} + 1)P_k - (c_k + 1)P_{k-1} = (1 - \epsilon_k)P_k$$

telescopes to

$$A(n) = (2 + \mathrm{wt}(m))A(m) + \sum_{\epsilon_k = 0} A(m + 2^k),$$

which is the claimed expression. □

Theorem (submask transform). Let $C(n) = \mathrm{A329369}(n)$. Then

$$A(n) = \sum_{k \subseteq_2 n} C(k), \qquad C(n) = \sum_{k \subseteq_2 n} (-1)^{\mathrm{wt}(n) - \mathrm{wt}(k)} A(k),$$

where $k \subseteq_2 n$ means each one-bit of k is a one-bit of n. In addition, for $L, r \ge 0$,

$$\mathrm{A008292}(L + 1, r + 1) = \sum_{\substack{0 \le k < 2^L \\ \mathrm{wt}(k) = r}} C(k).$$

Proof. Choose L with $n < 2^L$ and put $M = L + 1$. Let the allowable excedance set $S \subseteq \{1, \dots, M - 1\}$ contain $M - i$ exactly when bit $i - 1$ of n is one. Write $r = |S|$ and list its complement as $t_1 < \cdots < t_{M-r}$. A permutation with excedance set contained in S must assign restricted row t_j a column at most t_j. Processing these nested allowed sets in order gives exactly

$$r! \prod_{j=1}^{M-r} (t_j - j + 1)$$

permutations, the remaining r rows being unrestricted. Put $b_i = |S \cap \{1, \dots, i\}|$. The last product equals $\prod_{i=1}^{M-1}(1 + b_i) = A(n)$: the factors at $i \in S$ are $2, 3, \dots, r + 1$, whereas the terminal restricted factor at $i = M$ is $r + 1$ and the unrestricted-row factor is $r!$. Every exact excedance subset of S corresponds to one submask of n, giving the first identity. The second is Boolean-lattice Möbius inversion. The equivalent binomial-parity form follows by expanding $(1 + x)^n = \prod_{\text{bit } j \text{ of } n=1}(1 + x^{2^j})$ over $\mathbb{F}_2$. Finally, summing over all masks with r one-bits counts permutations of $L + 1$ elements with r excedances, namely the indicated Eulerian number. This last statement is conjecture 7 on the A329369 page. □

1.10.12 First occurrences in a prime-exponent encoding: A334109 and A334110

Sources: https://oeis.org/A334109 and https://oeis.org/A334110. The same first-occurrence conjecture appears in both entries.

Let p_i denote the ith prime. Define $f(2) = 0$, $f(p) = 1 + f(p-1)$ for odd primes, and extend f completely additively, with $f(1) = 0$. This is A329697: if p is the largest prime factor of a number which is not a power of two, then $f(n(p-1)/p) = f(n) - 1$, exactly its printed stopping recursion. Define

$$Q(e) = \prod_{\text{bit } j \text{ of } e=1} p_{j+1}, \qquad F(e) = f(Q(e)).$$

The displayed additive description of A334109 is

$$H(N) = \sum_i 2^{i-1} F(e_i), \qquad N = \prod_i p_i^{e_i}.$$

The sequence A334110 is $C(k) = Q(k)^2$.

Theorem. For every $k \geq 0$, the least positive integer N with $H(N) = k$ is $C(k)$.

Proof. For an integer $v \geq 0$, put $h(v) = \sum_{\text{bit } j \text{ of } v=1}(j+1)$. Binary carrying never increases this weight, since two bits of weight $j+1$ are replaced by one of weight $j+2 \leq 2(j+1)$. Thus h is subadditive for ordinary addition, and $h(v) \leq v$.

We have $f(n) \leq \log_2 n$. By complete additivity it suffices to check primes inductively: for odd p,

$$f(p) = 1 + f((p-1)/2) \leq \log_2(p-1) < \log_2 p,$$

while $p = 2$ is immediate. Bertrand's postulate implies $p_{j+1} < 2^{j+1}$, so $f(p_{j+1}) \leq j$ for $j \geq 1$. Consequently

$$\begin{aligned} h(F(e)) &\leq \sum_{\text{bit } j \text{ of } e=1} h(f(p_{j+1})) \\ &\leq \sum_{\substack{j \geq 1 \\ \text{bit } j \text{ of } e=1}} j \leq \sum_{\substack{j \geq 1 \\ \text{bit } j \text{ of } e=1}} 2^{j-1} = \lfloor e/2 \rfloor. \end{aligned}$$

The bit $j = 0$ contributes zero because $f(2) = 0$.

The function C is submultiplicative under addition: $C(a+b) \leq C(a)C(b)$. Indeed, a binary carry replaces p_i^4 by p_{i+1}^2, which does not increase the product since $p_{i+1} < 2p_i \leq p_i^2$. Bertrand also gives, for $v \geq 0$,

$$C(2^{i-1}v) = \prod_{\text{bit } j \text{ of } v=1} p_{i+j}^2 \leq p_i^{2h(v)},$$

because $p_{i+j} \leq p_i^{j+1}$ (use $p_{i+j} < 2^j p_i$ for $j > 0$). It follows that

$$C(H(N)) \leq \prod_i C(2^{i-1}F(e_i)) \leq \prod_i p_i^{2h(F(e_i))} \leq \prod_i p_i^{e_i} = N.$$

Conversely $F(2) = f(3) = 1$, so $H(C(k)) = k$ for every $k \geq 0$. The lower bound is therefore attained, proving minimality, including $k = 0$ where $C(0) = 1$. □

1.10.13 Row divisibility for functional digraphs: A216971

Source: https://oeis.org/A216971. Jon Perry's 2012 conjecture states that n divides every entry $T(n,k)$, where $T(n,k)$ counts functions on n labelled elements with exactly k nonrecurrent vertices whose images are recurrent. Here $n \geq 1$ and $0 \leq k \leq n-1$.

Theorem. One has $n \mid T(n,k)$ throughout the stated range.

Proof. Let $U(x)$ be the rooted-tree series, characterized by $U = xe^U$. The entry's bivariate exponential generating function is

$$\sum_{n\geq 1}\sum_{k=0}^{n-1} T(n,k)\frac{x^n}{n!}y^k = \frac{1}{1-xe^{yU(x)}} - 1 = \sum_{r\geq 1} x^r e^{ryU(x)}.$$

This identity also follows by viewing a function as a permutation of its r cyclic vertices, with rooted trees attached to those vertices; y marks each attached tree's root. At $k = 0$ the coefficient is $T(n,0) = n!$. For $k \geq 1$, put $m = n - r$. Lagrange inversion gives

$$[x^m]U(x)^k = \frac{k}{m}[u^{m-k}]e^{mu} = \frac{k\,m^{m-k-1}}{(m-k)!} \qquad (m \geq k).$$

Thus $T(n,k)$ is a sum, over $1 \leq r \leq n-k$, of terms

$$\frac{n!\,r^k m^{m-k-1}}{(k-1)!(m-k)!}, \qquad m = n - r.$$

For $m > k$, dividing this term by n leaves an integer: its factorial factor is

$$\frac{(n-1)!}{(k-1)!(m-k)!} = \binom{n-1}{k-1}\frac{(n-k)!}{(m-k)!} \in \mathbb{Z},$$

and the remaining powers have nonnegative exponents. At the endpoint $m = k$, the coefficient $[x^k]U^k$ equals one, so the divided term is $(n-1)!r^k/k!$, again an integer because $k \leq n-1$. Every summand is therefore divisible by n. □

1.10.14 Records of zeros in XOR triangles: A334592

Source: https://oeis.org/A334592. For a binary word of length L, form the triangle by replacing each successive pair by its XOR until a single entry remains. The entry counts zeros when the first row is the ordinary binary expansion of a positive integer n.

Theorem. The strict record indices of A334592 are exactly $n = 2^{L-1}$, $L \geq 1$; the record value is $L(L-1)/2$.

Proof. Every such triangle with a nonzero first row of length L contains at least L ones. This follows by induction on L. If its second row is nonzero, the triangle beneath the first row has at least $L-1$ ones, and the first row supplies another. If the second row is zero, the first row is constant; since it is nonzero, it already contains L ones. The assertion at $L = 1$ is immediate.

There are $L(L+1)/2$ positions in total, so at most $L(L-1)/2$ zeros. The word $10\cdots0$, representing 2^{L-1}, has one one in every row and attains this bound. It is the first integer with L binary digits. For $L \geq 2$ the bound exceeds $(L-1)(L-2)/2$, the maximum for shorter words; no later word of the same length can exceed it. This proves the assertion, with $n = 1$ as the initial record. □

1.10.15 Uniqueness for the XOR of consecutive squares: A224515 and A224511

Sources: https://oeis.org/A224515 and https://oeis.org/A224511. The former separately conjectures the existence and uniqueness of $k \geq 0$ with k^2 XOR $(k+1)^2 = (2n+1)^2$. Its existence assertion is explicitly acknowledged as proved in 2026; see Tom Adamczewski, *OEIS Open*, Appendix A, https://arxiv.org/abs/2608.11941, and the AlphaProof Nexus file linked from the entry. We count only the still separately labelled uniqueness assertion. The resulting permutation assertion in A224511 is an equivalent consequence and should not be counted as another independent problem. The existence argument is included for completeness and carries no priority claim.

Theorem. For every positive integer $S \equiv 1 \pmod 4$ there is exactly one $k \geq 0$ such that $k^2 \text{ XOR } (k+1)^2 = S$. In particular A224515's least index is its only index, and A224511 is a permutation of the positive odd integers.

Proof of uniqueness. Fix an odd $S = 2M + 1$. Define, for every signed integer k,

$$F(k) = k + (k^2 \text{ AND } S).$$

Only nonnegative squares enter the bitwise operation. The identity $u \text{ XOR } S = u + S - 2(u \text{ AND } S)$ gives

$$k^2 \text{ XOR } (k+1)^2 = S \quad \Longleftrightarrow \quad F(k) = M.$$

Suppose $k \neq l$ have the same parity, and put $t = v_2(k - l) \geq 1$. Since $k + l$ is even, $k^2 \equiv l^2 \pmod{2^{t+1}}$. Bitwise AND with a fixed mask preserves congruences modulo powers of two, so

$$F(k) - F(l) \equiv k - l \not\equiv 0 \pmod{2^{t+1}}.$$

Thus F is injective on each parity class of signed integers. If $F(l) = M$, then $-l - 1$ is also a root, because it simply interchanges the two squares in the XOR equation. Two different nonnegative roots of the same parity contradict injectivity directly. If their parities differ, reflect one: k and $-l - 1$ are distinct roots of the same parity, also impossible. This proves uniqueness for every odd target S.

Existence and the permutation consequence. Now assume $S \equiv 1 \pmod 4$, so M is even. Choose h with $2^h > S$. There is an even k with $0 \leq k < 2^h$ and $F(k) \equiv M \pmod{2^h}$: start with $k = 0$ modulo two, and successively choose whether to add 2^i at stage $i \geq 1$. Indeed,

$$(k + 2^i)^2 \equiv k^2 \pmod{2^{i+1}}, \qquad F(k + 2^i) \equiv F(k) + 2^i \pmod{2^{i+1}},$$

so exactly one choice matches the next target bit. Since $0 \leq k^2 \text{ AND } S \leq S < 2^h$, this solution satisfies $F(k) = M$ or $F(k) = M + 2^h$. In the latter case put $q = k - 2^h < 0$. The squares q^2 and k^2 agree modulo 2^h, and the mask S has no bits beyond this range, so $F(q) = F(k) - 2^h = M$. Reflecting gives the nonnegative solution $-q - 1$.

Every odd square is 1 modulo eight and hence qualifies for this existence theorem. Conversely, the XOR of consecutive squares is always odd. Existence and uniqueness therefore say that each positive odd square root is listed exactly once by A224511. □

1.11 Arithmetic-progression floors and symmetric-function recurrences

All sequence citations in this section refer to the primary OEIS entry at `https://oeis.org/A`*number*, inspected on 20 October 2026. The words "conjecture" and "empirical" were still present in the cited statements. No priority beyond these inspected entries is asserted.

1.11.1 One theorem for eight periodic symmetric-function entries

Let $c_1, \dots, c_L$ be nonzero numbers and repeat this block periodically. Write $A_n = e_{n-1}(c_1, \dots, c_n)$ for $n \ge 1$ and put $A_0 = 0$. Set $P = \prod_{j=1}^{L} c_j$, $S = \sum_{j=1}^{L} c_j^{-1}$, and $p_r = \prod_{j=1}^{r} c_j$, $s_r = \sum_{j=1}^{r} c_j^{-1}$, with $p_0 = 1, s_0 = 0$. Then, for $n = qL + r$, $0 \le r < L$,

$$A_{qL+r} = P^q p_r (qS + s_r), \qquad \sum_{n \ge 0} A_n x^n = \sum_{r=0}^{L-1} p_r x^r \left(\frac{s_r}{1 - Px^L} + \frac{SPx^L}{(1 - Px^L)^2} \right).$$

Consequently $A_n = 2PA_{n-L} - P^2 A_{n-2L}$ for $n \ge 2L$.

Proof. Each product in e_{n-1} omits exactly one factor, so $e_{n-1}(c_1, \dots, c_n) = (\prod c_j) \sum c_j^{-1}$. Split the product and reciprocal sum into full periods and a prefix. Summing $\sum q z^q = z/(1-z)^2$ proves the generating function. Its numerator has degree at most $2L - 1$, which proves the recurrence. □

This proves the conjectured recurrence and generating function in each of the following eight distinct entries. The first has an empirical generating function due to Colin Barker (2012) and a conjectured closed form due to Bruno Berselli; the other conjectures are due to R. J. Mathar (2013). In the last column $N(x)$ means that the generating function is $xN(x)/(1 - Px^L)^2$.

Entry	Block	P	$N(x)$
A203150	$(1, 2)$	2	$1 + 3x + x^2$
A203164	$(4, 1, 2, 3)$	24	$1 + 5x + 14x^2 + 50x^3 + 176x^4 + 80x^5 + 64x^6$
A203165	$(3, 4, 1, 2)$	24	$1 + 7x + 19x^2 + 50x^3 + 126x^4 + 432x^5 + 144x^6$
A203166	$(1, 2, 3, 4, 5)$	120	$1 + 3x + 11x^2 + 50x^3 + 274x^4 + 154x^5 + 188x^6 + 324x^7 + 576x^8$
A203167	$(2, 2, 1)$	4	$1 + 4x + 8x^2 + 12x^3 + 16x^4$
A203231	$(3, 1)$	3	$1 + 4x + 9x^2$
A203234	$(1, 1, 1, 2)$	2	$1 + 2x + 3x^2 + 7x^3 + 5x^4 + 3x^5 + x^6$
A203235	$(1, 1, 1, 3)$	3	$1 + 2x + 3x^2 + 10x^3 + 7x^4 + 4x^5 + x^6$

The table follows by finite substitution into the displayed formula, not by fitting initial terms. For A203150 the same theorem gives $A_{2q} = 3q2^{q-1}$ and $A_{2q+1} = (3q+2)2^{q-1}$; substituting even and odd n into Berselli's printed expression with $r = \sqrt{2}$ gives exactly these two formulas, so that conjecture is proved as well. Related entries A203160–A203163 already state their generating functions as facts, and A203233 explicitly records Robert Israel's 2017 verification; these were excluded from the count.

1.11.2 Two nonperiodic symmetric-function entries

A203157. For the triangular numbers $T_j = j(j+1)/2$,

$$A_n = e_{n-1}(T_1, \dots, T_n) = \frac{n!(n+1)!}{2^n} \sum_{j=1}^{n} \frac{2}{j(j+1)} = \frac{n(n!)^2}{2^{n-1}}.$$

The reciprocal sum telescopes to $2n/(n+1)$. Thus $2(n-1)A_n = n^3 A_{n-1}$ for $n \ge 2$, proving the recurrence conjectured by Mathar on 1 October 2016.

A203159. The same omitted-factor identity gives $A_n = 2^{n-1} n! H_n$ for the list $2, 4, \ldots, 2n$. Since $A_n - 2nA_{n-1} = 2^{n-1}(n-1)!$, subtracting $2(n-1)$ times the same equation at $n-1$ gives

$$A_n - (4n-2)A_{n-1} + 4(n-1)^2 A_{n-2} = 0 \quad (n \geq 3).$$

This proves Mathar's 1 October 2016 conjecture.

1.11.3 Three explicit floor recurrences

For a finite list with power sums p_j, Newton's identities give $e_2 = (p_1^2 - p_2)/2$ and $e_3 = (p_1^3 - 3p_1p_2 + 2p_3)/6$. Apply these identities to the indicated arithmetic progressions.

A024220. For the first $n+2$ positive integers congruent to 1 (mod 3),

$$\frac{e_3}{e_1} = \frac{n(n+1)(9n^2+21n-2)}{24} = Q(n).$$

The fractional part of $Q(n)$ is $1/3$ if $n \equiv 1 \pmod 3$, and zero otherwise. This is checked in the three residue classes; alternatively the numerator modulo 24 has these values. Thus $A_n = Q(n) - \frac{1}{3}\mathbf{1}_{n\equiv 1\ (3)}$ for every $n \geq 1$. Summation yields

$$\sum_{n\geq 1} A_n x^n = \frac{x(2+11x+7x^2+8x^3-x^4)}{(1+x+x^2)(1-x)^5}.$$

The denominator proves Mathar's conjectured signature $(4, -6, 5, -5, 6, -4, 1)$, valid for $n \geq 8$.

A024399. For the first $n+2$ positive integers congruent to 2 (mod 3),

$$\frac{e_3}{e_1} = \frac{n(n+1)(9n^2+33n+22)}{24} = Q(n).$$

Its fractional part has the same three-class rule as A024220. Hence

$$\sum_{n\geq 1} A_n x^n = \frac{x(5+11x+7x^2+5x^3-x^4)}{(1+x+x^2)(1-x)^5}.$$

This proves the other conjectured recurrence with the same signature.

A024385. For the first $n+1$ positive integers congruent to 1 (mod 4),

$$\frac{e_2}{e_1} = \frac{n(12n^2+8n-5)}{6(2n+1)} = n^2 + \frac{n-3}{6} + \frac{1}{2(2n+1)}.$$

For $n \geq 2$ the positive last term is less than $1/6$, while the preceding polynomial has fractional part in $\frac{1}{6}\mathbb{Z}$. The case $n = 1$ is checked directly. Therefore $A_n = n^2 + \lfloor (n-3)/6 \rfloor$ for all $n \geq 1$. Its generating function is

$$\frac{x^2(3+3x+x^2+2x^3+2x^4+2x^5-x^6)}{(1-x)^2(1-x^6)}.$$

This proves the conjectured signature $(2, -1, 0, 0, 0, 1, -2, 1)$, valid for $n \geq 9$.

1.11.4 Two empirical generating functions proved by finite corrections

A024390. Use the first $n+3$ numbers congruent to 1 (mod 4) and take e_4/e_3. Newton's identities through degree four give

$$\frac{e_4}{e_3} = Q(n) + R(n), \quad Q(n) = \frac{6n^2+9n-13}{12}, \quad R(n) = \frac{692n^2+3128n+2925}{60(2n+5)(4n^2+16n+9)}.$$

The fractional parts of $Q(n)$ for $n \equiv 1,2,3,0 \pmod 4$ are $1/6, 5/12, 2/3, 11/12$. For $n \geq 16$, $0 < R(n) < 1/12$: after clearing positive denominators the upper bound is equivalent to $20n^3 - 216n^2 - 1319n - 1350 > 0$, whose expansion at $n = 16+u$ is $20u^3+744u^2+7129u+4170 > 0$. Exact substitution at $1 \leq n \leq 15$ gives

$$A_n = \lfloor Q(n) \rfloor + \mathbf{1}_{n \in \{4,8,12\}}.$$

Summing this identity proves Colin Barker's empirical formula (2014):

$$\sum_{n \geq 1} A_n x^n = \frac{x^2(x^{16} - 2x^{15} + x^{14} + x^4 - 2x^2 - x - 2)}{(x-1)^3(x+1)(x^2+1)}.$$

A024403. For the first $n+3$ numbers congruent to 2 (mod 3), the ratio e_4/e_3 is

$$Q(n) + R(n), \quad Q(n) = \frac{3n^2+7n-4}{8}, \quad R(n) = \frac{153n^2+1002n+1600}{5(3n+10)(9n^2+51n+64)}.$$

All fractional parts of $Q(n)$ lie in $\{0, 1/4, 1/2, 3/4\}$. For $n \geq 3$, $0 < R(n) < 1/4$ follows by clearing denominators: $135n^3+603n^2-498n-3200 > 0$ expands at $n = 3+u$ as $135u^3+1818u^2+6765u+4378$. The two initial cases give $A_n = \lfloor Q(n) \rfloor + \mathbf{1}_{n \in \{1,2\}}$. Summation proves Barker's empirical generating function

$$\frac{x(x^2-x+1)(x^8-x^7-x^5+x^4-x^3-x-1)}{(x-1)^3(x^2+1)(x^4+1)}.$$

1.11.5 A024389: exact disproof of an empirical generating function

For $S = \{1, 5, \ldots, 4n+9\}$, Newton's identities give

$$\frac{e_4(S)}{e_2(S)} = \frac{n(n+1)(240n^4+1920n^3+4760n^2+3072n-1217)}{60(12n^2+56n+59)}.$$

At $n = 39$ this is $17588398646/20495$, with floor 858179. The coefficient of x^{39} in Barker's displayed empirical function is 858180. The latter assertion is an exact rational coefficient calculation: if $G = N/D$ is that function, generate $c_0, c_1, \ldots, c_{39}$ using $D_0 c_n = N_n - \sum_{j=1}^{\deg D} D_j c_{n-j}$. The accompanying verification script records the complete polynomial N, D and reproduces the contradiction with integer arithmetic. Thus the empirical generating function, as printed, is false.

1.11.6 A024388: index repair, counted separately

The entry's title specifies the first $n+2$ numbers $1, 5, 9, \ldots$, but its data and its cross-reference formula use the first $n+3$. At $n = 1$ the literal title gives $\lfloor 45/59 \rfloor = 0$; the displayed first term is 3. We do not silently conflate these. For the data-compatible interpretation one obtains

$$\frac{e_3}{e_2} = Q(n) + R(n), \quad Q(n) = \frac{18n^2+51n+11}{27}, \quad R(n) = \frac{2(118n+283)}{27(12n^2+56n+59)}.$$

The fractional part of $Q(n)$ is periodic modulo 9, with numerators $26, 23, 2, 17, 14, 20, 8, 5, 11$ over 27 in classes $1, \dots, 9$. For $n \geq 18$, $R(n) < 1/27$ because $4n^2 - 60n - 169 = 4(n-18)^2 + 84(n-18) + 47 > 0$. Checking $1 \leq n \leq 17$ exactly yields

$$A_n = \lfloor Q(n) \rfloor + \mathbf{1}_{n \in \{1,2,10\}}.$$

This identity proves the printed empirical generating function for the displayed data, once the title's index is repaired. This is listed as an index-repair result, separate from unqualified confirmations.

1.11.7 A173693: a ceiling preserves a finite recurrence

The source specifies $b_0, b_1, b_2, b_3, b_4 = 0, 0, 0, 0, 1$ and $b_n = b_{n-1} + b_{n-2} - b_{n-3} + b_{n-5}$, with $A_n = \lceil b_n/2 \rceil$. The parity sequence has the repeating block

$$0000110010011111011100010101101.$$

For a finite certificate, apply the recurrence modulo 2 through $n = 35$: $(b_{31}, \dots, b_{35}) \bmod 2 = (0, 0, 0, 0, 1)$, exactly the initial state. Determinism then proves period 31 for all subsequent terms. Writing $r_n = b_n \bmod 2$ gives $2A_n = b_n + r_n$. If E denotes backward shift, $1 - E - E^2 + E^3 - E^5$ annihilates b, and $1 - E^{31}$ annihilates r. Therefore their product annihilates A, proving Mathar's conjecture for $n \geq 36$:

$$\begin{aligned} A_n = {} & A_{n-1} + A_{n-2} - A_{n-3} + A_{n-5} + A_{n-31} \\ & - A_{n-32} - A_{n-33} + A_{n-34} - A_{n-36}. \end{aligned}$$

1.11.8 A227012: the harmonic mean between consecutive cubes

Let $M_n = (3n^2 - 3n + 1)/(H_{n^3} - H_{(n-1)^3})$. Then

$$\lfloor M_n \rfloor = \frac{2n^3 - 3n^2 + n + 2}{2} + \left\lfloor \frac{\max(0, n-3)}{4} \right\rfloor \qquad (n \geq 1).$$

This proves Ramaharo's 2018 conjecture and, by summation, the entry's conjectured ordinary generating function and seven-term recurrence. Here is an exact inequality certificate, not an asymptotic argument. Put $a = (n-1)^3, b = n^3, \Delta = b - a$ and $Q = (4n^3 - 6n^2 + 3n + 1)/4$. We prove $Q < M_n < Q + 1/4$ for $n \geq 3$.

The standard signed remainder bound for the digamma expansion gives

$$H_m = \log m + \gamma + \frac{1}{2m} - \frac{1}{12m^2} + \rho_m, \qquad 0 < \rho_m < \frac{1}{120m^4}.$$

See NIST DLMF, 5.11.2 and 5.11(ii), `https://dlmf.nist.gov/5.11`. Let $S = \sum_{j=1}^{4} (jn^j)^{-1}$ and $C = (1/b - 1/a)/2 - (1/b^2 - 1/a^2)/12$. The elementary positive logarithm series and its geometric remainder imply

$$L = 3S + C - \frac{1}{120a^4} < H_b - H_a < U = 3\left(S + \frac{1}{5n^4(n-1)}\right) + C + \frac{1}{120b^4}.$$

For a reproducible polynomial positivity certificate define

$$P_-(n) = 480n^{12}(n-1)^6(\Delta - QU), \quad P_+(n) = 480n^6(n-1)^{12}((Q + 1/4)L - \Delta).$$

Both are polynomials of degree 17. The ascending coefficient lists of $P_-(3+u)$ and $P_+(3+u)$ are given below; every coefficient is positive.

j	$[u^j]P_-(3+u)$	$[u^j]P_+(3+u)$
0	3559568768	21810535
1	22548059346	287558715
2	67082168268	1446297165
3	124564097140	4118075599
4	161774552436	7740002643
5	156012626130	10389087405
6	115733015648	10424077105
7	67475843205	8039537295
8	31310918742	4846668528
9	11632216336	2304171826
10	3460298664	865875390
11	819631074	256192530
12	152703988	59035670
13	21908256	10384770
14	2337384	1347990
15	174738	121770
16	8172	6840
17	180	180

Hence $\Delta/(Q+1/4) < L < H_b - H_a < U < \Delta/Q$ and the claimed bounds for M_n follow. Since $Q \in \frac{1}{4}\mathbb{Z}$, they imply $\lfloor M_n \rfloor = \lfloor Q \rfloor$ for $n \geq 3$. The cases $n = 1, 2$ give $1, 4$ directly. Rearrangement of $\lfloor Q \rfloor$ gives the stated conjectured formula. Explicitly the generating function is

$$\sum_{n\geq 1} \lfloor M_n \rfloor x^n = \frac{x(1 + x + 7x^2 + 6x^3 + 5x^4 + 5x^5 - 2x^7 + x^8)}{(1-x)^3(1-x^4)}.$$

Only the cube-block conjectures are proved here; the much broader arbitrary-polynomial assertion in the entry is not inferred from this case.

1.11.9 Five further polynomial-block harmonic means

For each of the following five polynomials, put

$$A_n = \left\lfloor \frac{g(n) - g(n-1)}{H_{g(n)} - H_{g(n-1)}} \right\rfloor, \qquad n \geq 1.$$

The entries A227013–A227017 conjecture that these are linear recurrence sequences and supply empirical generating functions. We prove their generating functions exactly, with the indexing qualifications stated below. A particularly compact form of the answers is

$$A_n = \lfloor Q(n) \rfloor + \mathbf{1}_{n \in C}. \tag{1.10}$$

Here are all the polynomials and exceptional sets:

$$\text{A227013}: \quad g(n) = n^4, \quad C = \{1, 2\}, \qquad Q(n) = n^4 - 2n^3 + \frac{5}{3}n^2 - \frac{2}{3}n + \frac{1}{2};$$

$$\text{A227014}: \quad g(n) = n^5, \quad C = \{1, 2\},$$
$$Q(n) = n^5 - \frac{5}{2}n^4 + \frac{35}{12}n^3 - \frac{15}{8}n^2 + \frac{85}{144}n + \frac{125}{288};$$

$$\text{A227015}: \quad g(n) = n^3 + n^2 + n + 1, \quad C = \{3\},$$
$$Q(n) = n^3 - \frac{1}{2}n^2 + \frac{3}{4}n + 1;$$

$$\text{A227016}: \quad g(n) = \frac{n(n+1)(n+2)}{6}, \quad C = \{1\},$$
$$Q(n) = \frac{1}{6}n^3 + \frac{1}{4}n^2 - \frac{1}{24}n + \frac{7}{16};$$

$$\text{A227017}: \quad g(n) = \frac{n(3n-1)}{2}, \quad C = \{1\}, \qquad Q(n) = \frac{3}{2}n^2 - 2n + 1.$$

Uniform rigorous bounds. Let $a = g(n-1)$, $b = g(n)$, $\Delta = b - a$, and

$$C_0 = \frac{1}{2}\left(\frac{1}{b} - \frac{1}{a}\right) - \frac{1}{12}\left(\frac{1}{b^2} - \frac{1}{a^2}\right).$$

We only use this notation when $a > 0$, handling the initial cases separately. Choose a rational lower approximation S to $\log(b/a)$ and a positive rational tail bound T so that $S < \log(b/a) < S + T$. For $g(n) = n^d$ use

$$S = d\sum_{j=1}^{K} \frac{1}{jn^j}, \qquad T = \frac{d}{(K+1)n^K(n-1)}.$$

This follows by expanding $-d\log(1-1/n)$ and bounding each remaining denominator j below by $K+1$. For the other three polynomials put $t = \Delta/(a+b)$ and use

$$S = 2\sum_{j=0}^{K-1} \frac{t^{2j+1}}{2j+1}, \qquad T = \frac{2t^{2K+1}}{(2K+1)(1-t^2)}.$$

Here $0 < t < 1$; this is the positive series for $2\operatorname{arctanh} t = \log(b/a)$ with the same geometric-tail argument. The signed harmonic-number bound used in A227012 now gives

$$L = S + C_0 - \frac{1}{120a^4} < H_b - H_a < U = S + T + C_0 + \frac{1}{120b^4}.$$

For each row in the next table, the exact polynomial certificates below prove

$$E_- := \Delta - (Q+\ell)U > 0, \qquad E_+ := (Q+h)L - \Delta > 0 \qquad (n \ge N).$$

The polynomials $a, b, \Delta, Q + \ell$ have positive constant terms and nonnegative coefficients after substituting $n = N + u$. Thus they are positive throughout the stated range. It follows in particular that $L > 0$, and division by positive quantities gives

$$Q(n) + \ell < \frac{\Delta}{H_b - H_a} < Q(n) + h.$$

The last column gives the period of the fractional part of Q.

Entry	N	K	ℓ	h	fractional period J
A227013	4	6	0	1/6	3
A227014	30	7	−1/288	1/288	144
A227015	4	3	0	1/4	4
A227016	3	3	−1/16	1/16	8
A227017	2	3	0	1/2	2

For A227013 the fractional parts belong to $\{1/2, 5/6\}$. For A227014 they are odd multiples of $1/288$. For A227015 they belong to $\{0, 1/4, 1/2, 3/4\}$. For A227016 they are odd multiples of $1/16$. For A227017 they belong to $\{0, 1/2\}$. In each case $\lfloor Q \rfloor \le Q + \ell$ and $Q + h \le \lfloor Q \rfloor + 1$. The strict inequalities for the harmonic mean therefore place it strictly between these consecutive integers, including when an endpoint of the bound is an integer. This proves $A_n = \lfloor Q(n) \rfloor$ for $n \ge N$. The stated periods can be checked globally by expanding $Q(n+J) - Q(n)$ in the integer-valued basis $\binom{n}{0}, \binom{n}{1}, \ldots$; all its coefficients are integers. This is a polynomial identity, not an extrapolation from observed terms.

Explicit polynomial positivity certificates. For each entry let $P_- = D_-E_-$ and $P_+ = D_+E_+$, using the following positive denominators. In the A227015 row abbreviate $B = 2n^3 - n^2 + 3n + 1$; in the A227017 row put $V = 3n^2 - 4n + 2$.

$$\begin{aligned}
\text{A227013}: &\quad D_- = 5040n^{16}(n-1)^8, \quad D_+ = 360n^8(n-1)^{16};\\
\text{A227014}: &\quad D_- = 120960n^{20}(n-1)^{10}, \quad D_+ = 120960n^{10}(n-1)^{20};\\
\text{A227015}: &\quad D_- = 3360n^2(n+1)^4(n^2+1)^4(n^2-2n+2)^2B^5,\\
&\quad D_+ = 480n^4(n+1)^2(n^2+1)^2(n^2-2n+2)^4B^5;\\
\text{A227016}: &\quad D_- = 560n^4(n-1)^2(n+1)^4(n+2)^4(2n+1)^5,\\
&\quad D_+ = 40n^4(n-1)^4(n+1)^4(n+2)^2(2n+1)^5;\\
\text{A227017}: &\quad D_- = 105n^4(n-1)^2(3n-4)^2(3n-1)^4V^4,\\
&\quad D_+ = 15n^2(n-1)^4(3n-4)^4(3n-1)^2V^5.
\end{aligned}$$

All factors are positive on the applicable ranges: in particular $n^2 - 2n + 2 = (n-1)^2 + 1$, $B = n^2(2n-1) + 3n + 1$, and $V = 3(n-2/3)^2 + 2/3$ are positive there. Exact expansion of $P_\pm(N+u)$ gives the following certificate. Each "minimum" is over every coefficient, including the constant coefficient; in particular no omitted coefficient is zero or negative.

Entry	$\deg P_-$	minimum coefficient	$\deg P_+$	minimum coefficient
A227013	23	1568	23	128
A227014	29	2100	29	2100
A227015	32	40320	32	5760
A227016	18	3360	18	240
A227017	17	2480058	21	2657205

The exact integer coefficient vectors, as well as the original numerator and denominator polynomials, are supplied in `oeis_more_means_certificates.json`. The companion `oeis_more_means_verification.py` reconstructs $S, T, L, U, E_\pm$ from the displayed definitions, checks the polynomial identities and all coefficients with exact rational arithmetic, and checks denominator positivity. Thus the table is a finite certificate for every real $n \geq N$, rather than a finite test of the sequence.

Finite initial cases. For $n < N$, exact evaluation gives, respectively,

$$(1, 6, 40), \qquad (2, 8, 26), \qquad (1, 2), \qquad (1)$$

for A227013, A227015, A227016, A227017. For A227014 the initial values through $n = 29$ are

$$\begin{aligned}
&(1, 10, 104, 543, 1883, 5102, 11717, 23906, 44626, 77735,\\
&\ 128110, 201769, 305989, 449428, 642243, 896212, 1224852,\\
&\ 1643541, 2169636, 2822595, 3624095, 4598154, 5771249,\\
&\ 7172438, 8833478, 10788947, 13076362, 15736301, 18812521).
\end{aligned}$$

These are certified by exact rational harmonic sums when $b \leq 10000$. For the larger finite endpoints the same arctanh bounds with $K = 100$ enclose $\Delta/(H_b - H_a)$ between two rational numbers having the same floor. This avoids floating-point assumptions. Subtracting $\lfloor Q(n) \rfloor$ gives exactly the exceptional sets in (1.10) and finishes its proof for every n.

Generating functions and literal source qualifications. Write $Q(n) = \sum_{k=0}^{d} q_k n^k$, let $\theta = x\,d/dx$, and set $f_r = Q(r) - \lfloor Q(r) \rfloor$. The proved formulas imply the exact rational function

$$G(x) = \sum_{n \geq 1} A_n x^n = \sum_{k=0}^{d} q_k \theta^k \frac{x}{1-x} - \frac{\sum_{r=1}^{J} f_r x^r}{1 - x^J} + \sum_{n \in C} x^n. \tag{1.11}$$

All quantities in this expression are explicitly specified above. Symbolic simplification gives exactly the empirical functions printed in the five source entries. The accompanying verifier compares the full rational functions, including the 151-coefficient numerator listed by Moses for A227014, and records an exact zero difference. Consequently a denominator for each is $(1-x)^d(1-x^J)$, with $(d,J) = (4,3),(5,144),(3,4),(3,8),(2,2)$. The respective numerator degrees are $9, 151, 10, 12, 5$, so the resulting recurrences hold from $n = 10, 152, 11, 13, 6$, respectively.

The source functions for A227013, A227014, A227016 and A227017 use $\sum_{n\geq 1} A_n x^{n-1}$; one multiplies them by x to obtain our convention. A227015 already uses $G(x)$. The prose recurrence signature in A227014 has sign errors; the source's polynomial denominator and the theorem instead give

$$\begin{aligned} A_n = {} & 5A_{n-1} - 10A_{n-2} + 10A_{n-3} - 5A_{n-4} + A_{n-5} \\ & + A_{n-144} - 5A_{n-145} + 10A_{n-146} - 10A_{n-147} + 5A_{n-148} - A_{n-149} \qquad (n \geq 152). \end{aligned}$$

Our result confirms the empirical generating function and establishes this corrected recurrence; it does not endorse the mistyped signature. The printed recurrence line in A227016 lacks an equality sign; the denominator above specifies its corrected recurrence. The starting ranges printed for several of these recurrences also precede the finite corrections, so the valid starting indices stated above must be used. The printed trigonometric polynomial in A227013 is shifted by two indices, and the printed closed form in A227017 is shifted by one index (with the exceptional first term still separate). Equation (1.10) consistently uses the definitions' offset $n = 1$. We make no assertion here about additional geometric-mean identities or the unrestricted polynomial conjecture in A227012.

1.12 Generating functions and exact recurrence certificates

This section records 53 distinct OEIS entries checked on 20 October 2026. There are 49 proved recurrence assertions, one entry whose recurrence is proved and whose separate limit assertion is refuted (A394578), one false recurrence (A027088), and two indexing repairs (A142243 and A059837). The two indexing repairs should be tabulated separately from substantive closures. The source pages still mark the targeted assertions as conjectural. That page status does not establish first-publication priority: the results here are independent certificates, and no submission or OEIS acceptance is claimed. The formulas explicitly described as seed premises are existing source formulas or definitions, not additional conjectures claimed to have been solved here.

For every entry, the certificate appendix prints the complete coefficient list $p_0(n), \dots, p_r(n)$, meaning

$$\sum_{j=0}^{r} p_j(n) a_{n-j} + c(n) = 0.$$

Here $c(n) = 0$ except in A027175 and A027182. A stated lower bound on n is part of the theorem. These are assertions about the actual sequence; isolated zeros of $p_0(n)$ do not create a problem because no division by $p_0(n)$ is used.

1.12.1 Why the certificates prove infinitely many cases

Let $\theta = x\,d/dx$. For an ordinary generating function $F(x) = \sum_{n\ge 0} a_n x^n$, extending any omitted initial indices by zero, define

$$\mathcal{L}_p = \sum_{j=0}^{r} x^j p_j(\theta + j).$$

Then coefficient extraction gives the exact identity

$$[x^n]\mathcal{L}_p F = \sum_{j=0}^{r} p_j(n) a_{n-j}. \tag{GF1}$$

Consequently, an identity $\mathcal{L}_p F = P(x)$, with P a polynomial of degree less than N, proves the recurrence for every $n \ge N$. For an exponential generating function $F = \sum a_n x^n/n!$, first replace $p_j(n)$ by

$$q_j(n) = p_j(n) \prod_{h=0}^{r-j-1} (n - j - h). \tag{GF2}$$

The coefficient in (GF1) is then the desired recurrence divided by $(n-r)!$. We use this conversion only where indicated.

Every radical certificate writes

$$x = f(v), \qquad F = \sum_i R_i(v)\sqrt{D_i(v)}.$$

The formal branch is the one specified by the source constant term. The identity can be checked using only rational arithmetic, since

$$\theta(R_i\sqrt{D_i}) = \frac{f(v)}{f'(v)}\left(R_i'(v) + \frac{R_i(v)D_i'(v)}{2D_i(v)}\right)\sqrt{D_i(v)}. \tag{GF3}$$

The appendix gives the exact resulting polynomial P. All coefficients of the nonconstant radical basis elements vanish. Thus the certificates are explicit rational-function identities, not a test of a finite prefix of the sequence. Clearing denominators converts each identity into a polynomial identity over the rationals. The accompanying script `oeis_gf_verify.py` performs precisely these operations using only the accompanying exact certificate JSON, without source-page snapshots.

1.12.2 Algebraic seed functions and elementary derivations

For A174193, A247173, A247102, A234269, A268600, A268601 and A244886, the ordinary generating function in the certificate is the entry's definition. For A108296 and A244884 it is an existing FORMULA statement. In particular, the A174193 result concerns its recurrence alone: the separately conjectural interpretation involving almost-avoiding permutations is not proved.

For A346074, multiplying the defining convolution by x^n and summing gives $F = (1-x)^{-1} + x^5F^2$. Solving this quadratic with $F(0) = 1$ gives the certified radical. For A268600 and A268601 use

$$\frac{1}{4-2\sqrt{1+4x}} = \frac{1}{3-4x} + \frac{\sqrt{1+4x}}{2(3-4x)}.$$

This explains the two radical basis elements in their certificates.

Write

$$M(x) = \frac{1-x-\sqrt{1-2x-3x^2}}{2x^2}.$$

For A244884, $F = x + (1-x)M$. For A257104 the existing continued fraction is

$$F = \cfrac{1}{1-x-\cfrac{x^2}{1-x-\cfrac{x^2}{1-x+x^2(1-M)}}}.$$

It also follows by the usual first-return decomposition at the restricted Motzkin heights. Substitution in (GF3) proves the printed recurrences with the bounds in the appendix. The bounds matter: the A244884 residual has degree four and the A257104 residual has degree six.

For A099038, reindex the existing binomial sum by $n = m + 2k$. The binomial coefficient extraction gives

$$F(x) = \mathrm{CT}_z \frac{1}{1-x(1+z)-x^2(1-z)/z} = \frac{1}{\sqrt{(1-x+x^2)^2-4x^3}}.$$

Indeed $\mathrm{CT}_z(A - Bz - C/z)^{-1} = (A^2-4BC)^{-1/2}$, by expanding the geometric series and matching equal powers of z.

For A337500 let $C = C(x^2) = (1-\sqrt{1-4x^2})/(2x^2)$. The generating function for unrestricted walks from 0 to displacement h is $(xC)^{|h|}/\sqrt{1-4x^2}$. One proof is to solve $W_h = \mathbf{1}_{h=0} + x(W_{h-1} + W_{h+1})$, selecting the formal small root for each half-line. Summing over $-3 \le h \le 3$ gives

$$F = \frac{1+2xC+2x^2C^2+2x^3C^3}{\sqrt{1-4x^2}}.$$

No restriction on intermediate positions is imposed.

1.12.3 Rational parametrizations from Lagrange inversion

For A127905 put $w = x(1+w+w^3)$. Lagrange inversion gives $a_n = n[x^n]w$, hence $F = xw'$. Setting $v = w$ gives

$$x = \frac{v}{1+v+v^3}, \qquad F = \frac{v(1+v+v^3)}{1-2v^3}.$$

For A392976 the definition itself gives $x = v((1-v^2)^2 - v)$ and $F = v/x$. For A186185 let $v = T(z)$, where $T = 1 + zT^3$ and $z = x/(1-x)$. Then

$$x = \frac{v-1}{v^3+v-1}, \qquad F = \frac{v^2}{v^2-v+1},$$

with $v = 1$ at $x = 0$. For A071264 the source gives $F = (1+xC^4)C^3$, so set $v = C$ and $x = (v-1)/v^2$, also about $v = 1$. For A220433 use the existing exponential generating function: $x = v - v^2 + v^3/6$, $F = v$, and apply (GF2).

For A045742, the accepted binomial sum is, with $m = n - 1$,

$$a_n = \frac{1}{m}\sum_{k=1}^{m-1} k\binom{m+k-1}{k}\binom{3m}{m-1-k}.$$

Using the derivative of $(1-u)^{-m}$, this becomes

$$a_n = [u^{m-1}]\frac{u}{1-u}\left(\frac{(1+u)^3}{1-u}\right)^m.$$

If $w = x(1+w)^3/(1-w)$, Lagrange inversion applied to an antiderivative of $u/(1-u)$ gives

$$F = x^2w'\frac{w}{1-w}, \qquad x = f(v) = \frac{v(1-v)}{(1+v)^3}, \qquad F = \frac{f(v)^2v}{(1-v)f'(v)}.$$

The exact boundary is $-2544x^3$; therefore the recurrence is proved for $n \geq 4$. The source conjecture does not prescribe a conflicting lower range.

For A392211 the accepted relation is

$$x = \frac{3 + 14F - 20F^2 - 3\sqrt{1+4F-4F^2}}{8}.$$

Set $\sqrt{1+4F-4F^2} = 1 + vF$. Then

$$F = \frac{4-2v}{v^2+4}, \qquad x = \frac{F(14-3v-20F)}{8},$$

about $v = 2$. Substitution into (GF3) gives the certificate in the appendix.

1.12.4 Products, binomial sums, and factorial-series transforms

A091539 has the existing product formula

$$a_n = \frac{1}{60}\prod_{j=0}^{n-1}(3j+2)\left(\prod_{j=0}^{n-1}3(j+1) - 3\prod_{j=0}^{n-1}(3j+1)\right).$$

The two product components have successive ratios $\rho(n) = 3n(3n-1)$ and $\rho(n) = (3n-1)(3n-2)$. For each ratio the exact identity

$$\sum_{j=0}^{r}\frac{p_j(n)}{\prod_{h=0}^{j-1}\rho(n-h)} = 0$$

holds. Multiplying by the corresponding product proves the recurrence.

The definition of A027170 and Pascal's identity give

$$T(n,k) = 2\binom{n+2}{k+1} + \binom{n}{k} - 4.$$

Therefore, writing $C_n = \binom{2n}{n}$,

$$\text{A027175: } a_n = C_n\frac{9n+5}{2(n+1)} - 4,$$

$$\text{A027182: } a_n = 4\binom{2n+4}{n+2} + 4\binom{2n+2}{n+1} + C_n + 16n + 40 - 72\,2^n.$$

For the second identity, symmetry of $T(n,k)$ turns the central self-convolution coefficient into $\sum_{k=0}^{n} T(n,k)^2$. Expanding the square and applying Vandermonde's identity yields the formula, including the boundary corrections from $k+1 = 1,\dots,n+1$. Now $C_n/C_{n-1} = 2(2n-1)/n$. Substitution in the respective recurrence separates a central-binomial component, a polynomial component, and a 2^n component. Each vanishes identically; the polynomial component includes the inhomogeneous term $c(n)$ printed in the appendix. The example value 120 on A027182 is inconsistent with its definition; the actual row at $n=2$ gives $5^2+10^2+5^2 = 150$, as in its data. This observation is not counted as a separate closure.

For A177258, A177259 and A177260 take $q = 2, 3, 4$, respectively, and use the established series

$$Q = 1 + x + x^q, \qquad F_q = Q^{-1} f(x/Q), \qquad f(z) = \sum_{k\geq 0} k! z^k.$$

This is an identity of formal series, so no analytic convergence is needed. Coefficient extraction proves $z^2 f' + (z-1)f + 1 = 0$. The chain rule then gives

$$x^2 Q F_q' + Q\big(-1 + x + (q-1)x^q\big)F_q + 1 - (q-1)x^q = 0.$$

Its coefficients are exactly

$$\begin{aligned} a_n = {} & (n-1)a_{n-1} + (n-1)a_{n-2} + (q-2)a_{n-q} \\ & + (n-1)a_{n-q-1} + (q-1)a_{n-2q}, \qquad n \geq 2q. \end{aligned}$$

When two indices coincide their terms are added, as in the printed vectors.

For A351933 differentiate its defining exponential generating function $F = \exp(x/(1-x^2/2))$. It satisfies

$$(1 - x^2/2)^2 F' - (1 + x^2/2)F = 0.$$

Taking $4(n-1)![x^{n-1}]$ gives the displayed five-term recurrence.

For A001722 and A345651 the accepted Stirling sums give, respectively,

$$a_n = [u^2](5+u)^{\overline{n+2}}, \qquad a_n = [u^4](4+u)^{\underline{n}}.$$

Factoring the common rising product $(5+u)^{\overline{n-1}}$ out of the first recurrence leaves exactly u^3. Thus its u^2 coefficient is zero. Factoring $(4+u)^{\underline{n-4}}$ out of the second leaves exactly u^4. For $n \geq 9$ that common falling product contains a factor u, so its product with u^4 has zero u^4 coefficient. The residual at $n = 8$ is 24, which explains the stated lower range.

1.12.5 Exact elimination of factorial and parity recurrences

Here is a reusable certificate rule. Fix n modulo 2, express each floor at the indices $m = n - j$ by its affine formula, and use the given short recurrence to express all a_{n-j} in a fixed finite list of base quantities. Normalize factorials by $(n-r)!$, using

$$\frac{(n-r+s)!}{(n-r)!} = \prod_{h=1}^{s} (n-r+h)$$

and the reciprocal formula for negative s. All coefficients are now rational functions of n. In each certificate the coefficient of every base quantity in $\sum p_j(n) a_{n-j}$ is identically zero, for both parities. The explicit substitution rules and base lists below completely specify the calculation; the appendix records its zero vectors.

For A121580 the existing factorial partial sum yields

$$a_m - a_{m-1} = \frac{(m-1)!(m^2 - m + 2)}{2}.$$

For A180195 the defining alternating sum yields

$$a_m = -a_{m-1} + (m-1)!(m^2 - m + 1).$$

For A121639 use the established identity $a_m = ma_{m-1} + (m-1)! - (m-3)!$, valid for $m \geq 4$. Each of these three cases uses the base list $a_{n-r}, (n-r)!$.

For A121751 put $e_m = m \bmod 2$. Its established recurrence is

$$a_m = 2\left\lfloor \frac{m-1}{2} \right\rfloor a_{m-1} + \left(1 - \left\lfloor \frac{m-1}{2} \right\rfloor \left\lfloor \frac{m-2}{2} \right\rfloor\right) a_{m-2}.$$

Use $\lfloor (m-1)/2 \rfloor = (m-2+e_m)/2$ and $\lfloor (m-2)/2 \rfloor = (m-2-e_m)/2$, with base list a_{n-r}, a_{n-r+1}.

For A121752 the accepted recurrence is

$$\begin{aligned} a_m = {} & (2m-3)a_{m-1} - (m-1)(m-3)a_{m-2} \\ & + (m-2)!\left(m\left\lfloor \frac{m}{2} \right\rfloor - (m-2)\left\lfloor \frac{m-1}{2} \right\rfloor - 1\right), \end{aligned}$$

with $a_1 = 1, a_2 = 2$. Define $b_m = a_m - ma_{m-1}$. Then

$$b_m = (m-3)b_{m-1} + f_m, \quad f_m = \begin{cases} (2m-3)(m-2)!, & m \text{ even}, \\ (m-2)(m-2)!, & m \text{ odd}. \end{cases}$$

The base values are $b_3 = 1, b_4 = 11$. Substituting each formula in the opposite parity step proves by induction

$$b_m = \begin{cases} (m-2)!(2m^2 - 3m + 2)/4, & m \geq 4 \text{ even}, \\ (m-3)!(m-1)^2(2m-5)/4, & m \geq 3 \text{ odd}. \end{cases}$$

Use $a_m = ma_{m-1} + b_m$ and base list $a_{n-r}, (n-r)!$.

For A121747 the accepted auxiliary sequence satisfies $d_1 = 1, d_2 = 0$, $d_{2m} = 3! + 5! + \cdots + (2m-1)!$, and $d_{2m+1} = -d_{2m}$. Equivalently, for $m \geq 2$,

$$d_m = -d_{m-1} + \mathbf{1}_{2|m}(m-1)!, \qquad a_m = ma_{m-1} - d_{m-1} + (m-1)!\left\lfloor \frac{m}{2} \right\rfloor.$$

The base list is $a_{n-r}, d_{n-r}, (n-r)!$.

For A121584 the entry gives $a = b + c$, with $b =$ A121580 and $c =$ A121582. The latter source's established recurrence is

$$c_m = \frac{2m-3}{m-2}c_{m-1} - \frac{m-1}{m-2}c_{m-2} + \frac{(m-1)!(m^2-3m+4)}{2}.$$

Together with the rule for b, it gives the base list $b_{n-r}, c_{n-r}, c_{n-r+1}, (n-r)!$. Every coefficient vanishes.

For A121696 write $f_m = m(m-2)!\lfloor (m-1)/2 \rfloor$, so the source recurrence is $a_m - a_{m-2} = f_m$. Adding at adjacent indices and checking the two parity classes proves, from $a_1 = 0, a_2 = 1$,

$$a_m + a_{m-1} = \left\lceil \frac{m}{2} \right\rceil (m-1)! \quad (m \geq 2).$$

The conjectured residual reduces to

$$f_n - n(n-2)(a_{n-2} + a_{n-3}) = n(n-2)!\left(\left\lfloor \frac{n-1}{2} \right\rfloor - \left\lceil \frac{n-2}{2} \right\rceil\right) = 0.$$

For A121553 use $a_n = n!(H_n + n(n-1)/4)$. For A187252 its existing exponential generating function is

$$F = -\frac{2x - 1 + e^{2x} + 4\log(1-x)}{4(1-x)}.$$

Hence for $n \geq 1$, $a_n = n!(H_n - 1/4 - E_n/4)$, where $E_n = \sum_{k=0}^{n} 2^k/k!$. Divide the proposed recurrence by $n!$, replace H_{n-j} by $H_n - \sum_{h=0}^{j-1} 1/(n-h)$, and replace E_{n-j} by $E_n - \sum_{h=0}^{j-1} 2^{n-h}/(n-h)!$. The resulting identity is zero in $\mathbb{Q}(n)[H_n]$ or $\mathbb{Q}(n)[H_n, E_n, 2^n/n!]$, respectively.

For A111942 the source product formula gives $a_n/a_{n-1} = -\lfloor n/2 \rfloor$ for $n \geq 2$. Dividing the target by a_{n-2} and inserting the two ratios proves $4a_n + 2a_{n-1} - (n-1)(n-2)a_{n-2} = 0$ separately for even and odd n.

1.12.6 Four fixed columns of A136123

The established bivariate ordinary generating function on https://oeis.org/A136123 is

$$G(x,t) = f(h(x,t)), \quad f(z) = \sum_{m \geq 0} m! z^m, \quad h = \frac{(1-t)x^2 - x}{(1-t)x^2 - 1}.$$

The source identifies its columns $k = 1, 2, 3, 4$ as A001277, A001278, A001279 and A001280. Set $F_0 = f(x/(1+x))$. The factorial-series equation already proved above gives

$$x^2(1+x)F_0' - F_0 + (1+x) = 0. \tag{GF4}$$

Put $K = h_t(1-h)/h^2$, $J = -h_t/h^2$, and write $K_i = [t^i]K$, $J_i = [t^i]J$. Differentiating in t gives $G_t = KG + J$. Therefore, if $[t^k]G = A_k F_0 + B_k$,

$$\begin{aligned} A_0 = 1, \quad B_0 = 0, \\ (k+1)A_{k+1} = \sum_{i=0}^{k} K_i A_{k-i}, \\ (k+1)B_{k+1} = \sum_{i=0}^{k} K_i B_{k-i} + J_k. \end{aligned}$$

This finite recursion derives, rather than guesses, the exact rational functions printed in the appendix. Apply the conjectured operator to $A_k F_0 + B_k$ and use (GF4) repeatedly. The coefficient of F_0 is zero, and the remaining polynomial is the displayed boundary. Thus all four recurrence assertions follow from the source generating function.

1.12.7 Three tribonacci triangles: a kernel proof

Let $T(n,k)$ be one of A027023 (type I), A027052 (type II), or A027082 (type III). The common interior rule is

$$T(n,k) = T(n-1,k-3) + T(n-1,k-2) + T(n-1,k-1).$$

Values outside a row are zero. The row at $n = 0$ is (1). Type I has left entries $(1,1,1)$ and rightmost entry 1; type II has left entries $(1,0,1)$ and rightmost entry 1. Type III has left entries $(1,1,1)$ and rightmost entry equal to the sum of the two rightmost entries of the preceding row. The row at $n = 1$ is interpreted with these boundary values, and each source's program supplies the same rule for the first overlapping boundary row. Explicitly, the rows at $n = 2$ are $(1,1,1,3,1)$, $(1,0,1,2,1)$, and $(1,1,1,3,2)$ for types I, II, and III.

Reverse each row:

$$U_n(z) = \sum_{m=0}^{2n} T(n, 2n-m) z^m, \quad B(x,z) = \sum_{n \geq 0} U_n(z) x^n, \quad B_j(x) = [z^j]B(x,z).$$

The kernel is $K = 1 - x(z + 1 + z^{-1})$. Its small root is $z = v$, with $x = v/(1 + v + v^2)$, and $v = x + O(x^2)$. For types I and II put

$$(\epsilon, L(z), \alpha) = \begin{cases} (1, z^2 - 1, 1), & \text{type I,} \\ (0, z^2 - z, 0), & \text{type II.} \end{cases}$$

Multiplication of the row polynomials, with the three left-edge terms and the rightmost term corrected, gives

$$KB = 1 + \epsilon x + \frac{xL(z)}{1 - xz^2} - xB_1 - \frac{x}{z(1 - x)}. \tag{GF5}$$

For completeness, the bulk operation is multiplication of U_{n-1} by $z+1+z^{-1}$. Its extraneous z^{-1} term contributes $-xB_0/z$; here $B_0 = (1-x)^{-1}$. Resetting the rightmost coefficient to 1 contributes $-xB_1$. The prescribed left boundary contributes the geometric sum $xL(z)/(1 - xz^2)$. The exceptional row $n = 1$ contributes ϵx. This accounts for every term in (GF5).

Substitute the small root, a valid formal-series substitution, to obtain

$$B_1 = \frac{1}{x} + \epsilon + \frac{L(v)}{1 - xv^2} - \frac{1}{v(1 - x)}.$$

Taking the z^1 and z^2 coefficients of (GF5) gives

$$B_2 = \frac{1 - x}{x} B_1 - B_0 - (\alpha - 1), \qquad B_3 = \frac{1 - x}{x} B_2 - B_1 - 1 - x(1 - \alpha - 1). \tag{GF6}$$

Only type II uses B_3 here, so its final term is zero. The target series are B_1 for A027050, B_2 for A027058, and B_3 for A027059.

For type III, denote the column series by C_j. The modified right boundary instead gives

$$KB = 1 + x + \frac{x(z^2 - 1)}{1 - xz^2} - \frac{xC_0}{z}. \tag{GF7}$$

The same bulk calculation applies; the rightmost column is now propagated by the rule instead of being reset to 1. Kernel substitution and extraction of the first two coefficients yield

$$\begin{aligned} C_0 &= \frac{v}{x}\left(1 + x + \frac{x(v^2 - 1)}{1 - xv^2}\right), \\ C_1 &= \frac{(1 - x)C_0 - 1}{x}, \\ C_2 &= \frac{1 - x}{x} C_1 - C_0. \end{aligned}$$

Thus A027088 has generating function C_1. A027089 starts at index 2, so its zero-extended generating function is $C_2 - x$, since the full column has an additional term at $n = 1$. These expressions are rational in v; the appendix prints their simplified forms and the exact certificates. Four conjectures are proved. A027088 is refuted as follows.

Its defining triangle gives $a_1, \ldots, a_7 = 1, 3, 6, 16, 41, 107, 283$. At $n = 7$, the printed coefficient vector is

$$(984, 4841, -3247, -5426, -2686, -1223, 213).$$

Its dot product with $(283, 107, 41, 16, 6, 3, 1)$ is 556944, not zero. There is also an eventual-recurrence obstruction: substituting $v = (1 - x - \sqrt{1 - 2x - 3x^2})/(2x)$ gives the radical form in the appendix. The printed operator leaves a nonzero coefficient of that radical. Since $1 - 2x - 3x^2 = (1 + x)(1 - 3x)$ is not a square in $\mathbb{Q}(x)$, the operator value is not rational and in particular is not a polynomial. By (GF1) it cannot vanish at all sufficiently large coefficient indices.

1.12.8 Interpolation and the two indexing repairs

The polynomial interpolating $(1,0)$ and $(k,1)$, $2 \le k \le n+1$, is uniquely

$$P_n(z) = 1 - \frac{\prod_{k=2}^{n+1}(k-z)}{n!}.$$

For A394578 evaluation at $z = n + 1/2$, with its specified scale, gives

$$a_n = \frac{4^n}{2} + (-1)^n \operatorname{Cat}_{n-1} \quad (n \ge 1), \qquad a_0 = 0,$$

and consequently

$$F = \frac{2x}{1-4x} + \frac{1-\sqrt{1+4x}}{2}.$$

Its recurrence follows from the exact certificate. Moreover, $\operatorname{Cat}_{n-1} \le 4^{n-1}/n$, so $a_n/(4^n/2) = 1 + O(1/n)$ and $a_{n+1}/a_n \to 4$. This disproves the distinct printed limit $1/4$.

For A396724 use the same interpolation polynomial with $z = -3/2$, and multiply by $-4^n/2$. The identity

$$nP_n(z) = (n+1-z)P_{n-1}(z) + z - 1$$

follows directly from its product. Subtracting four times the scaled relation one index earlier eliminates the geometric forcing and gives

$$na_n - (8n-4z)a_{n-1} + 16(n-z)a_{n-2} = 0.$$

Putting $z = -3/2$ proves the conjecture. A neighboring interpolation entry already containing a proof was not counted.

For A142243, the row sums are $s_n = [z^n](1+z)^{2n}/\sqrt{1-4z}$. Lagrange coefficient summation gives

$$S(x) = \frac{1+v}{(1-v)\sqrt{1-4v}}, \qquad x = \frac{v}{(1+v)^2}.$$

The actual sums at indices 0 through 3 are $1, 4, 20, 106$. The printed recurrence has residual -344 at $n = 3$. Replacing n by $n-1$ in all its polynomial coefficients gives the certified homogeneous operator identity $\mathcal{L}S = 0$ and proves the corrected recurrence for $n \ge 3$. This is recorded as an index repair.

For A059837 the actual offset is 1 and

$$a_n = (n-1)!\binom{n-1}{\lfloor (n-1)/2 \rfloor}.$$

The printed recurrence at $n = 3$ evaluates to $4 \cdot 4 - 6 \cdot 1 - 48 \cdot 1 = -38$. Its corrected version is

$$na_n - 2(n-1)a_{n-1} - 4(n-1)(n-2)^2 a_{n-2} = 0 \quad (n \ge 3).$$

Divide by a_{n-2} and use the exact ratio $a_n/a_{n-1} = (n-1)^2/\lfloor n/2 \rfloor$; both parity identities vanish. The originally printed formula applies to the reindexed sequence $b_n = a_{n+1}$. The source also discusses offset zero nearby, so this is conservatively classified as an index repair.

1.12.9 Complete recurrence coefficients and exact certificates

A174193 — proved

Source: https://oeis.org/A174193. Coefficients of the algebraic ordinary generating function in the entry definition; the separate permutation interpretation is outside this result. The certified

recurrence range is $n \geq 4$. Only the recurrence of the defined series is certified. The original coefficient polynomials are:

$$p_0(n) = -404489n^2 + 1057390n + 3732736$$

$$p_1(n) = 3147815n^2 - 8680082n - 14098754$$

$$p_2(n) = -4672878n^2 + 1304044n + 29029698$$

$$p_3(n) = -8358160n^2 + 84680184n - 162964424$$

$$p_4(n) = 10287712n^2 - 78361536n + 144300744$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{3x^2 - 1}{2x^3 - 2x^2}$$

$$D_0 = 1$$

$$R_1 = \frac{-4x^4 - 30x^3 + 4x^2 + 5x - 1}{32x^4 - 16x^3 + 2x^2}$$

$$D_1 = 1 - 4x$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 2708316x^3 + 2537736x^2 - 15245384x + 3732736$$

A346074 — proved

Source: https://oeis.org/A346074. Sequence determined by the convolution whose ordinary generating function satisfies F=1/(1-x)+x^5 F^2, with F(0)=1. The certified recurrence range is $n \geq 6$. The original coefficient polynomials are:

$$p_0(n) = n + 5$$

$$p_1(n) = -2n - 8$$

$$p_2(n) = n + 3$$

$$p_3(n) = 0$$

$$p_4(n) = 0$$

$$p_5(n) = 10 - 4n$$

$$p_6(n) = 4n - 12$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{1}{2x^5}$$

$$D_0 = 1$$

$$R_1 = -\frac{1}{2x^5}$$

$$D_1 = -\frac{4x^5}{1 - x} + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 5 - 4x$$

A247173 — proved

Source: https://oeis.org/A247173. Coefficients of the radical ordinary generating function specified in the entry definition. The certified recurrence range is $n \geq 5$. The original coefficient polynomials are:

$$p_0(n) = -3n^2 + n + 4$$
$$p_1(n) = 9n^2 - 3n - 4$$
$$p_2(n) = 18n^2 - 42n + 16$$
$$p_3(n) = -6n^2 + 2n + 16$$
$$p_4(n) = -15n^2 + 53n - 12$$
$$p_5(n) = -3n^2 + 13n - 4$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = -\frac{1}{2x^2 + 2x}$$
$$D_0 = 1$$
$$R_1 = \frac{1 - 2x}{2x^4 + 6x^3 - 10x^2 + 2x}$$
$$D_1 = x^4 + 4x^3 - 2x^2 - 4x + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = -4x^3 - 12x^2 + 4x + 4$$

A247102 — proved

Source: https://oeis.org/A247102. Coefficients of the radical ordinary generating function specified in the entry definition. The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = n + 1$$
$$p_1(n) = -2n - 5$$
$$p_2(n) = 21 - 24n$$
$$p_3(n) = 45 - 30n$$
$$p_4(n) = 18 - 9n$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{-2x - 1}{2x^2 + 2x}$$
$$D_0 = 1$$
$$R_1 = \frac{-3x - 1}{6x^4 + 18x^3 + 10x^2 - 2x}$$
$$D_1 = -3x^2 - 6x + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 9x^3 + 15x^2 + 6x + 2$$

A234269 — proved

Source: https://oeis.org/A234269. Coefficients of the radical ordinary generating function specified in the entry definition. The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = -2n^2 + 3n + 5$$
$$p_1(n) = -2n^2 + 8n$$
$$p_2(n) = 8n^2 - 20n + 12$$
$$p_3(n) = 16n^2 - 64n + 58$$
$$p_4(n) = 8n^2 - 44n + 56$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = -\frac{1}{2x}$$
$$D_0 = 1$$
$$R_1 = \frac{2x^2 - 1}{8x^4 + 8x^3 - 2x}$$
$$D_1 = -4x^3 - 4x^2 + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = x^2 (2x + 3)$$

A268600 — proved

Source: https://oeis.org/A268600. Coefficients of 1/(2 sqrt(1-4x))+1/(4-2 sqrt(1+4x)). The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = -3n^2 + 3n$$
$$p_1(n) = 40n^2 - 136n + 96$$
$$p_2(n) = 292 - 112n$$
$$p_3(n) = -640n^2 + 3840n - 5600$$
$$p_4(n) = 768n^2 - 4608n + 6720$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{1}{3 - 4x}$$
$$D_0 = 1$$
$$R_1 = \frac{1}{2 - 8x}$$
$$D_1 = 1 - 4x$$
$$R_2 = \frac{1}{6 - 8x}$$
$$D_2 = 4x + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 12x^2 (4x + 1)$$

A268601 — proved

Source: https://oeis.org/A268601. Coefficients of 1/(2 sqrt(1-4x))-1/(4-2 sqrt(1+4x)). The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = 3n^2 - 3n$$
$$p_1(n) = -40n^2 + 136n - 96$$
$$p_2(n) = 112n - 292$$
$$p_3(n) = 640n^2 - 3840n + 5600$$
$$p_4(n) = -768n^2 + 4608n - 6720$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = -\frac{1}{3 - 4x}$$
$$D_0 = 1$$
$$R_1 = \frac{1}{2 - 8x}$$
$$D_1 = 1 - 4x$$
$$R_2 = -\frac{1}{6 - 8x}$$
$$D_2 = 4x + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 12x^2 (4x + 1)$$

A108296 — proved

Source: https://oeis.org/A108296. Sequence with the ordinary generating function displayed in the certificate, as listed in its FORMULA section. The certified recurrence range is $n \geq 5$. The original coefficient polynomials are:

$$p_0(n) = n + 3$$
$$p_1(n) = -n - 2$$
$$p_2(n) = -2n - 2$$
$$p_3(n) = 6 - 2n$$
$$p_4(n) = n + 1$$
$$p_5(n) = 3n - 6$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{1}{2x^3}$$
$$D_0 = 1$$
$$R_1 = \frac{1}{2x^5 - 2x^3}$$
$$D_1 = -3x^4 - 4x^3 - 2x^2 + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = x + 3$$

A244884 — proved

Source: https://oeis.org/A244884. Coefficients of x+(1-x)M(x), where M is the Motzkin generating function. The certified recurrence range is $n \geq 5$. The original coefficient polynomials are:

$$p_0(n) = n + 2$$
$$p_1(n) = -3n - 1$$
$$p_2(n) = -n$$
$$p_3(n) = 3n - 9$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{2x^3 + x^2 - 2x + 1}{2x^2}$$
$$D_0 = 1$$
$$R_1 = \frac{x - 1}{2x^2}$$
$$D_1 = -3x^2 - 2x + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 3x^4 - 3x^3 - 5x^2 - x + 2$$

A244886 — proved

Source: https://oeis.org/A244886. Coefficients of the algebraic generating function in the definition. The certified recurrence range is $n \geq 6$. The original coefficient polynomials are:

$$p_0(n) = n$$
$$p_1(n) = 3 - 5n$$
$$p_2(n) = 2n$$
$$p_3(n) = 13n - 30$$
$$p_4(n) = -3$$
$$p_5(n) = 21 - 8n$$
$$p_6(n) = 9 - 3n$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{-2x^2 - 3x + 1}{2x^4 + 4x^3 - 2x^2 - 6x + 2}$$
$$D_0 = 1$$
$$R_1 = \frac{1}{2x^4 + 4x^3 - 2x^2 - 6x + 2}$$
$$D_1 = -3x^2 - 2x + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = x\left(3x^3 + 3x^2 + x - 1\right)$$

A257104 — proved

Source: https://oeis.org/A257104. Restricted Motzkin paths described by the continued fraction used in the proof. The certified recurrence range is $n \geq 7$. The original coefficient polynomials are:

$$p_0(n) = 2 - n$$
$$p_1(n) = 7n - 17$$
$$p_2(n) = 34 - 14n$$
$$p_3(n) = n + 22$$
$$p_4(n) = 16n - 89$$
$$p_5(n) = 23 - 4n$$
$$p_6(n) = 3n - 15$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{2x^4 - 3x^3 + 9x^2 - 8x + 2}{2x^4 - 4x^3 + 14x^2 - 10x + 2}$$
$$D_0 = 1$$
$$R_1 = -\frac{x^2}{2x^4 - 4x^3 + 14x^2 - 10x + 2}$$
$$D_1 = -3x^2 - 2x + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = \left(x^2 + 3x - 1\right)\left(3x^4 - 6x^3 + 4x^2 + 3x - 2\right)$$

A099038 — proved

Source: https://oeis.org/A099038. Diagonal binomial/Krawtchouk sum from the entry; its constant-term representation is derived below. The certified recurrence range is $n \geq 5$. The original coefficient polynomials are:

$$p_0(n) = n$$
$$p_1(n) = -n$$
$$p_2(n) = n$$
$$p_3(n) = 3 - 3n$$
$$p_4(n) = 13 - 5n$$
$$p_5(n) = n - 3$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = 0$$
$$D_0 = 1$$
$$R_1 = \frac{1}{x^4 - 6x^3 + 3x^2 - 2x + 1}$$
$$D_1 = -4x^3 + \left(x^2 - x + 1\right)^2$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 0$$

A337500 — proved

Source: https://oeis.org/A337500. Walks with steps +1 and -1 whose final displacement has absolute value at most 3; intermediate positions are unrestricted. The certified recurrence range is $n \geq 3$. The original coefficient polynomials are:

$$p_0(n) = -n^2 + n + 12$$

$$p_1(n) = 2n^2 - 4n - 22$$

$$p_2(n) = 4n^2 - 8n + 4$$

$$p_3(n) = -8n^2 + 24n - 16$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{-x - 1}{x^3}$$

$$D_0 = 1$$

$$R_1 = \frac{x^3 + 2x^2 - x - 1}{4x^5 - x^3}$$

$$D_1 = 1 - 4x^2$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 12$$

A127905 — proved

Source: https://oeis.org/A127905. a_n is the coefficient of u^(n-1) in (1+u+u^3)^n. The certified recurrence range is $n \geq 3$. The original coefficient polynomials are:

$$p_0(n) = 4n^3 - 6n^2 + 2$$

$$p_1(n) = -12n^3 + 24n^2 - 10n$$

$$p_2(n) = 12n^3 - 30n^2 + 18n$$

$$p_3(n) = -31n^3 + 93n^2 - 62n$$

For (GF3), the exact input is

$$x = \frac{t}{t^3 + t + 1}$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{t\,(t^3 + t + 1)}{1 - 2t^3}$$

$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 0$$

A392976 — proved

Source: https://oeis.org/A392976. The compositional inverse w of t((1-t^2)^2-t), divided by x, supplies the ordinary generating function. The certified recurrence range is $n \geq 5$. The original coefficient polynomials are:

$$\begin{aligned} p_0(n) = {} & 23495312375279n^5 - 97473596443030n^4 \\ & + 77470631009665n^3 + 97473596443030n^2 \\ & - 100965943384944n \end{aligned}$$

$$\begin{aligned} p_1(n) = {} & -73180721424774n^5 + 490203969890442n^4 \\ & - 1274195207740074n^3 + 1488868655358738n^2 \\ & - 631696696084332n \end{aligned}$$

$$\begin{aligned} p_2(n) = {} & -406745978528262n^5 + 2710034017555110n^4 \\ & - 6497864886329538n^3 + 6606177703000266n^2 \\ & - 2374863833644560n - 36737022053016 \end{aligned}$$

$$\begin{aligned} p_3(n) = {} & -384334715888720n^5 + 1886690791154422n^4 \\ & + 2005293720367118n^3 - 26729399150503384n^2 \\ & + 5317941580070088 0n - 33371926512356880 \end{aligned}$$

$$\begin{aligned} p_4(n) = {} & 78200611484375n^5 - 2829824426102500n^4 \\ & + 24949288577136625n^3 - 93338391900041900n^2 \\ & + 160425630781509240n - 104626628147875680 \end{aligned}$$

$$\begin{aligned} p_5(n) = {} & 140685548031250n^5 - 2487209724443750n^4 \\ & + 17499620703856250n^3 - 61230529923966250n^2 \\ & + 106501850375442900n - 73632743322117840 \end{aligned}$$

For (GF3), the exact input is

$$x = t\left(-t + \left(1 - t^2\right)^2\right)$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{1}{-t + (1 - t^2)^2}$$
$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 0$$

A186185 — proved

Source: https://oeis.org/A186185. Coefficients of 1/(1-zT(z)), with z=x/(1-x) and T=1+zT^3. The certified recurrence range is $n \geq 7$. The original coefficient polynomials are:

$$p_0(n) = 4n^2 - 6n + 2$$

$$p_1(n) = 9n^2 - 127n + 170$$

$$p_2(n) = -570n^2 + 3680n - 5590$$

$$p_3(n) = 2579n^2 - 18451n + 32892$$

$$p_4(n) = -4899n^2 + 39585n - 80634$$
$$p_5(n) = 4272n^2 - 40026n + 93330$$
$$p_6(n) = -1395n^2 + 15345n - 41850$$

For (GF3), the exact input is

$$x = \frac{t-1}{t^3+t-1}$$

with $F = \sum_i R_i\sqrt{D_i}$ and

$$R_0 = \frac{t^2}{t^2-t+1}$$
$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = (2-2x)\left(207x^3 - 243x^2 + 27x + 1\right)$$

A220433 — proved

Source: https://oeis.org/A220433. The exponential generating function is the inverse under composition of t-t^2+t^3/6. The certified recurrence range is $n \geq 3$. The original coefficient polynomials are:

$$p_0(n) = 4$$
$$p_1(n) = 18 - 12n$$
$$p_2(n) = -9n^2 + 36n - 35$$

For (GF3), the exact input is

$$x = \frac{t^3}{6} - t^2 + t$$

with $F = \sum_i R_i\sqrt{D_i}$ and

$$R_0 = t$$
$$D_0 = 1$$

Use the exponential conversion (GF2) before applying the operator. The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 2x^2$$

A071264 — proved

Source: https://oeis.org/A071264. Coefficients of (1+xC^4)C^3, where C=1+xC^2 is the Catalan series. The certified recurrence range is $n \geq 3$. The original coefficient polynomials are:

$$p_0(n) = 8587n^3 + 114480n^2 + 439415n + 370002$$
$$p_1(n) = 8587n^3 - 377960n^2 - 2268967n - 1091596$$
$$p_2(n) = -171740n^3 + 143858n^2 + 464342n - 246668$$

For (GF3), the exact input is

$$x = \frac{t-1}{t^2}$$

with $F = \sum_i R_i\sqrt{D_i}$ and

$$R_0 = t^3\left(t^2(t-1)+1\right)$$
$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 370002$$

A392211 — proved

Source: https://oeis.org/A392211. Algebraic series F specified by x=(3+14F-20F^2-3 sqrt(1+4F-4F^2))/8 and F(0)=0. The certified recurrence range is $n \geq 6$. The original coefficient polynomials are:

$$p_0(n) = 30632912640n^4 - 215110974720n^3 + 430902535680n^2 - 246424473600n$$

$$\begin{aligned} p_1(n) = &-61010825216n^4 + 756124952576n^3 \\ &- 3128904705856n^2 + 5009001827776n \\ &- 2575211249280 \end{aligned}$$

$$\begin{aligned} p_2(n) = &-1127075797485n^4 + 11295798763110n^3 \\ &- 40811553201975n^2 + 64043018252310n \\ &- 37173001041840 \end{aligned}$$

$$\begin{aligned} p_3(n) = &\ 2244769385584n^4 - 34554389907976n^3 \\ &+ 195408589554464n^2 - 483915433738736n \\ &+ 244700475929080 \end{aligned}$$

$$\begin{aligned} p_4(n) = &\ 765822816000n^4 - 9972711264000n^3 \\ &+ 48002376840000n^2 - 102102215016000n \\ &+ 82139932008000 \end{aligned}$$

$$\begin{aligned} p_5(n) = &-1525270630400n^4 + 28054747596800n^3 \\ &- 191649301734400n^2 + 575634929036800n \\ &- 640491407424000 \end{aligned}$$

For (GF3), the exact input is

$$x = \frac{(4-2t)\left(-3t - \frac{80-40t}{t^2+4} + 14\right)}{8t^2+32}$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{4-2t}{t^2+4}$$
$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = -4200x^3\,(36148218x - 40772785)$$

A142243 — proved/corrected

Source: https://oeis.org/A142243. Row sums of T(n,k)=binom(2n,k) binom(2n-2k,n-k), with rows indexed from n=0. The certified recurrence range is $n \geq 3$. This entry is an indexing repair and is excluded from the substantive closure count. The original coefficient polynomials are:

$$p_0(n) = 4n^2 + 6n + 2$$

$$p_1(n) = -81n^2 + 19n - 8$$

$$p_2(n) = 510n^2 - 770n + 300$$

$$p_3(n) = -1000n^2 + 2500n - 1500$$

The corrected coefficient vector is obtained by $n \mapsto n-1$ in every p_j. For (GF3), the exact input is

$$x = \frac{t}{(t+1)^2}$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = 0$$
$$D_0 = 1$$
$$R_1 = \frac{t+1}{(1-4t)\,(1-t)}$$
$$D_1 = 1-4t$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 0$$

A091539 — proved

Source: https://oeis.org/A091539. The product formula in the entry, equivalently a fixed linear combination of two hypergeometric products. The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = 1$$
$$p_1(n) = -18n^2 + 21n - 5$$
$$p_2(n) = 81n^4 - 351n^3 + 531n^2 - 321n + 60$$
$$\rho_0(n) = 3n\,(3n-1)$$
$$\rho_1(n) = (3n-2)\,(3n-1)$$

The normalized recurrence residual is identically zero for each component.

A027175 — proved

Source: https://oeis.org/A027175, https://oeis.org/A027170. The near-central entry T(2n-1,n-1) of the Pascal-derived array A027170. The certified recurrence range is $n \geq 3$. The original coefficient polynomials are:

$$p_0(n) = 2n + 2$$
$$p_1(n) = -7n - 5$$
$$p_2(n) = 6 - 4n$$
$$c(n) = 12 - 36n$$

In $a_n = \binom{2n}{n} R(n) + V(n) + E2^n$, the exact components are

$$R(n) = \frac{9n+5}{2n+2}$$
$$V(n) = -4$$
$$E = 0$$

The central-binomial, polynomial (including $c(n)$), and exponential residuals are all exactly zero.

A027182 — proved

Source: https://oeis.org/A027182, https://oeis.org/A027170. The central self-convolution coefficient of row n of A027170; symmetry makes it the sum of squares of that row. The certified recurrence range is $n \geq 6$. The original coefficient polynomials are:

$$p_0(n) = 2n + 4$$
$$p_1(n) = -17n - 23$$
$$p_2(n) = 47n + 36$$
$$p_3(n) = -44n - 34$$
$$p_4(n) = 80 - 4n$$
$$p_5(n) = 16n - 72$$
$$c(n) = 288n - 1128$$

In $a_n = \binom{2n}{n} R(n) + V(n) + E2^n$, the exact components are

$$R(n) = \frac{81n^2 + 171n + 66}{(n+1)(n+2)}$$
$$V(n) = 16n + 40$$
$$E = -72$$

The central-binomial, polynomial (including $c(n)$), and exponential residuals are all exactly zero.

A177258 — proved

Source: https://oeis.org/A177258. The adjacent-cycle derangement sequence with parameter q=2 and the listed double factorial sum. The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = 1$$
$$p_1(n) = 1 - n$$
$$p_2(n) = 1 - n$$
$$p_3(n) = 1 - n$$
$$p_4(n) = -1$$

The exact identity is derived in the corresponding proof subsection above; all displayed residuals in the machine-readable certificate are zero.

A177259 — proved

Source: https://oeis.org/A177259. The adjacent-cycle derangement sequence with parameter q=3 and the listed double factorial sum. The certified recurrence range is $n \geq 6$. The original coefficient polynomials are:

$$p_0(n) = 1$$
$$p_1(n) = 1 - n$$
$$p_2(n) = 1 - n$$
$$p_3(n) = -1$$
$$p_4(n) = 1 - n$$
$$p_5(n) = 0$$
$$p_6(n) = -2$$

The exact identity is derived in the corresponding proof subsection above; all displayed residuals in the machine-readable certificate are zero.

A177260 — proved

Source: https://oeis.org/A177260. The adjacent-cycle derangement sequence with parameter q=4 and the listed double factorial sum. The certified recurrence range is $n \geq 8$. The original coefficient polynomials are:

$$p_0(n) = 1$$

$$p_1(n) = 1 - n$$

$$p_2(n) = 1 - n$$

$$p_3(n) = 0$$

$$p_4(n) = -2$$

$$p_5(n) = 1 - n$$

$$p_6(n) = 0$$

$$p_7(n) = 0$$

$$p_8(n) = -3$$

The exact identity is derived in the corresponding proof subsection above; all displayed residuals in the machine-readable certificate are zero.

A351933 — proved

Source: https://oeis.org/A351933. Coefficients, in exponential generating function convention, of exp(x/(1-x^2/2)). The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = 4$$

$$p_1(n) = -4$$

$$p_2(n) = -4n^2 + 12n - 8$$

$$p_3(n) = -2n^2 + 6n - 4$$

$$p_4(n) = n^4 - 10n^3 + 35n^2 - 50n + 24$$

The exact identity is derived in the corresponding proof subsection above; all displayed residuals in the machine-readable certificate are zero.

A001722 — proved

Source: https://oeis.org/A001722. The accepted Stirling-number sum, equal to [u^2](5+u) rising(n+2). The certified recurrence range is $n \geq 3$. The original coefficient polynomials are:

$$p_0(n) = 1$$

$$p_1(n) = -3n - 15$$

$$p_2(n) = 3n^2 + 27n + 61$$

$$p_3(n) = -n^3 - 12n^2 - 48n - 64$$

After removing the common product specified in the proof, the exact remaining polynomial is

$$Q(u) = u^3$$

A345651 — proved

Source: https://oeis.org/A345651. The signed-Stirling sum, equal to [u^4](4+u) falling n. The certified recurrence range is $n \geq 9$. The original coefficient polynomials are:

$$p_0(n) = 1$$

$$p_1(n) = 4n - 26$$

$$p_2(n) = 6n^2 - 84n + 295$$

$$p_3(n) = 4n^3 - 90n^2 + 676n - 1695$$

$$p_4(n) = n^4 - 32n^3 + 384n^2 - 2048n + 4096$$

After removing the common product specified in the proof, the exact remaining polynomial is

$$Q(u) = u^4$$

A111942 — proved

Source: https://oeis.org/A111942. The signed product (-1)^(n-1) floor((n-1)/2)! floor(n/2)! for n>=1. The certified recurrence range is $n \geq 3$. The original coefficient polynomials are:

$$p_0(n) = 4$$

$$p_1(n) = 2$$

$$p_2(n) = -n^2 + 3n - 2$$

The exact identity is derived in the corresponding proof subsection above; all displayed residuals in the machine-readable certificate are zero.

A059837 — proved/corrected

Source: https://oeis.org/A059837. With the entry offset 1, a_n=(n-1)! binom(n-1,floor((n-1)/2)). The certified recurrence range is $n \geq 3$. This entry is an indexing repair and is excluded from the substantive closure count. The original coefficient polynomials are:

$$p_0(n) = n + 1$$

$$p_1(n) = -2n$$

$$p_2(n) = -4n^3 + 8n^2 - 4n$$

$$\widetilde{p}_0(n) = n$$

$$\widetilde{p}_1(n) = 2 - 2n$$

$$\widetilde{p}_2(n) = (4 - 4n)\,(n - 2)^2$$

The original residual at $n = 3$ is -38; both parity residuals of the corrected formula are zero.

A394578 — proved/corrected

Source: https://oeis.org/A394578. Scaled evaluation at z=n+1/2 of the polynomial interpolating (1,0) and (k,1) for 2<=k<=n+1. The certified recurrence range is $n \geq 3$. The original coefficient polynomials are:

$$p_0(n) = n$$
$$p_1(n) = 2 - 4n$$
$$p_2(n) = 48 - 16n$$
$$p_3(n) = 64n - 224$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = -\frac{1}{8x - 2}$$
$$D_0 = 1$$
$$R_1 = -\frac{1}{2}$$
$$D_1 = 4x + 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = x\,(12x + 1)$$

The separate limit assertion is false: $a_{n+1}/a_n \to 4$, as proved above.

A396724 — proved

Source: https://oeis.org/A396724. The same interpolation polynomial evaluated at z=-3/2 and multiplied by -4^n/2. The certified recurrence range is $n \geq 2$. The original coefficient polynomials are:

$$p_0(n) = n$$
$$p_1(n) = -8n - 6$$
$$p_2(n) = 16n + 24$$

The exact identity is derived in the corresponding proof subsection above; all displayed residuals in the machine-readable certificate are zero.

A121580 — proved

Source: https://oeis.org/A121580. First-column cell count for deco polyominoes, with accepted formula one half of sum_{j=0}^{n+1} j!, minus n!. The certified recurrence range is $n \geq 5$. The original coefficient polynomials are:

$$p_0(n) = 1$$
$$p_1(n) = -n - 4$$
$$p_2(n) = 3n + 3$$
$$p_3(n) = 6 - 4n$$
$$p_4(n) = 2n - 6$$

Using the exact substitution rules in the factorial/parity section, the coefficient residual vectors are

n mod 2	residual vector
0	$(0, 0)$
1	$(0, 0)$

A180195 — proved

Source: https://oeis.org/A180195. The alternating factorial partial sum in the definition. The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = 1$$

$$p_1(n) = -n - 1$$

$$p_2(n) = -4$$

$$p_3(n) = n - 2$$

Using the exact substitution rules in the factorial/parity section, the coefficient residual vectors are

n mod 2	residual vector
0	$(0, 0)$
1	$(0, 0)$

A121639 — proved

Source: https://oeis.org/A121639. The deco-polyomino statistic with the accepted recurrence a_m=m a_(m-1)+(m-1)!-(m-3)!. The certified recurrence range is $n \geq 7$. The original coefficient polynomials are:

$$p_0(n) = 1$$

$$p_1(n) = 1 - 2n$$

$$p_2(n) = n^2 - 2n$$

$$p_3(n) = 2n - 7$$

$$p_4(n) = -n^2 + 8n - 15$$

Using the exact substitution rules in the factorial/parity section, the coefficient residual vectors are

n mod 2	residual vector
0	$(0, 0)$
1	$(0, 0)$

A121751 — proved

Source: https://oeis.org/A121751. The zero column of A121697, satisfying the established second-order floor recurrence. The certified recurrence range is $n \geq 7$. The original coefficient polynomials are:

$$p_0(n) = 256$$

$$p_1(n) = -384$$

$$p_2(n) = -128n^2 + 640n - 1072$$

$$p_3(n) = 128n^2 - 864n + 1392$$

$$p_4(n) = 16n^4 - 224n^3 + 968n^2 - 1216n - 84$$

$$p_5(n) = -8n^4 + 152n^3 - 996n^2 + 2588n - 2216$$

$$p_6(n) = n^4 - 24n^3 + 207n^2 - 756n + 992$$

Using the exact substitution rules in the factorial/parity section, the coefficient residual vectors are

n mod 2	residual vector
0	$(0, 0)$
1	$(0, 0)$

A121747 — proved

Source: https://oeis.org/A121747. The first moment of A121745, with the established auxiliary sequence d_m in its recurrence. The certified recurrence range is $n \geq 6$. The original coefficient polynomials are:

$$p_0(n) = 38 - 9n$$

$$p_1(n) = 9n^2 - 57n + 81$$

$$p_2(n) = 9n^3 - 37n^2 - 24n + 127$$

$$p_3(n) = -9n^4 + 102n^3 - 392n^2 + 525n - 85$$

$$p_4(n) = -28n^4 + 307n^3 - 1179n^2 + 1767n - 711$$

$$p_5(n) = -19n^4 + 224n^3 - 925n^2 + 1512n - 720$$

Using the exact substitution rules in the factorial/parity section, the coefficient residual vectors are

$n \bmod 2$	residual vector
0	$(0, 0, 0)$
1	$(0, 0, 0)$

A121584 — proved

Source: https://oeis.org/A121584, https://oeis.org/A121580, https://oeis.org/A121582. The sum of the deco-polyomino cell counts A121580 and A121582. The certified recurrence range is $n \geq 7$. The original coefficient polynomials are:

$$p_0(n) = 14 - 14n$$

$$p_1(n) = 14n^2 + 1731n - 6995$$

$$p_2(n) = -1731n^2 + 1440n + 21729$$

$$p_3(n) = 5562n^2 - 23904n + 12008$$

$$p_4(n) = -5987n^2 + 36181n - 54220$$

$$p_5(n) = 2142n^2 - 15434n + 27464$$

Using the exact substitution rules in the factorial/parity section, the coefficient residual vectors are

$n \bmod 2$	residual vector
0	$(0, 0, 0, 0)$
1	$(0, 0, 0, 0)$

A121696 — proved

Source: https://oeis.org/A121696. The sequence with a_1=0,a_2=1 and a_m-a_(m-2)=m(m-2)! floor((m-1)/2). The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = 1$$

$$p_1(n) = 0$$

$$p_2(n) = -n^2 + 2n - 1$$

$$p_3(n) = -n^2 + 2n$$

The exact identity is derived in the corresponding proof subsection above; all displayed residuals in the machine-readable certificate are zero.

A121553 — proved

Source: https://oeis.org/A121553. The accepted harmonic-number expression n! (H_n+n(n-1)/4). The certified recurrence range is $n \geq 5$. The original coefficient polynomials are:

$$p_0(n) = 1$$

$$p_1(n) = -2n - 3$$

$$p_2(n) = n^2 + 4n - 3$$

$$p_3(n) = -2n^2 + 2n + 6$$

$$p_4(n) = 2n^2 - 12n + 18$$

The exact identity is derived in the corresponding proof subsection above; all displayed residuals in the machine-readable certificate are zero.

A187252 — proved

Source: https://oeis.org/A187252. The first moment of A187250, with the exponential generating function stated in the proof. The certified recurrence range is $n \geq 5$. The original coefficient polynomials are:

$$p_0(n) = 1$$

$$p_1(n) = -2n - 3$$

$$p_2(n) = n^2 + 6n - 7$$

$$p_3(n) = -4n^2 + 8n + 4$$

$$p_4(n) = 4n^2 - 24n + 36$$

The exact identity is derived in the corresponding proof subsection above; all displayed residuals in the machine-readable certificate are zero.

A001277 — proved

Source: https://oeis.org/A001277, https://oeis.org/A136123. Column k=1 of the permutation triangle A136123. The certified recurrence range is $n \geq 5$. The original coefficient polynomials are:

$$p_0(n) = 1$$

$$p_1(n) = -n$$

$$p_2(n) = 0$$

$$p_3(n) = n - 1$$

The derivation from A136123 gives $F = AF_0 + B$.

$$A = -\frac{1}{(x-1)(x+1)}$$

$$B = \frac{1}{x-1}$$

Reduction using (GF4) gives $\mathcal{L}F = P(x)$ and zero coefficient of F_0, with

$$P(x) = x^2$$

A001278 — proved

Source: https://oeis.org/A001278, https://oeis.org/A136123. Column k=2 of the permutation triangle A136123. The certified recurrence range is $n \geq 10$. The original coefficient polynomials are:

$$p_0(n) = 1$$
$$p_1(n) = -n - 3$$
$$p_2(n) = 3n - 6$$
$$p_3(n) = 2n - 2$$
$$p_4(n) = 17 - 6n$$
$$p_5(n) = 5 - n$$
$$p_6(n) = 3n - 12$$

The derivation from A136123 gives $F = AF_0 + B$.

$$A = \frac{1 - 3x}{2\,(x-1)^2\,(x+1)}$$

$$B = \frac{2x^2 + 2x - 1}{2\,(x-1)^2\,(x+1)}$$

Reduction using (GF4) gives $\mathcal{L}F = P(x)$ and zero coefficient of F_0, with

$$P(x) = x^4\,(3x+1)$$

A001279 — proved

Source: https://oeis.org/A001279, https://oeis.org/A136123. Column k=3 of the permutation triangle A136123. The certified recurrence range is $n \geq 12$. The original coefficient polynomials are:

$$p_0(n) = 18 - 3n$$
$$p_1(n) = 3n^2 - 23n + 67$$
$$p_2(n) = 5n^2 - 58n + 179$$
$$p_3(n) = n^2 - 51n + 128$$
$$p_4(n) = -17n^2 + 183n - 364$$
$$p_5(n) = -4n^2 + 60n - 176$$
$$p_6(n) = 12n^2 - 108n + 240$$

The derivation from A136123 gives $F = AF_0 + B$.

$$A = \frac{(1 - 2x)\,(6x - 1)}{6\,(x-1)^3\,(x+1)}$$

$$B = \frac{6x^4 + 12x^3 - 2x^2 - 6x + 1}{6\,(x-1)^3\,(x+1)^2}$$

Reduction using (GF4) gives $\mathcal{L}F = P(x)$ and zero coefficient of F_0, with

$$P(x) = 0$$

A001280 — proved

Source: https://oeis.org/A001280, https://oeis.org/A136123. Column k=4 of the permutation triangle A136123. The certified recurrence range is $n \geq 13$. The original coefficient polynomials are:

$$p_0(n) = n^2 - 17n + 72$$

$$p_1(n) = -n^3 + 15n^2 - 52n - 18$$

$$p_2(n) = 2n^3 - 33n^2 + 143n - 118$$

$$p_3(n) = 5n^2 - 27n + 16$$

$$p_4(n) = -2n^3 + 24n^2 - 94n + 120$$

$$p_5(n) = n^3 - 12n^2 + 47n - 60$$

The derivation from A136123 gives $F = AF_0 + B$.

$$A = \frac{-60x^3 + 60x^2 - 15x + 1}{24\,(x-1)^4\,(x+1)}$$

$$B = \frac{24x^6 + 72x^5 + 12x^4 - 66x^3 - 19x^2 + 12x - 1}{24\,(x-1)^4\,(x+1)^3}$$

Reduction using (GF4) gives $\mathcal{L}F = P(x)$ and zero coefficient of F_0, with

$$P(x) = 0$$

A045742 — proved

Source: https://oeis.org/A045742. The established binomial sum for the face-count sequence: m=n-1 and the summation formula given below. The certified recurrence range is $n \geq 4$. The original coefficient polynomials are:

$$p_0(n) = 523n^3 - 2516n^2 + 1993n$$

$$p_1(n) = -4554n^3 + 40668n^2 - 112854n + 76740$$

$$p_2(n) = -56484n^3 + 328212n^2 - 505608n + 150240$$

$$p_3(n) = 491832n^3 - 4883976n^2 + 16099344n - 17614080$$

For (GF3), the exact input is

$$x = \frac{t\,(1-t)}{(t+1)^3}$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{t^3\,(1-t)}{(t+1)^6\left(-\frac{3t(1-t)}{(t+1)^4} - \frac{t}{(t+1)^3} + \frac{1-t}{(t+1)^3}\right)}$$

$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = -2544x^3$$

A121752 — proved

Source: https://oeis.org/A121752. The first moment of A121697, satisfying the established inhomogeneous second-order recurrence. The certified recurrence range is $n \geq 7$. The original coefficient polynomials are:

$$p_0(n) = 1223 - 460n$$

$$p_1(n) = 460n^2 + 460n - 5569$$

$$p_2(n) = 460n^3 - 3826n^2 + 7853n + 1313$$

$$p_3(n) = -460n^4 + 1840n^3 + 5501n^2 - 31045n + 31726$$

$$p_4(n) = 1223n^4 - 13205n^3 + 45787n^2 - 51389n + 558$$

$$p_5(n) = -852n^4 + 14150n^3 - 85636n^2 + 224000n - 213312$$

Using the exact substitution rules in the factorial/parity section, the coefficient residual vectors are

n mod 2	residual vector
0	$(0, 0)$
1	$(0, 0)$

A027050 — proved

Source: https://oeis.org/A027050, https://oeis.org/A027023. T(n,2n-1) in the tribonacci triangle A027023. The certified recurrence range is $n \geq 12$. The original coefficient polynomials are:

$$p_0(n) = 1 - n$$

$$p_1(n) = 6n - 9$$

$$p_2(n) = 10 - 7n$$

$$p_3(n) = 38 - 8n$$

$$p_4(n) = 5n - 23$$

$$p_5(n) = 2n - 5$$

$$p_6(n) = 3n - 12$$

For (GF3), the exact input is

$$x = \frac{t}{t^2 + t + 1}$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{-3t^4 - 3t^3 - 3t^2 - t}{t^5 - t^4 - 2t^2 - t - 1}$$

$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = 2x^3 \left(3x^3 + x^2 - 3x + 3\right)$$

A027058 — proved

Source: https://oeis.org/A027058, https://oeis.org/A027052. T(n,2n-2) in the tribonacci triangle A027052. The certified recurrence range is $n \geq 12$. The original coefficient polynomials are:

$$p_0(n) = n$$
$$p_1(n) = 4 - 7n$$
$$p_2(n) = 13n - 10$$
$$p_3(n) = n - 34$$
$$p_4(n) = 84 - 13n$$
$$p_5(n) = 3n - 32$$
$$p_6(n) = 6 - n$$
$$p_7(n) = 3n - 18$$

For (GF3), the exact input is

$$x = \frac{t}{t^2 + t + 1}$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{-2t^5 - 3t^4 - 3t^3 - t^2 - t}{t^5 - t^4 - 2t^2 - t - 1}$$
$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = x\left(3x^6 - 9x^4 - 24x^3 + 21x^2 - 8x + 1\right)$$

A027059 — proved

Source: https://oeis.org/A027059, https://oeis.org/A027052. T(n,2n-3) in the tribonacci triangle A027052, with offset 2. The certified recurrence range is $n \geq 12$. The original coefficient polynomials are:

$$p_0(n) = n + 1$$
$$p_1(n) = -8n - 2$$
$$p_2(n) = 19n - 5$$
$$p_3(n) = -6n - 30$$
$$p_4(n) = 123 - 21n$$
$$p_5(n) = 8n - 58$$
$$p_6(n) = n + 1$$
$$p_7(n) = 6n - 30$$

For (GF3), the exact input is

$$x = \frac{t}{t^2 + t + 1}$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{-2t^6 - 4t^5 - 2t^4 - 2t^3}{t^5 - t^4 - 2t^2 - t - 1}$$
$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = -2x^3\left(12x^4 + 7x^3 - 18x^2 + 19x - 4\right)$$

A027088 — refuted

Source: https://oeis.org/A027088, https://oeis.org/A027082. T(n,2n-1) in the tribonacci triangle A027082, with offset 1. The original coefficient polynomials are:

$$p_0(n) = 54n^2 - 201n - 255$$

$$p_1(n) = 292n^2 - 1316n - 255$$

$$p_2(n) = -218n^2 + 1213n - 1056$$

$$p_3(n) = -496n^2 + 3332n - 4446$$

$$p_4(n) = -50n^2 - 497n + 3243$$

$$p_5(n) = -52n^2 + 80n + 765$$

$$p_6(n) = 66n^2 - 645n + 1494$$

For (GF3), the exact input is

$$x = x$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = \frac{x^4 - 2x^3 - 4x^2 - 6x + 3}{2x\left(x^3 + x^2 + 3x - 1\right)}$$

$$D_0 = 1$$

$$R_1 = \frac{(x-1)\left(x^2+3\right)}{2x\left(x^3 + x^2 + 3x - 1\right)}$$

$$D_1 = -3x^2 - 2x + 1$$

At $n = 7$ the exact recurrence residual is 556944. Moreover $\mathcal{L}_p F = U_0 + U_1\sqrt{1 - 2x - 3x^2}$, with the following nonzero U_1; this proves the eventual obstruction. Write $U_1 = N_1/D_1^*$.

$$\begin{aligned} N_1 = &-2295x^{12} - 9180x^{11} \\ &- 30030x^{10} - 23112x^9 \\ &- 73269x^8 - 77880x^7 \\ &- 106692x^6 - 42480x^5 \\ &+ 36015x^4 + 13428x^3 \\ &+ 3474x^2 - 5160x \\ &+ 765 \end{aligned}$$

$$D_1^* = (x+1)^2 (3x-1)^2 \left(x^3 + x^2 + 3x - 1\right)^3$$

A027089 — proved

Source: https://oeis.org/A027089, https://oeis.org/A027082. T(n,2n-2) in the tribonacci triangle A027082, with offset 2. The certified recurrence range is $n \geq 12$. The original coefficient polynomials are:

$$p_0(n) = -244n^2 + 577n + 2130$$

$$p_1(n) = 1266n^2 - 3843n - 5109$$

$$p_2(n) = -718n^2 + 2422n - 2840$$

$$p_3(n) = -2348n^2 + 12638n - 14820$$

$$p_4(n) = -760n^2 + 2173n + 1616$$

$$p_5(n) = -502n^2 + 2245n - 1123$$
$$p_6(n) = 138n^2 - 1092n + 2010$$

For (GF3), the exact input is

$$x = \frac{t}{t^2 + t + 1}$$

with $F = \sum_i R_i \sqrt{D_i}$ and

$$R_0 = -\frac{t}{t^2 + t + 1} + \frac{-3t^5 - 3t^4 - 4t^3 - t^2 - t}{t^3 - t^2 - t - 1}$$
$$D_0 = 1$$

The exact operator identity is $\mathcal{L}F = P(x)$, with every nonconstant radical coefficient zero, where

$$P(x) = -x^2 \left(1128x^5 - 5725x^4 - 6519x^3 - 1711x^2 - 3081x - 2308\right)$$

1.13 Validation and prior-work exclusions

Structural and arithmetic validation. The quadratic-character proof was independently reviewed, including its primitive reduction, congruences, and exceptional discriminants. Exact direct evaluation checked the divisibility and nonvanishing normalizations on 300 admissible discriminants up to 600. Matrices constructed from the self-similarity condition independently matched ordinary coin-change counts in nine binary rows and seven ternary rows. The universal square formula was checked as a polynomial identity in its parameter through index 23, in addition to the characteristic-polynomial proof.

The arithmetic verifier independently enumerated partitions, permutations, digital representations, and divisor sums within its documented bounds. The prime-exponent first-occurrence theorem was checked against every index up to 65537; its proof establishes the bound for all indices. For A216971, every function on up to six labelled elements was enumerated. All binary words through length 16 were checked for A334592. The consecutive-square XOR map was checked for injectivity through index 200000, and its construction was checked for every target congruent to 1 modulo 4 below 65536.

Exact certificate validation. The floor verifiers reconstruct their rational bounds and integer polynomial coefficients, including finite exceptional indices. They compare complete rational generating functions. The generating-function verifier applies the recorded differential operators with exact arithmetic, retaining the boundary polynomial and the correct ordinary or exponential normalization. Each counterexample is evaluated with exact integers or rational numbers. Program outputs are included so that the validation can be repeated.

Examples excluded because of earlier work. These exclusions illustrate why an old conjecture label alone was insufficient:

- A111967: a linked 2021 proof already settles the relevant matrix row-sum assertion; related c=2 copies were not counted.
- A054120: the first McGarvey assertion was proved by Hendel in 2015; only the second assertion is selected here.
- A224515: existence was already proved in 2026; uniqueness is selected.
- A113249: Israel's 2017 proof covers m=3; only the universal parameter assertion is selected, with all its copies counted once.
- A024219, A024401, A075002, A105696, and the inspected A220119 divisibility assertion had recorded prior resolutions, including updates in 2026.
- A203233, A269027, A155828, A001969, A008364, A008365, and A008486 also had prior proofs or refutations and were not included as new selections.

The complete audit scope and its access limitation are recorded in `oeis_open_prior_work_audit.json`. This dossier makes no claim that all relevant literature has been searched exhaustively.

Standard identities used. The quadratic character proof uses the classical quadratic Gauss sum, Dirichlet Euler product, and Fourier expansion of the second Bernoulli polynomial. Reference sections are NIST DLMF https://dlmf.nist.gov/27.10, https://dlmf.nist.gov/25.15, and https://dlmf.nist.gov/24.8. Other standard tools used explicitly in the proofs are Lagrange inversion, Newton's symmetric-function identities, the finite q-binomial theorem, and Bertrand's postulate. The report claims no novelty for these tools.

Part II

Exact sums, prime-power matrices, connected arrays, and recurrences

Scope, counting, and how to use the packet

This second round provides 101 selected conjecture or empirical assertion groups, with no counted entry repeated from the first round. The headline total consists of 100 proof-only groups and one mixed group, A004737: its proposed matrix property has a small exact counterexample, while its normalized Gram-matrix pattern is proved for every size. The cumulative total under the same counting convention is 208 selected groups.

The unit being counted. Each row of the catalogue is one entry-level selection. Multiple related conjectures in one entry count once. Fixed-width or fixed-color assertions on different entries count separately even when one general theorem proves them together. Copied universal assertions and aliases receive no extra count. Thus 22 connected-array selections share three methods; they are 22 source assertions, not 22 unrelated mathematical discoveries.

Selected scope is explicit. Several entries contain further questions that are outside the selected result. For example, A376336 is resolved only for existence of a tau-zero modulo each modulus, not for its proposed quadratic bound. A290815 is resolved for the odd-modulus reciprocal-square equivalence, not for the general-exponent extension. The source wording, formal statement, applicable range, and proof are tied together in the ledger and the following sections.

Qualifications do not add to the total. The normalized transform theorem for A153197 is included as an uncounted note: its normalization is essential, and the source also records established fixed-point formulas. Index-only or definition-only repairs, such as A395454 and A178113, are excluded. The independent divisibility proof found for A098275 was excluded after identifying an earlier correct proof by Roberto Blanco Gómez, submitted on 14 October 2026: https://aixiv.online/abs/2610.01676.

What is verified. The infinite assertions are proved by algebra, induction, bijections, global inequalities, or complete finite-state certificates. An initial segment agreeing with a guessed recurrence is never the proof. In the floor family, the finite state space is exhaustive, and exact matrix or rational-function identities imply every subsequent coefficient. In the generating-function family, exact differential identities imply the stated recurrence beyond an explicitly checked boundary polynomial. Finite numerical or rational comparisons are labeled supplementary regression checks.

Source audit. The inspected entries still mark the selected statements conjectural or empirical, with the qualifications stated here. Later confirmations in the source text and identified linked literature were removed from the count. A second check compared all selected IDs with the 97 distinct entry IDs appearing in the 100-claim Appendix A of Tom Adamczewski's *OEIS Open*; there is no overlap. That appendix covers the 100 costliest of 153 combined-model resolutions, not every result in its repositories or every published OEIS proof. The audit therefore does not establish worldwide novelty. Its exact scope is retained in `r2_prior_work_audit.json`.

Files. The supplied `oeis_hunt_round2.tex` is standalone and contains this complete report. The ZIP includes the ledger, proof fragments, full certificates, the programs that check them, and the validation outputs. The report can be recompiled with pdfLaTeX; verification uses Python and SymPy. See `README_ROUND2.txt`. No OEIS edit or submission is represented by this packet.

2.1 Catalogue of the selected assertions

P: proved. **P/R**: some selected assertions proved and another refuted in the same entry. Each entry ID links to its primary source; the following sections contain the proof and exact scope.

Structural identities and exact sums (20)

No.	Entry	Result	Selected assertion
1	A350900	P	Double-gcd convolution for arbitrary arithmetic functions
2	A347293	P	Residue-product gcd convolution and multiplicativity
3	A130540	P	Even-column and alternating row sums
4	A001222	P	Complex two-parameter convolution identity
5	A265649	P	Both finite sums for the generalized Bessel triangle
6	A097724	P	Chebyshev translation coefficients
7	A102086	P	The row-polynomial recurrence
8	A379907	P	General Riordan convolution and column recurrence
9	A380191	P	Central-binomial weighted columns
10	A004737	P/R	E-matrix refutation and exact Gram-matrix formula
11	A005773	P	Order-comparable coprime divisors and Motzkin meanders
12	A376182	P	The triangular enumeration is a permutation
13	A378277	P	Alternating harmonic Fibonacci row sums
14	A349802	P	Exact Euler transforms of binary Lyndon columns
15	A163774	P	A single binomial sum for the defining double sum
16	A152920	P	Binomially weighted rows and A193844
17	A105794	P	Factorial and Bernoulli weighted sums
18	A132393	P	The full Vandermonde-squared nested-sum identity
19	A397048	P	The first signed-Stirling convolution conjecture
20	A112302	P	Limit of the Lerch-transcendent recurrence

Arithmetic and congruences (11)

No.	Entry	Result	Selected assertion
21	A334595	P	XOR-triangle right-boundary records and value one
22	A133910	P	Divisibility by the index
23	A051283	P	The A346418 exponent characterization
24	A354995	P	The nonzero positions of a unitary-divisor difference
25	A376336	P	Existence of a zero of tau modulo every modulus
26	A087726	P	All prime-power counts of square-zero matrices
27	A322327	P	The general parameter multiplicative function
28	A033955	P	Strict bound for the sum of prime remainders
29	A166306	P	Nondivisibility by 3 of the Bernoulli-prime sums
30	A234307	P	Alternating gcd-sum identity
31	A290815	P	Odd-modulus reciprocal-square characterization

Connected arrays and floor identities (28)

No.	Entry	Result	Selected assertion
32	A166756	P	Empirical connected-coloring generating function
33	A166757	P	Empirical connected-coloring generating function
34	A166758	P	Empirical connected-coloring generating function
35	A166759	P	Empirical connected-coloring generating function
36	A166760	P	Empirical connected-coloring generating function
37	A166762	P	The ninth-order connected-coloring recurrence
38	A166776	P	Empirical connected-coloring generating function
39	A166777	P	Empirical connected-coloring generating function
40	A166778	P	Empirical connected-coloring generating function
41	A166779	P	Empirical connected-coloring generating function
42	A166780	P	Empirical connected-coloring generating function
43	A166781	P	Empirical connected-coloring generating function
44	A166782	P	Empirical connected-coloring generating function
45	A166783	P	Empirical connected-coloring generating function
46	A166786	P	Empirical connected-coloring generating function
47	A166796	P	Empirical connected-coloring generating function
48	A166797	P	Empirical connected-coloring generating function
49	A166798	P	Empirical connected-coloring generating function
50	A166799	P	Empirical connected-coloring generating function
51	A166800	P	Empirical connected-coloring generating function
52	A166801	P	Empirical connected-coloring generating function
53	A166802	P	Empirical connected-coloring generating function
54	A024172	P	The empirical symmetric-function floor series
55	A024180	P	The empirical shifted symmetric-function floor series
56	A184534	P	Cube-root fractional-part reciprocal series
57	A002620	P	Exponential floor identity with parameter 4
58	A001840	P	Exponential floor identity with parameter 6
59	A227018	P	Quartic-binomial harmonic means and their series

Generating functions and recurrences (42)

No.	Entry	Result	Selected assertion
60	A089354	P	The specified polynomial-coefficient recurrence
61	A247471	P	The specified polynomial-coefficient recurrence
62	A135925	P	The specified polynomial-coefficient recurrence
63	A218045	P	The specified polynomial-coefficient recurrence
64	A006343	P	The specified polynomial-coefficient recurrence
65	A158826	P	The specified polynomial-coefficient recurrence
66	A075045	P	The specified polynomial-coefficient recurrence
67	A082397	P	The specified polynomial-coefficient recurrence
68	A190160	P	The specified polynomial-coefficient recurrence
69	A001723	P	The specified polynomial-coefficient recurrence
70	A001724	P	The specified polynomial-coefficient recurrence
71	A027051	P	The specified polynomial-coefficient recurrence
72	A027027	P	The specified polynomial-coefficient recurrence
73	A027028	P	The specified polynomial-coefficient recurrence

No.	Entry	Result	Selected assertion
74	A027056	P	The specified polynomial-coefficient recurrence
75	A027087	P	The specified polynomial-coefficient recurrence
76	A025258	P	The specified polynomial-coefficient recurrence
77	A157125	P	The specified polynomial-coefficient recurrence
78	A186335	P	The specified polynomial-coefficient recurrence
79	A133158	P	The specified polynomial-coefficient recurrence
80	A157021	P	The specified polynomial-coefficient recurrence
81	A173993	P	The specified polynomial-coefficient recurrence
82	A141353	P	The specified polynomial-coefficient recurrence
83	A078999	P	The specified polynomial-coefficient recurrence
84	A000159	P	The specified polynomial-coefficient recurrence
85	A000181	P	The specified polynomial-coefficient recurrence
86	A000185	P	The specified polynomial-coefficient recurrence
87	A080894	P	The specified polynomial-coefficient recurrence
88	A272261	P	The specified polynomial-coefficient recurrence
89	A144575	P	The specified polynomial-coefficient recurrence
90	A053532	P	The specified polynomial-coefficient recurrence
91	A250917	P	The specified polynomial-coefficient recurrence
92	A292461	P	The specified polynomial-coefficient recurrence
93	A126089	P	The specified polynomial-coefficient recurrence
94	A108629	P	The specified polynomial-coefficient recurrence
95	A239230	P	The specified polynomial-coefficient recurrence
96	A032115	P	The specified polynomial-coefficient recurrence
97	A168452	P	The specified polynomial-coefficient recurrence
98	A005573	P	Terminating hypergeometric formula
99	A368633	P	Odd coefficients occur exactly at indices 2^k-1
100	A027116	P	Rational series and the constant-coefficient recurrence
101	A027117	P	Rational series and the constant-coefficient recurrence

2.2 Structural identities, transforms, and exact sums

This section resolves 20 counted entry-level selections and includes one uncounted normalization note. Unless a different range is stated, triangular arrays have indices $0 \le k \le n$. Every source is the corresponding linked OEIS entry, inspected on 20 October 2026. The statements below specify the selected assertions; unrelated conjectures in the same entry are not implicitly included.

2.2.1 A350900: a double-gcd convolution

For an arbitrary arithmetic function f, put

$$A_f(n) = \sum_{i,k=1}^{n} f\left(\frac{\gcd(i,n)}{\gcd(i,k,n)}\right), \qquad C(n) = \sum_{d|n} d\varphi(d).$$

The two selected conjectures follow from the identity

$$\boxed{A_f = (f\varphi) * C.}$$

Here multiplication inside $f\varphi$ is pointwise and $*$ is Dirichlet convolution. To prove it, fix $d = \gcd(i,n)$; there are $\varphi(n/d)$ choices of i. For $e \mid d$, exactly $(n/d)\varphi(e)$ choices of $k \in [1,n]$ satisfy $d/\gcd(d,k) = e$. Consequently

$$A_f(n) = \sum_{d|n} \frac{n}{d}\varphi(n/d) \sum_{e|d} f(e)\varphi(e).$$

Associativity gives $(\mathrm{id}\ \varphi) * 1 * (f\varphi)$, which is the displayed result. If f is multiplicative, all three factors are multiplicative. Taking $f(n) = n$ gives the stated row-sum convolution A002618∗A057660. The auxiliary identity defining C is already recorded in A057660; it is an input, not a new selection.

2.2.2 A347293: a residue-product convolution

For any arithmetic function f, including nonmultiplicative ones, the selected formula is

$$\boxed{\sum_{a,b \bmod n} f(\gcd(1+ab,n)) = \sum_{d|n} (n/d)^2 \varphi(d)(\mu * f)(d).}$$

Indeed $f(m) = \sum_{d|m}(\mu * f)(d)$. For each $d \mid n$, the congruence $ab \equiv -1 \pmod d$ has $\varphi(d)$ solutions modulo d: a must be a unit and then b is determined. Each solution has $(n/d)^2$ lifts modulo n. Interchanging finite sums proves the formula. Thus the function is $\mathrm{id}^{\cdot 2} * (\varphi(\mu * f))$, and is multiplicative when f is. For $f = \mathrm{id}$, this becomes $\mathrm{id}^{\cdot 2} * \varphi^2$, giving the proposed identification with A127473. The argument includes $n = 1$.

2.2.3 A130540: even and alternating row sums

The array is $T(n,k) = \sigma(n/k)$ if $k \mid n$, and zero otherwise, for $1 \le k \le n$. Let $R(n) = \sum_{d|n} \sigma(d)$, and set $R(t) = 0$ when t is not a positive integer. The even-column sum is exactly

$$E(n) = \sum_{2j|n} \sigma(n/(2j)) = R(n/2).$$

This proves the second selected conjecture, including the zero values for odd n. The alternating sum, with a positive sign at odd k, is

$$R(n) - 2R(n/2) = \sum_{d|n} \sigma_{\mathrm{odd}}(d),$$

because $\sigma_{\mathrm{odd}}(d) = \sigma(d) - 2\sigma(d/2)$. The last expression is the defining divisor sum for A288417, proving the third conjecture. The separate identification of the odd-column sum with A327096 is not counted or asserted here; only its elementary formula $R(n) - R(n/2)$ follows from this proof.

2.2.4 A001222: the two-parameter convolution

Let Ω count prime factors with multiplicity and ω count distinct prime factors. For all complex x, y and positive integers n,

$$(x+y)^{\Omega(n)} = \sum_{d|n} x^{\Omega(d)} (x+y)^{\Omega(n/d)-\omega(n/d)} y^{\omega(n/d)}.$$

Use $0^0 = 1$. Both sides define multiplicative functions of n, so it suffices to take $n = p^e$. The right side is

$$x^e + y \sum_{i=0}^{e-1} x^i (x+y)^{e-i-1} = (x+y)^e.$$

This is a polynomial telescoping identity, so no division by x, y, or $x+y$ is required. It therefore includes all degenerate parameter values. The source already credits the normalized case $x+y=1$ to earlier work; the selected assertion is the displayed general-parameter version, with that antecedent explicitly retained.

2.2.5 A265649: two finite sums

Write $p_n(x) = \sum_k T(n,k) x^k$, where $T(0,0) = 1$ and $T(n,k) = T(n-1,k-1) + (2n-1+k)T(n-1,k)$. Its exponential generating function is

$$E(z,x) = \sum_{n \geq 0} p_n(x) \frac{z^n}{n!} = g \exp(x(g-1)), \qquad g = (1-2z)^{-1/2}.$$

For example, the recurrence gives $p_{n+1} = (2n+1+x)p_n + x p_n'$, and direct differentiation verifies the corresponding differential equation and the initial value $E(0,x) = 1$. Hence $T(n,k) = \frac{n!}{k!}[z^n] g(g-1)^k$.

First sum. For $0 \leq k \leq n$,

$$T(n,k) = 2^{n-k}(n-k)! \binom{n}{k} \sum_{j=0}^{n-k} (-1/4)^j \binom{2j+k}{j} \binom{n}{j+k}.$$

After division by $n!/k!$ and summation in n, the proposed right side becomes

$$\frac{z^k}{(1-2z)^{k+1}} \sum_{j \geq 0} \binom{2j+k}{j} \left(\frac{-z}{2(1-2z)} \right)^j.$$

The Catalan identity $\sum_{j \geq 0} \binom{2j+k}{j} w^j = C(w)^k / \sqrt{1-4w}$, with $C(w) = 2/(1+\sqrt{1-4w})$, follows from the binomial theorem (or Lagrange inversion). Here $\sqrt{1-4w} = g$. Substitution, using $2zg^2 = g^2 - 1$, reduces the last display to $g(g-1)^k$, as required.

Second sum. Let $B(m,j)$ be the reverse Bessel coefficients A001497, and let $L(m,k)$ be the signed Laguerre coefficients A021009. For $n \geq 1$,

$$T(n,k) = (-1)^k \sum_{j=0}^{n-1} B(n-1,j) L(j+1,k).$$

Put $R_m(x) = m! L_m(-x) = \sum_k m! \binom{m}{k} x^k / k!$. Then

$$\sum_{m \geq 0} R_m(x) \frac{y^m}{m!} = \frac{1}{1-y} \exp\left(\frac{xy}{1-y} \right), \qquad \sum_{m \geq 0} \sum_j B(m,j) y^j \frac{z^m}{m!} = g \exp(y(1-1/g)).$$

Consequently the derivative, with respect to z, of the exponential generating function of the proposed rows is

$$g\sum_{j\ge 0} R_{j+1}(x)\frac{(1-1/g)^j}{j!} = g^3(1+xg)e^{x(g-1)} = E_z(z,x).$$

This proves all rows $n \ge 1$. The empty sum at $n = 0$ is not claimed to equal $T(0,0)$; its initial row is stated separately.

2.2.6 A097724: Chebyshev translation

Let $u_n(x) = U_n(x/2)$, so that $u_0 = 1$, $u_1 = x$, and $u_n = xu_{n-1} - u_{n-2}$. The selected connection-coefficient conjecture is

$$\boxed{u_n(x+1) = \sum_{k=0}^{n} T(n,k)u_k(x).}$$

In this orientation it agrees with the examples in the entry, whose description reverses a row when listing the basis in descending degree. The known column generating functions are $G(z)(zG(z))^k$, where

$$z^2G^2 - (1 - z + z^2)G + 1 = 0, \qquad G(0) = 1.$$

Therefore

$$\sum_{n\ge 0}\sum_{k} T(n,k)u_k(x)z^n = \frac{G}{1 - xzG + z^2G^2} = \frac{1}{1-(x+1)z+z^2}.$$

The final expression is the generating function of $u_n(x+1)$. The related entry A191579 adds no separate count.

2.2.7 A102086: rows of a self-squaring triangle

The lower triangular matrix T has diagonal $T(n,n) = n+1$ and $T(n,k) = (T^2)(n-1,k)$ for $k < n$. If L is the one-step downward shift and $D = \mathrm{diag}(1,2,3,\ldots)$, this is precisely

$$T = D + LT^2.$$

The conjectured polynomial recurrence is $P(n,n) = x^n$, $P(n,0) = 1$, and $P(n,k) = P(n,k-1) + (k+1)P(n-1,k)$ for $0 < k < n$. We prove that $P(n+1,n) = \sum_j T(n,j)x^j$.

For $n \ge k$, define

$$Q(n,k) = \sum_{j=0}^{k}(T^{n-k})(k,j)x^j.$$

Clearly $Q(k,k) = x^k$ and $Q(n,0) = 1$. Multiplying the matrix equation on the right by T^{m-1} gives $T^m = DT^{m-1} + LT^{m+1}$. Its row k, with $m = n-k \ge 1$, says

$$Q(n,k) = (k+1)Q(n-1,k) + Q(n,k-1).$$

Uniqueness in the triangular recurrence gives $Q = P$, and the result follows by taking $n - k = 1$. The same proof works for any diagonal $D = \mathrm{diag}(1,d_1,d_2,\ldots)$, with $k+1$ replaced by d_k. The source links an earlier proof of a different column-vector algorithm; the selected target here is its later row-polynomial assertion.

2.2.8 A379907: a Riordan convolution and column relation

For ordinary Riordan arrays $A = (g, f)$ and $B = (u, v)$, meaning $A(n, k) = [t^n]g(t)f(t)^k$ and similarly for B, the general selected identity is

$$\sum_{i=0}^{n-k} A(n-k, i)B(k+i, k) = [t^n]g(t)u(f(t)) \left(\frac{t\, v(f(t))}{f(t)}\right)^k.$$

Since $B(k+i, k) = [s^i]u(s)(v(s)/s)^k$, substitution in $g(t)f(t)^i$ proves the identity immediately. The quotients are well-defined formal series because f and v have order one.

For the particular array in this entry, put $G = C(t/(1+t))/(1+t)$ and $F = tC(t/(1+t))$, so $T = (G, F)$. The Catalan equation yields $F^2 = (1+t)F - t(1+t)$. Multiplication by GF^{k-2} and coefficient extraction prove, for $k \geq 2$,

$$T(n, k) = T(n, k-1) + T(n-1, k-1) - T(n-1, k-2) - T(n-2, k-2),$$

where entries outside the triangle are zero. Both assertions belong to one counted selection.

2.2.9 A380191: central-binomial weights

Let $D = 1 + xD^3$ with $D(0) = 1$. The established Riordan form of the source array is $T = (2 - D, xD)$. For each $k \geq 0$, the generating function of the selected weighted column sum is

$$\sum_{n \geq k} \sum_{i \geq 0} \binom{2i}{i} T(n, k+i)x^n = \frac{(2-D)(xD)^k}{\sqrt{1-4xD}}.$$

Since $xD = (D-1)/D^2$ and the square root has constant term one, $\sqrt{1-4xD} = (2-D)/D$. The result is $D(x)(xD(x))^k$, exactly the column generating function of A110616. Thus the identification holds for every $n \geq k \geq 0$.

2.2.10 A004737: a false matrix claim and a true Gram formula

Let $q \geq 1$, $\theta_a = (2a-1)\pi/(2q)$, and

$$E_{a,j} = \frac{\sin(j\theta_a)}{\sin\theta_a} = U_{j-1}(\cos\theta_a), \qquad 1 \leq a, j \leq q.$$

The source's proposed integrality and symmetry of E fail already at $q = 2$:

$$E = \begin{pmatrix} 1 & \sqrt{2} \\ 1 & -\sqrt{2} \end{pmatrix}.$$

However, the normalized Gram matrix has the exact formula

$$\boxed{\frac{1}{q}(E^T E)_{ij} = \begin{cases} \min(i, j), & i \equiv j \pmod 2, \\ 0, & i \not\equiv j \pmod 2. \end{cases}}$$

To prove it, the average over $a = 1, \ldots, q$ of $\cos(\ell\theta_a)$ is zero for $1 \leq \ell \leq 2q-2$: odd ℓ cancel in symmetric pairs, and even ℓ give a geometric sum. Hence the average of $U_r(\cos\theta_a)$ is one for even r and zero for odd r, for $0 \leq r \leq 2q-2$. Apply the product identity

$$U_{i-1}(t)U_{j-1}(t) = \sum_{r=0}^{\min(i,j)-1} U_{i+j-2-2r}(t).$$

All summands have the same parity, proving the formula. In particular the Gram matrix is integral and symmetric, and its even antidiagonals give the stated increasing-then-decreasing integer pattern. This entry is one mixed proof/refutation selection.

2.2.11 A153197: formal convergence of alternating transforms

Uncounted qualification. The source also records an explicit iteration program and established fixed-point formulas. The precise normalization below is mathematically useful, but the relationship between the original wording and its broader conjectural extension is not sharp enough to count this as another settled open assertion.

We use the prepending convention described with the source conjecture. For a formal power series F, define

$$(\mathcal{B}F)(x) = \frac{1}{1-x}F\left(\frac{x}{1-x}\right), \qquad (\mathcal{H}G)(x) = \frac{1}{1-x-x^2G(x)}.$$

Here $\mathcal{B}$ is the binomial transform. To see the normalization of $\mathcal{H}$, prepend 1 to G to get $A = 1+xG$, apply the invert transform $A/(1-xA)$, and prepend 1 once more; the result is $1+xA/(1-xA) = \mathcal{H}G$.

The x-adic order of $\mathcal{B}F - \mathcal{B}\widetilde{F}$ equals that of $F - \widetilde{F}$. Direct subtraction shows that $\mathcal{H}$ raises the order of a difference by two. Therefore $\mathcal{H}\mathcal{B}$ is a strict contraction in the complete ring of formal power series. Its iterates from any initial series converge coefficient by coefficient to a unique fixed point F_*. The interleaved iterates converge to $G_* = \mathcal{B}F_*$. This proves the selected convergence assertion, and in fact removes the need for the initial Hankel-transform condition.

For identification, the Jacobi continued fraction

$$F_* = \cfrac{1}{1-x-\cfrac{x^2}{1-2x-\cfrac{x^2}{1-3x-\cdots}}}$$

has diagonal coefficients $1, 2, 3, \ldots$ and all off-diagonal squared coefficients equal to one. The binomial transform adds one to every diagonal coefficient; this follows by substituting $x/(1-x)$ in each level of the fraction and canceling $1-x$. Applying $\mathcal{H}$ then puts back the first level with coefficient one, so this fraction is the fixed point just proved unique. It is A006789; its binomial transform is the defining series for A153197. The source's less precise extension to arbitrary Hankel transforms is outside this selection.

2.2.12 A005773: universally ordered coprime divisors

We give a precise version of the source's empirical comparison assertion: an inequality is deducible from prime order alone when it holds for every increasing list of primes $p_1 < \cdots < p_n$. For disjoint index sets I, J, the coprime divisors $d_I = \prod_{i\in I} p_i$ and $d_J = \prod_{j\in J} p_j$ satisfy $d_I \le d_J$ for every such list if and only if

$$|J \cap \{r, \ldots, n\}| \ge |I \cap \{r, \ldots, n\}| \quad (1 \le r \le n).$$

Sufficiency follows by matching each index of I, starting with the largest, to a distinct index of J that is at least as large. Extra factors in J are positive integers greater than one.

For necessity, suppose a suffix has excess $\delta > 0$ indices from I. Fix the earlier primes. By repeated application of Bertrand's theorem, the remaining m primes can all be chosen between X and $2^m X$ for arbitrarily large X. The ratio d_I/d_J is then at least a fixed positive constant times $X^\delta/2^{m|J\cap\{r,\ldots,n\}|}$, and eventually exceeds one, a contradiction.

Read indices in order $n, n-1, \ldots, 1$, writing an up-step for J, a down-step for I, and a horizontal step for neither. The displayed condition is precisely that every initial height is nonnegative. These are Motzkin meanders of length n, counted by A005773$(n+1)$ with the source's stated length-to-index shift. The empty pair gives the initial case. Other assertions attached to this entry are not included.

2.2.13 A376182: the triangular enumeration is a permutation

For $n \geq 1$ set

$$C_n = \frac{2n^2 + 4n + 1 - (-1)^n}{4}, \qquad T(n,k) = \begin{cases} C_n - k, & k \text{ odd}, \\ C_n - 2n + k, & k \text{ even}. \end{cases}$$

Every C_n is even. In row n, the odd entries form the block of odd integers from $C_n - 2\lceil n/2 \rceil + 1$ to $C_n - 1$. The even entries form the block from $C_n - 2n + 2$ to $C_n - 2\lceil n/2 \rceil$, when this block is nonempty. Since

$$C_n - C_{n-1} = 2\lceil n/2 \rceil, \qquad C_n - C_{n-2} = 2n,$$

each odd block starts two after the preceding odd block ends, and the same is true of the nonempty even blocks. The first odd value is one and the first even value is two; the block endpoints tend to infinity. Thus every positive integer occurs exactly once, proving the conjecture.

2.2.14 A378277: alternating harmonic Fibonacci rows

Take $F_0 = 0, F_1 = 1$ and $F_{-1} = 1$. The denominator array has $D(n,k) = F_k F_{k+2}$ for $k < n$, and $D(n,n) = F_n F_{n+1}$. The selected identity is

$$H_n := \sum_{k=1}^{n} \frac{(-1)^{k-1}}{D(n,k)} = \frac{F_{n-2}}{F_{n+1}}, \qquad n \geq 1.$$

The initial case is $H_1 = 1$. For $n \geq 2$, subtraction of consecutive rows gives

$$H_n - H_{n-1} = \frac{2(-1)^{n-1}}{F_n F_{n+1}}.$$

The proposed right side has the same difference, because $F_n F_{n-2} - F_{n+1} F_{n-3} = 2(-1)^{n-1}$. This last identity follows from the Fibonacci recurrence and Cassini's identity, including $n = 2$ with the stated negative-index convention. Induction proves the claim.

2.2.15 A349802: exact Euler transforms of Lyndon columns

Let $T(n,k)$ count binary Lyndon words of length n beginning with exactly k zeros, and let $E_k(x) = \prod_{n \geq 1} (1 - x^n)^{-T(n,k)}$ for $k \geq 1$. The precise form of the source's Fibonacci and higher-bonacci assertion is

$$\boxed{E_1(x) = \frac{1}{1 - x - x^2}, \qquad E_k(x) = \frac{1 - x - \cdots - x^k}{1 - x - \cdots - x^{k+1}} \quad (k \geq 2).}$$

In particular, for $k \geq 2$ the series is $1 + x^{k+1}/(1 - x - \cdots - x^{k+1})$; this states its initial terms as well as its higher-bonacci tail, avoiding an unspecified shift.

Every nonconstant binary Lyndon word starts at a longest cyclic run of zeros. Decompose its cyclic word into blocks $0^j 1$. Primitivity is preserved by this block encoding, and the blocks are a weighted alphabet with weights $1, 2, \ldots, k+1$. Beginning with exactly k zeros is equivalent to having maximum block weight $k+1$.

For any weighted alphabet of inventory $S(x)$, unique factorization of words into a nonincreasing product of Lyndon words gives

$$\prod_{\text{primitive necklaces } w} (1 - x^{|w|})^{-1} = \frac{1}{1 - S(x)}.$$

This identity can alternatively be obtained by taking logarithms and partitioning words by their primitive period. Dividing the identity for weights $1, \ldots, k+1$ by the one for weights $1, \ldots, k$ proves the second formula. For $k = 1$, the one-letter word 0 contributes the additional factor $(1 - x)^{-1}$, giving the first formula. The identity and its normalization are the counted result; the constant term is not silently identified with a different convention for higher-bonacci numbers.

2.2.16 A163774: collapsing the binomial sum

Using the established even swing-number identity $\mathrm{swing}(2i) = \binom{2i}{i}$, the definition is

$$a(n) = \sum_{k=0}^{n} \sum_{i=k}^{n} (-1)^{n-i} \binom{n-k}{n-i} \binom{2i}{i}.$$

The finite sums can be interchanged. The hockey-stick identity gives $\sum_{k=0}^{i} \binom{n-k}{n-i} = \binom{n+1}{i}$. Therefore

$$\boxed{a(n) = \sum_{i=0}^{n} (-1)^{n-i} \binom{n+1}{i} \binom{2i}{i},}$$

which is the selected conjecture for every $n \geq 0$.

2.2.17 A152920: weighted rows

The array is $T(N,k) = (2N-k)2^{k-1}$, with its $k=0$ value understood as N. Define

$$B(N,p) = \frac{1}{N+p+1} \sum_{k=0}^{N} \binom{p+k}{p} T(N,k).$$

The source conjectures $B(n+1,p) = A(n,p)$, where $A(n,p) = \texttt{A193844}(n+p,n)$. The known bivariate generating function of the latter triangle, $1/((1-xy)(1-x-2xy))$, gives for fixed $p \geq 0$

$$A_p(x) = \sum_{n \geq 0} A(n,p) x^n = \frac{1}{(1-x)(1-2x)^{p+1}}.$$

Indeed substitute $x = u, y = v/u$ and extract $[u^p]$. Put $N = k + r$ in the unnormalized sum, and use the negative-binomial series and its derivative. The result is

$$\begin{aligned} \sum_{N \geq 0} \sum_{k=0}^{N} \binom{p+k}{p} (2N-k) 2^{k-1} x^N &= \frac{x((p+2) - (p+3)x)}{(1-x)^2 (1-2x)^{p+2}} \\ &= x((p+2)A_p(x) + xA_p'(x)). \end{aligned}$$

The coefficient of x^N is $(N+p+1)A(N-1,p)$ for $N \geq 1$. Division by the nonzero factor proves the result.

2.2.18 A105794: factorial and Bernoulli sums

The defining connection formula is $(x-1)^n = \sum_k T(n,k) x_{\underline{k}}$, equivalently the column exponential generating function is $e^{-z}(e^z-1)^k/k!$. The two selected 2025 conjectures follow directly from this representation.

First, set $x = -m-1$ for any integer $m \geq 0$. Since $(-m-1)_{\underline{k}} = (-1)^k (k+m)!/m!$, one obtains

$$\sum_{k=0}^{n} (-1)^{n-k} T(n,k) \frac{(k+m)!}{m!} = (m+2)^n.$$

Second, with Bernoulli convention $B_1 = -1/2$,

$$\sum_{k=1}^{n} (-1)^k T(n,k) \frac{(k-1)!}{k+1} = (-1)^n - B_n.$$

For the proof, the exponential generating function of the left side is

$$e^{-z} \sum_{k \geq 1} \frac{(-1)^k (e^z-1)^k}{k(k+1)} = e^{-z} - \frac{z}{e^z - 1}.$$

Here we used $\sum_{k \geq 1} (-1)^k w^k/(k(k+1)) = 1 - (1+w)\log(1+w)/w$. Coefficient extraction proves the result, including $n = 0$.

2.2.19 A132393: a Vandermonde-squared multisum

Let $c(a,b)$ denote unsigned Stirling numbers of the first kind. The selected May 2026 conjecture is, for integers $n, m \geq 0$,

$$c(n+m+1, m+1) = (n+m)! \prod_{i=1}^{m} (n+i)^i \times \sum_{n \geq j_1 \geq \cdots \geq j_m \geq 0} (-1)^{nm+\sum_t j_t} \prod_{1 \leq q < p \leq m} (j_q - j_p + p - q)^2 \times \prod_{t=1}^{m} \frac{(n+t-1)_{\underline{j_t}}}{(n-j_t+t)^{m+1}(j_t+m-t)!}.$$

All products and the nested sum for $m = 0$ have value one, giving the correct $n!$; below assume $m \geq 1$ and put $N = n + m$.

Set $r_t = j_t + m - t$ and $s_t = N - r_t$. This is a bijection from the summation region to $1 \leq s_1 < \cdots < s_m \leq N$. The squared product becomes the squared Vandermonde $V(s)^2$, and the sign becomes $(-1)^{m(m+1)/2+\sum s_t}$. The factorial identity

$$\prod_{i=1}^{m} (n+i)^i \prod_{t=1}^{m} (n+t-1)! = (N!)^m$$

and $\binom{N-1}{s-1} = \frac{s}{N}\binom{N}{s}$ reduce the proposed right side exactly to

$$N!(-1)^{m(m+1)/2} \sum_{1 \leq s_1 < \cdots < s_m \leq N} V(s)^2 \prod_{t=1}^{m} \frac{(-1)^{s_t}\binom{N}{s_t}}{s_t^m}.$$

By Cauchy–Binet, the sum is $\det H$, where

$$H_{a,b} = \sum_{s=1}^{N} (-1)^s \binom{N}{s} s^{a+b-m}, \qquad 0 \leq a, b < m.$$

Let h_j be the complete homogeneous symmetric function of degree j in $1, 1/2, \ldots, 1/N$, with $h_0 = 1$ and $h_j = 0$ for $j < 0$. Partial fractions give

$$\prod_{r=1}^{N} (1 - z/r)^{-1} = \sum_{s=1}^{N} \frac{(-1)^{s-1}\binom{N}{s}}{1 - z/s}.$$

Thus $H_{a,b} = -h_{m-a-b}$ whenever $m - a - b \geq 0$. When the exponent $a + b - m$ is positive, it is at most $m - 2 < N$; the corresponding binomial finite difference is zero, giving the same formula with $h_j = 0$. Reverse the m columns. The resulting entries are $-h_{1-a+b}$, hence

$$\det H = (-1)^{m(m+1)/2} \det[h_{1-a+b}]_{a,b=0}^{m-1} = (-1)^{m(m+1)/2} e_m(1, 1/2, \ldots, 1/N).$$

The last determinant identity is the single-column Jacobi–Trudi identity; it also follows recursively by multiplying $\sum h_j z^j$ by $\sum (-1)^j e_j z^j$ to obtain one. Finally $N! e_m(1, 1/2, \ldots, 1/N) = c(N+1, m+1)$, from $t \prod_{r=1}^{N} (t + r)$. This proves the full multisum, including all boundary cases stated above.

2.2.20 A397048: a Stirling convolution

The defining triangle is

$$T(n,k) = \frac{1}{k!} \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} (n+i)^n.$$

We prove the first conjecture only:

$$T(n,k)=\sum_{i=0}^{n-k} s(n,k+i)S(n+k+i,n+k),$$

where s is the signed first-kind Stirling triangle A048994 and S is the second-kind triangle A048993. The elementary and complete symmetric representations of these numbers turn the right side into

$$[z^{n-k}]\frac{\prod_{j=1}^{n-1}(1-jz)}{\prod_{j=1}^{n+k}(1-jz)}=h_{n-k}(n,n+1,\dots,n+k).$$

For distinct $x_0,\dots,x_k$, partial fractions of $\prod_i(1-x_iz)^{-1}$ give

$$h_{n-k}(x_0,\dots,x_k)=\sum_{i=0}^{k}\frac{x_i^n}{\prod_{j\neq i}(x_i-x_j)}.$$

At $x_i=n+i$, the denominator is $(-1)^{k-i}i!(k-i)!$, exactly the defining finite difference. The case $n=k=0$ is checked separately as $1=1$. The source's other conjectures are not needed for this selection.

2.2.21 A112302: a Lerch-transcendent limit

Define $L_0=1$ and

$$nL_n=\sum_{k=0}^{n-1}\Phi(1/2,n-k,1)L_k,\qquad n\geq 1,$$

where $\Phi(1/2,r,1)=\sum_{j\geq 0}2^{-j}(j+1)^{-r}$. The selected 2016 conjecture states that L_n tends to Somos's quadratic recurrence constant $\prod_{j\geq 1}j^{2^{-j}}$.

Logarithmic differentiation of $F(x)=\sum_{n\geq 0}L_nx^n$ gives, for $|x|<1$,

$$F(x)=\prod_{j\geq 0}\left(1-\frac{x}{j+1}\right)^{-2^{-j}}=\frac{H(x)}{1-x}.$$

The product H, obtained by restricting to $j\geq 1$, converges analytically on $|x|<2$ and has nonnegative coefficients h_r. Thus $L_n=\sum_{r=0}^{n}h_r$ and $L_n\uparrow H(1)$. Finally

$$\begin{aligned}\log H(1)&=\sum_{j\geq 1}2^{-j}\log\frac{j+1}{j}\\&=\sum_{j\geq 1}2^{-j}\log j,\end{aligned}$$

by shifting the absolutely convergent series. This is the logarithm of the proposed constant. The proof also supplies the explicit error bound $0\leq H(1)-L_n\leq\rho^{-n-1}H(\rho)$ for every $1<\rho<2$. The product representation of the constant was already known; the selected result is convergence of the particular Lerch recurrence.

2.3 Arithmetic and congruences

2.3.1 A334595: right boundaries of XOR triangles

Source: https://oeis.org/A334595. Peter Kagey's entry lists the two assertions below as conjectures.

Theorem 2.3.1. *For the binary value $a(n)$ obtained by reading the right boundary of the XOR triangle of n, $a(n) = 1$ exactly when n is a power of two. The strict record positions are 1 and $2^k + 1$ for $k \geq 1$.*

Proof. Let the input word have length L and leading digit 1. Reverse it, obtaining digits $x_0, \ldots, x_{L-1}$. The successive right-boundary digits, read from top to bottom, are

$$y_j = \sum_{i=0}^{j} \binom{j}{i} x_i \pmod 2, \qquad 0 \leq j < L.$$

The lower Pascal matrix $P = (\binom{j}{i})$ is an involution over $\mathbb{F}_2$, since its square has (j, i) entry $\binom{j}{i} 2^{j-i}$, equal to 1 for $j = i$ and 0 otherwise. Thus the boundary map is one-to-one on words of length L, including words with leading zeros.

The input $10\cdots0$ has boundary $00\cdots01$, hence value 1. Invertibility proves that no other length-L input gives this value. The all-ones boundary has inverse $x = (1, 0, \ldots, 0)$: this follows from $\sum_{i=0}^{j} \binom{j}{i} = 2^j$. For $L \geq 2$ the corresponding unreversed input has leading digit zero and is inadmissible. Therefore every admissible boundary has value at most $2^L - 2$.

The input $10\cdots01$ is the sum over $\mathbb{F}_2$ of $10\cdots0$ and $00\cdots01$. Their boundaries are respectively $00\cdots01$ and $11\cdots11$, so its boundary is $11\cdots10$, of value $2^L - 2$. The first input of length L is 2^{L-1} and has value 1; the second, $2^{L-1} + 1$, attains this length's maximum. All inputs of shorter length have value at most $2^{L-1} - 2$, except the single length-one value 1, which is still smaller when $L = 2$. Thus each second input is a strict record and no subsequent input of the same length is. The initial input 1 is the remaining record. □

2.3.2 A133910: divisibility by the index

Source: https://oeis.org/A133910. The target is Ridouane Oudra's January 2026 conjecture. We use Hari Rajesh's October 2025 product formula explicitly recorded there.

For $n > 1$ write $n = \prod_p p^{e_p}$ and put

$$P(n) = \max_{p \mid n} p^{e_p}.$$

The set A051283 consists of $n > 1$ such that $n > P(n)^2$.

Theorem 2.3.2. *For A133910, $n \mid a(n)$ if and only if $n = 1$ or $n \in$ A051283.*

Proof. The recorded formula is

$$a(n) = \prod_{p \mid n} p^{\lfloor \log_p n \rfloor - e_p}.$$

Consequently $n \mid a(n)$ is equivalent to $\lfloor \log_p n \rfloor \geq 2e_p$ for every $p \mid n$, or to $n \geq P(n)^2$. Equality is impossible for $n > 1$: if $P(n) = p^{e_p}$ and $n = P(n)^2 = p^{2e_p}$, the exact exponent of p in n would be both e_p and $2e_p$. Therefore the condition is $n > P(n)^2$. At $n = 1$, $a(1) = 1$. □

2.3.3 A051283: an exponent in an initial least common multiple

Sources: https://oeis.org/A051283 and https://oeis.org/A346418. Only the characterization using A346418 is selected here. The other binomial-gcd conjecture has an acknowledged 2026 proof and is excluded.

Theorem 2.3.3. *The members of A051283 are exactly the integers n for which* A346418$(n) > 1$.

Proof. For $n > 1$, A346418 is the greatest exponent r such that n^r divides $L_n = \operatorname{lcm}(1, \ldots, n)$. Hence its value is greater than one exactly when $n^2 \mid L_n$. Since $v_p(L_n) = \lfloor \log_p n \rfloor$, this is equivalent to $2e_p \le \lfloor \log_p n \rfloor$ for every exact prime-power factor p^{e_p} of n. As in the preceding proof, the condition is $n > P(n)^2$. The defined initial value of A346418 is 1, so $n = 1$ belongs to neither set. □

2.3.4 A354995: when the unitary-divisor difference vanishes

Sources: https://oeis.org/A354995 and https://oeis.org/A354933. The latter entry identifies its value $U(n)$ as the least unitary divisor of n at least $\sqrt{n}$. Put $P(1) = U(1) = 1$.

Theorem 2.3.4. *The difference $U(n) - P(n)$ is positive exactly at the members of A051283, and is zero elsewhere. Thus its nonzero positions are exactly the positions conjectured in A354995.*

Proof. Every unitary divisor is a product of whole exact prime-power blocks of n. If $n > P(n)^2$, then every divisor at least $\sqrt{n}$ is greater than $P(n)$, giving $U(n) > P(n)$. If $n < P(n)^2$, the block $P(n)$ itself is a unitary divisor larger than $\sqrt{n}$, so $U(n) \le P(n)$. A unitary divisor $d < P(n)$ cannot contain that block, and therefore divides $n/P(n) < \sqrt{n}$. Thus none of these smaller divisors is eligible, and $U(n) = P(n)$. Equality $n = P(n)^2$ is impossible for $n > 1$ by the exact-exponent argument above. The initial difference is 0. □

2.3.5 A376336: existence of a zero of tau modulo every modulus

Sources: https://oeis.org/A376336 and https://oeis.org/A000594.

Only existence is proved; the separate proposed bound $a(n) \le n^2$ remains outside this result. The standard Hecke recurrence used below follows from Mordell's theorem on Ramanujan's tau function, as referenced in A000594.

Theorem 2.3.5. *For every positive integer N there is a positive integer m for which $N \mid \tau(m)$.*

Proof. The case $N = 1$ is immediate. Choose a prime p not dividing N. Set $u_r = \tau(p^r)$, so $u_0 = 1$, $u_1 = \tau(p)$, and

$$u_r = \tau(p)u_{r-1} - p^{11}u_{r-2} \quad (r \ge 2).$$

The matrix $M = \left(\begin{smallmatrix} \tau(p) & -p^{11} \\ 1 & 0 \end{smallmatrix}\right)$ has determinant p^{11}, a unit modulo N. It therefore belongs to the finite group $\mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$ and has some finite order h. Since its lower-left entry is 1, $M \ne I$ modulo N, so $h \ge 2$.

For every $r \ge 1$, recurrence and induction give

$$M^r \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} u_r \\ u_{r-1} \end{pmatrix} \pmod{N}.$$

Taking $r = h$ gives $u_{h-1} \equiv 0 \pmod{N}$. The integer $m = p^{h-1}$ has the required property, proving the existence of a least such positive m by well-ordering. □

2.3.6 A087726: square-zero matrices over prime-power rings

Source: https://oeis.org/A087726. The prime-modulus case was already proved by Manfred Scheucher in 2015; the proof below covers every exponent and the full squarefree characterization.

Theorem 2.3.6. *Let $A(N)$ count matrices $X \in M_2(\mathbb{Z}/N\mathbb{Z})$ satisfying $X^2 = 0$. For every prime p and $k \geq 1$,*

$$A(p^k) = p^{2k}\left(1 + \left\lfloor \frac{k}{2} \right\rfloor (1 - p^{-2})\right).$$

Equivalently, this is Robert Israel's displayed formula

$$\frac{p^{2k-2}}{4}(1 + 3p^2 + 2k(p^2 - 1) + (-1)^k(p^2 - 1)).$$

Moreover $A(N) = N^2$ exactly when N is squarefree, including $N = 1$.

Proof. Fix $R = \mathbb{Z}/p^k\mathbb{Z}$. Call a matrix primitive if its reduction modulo p is nonzero. A primitive square-zero matrix reduces to a nonzero nilpotent map on the two-dimensional vector space over $\mathbb{F}_p$. There is a vector $\bar{v}$ for which $(X\bar{v}, \bar{v})$ is a basis. Indeed, choose $\bar{v}$ outside the one-dimensional kernel; the nonzero vector $X\bar{v}$ lies in that kernel and is independent of $\bar{v}$.

Lift $\bar{v}$ to $v \in R^2$. The columns (Xv, v) have unit determinant, hence form an R-basis. In this basis X has matrix $J = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Therefore all primitive square-zero matrices form exactly the conjugacy orbit of J.

The invertible matrices commuting with J are precisely $\left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right)$ with a a unit and b arbitrary. Their number is $p^{2k-1}(p - 1)$. Since

$$|\mathrm{GL}_2(R)| = p^{4(k-1)}(p^2 - 1)(p^2 - p),$$

the number of primitive solutions is $p^{2k-2}(p^2 - 1)$.

For $k \geq 2$, a nonprimitive matrix is uniquely $X = pY$ with Y taken modulo p^{k-1}. Its square vanishes modulo p^k exactly when Y^2 vanishes modulo p^{k-2}. Each matrix modulo p^{k-2} has p^4 lifts modulo p^{k-1}. With $A(p^0) = 1$, we obtain

$$A(p^k) = p^{2k-2}(p^2 - 1) + p^4 A(p^{k-2}).$$

For $k = 1$ there are $p^2 - 1$ primitive solutions and the zero matrix, so $A(p) = p^2$. Division by p^{2k} and induction on each parity of k gives the stated formula.

The Chinese remainder theorem identifies the matrix ring modulo a product of coprime moduli with the product of their matrix rings, preserving $X^2 = 0$. Hence A is multiplicative. Each ratio $A(p^k)/p^{2k}$ is 1 for $k = 1$ and strictly greater than 1 for $k \geq 2$. Their product equals 1 exactly for squarefree N. □

2.3.7 A322327: the general parameter statement

Source: https://oeis.org/A322327. The general-k assertion is explicitly conjectured; the $k = 2$ instance is already recorded as a formula.

Theorem 2.3.7. *Let $k \in \mathbb{Z}$ be fixed and let $E(n)$ be the product of the positive exponents in the prime factorization of n, with $E(1) = 1$. Then $f_k(n) = E(n)k^{\omega(n)}$, using $0^0 = 1$, is multiplicative and $f_k(p^e) = ke$ for $e > 0$.*

Proof. For $n = \prod_{j=1}^{r} p_j^{e_j}$ one has

$$f_k(n) = \left(\prod_{j=1}^{r} e_j\right) k^r = \prod_{j=1}^{r} (ke_j).$$

The formula holds for the empty product at $n = 1$, including $k = 0$. For coprime integers the lists of prime powers are disjoint, so the displayed product for their product is the product of their individual values. A single prime-power factor gives ke. □

2.3.8 A033955: a strict bound for the sum of prime remainders

Source: https://oeis.org/A033955. The target is Alonso del Arte's July 2026 conjecture. We use Jitsuro Nagura, "On the interval containing at least one prime number," *Proceedings of the Japan Academy* 28 (1952), 177–181, https://doi.org/10.3792/pja/1195570997. That theorem provides a prime in $(x, 6x/5)$ whenever $x \geq 25$.

Theorem 2.3.8. *For the nth prime $q = p_n$, set $S(q) = \sum_{p<q,\ p\ prime}(q \bmod p)$. If $n > 6$, then $S(q) > q$. The equality $S(p_n) = p_n$ occurs only for $n = 6$.*

Proof. Assume $q \geq 79$. Apply Nagura's theorem at $x = q/3, q/2, 3q/5, 18q/25$. This produces four distinct primes in the disjoint intervals

$$\left(\frac{q}{3}, \frac{2q}{5}\right), \quad \left(\frac{q}{2}, \frac{3q}{5}\right), \quad \left(\frac{3q}{5}, \frac{18q}{25}\right), \quad \left(\frac{18q}{25}, \frac{108q}{125}\right).$$

The first corresponding quotient is 2 and the other three are 1. Their remainders are respectively greater than $q/5, 2q/5, 7q/25, 17q/125$. All other remainders are nonnegative. Consequently

$$S(q) > \left(\frac{1}{5} + \frac{2}{5} + \frac{7}{25} + \frac{17}{125}\right) q = \frac{127}{125} q > q.$$

The primes $17 \leq q < 79$ and their exact sums are

q	17	19	23	29	31	37	41	43
$S(q)$	18	27	29	46	56	70	74	88

q	47	53	59	61	67	71	73
$S(q)$	98	134	147	171	200	217	252.

Each exceeds q. For the first six primes the sums are $0, 1, 3, 4, 8, 13$, establishing the stated unique equality. These finite sums can be checked by the supplied exact verifier; the table exhausts the range not covered by the theorem. □

2.3.9 A166306: the empirical residue restriction

Source: https://oeis.org/A166306. Only the assertion about residues modulo 9 is selected.

Theorem 2.3.9. *No term of A166306 is divisible by 3. Equivalently its residue modulo 9 always lies in $\{1, 2, 4, 5, 7, 8\}$.*

Proof. The first term corresponds to B_1 and is 1. For an even Bernoulli index $2m$, $m \geq 1$, let

$$\mathcal{P} = \{p \text{ prime} : p - 1 \mid 2m\}, \qquad D = \prod_{p \in \mathcal{P}} p.$$

By the definition using the von Staudt–Clausen primes, the term is $\sum_{p \in \mathcal{P}} D/p$. The prime 3 belongs to $\mathcal{P}$, and D is squarefree. Modulo 3 every summand except $D/3$ vanishes; $D/3$ is not divisible by 3. Thus the sum is not divisible by 3. □

2.3.10 A234307: an alternating gcd identity

Source: https://oeis.org/A234307. Peter Bala's January 2024 assertion is proved below.

Theorem 2.3.10. *For every $n \geq 1$,*

$$\sum_{i=1}^{n} \gcd(2n-i, i) = \frac{1}{4} \sum_{k=1}^{4n} (-1)^k \gcd(k, 8n).$$

Proof. Write $G(m) = \sum_{k=1}^{m} \gcd(k, m)$. Pairing k with $2n-k$ gives

$$\sum_{i=1}^{n} \gcd(2n-i, i) = \frac{G(2n) - n}{2}.$$

The full alternating sum at modulus $8n$ is

$$H = \sum_{k=1}^{8n} (-1)^k \gcd(k, 8n) = 4G(4n) - G(8n),$$

because the even-index contribution is $2G(4n)$. If T is the half sum from 1 through $4n$, reflection about $4n$ shows $H = 2T + 4n$: the two endpoint terms are $4n$ and $8n$.

The formula $G(m) = \sum_{d|m} d\varphi(m/d)$ proves that G is multiplicative. Directly,

$$G(2^r) = (r+2)2^{r-1} \qquad (r \geq 1).$$

Writing $2n = 2^r u$ with u odd and $r \geq 1$, this yields

$$4G(4n) - G(8n) = 4G(2n).$$

Therefore $T = 2G(2n) - 2n$, and $T/4 = (G(2n) - n)/2$, as required. □

2.3.11 A290815: the odd-modulus reciprocal-square characterization

Source: https://oeis.org/A290815. This proves the empirical odd-term characterization of August 2019. It does not resolve the separate conjecture for arbitrary exponents.

Theorem 2.3.11. *For every odd integer $m \geq 1$, let*

$$S_j(m) = \sum_{\substack{1 \leq k \leq m \\ (k,m)=1}} \frac{1}{k^j}.$$

Then $m^3 \mid \mathrm{num}(S_1(m))$ if and only if $m^2 \mid \mathrm{num}(S_2(m))$, where the numerator is in lowest terms.

Proof. For $m = 1$ both divisibilities are automatic. Assume $m > 1$ is odd. All denominators are coprime to m, so congruences can be computed in the localization of $\mathbb{Z}$ obtained by inverting integers coprime to m. Divisibility of a reduced numerator by m^r is exactly congruence to zero modulo m^r in this ring.

The substitution $k \mapsto m - k$ permutes the reduced residues. Modulo m, it gives $S_3 \equiv -S_3$, so $S_3 \equiv 0 \pmod{m}$ because 2 is a unit. Expansion of $(m-k)^{-1}$ modulo m^3 gives

$$\frac{1}{m-k} \equiv -\frac{1}{k} - \frac{m}{k^2} - \frac{m^2}{k^3} \pmod{m^3}.$$

Summing and again using the permutation gives

$$2S_1 \equiv -mS_2 - m^2 S_3 \equiv -mS_2 \pmod{m^3}.$$

Since 2 is invertible, $S_1 \equiv 0 \pmod{m^3}$ is equivalent to $S_2 \equiv 0 \pmod{m^2}$, proving both directions. □

2.4 Second round: connected arrays and exact floor identities

This section resolves 28 distinct entry-level assertions, all disjoint from the first-round ledger. The primary sources are the current OEIS entries at `https://oeis.org/`*A-number*, inspected on 20 October 2026. The selected formulas remain explicitly conjectural or empirical in those entries. The source status is not a claim of global publication priority. We exclude A166761, which records Howroyd's 2024 confirmation, and nearby entries whose formulas are already stated as established or which contain no conjectural assertion.

2.4.1 Five connected-coloring formulas from a two-state transfer matrix

For $q \geq 1$, let $a_q(n)$ count $n \times 2$ arrays over q labeled colors, using every color, in which the cells of each color form one connected region under horizontal and vertical adjacency. Define

$$B(x,t) = \frac{xt(1+t+(t-1)x)}{(1-x)^2 - xt(x+3) - xt^2(x+1)}.$$

Then the complete ordinary generating function is

$$\sum_{n\geq 1} a_q(n)x^n = q![t^q]B(x,t). \tag{2.1}$$

Taking $q = 3,4,5,6,7$ proves the empirical generating functions in the five entries A166756–A166760.

Proof. First forget the names of the colors, and give each monochromatic region a new name at its first appearance. Mark its introduction by t. Once a color has disappeared from the last processed row, connectivity prevents it from reappearing. The last row has one of two types: M (both cells the same color) or D (two different colors). In width two, each active color has a single connected component.

For an M row, the next M row either retains its color or starts a new one, giving $1+t$. A next D row can retain the old color in either position and introduce one new color, or introduce two new colors, giving $2t+t^2$. For a D row, a next M row can retain either of its two colors or introduce a new one, giving $2+t$. A next D row can retain both colors in their original positions, retain exactly one in its original position and introduce a new one, or introduce two new colors, giving $1+2t+t^2$. A color moved only diagonally would be disconnected and is excluded. Thus the exact transfer matrix is

$$T(t) = \begin{pmatrix} 1+t & 2t+t^2 \\ 2+t & 1+2t+t^2 \end{pmatrix}.$$

The first row contributes (t,t^2), and both frontier types are accepting. Therefore the generating function for the unnamed region partitions is $x(t,t^2)(I-xT(t))^{-1}(1,1)^{\mathsf{T}} = B(x,t)$. A partition into exactly q nonempty regions has exactly $q!$ labelings by the required colors, proving (2.1). □

For a directly reproducible coefficient extraction, write $b_q(x) = [t^q]B(x,t)$. Then

$$b_1 = \frac{x}{1-x}, \qquad b_2 = \frac{x(1+3x)}{(1-x)^3}, \qquad b_q = \frac{x(x+3)b_{q-1} + x(x+1)b_{q-2}}{(1-x)^2} \quad (q \geq 3).$$

The following table gives the resulting source functions in the form

$$G_q(x) = \frac{q!x^h \sum_{j=0}^{m} v_j x^j}{(1-x)^{2q-1}}.$$

All coefficient lists are ascending, starting with v_0.

Entry	q	h	$(v_0, \ldots, v_m)$
A166756	3	2	$4, 9, 2, 1$
A166757	4	2	$1, 14, 27, 13, 8, 1$
A166758	5	3	$7, 48, 84, 60, 43, 12, 2$
A166759	6	3	$1, 34, 163, 273, 246, 196, 83, 25, 3$
A166760	7	4	$10, 144, 552, 917, 958, 827, 452, 187, 44, 5$

2.4.2 Sixteen formulas from deleting repeated rows

Consider arrays whose rows are lexicographically nondecreasing. Some of the entries additionally require the columns to be lexicographically nondecreasing. In each case all prescribed colors must occur and each color must be connected.

Let c_j be the number of valid arrays with exactly j distinct rows, with those rows in strictly increasing order, and no repeated row. The following identity holds for every positive integer n:

$$a(n) = \sum_{j \geq 1} c_j \binom{n-1}{j-1}, \qquad \sum_{n \geq 1} a(n) x^n = \sum_{j \geq 1} c_j \left(\frac{x}{1-x} \right)^j . \tag{2.2}$$

The sums are finite because there are only q^w possible rows for width w and q colors.

Proof. Equal rows occur consecutively. Contracting vertical edges between two identical consecutive rows deletes one copy and preserves the connected components of every color. It also preserves which colors occur, the row ordering, and every lexicographic comparison between columns. Conversely, duplicating a row preserves the same properties. Thus a valid array with j distinct rows is uniquely a valid strict-row array and a choice of j positive multiplicities summing to n. There are $\binom{n-1}{j-1}$ such choices. Summing their generating functions proves both identities. □

Exact finite computation of c_j. For completeness, the finite enumeration used in the coefficient table is specified here. A state consists of the last row r, a partition π of its positions into connected components in the processed grid, the set U of colors already used, and (when needed) one bit for each adjacent pair of columns indicating whether the lexicographic comparison has already become strict. A partition block contains positions of only one color. Colors in U absent from r have been completed.

To append a proposed row r', perform these exact steps:

1. Require $r' > r$ for the strict-row enumeration. For each unsettled column comparison, reject if the new left entry exceeds the right entry, and mark it settled if the left entry is smaller.
2. Reject a color that was completed earlier. Form the disjoint union of the old frontier components and the new cells. Join vertically adjacent equal colors and horizontally adjacent equal colors in r'.
3. If an old component has no representative in the new row, its color must have no other component and must be absent from the new row. Otherwise reject. Such a color is now completed.
4. Restrict the resulting component partition to r' and update U. At the end accept exactly when every color has occurred and every color still in the frontier has a single component.

These rules are equivalent to extendible monochromatic connectivity: a component that has left the frontier can never join a future cell, while all components still meeting the frontier remain available to join. Consequently the state paths are in bijection with the arrays being counted.

Every transition strictly increases the row, so the state graph is acyclic. Starting with polynomial 1 at the empty state, add z times each state's polynomial to all its successors, in increasing row order. The sum over accepting states is exactly $\sum c_j z^j$. This algorithm terminates after at most q^w rows and establishes that every unlisted coefficient is zero. It does not infer a formula from an initial segment of the infinite sequence.

The table gives the complete nonzero coefficient vectors, with h the first listed index. All c_j for $j < h$ or beyond the list are zero. "Columns" means that column ordering is required in addition to row ordering. Substitution into (2.2) gives exactly the empirical generating function in each of these sixteen entries.

Entry	w	q	Columns	h	$(c_h, c_{h+1}, \ldots)$
A166776	2	3	no	2	$12, 27, 18, 4$
A166777	2	4	no	2	$12, 92, 175, 138, 52, 8$
A166778	2	5	no	3	$160, 765, 1306, 1096, 516, 136, 16$
A166779	2	6	no	3	$120, 1920, 6866, 10823, 9332, 4928, 1656, 336, 32$
A166780	2	7	no	4	$2730, 22666, 66234, 97749, 85222, 48204, 18544, 4848, 800, 64$
A166781	3	2	no	1	$4, 13, 14, 6$
A166782	3	3	no	1	$6, 84, 280, 443, 392, 192, 44$
A166783	3	4	no	2	$252, 2120, 7143, 13448, 16006, 12584, 6492, 2056, 320$
A166786	4	2	no	1	$6, 25, 56, 68, 36, 8$
A166796	2	3	yes	2	$6, 14, 9, 2$
A166797	2	4	yes	2	$8, 51, 92, 70, 26, 4$
A166798	2	5	yes	3	$105, 452, 722, 576, 262, 68, 8$
A166799	2	6	yes	3	$90, 1282, 4234, 6262, 5109, 2578, 840, 168, 16$
A166800	2	7	yes	4	$2016, 15421, 42152, 58628, 48404, 26088, 9652, 2456, 400, 32$
A166801	3	3	yes	1	$1, 25, 83, 118, 92, 40, 8$
A166802	3	4	yes	2	$67, 586, 1950, 3528, 3973, 2924, 1401, 410, 58$

The finite graphs have between 11 and 1437 reachable states. The full vectors and graph sizes are in `r2_floor_hardin_certificates.json`; the accompanying verifier reconstructs the graphs from the preceding rules and checks every rational-function identity. It also compares the frontier implementation to independent whole-grid connectivity searches on small arrays, as an implementation audit.

2.4.3 A166762: an exact 67-state recurrence certificate

Here the width is three, there are three colors, and no row or column ordering is imposed. Use the same frontier states and connectivity rules, but permit every next row. There are exactly 67 reachable states and 627 transitions, including the empty starting state indexed by 0. Let T be the integer adjacency matrix with rows as source states, and let b be the column vector whose entry is 1 precisely at an accepting state. Each path specifies its rows uniquely, so

$$a(n) = e_0^{\mathsf{T}} T^n b.$$

The exact finite matrix certificate is

$$(T - I)^7 (T - 2I)^2 b = 24 e_0, \qquad T e_0 = 0. \tag{2.3}$$

All 67 states, their successor lists, the accepting vector, and the residual vector are supplied in `r2_floor_hardin_automaton.json`. The verifier regenerates every state and transition, and verifies (2.3) using integer vector operations. The second identity also follows directly because no transition returns to the empty state.

Expanding $(z-1)^7(z-2)^2$ and applying $e_0^{\mathsf{T}} T^{n-9}$ to the first identity proves, for every $n \geq 10$,

$$\begin{aligned} a(n) = {} & 11a(n-1) - 53a(n-2) + 147a(n-3) - 259a(n-4) \\ & + 301a(n-5) - 231a(n-6) + 113a(n-7) - 32a(n-8) + 4a(n-9). \end{aligned}$$

This is the entry's empirical recurrence. The same construction gives the initial values

$$(a(1), \ldots, a(9)) = (6, 174, 1548, 8100, 32208, 108456, 327042, 912330, 2403570).$$

2.4.4 Two arithmetic-progression floor generating functions

A024172. For $S = \{1, 2, \ldots, n\}$ and $n \geq 2$, Newton's identities yield

$$\frac{e_3(S)}{e_2(S)} = \frac{n(n-2)(n+1)}{2(3n+2)} = Q(n) + \frac{8}{27(3n+2)}, \qquad Q(n) = \frac{n^2}{6} - \frac{5n}{18} - \frac{4}{27}.$$

The denominator is positive. Since $Q(n+9) - Q(n) = 3n+11$, its fractional part has period nine. For $n \equiv 1, \ldots, 9 \pmod 9$ its numerators over 27 are $20, 26, 14, 11, 17, 5, 2, 8, 23$. For $n \geq 3$ the positive remainder is less than $1/27$. The case $n = 2$ gives a correction of one. Thus

$$a(n) = \lfloor Q(n) \rfloor + \mathbf{1}_{n=2} \qquad (n \geq 2).$$

Summing the polynomial part and the nine fractional residues proves Barker's 2014 empirical generating function:

$$\sum_{n \geq 2} a(n) x^n = \frac{x^4(x^4 - x^3 + x^2 - x + 1)(x^5 - x^3 - x^2 - x - 1)}{(x-1)^3(x^2 + x + 1)(x^6 + x^3 + 1)}.$$

A024180. For $S = \{2, 3, \ldots, n+3\}$, Newton's identities give

$$\frac{e_3(S)}{e_2(S)} = \frac{n(n+5)(n^2 + 9n + 22)}{2(3n^2 + 29n + 72)} = Q(n) + R(n),$$

$$Q(n) = \frac{n^2}{6} + \frac{13n}{18} + \frac{5}{27}, \qquad R(n) = -\frac{8(8n+45)}{27(3n^2 + 29n + 72)}.$$

Again all denominators are positive on $n \geq 1$. The identity $Q(n+9) - Q(n) = 3n + 20$ proves period nine for the fractional part, whose numerators over 27 are $2, 8, 23, 20, 26, 14, 11, 17, 5$. For $n \geq 7$, $-2/27 < R(n) < 0$: the nontrivial inequality is equivalent to

$$n^2 - n - 36 = (n-7)^2 + 13(n-7) + 6 > 0.$$

Checking $n = 1, \ldots, 6$ gives only the correction -1 at $n = 1$. Consequently $a(n) = \lfloor Q(n) \rfloor - \mathbf{1}_{n=1}$ for every $n \geq 1$, and exact summation proves Barker's empirical function

$$\sum_{n \geq 1} a(n) x^n = -\frac{x^2(x^{10} - 2x^9 + x^7 + x^4 + x^2 - x + 2)}{(x-1)^3(x^2 + x + 1)(x^6 + x^3 + 1)}.$$

2.4.5 A184534: reciprocal fractional parts of cube roots

Let $a(n) = \lfloor 1/\{(n^3 + 4)^{1/3}\} \rfloor$ for $n \geq 1$. The exact formula is

$$a(n) = \left\lfloor \frac{3n^2}{4} \right\rfloor + \mathbf{1}_{n=1} + \mathbf{1}_{n=3}. \tag{2.4}$$

Put $\delta = (n^3 + 4)^{1/3} - n$. Since $n^3 < n^3 + 4 < (n+1)^3$, this is the fractional part in the definition. Expanding the cube gives $0 < \delta < 4/(3n^2)$, and rationalization gives

$$\frac{1}{\delta} = \frac{3n^2}{4} + \frac{3n\delta + \delta^2}{4}.$$

For $n \geq 5$, the positive residual is smaller than

$$\frac{1}{n} + \frac{4}{9n^4} \leq \frac{1}{5} + \frac{4}{9 \cdot 5^4} < \frac{1}{4}.$$

The fractional part of $3n^2/4$ is either 0 or 3/4, so its floor is unchanged. For $n = 1, 2, 3, 4$, comparing $n^3 + 4$ to the rational cubes $(n + 1/(v + 1))^3$ and $(n + 1/v)^3$, with $v = 1, 3, 7, 12$, respectively, gives the exact remaining values. This proves (2.4). Its generating function is exactly Barker's empirical one:

$$\frac{3x(1+x)}{4(1-x)^3} - \frac{3x}{4(1-x^2)} + x + x^3 = \frac{x(x^6 - 2x^5 + x^4 - x^2 - x - 1)}{(x-1)^3(x+1)}.$$

The separate empirical closed form printed in the entry uses a shifted index; the count here is for the generating-function assertion.

2.4.6 A002620 and A001840: two exponential floor identities

For every real $t > 0$ define

$$L(t) = \frac{t \coth t - 1}{t^2}.$$

We first prove the global inequalities

$$\frac{1}{3 + t^2} < L(t) < \frac{1}{3}. \tag{2.5}$$

The coefficients of t^{2k+1}, $k \geq 1$, in the two entire functions

$$(1 + t^2/3) \sinh t - t \cosh t, \qquad (3 + t^2)t \cosh t - (3 + 2t^2) \sinh t$$

are respectively

$$\frac{4k(k-1)}{3(2k+1)!}, \qquad \frac{8k^2(k-1)}{(2k+1)!}.$$

They vanish for $k = 1$ and are strictly positive for $k \geq 2$; the lower coefficients also vanish. The convergent series are therefore positive for $t > 0$. Dividing by positive $\sinh t$ and rearranging proves (2.5).

A002620. The selected empirical identity of Forberg (2013) is

$$\left\lfloor \frac{n}{e^{4/n} - 1} \right\rfloor = \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \qquad (n \geq 1).$$

For $n \geq 2$ put $t = 2/n \leq 1$. Then

$$\frac{n}{e^{4/n} - 1} = \frac{n^2}{4} - \frac{n}{2} + L(t), \qquad \frac{1}{4} < L(t) < \frac{1}{3}.$$

The base polynomial has fractional part 0 for even n and 3/4 for odd n. In either case adding $L(t)$ gives the same floor as adding 1/4, which is the claimed right-hand side. For $n = 1$, $0 < 1/(e^4 - 1) < 1$, so the identity also holds.

A001840. Here Forberg's selected empirical identity is

$$\left\lfloor \frac{n+3}{e^{6/(n+3)} - 1} \right\rfloor = \left\lfloor \frac{(n+1)(n+2)}{6} \right\rfloor \qquad (n \geq 0).$$

Set $m = n + 3 \geq 3$ and $t = 3/m \leq 1$. The left-hand expression before taking its floor equals

$$\frac{m^2}{6} - \frac{m}{2} + \frac{3}{2}L(t), \qquad \frac{3}{8} < \frac{3}{2}L(t) < \frac{1}{2}.$$

The fractional part of $m^2/6 - m/2$ is either 0 or 2/3. Adding any number in $(3/8, 1/2)$ consequently gives the same floor as adding 1/3. This is exactly the right-hand side, proving the identity for every permitted index. No other conjecture appearing in these two entries is included in this result.

2.4.7 A227018: harmonic means between consecutive binomial values

Put $g(n) = n(n+1)(n+2)(n+3)/24$ and

$$M_n = \frac{g(n) - g(n-1)}{H_{g(n)} - H_{g(n-1)}}, \qquad Q(n) = \frac{n^4}{24} + \frac{n^3}{6} + \frac{11n^2}{72} - \frac{n}{36} + \frac{119}{270}.$$

We prove, for every $n \geq 1$,

$$\lfloor M_n \rfloor = \lfloor Q(n) \rfloor + \mathbf{1}_{n \in \{1,2,5,8\}}. \tag{2.6}$$

This proves the entry's conjectured generating function.

Let $a = g(n-1)$, $b = g(n)$, $\Delta = b - a$, and $t = 2/(n+1)$. For $n \geq 2$ all these denominators are nonzero and $0 < t < 1$. Since $b/a = (n+3)/(n-1)$, $\log(b/a) = 2\operatorname{arctanh} t$. Define the explicit rational quantities

$$S = 2\sum_{j=0}^{3} \frac{t^{2j+1}}{2j+1}, \qquad R = \frac{2t^9}{9(1-t^2)}, \qquad C = \frac{1}{2}\left(\frac{1}{b} - \frac{1}{a}\right) - \frac{1}{12}\left(\frac{1}{b^2} - \frac{1}{a^2}\right).$$

The positive logarithm series gives $S < \log(b/a) < S + R$. The signed harmonic-number remainder bound is

$$H_m = \log m + \gamma + \frac{1}{2m} - \frac{1}{12m^2} + \rho_m, \qquad 0 < \rho_m < \frac{1}{120m^4}.$$

This is the positive-real signed remainder in the digamma expansion; see NIST DLMF 5.11.2 and 5.11(ii), `https://dlmf.nist.gov/5.11`. It follows that

$$L = S + C - \frac{1}{120a^4} < H_b - H_a < U = S + R + C + \frac{1}{120b^4}.$$

The following exact polynomial certificate proves the required bounds throughout the infinite range $n \geq 13$:

$$P_-(n) = 42525n^4(n-1)^2(n+1)^7(n+2)^4(n+3)^4(\Delta - QU),$$
$$P_+(n) = 945n^4(n-1)^4(n+1)^7(n+2)^4(n+3)^2((Q + 1/270)L - \Delta).$$

The two expressions are polynomials of degrees 18 and 20. Their complete ascending coefficients after substituting $n = 13 + u$ are in the following table. All coefficients within each polynomial's degree are positive.

j	$[u^j]P_-(13+u)$	$[u^j]P_+(13+u)$
0	5583594738457401551083929 6	1096587600955457474396 16
1	7097331668975690730179788 8	2651018952239211567513 60
2	4258971359715335078277734 4	2524563030461511763353 60
3	1603163394515848668802560 0	1387689830767557663989 76
4	4241979336413711433369600	5122072693992641131468 8
5	8378621650535013555097 60	1373734795018692809088 0
6	1280260645000981068248 32	2805670927599765213312
7	1547405355543382940140 8	4496010463785474403 20
8	1499750431545219474368	57646663206762721184
9	117430034768907000512	5989088391494318208
10	7445872019338020336	507962140667685096
11	381415588809560224	35285154142911672
12	15667255218382072	2006166896915838
13	509062269622464	92930287845888
14	12796039020312	3475232824328
15	240104119968	103354758768
16	3167005416	2388148548
17	26204160	41330304
18	102360	504096
19	–	3864
20	–	14

All displayed denominator factors are positive for $n \geq 13$, as are Q and Δ. Hence $\Delta - QU > 0$ and $(Q + 1/270)L - \Delta > 0$. In particular $L > 0$, and division by positive quantities gives the rigorous enclosure

$$Q(n) < M_n < Q(n) + \frac{1}{270} \qquad (n \geq 13).$$

The identity

$$Q(n+36) - Q(n) = 6n^3 + 342n^2 + 8435n + 77957$$

proves period 36 for the fractional part. Its numerators over 270, in residue classes $1, \ldots, 36$, are

$$209, 269, 164, 29, 269, 209, 254, 269, 119, 209, 134, 29,$$
$$29, 269, 74, 119, 269, 119, 74, 269, 29, 29, 134, 209,$$
$$119, 269, 254, 209, 269, 29, 164, 269, 209, 119, 134, 119.$$

They are at most 269, so the strict enclosure proves equality of floors. Exact rational harmonic sums for $n = 1, \ldots, 12$ give

$$1, 3, 9, 24, 51, 95, 164, 266, 407, 598, 850, 1174,$$

which differ from $\lfloor Q(n) \rfloor$ precisely by the four corrections in (2.6).

Finally, if $Q(n) = \sum_{k=0}^{4} q_k n^k$, $f_r = \{Q(r)\}$, and $\theta = x\, d/dx$, the exact generating function is

$$G(x) = \sum_{k=0}^{4} q_k \theta^k \frac{x}{1-x} - \frac{\sum_{r=1}^{36} f_r x^r}{1 - x^{36}} + x + x^2 + x^5 + x^8.$$

This supplies an explicit reconstructible rational expression for every coefficient. Its denominator is $(1-x)^4(1-x^{36})$, and exact simplification yields the entry's conjectured function multiplied by x, because the source uses $\sum_{n \geq 1} a(n) x^{n-1}$. The full source numerator and the exact identity check are in `r2_floor_harmonic_certificate.json` and its companion verifier. This result concerns the specific binomial-block assertion, not an unrestricted polynomial generalization.

2.5 Round 2: generating functions and exact recurrence proofs

This section proves explicitly conjectural assertions attached to 42 distinct OEIS entries, using pages checked on 20 October 2026. All entries are disjoint from the first-round ledger. Forty entries concern a recurrence (in three cases a generating-function assertion is also proved); A005573 concerns a hypergeometric evaluation, and A368633 concerns parity. Each entry counts once. Existing generating functions and finite sums are identified as source premises; their presence is not presented as a new result. No first-publication priority, external submission, or OEIS acceptance is claimed.

The on-page audit removed A001712: its 2020 continuation explicitly derives the recurrence despite the surviving conjecture label. The normalization correction in A065919 is also excluded. Unrelated conjectures on a retained page are outside the asserted scope unless expressly named below.

2.5.1 Exact certificates and their logical meaning

Write $\theta = x\,d/dx$. For $F = \sum_{n\ge 0} a_n x^n$, with omitted initial indices extended by zero, define

$$\mathcal{L}_p = \sum_{j=0}^{r} x^j p_j(\theta + j).$$

Then, solely by coefficient extraction,

$$[x^n]\mathcal{L}_p F = \sum_{j=0}^{r} p_j(n) a_{n-j}. \tag{R2-G1}$$

An exact identity $\mathcal{L}_p F = P(x)$, where P is a polynomial of degree below N, proves the printed recurrence for all $n \ge N$. The bounds in the individual records are part of the conclusions. No division by a leading recurrence coefficient is used. If the entry gives no lower bound, we state an explicit sufficient bound; in particular, A135925 requires $n \ge 14$, A239230 requires $n \ge 8$, and A032115 requires $n \ge 6$.

For an exponential generating function $F = \sum a_n x^n/n!$, set

$$q_j(n) = p_j(n) \prod_{h=0}^{r-j-1} (n - j - h). \tag{R2-G2}$$

For $n \ge r$, the coefficient of x^n in $\mathcal{L}_q F$ is the desired recurrence divided by $(n-r)!$. Thus a polynomial boundary works in the exponential convention as well.

For the rational and radical certificates write

$$x = f(t), \qquad F = \sum_i R_i(t)\sqrt{D_i(t)}.$$

The specified formal branch selects the series at $x = 0$. Ordinary differentiation gives the exact reduction rule

$$\theta(R_i\sqrt{D_i}) = \frac{f(t)}{f'(t)}\left(R_i'(t) + \frac{R_i(t)D_i'(t)}{2D_i(t)}\right)\sqrt{D_i(t)}. \tag{R2-G3}$$

The per-entry certificate prints all p_j, all seed functions, and the resulting boundary polynomial. The nonrational basis coefficients are zero. These identities can be checked by clearing denominators in the rational function field. They do not rely on fitting or testing finitely many terms. The standalone script `r2_gf_verify.py` performs those exact operations from `r2_gf_certificates.json`, without source-page snapshots or network access.

2.5.2 Elementary generating-function derivations

For A025258, let $F = \sum_{n\ge1} a_n x^n$. The defining convolution gives

$$F - x + x^2 - x^3 = F(F - x + x^2), \qquad F = x\left(1 - x + \frac{x^2}{1-F}\right).$$

The formal root with $F(0) = 0$ is the radical printed in its certificate. This also agrees with the existing December 2025 formula of Seiichi Manyama.

For A075045, use Manyama's existing finite sum

$$a_n = \sum_{k=0}^{n} 2^{n-k}\binom{3n+4}{k} = [u^n]\frac{(1+u)^{3n+4}}{1-2u}.$$

The formal change of variable $t = x\phi(t)$ in coefficient residues gives the diagonal Lagrange identity

$$\sum_{n\ge0}[u^n]\psi(u)\phi(u)^n x^n = \frac{\psi(t)}{1 - x\phi'(t)}.$$

Indeed, substitute $x = u/\phi(u)$ in the residue computing each coefficient; its Jacobian is precisely $1 - x\phi'(u)$, which cancels the displayed denominator. Here $\phi(u) = (1+u)^3$, $\psi(u) = (1+u)^4/(1-2u)$, so

$$x = \frac{t}{(1+t)^3}, \qquad F = \frac{(1+t)^5}{(1-2t)^2}.$$

Putting $g = t/(1+t)$ gives $g(1-g)^2 = x$ and $F = (3g-1)^{-2}(1-g)^{-3}$, proving the separately empirical generating function as well as the recurrence. That empirical function is not a premise.

For A082397 put $h_k = \binom{k}{\lfloor (k-1)/2 \rfloor}$, taking $h_0 = 0$, and $C(z) = (1 - \sqrt{1-4z})/(2z)$. Splitting even and odd indices gives

$$h_{2m} = \binom{2m}{m} - C_m, \qquad h_{2m+1} = 2\binom{2m}{m} - C_m,$$

and hence

$$H(z) = \sum_{k\ge0} h_k z^k = \frac{1+2z}{\sqrt{1-4z^2}} - (1+z)C(z^2).$$

In the accepted alternating binomial sum, summing first over $n \ge k$ uses $\sum_{n\ge k}\binom{n+1}{k+1}x^n = x^k/(1-x)^{k+2}$. Consequently $F(x) = -H(-x/(1-x))/(1-x)^2$, the certified radical.

For A186335 use its existing double sum. Let $m = n-k-j$ and $\ell = k-j-m$; the nonzero terms have $j, m, \ell \ge 0$, with $n = 2j + 2m + \ell$. Thus

$$F = \sum_{j,m,\ell\ge0}\binom{m+\ell}{m}\binom{j+m+\ell}{j}\binom{2m}{m}x^{2j+2m+\ell}.$$

Summing in j, then ℓ, with the binomial series gives

$$F = \frac{1}{1-x-x^2}\sum_{m\ge0}\binom{2m}{m}\left(\frac{x^2}{1-x-x^2}\right)^m = \frac{1}{\sqrt{(1-x-x^2)(1-x-5x^2)}}.$$

For A133158, the defining twice-binomial transform implies $F(x) = (1-2x)^{-1}B(x/(1-2x))$. The existing formula in A026641 is

$$B(z) = \frac{2}{3\sqrt{1-4z} - 1 + 4z}.$$

Substitution and rationalization yield the printed certificate. For A157021 the defining binomial sum gives

$$F(x) = \frac{1-2x+2x^2}{(1-x)^3} C\left(\frac{x^4}{(1-x)^4}\right):$$

indeed $\sum_{k\ge 0} C_{\lfloor k/2\rfloor} z^{2k} = (1+z^2)C(z^4)$, followed by the usual binomial transform $z = x/(1-x)$. For A141353, the definition immediately gives $F = C(x) + 2x/(1-2x)$.

2.5.3 Rational parameters for nested and inverse series

The remaining substitutions in this group are consequences of existing source formulas. For A089354 set $t = g$ in $g = 1 + xg^3$, $F = g/(1+xg^2)$; this gives $x = (t-1)/t^3$, $F = t^2/(2t-1)$, about $t = 1$. For A218045 use Kruchinin's November 2014 equation

$$F = xG, \qquad G = x\left(1 + \frac{G(G+1)}{1-G}\right)^2 = x\left(\frac{1+G^2}{1-G}\right)^2.$$

Putting $t = G$ gives $x = t(1-t)^2/(1+t^2)^2$, $F = xt$, about $t = 0$. The later exponential formula on that page is inconsistent with its zero constant coefficient; it is not used.

For A006343, Bala's June 2015 inverse-series formula is $F = w/x$, $x = w/(1 + w^2C(w))$. Setting $t = C(w)$, so $w = (t-1)/t^2$, gives

$$x = \frac{t(t-1)}{t^3 + (t-1)^2}, \qquad F = 1 + \frac{(t-1)^2}{t^3}, \qquad t(0) = 1.$$

For A158826, composition of inverses shows that the third iterate of $xC(x)$ is the inverse of the third iterate of $t(1-t)$. Therefore

$$x = t - 3t^2 + 6t^3 - 9t^4 + 10t^5 - 8t^6 + 4t^7 - t^8, \qquad F = t,$$

with $t(0) = 0$, exactly as the existing source formula states.

For A078999 use Irvine's July 2025 relation

$$a_n = \tfrac{1}{2}(A078995(n+1) - (5n+6)A002293(n+1)).$$

The accepted generating functions are $t = 1 + xt^4$ for A002293 and $H = t^2/(3t-4)^2$ for A078995 (van Hoeij, May 2013). Shifting the indices in the relation gives

$$F = \frac{H - t - 5xt'}{2x}, \qquad x = \frac{t-1}{t^4}, \qquad t(0) = 1.$$

For A239230 define

$$v = \sqrt{1-4x}, \qquad w = \sqrt{5+2v-3v^2}, \qquad B = \frac{-w+v+5}{4}.$$

Its defining generating function is $F = -xB'/B$. The conic is rationally parametrized by the line $w = 2 + t(v-1)$:

$$v = 1 - \frac{4(t+1)}{t^2+3}, \qquad w = 2 - \frac{4t(t+1)}{t^2+3}, \qquad x = \frac{1-v^2}{4}.$$

The branch at $t = -1$ has $v = 1, w = 2, x = 0$, and $dx/dt \neq 0$. Consequently $F = -xB_t/(x_tB)$ is the certified rational function.

2.5.4 Stirling sums as coefficients of rising products

Write $(z)^{\overline{m}} = \prod_{i=0}^{m-1}(z+i)$. For A001723 and A001724 the source formula is

$$a_n = \sum_{k=0}^{n}(-1)^{n+k}\binom{k+d}{d}5^k s(n+d,k+d), \qquad d=3,4,$$

where s is the signed Stirling number of the first kind. Expanding the rising product, the two signs cancel and give

$$a_n = [u^d](5+u)^{\overline{n+d}}.$$

For either printed coefficient list, with $r = d+1$, direct polynomial multiplication gives the finite identity

$$\sum_{j=0}^{r} p_j(n)\prod_{h=j}^{r-1}(u+5+n+d-1-h) = u^{d+1}. \tag{R2-G4}$$

For $n \geq r$, factor the common product $(5+u)^{\overline{n-1}}$ from the recurrence. Its remaining factor is u^{d+1}, so its coefficient of u^d is zero. This proves the two recurrences for every stated index.

2.5.5 The three reversed tribonacci triangles

Let the triangle be A027023 (type I), A027052 (type II), or A027082 (type III), with rows indexed from $n = 0$. All have the interior rule

$$T(n,k) = T(n-1,k-3) + T(n-1,k-2) + T(n-1,k-1),$$

and values outside a row are zero. The initial row is (1). Types I and II have left triples $(1,1,1)$, $(1,0,1)$, respectively, and rightmost entry 1. Type III has left triple $(1,1,1)$, while its rightmost entry sums the two rightmost entries of the preceding row. The row at $n = 1$ is respectively $(1,1,1)$, $(1,0,1)$, $(1,1,1)$. At $n = 2$ the rows are $(1,1,1,3,1)$, $(1,0,1,2,1)$, and $(1,1,1,3,2)$, fixing the overlapping boundary conventions.

Reverse each row and put

$$U_n(z) = \sum_{m=0}^{2n} T(n,2n-m)z^m, \qquad B(x,z) = \sum_{n\geq 0} U_n(z)x^n, \qquad B_j = [z^j]B.$$

The bulk multiplication is $z+1+z^{-1}$. Correcting its extraneous z^{-1} coefficient, its reset rightmost coefficient, and its three leftmost coefficients gives, for types I and II,

$$(1-x(z+1+z^{-1}))B = 1+\epsilon x + \frac{xL(z)}{1-xz^2} - xB_1 - \frac{x}{z(1-x)}, \tag{R2-G5}$$

where $(\epsilon, L) = (1, z^2-1)$ or $(0, z^2-z)$, respectively. Here $B_0 = 1/(1-x)$; its unwanted shift produces the last term. The right-edge reset produces $-xB_1$; the left-edge corrections form the geometric series $xL(z)/(1-xz^2)$; the exceptional first row produces ϵx. Thus every correction follows from the definition.

Let $t = x + O(x^2)$ be the small root of the kernel, so $x = t/(1+t+t^2)$. Substitution into (R2-G5) gives

$$B_1 = \frac{1}{x} + \epsilon + \frac{L(t)}{1-xt^2} - \frac{1}{t(1-x)}.$$

For type I, coefficient extraction at z^1, z^2, z^3 gives

$$B_2 = \frac{1-x}{x}B_1 - B_0, \quad B_3 = \frac{1-x}{x}B_2 - B_1 - 1 + x, \quad B_4 = \frac{1-x}{x}B_3 - B_2.$$

Therefore the target functions are $B_2 - x$ for A027051, B_3 for A027027, and B_4 for A027028. The subtraction in A027051 removes the column's extra $n = 1$ term, absent from its specified offset. Type II's B_1 gives A027056.

For type III the rightmost coefficient propagates instead of being reset, and the same row calculation gives

$$(1 - x(z + 1 + z^{-1}))B = 1 + x + \frac{x(z^2 - 1)}{1 - xz^2} - \frac{xB_0}{z}.$$

Kernel substitution therefore gives the target A027087 as

$$B_0 = \frac{t}{x}\left(1 + x + \frac{x(t^2 - 1)}{1 - xt^2}\right).$$

All five expressions are rational in t; (R2-G3) proves their recurrences.

2.5.6 Two central diagonals in A027113

This fourth triangle has left boundary $(1, 1, 2)$, the same interior rule, and rightmost entry equal to the sum of the two rightmost entries of the preceding row. Let $c_k = [x^k](1 - x - x^2 - x^3)^{-1}$. Induction using the row rule proves $T(n, k) = c_k$ whenever $0 \le k \le n+1$, with the initial row treated separately. Put

$$D_j(x) = \sum_{n \ge j} T(n, n + j)x^n \quad (j \ge 0).$$

The first two functions are $D_0 = C$, $D_1 = (C - 1 - x)/x$, where $C = (1 - x - x^2 - x^3)^{-1}$. For $n \ge j + 1$, the defining rule says $d_j(n) = d_j(n - 1) + d_{j-1}(n - 1) + d_{j-2}(n - 1)$. Using $d_2(2) = 3, d_3(3) = 7, d_4(4) = 16$, from the initial rows, yields

$$\begin{aligned}(1 - x)D_2 &= x(D_0 + D_1) - x,\\ (1 - x)D_3 &= x(D_2 + D_1) - 2x^2,\\ (1 - x)D_4 &= x(D_3 + D_2) - 3x^3.\end{aligned}$$

Solving gives precisely Barker's two conjectural rational generating functions:

$$D_3 = \frac{x^3(7 + 2x + x^2 - 2x^3)}{(1 - x)^2(1 - x - x^2 - x^3)},$$

$$D_4 = \frac{x^4(16 - 5x - x^2 - 5x^3 + 3x^4)}{(1 - x)^3(1 - x - x^2 - x^3)}.$$

Multiplying by the denominators proves the paired constant-coefficient recurrences, with the source's bounds $n > 7$, $n > 9$, respectively.

2.5.7 Fixed diagonals of the menage hit polynomials

The existing Jovovic bivariate generating function in A058087 is

$$P(x, y) = (1 - x(y - 1)) \sum_{m \ge 0} \frac{m!(xy)^m}{(1 + x(y - 1))^{2m+1}}.$$

Its coefficient is $[x^n y^k]P = T(n, k)$, with row coefficients in descending-power order. Replacing (x, y) by $(xt, 1/t)$ reverses that order. Write

$$H(x, t) = \frac{1 - x(1 - t)}{1 + x(1 - t)} f\left(\frac{x}{(1 + x(1 - t))^2}\right), \qquad f(z) = \sum_{m \ge 0} m! z^m.$$

Then $[t^d]H$ gives A000159, A000181, and A000185 for $d = 3, 4, 5$. Everything here is formal; no convergence of the factorial series is asserted. The factorial identity gives

$$z^2 f'(z) + (z-1)f(z) + 1 = 0. \tag{R2-G6}$$

For $h = x/(1 + x(1-t))^2$, put

$$K = \frac{h_t(1-h)}{h^2}, \qquad J = -\frac{h_t}{h^2}, \qquad f(h) = \sum_{k\geq 0}(\alpha_k F_0 + \beta_k)t^k,$$

where $F_0 = f(x/(1+x)^2)$, $\alpha_0 = 1, \beta_0 = 0$. Equation (R2-G6) gives the finite rational recursion

$$(k+1)\alpha_{k+1} = \sum_{i=0}^{k}[t^i]K\,\alpha_{k-i},$$
$$(k+1)\beta_{k+1} = \sum_{i=0}^{k}[t^i]K\,\beta_{k-i} + [t^k]J.$$

Multiply by the rational prefactor of H. This derives the printed $F = A_d F_0 + B_d$. For $h_0 = x/(1+x)^2$, differentiating (R2-G6) also gives

$$F_0' = UF_0 + V, \qquad U = \frac{h_0'(1-h_0)}{h_0^2}, \qquad V = -\frac{h_0'}{h_0^2}.$$

The exact operator calculation on the basis $(F_0, 1)$ gives zero coefficient of F_0 and the stated polynomial boundary. The three recurrence assertions follow. Their separate prime-divisibility claims are not included.

2.5.8 Exponential and symmetric-square differential spaces

For A080894, A272261, A144575, A053532 and A250917, the EGF F is specified in its individual record. Its logarithmic derivative has the form $F'/F = \ell_0 + \ell_1 y$, $y^2 = D(x)$. Writing $\theta^k F = F(A_k + B_k y)$, with $(A_0, B_0) = (1, 0)$, gives

$$\begin{aligned} A_{k+1} &= x(A_k' + \ell_0 A_k + D\ell_1 B_k), \\ B_{k+1} &= x\left(B_k' + \frac{D'}{2D}B_k + \ell_1 A_k + \ell_0 B_k\right). \end{aligned} \tag{R2-G7}$$

Applying the coefficients q_j from (R2-G2) gives exactly zero in both components. This proves all five recurrences. In A053532 there is an especially short direct calculation: $F'/F = x^3/(2(1-x))$, so $2(1-x)F' = x^3 F$; equating coefficients yields the printed assertion.

For A032115, Howroyd's existing EGF is

$$F = 1 + \frac{e^{-x}}{2(1-2x)} - \frac{1-2x}{2}e^x.$$

On $R(x)e^{cx}$, Euler differentiation replaces R by $x(R' + cR)$. The two exponential components vanish under the adjusted operator, and the rational component gives $2x^5$. This proves the recurrence for $n \geq 6$; a universal assertion beginning at $n = 5$ would fail there.

For A168452 let $G = \sum_{n\geq 0} C_n C_{n+1} x^n$; its defining self-convolution gives $F = G^2$. The coefficient ratio

$$\frac{C_n C_{n+1}}{C_{n-1}C_n} = \frac{4(2n-1)(2n+1)}{(n+1)(n+2)}$$

implies

$$\big((\theta+1)(\theta+2) - 4x(2\theta+1)(2\theta+3)\big)G = 2.$$

Thus $G'' = AG + BG' + C$, with the three rational coefficients printed below. The six functions $G^2, GG', (G')^2, G, G', 1$ form a differential space. For a polynomial expression $R(x, G, H)$, $H = G'$, its Euler derivative is

$$x\,(\partial_x R + H\partial_G R + (AG + BH + C)\partial_H R)\,. \tag{R2-G8}$$

The conjectured operator applied to G^2 reduces to $72 - 1152x$, with all other basis coefficients zero. Hence the recurrence holds for $n \geq 4$.

2.5.9 The hypergeometric evaluation and the parity conjecture

For A005573, the proposed evaluation is

$$a_n = 6^n\, {}_2F_1\left(\frac{1}{2}, -n; 2; \frac{2}{3}\right), \qquad n \geq 0.$$

The series terminates at $k = n$. The identities

$$\frac{(-n)_k}{k!} = (-1)^k \binom{n}{k}, \qquad \frac{(1/2)_k}{(2)_k} = \frac{C_k}{4^k}$$

turn its k-th summand into $(-1)^k \binom{n}{k} C_k 6^{n-k}$, exactly the existing finite-sum formula. This proves the evaluation term by term.

For A368633, the defining equation is

$$F(x) = 1 + 2xF(x)^2 - xF(-x)^2.$$

It uniquely determines integral coefficients: the coefficient of degree $n > 0$ only uses earlier coefficients on its right-hand side. Modulo 2 it becomes

$$F(x) = 1 + xF(x)^2 = 1 + xF(x^2).$$

Here $-x = x$ in characteristic 2 and the Frobenius identity applies coefficientwise. Therefore a_0 is odd, a_{2m} is even for $m \geq 1$, and $a_{2m+1} \equiv a_m \pmod 2$. Repeatedly applying this recursion shows

$$F(x) \equiv \sum_{k \geq 0} x^{2^k - 1} \pmod 2.$$

Thus precisely the indices $n = 2^k - 1$, including $n = 0$, have odd coefficients. The page's separate D-finite conjecture is not used or counted.

2.5.10 Per-entry statements and exact certificates

In the following records, every displayed coefficient vector means $\sum_{j=0}^{r} p_j(n) a_{n-j} = 0$ in the stated range. All coefficients and boundaries are exact integers or rational functions. For a radical certificate, the ordinary convention is used unless the exponential convention is stated; $D_0 = 1$ denotes its rational component. Each source URL identifies the current conjecture and the seed premise, and additional seed URLs are listed where required.

2.5.11 A089354

Source: https://oeis.org/A089354. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 3$. Its complete coefficient list is:

$$p_0(n) = 128n^2 + 64n$$

$$p_1(n) = -1136n^2 + 1640n - 528$$
$$p_2(n) = 1760n^2 - 7802n + 7848$$
$$p_3(n) = 513n^2 - 2052n + 1995$$

The source premise is the existing implicit GF, with the rational substitution derived above. The branch is $t(0) = 1$. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$
$$x = \frac{t-1}{t^3}.$$
$$R_0 = \frac{t^2}{2t-1}.$$
$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -4x\,(19x + 6)\,.$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.12 A247471

Source: https://oeis.org/A247471. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 8$. Its complete coefficient list is:

$$p_0(n) = n + 3$$
$$p_1(n) = -3n - 3$$
$$p_2(n) = 0$$
$$p_3(n) = n - 9$$
$$p_4(n) = 4n - 18$$
$$p_5(n) = n - 9$$
$$p_6(n) = 9 - n$$

Use the existing Deutsch formula $F = x^3(1-x-x^2)g^2$, $g = 1 + xg + x^2g + x^3g^2$. The formal solution is $g = (1 - x - x^2 - \sqrt{(1-x-x^2)^2 - 4x^3})/(2x^3)$, with $g(0) = 1$. Substitution gives the displayed coefficients. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$
$$x = x.$$
$$R_0 = -\frac{(x^2 + x - 1)\,(x^4 - x^2 - 2x + 1)}{2x^3}.$$
$$D_0 = 1.$$

$$R_1 = -\frac{\left(x^2+x-1\right)^2}{2x^3}.$$
$$D_1 = \left(x^2-3x+1\right)\left(x^2+x+1\right).$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -2x^3\left(x^2-x+3\right)\left(x^2+x-1\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.13 A135925

Source: https://oeis.org/A135925. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 14$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= 30715031990396n^2 - 433554237731400n \\ &\quad + 402839205741004 \\ p_1(n) &= -291262905844111n^2 + 4745710679447853n \\ &\quad - 12935393149282124 \\ p_2(n) &= 805570866653979n^2 - 14936203714114791n \\ &\quad + 60755127774344160 \\ p_3(n) &= -393057731655406n^2 + 8294532445122504n \\ &\quad - 43776275319524372 \\ p_4(n) &= -539125155360312n^2 + 12669158483525292n \\ &\quad - 74126175519017400 \end{aligned}$$

The seed is the entry definition. Put $s = \sqrt{1-4x}$; the defining function is the product $(x^2+x-1)(1+2x+s)(1-5x+2x^2+(3x-1)s)^2(-1+3x+2x^2+(1-x)s)$. Reducing modulo $s^2 = 1-4x$ gives the two coefficients below. Its polynomial boundary explains the lower bound 14. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$
$$x = x.$$
$$\begin{aligned} R_0 &= 16x^9 - 160x^8 \\ &\quad - 824x^7 + 1024x^6 \\ &\quad + 1240x^5 - 2152x^4 \\ &\quad + 1056x^3 - 216x^2 \\ &\quad + 16x \end{aligned}$$
$$D_0 = 1.$$
$$\begin{aligned} R_1 &= 48x^8 + 72x^7 \\ &\quad - 640x^6 - 72x^5 \\ &\quad + 1016x^4 - 720x^3 \\ &\quad + 184x^2 - 16x \end{aligned}$$
$$D_1 = 1 - 4x.$$

Applying (R2-G3) gives the exact identity

$$\begin{aligned}\mathcal{L}_p F = &-8628263825301312x^{13} - 56627815207112768x^{12}\\ &- 82029355854002112x^{11} + 107522236010828544x^{10}\\ &+ 267411638401515712x^{9} + 26860851665524032x^{8}\\ &- 96203187527862528x^{7} + 17483921103890240x^{6}\end{aligned}$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.14 A218045

Source: https://oeis.org/A218045. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 5$. Its complete coefficient list is:

$$\begin{aligned}p_0(n) &= 2n^3 - 5n^2\\ &\quad + 3n\\ p_1(n) &= -22n^3 + 100n^2\\ &\quad + 70 - 148n\\ p_2(n) &= -24n^3 + 336n^2\\ &\quad + 1400 - 1272n\\ p_3(n) &= 512n^3 - 5696n^2\\ &\quad + 21024n - 25760\\ p_4(n) &= 512n^3 - 6656n^2\\ &\quad + 28544n - 40320\end{aligned}$$

Use only Kruchinin's accepted implicit GF described above. The parameter is $t = G$, with $t(0) = 0$. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = \frac{t\,(t-1)^2}{(t^2+1)^2}.$$

$$R_0 = \frac{t^2\,(t-1)^2}{(t^2+1)^2}.$$

$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = 2x^2\left(16x^2 - 16x + 1\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.15 A006343

Source: https://oeis.org/A006343. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 5$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= -n^2 - 2n \\ &\quad + 3 \\ p_1(n) &= 11n^2 - 2n \\ &\quad - 45 \\ p_2(n) &= -37n^2 + 82n \\ &\quad + 87 \\ p_3(n) &= 29n^2 - 125n \\ &\quad + 78 \\ p_4(n) &= 61n^2 - 289n \\ &\quad + 318 \\ p_5(n) &= -155n^2 + 1085n \\ &\quad - 1860 \end{aligned}$$

Use Bala's established inverse-series formula and the substitution derived above, with $t(0) = 1$. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = \frac{t\,(t-1)}{t^3 + t^2 - 2t + 1}.$$

$$R_0 = \frac{t^3 + t^2 - 2t + 1}{t^3}.$$

$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = 98x^2 - 36x + 3.$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.16 A158826

Source: https://oeis.org/A158826. Offset 1.

Conclusion. The selected printed recurrence holds for every integer $n \geq 6$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= 1053n^4 - 6318n^3 \\ &\quad + 11583n^2 - 6318n \\ p_1(n) &= -22824n^4 + 186156n^3 \\ &\quad - 546336n^2 + 678276n \\ &\quad - 295272 \end{aligned}$$

$$\begin{aligned}p_2(n) &= 191184n^4 - 1999920n^3\\ &\quad + 7835448n^2 - 13605720n\\ &\quad + 8810064\end{aligned}$$

$$\begin{aligned}p_3(n) &= -769024n^4 + 9966464n^3\\ &\quad - 48753200n^2 + 106646584n\\ &\quad - 87964536\end{aligned}$$

$$\begin{aligned}p_4(n) &= 1470464n^4 - 23144448n^3\\ &\quad + 137212672n^2 - 363259008n\\ &\quad + 362432208\end{aligned}$$

$$\begin{aligned}p_5(n) &= -1048576n^4 + 19922944n^3\\ &\quad - 141787136n^2 + 447954944n\\ &\quad - 530100480\end{aligned}$$

The seed is the inverse of the third iterate of $t(1-t)$, as derived above. Its local branch has $t(0) = 0$. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = -t\,(t-1)\left(t^2 - t + 1\right)\left(t^4 - 2t^3 + 2t^2 - t + 1\right).$$

$$R_0 = t.$$

$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = 120x^4\,(16x - 29).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.17 A075045

Source: https://oeis.org/A075045. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 2$. Its complete coefficient list is:

$$\begin{aligned}p_0(n) &= 112n^3 + 296n^2\\ &\quad + 208n + 24\end{aligned}$$

$$\begin{aligned}p_1(n) &= -1512n^3 - 2862n^2\\ &\quad - 1320n - 66\end{aligned}$$

$$\begin{aligned}p_2(n) &= 5103n^3 + 5832n^2\\ &\quad - 567n - 648\end{aligned}$$

The diagonal Lagrange argument above proves the generating function from the existing finite sum. The branch has $t(0) = 0$. This also proves the empirical GF separately printed in the entry. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = \frac{t}{(t+1)^3}.$$
$$R_0 = \frac{(t+1)^5}{(2t-1)^2}.$$
$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = 24.$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.18 A082397

Source: https://oeis.org/A082397. Offset 1.

Conclusion. The selected printed recurrence holds for every integer $n \geq 4$. Its complete coefficient list is:

$$p_0(n) = n + 2$$
$$p_1(n) = -3n - 2$$
$$p_2(n) = -n$$
$$p_3(n) = 3n$$

The alternating binomial-sum derivation above identifies the GF. Only the recurrence is targeted; the separate partial-sum claim involving A342912 remains outside this packet. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$
$$x = x.$$
$$R_0 = \frac{2x-1}{2x^2\,(x-1)}.$$
$$D_0 = 1.$$
$$R_1 = \frac{1}{2x^2\,(x-1)\,(x+1)}.$$
$$D_1 = -\,(x+1)\,(3x-1)\,.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = 3x.$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.19 A190160

Source: https://oeis.org/A190160. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 9$. Its complete coefficient list is:

$$p_0(n) = n + 2$$
$$p_1(n) = -5n - 6$$
$$p_2(n) = 8n + 6$$

$$p_3(n) = -2n - 9$$
$$p_4(n) = 25 - 8n$$
$$p_5(n) = 6n - 12$$
$$p_6(n) = 9n - 63$$
$$p_7(n) = 96 - 15n$$
$$p_8(n) = 7n - 49$$
$$p_9(n) = 8 - n$$

The existing implicit equation is $G = 1+xG+x^2G(x+(1-x)^2(G-xG-1))/(1-x)$. Multiplication by $1-x$ gives $AG^2+BG+C=0$, where $A = x^2(1-x)^3$, $B = x^3 - x^2(1-x)^2 - (1-x)^2$, and $C = 1-x$. The root $G = (-B-\sqrt{B^2-4AC})/(2A)$, with $G(0)=1$, is the certificate. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$
$$x = x.$$
$$R_0 = -\frac{x^4 - 3x^3 + 2x^2 - 2x + 1}{2x^2\,(x-1)^3}.$$
$$D_0 = 1.$$
$$R_1 = \frac{1}{2x^2\,(x-1)^3}.$$
$$D_1 = \left(x^4 - x^3 - 2x^2 + 1\right)\left(x^4 - 5x^3 + 6x^2 - 4x + 1\right).$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -2\,(x-1)\left(x^5 - 3x^4 + 4x^3 + 2x^2 - 3x + 1\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.20 A001723

Source: https://oeis.org/A001723. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 4$. Its complete coefficient list is:

$$p_0(n) = 1$$
$$p_1(n) = -4n - 22$$
$$p_2(n) = 6n^2 + 60n + 151$$
$$p_3(n) = -4n^3 - 54n^2 - 244n - 369$$
$$p_4(n) = n^4 + 16n^3 + 96n^2 + 256n + 256$$

Use the rising-product proof (R2-G4), with $d=3$, $r=4$. The finite polynomial identity has residual u^4; multiplying by $(5+u)^{\overline{n-1}}$ and extracting $[u^3]$ proves the recurrence.

2.5.21 A001724

Source: https://oeis.org/A001724. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 5$. Its complete coefficient list is:

$$\begin{aligned}
p_0(n) &= 1 \\
p_1(n) &= -5n - 30 \\
p_2(n) &= 10n^2 + 110n \\
&\quad + 305 \\
p_3(n) &= -10n^3 - 150n^2 \\
&\quad - 755n - 1275 \\
p_4(n) &= 5n^4 + 90n^3 \\
&\quad + 610n^2 + 1845n \\
&\quad + 2101 \\
p_5(n) &= -n^5 - 20n^4 \\
&\quad - 160n^3 - 640n^2 \\
&\quad - 1280n - 1024
\end{aligned}$$

Use the rising-product proof (R2-G4), with $d = 4$, $r = 5$. The finite polynomial identity has residual u^5; multiplying by $(5+u)^{\overline{n-1}}$ and extracting $[u^4]$ proves the recurrence.

2.5.22 A027051

Source: https://oeis.org/A027051. Additional established seed sources: https://oeis.org/A027023. Offset 2.

Conclusion. The selected printed recurrence holds for every integer $n \geq 9$. Its complete coefficient list is:

$$\begin{aligned}
p_0(n) &= n \\
p_1(n) &= 5 - 7n \\
p_2(n) &= 13n - 18 \\
p_3(n) &= n - 13 \\
p_4(n) &= 64 - 13n \\
p_5(n) &= 3n - 25 \\
p_6(n) &= 2 - n \\
p_7(n) &= 3n - 15
\end{aligned}$$

The reversed-row kernel derivation above identifies this column GF from its defining triangle. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = \frac{t}{t^2 + t + 1}.$$

$$R_0 = \frac{\mathcal{N}}{\mathcal{D}}.$$

$$\begin{aligned}\mathcal{N} &= -4t^7 - 8t^6 \\ &\quad - 11t^5 - 9t^4 \\ &\quad - 5t^3 - t^2\end{aligned}$$

$$\begin{aligned}\mathcal{D} &= t^7 - 3t^4 \\ &\quad - 3t^3 - 4t^2 \\ &\quad - 2t - 1\end{aligned}$$

$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -x^2 \left(9x^6 + x^5 - 20x^4 - 2x^3 + 29x^2 + x - 2\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.23 A027027

Source: https://oeis.org/A027027. Additional established seed sources: https://oeis.org/A027023. Offset 2.

Conclusion. The selected printed recurrence holds for every integer $n \geq 9$. Its complete coefficient list is:

$$p_0(n) = n + 1$$

$$p_1(n) = -8n - 1$$

$$p_2(n) = 19n - 14$$

$$p_3(n) = -6n - 2$$

$$p_4(n) = 89 - 21n$$

$$p_5(n) = 8n - 45$$

$$p_6(n) = n - 4$$

$$p_7(n) = 6n - 24$$

The reversed-row kernel derivation above identifies this column GF from its defining triangle. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = \frac{t}{t^2 + t + 1}.$$

$$R_0 = \frac{\mathcal{N}}{\mathcal{D}}.$$

$$\begin{aligned}\mathcal{N} &= -4t^8 - 8t^7 \\ &\quad - 12t^6 - 11t^5 \\ &\quad - 7t^4 - 3t^3 \\ &\quad - t^2\end{aligned}$$

$$\begin{aligned}\mathcal{D} &= t^7 - 3t^4 \\ &\quad - 3t^3 - 4t^2 \\ &\quad - 2t - 1\end{aligned}$$

$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = x^2 \left(24x^6 - 15x^5 - 35x^4 + 4x^3 + 8x^2 - 13x + 3\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.24 A027028

Source: https://oeis.org/A027028. Additional established seed sources: https://oeis.org/A027023. Offset 2.

Conclusion. The selected printed recurrence holds for every integer $n \geq 8$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= -n^2 + 4n \\ &\quad + 12 \\ p_1(n) &= 6n^2 - 27n \\ &\quad - 33 \\ p_2(n) &= -7n^2 + 31n \\ p_3(n) &= -8n^2 + 62n \\ &\quad - 114 \\ p_4(n) &= 5n^2 - 38n \\ &\quad + 84 \\ p_5(n) &= 2n^2 - 11n \\ &\quad + 15 \\ p_6(n) &= 3n^2 - 21n \\ &\quad + 36 \end{aligned}$$

The reversed-row kernel derivation above identifies this column GF from its defining triangle. The ordinary generating function has certificate

$$\begin{aligned} F &= \sum_{i=0}^{0} R_i \sqrt{D_i}. \\ x &= \frac{t}{t^2 + t + 1}. \\ R_0 &= \frac{\mathcal{N}}{\mathcal{D}}. \\ \mathcal{N} &= -4t^9 - 8t^8 \\ &\quad - 12t^7 - 12t^6 \\ &\quad - 7t^5 - 5t^4 \\ &\quad - t^3 - t^2 \\ \mathcal{D} &= t^7 - 3t^4 \\ &\quad - 3t^3 - 4t^2 \\ &\quad - 2t - 1 \\ D_0 &= 1. \end{aligned}$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -x^2 (x - 1) \left(36x^4 + 3x^3 - 2x^2 - 29x + 16\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.25 A027056

Source: https://oeis.org/A027056. Additional established seed sources: https://oeis.org/A027052. Offset 1.

Conclusion. The selected printed recurrence holds for every integer $n \geq 7$. Its complete coefficient list is:

$$p_0(n) = 1 - n$$
$$p_1(n) = 6n - 8$$
$$p_2(n) = 3 - 7n$$
$$p_3(n) = 52 - 8n$$
$$p_4(n) = 5n - 29$$
$$p_5(n) = 2n - 4$$
$$p_6(n) = 3n - 15$$

The reversed-row kernel derivation above identifies this column GF from its defining triangle. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = \frac{t}{t^2 + t + 1}.$$
$$R_0 = -\frac{2t^2\left(t^2 + t + 1\right)}{\left(t^2 + 1\right)\left(t^3 - t^2 - t - 1\right)}.$$
$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -2x^2\left(x - 1\right)\left(5x - 1\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.26 A027087

Source: https://oeis.org/A027087. Additional established seed sources: https://oeis.org/A027082. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 7$. Its complete coefficient list is:

$$p_0(n) = 3n$$
$$p_1(n) = 9 - 15n$$
$$p_2(n) = 7n - 4$$
$$p_3(n) = 25n - 67$$
$$p_4(n) = 17n - 44$$
$$p_5(n) = 19n - 97$$
$$p_6(n) = 5n - 32$$
$$p_7(n) = 3n - 21$$

The reversed-row kernel derivation above identifies this column GF from its defining triangle. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = \frac{t}{t^2 + t + 1}.$$

$$R_0 = -\frac{(t+1)\left(2t^2 + t + 1\right)}{t^3 - t^2 - t - 1}.$$

$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = x\,(x+1)^2 \left(x^3 - 13x^2 + 7x - 3\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.27 A025258

Source: https://oeis.org/A025258. Offset 1.

Conclusion. The selected printed recurrence holds for every integer $n \geq 6$. Its complete coefficient list is:

$$p_0(n) = -n$$

$$p_1(n) = 2n - 3$$

$$p_2(n) = 9 - 3n$$

$$p_3(n) = 6n - 27$$

$$p_4(n) = 6 - n$$

The convolution argument above derives the radical from the definition; the same expression is an established Manyama formula on the page. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$

$$x = x.$$

$$R_0 = -\frac{x^2 - x - 1}{2}.$$

$$D_0 = 1.$$

$$R_1 = -\frac{1}{2}.$$

$$D_1 = x^4 - 6x^3 + 3x^2 - 2x + 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -x\left(x^4 - x^3 + 6x^2 - 3x + 1\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.28 A157125

Source: https://oeis.org/A157125. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 5$. Its complete coefficient list is:

$$p_0(n) = n^2 + n - 6$$

$$p_1(n) = -4$$

$$p_2(n) = 2n^2 - 2n - 8$$

$$p_3(n) = 4n^2 - 14n + 4$$

$$p_4(n) = n^2 - 3n - 4$$

$$p_5(n) = 4n^2 - 18n + 14$$

The accepted GF is $(1-x)(\sqrt{1+x^2+4x^3} - \sqrt{1+x^2})/(2x^3\sqrt{1+x^2})$. Multiplying the quotient of radicals by $\sqrt{1+x^2}$ gives the single-radical certificate. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$

$$x = x.$$

$$R_0 = \frac{x-1}{2x^3}.$$

$$D_0 = 1.$$

$$R_1 = -\frac{x-1}{2x^3\left(x^2+1\right)}.$$

$$D_1 = \left(x^2+1\right)\left(4x^3+x^2+1\right).$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -2\left(x^3+3\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.29 A186335

Source: https://oeis.org/A186335. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 4$. Its complete coefficient list is:

$$p_0(n) = n$$

$$p_1(n) = 1 - 2n$$

$$p_2(n) = 5 - 5n$$

$$p_3(n) = 6n - 9$$

$$p_4(n) = 5n - 10$$

The triple reindexing and binomial-series summation above derive this GF directly from the established double sum. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$

$$x = x.$$

$$R_0 = 0.$$

$$D_0 = 1.$$

$$R_1 = \frac{1}{(x^2 + x - 1)\,(5x^2 + x - 1)}.$$

$$D_1 = \left(x^2 + x - 1\right)\left(5x^2 + x - 1\right).$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = 0.$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.30 A133158

Source: https://oeis.org/A133158. Additional established seed sources: https://oeis.org/A026641. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 3$. Its complete coefficient list is:

$$p_0(n) = 2n$$

$$p_1(n) = 12 - 19n$$

$$p_2(n) = 48n - 66$$

$$p_3(n) = 72 - 36n$$

The second binomial-transform calculation above derives the function from the established GF of A026641. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$

$$x = x.$$

$$R_0 = -\frac{1}{2\,(3x - 2)}.$$

$$D_0 = 1.$$

$$R_1 = \frac{3}{2\,(3x - 2)\,(6x - 1)}.$$

$$D_1 = (2x - 1)\,(6x - 1).$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -x.$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.31 A157021

Source: https://oeis.org/A157021. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 5$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= n^3 - 7n^2 \\ &\quad + 128 - 12n \\ p_1(n) &= -5n^3 + 41n^2 \\ &\quad - 26n - 360 \\ p_2(n) &= 10n^3 - 94n^2 \\ &\quad + 224n + 128 \\ p_3(n) &= -10n^3 + 106n^2 \\ &\quad + 336 - 364n \\ p_4(n) &= n^3 - 7n^2 \\ &\quad + 4n + 32 \\ p_5(n) &= 3n^3 - 39n^2 \\ &\quad + 174n - 264 \end{aligned}$$

The target is the second, shorter conjectural recurrence on the page (the one with five backward shifts). Its seed GF follows from the defining binomial sum as shown above. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$

$$x = x.$$

$$R_0 = -\frac{(x-1)\left(2x^2 - 2x + 1\right)}{2x^4}.$$

$$D_0 = 1.$$

$$R_1 = \frac{2x^2 - 2x + 1}{2x^4\,(x-1)}.$$

$$D_1 = -\left(x^2 + 2x - 1\right)\left(3x^2 - 2x + 1\right).$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -16\left(2x^3 - 10x^2 + 15x - 8\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.32 A173993

Source: https://oeis.org/A173993. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 6$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= -8n - 2 \\ p_2(n) &= 19n - 5 \\ p_3(n) &= -36 \end{aligned}$$

$$p_4(n) = 208 - 56n$$
$$p_5(n) = 68n - 286$$
$$p_6(n) = 120 - 24n$$

The seed is the existing GF $(\sqrt{(1-2x)(1-2x-4x^2+4x^3)} - 2x^2 + 4x - 1)/(2x(1-4x+3x^2))$. The target is its recurrence only. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$
$$x = x.$$
$$R_0 = -\frac{2x^2 - 4x + 1}{2x\,(x-1)\,(3x-1)}.$$
$$D_0 = 1.$$
$$R_1 = \frac{1}{2x\,(x-1)\,(3x-1)}.$$
$$D_1 = -\,(2x-1)\left(4x^3 - 4x^2 - 2x + 1\right).$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = (x-1)\left(10x^3 - 10x^2 + 5x - 1\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.33 A141353

Source: https://oeis.org/A141353. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 4$. Its complete coefficient list is:

$$p_0(n) = n + 1$$
$$p_1(n) = 2 - 8n$$
$$p_2(n) = 20n - 28$$
$$p_3(n) = 40 - 16n$$

The definition gives $F = C(x) + 2x/(1-2x)$, with C the Catalan series and the constant term fixed at 1. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$
$$x = x.$$
$$R_0 = -\frac{4x^2 - 2x + 1}{2x\,(2x-1)}.$$
$$D_0 = 1.$$
$$R_1 = -\frac{1}{2x}.$$
$$D_1 = 1 - 4x.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = 8x^3 - 12x^2 + 1.$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.34 A078999

Source: https://oeis.org/A078999. Additional established seed sources: https://oeis.org/A078995, https://oeis.org/A002293. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 3$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) = &-2937742094763n^4 - 6138004010652n^3 \\ &- 461143675071n^2 + 5117093595162n \\ &+ 2377975354344 \end{aligned}$$

$$\begin{aligned} p_1(n) = &\ 39322791505152n^4 + 130827262124736n^3 \\ &- 77390801861328n^2 - 57276849171240n \\ &- 8278877159568 \end{aligned}$$

$$\begin{aligned} p_2(n) = &\ 46619105624064n^4 - 1454666767368192n^3 \\ &+ 2115306373226496n^2 - 874201548149760n \\ &+ 84277928590848 \end{aligned}$$

$$\begin{aligned} p_3(n) = &-1473031998275584n^4 + 7263078429229056n^3 \\ &- 12705856454918144n^2 + 9076166541705216n \\ &- 2081737996369920 \end{aligned}$$

The source cross-sequence formula and the two established GFs give the rational substitution derived above, about $t = 1$. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = \frac{t-1}{t^4}.$$

$$R_0 = \frac{t^5\,(3t-2)}{(3t-4)^2}.$$

$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -216\,(6397991648x - 11009145159)\,.$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.35 A000159

Source: https://oeis.org/A000159. Additional established seed sources: https://oeis.org/A058087. Offset 3.

Conclusion. The selected printed recurrence holds for every integer $n \geq 9$. Its complete coefficient list is:

$$p_0(n) = 2082154 - 504614n$$

$$\begin{aligned} p_1(n) = &\ 504614n^2 - 3362985n \\ &+ 5118150 \end{aligned}$$

$$\begin{aligned} p_2(n) = &\ 1280831n^2 - 7397886n \\ &+ 6461565 \end{aligned}$$

$$p_3(n) = 746598n^2 - 2913543n - 1336090$$
$$p_4(n) = -405481n^2 + 6175011n - 15469320$$
$$p_5(n) = -375862n^2 + 4098537n - 8846430$$
$$p_6(n) = 1121260 - 375862n$$

Use the fixed-diagonal construction above with $d = 3$. In its notation,

$$F = AF_0 + B, \qquad F_0 = f\left(\frac{x}{(1+x)^2}\right).$$

$$A = -\frac{2\left(2x^5 - 3x^4 - 5x^3 - 4x^2 - 3x - 2\right)}{3\left(x+1\right)^2}.$$
$$B = \frac{2\left(2x^5 - x^4 - 8x^3 - 7x^2 - 5x - 2\right)}{3\left(x+1\right)^2}.$$

The coefficient of F_0 in $\mathcal{L}_p F$ is zero, while its rational part is

$$\mathcal{L}_p F = 4x^3\left(379025x^2 - 2587x + 284156\right).$$

The factorial-series differential equation therefore proves the recurrence.

2.5.36 A000181

Source: https://oeis.org/A000181. Additional established seed sources: https://oeis.org/A058087. Offset 4.

Conclusion. The selected printed recurrence holds for every integer $n \geq 8$. Its complete coefficient list is:

$$p_0(n) = 25332n^4 - 428952n^3 + 2720193n^2 - 7656417n + 8070516$$
$$p_1(n) = -25332n^5 + 453232n^4 - 3030389n^3 + 9016781n^2 + 1452000 - 10446214n$$
$$p_2(n) = -24280n^5 + 333424n^4 - 1619342n^3 + 3451376n^2 + 1805760 - 3470928n$$
$$p_3(n) = 1052n^5 - 40656n^4 + 266063n^3 - 549153n^2 + 49850n + 655200$$
$$p_4(n) = 1052n^4 - 7960n^3 + 9379n^2 + 46459n - 94080$$

Use the fixed-diagonal construction above with $d = 4$. In its notation,

$$F = AF_0 + B, \qquad F_0 = f\left(\frac{x}{(1+x)^2}\right).$$

$$A = -\frac{4x^6 - 12x^5 - 13x^4 - 6x^3 - 3x^2 - 4x - 4}{6\,(x+1)^2}.$$

$$B = \frac{4x^7 - 4x^6 - 33x^5 - 40x^4 - 20x^3 - 15x^2 - 12x - 4}{6\,(x+1)^3}.$$

The coefficient of F_0 in $\mathcal{L}_p F$ is zero, while its rational part is

$$\mathcal{L}_p F = 0.$$

The factorial-series differential equation therefore proves the recurrence.

2.5.37 A000185

Source: https://oeis.org/A000185. Additional established seed sources: https://oeis.org/A058087. Offset 5.

Conclusion. The selected printed recurrence holds for every integer $n \geq 9$. Its complete coefficient list is:

$$\begin{aligned}
p_0(n) &= 482n^4 - 9603n^3 \\
&\quad + 71606n^2 - 236835n \\
&\quad + 293150 \\
p_1(n) &= -482n^5 + 10099n^4 \\
&\quad - 79756n^3 + 285961n^2 \\
&\quad + 149292 - 426904n \\
p_2(n) &= -496n^5 + 8178n^4 \\
&\quad - 50261n^3 + 150426n^2 \\
&\quad + 192618 - 243765n \\
p_3(n) &= -14n^5 - 49n^4 \\
&\quad - 619n^3 + 13174n^2 \\
&\quad + 61248 - 51690n \\
p_4(n) &= -14n^4 + 321n^3 \\
&\quad - 2338n^2 + 6939n \\
&\quad - 7308
\end{aligned}$$

Use the fixed-diagonal construction above with $d = 5$. In its notation,

$$F = AF_0 + B, \qquad F_0 = f\left(\frac{x}{(1+x)^2}\right).$$

$$A = -\frac{4x^7 - 20x^6 - 9x^5 + 10x^4 + 15x^3 + 9x^2 - 4}{15\,(x+1)^2}.$$

$$B = \frac{4x^9 - 8x^8 - 57x^7 - 69x^6 + 2x^5 + 58x^4 + 34x^3 - 3x^2 - 12x - 4}{15\,(x+1)^4}.$$

The coefficient of F_0 in $\mathcal{L}_p F$ is zero, while its rational part is

$$\mathcal{L}_p F = 0.$$

The factorial-series differential equation therefore proves the recurrence.

2.5.38 A080894

Source: https://oeis.org/A080894. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 3$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= 3 - 2n \\ p_1(n) &= -2n^3 + 9n^2 \\ &\quad + 1 - 9n \\ p_2(n) &= 4n^4 - 14n^3 \\ &\quad + 11n^2 + 5n \\ &\quad - 6 \\ p_3(n) &= 6n^5 - 51n^4 \\ &\quad + 162n^3 - 237n^2 \\ &\quad + 156n - 36 \end{aligned}$$

The defining or established EGF is

$$F = \frac{e^{\frac{1}{2x}} e^{-\frac{\sqrt{-3x^2-2x+1}}{2x}}}{e^{\frac{1}{2}}}.$$

For the reduction (R2-G7), its exact logarithmic derivative is $F'/F = \ell_0 + \ell_1\sqrt{D}$, with

$$\begin{aligned} D &= -(x+1)(3x-1). \\ \ell_0 &= -\frac{1}{2x^2}. \\ \ell_1 &= \frac{x-1}{2x^2(x+1)(3x-1)}. \end{aligned}$$

The adjusted operator has zero residual in both components: $\mathcal{L}_q F = F(0 + 0\sqrt{D}) = 0$. Thus (R2-G2) proves the recurrence.

2.5.39 A272261

Source: https://oeis.org/A272261. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 2$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= 1 \\ p_1(n) &= 2 - 4n \\ p_2(n) &= -4 \end{aligned}$$

The defining or established EGF is

$$F = \frac{e^{\sqrt{1-4x}}}{e\sqrt{1-4x}}.$$

For the reduction (R2-G7), its exact logarithmic derivative is $F'/F = \ell_0 + \ell_1\sqrt{D}$, with

$$\begin{aligned} D &= 1 - 4x. \\ \ell_0 &= -\frac{2}{4x-1}. \\ \ell_1 &= \frac{2}{4x-1}. \end{aligned}$$

The adjusted operator has zero residual in both components: $\mathcal{L}_q F = F(0 + 0\sqrt{D}) = 0$. Thus (R2-G2) proves the recurrence.

2.5.40 A144575

Source: https://oeis.org/A144575. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 4$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= 1 \\ p_1(n) &= 3 - 2n \\ p_2(n) &= -3n^2 + 12n \\ &\quad - 10 \\ p_3(n) &= 9 - 6n \\ p_4(n) &= -9n^2 + 36n \\ &\quad - 27 \end{aligned}$$

The defining or established EGF is

$$F = ee^{-\sqrt{-3x^2-2x+1}}.$$

For the reduction (R2-G7), its exact logarithmic derivative is $F'/F = \ell_0 + \ell_1\sqrt{D}$, with

$$D = -(x+1)(3x-1).$$

$$\begin{aligned} \ell_0 &= 0. \\ \ell_1 &= -\frac{3x+1}{(x+1)(3x-1)}. \end{aligned}$$

The adjusted operator has zero residual in both components: $\mathcal{L}_q F = F(0 + 0\sqrt{D}) = 0$. Thus (R2-G2) proves the recurrence.

2.5.41 A053532

Source: https://oeis.org/A053532. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 4$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) &= 2 \\ p_1(n) &= 2 - 2n \\ p_2(n) &= 0 \\ p_3(n) &= 0 \\ p_4(n) &= -n^3 + 6n^2 \\ &\quad + 6 - 11n \end{aligned}$$

The defining or established EGF is

$$F = \frac{e^{-\frac{x}{2}} e^{-\frac{x^2}{4}} e^{-\frac{x^3}{6}}}{\sqrt{1-x}}.$$

For the reduction (R2-G7), its exact logarithmic derivative is $F'/F = \ell_0 + \ell_1\sqrt{D}$, with

$$\begin{aligned} D &= 1. \\ \ell_0 &= -\frac{x^3}{2(x-1)}. \\ \ell_1 &= 0. \end{aligned}$$

The adjusted operator has zero residual in both components: $\mathcal{L}_q F = F(0 + 0\sqrt{D}) = 0$. Thus (R2-G2) proves the recurrence.

2.5.42 A250917

Source: https://oeis.org/A250917. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 5$. Its complete coefficient list is:

$$\begin{aligned}
p_0(n) &= 2 \\
p_1(n) &= 20 - 11n \\
p_2(n) &= n^3 + 9n^2 \\
&\quad + 164 - 116n \\
p_3(n) &= -4n^4 + 35n^3 \\
&\quad + n^2 - 317n \\
&\quad + 342 \\
p_4(n) &= -36n^4 + 408n^3 \\
&\quad - 1782n^2 + 3702n \\
&\quad - 3168 \\
p_5(n) &= 24n^4 - 396n^3 \\
&\quad + 2424n^2 - 6516n \\
&\quad + 6480
\end{aligned}$$

The defining or established EGF is

$$F = e^{\frac{1}{2x^2}} e^{-\frac{3}{2x}} e^{-\frac{\sqrt{1-4x}}{2x^2}} e^{\frac{\sqrt{1-4x}}{2x}}.$$

For the reduction (R2-G7), its exact logarithmic derivative is $F'/F = \ell_0 + \ell_1\sqrt{D}$, with

$$\begin{aligned}
D &= 1 - 4x. \\
\ell_0 &= \frac{3x-2}{2x^3}. \\
\ell_1 &= -\frac{2x^2 - 7x + 2}{2x^3\,(4x-1)}.
\end{aligned}$$

The adjusted operator has zero residual in both components: $\mathcal{L}_q F = F(0 + 0\sqrt{D}) = 0$. Thus (R2-G2) proves the recurrence.

2.5.43 A292461

Source: https://oeis.org/A292461. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 6$. Its complete coefficient list is:

$$\begin{aligned}
p_0(n) &= n \\
p_1(n) &= 3 - 2n \\
p_2(n) &= 3 - n \\
p_3(n) &= 9 - 2n \\
p_4(n) &= n - 6
\end{aligned}$$

The seed is exactly the ordinary GF in the entry definition. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$

$$x = x.$$

$$R_0 = -\frac{x^2 + x - 1}{2}.$$

$$D_0 = 1.$$

$$R_1 = \frac{1}{2}.$$

$$D_1 = \left(x^2 - 3x + 1\right)\left(x^2 + x + 1\right).$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = x^3\left(x^2 - x + 3\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.44 A126089

Source: https://oeis.org/A126089. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 1$. Its complete coefficient list is:

$$p_0(n) = 4 - n$$

$$p_1(n) = 4n^2 - 22n + 30$$

The seed is exactly the exponential GF in the entry definition. The target is its conjectural first-order recurrence, rather than its other already unqualified recurrence. The exponential generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$

$$x = x.$$

$$R_0 = 0.$$

$$D_0 = 1.$$

$$R_1 = 1 - 2x.$$

$$D_1 = 1 - 4x.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_q F = 0.$$

Every remaining radical coefficient is zero. Equations (R2-G1) and (R2-G2) prove the stated conclusion.

2.5.45 A108629

Source: https://oeis.org/A108629. Offset 1.

Conclusion. The selected printed recurrence holds for every integer $n \geq 6$. Its complete coefficient list is:

$$p_0(n) = n$$
$$p_1(n) = 3 - n$$
$$p_2(n) = 12 - 4n$$
$$p_3(n) = 30 - 9n$$
$$p_4(n) = 27 - 7n$$
$$p_5(n) = 18 - 4n$$

The seed is the ordinary GF in the entry definition. Its zero constant coefficient agrees with the index-1 offset. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$

$$x = x.$$
$$R_0 = \frac{x+1}{2\left(x^2+x+1\right)}.$$
$$D_0 = 1.$$
$$R_1 = -\frac{1}{2\left(x^2+x+1\right)}.$$
$$D_1 = -4x^3 - 3x^2 - 2x + 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = x\left(x^3 + 3x^2 + x + 1\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.46 A239230

Source: https://oeis.org/A239230. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 8$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) = {} & 17521226928n^4 - 141607619832n^3 \\ & + 342315933792n^2 - 298512866472n \\ & + 80283325584 \end{aligned}$$

$$\begin{aligned} p_1(n) = {} & -108516679788n^4 + 1055343845440n^3 \\ & - 3529733053572n^2 + 4962562394720n \\ & - 2512676462064 \end{aligned}$$

$$\begin{aligned} p_2(n) = {} & -57647707646n^4 + 974519385068n^3 \\ & - 6210429875914n^2 + 17628548676196n \\ & - 18287840662872 \end{aligned}$$

$$\begin{aligned}p_3(n) &= 1104332263320n^4 - 16688474493280n^3\\ &\quad + 99299523282780n^2 - 277054887180164n\\ &\quad + 303328616888592\end{aligned}$$

$$\begin{aligned}p_4(n) &= -587491214125n^4 + 9941738070620n^3\\ &\quad - 75070680472775n^2 + 281912285021344n\\ &\quad - 413197788157152\end{aligned}$$

$$\begin{aligned}p_5(n) &= -2373849695822n^4 + 52217689535398n^3\\ &\quad - 423872944767808n^2 + 1515169459097264n\\ &\quad - 2019443387466624\end{aligned}$$

$$\begin{aligned}p_6(n) &= 2628244359392n^4 - 56698793715560n^3\\ &\quad + 459016033763548n^2 - 1650515137948516n\\ &\quad + 2220852187679664\end{aligned}$$

$$\begin{aligned}p_7(n) &= -1193212957536n^4 + 26793378151488n^3\\ &\quad - 224248504659624n^2 + 828694144073232n\\ &\quad - 1140184384447680\end{aligned}$$

The conic parametrization above follows from the defining nested radical. The local parameter is $t = -1$ at $x = 0$; both square roots have their positive constant terms. The target is the recurrence, not the separate double-sum conjecture. The ordinary generating function has certificate

$$F = \sum_{i=0}^{0} R_i \sqrt{D_i}.$$

$$x = \frac{2\,(t-1)^2\,(t+1)}{(t^2+3)^2}.$$

$$R_0 = \frac{4t\,(t-1)\,(t+1)}{(t+3)\,(t^2+1)\,(t^2-4t-1)}.$$

$$D_0 = 1.$$

Applying (R2-G3) gives the exact identity

$$\begin{aligned}\mathcal{L}_p F &= 1138854670092x^7 - 801911343684x^6\\ &\quad - 1058136530640x^5 - 414143562744x^4\\ &\quad + 103948668000x^3\end{aligned}$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.47 A032115

Source: https://oeis.org/A032115. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 6$. Its complete coefficient list is:

$$p_0(n) = 1$$

$$p_1(n) = -2n - 2$$

$$p_2(n) = 4n - 4$$

$$p_3(n) = 2$$

$$p_4(n) = 11 - 4n$$
$$p_5(n) = 2n - 8$$

The established EGF and the componentwise derivative rule are derived above. The exact certificate is

$$\mathcal{L}_q F = 2x^5.$$

Its boundary is why the statement begins at 6.

2.5.48 A168452

Source: https://oeis.org/A168452. Offset 0.

Conclusion. The selected printed recurrence holds for every integer $n \geq 4$. Its complete coefficient list is:

$$\begin{aligned} p_0(n) = {} & 3n^5 + 33n^4 \\ & + 135n^3 + 255n^2 \\ & + 222n + 72 \end{aligned}$$
$$\begin{aligned} p_1(n) = {} & -128n^5 - 784n^4 \\ & - 1588n^3 - 1232n^2 \\ & - 300n \end{aligned}$$
$$\begin{aligned} p_2(n) = {} & 1536n^5 + 2304n^4 \\ & - 1344n^3 - 2688n^2 \\ & + 288 - 1536n \end{aligned}$$
$$\begin{aligned} p_3(n) = {} & -12288n^4 + 33792n^3 \\ & + 23040 - 49152n \end{aligned}$$
$$\begin{aligned} p_4(n) = {} & -65536n^5 + 655360n^4 \\ & - 2539520n^3 + 4751360n^2 \\ & + 1474560 - 4276224n \end{aligned}$$

Use the Catalan-product ODE and the six-dimensional calculation (R2-G8) above. Its coefficients in $G'' = AG + BG' + C$ are

$$A = -\frac{2\,(6x-1)}{x^2\,(16x-1)}.$$
$$B = -\frac{4\,(12x-1)}{x\,(16x-1)}.$$
$$C = -\frac{2}{x^2\,(16x-1)}.$$

The exact six-component residual is

$$(0, 0, 0, 0, 0, 72 - 1152x).$$

Hence $\mathcal{L}_p(G^2) = 72 - 1152x$, proving the recurrence.

2.5.49 A005573

Source: https://oeis.org/A005573. Offset 0.

The assertion is the terminating hypergeometric evaluation proved term by term above, for every integer $n \geq 0$. Its proof needs no recurrence certificate.

2.5.50 A368633

Source: https://oeis.org/A368633. Offset 0.

The assertion is the complete parity characterization proved above, for every integer $n \geq 0$. Its proof uses the defining functional equation over $\mathbb{F}_2[[x]]$.

2.5.51 A027116

Source: https://oeis.org/A027116. Additional established seed sources: https://oeis.org/A027113. Offset 3.

Conclusion. The selected printed recurrence holds for every integer $n \geq 8$. Its complete coefficient list is:

$$p_0(n) = 1$$
$$p_1(n) = -3$$
$$p_2(n) = 2$$
$$p_3(n) = 0$$
$$p_4(n) = 1$$
$$p_5(n) = -1$$

The central-diagonal derivation above proves both the conjectural GF and this recurrence. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$
$$x = x.$$
$$R_0 = \frac{x^3 \left(2x^3 - x^2 - 2x - 7\right)}{(x-1)^2 \left(x^3 + x^2 + x - 1\right)}.$$
$$D_0 = 1.$$
$$R_1 = 0.$$
$$D_1 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = -x^3 \left(2x^3 - x^2 - 2x - 7\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.5.52 A027117

Source: https://oeis.org/A027117. Additional established seed sources: https://oeis.org/A027113. Offset 4.

Conclusion. The selected printed recurrence holds for every integer $n \geq 10$. Its complete coefficient list is:

$$p_0(n) = 1$$
$$p_1(n) = -4$$
$$p_2(n) = 5$$
$$p_3(n) = -2$$

$$p_4(n) = 1$$
$$p_5(n) = -2$$
$$p_6(n) = 1$$

The central-diagonal derivation above proves both the conjectural GF and this recurrence. The ordinary generating function has certificate

$$F = \sum_{i=0}^{1} R_i \sqrt{D_i}.$$
$$x = x.$$
$$R_0 = \frac{x^4 \left(3x^4 - 5x^3 - x^2 - 5x + 16\right)}{(x-1)^3 \left(x^3 + x^2 + x - 1\right)}.$$
$$D_0 = 1.$$
$$R_1 = 0.$$
$$D_1 = 1.$$

Applying (R2-G3) gives the exact identity

$$\mathcal{L}_p F = x^4 \left(3x^4 - 5x^3 - x^2 - 5x + 16\right).$$

Every remaining radical coefficient is zero. Equations (R2-G1) prove the stated conclusion.

2.6 Validation, exclusions, and provenance

Independent mathematical review. The full A132393 proof was independently checked through its change of variables, factorial and sign cancellations, moment determinant, and final symmetric-function identity. The A349802, A005773, and A152920 derivations received separate checks of their normalization and indexing. The A153197 contraction proof is sound under the displayed normalization and remains uncounted.

The connected-array transition rules were independently reviewed against the whole-grid definition. A second reconstruction checked all 21 rational generating-function identities, the 67-state/627-edge matrix certificate for A166762, and independent small-grid enumerations. The arithmetic arguments were reviewed directly, including the prime-power square-zero conjugacy count and the odd-modulus localization argument. The certificate engine retains exact rational arithmetic and the ordinary/exponential normalization explicitly.

Reproducible checks. The structural packet includes exact finite checks for all 21 written arguments, of which 20 are counted. These checks supplement the proofs. The arithmetic verifier covers all 11 counted results and labels the additional A098275 rediscovery separately. The three floor programs reconstruct all 28 all-index certificates. The generating-function program checks all retained operator, product, hypergeometric, and parity certificates directly from JSON. The verification bundle records the actual outputs and includes a one-command runner.

Some prior-resolution exclusions. The following are examples of candidates removed after reading beyond the surviving conjecture label:

- A001712: the later Hadjicostas continuation already proves the selected recurrence.
- A025273: the entry credits Israel's 2016 resolution.
- A032123, A348410, and A001711: linked 2026 proofs already address the selected targets.
- A166761: Howroyd's 2024 confirmation is recorded in the source.
- A059297, A059259, and A100631: later source material already establishes the contemplated claims.
- A037223: the proposed universal statement already has a recorded counterexample.
- A098275: the prior 14 October 2026 paper proves the same divisibility statement, so our different proof adds no counted target.

The A065919 exponential/ordinary generating-function normalization issue was likewise kept outside the count. The same policy applies to the index and definition discrepancies noted earlier.

Standard references and existing inputs. Each proof identifies its entry-specific established formulas. Classical tools such as Cauchy–Binet, the symmetric-function determinant identity, Lagrange inversion, the Chinese remainder theorem, and Bertrand's theorem are used as standard results. The harmonic remainder enclosure uses NIST DLMF https://dlmf.nist.gov/5.11. The prime-interval estimate is Nagura's original 1952 theorem, cited in the relevant proof. The extra published-results comparison is documented against https://arxiv.org/abs/2608.11941.

The resulting dossier is a reviewable mathematical and computational packet. Public submission, editorial acceptance, exhaustive literature priority, and proof-assistant formalization are separate steps and are not claimed here.

Part III

Pell zeros, modular periods, formal congruences, and transfer certificates

Round overview

We assemble a final third-round dossier containing 100 selected OEIS assertion packages. Eighteen are retained and audited from the supplied provisional manuscript, and 82 are added here. The outcomes are 95 positive proofs, four refutations, and one entry with both a refutation and a complete replacement congruence. The arguments include an explicit infinite Pell family of zero partial sums, prime divisibility in finite algebraic rings, a general modular-period theorem for integral Stirling transforms, finite differential quotients, exact formal-series congruences, arithmetic restrictions at twin-prime centers, and definition-derived transfer certificates for 41 empirical array recurrences. Three universal prime-divisibility conjectures are refuted by complete zero-free residue cycles. A fourth pure-period conjecture fails modulo 27. Every counted entry is assigned a precise target, source, proof location, and reproducible check. The word "additional" describes this project's provenance and does not assert historical priority.

Scope and counting

What the total means. One catalogue row is one specified assertion package associated with a distinct OEIS identifier. Several assertions in the same entry count once; an alias, a copied universal assertion, or an uncounted corollary adds nothing. Different width or color parameters on different source entries count separately even when one theorem proves them together. Thus the 41 array rows share a transfer method; they do not represent 41 unrelated mathematical methods. The 100 selections comprise 18 inherited and 82 additional packages. They have no identifier overlap with the previous 208 selections, giving a project total of 308 under this convention.

Relation to the provisional manuscript. The inherited part consists of six elementary/formal-series packages and twelve algebraic-power coefficient packages. Their full arguments are included below, so this edition is readable without the earlier file. The provisional manuscript's proposed 100-entry candidate queue is not used as proof evidence. The three constant-coefficient existence questions left unresolved there are now refuted, at the primes 71, 241, and 167. The transferred congruence for A396845 remains an uncounted consequence of the general theorem.

Positive, negative, and mixed outcomes. The four refuted selections are A188570, A377109, A377113, and A000364. The mixed entry is A396844: its proposed mod-16 pattern fails at index 17, and a complete replacement law is proved. Accordingly, "100 resolved" does not mean that 100 proposed statements are true.

The selected scope is part of each result. For example, A348295 is selected for infinitely many zeros, not its separately treated unboundedness question. A396803, A397241, A396808, and A397356 are selected for the stated parity assertions; other congruences or asymptotic questions on those pages are not included. A378580 is selected for its mod-4 assertion, not the question about convergence radius. The A000364 counterexample concerns a pure period dividing $\varphi(m)$ from index 1, not eventual periodicity. Ancillary erroneous small-prime examples in the algebraic-power entries are explicitly separated from the universal infinitude and eventual-gap statements proved here.

What is a proof and what is a check. All-index conclusions follow from the written algebraic, combinatorial, or arithmetic arguments, or from a proved reduction to a complete finite certificate. In the transfer family every state and every transition is enumerated from the definition, and a complete integer vector identity proves the recurrence for all subsequent indices. The three zero-free prime certificates cover an entire proved period. Other finite-prefix calculations are supplementary checks of transcription, boundaries, and implementations. No claim of proof-assistant formalization or external refereeing is made.

Source and prior-work audit. Primary entries and relevant linked material were checked on 20 October 2026. The separate audit file records comparison with all 100 costliest claims in Appendix A of the OEIS Open paper. The only identifier overlap is A348295; that paper's result is unboundedness, whereas the selected assertion here is the infinite zero set. This is a claim-level distinction, not an assertion that the identifier is absent from the paper. The audit is limited to the stated appendix, inspected linked work, and targeted searches, rather than an exhaustive worldwide literature review. Known results used as inputs are credited. For example, the Catwalk binomial enumeration is due to Guy, and the earlier recurrences of Sela Fried and Robert Israel are not recounted as new results here. A141117 was excluded after locating its linked 2026 proof; A002050 is an uncounted known consequence of Barsky's congruences. This dossier does not report an OEIS edit, submission, or editorial acceptance.

Files. The standalone source is `oeis_hunt_round3.tex`. The portable bundle contains the complete 100-row ledger, `oeis_hunt_round3_ledger.json`, all proof fragments, four verifiers, 90 complete transfer graphs, exact counterexample certificates, audit notes, and `run_all.py`.

Catalogue of the 100 selected packages

P = proved; R = refuted; M = mixed proof and refutation. I = inherited from the supplied manuscript; N = additional this round. The page column links to the proof. Each identifier links to its primary OEIS record. Full target qualifications are given in the proof and ledger.

No.	Entry	Result	Origin	Selected target	Page
1	A348295	P	I	Infinite Pell family of zero partial sums	139
2	A014865	P	I	Oddness, restricted prime factors, and compositeness	140
3	A048868	P	I	Prime-majority reduced residues force an even modulus	140
4	A131418	P	I	Digit-sum equality forces divisibility by 9	140
5	A396843	P	I	Every coefficient is odd; full mod-9 law included	140
6	A396844	M	I	Mod-16 counterexample and complete replacement law	140
7	A188571	P	I	Infinite prime divisibility and eventual zero-gap periods	146
8	A188572	P	I	Infinite prime divisibility and eventual zero-gap periods	146
9	A188573	P	I	Infinite prime divisibility and eventual zero-gap periods	146
10	A377110	P	I	Infinite prime divisibility and eventual zero-gap periods	146
11	A377111	P	I	Infinite prime divisibility and eventual zero-gap periods	146
12	A377112	P	I	Infinite prime divisibility and eventual zero-gap periods	146
13	A377114	P	I	Infinite prime divisibility and eventual zero-gap periods	146
14	A377115	P	I	Infinite prime divisibility and eventual zero-gap periods	146
15	A377116	P	I	Infinite prime divisibility and eventual zero-gap periods	146
16	A377117	P	I	Infinite prime divisibility and eventual zero-gap periods	147
17	A377118	P	I	Infinite prime divisibility and eventual zero-gap periods	147
18	A377119	P	I	Infinite prime divisibility and eventual zero-gap periods	147
19	A195194	P	N	Nested-reversion parity: power-of-two indices	142
20	A397241	P	N	All coefficients of the diagonal relation are odd	142
21	A396808	P	N	Parity support at zero and powers of two	142
22	A397356	P	N	Parity support where n+1 is a power of two	143
23	A396803	P	N	Iterated exponential composition: a(n) = n mod 2	143
24	A368634	P	N	Odd coefficients exactly at n = 2^j-1	144
25	A393172	P	N	Complete mod-4 support at powers of two	144
26	A389536	P	N	Complete mod-4 support at n = 2^j+1	145
27	A378580	P	N	Complete mod-4 support at square indices	145
28	A188570	R	N	No term divisible by 71: universal-prime claim false	148
29	A377109	R	N	No term divisible by 241: universal-prime claim false	148
30	A377113	R	N	No term divisible by 167: universal-prime claim false	148
31	A004123	P	N	Stirling-transform period dividing Carmichael lambda(m)	150
32	A306082	P	N	Stirling-transform period dividing Carmichael lambda(m)	150
33	A006531	P	N	Stirling-transform period dividing Carmichael lambda(m)	150
34	A167137	P	N	Stirling-transform period dividing Carmichael lambda(m)	150
35	A080253	P	N	Stirling-transform period dividing Carmichael lambda(m)	150
36	A305550	P	N	Stirling-transform period dividing Carmichael lambda(m)	150
37	A354242	P	N	Stirling-transform period dividing Carmichael lambda(m)	150
38	A301921	P	N	Stirling-transform period dividing Carmichael lambda(m)	150
39	A320352	P	N	Stirling-transform period dividing Carmichael lambda(m)	150

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No.	Entry	Result	Origin	Selected target	Page
40	A354253	P	N	Stirling-transform period dividing Carmichael lambda(m)	150
41	A005046	P	N	Eventual modular periodicity for even-block partitions	151
42	A006154	P	N	Eventual modular periodicity for ordered odd blocks	151
43	A121860	P	N	Factorial divisor sum is eventually 2 modulo m	152
44	A258899	P	N	Explicit modular tail 2+2^n and eventual period	152
45	A258903	P	N	Explicit modular tail 3*2^n+2*3^n and eventual period	152
46	A000364	R	N	Pure period dividing phi(m) fails at m = 27	153
47	A394757	P	N	Twin-prime centers: gcd is 2 or 3; every term is 18-divisible	154
48	A394399	P	N	Gcd-3 centers: twice-square form and set relations	154
49	A394815	P	N	Gcd-2 centers are 36-divisible; disjoint-union identity	154
50	A048702	P	N	Both empirical formulas for even binary palindromes	155
51	A230089	P	N	Complex-power formula and rational generating function	155
52	A389425	P	N	Unit gaps have density one in the sequence index	156
53	A092278	P	N	Empirical generating function for floor((3n+4)/16)	156
54	A005558	P	N	The displayed lattice-walk recurrence(s), column y = 0	157
55	A005559	P	N	The displayed lattice-walk recurrence(s), column y = 1	157
56	A005560	P	N	The displayed lattice-walk recurrence(s), column y = 2	157
57	A005561	P	N	The displayed lattice-walk recurrence(s), column y = 3	157
58	A005562	P	N	The displayed lattice-walk recurrence(s), column y = 4	157
59	A205355	P	N	Empirical array recurrence of order 4	160
60	A205356	P	N	Empirical array recurrence of order 5	161
61	A205357	P	N	Empirical array recurrence of order 12	161
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63	A206377	P	N	Empirical array recurrence of order 5	161
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66	A206380	P	N	Empirical array recurrence of order 20	162
67	A209709	P	N	Empirical array recurrence of order 6	162
68	A209729	P	N	Empirical array recurrence of order 3	162
69	A209807	P	N	Empirical array recurrence of order 4	162
70	A209808	P	N	Empirical array recurrence of order 7	163
71	A209809	P	N	Empirical array recurrence of order 14	163
72	A209810	P	N	Empirical array recurrence of order 33	163
73	A209811	P	N	Empirical array recurrence of order 91	163
74	A209822	P	N	Empirical array recurrence of order 3	164
75	A209823	P	N	Empirical array recurrence of order 5	164
76	A209824	P	N	Empirical array recurrence of order 9	165
77	A209825	P	N	Empirical array recurrence of order 21	165
78	A209826	P	N	Empirical array recurrence of order 51	165
79	A209841	P	N	Empirical array recurrence of order 4	165
80	A209842	P	N	Empirical array recurrence of order 6	166
81	A209843	P	N	Empirical array recurrence of order 10	166
82	A209844	P	N	Empirical array recurrence of order 24	166
83	A209845	P	N	Empirical array recurrence of order 59	166

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No.	Entry	Result	Origin	Selected target	Page
84	A209890	P	N	Empirical array recurrence of order 2	167
85	A209891	P	N	Empirical array recurrence of order 3	167
86	A209892	P	N	Empirical array recurrence of order 9	167
87	A209893	P	N	Empirical array recurrence of order 23	167
88	A209894	P	N	Empirical array recurrence of order 67	167
89	A209962	P	N	Empirical array recurrence of order 6	168
90	A209963	P	N	Empirical array recurrence of order 10	168
91	A209964	P	N	Empirical array recurrence of order 18	168
92	A209965	P	N	Empirical array recurrence of order 44	168
93	A210045	P	N	Empirical array recurrence of order 4	169
94	A210046	P	N	Empirical array recurrence of order 6	169
95	A210047	P	N	Empirical array recurrence of order 14	169
96	A210048	P	N	Empirical array recurrence of order 32	169
97	A210088	P	N	Empirical array recurrence of order 4	170
98	A210089	P	N	Empirical array recurrence of order 6	170
99	A210091	P	N	Empirical array recurrence of order 22	170
100	A368628	P	N	Odd coefficients exactly at n = (4^j-1)/3	171

3.1 Inherited arguments: Pell zeros and elementary obstructions

3.1.1 A348295: an infinite Pell family of zeros

Put $\alpha = \sqrt{2} - 1$ and

$$S(N) = \sum_{k=1}^{N} (-1)^{\lfloor k\alpha \rfloor}, \qquad S(0) = 0.$$

Let Pell numbers satisfy $P_0 = 0$, $P_1 = 1$, $P_{j+2} = 2P_{j+1} + P_j$.

Theorem 3.1.1. *For each integer $m \geq 1$,*

$$\boxed{S(2P_{2m}) = 0.}$$

In particular, the OEIS zero-index sequence A348299 is infinite; the "infinitely many zeros" conjecture in A348295 holds.

Proof. Set $p = P_{2m}$, $q = P_{2m-1}$, and $\delta = (\sqrt{2} - 1)^{2m}$. Standard Pell identities give

$$p\alpha = q - \delta, \quad \gcd(p, q) = 1, \quad p \text{ even}, \quad q \text{ odd}.$$

Using $P_{2m} = ((1 + \sqrt{2})^{2m} - (\sqrt{2} - 1)^{2m})/(2\sqrt{2})$,

$$2p\delta = \frac{1 - (\sqrt{2} - 1)^{4m}}{\sqrt{2}} < 1, \quad \text{hence} \quad 2\delta < \frac{1}{p}.$$

For every $1 \leq k < 2p$, $k \neq p$, the fractional part of kq/p is r/p for some $1 \leq r < p$. Since $0 < k\delta/p < 2\delta < 1/p$, subtraction produces no integer carry, and

$$\{k\alpha\} = \frac{r}{p} - \frac{k\delta}{p} \leq 1 - \frac{1}{p} < 1 - 2\delta.$$

Therefore, using $2p\alpha = 2q - 2\delta$,

$$\lfloor (2p - k)\alpha \rfloor = 2q - 1 - \lfloor k\alpha \rfloor.$$

Thus the terms indexed by k and $2p - k$ have opposite signs and cancel, apart from the central index $k = p$. At $k = p$ the sign is $(-1)^{q-1} = +1$, while the final term at $k = 2p$ has sign $(-1)^{2q-1} = -1$. The total is zero. □

Remark 3.1.2. The first positive witnesses are $4, 24, 140, 816, 4756, 27720, 161564, \ldots$. A separate 2026 OEIS Open result addresses the unboundedness conjecture of A348295; we do not count that published result as ours. Combining unboundedness with the infinite zero family implies every fixed nonnegative value is visited infinitely often.

3.1.2 A014865: a general least-prime-factor obstruction

Let $b \geq 2$ be an integer, $s_1 = 1$, and $s_n = bs_{n-1} + n$ for $n \geq 2$. The identity

$$(b - 1)^2 s_n = b^{n+1} - (n + 1)b + n = b(b^n - 1) - n(b - 1) \tag{3.1}$$

holds for every $n \geq 1$.

Theorem 3.1.3. *If $n > 1$ and $n \mid s_n$, then the least prime factor p of n divides $b(b - 1)$.*

Proof. Let p be the least prime factor of n. Reducing (3.1) modulo p and using $p \mid n, s_n$ gives $b(b^n - 1) \equiv 0 \pmod{p}$. If $p \mid b$, the conclusion holds. Otherwise $b^n \equiv 1 \pmod{p}$, so the order of b modulo p divides both n and $p - 1$. As p is the least prime factor of n, one has $\gcd(n, p - 1) = 1$. The order is one, yielding $b \equiv 1 \pmod{p}$, hence $p \mid b - 1$. □

Theorem 3.1.4. *For $b = 15$, every n satisfying $n \mid s_n$ is odd. Moreover, either $n = 1$ or n has a factor in $\{3, 5, 7\}$; thus every such $n > 7$ is composite.*

Proof. If n is even and $a = v_2(n) \geq 1$, the lifting-the-exponent formula gives

$$v_2(15^n - 1) = v_2(14) + v_2(16) + v_2(n) - 1 = a + 4.$$

Using (3.1) with $(b-1)^2 = 196$, the numerator $15(15^n - 1) - 14n$ has valuation exactly $a+1$, because the first term has valuation $a + 4$ and the second valuation $a + 1$. Therefore

$$\boxed{v_2(s_n) = a - 1 < v_2(n)},$$

which precludes $n \mid s_n$. So n is odd. By Theorem 3.1.3 its smallest prime factor must divide $15 \cdot 14 = 2 \cdot 3 \cdot 5 \cdot 7$; oddness eliminates 2. Any prime term must consequently be 3, 5, or 7. □

Remark 3.1.5. This proves all three assertions displayed under A014865. The general obstruction also applies to the related divisor-recursion sequences A014866 ($b = 16$) and A014868 ($b = 18$), but these latter applications are not counted as settled conjectures.

3.1.3 A048868 and A131418: two short structural proofs

Theorem 3.1.6. *If the reduced residue system modulo n contains more primes than nonprimes, then n is even (A048868).*

Proof. For odd $n \geq 3$, the totatives in $\{1, \ldots, n-1\}$ consist of exactly $\varphi(n)/2$ odd and $\varphi(n)/2$ even integers, under the involution $r \mapsto n - r$. Among even totatives, only 2 can be prime, and among odd totatives, 1 is certainly nonprime. Thus the reduced residue system has at least $(\varphi(n)/2 - 1) + 1 = \varphi(n)/2$ nonprimes. Its number of primes is at most $\varphi(n)/2$, contrary to a strict prime majority. The case $n = 1$ is immediate. □

Theorem 3.1.7. *Let $s_b(n)$ denote the base-b digit sum for $b \geq 2$. If $s_b(n) = s_b(n + s_b(n))$, then $(b-1) \mid n$. In particular every term of A131418 is divisible by 9.*

Proof. Since $s_b(t) \equiv t \pmod{b-1}$, the hypothesis implies $n \equiv n + s_b(n) \pmod{b-1}$. Hence $s_b(n) \equiv 0$, whence $n \equiv 0 \pmod{b-1}$. □

3.1.4 A396843 and A396844: self-composed generating functions

For an integer $c \geq 2$, let $A_c(x) = \sum_{n \geq 1} a_c(n) x^n \in \mathbb{Z}[[x]]$ be the solution with $a_c(1) = 1$ of

$$A_c(xA_c(x) - cxA_c(x)^2) = x^2. \tag{3.2}$$

Such a solution exists uniquely: for $n \geq 2$, the coefficient of x^{n+1} on the left is $a_c(n)$ plus an integer polynomial in $a_c(1), \ldots, a_c(n-1)$. Thus its vanishing determines $a_c(n)$ successively.

Theorem 3.1.8. *For each integer $c \geq 2$, one has in $\mathbb{Z}[[x]]$,*

$$\boxed{A_c(x) \equiv x + c \sum_{j \geq 0} (-1)^j x^{2^j + 1} \pmod{c^2}.} \tag{3.3}$$

Moreover, when c is odd, every coefficient $a_c(n)$ is odd.

Proof. Modulo c, Equation (3.2) reduces to $A(xA) = x^2$. The series $A = x$ solves this equation and uniqueness gives $A_c \equiv x \pmod{c}$. Write $A_c = x + cH$ with $H \in x^2\mathbb{Z}[[x]]$. Working modulo c^2,

$$xA_c - cxA_c^2 \equiv x^2 + c(xH - x^3),$$

so composition in (3.2) yields

$$xH - x^3 + H(x^2) \equiv 0 \pmod{c}.$$

Equivalently $H(x) \equiv x^2 - x^{-1}H(x^2) \pmod{c}$. Iteration, or coefficient induction, gives the unique solution $H(x) \equiv \sum_{j\geq 0}(-1)^j x^{2^j+1} \pmod{c}$. This proves (3.3).

If c is odd, reduce (3.2) modulo 2. There $c \equiv 1$. The all-ones series $U(x) = x/(1+x)$ satisfies

$$U(xU + xU^2) = U\left(\frac{x^2}{1+x^2}\right) = x^2 \quad \text{in } \mathbb{F}_2[[x]].$$

Uniqueness gives $A_c \equiv U \pmod{2}$, meaning every coefficient is odd. □

Corollary 3.1.9 (A396843, $c = 3$)**.** *All terms of A396843 are odd. In addition, for $n \geq 2$,*

$$a_3(n) \equiv \begin{cases} 3(-1)^j \pmod 9, & n = 2^j + 1, \\ 0 \pmod 9, & \text{otherwise.} \end{cases}$$

This supplies a full modular classification, including $a_3(5) \equiv 3 \pmod 9$. The source already records a July 2026 counterexample to its older mod-9 conjecture; that refutation is not counted again. The counted target for this entry is the all-odd conjecture.

Corollary 3.1.10 (A396844, $c = 4$: a refutation and replacement)**.** *For $n \geq 2$,*

$$a_4(n) \equiv \begin{cases} 4(-1)^j \pmod{16}, & n = 2^j + 1, \\ 0 \pmod{16}, & \text{otherwise.} \end{cases}$$

The OEIS conjecture that indices $n > 8$ not of the form $2 \cdot 4^k + 1$ give multiples of 16 is false:

$$\boxed{n = 17 = 2^4 + 1, \quad a_4(17) = 113101684573952804 \equiv 4 \pmod{16}.}$$

In fact every index $n = 2^{2j}+1$ with $j \geq 2$ is such a counterexample; the conjecture omits the residue-4 branch.

Corollary 3.1.11 (A396845, $c = 5$)**.** *Every coefficient is odd, and for $n \geq 2$ the complete residue modulo 25 is $5(-1)^j$ if $n = 2^j + 1$, and 0 otherwise. This consequence is retained from the supplied manuscript but is not counted as a selected open conjecture.*

Provenance and scope. These six counted entry-level selections are retained from the supplied exploratory manuscript, after rechecking the proofs and the current OEIS statements. A348299 is an alias for the zero-index question and adds no count. The published OEIS Open result for A348295 concerns unboundedness (Conjecture 1), whereas the selected Pell construction proves infinitely many zeros (Conjecture 2). The auxiliary applications A014866, A014868, and A396845 add no count.

3.2 New coefficient congruences

All congruences between ordinary generating functions in this section are coefficientwise. An exponential generating function is explicitly identified as such; its integer coefficients are never reduced modulo a prime by illegally inverting factorials.

3.2.1 A195194: nested reversions

Write $R_k(x) = \mathrm{Rev}(x - kxR_{k+1}(x))$, where Rev denotes compositional inversion and $R_k = x + O(x^2)$. The coefficients of any fixed degree stabilize as the nesting depth increases: multiplication by x raises the order of a change by one, and inversion preserves the order of a change. The sequence A195194 is $[x^n]R_1(x)$, $n \geq 1$.

Theorem 3.2.1. *The term of A195194 at index n is odd if and only if n is a power of 2.*

Proof. Reduction of compositional inverses commutes with reduction modulo 2. The factor 2 gives $R_2(x) \equiv x$, and consequently

$$R_1(x) \equiv \mathrm{Rev}(x + x^2) = \sum_{j \geq 0} x^{2^j} \pmod 2.$$

The last identity follows by squaring: if $S = \sum_{j \geq 0} x^{2^j}$, then $S + S^2 = x$. Its linear coefficient is 1, so it is the unique compositional inverse. □

3.2.2 A397241: a diagonal-power relation

Here $A(x) = \sum_{n \geq 0} a(n)x^n$, $a(0) = a(1) = 1$, and the defining relation is

$$n[x^n]A(x)^n = (n-1)[x^n]A(x)^{n+1} \qquad (n > 1).$$

Theorem 3.2.2. *Every coefficient of A397241 is odd.*

Proof. For a series with constant term 1, its coefficient $a(n)$ occurs in $[x^n]A^k$ with multiplier k. Thus the difference of the two sides has multiplier $n^2 - (n-1)(n+1) = 1$ on $a(n)$. The initial coefficients therefore determine a unique series over every commutative ring.

We show that $U(x) = (1-x)^{-1}$ satisfies the relation over $\mathbb{F}_2$. If n is even, the required vanishing is $\binom{2n}{n} \equiv 0 \pmod 2$. If $n > 1$ is odd, it is $\binom{2n-1}{n} \equiv 0 \pmod 2$. Indeed, with $s_2(n)$ denoting the number of 1's in the binary expansion, the factorial valuation formula gives

$$v_2\binom{2n}{n} = s_2(n), \qquad v_2\binom{2n-1}{n} = s_2(n) - 1.$$

For an odd $n > 1$, $s_2(n) \geq 2$. Uniqueness now gives $A \equiv U$. Only the parity conjecture is selected here; the source's mod-3, mod-4, and asymptotic assertions are not included in this count. □

3.2.3 A396808: another diagonal-power relation

The definition, again with $a(0) = a(1) = 1$, is

$$(n+1)[x^n]A(x)^{n+1} = n[x^n]A(x)^{n+2} \qquad (n > 1).$$

Theorem 3.2.3. *For $n > 0$, the term of A396808 at index n is odd if and only if n is a power of 2.*

Proof. The coefficient multiplier is $(n+1)^2 - n(n+2) = 1$, so reduction modulo 2 still gives a unique solution. Let

$$R(x) = \frac{1 + \sqrt{1 - 4x}}{2} = 1 - \sum_{n \geq 1} C_{n-1}x^n,$$

where C_j are the Catalan numbers. Lagrange inversion, applied to $1 - R = x/(1 - (1 - R))$, gives, for integers $k > n \geq 1$,

$$[x^n]R(x)^k = (-1)^n \frac{k}{k-n}\binom{k-n}{n}.$$

For $n > 2$, both coefficients with $k = n+1, n+2$ are zero. For $n = 2$, they are 0 and 2, respectively; the defining relation therefore holds modulo 2 also at this boundary. Moreover $R \equiv 1+x \pmod{x^2, 2}$, so uniqueness gives $A \equiv R$. Writing $S = 1 - R$, the equation $R^2 - R + x = 0$ reduces to $S^2 + S = x$; hence $S = \sum_{j\geq 0} x^{2^j}$. This proves the exact parity pattern. The separate mod-3 conjectures and asymptotic limit are not claimed here. □

3.2.4 A397356: a reciprocal diagonal relation

For A397356, $A(0) = A'(0) = 1$ and

$$[x^n]A(x)^{-n^2} = [x^n]A(x)^{-(n^2-1)} \qquad (n > 1).$$

Theorem 3.2.4. *The coefficient at index $n \geq 0$ is odd if and only if $n+1$ is a power of 2.*

Proof. The coefficient $a(n)$ has multiplier -1 in the difference, so this system is triangular with a unique solution modulo 2. Let $C = 1 + xC^2$ be the usual Catalan generating function. Substituting it into that difference gives

$$[x^n](1 - C)C^{-n^2} = -[x^{n-1}]C^{\,2-n^2}.$$

If n is even, the exponent is even; modulo 2 its series has only even powers, while $n - 1$ is odd.

If $n > 1$ is odd, the Lagrange coefficient formula yields

$$[x^{n-1}]C^{\,2-n^2} = \frac{2-n^2}{n(2-n)}\binom{n(2-n)}{n-1} = \frac{2-n^2}{n(2-n)}\binom{n^2-n-2}{n-1}.$$

The prefactor has odd numerator and odd denominator, so it is enough to show that the last binomial coefficient is even. Put $t = v_2(n-1) \geq 1$. The bit at position t of $n-1$ is 1, whereas

$$n^2 - n - 2 = n(n-1) - 2 \equiv 2^t - 2 \pmod{2^{t+1}}$$

has bit 0 there. The parity criterion for binomial coefficients therefore gives the required evenness. Thus $A \equiv C \pmod 2$. Finally $C = 1 + xC^2$ gives

$$C(x) \equiv \sum_{j\geq 0} x^{2^j-1} \pmod 2,$$

which proves the assertion. The source's additional mod-3 conjecture remains outside the selected scope. □

3.2.5 A396803: exponential generating functions and parity

Let $\ell \geq 0$ and write $A^{\circ\ell}$ for the ℓ-fold compositional iterate, with $A^{\circ 0}(x) = x$. Consider

$$A(x) = x\exp(A^{\circ\ell}(x)) = \sum_{n\geq 1} a_\ell(n)\frac{x^n}{n!}.$$

Its coefficients are integers: composition and exponential of integer exponential series use integer Bell polynomials, and multiplication by x multiplies the n-th coefficient by n. The same operations give coefficientwise existence and uniqueness, since the coefficient of degree n on the right depends only on coefficients of A of degrees less than n.

Lemma 3.2.5. *If the integer sequence $b_1, b_2, \ldots$ satisfies $b_1 = 1$ and b_{2j} is even for every $j \geq 1$, then the complete Bell polynomial $B_{2m}(b_1, \ldots, b_{2m})$ is odd.*

Proof. Reduce the block weights modulo 2. It suffices to count set partitions of $2m$ labels into allowed odd-sized blocks; the allowed sizes are those j with b_j odd. Act on the labels by a fixed-point-free involution consisting of m prescribed transpositions. Partitions not fixed by this involution pair off.

In a fixed partition, no odd-sized block can be fixed, since a fixed block would be a union of transposition pairs. Blocks therefore come in exchanged pairs. A pair of blocks of size r occupies exactly r of the prescribed label pairs, one label from each in each block. For a prescribed group of r pairs there are 2^{r-1} ways to choose these two unordered blocks. If $r > 1$, that factor is even. The sole odd contribution has $r = 1$ for every block pair, namely the all-singleton partition, which is allowed because $b_1 = 1$. □

Theorem 3.2.6. *For every integer $\ell \geq 0$ and every $n \geq 1$,*

$$a_\ell(n) \equiv n \pmod 2.$$

In particular, this proves the parity conjecture in A396803, where $\ell = 3$.

Proof. Write $B = A^{\circ \ell}$, with exponential coefficients b_j. The equation immediately gives

$$a_\ell(n) = n\, B_{n-1}(b_1, \ldots, b_{n-1}),$$

so every even-indexed coefficient of A is even. Composition preserves this property for exponential series with zero constant term: in a Bell-polynomial expansion at an even index, an even-sized block contributes an even inner coefficient, while if all blocks have odd size, their number is even and the outer coefficient is even. Consequently all b_{2j} are even and $b_1 = 1$. For odd n, Lemma 3.2.5 makes the displayed Bell polynomial odd. This proves both directions. The selected claim is parity; other congruences printed in the OEIS entry are not silently included. □

3.2.6 A368634: a parity-sensitive quadratic equation

Let

$$A(x) = 1 + \frac{x}{2}(3A(x)^2 - A(-x)^2) = \sum_{n \geq 0} a(n) x^n.$$

Theorem 3.2.7. *For A368634, $a(n)$ is odd exactly when $n = 2^j - 1$, where $j \geq 0$.*

Proof. For $n \geq 1$, comparing coefficients gives the integer recurrence

$$a(n) = \begin{cases} 1, & n \text{ odd}, \\ 2, & n \text{ even} \end{cases} \sum_{i=0}^{n-1} a(i) a(n-1-i).$$

Thus positive even indices give even values. For odd n, the off-diagonal convolution terms pair off modulo 2, leaving $a(n) \equiv a((n-1)/2)$. Starting from $a(0) = 1$, repeated application gives precisely the stated indices. □

3.2.7 A393172: a complete congruence modulo four

The defining equation is $A(x - x^2) = x + A(x)^2$, with $A(x) = x + O(x^2)$.

Theorem 3.2.8. *For A393172,*

$$A(x) \equiv x + 2 \sum_{j \geq 1} x^{2^j} \pmod 4.$$

Thus, for $n > 1$, $a(n) \equiv 2 \pmod 4$ exactly at powers of 2, and $a(n) \equiv 0 \pmod 4$ otherwise.

Proof. The equation is triangular with coefficient 1 on each new coefficient of A. Modulo 2, its solution is $A = x$. Write $A = x + 2H$. Reduction modulo 4 gives

$$x - x^2 + 2H(x - x^2) \equiv x + x^2 \pmod 4,$$

hence $H(x + x^2) = x^2$ in $\mathbb{F}_2[[x]]$. The inverse of $x + x^2$ is $S = \sum_{j\geq 0} x^{2^j}$, so $H = S^2 = \sum_{j\geq 1} x^{2^j}$ modulo 2. □

3.2.8 A389536: a shifted dyadic support

For A389536, $A = x + O(x^2)$ satisfies $xA(x) = A(x^2 + 2x^3)$.

Theorem 3.2.9.

$$A(x) \equiv x + 2\sum_{j\geq 0} x^{2^j+1} \pmod 4.$$

In particular, for $n > 1$, $a(n) \equiv 2 \pmod 4$ exactly when $n = 2^j + 1$, and is divisible by 4 at every other index.

Proof. For $n > 1$, the coefficient $a(n)$ is determined by coefficients of A with indices at most $\lfloor (n + 1)/2 \rfloor < n$. Modulo 2, $xA = A(x^2)$ forces $A = x$. Put $A = x + 2H$. Modulo 4,

$$x^2 + 2xH(x) = x^2 + 2x^3 + 2H(x^2),$$

so $H = x^2 + x^{-1}H(x^2)$ modulo 2. Iterating this order-increasing relation gives the asserted sum. □

3.2.9 A378580: a theta-series congruence

The definition is

$$A(x/A(x)) = 1 + 2\sum_{j\geq 1} x^{j^2}, \qquad A(0) = 1.$$

Theorem 3.2.10. *For A378580,*

$$A(x) \equiv 1 + 2\sum_{j\geq 1} x^{j^2} \pmod 4.$$

Thus a positive-indexed term is 2 modulo 4 exactly at square indices and is 0 modulo 4 elsewhere.

Proof. The coefficient of degree n in the left side depends on the new coefficient $a(n)$ with multiplier 1; hence the equation defines an integer series recursively. Since $x/A(x)$ has linear coefficient 1, composition with it is invertible modulo 2. The defining equation therefore first gives $A \equiv 1 \pmod 2$. Write $A = 1 + 2H$. Then $x/A(x) \equiv x \pmod 2$, and hence

$$A(x/A(x)) = 1 + 2H(x/A(x)) \equiv 1 + 2H(x) \pmod 4.$$

Comparing with the defining theta series proves the result. The separate question about the radius of convergence is not selected. □

3.3 Prime divisibility and modular periodicity

This section settles 31 selected entry-level assertions: 12 inherited from the supplied draft and 19 additional assertions. There are 27 proofs and four refutations. Each OEIS identifier below links to the primary record checked on 20 October 2026. "Additional" describes this project's provenance; it is not a claim of worldwide mathematical priority. The repeated general conjecture about $G(e^x - 1)$ is proved once and contributes no count beyond the ten distinct sequence-specific assertions to which it is applied. The established consequence for A002050 is explicitly excluded from the count.

3.3.1 Twelve inherited assertions about algebraic powers

The selected assertion in each of the following twelve records is that every prime divides infinitely many terms and that the successive gaps between all such indices are eventually periodic:

power	$[\sqrt{2}]$	$[\sqrt{3}]$	$[\sqrt{6}]$
$(1+\sqrt{2}+\sqrt{3})^n$	A188571	A188572	A188573
$(2+\sqrt{2}+\sqrt{3})^n$	A377110	A377111	A377112
$(3+\sqrt{2}+\sqrt{3})^n$	A377114	A377115	A377116

The remaining three entries are the coefficients of $1, t, t^2$ in $(t+t^2)^n$, where $t^3 = 2$:

coefficient	1	t	t^2
entry	A377117	A377118	A377119.

The assertion is about the infinite tail, including the full zero-index set, not merely one infinite subsequence of zeros. The finite-state argument below supplies the required statement about the full set.

Lemma 3.3.1. *Let a_n be an integer sequence that is eventually periodic modulo a prime p. If $p \mid a_n$ for infinitely many n, then the increasing enumeration of all these indices has eventually periodic successive gaps.*

Proof. Choose N, T such that $a_{n+T} \equiv a_n \pmod{p}$ for $n \geq N$. Let $r_1 < \cdots < r_s$ be the zero indices in $[N, N+T)$. Infinitely many zeros imply $s \geq 1$. After the finite initial portion, the zero set is exactly

$$\{r_i + kT : 1 \leq i \leq s,\ k \geq 0\}.$$

Its repeating gap block is $(r_2 - r_1, \ldots, r_s - r_{s-1}, T + r_1 - r_s)$, interpreted as (T) if $s = 1$. □

Theorem 3.3.2. *For $c = 1, 2, 3$, every prime divides infinitely many terms of each of the three nonconstant coefficient sequences in $(c+\sqrt{2}+\sqrt{3})^n$. Their full zero-index sets have eventually periodic gaps.*

Proof. Work in the free rank-four ring

$$R = \mathbb{Z}[u, v]/(u^2 - 2, v^2 - 3), \qquad \beta_c = c + u + v,$$

with basis $1, u, v, uv$. These coordinates agree with the indicated real algebraic coefficients. The determinant of multiplication by β_c is its norm

$$N(\beta_c) = \prod_{\epsilon,\delta \in \{1,-1\}} (c + \epsilon\sqrt{2} + \delta\sqrt{3}) = c^4 - 10c^2 + 1.$$

Its values for $c = 1, 2, 3$ are respectively $-8, -23, -8$. Whenever this determinant is nonzero modulo p, multiplication is invertible on R/pR; in particular β_c is a unit there.

For $p \geq 5$, Fermat's theorem gives $u^{p^2} = u$ and $v^{p^2} = v$: for example $u^{p^2} = 2^{(p^2-1)/2}u = u$, since $(p^2-1)/2$ is divisible by $p-1$. Frobenius therefore gives $\beta_c^{p^2} = \beta_c$. If also $p \nmid N(\beta_c)$, cancellation yields

$$\beta_c^{p^2-1} = 1 \quad \text{in } R/pR. \tag{3.4}$$

Each nonconstant coefficient is consequently zero at every positive multiple of p^2-1.

Here are all exceptional primes, including the nonunit cases. Modulo 2,

$$\beta_1^2 = \beta_3^2 = 0, \qquad \beta_2^2 = 1.$$

Thus the first and third families are identically zero after $n = 1$, and the second family has zeros at every positive even index. Modulo 3 one has $\beta_c^3 = c - u$. For $c = 1, 2$ this is a nonzero element of the field $\mathbb{F}_3[u]/(u^2+1)$, so

$$\beta_1^{24} = \beta_2^{24} = 1.$$

For $c = 3$, $(\beta_3^3)^4 = (-u)^4 = 1$, giving $\beta_3^{12} = 1$. These identities again supply infinitely many simultaneous zeros of the three nonconstant coefficients.

The only remaining case is $c = 2$, $p = 23$. Since $2^{11} = 3^{11} = 1 \pmod{23}$, Frobenius now gives

$$\beta_2^{23} = \beta_2, \qquad \beta_2^{n+22} = \beta_2^n \quad (n \geq 1).$$

Direct integer expansions give

$$[u]\beta_2^3 = 23, \qquad [v]\beta_2^5 = 529 = 23^2, \qquad [uv]\beta_2^1 = 0.$$

These yield zeros at the positive indices congruent to $3, 5, 1$ modulo 22, respectively. No cancellation of the nonunit β_2 has been used.

Finally, for any fixed p the sequence β_c^n is an orbit under multiplication by β_c in the finite ring R/pR. Every such orbit is eventually periodic. Each coordinate is therefore eventually periodic; the infinite zero families already proved and Lemma 3.3.1 prove the assertion about all successive gaps. □

Theorem 3.3.3. *Every prime divides infinitely many terms of each coefficient sequence of $(t+t^2)^n$ in the basis $1, t, t^2$, with $t^3 = 2$. The associated zero-index gaps are eventually periodic.*

Proof. In $T = \mathbb{Z}[t]/(t^3-2)$ put $\eta = t + t^2$. Exact multiplication gives

$$\eta^3 = 6(1 + t + t^2) = 6(1 + \eta).$$

For $p = 2$ or 3 we have $\eta^3 = 0$ in T/pT, so every coefficient vanishes for all $n \geq 3$.

Let $p \geq 5$. The identity $\eta(\eta^2 - 6) = 6$ shows that η is a unit. The polynomial $t^3 - 2$ is squarefree modulo p, since it has no common root with $3t^2$. The algebra T/pT is therefore a product of finite fields of degrees among $1, 2, 3$. Each such degree divides 6, so the p^6-power map is the identity on every factor. Thus

$$\eta^{p^6} = \eta, \qquad \eta^{p^6-1} = 1.$$

Writing $L = p^6 - 1$, the two nonconstant coefficients vanish at every positive multiple of L. The constant coefficient of η is zero, so the constant coefficient vanishes at every $1 + kL$, $k \geq 0$. All three zero sets are infinite. Periodicity of the powers in the finite ring, followed by Lemma 3.3.1, proves the gap claim. □

Remark 3.3.4 (Source scope and a finite initial discrepancy)**.** The displayed data of A377117 have value 1 at $n = 1$, whereas the coefficient definition gives $[1]\eta = 0$. This is precisely a one-term difference, as can be checked from the printed generating function. The true constant-coefficient generating function is

$$\frac{1 - 2x^2}{1 - 6x^2 - 6x^3},$$

whereas the printed one is

$$\frac{1+x-2x^2-6x^3-6x^4}{1-6x^2-6x^3} = \frac{1-2x^2}{1-6x^2-6x^3} + x.$$

Both tails therefore satisfy the selected assertion. This finite correction is not a separate counted result.

Some ancillary short-period examples printed in the source comments do not agree with the definitions. For example, modulo 2 the nonconstant coefficients for $c = 1$ vanish at every $n \geq 2$, so the eventual zero-index gap is 1. The twelve results above settle the universal infinitude and eventual-gap assertions; they do not validate the source's separate numerical period lists. The two-page Sela Fried note linked from A188571, dated 1 January 2025 proves the already known recurrence and generating function, not the modular assertion selected here. Those earlier recurrence and generating-function results are not counted.

3.3.2 Three constant-coefficient conjectures refuted by complete cycles

Theorem 3.3.5. *Let $C_c(n) = [1](c+u+v)^n$ in the ring above. Then*

entry	c	p	*conclusion for every* $n \geq 0$
A188570	1	71	$C_1(n) \not\equiv 0 \pmod{71}$
A377109	2	241	$C_2(n) \not\equiv 0 \pmod{241}$
A377113	3	167	$C_3(n) \not\equiv 0 \pmod{167}$.

Consequently each entry's conjecture that every prime divides infinitely many terms is false.

Proof. For the three respective primes, choose square roots $r^2 = 2$ and $s^2 = 3$ as follows. All four entries in the last column are nonzero.

c	p	r	s	$(c+r+s, c+r-s, c-r+s, c-r-s) \bmod p$
1	71	12	28	$(41, 56, 17, 32)$
2	241	22	56	$(80, 209, 36, 165)$
3	167	13	62	$(78, 121, 52, 95)$

Adding the four evaluations under $(u, v) = (\pm r, \pm s)$ cancels all nonconstant coordinates. Hence, if the last column is (w_1, w_2, w_3, w_4),

$$C_c(n) \equiv 4^{-1} \sum_{i=1}^{4} w_i^n \pmod{p}. \tag{3.5}$$

By Fermat's theorem the right-hand side has period $p-1$ for all $n \geq 0$. The complete residue lists in the following certificate, covering $0 \leq n < p-1$, contain no zero. This proves the claim for every n, not merely for a tested finite initial segment.

The lists can be checked directly from (3.5), or from the integer coefficient update

$$(a, b, d, e) \longmapsto (ca + 2b + 3d,\ a + cb + 3e,\ a + cd + 2e,\ b + d + ce) \tag{3.6}$$

starting at $(1, 0, 0, 0)$ and reducing modulo p at each step. The machine-readable certificate includes all four coordinates at every step and verifies return to the initial vector after $p-1$ steps. Its independent four-power check uses (3.5). □

Complete zero-free residue certificates. Each line is labeled by the index of its first value; values continue in increasing order of n. The three blocks contain respectively 70, 240, and 166 residues. Every listed value is nonzero.

A188570 modulo 71 (period 70).

```
  0:   1,   1,   6,  16,   9,  12,  18,  33,  13,  63,  62,  59
 12:   6,  56,  13,  13,  37,  14,  29,  39,  60,  44,  24,  50
 24:   1,  30,  13,  59,  29,  29,  31,  35,  32,   4,  45,  35
 36:  15,   9,  38,  15,  46,  63,   3,   3,  23,  63,  36,  52
 48:  25,  23,   3,  49,  40,  66,  19,  21,  63,  63,  36,  53
 60:  65,  45,  22,   7,  58,  55,  19,  63,  54,  36
```

A377109 modulo 241 (period 240).

```
  0:   1,   2,   9,  38, 185, 199, 110, 139,  66, 110, 169,  71
 12:  45,  62,  52,  98, 113,  60,  33,  51, 137,  52,  54,  22
 24: 227, 103, 158, 199, 162,  97,  68,  57,  44,  36,  57, 188
 36: 225,  22, 207,  16,  60,  70,  15, 233,  64, 186,  37, 129
 48:  16, 135, 193,  83, 142,  89,  89, 239,  87, 131, 206,  36
 60:  44,   8,  42, 218, 176, 132, 224, 176,  59, 216,  67, 124
 72: 165, 160,   1,  23,  34,   7, 142,  90, 181, 177, 223, 169
 84:  13,  25, 180,  53, 172,  10, 183, 203,  46, 148, 233, 237
 96: 195,  23, 194,  60, 137, 197,  25, 136, 144, 205,  75, 188
108: 198, 175,  54, 240, 221,  73, 186, 121, 212, 193, 199, 218
120: 121,  40,  13, 216, 153, 221, 125, 204, 187,  72, 165, 134
132:  77,  40,  37,  33, 233,  98,  37, 229, 105,  74,  69,  87
144: 107,  65, 154,  21, 194,  75,  53, 233, 164,  74,  61, 125
156: 193,  44, 222,  81, 181,  32,  11,  55,  96, 164,  22,  64
168: 136, 173, 197,  20, 110, 111, 104,  63, 208,  93, 202,  99
180:  76, 227,  27, 153,  55, 170, 228, 113,  27, 238,  82, 189
192:  45, 122, 238,  86,  66, 226, 127,  25,  60, 215, 227, 106
204: 222,  47, 195, 118,  52, 213, 179,  25,  78, 126, 218, 172
216:  74,  61, 198, 238, 105, 219,  40, 201,  24, 167,  71,  10
228: 230, 153,  39, 175, 100, 111, 190,  58, 180, 215, 214,  42
```

A377113 modulo 167 (period 166).

```
  0:   1,   3,  14,  72,  66, 157,  43,  24, 114, 125, 151, 139
 12:  99, 147, 111,  60,  10, 143,  34,  86, 134,  99,  26,  70
 24:   9,  70,   4, 131, 152, 152, 120,  90, 137, 151,  62,  60
 36: 157,  88,  29, 150, 159,  76,  12, 121, 166,  23,  45, 114
 48: 150, 130, 124,  39, 114, 131,  88,  74,  41,   3, 150,  53
 60:  19, 110,  53, 138,  80, 149, 139,  56,  10,   9, 128,  64
 72: 157, 107, 142,  34,  51, 108, 150, 102,  49, 156, 134,  83
 84: 141,  42,  24,  27,  74,  19, 130, 150,  89,  25, 115,  74
 96:  78,  60,  90, 104, 124,  42,  92,  28, 131,  82, 139,  63
108: 125,  44, 166,  47,  46, 124,  42, 137, 104,  65, 110,  39
120:  81,  46,  74,  58, 129,  77, 127, 116,  31, 143,  89,  31
132:  61, 108, 144, 152, 158, 145,  63,   3, 144, 102,  45, 149
144:  11,  59,  54, 106, 146, 152,  85,  17,  85,  57, 110,  22
156:   9,  33,  99,  10,  92,   2,  87, 108, 137,  82
```

3.3.3 One general Stirling theorem and ten new applications

Write $S(n,k)$ for Stirling numbers of the second kind. For $m \geq 2$ define

$$e(m) = \max_{p \mid m} v_p(m), \qquad \lambda(m) = \exp((\mathbb{Z}/m\mathbb{Z})^\times),$$

where λ is the Carmichael function. It divides $\varphi(m)$.

Theorem 3.3.6. *For any integer sequence $(b_k)_{k\ge 0}$, put*

$$A_n = \sum_{k=0}^{n} b_k\, k! S(n,k).$$

For every $m \ge 2$ and every $n \ge e(m)$,

$$A_{n+\lambda(m)} \equiv A_n \pmod m. \tag{3.7}$$

Equivalently, every exponential generating function of the form $G(e^x-1)$ with $G(t) \in \mathbb{Z}[[t]]$ has this eventual period. In particular its least eventual period divides $\varphi(m)$.

Proof. Since $m \mid k!$ for $k \ge m$, and $S(n,k) = 0$ for $k > n$,

$$A_n \equiv \sum_{k=0}^{m-1} b_k\, k! S(n,k) = \sum_{k=0}^{m-1} b_k \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} j^n \pmod m.$$

Let $p^a \| m$. If $p \mid j$ and $n \ge e(m) \ge a$, then j^n and $j^{n+\lambda(m)}$ both vanish modulo p^a. If $p \nmid j$, then $j^{\lambda(m)} \equiv 1 \pmod{p^a}$ because $\lambda(p^a) \mid \lambda(m)$. Thus every summand has the stated period modulo every prime power dividing m. The Chinese remainder theorem proves (3.7).

The familiar identity $(e^x-1)^k = \sum_{n\ge 0} k! S(n,k) x^n/n!$ proves the equivalent generating-function formulation by coefficient comparison. A least eventual period divides any eventual period, and $\lambda(m) \mid \varphi(m)$. Modulo 1 the assertion is immediate. □

Each of the following ten source records explicitly proposes eventual periodicity with least period dividing $\varphi(m)$. Their known definitions or formulas give the indicated integral weights, so Theorem 3.3.6 proves each selected assertion with the stronger period $\lambda(m)$.

Entry	Weights b_k, or their ordinary generating function $G(t)$
A004123	$b_k = 2^k$, so $G(t) = (1-2t)^{-1}$. The displayed OEIS offset is 1: its $a(n)$ is A_{n-1}.
A306082	b_k counts partitions of k into squares; $G(t) = \prod_{r\ge 1}(1-t^{r^2})^{-1}$.
A006531	$b_0 = 1$ and $b_k = M_{k-1}$ for $k \ge 1$, where M_j is the jth Motzkin number.
A167137	b_k is the ordinary partition number; $G(t) = \prod_{r\ge 1}(1-t^r)^{-1}$.
A080253	$G(t) = (1+t)/(1-2t-t^2)$, so $b_0 = 1$, $b_1 = 3$, $b_k = 2b_{k-1} + b_{k-2}$.
A305550	b_k counts partitions into distinct parts; $G(t) = \prod_{r\ge 1}(1+t^r)$.
A354242	$b_k = \binom{2k}{k}$, so $G(t) = (1-4t)^{-1/2}$.
A301921	$G(t) = \cfrac{1}{1-\cfrac{t}{1-\cfrac{t^2}{1-\cfrac{t^3}{1-\cdots}}}}$.
A320352	$b_k = F_k$ with $F_0 = 0$, $F_1 = 1$; $G(t) = t/(1-t-t^2)$.
A354253	$b_k = 2^k\binom{2k}{k}$, so $G(t) = (1-8t)^{-1/2}$.

For the continued fraction, the coefficient of any fixed degree is unchanged after sufficiently deep truncation, because successive numerators have strictly increasing positive degrees. Every finite truncation is obtained by reciprocating series with constant term 1 over $\mathbb{Z}[[t]]$; hence every coefficient of its formal limit is an integer. This justifies the hypothesis for A301921 without appealing to a numerically fitted series. The infinite products likewise have only finitely many relevant factors at any fixed degree.

The index threshold is $n \ge e(m)$ for the nine zero-offset sequences and $n \ge e(m)+1$ in the displayed indexing of A004123. For A080253, for example, the source exponential generating function

$e^x/(2-e^{2x})$ becomes $(1+t)/(1-2t-t^2)$ upon setting $t=e^x-1$, as required. The formulas used for the weights are existing identities on the cited pages; the new selected targets here are their modular conjectures.

Remark 3.3.7 (A deliberately uncounted known consequence)**.** The same proof applies to A002050, with $G(t)=(t+t^2)/(1-t)$. However, for $n>0$ this sequence is $2\,\mathtt{A000670}(n)-1$, and the primary record for A000670 already cites Barsky's congruence with period $(p-1)p^{h-1}$ modulo p^h for $n\geq h$. Combining these known congruences by the Chinese remainder theorem immediately gives a period dividing $\varphi(m)$ for A002050. It contributes zero to the 31-entry count.

3.3.4 Two eventual-periodicity proofs from polynomial differential equations

Lemma 3.3.8. *Let $D=\sum_{i=1}^{d}Q_i(X_1,\dots,X_d)\partial_{X_i}$ be a polynomial derivation over $\mathbb{Z}$, let H be any integer polynomial, and let $\mathcal{L}=D+H$ act on polynomials, with H acting by multiplication. Fix an integer point $\boldsymbol{c}=(c_1,\dots,c_d)$ and an integer polynomial P. For every positive modulus m, the sequence*

$$(\mathcal{L}^nP)(\boldsymbol{c}) \pmod m$$

is eventually periodic.

Proof. For $m\geq 2$, work in

$$Q=(\mathbb{Z}/m\mathbb{Z})[X_1,\dots,X_d]/(X_1^m-c_1^m,\dots,X_d^m-c_d^m).$$

The ideal is stable under D, since the derivative of each generator is $mX_i^{m-1}Q_i=0$ modulo m. It is also stable under multiplication by H. Consequently $\mathcal{L}$ is a well-defined endomorphism of the finite additive module Q. The quotient has basis the m^d monomials whose individual exponents are less than m, and therefore has m^{m^d} elements. The orbit of P is eventually periodic by the pigeonhole principle and deterministic iteration. Evaluation at $\boldsymbol{c}$ is well-defined on the quotient, and yields the stated eventual period. Modulo 1 there is nothing to prove. □

Theorem 3.3.9. *The sequence A005046 is eventually periodic modulo every positive integer.*

Proof. This sequence counts partitions of a $2n$-element set into blocks of even size and has

$$F(x)=\exp(\cosh x-1)=\sum_{n\geq 0}a(n)\frac{x^{2n}}{(2n)!}.$$

Put $X=\sinh x$ and $Y=\cosh x$. Since $X'=Y$, $Y'=X$ and $F'=XF$, induction gives

$$F^{(j)}(x)=P_j(X,Y)F(x),\qquad P_0=1,\qquad P_{j+1}=(Y\partial_X+X\partial_Y+X)P_j.$$

All P_j are integer polynomials, and $F(0)=1$, $(X(0),Y(0))=(0,1)$. Apply Lemma 3.3.8 with the ideal (X^m,Y^m-1). It shows that the full derivative sequence $F^{(j)}(0)$ is eventually periodic modulo m. Every fixed-step subsequence of an eventually periodic sequence is eventually periodic: if a full-sequence period is T, then advancing n by T advances the derivative index $2n$ by $2T$. The even derivative subsequence $a(n)=F^{(2n)}(0)$ therefore has the required property. □

Theorem 3.3.10. *The sequence A006154 is eventually periodic modulo every positive integer.*

Proof. The exponential generating function of ordered set partitions into odd-sized blocks is $F(x)=1/(1-\sinh x)$. Put $X=\sinh x$, $Y=\cosh x$ and $Z=F(x)$. Then

$$X'=Y,\qquad Y'=X,\qquad Z'=YZ^2,\qquad (X(0),Y(0),Z(0))=(0,1,1).$$

For the integer polynomial derivation

$$D = Y\partial_X + X\partial_Y + YZ^2\partial_Z,$$

the chain rule gives $a(n) = (D^n Z)(0,1,1)$. Lemma 3.3.8, with $H = 0$ and ideal $(X^m, Y^m - 1, Z^m - 1)$, proves eventual periodicity. □

The selected source conjectures for these two entries ask for existence of an eventual period. The proof gives existence for every modulus, without asserting a period dividing $\varphi(m)$ or a sharp bound on its length.

3.3.5 Three factorial divisor sums with explicit modular tails

Theorem 3.3.11. *For positive integers r, s define, for $n \geq 1$,*

$$B_{r,s}(n) = \sum_{d|n} r^d s^{n/d} \frac{n!}{d!(n/d)!}.$$

For every positive integer m and all $n \geq 2m + 4$,

$$B_{r,s}(n) \equiv rs^n + sr^n \pmod{m}. \tag{3.8}$$

In particular its least eventual period divides $\lambda(m)$, with the convention $\lambda(1) = 1$.

Proof. The case $m = 1$ is immediate, so assume $m \geq 2$. The terms $d = 1$ and $d = n$ give rs^n and sr^n, respectively. For any other divisor put $e = n/d$; then $d, e \geq 2$ and $de = n$. The inequality $(d-2)(e-2) \geq 0$ implies

$$d + e \leq \frac{n}{2} + 2, \qquad N = n - d - e \geq \frac{n}{2} - 2 \geq m.$$

Thus the corresponding factorial quotient can be written as

$$\frac{n!}{d!e!} = \binom{n}{d, e, N} N!,$$

which is an integer multiple of m. All nonendpoint terms vanish modulo m, proving (3.8).

For any integer q, the proof of Theorem 3.3.6 also showed directly that $q^{n+\lambda(m)} \equiv q^n \pmod{m}$ once $n \geq e(m)$. Since $2m + 4 \geq e(m)$, the displayed tail has period $\lambda(m)$. □

This settles exactly the following three printed modular conjectures:

entry	r	s	$a(n) \pmod{m}$ for $n \geq 2m+4$
A121860	1	1	2
A258899	2	1	$2 + 2^n$
A258903	2	3	$3 \cdot 2^n + 2 \cdot 3^n$.

The first source asks for eventual constancy equal to 2 for $m \geq 3$; the proof supplies it for every modulus. The latter two ask for an eventual period dividing $\varphi(m)$, which follows from the stronger period $\lambda(m)$. The divisor-sum identities are the existing source formulas, not conjectures inferred from an initial numerical list.

3.3.6 A pure-period conjecture for Euler numbers fails modulo 27

The selected May 2023 conjecture in A000364 concerns the sequence starting at index 1: it asks for a *pure* period dividing $\varphi(m)$ for every modulus m. This starting-index requirement is essential to the following refutation.

Theorem 3.3.12. *For the secant numbers $a(n) = |E_{2n}|$ of A000364, the sequence $(a(n))_{n\geq 1}$ has no period dividing $\varphi(27) = 18$ modulo 27. Hence the stated all-moduli pure-period conjecture is false.*

Proof. From

$$\left(\sum_{n\geq 0} a(n)\frac{x^{2n}}{(2n)!}\right)\cos x = 1$$

one obtains $a(0) = 1$ and the exact recurrence

$$a(n) = \sum_{k=0}^{n-1}(-1)^{n-k+1}\binom{2n}{2k}a(k) \qquad (n \geq 1). \tag{3.9}$$

Its residues modulo 27 at indices $0, 1, \ldots, 19$ are

$$1, 1, 5, 7, 8, 4, 11, 1, 14, 25, 17, 22, 20, 19, 23, 16, 26, 13, 2, 10.$$

In particular,

$$a(1) = 1, \qquad a(19) = 23489580527043108252017828576198947741 \equiv 10 \pmod{27}.$$

Any pure period T dividing 18 from index 1 would imply $a(19) \equiv a(1) \pmod{27}$, by iterating the period $18/T$ times. The two residues contradict that implication. □

This refutes only the stated pure-period claim. It does not refute eventual periodicity or the established odd-prime theorem of Knuth and Buckholtz, which the source separately credits. The recurrence and the 20-value residue certificate above make the counterexample exact; no approximation to the Euler numbers is used.

3.3.7 Reproducible checks and what they certify

The standard-library program `r3_modular_verify.py` writes `r3_modular_certificates.json`. Its complete cycle certificates prove the three zero-free-prime claims by the finite reduction already proved above, and its exact secant-number recurrence certifies the fourth counterexample. For the universal positive statements, the written arguments supply the quantifiers; numerical checks merely guard the definitions, exceptional primes, offsets, and algebraic identities. The checks include the biquadratic exceptional cases, cubic powers for primes below 300, Stirling representations and congruences through modulus 40, both polynomial operators through derivative 50 and modulus 12, and factorial-tail identities through modulus 30. All five verifier components pass. The accompanying entry ledger records all 31 counted identifiers and the uncounted A002050; it verifies disjointness from the 208 prior counted identifiers supplied to this round.

3.4 Twelve arithmetic and lattice-walk assertion groups

This section settles the specified conjectural formulas or comments in twelve distinct OEIS entries. Several assertions in the same entry count once. All source pages were checked on 20 October 2026. The lattice-walk binomial enumeration is an established input, not a claimed new result; the selected targets there are the recurrences still labeled conjectural.

3.4.1 A394757, A394399, A394815: twin-prime centers and divisor sums

Let $\sigma(n) = \sum_{d|n} d$. Call n a twin-prime center when both $n-1$ and $n+1$ are prime. The three entries respectively impose $\gcd(n, \sigma(n))$ prime, equal to 3, and equal to 2.

Theorem 3.4.1. *If n is a twin-prime center and $g = \gcd(n, \sigma(n))$ is prime, then $g \in \{2, 3\}$ and $18 \mid n$. More precisely:*

1. *if $g = 3$, then $n = 2m^2$ for an integer m divisible by 3;*
2. *if $g = 2$, then both $v_2(n)$ and $v_3(n)$ are positive even integers, and in particular $36 \mid n$.*

Consequently A394757 is the disjoint union of A394399 and A394815. Also, A394399 is precisely the intersection of the twin-prime centers, the numbers having prime $\gcd(n, \sigma(n))$, and the numbers of the form $2m^2$.

Proof. The center $n = 4$ gives $\gcd(4, 7) = 1$, so it does not occur here. Every other twin-prime center is a multiple of 6: both surrounding primes are odd, and one of $n-1, n+1$ would be divisible by 3 unless $3 \mid n$. In what follows n is therefore even and divisible by 3.

The standard parity criterion for the divisor sum follows directly from its prime factorization: $\sigma(n)$ is odd if and only if the exponent of every odd prime in n is even. Equivalently, n is a square or twice a square. If g is odd, the evenness of n forces $\sigma(n)$ to be odd. A square center greater than 4 is impossible because $n - 1 = (\sqrt{n} - 1)(\sqrt{n} + 1)$ would be composite. Hence $n = 2m^2$. Now $a = v_2(n)$ is odd, so

$$\sigma(2^a) = 2^{a+1} - 1 \equiv 0 \pmod 3.$$

Since $3 \mid n$, the prime g is divisible by 3, and thus $g = 3$. Moreover $3 \mid m$, giving $18 \mid n$.

It remains to consider $g = 2$. Put $a = v_2(n) \geq 1$ and $b = v_3(n) \geq 1$. If a were odd, the same factor $\sigma(2^a)$ would make $3 \mid \gcd(n, \sigma(n))$, a contradiction. Thus a is positive and even, and $4 \mid n$. If b were odd, then

$$\sigma(3^b) = 1 + 3 + \cdots + 3^b \equiv 0 \pmod 4,$$

by grouping successive pairs. Multiplicativity of σ would imply $4 \mid \sigma(n)$ and hence $4 \mid g$, again a contradiction. Therefore b is positive and even, proving $36 \mid n$.

The disjoint-union statement now follows from the only possible prime values 2 and 3. For the stated intersection, the forward inclusion has just been proved. Conversely, an even number $n = 2m^2$ has odd $\sigma(n)$; its gcd cannot equal 2, so a prime gcd at a twin-prime center must be 3. □

Exact source scope. This proves both displayed conjectures of A394757, the twice-a-square conjecture and conjectural intersection of A394399, and both the divisibility and disjoint-union assertions of A394815. The checked entries link only numerical tables; they do not contain an earlier proof of these selected claims.

3.4.2 A048702: two formulas for even binary palindromes

Let $a(0) = 0$, and let $a(n)$ be the nth positive even-length binary palindrome divided by 3, in increasing order.

Theorem 3.4.2. *For $n \geq 1$, put $\ell = \lfloor \log_2 n \rfloor$. Then*

$$a(n) = 2^{\ell} \sum_{i=1}^{n} 2^{-v_2(i)}, \tag{3.10}$$

$$= n2^{\ell} - \sum_{j=1}^{\ell} 2^{\ell-j} \left\lfloor \frac{n}{2^j} \right\rfloor. \tag{3.11}$$

Proof. Write $L = \ell + 1$ and $n = \sum_{t=0}^{L-1} b_t 2^t$ with $b_t \in \{0, 1\}$. The palindrome having n as its first half is

$$P(n) = 2^L n + \sum_{t=0}^{L-1} b_t 2^{L-1-t}.$$

These are all positive even-length binary palindromes, in increasing order: comparison first by length and then by the first half gives the same order as comparison of n. Thus $a(n) = P(n)/3$.

For every integer $v \geq 0$, $2^{-v} = 1 - \sum_{j=1}^{v} 2^{-j}$. Summing this identity over $i \leq n$ shows that the right sides of (3.10) and (3.11) agree. In the latter expression, the contribution of a single bit $b_t = 1$ is

$$2^{L-1} \left(2^t - \sum_{j=1}^{t} 2^{t-2j} \right) = \frac{2^{L+t} + 2^{L-1-t}}{3}.$$

Summing over the bits therefore gives exactly $P(n)/3$. □

Exact source scope. The two displayed conjectures in A048702 are proved for their positive-index domain. The existing 2019 comment proving divisibility of even binary palindromes by 3 is acknowledged; it does not establish these two summation identities. The separate unproved Maple algorithms are not included in the selected assertion group.

3.4.3 A230089: a residue-class formula

The sequence is defined for $n \geq 1$ by $a(n) = n$ for odd n, $a(n) = 2$ for $n \equiv 2 \pmod 4$, and $a(n) = 4$ for $n \equiv 0 \pmod 4$.

Theorem 3.4.3. *For $n \geq 1$,*

$$a(n) = \frac{3 + 3(-1)^n + (-i)^n + i^n + n - (-1)^n n}{2}.$$

Its ordinary generating function, starting at index 1, is

$$\sum_{n \geq 1} a(n) x^n = -\frac{x(4x^5 - x^4 - 2x^3 - 2x^2 - 2x - 1)}{(x-1)^2 (x+1)^2 (x^2+1)}.$$

Proof. For odd n, the two imaginary powers cancel, the constant terms cancel, and the numerator is $2n$. For $n \equiv 0 \pmod 4$ the numerator is $6 + 2 = 8$; for $n \equiv 2 \pmod 4$ it is $6 - 2 = 4$. This proves the asserted expression in every residue class. Summing the defining classes gives

$$\sum_{n \geq 1} a(n) x^n = \frac{x(1 + x^2)}{(1 - x^2)^2} + \frac{2x^2 + 4x^4}{1 - x^4},$$

which simplifies to the displayed rational function. □

The selected source assertion is the conjectural closed form and its accompanying generating function in A230089; both are counted together as one entry.

3.4.4 A389425: density one of unit gaps

Let $s_1 < s_2 < \cdots$ enumerate the positive integers k for which $6k-5$ and $6k-1$ are both composite (A199716). The sequence A389425 is $a(j) = s_{j+1} - s_j$.

Theorem 3.4.4. *The set* $\{j \geq 1 : a(j) = 1\}$ *has natural density* 1.

Proof. We first recall, with an elementary justification, that the primes have natural density zero. For any fixed finite set $\mathcal{P}$ of primes, all primes outside $\mathcal{P}$ avoid multiples of its elements. The density of integers avoiding these multiples is $\prod_{p\in\mathcal{P}}(1-1/p)$. Taking $\mathcal{P}$ to be the primes up to Y, these products tend to zero: their reciprocals satisfy

$$\prod_{p\leq Y}(1-1/p)^{-1} = \sum_{\substack{m\geq 1\\ p\mid m\Rightarrow p\leq Y}} \frac{1}{m} \geq \sum_{m\leq Y}\frac{1}{m} \longrightarrow \infty.$$

Thus $\pi(X) = o(X)$, without requiring the prime number theorem.

Let $S = \{s_1, s_2, \ldots\}$ and $E(X) = \#(\{1,\ldots,\lfloor X\rfloor\}\setminus S)$. Every excluded $k > 1$ has at least one of $6k-5, 6k-1$ prime; the exceptional possibility that one is 1 contributes at most one. Consequently

$$E(X) \leq 2\pi(6X) + 1 = o(X).$$

It follows that S has density one and $s_N \sim N$. For every $j < N$ with $s_{j+1} - s_j > 1$, the integer $s_j + 1$ is excluded from S. These witnesses are distinct and at most s_N. Hence

$$\#\{j < N : a(j) > 1\} \leq E(s_N) = o(s_N) = o(N),$$

which is exactly the asserted natural-density statement. □

This supplies a proof of the explicitly conjectural frequency assertion in A389425. It is a statement about density in the sequence index, which is why the passage from s_N to N is included.

3.4.5 A092278: the empirical floor generating function

Theorem 3.4.5. *For* $a(n) = \lfloor (3n+4)/16\rfloor$, $n \geq 0$,

$$\sum_{n\geq 0} a(n)x^n = \frac{x^4 + x^{10} + x^{15}}{(1-x)(1-x^{16})} = \frac{x^4(x^{11}+x^6+1)}{x^{17}-x^{16}-x+1}.$$

Proof. One has $a(0) = 0$ and $a(n+16) = a(n)+3$. The increment $a(n)-a(n-1)$, for $n \geq 1$, is 1 exactly in the classes $n \equiv 4, 10, 15 \pmod{16}$ and is 0 in all other classes. Therefore, if $G(x) = \sum_{n\geq 0} a(n)x^n$, then

$$(1-x)G(x) = \frac{x^4 + x^{10} + x^{15}}{1-x^{16}}.$$

This identity in formal power series proves the claimed expression. □

The target is the generating function labeled empirical in A092278; this is one assertion group.

3.4.6 A005558–A005562: five lattice-walk recurrence claims

Let $w_y(n)$ be column y of A052174, with $y \geq 0$ and $n \geq y$. The established enumeration in R. K. Guy, *Catwalks, Sandsteps and Pascal Pyramids*, J. Integer Sequences 3 (2000), Article 00.1.6, gives the following two formulas, also recorded in A052174. With $\binom{n}{-1} = 0$,

$$w_y(y+2k) = \binom{y+2k+1}{k}\left(\binom{y+2k}{k} - \binom{y+2k}{k-1}\right),$$
$$w_y(y+2k+1) = \binom{y+2k+2}{k}\binom{y+2k+1}{k+1} - \binom{y+2k+2}{k+1}\binom{y+2k+1}{k-1}.$$

Both formulas hold for $k \geq 0$. The relevant OEIS offsets are $n = y$, not $n = 0$ for every column. In particular $w_y(y) = 1$ and $w_y(y+1) = y+1$.

Theorem 3.4.6. *For all integers $y \geq 0$ and $n \geq y+2$,*

$$\begin{aligned}&(n-y+1)(n-y)(2n+1)(n+y+3)(n+y+2)w_y(n)\\ &\quad - 4n(n+1)(2n^2+4n+2y^2+4y+3)w_y(n-1)\\ &\quad - 16n^2(n-1)(2n+3)(n+1)w_y(n-2) = 0.\end{aligned} \tag{3.12}$$

Its instances $y = 0, 1, 2, 3, 4$ prove the selected recurrence conjectures in A005558, A005559, A005560, A005561, and A005562, respectively.

Proof. Writing the known binomial expressions as factorials gives

$$E_y(k) := w_y(y+2k) = \frac{(y+1)(y+2k+1)!(y+2k)!}{k!^2(y+k+1)!^2},$$
$$O_y(k) := w_y(y+2k+1) = \frac{(y+1)(y+2k+2)!(y+2k+1)!}{k!(k+1)!(y+k+2)!(y+k+1)!}.$$

These are positive. The adjacent ratios reduce to

$$\frac{w_y(n-1)}{w_y(n)} = \frac{u}{n(n+1)}, \qquad \frac{w_y(n-2)}{w_y(n)} = \frac{uv}{n^2(n^2-1)},$$

where the exact values are

n	u	v
$y+2k$	$k(y+k+1)$	$k(y+k+1)$
$y+2k+1$	$(k+1)(y+k+2)$	$k(y+k+1)$.

In the stated range $n \geq y+2$, each denominator is nonzero. After division by $w_y(n)$, the left side of (3.12) becomes

$$(n-y+1)(n-y)(2n+1)(n+y+3)(n+y+2) - 4(2n^2+4n+2y^2+4y+3)u - 16(2n+3)uv.$$

Substituting either row of the table and $n = y+2k$ or $n = y+2k+1$ gives the zero polynomial in k, y. This is a polynomial identity, not a finite-term inference. The verifier independently expands both polynomials over $\mathbb{Z}[k, y]$ and records zero residuals. □

Correspondence with the five printed recurrences. For $y = 0$, divide (3.12) by $n(n+1)$ to obtain

$$(2n+1)(n+3)(n+2)w_0(n) - 4(2n^2+4n+3)w_0(n-1) - 16n(2n+3)(n-1)w_0(n-2) = 0 \quad (n \geq 2).$$

For $y = 1$, divide by n and multiply by -1; the result is the recurrence printed in A005559, for $n \geq 3$. For $y = 2, 3, 4$ the formula already has the printed normalization; the constants $2y^2+4y+3$ are $19, 33, 51$, and the valid ranges are $n \geq 4, 5, 6$, respectively. Thus the source offsets and boundary conditions agree, rather than merely matching a shifted numerical list.

Theorem 3.4.7. *The second recurrence conjectured in A005558 also holds:*

$$(n+3)(n+2)w_0(n) - 4(n^2+3n+1)w_0(n-1) \\ + 16(-n^2+n+1)w_0(n-2) + 64(n-1)(n-2)w_0(n-3) = 0 \qquad (n \geq 3).$$

Proof. Use the ratio table above with $y = 0$. To obtain the third ratio, multiply $w_0(n-2)/w_0(n)$ by $w/((n-2)(n-1))$, where $w = k(k-1)$ for $n = 2k$ and $w = k(k+1)$ for $n = 2k+1$. Substitution into the displayed recurrence, followed by multiplication by $n^2(n-1)(n+1)$, yields in both parity cases the polynomial identity

$$0 = (n+3)(n+2)n^2(n-1)(n+1) \\ - 4(n^2+3n+1)un(n-1) + 16(-n^2+n+1)uv + 64uvw.$$

The identity is verified by inserting the stated values of n, u, v, w; all denominators canceled above are nonzero for $n \geq 3$. □

Prior-work and counting boundary. The binomial enumeration and existing special-function generating functions are credited to their earlier sources. This section provides explicit algebraic proofs of Mathar's recurrence conjectures dated 2 April 2017, which remain labeled conjectural on the five checked pages. It makes no claim of a new lattice-walk enumeration or of bibliographic priority for consequences of the known formulas. The two recurrences in A005558 count as one entry; its other conjecturally worded formulas are outside the selected target.

Reproducible certificates. The files `r3_arith_verify.py` and `r3_arith_certificates.json` contain exact symbolic certificates for the two general parity identities and the two third-order parity identities. They also check the complete residue periods for A230089 and A092278. Finite arithmetic and walk checks are recorded separately as implementation checks, not as substitutes for the proofs above. Source URLs, original assertion scopes, offsets, and count decisions are preserved in `r3_arith_results.json`.

3.5 Exact transfer certificates for 41 array recurrences

The 41 claims in this section are empirical recurrences explicitly printed on the cited OEIS pages, checked on 20 October 2026. They concern different widths, color bounds, or local constraints. They are counted by distinct entry, not as 41 unrelated general methods. The proof below applies to all indices and is followed by the complete recurrence signatures. Finite prefix agreement is used only to check definitions and indexing.

3.5.1 A definition-derived transfer theorem

Let $w \geq 2$. A row is a word of length w over a fixed alphabet of j colors. Its canonical form replaces its first distinct color by 0, its next new color by 1, and so on. Let $\mathcal{R}_{j,w}$ be the set of canonical rows with at most j distinct colors. A canonical row r with $k(r)$ distinct colors has

$$m_j(r) = (j)_{k(r)} = j(j-1)\cdots(j-k(r)+1)$$

actual labelings. Suppose the allowed relation $C(r,v)$ between consecutive rows is invariant under simultaneous color permutations. Define

$$T_{j,C}(r,s) = \#\{v \in \{0,\dots,j-1\}^w : \mathrm{can}(v) = s,\ C(r,v)\}.$$

This is a complete finite matrix: each possible next row is examined once. If $L_j(n)$ is the number of labeled arrays with $n+1$ rows, then

$$L_j(n) = m_j^{\mathsf{T}} T_{j,C}^n \mathbf{1}. \tag{3.13}$$

Indeed, the number of legal completions of an actual row depends only on its canonical form. Induction on the number of remaining rows gives the right side; the weights m_j count all choices of the first row.

For an array using exactly q colors, renaming colors in their order of first appearance gives one canonical array per $q!$ labelings. Inclusion– exclusion therefore gives

$$U_q^{=}(n) = \sum_{j=1}^{q} \frac{(-1)^{q-j}}{j!(q-j)!} L_j(n). \tag{3.14}$$

An array with at most q colors has a unique actual number of colors, so

$$U_q^{\leq}(n) = \sum_{j=1}^{q} \left(\frac{1}{j!} \sum_{h=0}^{q-j} \frac{(-1)^h}{h!} \right) L_j(n). \tag{3.15}$$

These are precisely the "new values introduced in row major order" conditions in the definitions. No assumption that all permitted colors occur is made unless the entry explicitly requires it.

For a 2×2 block $\left(\begin{smallmatrix} u & v \\ w & t \end{smallmatrix}\right)$, the local rules used below are:

- D_2: $\#\{u,v,w,t\} = 2$;
- D_{12}: $\#\{u,v,w,t\} \leq 2$;
- H_h: $\mathbf{1}_{u=t} + \mathbf{1}_{v=w} = h$, for a fixed $h \in \{0,1,2\}$.

For D_2 and D_{12} the condition holds at every adjacent column pair. For the diagonal-equality family the same value h is used throughout the array, so the count is the sum of the three H_h counts. Since $n \geq 1$ and $w \geq 2$, every array has a 2×2 block and these three cases are disjoint. This explains the source wording "the same number of equal diagonal or antidiagonal elements."

The two edge-statistic entries use the actual numeric labels, so color permutations are not taken. Their states are all $4^2 = 16$ rows, initial weights are all one, and the final count is divided by 4 as specified in their definitions. For edge sums the four values $u+v, v+t, t+w, w+u$ must be distinct; for clockwise edge differences the four values $v-u, t-v, w-t, u-w$ must be distinct.

3.5.2 The finite identities that certify every index

For a displayed signature $(c_1, \ldots, c_d)$ put

$$R(z) = z^d - \sum_{i=1}^{d} c_i z^{d-i}, \qquad Q(x) = 1 - \sum_{i=1}^{d} c_i x^i.$$

For every transfer component of every entry listed below, the supplied integer certificate verifies the vector identity

$$\boxed{TR(T)\mathbf{1} = 0.} \tag{3.16}$$

There is no floating point arithmetic and no inferred recurrence in this check. The state set and every matrix entry are obtained from the explicit finite definitions above. For $n \geq d+1$, multiplying (3.16) by $m^{\mathsf{T}} T^{n-d-1}$ gives

$$m^{\mathsf{T}} T^n \mathbf{1} = \sum_{i=1}^{d} c_i m^{\mathsf{T}} T^{n-i} \mathbf{1}.$$

The linear combinations (3.14) and (3.15), and the disjoint sum over h, preserve this identity. Thus the precise all-index conclusion for each entry is

$$\boxed{a(n) = \sum_{i=1}^{d} c_i a(n-i) \quad (n \geq d+1).} \tag{3.17}$$

The source index begins at $n = 1$; this range uses only defined terms.

The complete integer matrices, their initial weights and canonical rows are in `r3_gf_transition_matrices.json`. The verifier reconstructs every matrix from its combinatorial predicate, compares it with that file, and recomputes (3.16). This is a reproducible finite proof certificate for the universal recurrence, rather than a test of an arbitrarily long initial segment. The files `r3_gf_array_certificates.json` and `r3_gf_verify.py` record the signatures and checks.

For $F(x) = \sum_{n \geq 1} a(n) x^n$, this also gives $QF = P$ with $\deg P \leq d$. The polynomial P is computed from the first d values obtained directly from (3.13). Seventeen correctly indexed empirical generating functions in the source entries agree exactly with this construction. The separately printed generating function in A209962 needs an additional factor x under this indexing; its empirical recurrence is proved and counted, while the indexing correction adds no count.

3.5.3 Individual recurrence signatures

In the following entries, D is the sum of the dimensions of the transfer components with nonzero coefficients. All of their component identities (3.16) hold. The coefficient list gives $c_1, c_2, \ldots, c_d$ in that order. Each entry constitutes one counted assertion group.

3.5.3.1 A205355

Arrays with $n+1$ rows, width $w = 3$, at most 3 colors up to order of first appearance, with rule H_0, H_1, H_2.

The recurrence (3.17) holds with $d = 4$, for $n \geq 5$. The certificate has $D = 18$ states. Source: https://oeis.org/A205355.

$$(c_1, \ldots, c_d) = (15, -76, 146, -84).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.2 A205356

Arrays with $n+1$ rows, width $w=4$, at most 3 colors up to order of first appearance, with rule H_0, H_1, H_2.

The recurrence (3.17) holds with $d=5$, for $n \geq 6$. The certificate has $D=45$ states. Source: https://oeis.org/A205356.

$$(c_1, \ldots, c_d) = (23, -187, 645, -912, 432).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.3 A205357

Arrays with $n+1$ rows, width $w=5$, at most 3 colors up to order of first appearance, with rule H_0, H_1, H_2.

The recurrence (3.17) holds with $d=12$, for $n \geq 13$. The certificate has $D=126$ states. Source: https://oeis.org/A205357.

```
41, -634, 4322, -6905, -79071, 489544, -808128, -1807684, 9900708, -17129232, 13584240,
-4147200
```

3.5.3.4 A205358

Arrays with $n+1$ rows, width $w=6$, at most 3 colors up to order of first appearance, with rule H_0, H_1, H_2.

The recurrence (3.17) holds with $d=19$, for $n \geq 20$. The certificate has $D=369$ states. Source: https://oeis.org/A205358.

```
72, -2111, 31963, -244721, 393930, 9129842, -78749065, 200421324, 702551851,
-6318391361, 15148890613, 7356872888, -116533400840, 228856999872, -38635874496,
-523003405824, 885155081472, -617973276672, 165112971264
```

3.5.3.5 A206377

Arrays with $n+1$ rows, width $w=3$, at most 4 colors up to order of first appearance, with rule H_0, H_1, H_2.

The recurrence (3.17) holds with $d=5$, for $n \geq 6$. The certificate has $D=30$ states. Source: https://oeis.org/A206377.

$$(c_1, \ldots, c_d) = (36, -377, 1398, -2064, 1008).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.6 A206378

Arrays with $n+1$ rows, width $w=4$, at most 4 colors up to order of first appearance, with rule H_0, H_1, H_2.

The recurrence (3.17) holds with $d=6$, for $n \geq 7$. The certificate has $D=72$ states. Source: https://oeis.org/A206378.

$$(c_1, \ldots, c_d) = (74, -1401, 9064, -21992, 21312, -7056).$$

3.5.3.7 A206379

Arrays with $n+1$ rows, width $w=5$, at most 4 colors up to order of first appearance, with rule H_0, H_1, H_2.

The recurrence (3.17) holds with $d=11$, for $n \geq 12$. The certificate has $D=204$ states. Source: https://oeis.org/A206379.

```
160, -5537, 25556, 1495902, -23370552, 81849864, 544237824, -5119960320, 15199301376,
-19180578816, 8497004544
```

3.5.3.8 A206380

Arrays with $n+1$ rows, width $w=6$, at most 4 colors up to order of first appearance, with rule H_0, H_1, H_2.

The recurrence (3.17) holds with $d=20$, for $n \geq 21$. The certificate has $D=660$ states. Source: https://oeis.org/A206380.

```
394, -38562, 1246392, 6334675, -1215511426, 25096715040, -111433495776, -2633554263696,
38447016473440, -124529929487104, -973109346072576, 8528195948079360,
-13461810857735680, -67310021223342080, 238037998037753856, 44544280591749120,
-947327372288188416, 643157082054918144, 987246769084563456, -892353207989698560
```

3.5.3.9 A209709

One quarter of the labeled arrays on colors $0,1,2,3$, width $w=2$, with distinct clockwise edge differences.

The recurrence (3.17) holds with $d=6$, for $n \geq 7$. The certificate has $D=16$ states. Source: https://oeis.org/A209709.

$$(c_1, \ldots, c_d) = (2, 34, 30, -76, -24, 16).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.10 A209729

One quarter of the labeled arrays on colors $0,1,2,3$, width $w=2$, with distinct edge sums.

The recurrence (3.17) holds with $d=3$, for $n \geq 4$. The certificate has $D=16$ states. Source: https://oeis.org/A209729.

$$(c_1, \ldots, c_d) = (3, 14, 4).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.11 A209807

Arrays with $n+1$ rows, width $w=2$, at most 4 colors up to order of first appearance, with rule D_2.

The recurrence (3.17) holds with $d=4$, for $n \geq 5$. The certificate has $D=5$ states. Source: https://oeis.org/A209807.

$$(c_1, \ldots, c_d) = (8, -7, -28, -12).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.12 A209808

Arrays with $n+1$ rows, width $w = 3$, at most 4 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d = 7$, for $n \geq 8$. The certificate has $D = 10$ states. Source: https://oeis.org/A209808.

$$(c_1, \ldots, c_d) = (17, -68, -56, 397, -181, -264, 108).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.13 A209809

Arrays with $n+1$ rows, width $w = 4$, at most 4 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d = 14$, for $n \geq 15$. The certificate has $D = 24$ states. Source: https://oeis.org/A209809.

```
26, -68, -1905, 6356, 40342, -117881, -302966, 604526, 940543, -1163096, -1137756,
834912, 402624, -124416
```

3.5.3.14 A209810

Arrays with $n+1$ rows, width $w = 5$, at most 4 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d = 33$, for $n \geq 34$. The certificate has $D = 68$ states. Source: https://oeis.org/A209810.

```
66, -1211, -3327, 287711, -1650227, -18911756, 200787144, 249811709, -8470618231,
12934099929, 169407260904, -526263544629, -1680919496686, 8328865779706, 6270909877563,
-70313845544213, 27029214503869, 333056045453117, -377395472742367, -825772616226370,
1585680300164445, 703085339056680, -3069417122224407, 900452612086569, 2425264270015320,
-1808654163691437, -347830308417366, 652709084826024, -96050625683220, -59518267835508,
14239281350520, 1128398046480, -309936391200
```

3.5.3.15 A209811

Arrays with $n+1$ rows, width $w = 6$, at most 4 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d = 91$, for $n \geq 92$. The certificate has $D = 220$ states. Source: https://oeis.org/A209811.

```
107, -1878, -137789, 4495053, 60979281, -3382378658, -5378727169, 1356204907007,
-5005172801248, -338085898941018, 2348610841442425, 56536290959562784,
-541259839742806307, -6608801096459681129, 81395189596556559983, 548394351778255652595,
-877020702450115986552 9, -3166697358686804181 2990, 710376576458807557547216,
112892419348474615188 1180, -44551314523662218100563425, -72632663219794539323 60270,
22063760310451695899727581 06, -20263534076869103577307440 32,
-87454472438177627226033574345, 15714135741372589808688066 2843,
27990172393993719770205927882 97, -71782942287842807405647819 01801,
-72691433711711173410574983496660, 23944361725412626220984854839 6745,
15332574314970691665911492398293 07, -62219224885960660277515492173 59670,
```

-261591750171839624020681628147585 01, 12962757340036771629706665513689833 1,
35667119117064014532314793705192080 0, -21983517618762852733023844972264201 97,
-37773106665530599033228986214303595 69, 30602786767129664524288240851030652355,
28834501346077162987374441766139145544, -35120397175342220968611557789170978 5746,
-11670142826092897953214263266820000 5284, 33273536608641630711426000265220294 60808,
-56001764461399450702739420645532519 7980, -25995896123168019862614781805551174366491,
15575168550871264099563278351155746483043, 16687253383424768022633316681047300 7305778,
-16558363191934771524045327073378780030 0967,
-87407145396801466678896737299858517243 1989,
12062351283089248579408195226565018389 18602,
36921614512014154264882743812012492585 74504,
-66658587972053597806343413108435096851 36667,
-12319864530826648929075899837324201902878036,
28901545602486379348422349423810863556824093,
31170391469749070397533351890857301477542909,
-99502674405964027780060847135394733515338115,
-53857770398272002266481193998251821786322739,
27248277634904656177990521633422522603 6875695,
37540447794350619447986912229887791121752431,
-59041536769410324573990683672213872171 1566806,
11068626659966646704755353931181023501 4535318,
10001129283725315722128177683266323010938 22884,
-47336305451767498658602893555294930219 2543622,
-12960492227940510731043696600024519053773 15569,
98306999177987341574897790061290524956234 6419,
12345907594379866247435955688347443195572 69623,
-13673166460069056691944958998035550960310 34679,
-79032943041429220182666949684054738313241 5040,
13524585059399041898932286371655714817765 53934,
24244048682529740369785490355875802577519 6236,
-95559606488305541871733429099533383684808 8796,
91237272508018656852555859600497039867416040,
47036498098692249522170129738925052064520 3036,
-14941461156172283449700296790608348112521 8504,
-15161219493090938989522591379635800979882 9200,
83459176699483481099731422805986592523856192,
27203058188450643926088113201093410799384736,
-26266525941365495565985041894404510362335360,
-83532818757082608955401052435722129454 9056,
48219499638073868957874590039610427250668 80,
-67911290359047709880021536604665865370220 8,
-47864616976578512517040300724066102066726 4, 13438835015753952938318676472472544022 1184,
19845613671413486485308605949798443768832, -10375428852018050542986066625716287592960,
11838559628316947915103595163878055731 2, 34224851875040630324667313737710059929 6,
-26440425803133463527797481814195814400, -42340516458139532346376479446413393 92,
45495349108235136710480672412165734 4, 13225918069327774097709415191674880,
-17389000867899553664826345416294 40

3.5.3.16 A209822

Arrays with $n+1$ rows, width $w = 2$, at most 3 colors up to order of first appearance, with rule D_{12}.

The recurrence (3.17) holds with $d = 3$, for $n \geq 4$. The certificate has $D = 3$ states. Source: https://oeis.org/A209822.

$$(c_1, \ldots, c_d) = (6, -3, -2).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.17 A209823

Arrays with $n+1$ rows, width $w = 3$, at most 3 colors up to order of first appearance, with rule D_{12}.

The recurrence (3.17) holds with $d = 5$, for $n \geq 6$. The certificate has $D = 6$ states. Source: https://oeis.org/A209823.

$$(c_1, \ldots, c_d) = (12, -16, -6, 13, -2).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.18 A209824

Arrays with $n+1$ rows, width $w = 4$, at most 3 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d = 9$, for $n \geq 10$. The certificate has $D = 15$ states. Source: https://oeis.org/A209824.

$$(c_1, \ldots, c_d) = (24, -62, -226, 445, 475, -811, 29, 147, -20).$$

3.5.3.19 A209825

Arrays with $n+1$ rows, width $w = 5$, at most 3 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d = 21$, for $n \geq 22$. The certificate has $D = 42$ states. Source: https://oeis.org/A209825.

```
60, -839, 1399, 31874, -122506, -318771, 1769870, 389301, -8009315, 3097168, 13833529,
-8760492, -9461275, 7615892, 2191722, -2581040, 92008, 286200, -54000, -1264, 480
```

3.5.3.20 A209826

Arrays with $n+1$ rows, width $w = 6$, at most 3 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d = 51$, for $n \geq 52$. The certificate has $D = 123$ states. Source: https://oeis.org/A209826.

```
138, -5291, 41948, 1246056, -19527125, -94011205, 2626583096, 1511468746, -177763031682,
138302103902, 7019042194083, -7969961832191, -174075869696471, 186648296877104,
2832030056843165, -2415279667981283, -31006473277373149, 19111371400840450,
232526299989981431, -99505929279539438, -1213858407562141252, 370715377868919185,
4475559340570646909, -1096935236489998102, -11786638113405890713, 2781294795279031273,
22309407784237294666, -5920732121027419916, -30324789854047144258, 9761270047165567745,
29307097858909263706, -11646482334059398589, -19688084504632376578, 9649239854975506279,
8805727983122157768, -5380377784965746096, -2398306480372906352, 1943991227752730806,
304993982412468654, -428499039605489966, 11979961616427444, 51791333004986360,
-7840293455999272, -2807693417208336, 702780093944768, 47980140467616, -22670306024704,
380884158080, 247814909440, -6850234368, -945684480
```

3.5.3.21 A209841

Arrays with $n+1$ rows, width $w = 2$, at most 4 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d = 4$, for $n \geq 5$. The certificate has $D = 5$ states. Source: https://oeis.org/A209841.

$$(c_1, \ldots, c_d) = (11, -30, 4, 16).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.22 A209842

Arrays with $n+1$ rows, width $w=3$, at most 4 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=6$, for $n \geq 7$. The certificate has $D=10$ states. Source: https://oeis.org/A209842.

$$(c_1, \ldots, c_d) = (22, -127, 90, 288, -400, 128).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.23 A209843

Arrays with $n+1$ rows, width $w=4$, at most 4 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=10$, for $n \geq 11$. The certificate has $D=24$ states. Source: https://oeis.org/A209843.

```
43, -459, -321, 13084, -11850, -84560, 143072, 6528, -98304, 32768
```

3.5.3.24 A209844

Arrays with $n+1$ rows, width $w=5$, at most 4 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=24$, for $n \geq 25$. The certificate has $D=68$ states. Source: https://oeis.org/A209844.

```
101, -3119, 23301, 332146, -4841510, 189458, 228888364, -753564284, -2816048344,
16949078144, -1771169504, -117752841744, 156873217312, 244896903072, -582512363712,
-47046663680, 765499169280, -294034673664, -383848898560, 236348588032, 55918985216,
-46495170560, -1491075072, 1811939328
```

3.5.3.25 A209845

Arrays with $n+1$ rows, width $w=6$, at most 4 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=59$, for $n \geq 60$. The certificate has $D=220$ states. Source: https://oeis.org/A209845.

```
223, -16175, 333859, 9171561, -425242479, 693710971, 161486863949, -1475551022652,
-27865453094464, 411423026735736, 2219874285551176, -55658326952391880,
-36672282912577248, 4395265143027791120, -7555217898067289496, -220424722751437565016,
708740451169041114912, 7370084221594588613424, -32339860483036642598592,
-168530209053002435663904, 936812779773450938868224, 2642855316304854396118720,
-1871548856763089445065459 2, -2747217254645182050245 8112, 2680228586767502415616732 6,
162950499399519898498751232, -280876520796805086394545 8688, -4947170980397186518717 6448,
217746730816080465843750297 60, -899822740267841709733347 8400,
-125431619657959886331737444352, 936877528473287910955500707 84,
536479435921838898006754967552, -549614260424646029255096614912,
-169339869214809045177789317 1200, 216642577109363794512575260 2624,
389364954277624122734231735 5008, -602966443657055468616609064 5504,
-636104371106098978307733349 9904, 120400571863039715550997774 33600,
701691902256461660658451297 0752, -172592915765355079770889896 59136,
-457092555782772960485703260 5696, 175871407350695804707746359 54176,
764365413232018874812935962624, -124912427050257070870311773 14304,
140821028709813014402890203 1360, 599755191980563504675251080 3968,
-132951597569672541489416948 9408, -186385726563817967802006896 6400,
547108429772846613377035796480, 353189996395406454860564922368,
-11780125469550700305181992 5504, -37321328417622966875887501312,
127581800057248562343984496 64, 187023458932745873621463859 2,
-59376119211166377981876633 6, -25884281547576941708574720, 774058527826110232657920 0
```

3.5.3.26 A209890

Arrays with $n+1$ rows, width $w=2$, at most 3 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d=2$, for $n \geq 3$. The certificate has $D=3$ states. Source: https://oeis.org/A209890.

$$(c_1, \ldots, c_d) = (4, 4).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.27 A209891

Arrays with $n+1$ rows, width $w=3$, at most 3 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d=3$, for $n \geq 4$. The certificate has $D=6$ states. Source: https://oeis.org/A209891.

$$(c_1, \ldots, c_d) = (8, 4, -16).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.28 A209892

Arrays with $n+1$ rows, width $w=4$, at most 3 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d=9$, for $n \geq 10$. The certificate has $D=15$ states. Source: https://oeis.org/A209892.

$$(c_1, \ldots, c_d) = (14, 31, -366, -210, 2564, 40, -4096, -96, 1152).$$

3.5.3.29 A209893

Arrays with $n+1$ rows, width $w=5$, at most 3 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d=23$, for $n \geq 24$. The certificate has $D=42$ states. Source: https://oeis.org/A209893.

```
37, -224, -2979, 29449, 46557, -1073860, 1135122, 15431403, -33807620, -96498020,
305903160, 230102448, -1252267328, 63674496, 2499686528, -1105423872, -2340302336,
1583680512, 822790144, -753893376, -7176192, 78643200, -10485760
```

3.5.3.30 A209894

Arrays with $n+1$ rows, width $w=6$, at most 3 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d=67$, for $n \geq 68$. The certificate has $D=123$ states. Source: https://oeis.org/A209894.

```
59, 104, -42972, 225150, 13058670, -99533244, -2197387878, 19875267460, 229851257770,
-2366141610633, -15894063929471, 186716781277733, 749911518956427, -10357596907410544,
-24242602552629694, 419545423349562323, 513419955614790263, -12743636649243029327,
-5588724654850929951, 295948232414959255193, -410301288672637732 57,
-5330465080579574799097, 309653561696217 0688023, 752459000339790902 63854,
```

```
-70343205102961136452140, -838515904703767168682000, 1033530787470818142569456,
74081914835217214973258888, -11102771711988258151207232, -51956877639237774786347264,
91197930041539107221974784, 28863700536486104426813542 4, -58491844933220440853197 5168,
-12613919197822705629676462 08, 29601050913710208698767933 44,
42741729575374366055063224 32, -11873183283623499865326174208,
-10900689087152926975005589504, 37756515336414977978883178496,
19498582274969148853931212800, -94868879756767967625610788864,
-18957098993963844384982564864, 18701408829824552098159027 8144,
-10916938203208559688348073984, -28586079323961437963887732 3264,
79644827524571502161338302464, 33266579649591874060156718 2848,
-15969802048128245481341347 4304, -28618217663362412943752822 7840,
19680426646818956459063744 9216, 17267669618604358165743848 6528,
-16441414467934049821604079 2064, -64875360508321568620864864256,
93966732830765490489645858816, 89844170884962182465218150 40,
-35640178748572809806303199232, 40055036151908527158622945 28,
83375171330606403651350036 48, -23502111621872703221855682 56,
-10122173188631174348579799 04, 49480075574014644396123750 4, 26276649640154457035505664,
-44578672501261648598663168, 52843427152673232524410 88, 10837455477976913066065 92,
-28444630762960697740492 8, 17414014635957968437248
```

3.5.3.31 A209962

Arrays with $n+1$ rows, width $w=2$, exactly 4 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=6$, for $n \geq 7$. The certificate has $D=7$ states. Source: https://oeis.org/A209962.

$$(c_1, \ldots, c_d) = (16, -83, 132, 56, -88, -32).$$

The source's separate generating function omits a factor x. With the stated offset the correct version is

$$F(x) = \frac{9x^3(1+2x)}{(1-x)(1-4x)(1-5x-2x^2)(1-6x-4x^2)}.$$

The initial values $a(1)=a(2)=0$, $a(3)=9$ explain the indexing issue. The count for this entry is the proved empirical recurrence, not a separate claim for the correction.

3.5.3.32 A209963

Arrays with $n+1$ rows, width $w=3$, exactly 4 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=10$, for $n \geq 11$. The certificate has $D=15$ states. Source: https://oeis.org/A209963.

```
33, -374, 1586, -1093, -4559, 7212, -420, -4336, 2208, -256
```

3.5.3.33 A209964

Arrays with $n+1$ rows, width $w=4$, exactly 4 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=18$, for $n \geq 19$. The certificate has $D=38$ states. Source: https://oeis.org/A209964.

```
66, -1487, 11648, 14141, -454021, 667500, 5458068, -11933206, -23623289, 65454499,
26595728, -134617798, 33919512, 79370592, -38830848, -7503360, 6127616, -655360
```

3.5.3.34 A209965

Arrays with $n+1$ rows, width $w=5$, exactly 4 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=44$, for $n \geq 45$. The certificate has $D=109$ states. Source: https://oeis.org/A209965.

```
160, -9858, 286721, -3505459, -7704489, 640517305, -4759487502, -21219049490,
420859791361, -878171254221, -12643232573808, 66259330277918, 122870759491994,
-1553376671736484, 1219539182999473, 17359779478680752, -39146051553201926,
-91438263542603406, 373898684344261096, 116854653298320024, -1799083158352232088,
998160450833133144, 4801216485839890224, -5451035382131524304, -7054956173126826080,
12732134852398029184, 4672289972159061760, -16728156971141573696, 1059265242914046976,
13059429141341396224, -4207097515631301632, -5978493921676298752, 3137966625793618432,
1468906693869291520, -1139164366776567808, -134802883891748864, 213153759748931584,
-11638547304415232, -18765649522720768, 2857764298162176, 564292134895616,
-120413963157504, -2136276467712, 869730877440
```

3.5.3.35 A210045

Arrays with $n+1$ rows, width $w=2$, exactly 3 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d=4$, for $n \geq 5$. The certificate has $D=5$ states. Source: https://oeis.org/A210045.

$$(c_1, \ldots, c_d) = (7, -6, -20, -8).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.36 A210046

Arrays with $n+1$ rows, width $w=3$, exactly 3 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d=6$, for $n \geq 7$. The certificate has $D=10$ states. Source: https://oeis.org/A210046.

$$(c_1, \ldots, c_d) = (14, -41, -66, 100, 56, -32).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.37 A210047

Arrays with $n+1$ rows, width $w=4$, exactly 3 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d=14$, for $n \geq 15$. The certificate has $D=23$ states. Source: https://oeis.org/A210047.

```
24, -89, -977, 3094, 13063, -28268, -71414, 90536, 163472, -114928, -148256, 54080,
35328, -9216
```

3.5.3.38 A210048

Arrays with $n+1$ rows, width $w=5$, exactly 3 colors up to order of first appearance, with rule D_2. The recurrence (3.17) holds with $d=32$, for $n \geq 33$. The certificate has $D=58$ states. Source: https://oeis.org/A210048.

```
58, -992, 1114, 104383, -609244, -3158300, 31872941, 14255768, -695247797, 851352494,
7722215712, -17461446135, -44607174085, 153341052250, 107386696583, -724717887559,
133505885446, 1890011566228, -1409205824752, -2511295843328, 3234462976416,
1147782485632, -3211415530368, 620083812608, 1267243773440, -658602110976, -85186113536,
120800202752, -16266002432, -5194055680, 1509949440, -104857600
```

3.5.3.39 A210088

Arrays with $n+1$ rows, width $w=2$, exactly 3 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=4$, for $n \geq 5$. The certificate has $D=5$ states. Source: https://oeis.org/A210088.

$$(c_1, \ldots, c_d) = (10, -27, 10, 8).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.40 A210089

Arrays with $n+1$ rows, width $w=3$, exactly 3 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=6$, for $n \geq 7$. The certificate has $D=10$ states. Source: https://oeis.org/A210089.

$$(c_1, \ldots, c_d) = (20, -112, 122, 61, -106, 16).$$

The empirical generating function on the cited page is also obtained from $QF = P$ with the exact boundary polynomial stored in the certificate.

3.5.3.41 A210091

Arrays with $n+1$ rows, width $w=5$, exactly 3 colors up to order of first appearance, with rule D_{12}. The recurrence (3.17) holds with $d=22$, for $n \geq 23$. The certificate has $D=58$ states. Source: https://oeis.org/A210091.

```
92, -2759, 28247, -12894, -1142474, 3601421, 11970542, -56246539, -20466947, 259395248,
-85275847, -451433420, 270874469, 310376692, -241516822, -72716144, 82685288, -2658056,
-9212400, 1726736, 40928, -15360
```

3.6 A368628: the exact parity support

Let $A(x) = \sum_{n\ge 0} a(n)x^n$ be the unique formal series with $A(0) = 1$ satisfying

$$A(x) = 1 + \frac{x}{2}(A(x)^2 - A(-x)^2) + \frac{x}{2}(A(x)^4 + A(-x)^4).$$

The right side determines successive coefficients from earlier ones, so existence and uniqueness are coefficientwise. The divisions by 2 extract odd and even coefficients and preserve integrality. The explicitly conjectured claim on https://oeis.org/A368628 is

$$\boxed{a(n) \text{ is odd} \quad \Longleftrightarrow \quad n = \frac{4^k - 1}{3} \text{ for some integer } k \ge 0.}$$

To prove it, write $A = E + O$ for its even and odd parts. Taking these parts in the defining equation gives integral equations

$$E = 1 + 2xEO, \qquad O = x(E^4 + 6E^2O^2 + O^4).$$

In $\mathbb{F}_2[[x]]$, therefore, $E = 1$ and $O = x(1 + O^4)$. Since $A = 1 + O$, this is $A = 1 + xA^4 = 1 + xA(x^4)$. Iterating the last equation in the x-adic topology gives the unique solution

$$A(x) \equiv \sum_{k\ge 0} x^{1+4+\cdots+4^{k-1}} = \sum_{k\ge 0} x^{(4^k-1)/3} \pmod 2.$$

The exponents are distinct, proving both directions of the claim, including $n = 0$. The algebraic equation and order-68 recurrence credited to Robert Israel in July 2026 are prior results and are not counted here; this entry counts only the parity conjecture. The accompanying check computes exact coefficients from the original defining equation, independently of this modulo-two simplification.

3.7 Reproduction and certificate inventory

The accompanying archive is self-contained for checking the mathematical certificates: after extraction, run the following command in its directory.

```
python3 run_all.py
```

Python 3.10 or later and SymPy are required; the recorded environment used Python 3.12.14 and SymPy 1.14.0. The root and modular verifiers use only the Python standard library. The arithmetic and transfer verifiers use SymPy for exact polynomial or rational-function identities. Verification reads no websites and does not need the earlier rounds or the provisional file. The requirements file records the tested SymPy version.

Component	Count	Certificate and role
Root arguments	15	Formal-series calculations and exact arithmetic checks supplement the written all-index proofs.
Modular arguments	31	Complete residue cycles and an exact Euler-number recurrence certify four refutations; the general positive results have written proofs, with checks for representations and exceptional primes.
Arithmetic arguments	12	Exact zero polynomial identities certify the walk recurrences. Complete residue checks certify the short periodic formulas; bounded arithmetic computations are supplementary.
Transfer and parity	42	41 complete matrix identities certify the empirical recurrences. The separate parity theorem has a formal proof; its exact initial coefficients check the implementation.
Total	100	95 proved, 4 refuted, 1 mixed.

Transfer data. The 90 distinct complete graphs in `r3_gf_transition_matrices.json` are reconstructed from their predicates and compared entry by entry. Each graph has at most 187 states; the largest sum of component dimensions for an entry is 660. The recurrence of A209811 has order 91. The certificate test is $TR(T)\mathbf{1} = 0$ in every component; it is not a fitted scalar recurrence. Initial values and the explicitly specified starting index complete the statement for each sequence.

Review and reproducibility limits. The packet includes independent mathematical cross-checks of the inherited elementary arguments, the new root congruences, the arithmetic core, the transfer theorem, and the modular-period theorems. One off-by-one error in an intermediate binomial transformation for A397356 was corrected during review; the final expression and its exact regression check agree. These reviews improve reliability but are not external peer review. The archive contains the actual programs and finite data so that a reader can inspect or reproduce the computational portions of the proofs.

Rebuilding this document. The supplied `.tex` contains the entire document; it does not depend on external proof-fragment inputs. Compile it twice with pdfLaTeX after installing the standard packages listed in its preamble. The optional builder `r3_build_dossier.py` regenerates the standalone source and ledger from the included component records and fragments.

Part IV

Modular obstructions, arithmetic, determinants, and recurrence certificates

Round overview

This fourth-round dossier resolves 100 selected assertion packages on 100 distinct OEIS entries, disjoint from the 308 entries counted in the preceding three rounds. There are 97 proof-only packages, two refutations, and one boundary correction. The last concerns A243642: its unqualified order-six recurrence has residual 24 at the first admissible index $n = 6$, and holds for every $n \geq 7$. Since the source specifies no starting index, this is recorded as a boundary correction rather than an unqualified refutation of an explicitly stated lower bound.

The work comprises 14 packages on congruences and formal series, nine on arithmetic and error estimates, 35 on generating functions, and 42 on locally constrained arrays. The two refutations are particularly explicit: A126156 satisfies $a(n) \equiv 1 + 3n(n-1) \pmod 4$, which rules out the proposed eventual period dividing $\varphi(4)$; the recurrence proposed for A190363 fails at $n = 161$ and is also incompatible with its irrational limiting slope.

The 42 array recurrences follow from complete integer transfer-matrix identities. The generating-function recurrences follow from exact differential identities with all boundary terms retained. Other statements have written all-index arguments, accompanied by supplementary checks. The verification archive includes the matrices, symbolic certificates, component proofs, source ledger and code needed to repeat these checks without network access.

What the count means

One catalogue row is one selected assertion package on one A-number. Several claims proved together on the same entry add one count. Different entries in a parameter family may use the same theorem; in particular, the 42 array results use a common transfer construction. The catalogue therefore does not represent 100 unrelated proof methods.

The selected primary-page assertions are explicitly conjectural, empirical, or described as probable. An entry can also contain established formulas and other unresolved questions. Only the assertion specified in the catalogue and proof is included. Established inputs are credited; proving a selected comment does not resolve every question on its page. Examples of deliberately limited scope are the reciprocal-square comment in A320192, part (ii) in A228884, the 2019 divisor-gap assertion in A258409, and the mod-3 statements in A368630 and A368631.

The source audit used primary pages consulted on 20 October 2026, relevant linked work and targeted searches. Previously proved candidates were excluded when found. The separate comparison with the OEIS Open paper covers only its stated Appendix A of 100 costliest claims, not every result in that project. Some retrieved pages may be cached; a consultation date does not certify the site's instantaneous editorial state. This dossier asserts the mathematical results given here, and does not certify worldwide priority or OEIS editorial acceptance. No OEIS submission or edit is part of this dossier.

The reported reviews are independent calculations and readings by other agents in the same assistant workflow. They are not external peer review or proof-assistant formalization. The exact finite certificates are part of the proofs; bounded sequence-prefix checks are identified separately.

4.1 Catalogue of the 100 selected entries

P: proved. **R**: refuted. **B**: boundary corrected with the corrected tail proved. The page reference leads to the individual argument or recurrence signature. Full definitions, attribution, formulas and scope notes are retained in `oeis_hunt_round4_ledger.json`.

No.	Entry	Selected assertion package	Result	Page
Congruences and formal series				
1	A000657	Eventual modular periods dividing Euler phi	P	176
2	A064618	Eventual modular periods for the squared-factorial transform	P	177
3	A126156	All-index mod-4 obstruction to the proposed period bound	R	177
4	A396099	Four mod-4 assertions for iterated composition	P	178
5	A378575	Every coefficient is 1 modulo 4	P	178
6	A378578	Odd-valued terms at binary indices with no adjacent ones	P	178
7	A380709	Binomial parity for a coefficientwise absolute-value equation	P	179
8	A380711	Binomial parity and its complete binary support	P	179
9	A369083	Parity of the signed quadratic functional equation	P	179
10	A361552	Half-coefficient parity equals the shifted partition sequence	P	180
11	A359914	Half-coefficient parity equals the partition sequence	P	180
12	A380056	All coefficients are 1 mod 4; multiples of four give 0 mod 5	P	181
13	A368630	Every exponential coefficient is 1 modulo 3	P	181
14	A368631	Every exponential coefficient is 1 modulo 3	P	181
Arithmetic and exact bounds				
15	A190363	Proposed recurrence fails at 161 and cannot hold eventually	R	182
16	A336838	Divisibility survives reduction of a shifted divisor mean	P	182
17	A320192	The uniquely closest reciprocal-square partial sum (comment only)	P	183
18	A111205	An explicit infinite family of the required composite numbers	P	184
19	A025060	Every integer has the signed distinct nonzero representation	P	184
20	A228884	Prime factors of the gcd determinant are below composite n	P	184
21	A228885	Two totient-product determinant formulas and the squarefree sign	P	185
22	A258409	Gcd of consecutive divisor gaps (2019 assertion only)	P	186
23	A176615	The square-root error term, with an explicit constant 3	P	186
Generating-function certificates				
24	A210496	Conjectured polynomial recurrence of order 8	P	190
25	A102318	Conjectured polynomial recurrence of order 5	P	190
26	A162481	Conjectured polynomial recurrence of order 4	P	191
27	A116391	Conjectured polynomial recurrence of order 5	P	191
28	A114464	Conjectured polynomial recurrence of order 5	P	192
29	A097332	Conjectured polynomial recurrence of order 3	P	192
30	A128057	Conjectured polynomial recurrence of order 2	P	193
31	A100095	Conjectured polynomial recurrence of order 5	P	193
32	A171851	Conjectured polynomial recurrence of order 9	P	194
33	A171847	Conjectured polynomial recurrence of order 10	P	195
34	A171849	Conjectured polynomial recurrence of order 10	P	195
35	A191792	Conjectured polynomial recurrence of order 8	P	196
36	A186338	Conjectured polynomial recurrence of order 4	P	196
37	A182899	Conjectured polynomial recurrence of order 8	P	197
38	A187256	Conjectured polynomial recurrence of order 4	P	198
39	A184120	Conjectured polynomial recurrence of order 4	P	198
40	A186940	Conjectured polynomial recurrence of order 5	P	199
41	A176610	Conjectured polynomial recurrence of order 5	P	199
42	A179648	Conjectured polynomial recurrence of order 4	P	200
43	A166228	Conjectured polynomial recurrence of order 3	P	200
44	A050168	Conjectured polynomial recurrence of order 3	P	201
45	A072100	Conjectured polynomial recurrence of order 3	P	201

Continued on the next page

No.	Entry	Selected assertion package	Result	Page
46	A176611	Conjectured polynomial recurrence of order 5	P	202
47	A176612	Conjectured polynomial recurrence of order 5	P	202
48	A176605	Conjectured polynomial recurrence of order 4	P	203
49	A176607	Conjectured polynomial recurrence of order 4	P	203
50	A045743	Conjectured polynomial recurrence of order 4	P	204
51	A308726	Conjectured polynomial recurrence of order 5	P	204
52	A285195	Conjectured polynomial recurrence of order 6	P	205
53	A243642	First-index exception and the corrected recurrence from n = 7	B	205
54	A243816	Conjectured polynomial recurrence of order 6	P	206
55	A000986	Conjectured polynomial recurrence of order 5	P	207
56	A121754	Conjectured polynomial recurrence of order 5	P	207
57	A004395	Conjectured polynomial recurrence of order 4	P	208
58	A052731	The factorial-weighted shift of A002026	P	209
Finite-state array certificates				
59	A210005	Empirical array recurrence of order 4	P	212
60	A210006	Empirical array recurrence of order 6	P	212
61	A210007	Empirical array recurrence of order 10	P	212
62	A210008	Empirical array recurrence of order 19	P	212
63	A210054	Empirical array recurrence of order 4	P	213
64	A210055	Empirical array recurrence of order 6	P	213
65	A210056	Empirical array recurrence of order 11	P	213
66	A210057	Empirical array recurrence of order 28	P	213
67	A210079	Empirical array recurrence of order 5	P	213
68	A210080	Empirical array recurrence of order 7	P	213
69	A210081	Empirical array recurrence of order 19	P	214
70	A210092	Empirical array recurrence of order 52	P	214
71	A210100	Empirical array recurrence of order 3	P	214
72	A210103	Empirical array recurrence of order 22	P	214
73	A210118	Empirical array recurrence of order 3	P	214
74	A210119	Empirical array recurrence of order 7	P	215
75	A210129	Empirical array recurrence of order 11	P	215
76	A210130	Empirical array recurrence of order 32	P	215
77	A210136	Empirical array recurrence of order 6	P	215
78	A210137	Empirical array recurrence of order 10	P	215
79	A210138	Empirical array recurrence of order 19	P	215
80	A210139	Empirical array recurrence of order 48	P	216
81	A210158	Empirical array recurrence of order 4	P	216
82	A210159	Empirical array recurrence of order 7	P	216
83	A210160	Empirical array recurrence of order 16	P	216
84	A210161	Empirical array recurrence of order 40	P	216
85	A210167	Empirical array recurrence of order 5	P	217
86	A210168	Empirical array recurrence of order 10	P	217
87	A210169	Empirical array recurrence of order 24	P	217
88	A210260	Empirical array recurrence of order 4	P	217
89	A210261	Empirical array recurrence of order 8	P	217
90	A210262	Empirical array recurrence of order 18	P	218
91	A210302	Empirical array recurrence of order 5	P	218
92	A210303	Empirical array recurrence of order 7	P	218
93	A210304	Empirical array recurrence of order 20	P	218
94	A210313	Empirical array recurrence of order 6	P	218
95	A210314	Empirical array recurrence of order 10	P	218
96	A210315	Empirical array recurrence of order 21	P	219
97	A210316	Empirical array recurrence of order 50	P	219
98	A210397	Empirical array recurrence of order 3	P	219
99	A210398	Empirical array recurrence of order 5	P	219
100	A210400	Empirical array recurrence of order 32	P	220

4.2 Congruences, formal series, and a persistent counterexample

All ordinary power series in this section are formal. Congruences of exponential generating functions refer to their *integer derivative sequences*, not to ordinary coefficients with factorial denominators. A conjecture is counted only at its explicitly named OEIS entry; auxiliary consequences add no count.

4.2.1 Finite Euler moments and the median Euler numbers

Write

$$\operatorname{sech} t = \sum_{r\ge 0} E_r \frac{t^r}{r!}; \qquad E_0 = 1, \quad E_2 = -1, \quad E_4 = 5.$$

These are the signed Euler numbers, and $E_{2j+1} = 0$.

Lemma 4.2.1. *For every odd positive integer M and every $r \ge 0$,*

$$E_r \equiv \sum_{j=0}^{M-1} (-1)^j (2j+1)^r \pmod M.$$

Proof. The finite geometric sum gives the exact identity

$$2\sum_{j=0}^{M-1} (-1)^j e^{(2j+1)t} = (1+e^{2Mt}) \operatorname{sech} t.$$

After taking r derivatives at zero, the right side is $2E_r$ plus $\sum_{h=1}^{r} \binom{r}{h} (2M)^h E_{r-h}$. Every summand in this last expression is divisible by $2M$. Divide the integer identity by 2. □

Theorem 4.2.2 (The selected periodicity conjecture in A000657)**.** *The median Euler numbers are eventually periodic modulo every positive integer m, with a period dividing $\varphi(m)$.*

Proof. We use two existing descriptions recorded in the entry: Kruchinin's Euler-number formula and the Jacobi continued fraction,

$$a_n = (-1)^n \sum_{k=0}^{n} \binom{n}{k} E_{n+k}, \qquad \sum_{n\ge 0} a_n x^n = \cfrac{1}{1 - b_0 x - \cfrac{\lambda_1 x^2}{1 - b_1 x - \cfrac{\lambda_2 x^2}{1 - b_2 x - \cdots}}},$$

where

$$b_i = 8i^2 + 4i + 1, \qquad \lambda_i = i^2(4i-3)(4i-1) \quad (i \ge 1).$$

Their use does not count as a new enumeration formula. For odd M, Lemma 4.2.1 and the binomial theorem imply

$$a_n \equiv \sum_{j=0}^{M-1} (-1)^j \big(-(2j+1)(2j+2)\big)^n \pmod M. \tag{R1}$$

Let $e(M) = \max_{p\mid M} v_p(M)$ for $M > 1$. Every scalar power sequence on the right is periodic with period $\lambda(M)$ after index $e(M)$: modulo p^α, a nonunit has become zero by index α, while a unit has period dividing the Carmichael exponent $\lambda(M)$. Chinese remaindering proves this assertion modulo M.

It remains to handle powers of 2, for which we prove eventual zero. Fix $a \ge 1$, and put $K = 2^{\lceil a/2 \rceil}$. Let T be the K by K integer tridiagonal matrix indexed by $0, \dots, K-1$, with diagonal b_i, superdiagonal 1, and subdiagonal $T_{i,i-1} = \lambda_i$. The continued-fraction path expansion gives

$$a_n \equiv (T^n)_{0,0} \pmod{2^a} \quad (n \ge 0). \tag{R2}$$

Indeed, every omitted excursion reaching height K must return across the edge of weight λ_K, and $2^a \mid \lambda_K$. Modulo 2, all diagonal entries are 1 and $\lambda_i \equiv i$. In consecutive pairs of indices, T is block upper triangular, with diagonal blocks $\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$. Thus its characteristic polynomial is X^K over $\mathbb{F}_2$. Cayley–Hamilton gives $T^K = 2U$ for an integer matrix U, so $T^{aK} = 2^a U^a$. Equation (R2) yields $a_n \equiv 0 \pmod{2^a}$ for $n \geq aK$.

Write $m = 2^a M$ with M odd. Combine this eventual zero with (R1). For $M > 1$ one may take the period $\lambda(M)$ and threshold $\max(e(M), a2^{\lceil a/2 \rceil})$, omitting the second argument when $a = 0$. For $M = 1$ take period 1. These periods divide $\varphi(m)$. The modulus 1 is immediate. This proves the stated conjecture, but makes no assertion about the separate exponential continued-fraction conjecture. □

4.2.2 A factorial Stirling transform

Theorem 4.2.3 (The selected conjecture in A064618)**.** *For*

$$a_n = \sum_{k=0}^{n} (k!)^2 S(n,k),$$

the reduction modulo $m \geq 2$ has period dividing $\lambda(m)$, hence dividing $\varphi(m)$, for every $n \geq e(m) = \max_{p|m} v_p(m)$.

Proof. This is a new entry-level application of the factorial Stirling-transform argument used in Round III. We reproduce the entire argument. More generally let c_k be any integer sequence and put $u_n = \sum_k c_k k! S(n,k)$. Terms with $k \geq m$ vanish modulo m. For each remaining k,

$$k!S(n,k) = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} j^n.$$

Each scalar j^n has the claimed eventual period by the prime-power argument in (R1). Sum this finite collection of congruences. Set $c_k = k!$. The modulus 1 is trivial. □

4.2.3 A period bound which fails forever

Theorem 4.2.4 (Refutation for A126156)**.** *Define a_n by*

$$\sqrt{\sec(\sqrt{2}\,t)} = \sum_{n \geq 0} a_n \frac{t^{2n}}{(2n)!}.$$

Then

$$a_n \equiv 1 + 3n(n-1) \pmod 4 \quad (n \geq 0). \tag{R3}$$

Consequently $a_{n+2} - a_n \equiv 2 \pmod 4$ for every $n \geq 0$. The entry's conjecture of eventual period dividing $\varphi(m)$ is false: for $m = 4$ no tail has period 1 or 2.

Proof. Set $y(t) = \tan(\sqrt{2}\,t)/\sqrt{2}$ and $f(t) = \sqrt{\sec(\sqrt{2}\,t)}$. Then $y' = 1 + 2y^2$ and $f' = yf$. Repeated differentiation therefore gives

$$a_n = (L^{2n}1)(0), \qquad L = (1+2y^2)\frac{d}{dy} + y.$$

In particular the coefficients being evaluated are integers. Direct expansion yields

$$L^2 = (1+2y^2)^2 \frac{d^2}{dy^2} + 6y(1+2y^2)\frac{d}{dy} + 1 + 3y^2.$$

This preserves the even polynomials. On writing $z = y^2$, its reduction modulo 4 is the operator

$$H : Q(z) \longmapsto (1 + 3z)Q(z) + 2Q'(z).$$

Put $v = 1 + 3z$. In expanding $H^n 1$, terms with two or more derivative operators have a factor 4. For $n \geq 2$ the remaining terms are

$$H^n 1 \equiv v^n + 2 \sum_{i=0}^{n-1} v^i \frac{d}{dz} v^{n-1-i} = v^n + 3n(n-1)v^{n-2} \pmod 4.$$

Evaluate at $z = 0$. The cases $n = 0, 1$ give 1 separately. Replacing n by $n + 2$ in (R3) gives the persistent difference 2. This is an all-index obstruction, not a finite-prefix disproof of eventual periodicity. The sequence does have the exact period 4 modulo 4. □

4.2.4 Iterated composition modulo four

Theorem 4.2.5 (All four selected congruences in A396099)**.** *Let $A = x + A^{\circ 2} A^{\circ 3}$, with $A = x + O(x^2)$. Its coefficients are all odd, and for $n = 3, 4, 5, 6, \ldots$ their residues modulo 4 repeat $1, 3, 3, 1$. Moreover,*

$$A^{\circ 2} \equiv x + 2x^2 \pmod 4, \qquad A - xA^{\circ 3} \equiv x + \frac{2x^3}{1 - x} \pmod 4.$$

Proof. The coefficient of degree n on the right of the defining equation depends only on coefficients of A of smaller degree: both factors in the product have positive order. Thus it defines a unique integral series, and the same uniqueness holds over $\mathbb{Z}/4\mathbb{Z}$. Consider the rational series

$$F = x + x^2 + \frac{x^3(1 + x)^3}{1 - x^4} = \frac{x + x^2 + x^3 + 3x^4 + 2x^5}{1 - x^4}, \qquad B = x + 2x^2.$$

Clearing denominators, or simply substituting these rational expressions, gives the following identities over $\mathbb{Z}/4\mathbb{Z}$:

$$F(F(x)) = B, \qquad F = x + B\,F(B), \qquad F - xF(B) = x + \frac{2x^3}{1 - x}. \tag{R4}$$

Every denominator used here has constant term 1. Hence these are valid formal-series identities, not evaluations away from poles. The first two identities show that F satisfies the defining equation modulo 4, because $F^{\circ 3} = F(B)$; uniqueness gives $A \equiv F$. Its displayed expansion and the third identity prove all the assertions. The accompanying exact verifier independently clears the three denominators in (R4) and checks every coefficient of the resulting polynomial numerators. □

Theorem 4.2.6 (The selected conjecture in A378575)**.** *For the unique series $A = x + xA^{\circ 5}$, every coefficient of positive degree is 1 modulo 4.*

Proof. The leading factor x makes the equation triangular in its coefficients. For $U = x/(1 - x)$ one has $U^{\circ j} = x/(1 - jx)$ by induction. Therefore $U^{\circ 5} = U$ modulo 4, and $x + xU^{\circ 5} = x + xU = U$ modulo 4. Uniqueness over $\mathbb{Z}/4\mathbb{Z}$ gives $A \equiv U$. The same proof works for iteration counts $j \equiv 1 \pmod 4$, but these unlisted specializations add no entry to the count. □

4.2.5 Removing signs and odd factors

Theorem 4.2.7 (The selected conjecture in A378578)**.** *Starting with $a_0 = 1$, obtain a_n by removing all factors of 3 from $[x^{n-1}]A(x)^3$. Then, for every $n \geq 0$,*

$$a_n \equiv \frac{1}{2n + 1} \binom{3n}{n} \pmod 2.$$

Proof. The coefficient used to define a_n is positive and involves only preceding terms, so the definition gives unique positive integers. Removing a power of 3 preserves parity. Thus the reduced series satisfies $A = 1 + xA^3$ over $\mathbb{F}_2$. The unique integral solution $T = 1 + xT^3$ has coefficients $[x^n]T = \binom{3n}{n}/(2n+1)$ by Lagrange inversion. Reduction and uniqueness prove the assertion. By Lucas's theorem this coefficient is odd precisely when the binary expansion of n contains no adjacent ones: this is the no-carry condition for $n + 2n$. The additional empirical run-length descriptions are not separately counted. □

Theorem 4.2.8 (The selected conjecture in A380709)**.** *For $A = 1 + x\,|1/A|^3$, where absolute value is taken coefficient by coefficient,*

$$a_n \equiv \binom{4n-1}{n} \pmod 2 \quad (n \geq 0).$$

Proof. The equation is triangular and integral. Taking an absolute value preserves parity, so $A = 1 + xA^{-3}$ modulo 2. The integral solution $B = 1 + xB^{-3}$ has, for $n \geq 1$,

$$[x^n]B = \frac{1}{n}\binom{-3n}{n-1} = \frac{(-1)^{n-1}}{4n-1}\binom{4n-1}{n}.$$

The denominator $4n - 1$ is odd, giving the required parity. Both constant terms are 1, with $\binom{-1}{0} = 1$. □

Theorem 4.2.9 (Both parity descriptions in A380711)**.** *For $A = 1 + xA\,|A^{-3}|$, the coefficients satisfy*

$$a_n \equiv \frac{1}{3n-1}\binom{3n-1}{n} \pmod 2 \quad (n \geq 0).$$

For $n > 0$, a_n is odd exactly when $n = 2j + 1$ and the binary expansion of j has no adjacent ones.

Proof. Reduction of the defining equation gives $A = 1 + xA^{-2}$ over $\mathbb{F}_2$. For the integral solution $B = 1 + xB^{-2}$, Lagrange inversion gives

$$[x^n]B = \frac{(-1)^{n-1}}{3n-1}\binom{3n-1}{n} \quad (n \geq 1).$$

This is an identity of integers, so reducing it modulo 2 does not require inverting the possibly even denominator $3n - 1$. At $n = 0$ the displayed quotient is -1, congruent to the constant coefficient 1.

For the second description let $T = 1 + tT^3$ over $\mathbb{F}_2$. Then $F = 1 + xT(x^2)$ satisfies $F = 1 + xF^{-2}$: indeed $T(x^2)(1 + x^2T(x^2)^2) = 1$. By uniqueness $A = F$ modulo 2. The ternary Catalan parity condition proved above identifies precisely the indicated indices. The entry's separate analytic radius and asymptotic observations are outside this target. □

4.2.6 An algebraic parity identity

Theorem 4.2.10 (The selected conjecture in A369083)**.** *For $A = 1 + x(5A(x)^2 - A(-x)^2)/4$ one has*

$$a_n \equiv \binom{4n+3}{n} \pmod 2 \quad (n \geq 0).$$

Proof. Write $A = E + O$, separating the even and odd degrees. The defining equation is equivalent to the integral triangular system

$$E = 1 + 3xEO, \qquad O = x(E^2 + O^2).$$

It proves integrality without dividing by 4 in a finite field. Modulo 2 we have $O = xA^2$ and hence

$$A = 1 + xA^2 + x^2A^3 + x^3A^4. \tag{R5}$$

The constant term 1 solution of (R5) is unique. Let $T = 1 + xT^4$ and put $B = T^3$. In characteristic 2, $1 + xB = T^{-1}$, so $B(1 + xB)^3 = 1$, which is (R5). Lagrange inversion gives

$$[x^n]T^3 = \frac{3}{4n+3}\binom{4n+3}{n}.$$

Both 3 and $4n + 3$ are odd. Therefore these coefficients have exactly the parity asserted. □

4.2.7 Two implicit pentagonal series

Let $g_j = j(3j + 1)/2$, and let

$$P(x) = \prod_{k\geq 1}(1 - x^k) = \sum_{j\in\mathbb{Z}}(-1)^j x^{g_j}.$$

The last equality is Euler's pentagonal identity; $1/P(x) = \sum p(n)x^n$ is the partition generating function.

Theorem 4.2.11 (The selected parity claims in A361552 and A359914)**.** *Let A and B, respectively, be the constant-term-1 solutions*

$$2x = \sum_{j\in\mathbb{Z}} x^{g_j}(A^{3j} - A^{-3j-1}), \tag{R6}$$

$$2 = \sum_{j\in\mathbb{Z}}(-1)^j x^{j(3j-1)/2}B^{3j}(1 + x^jB). \tag{R7}$$

Then

$$A \equiv 1 + \frac{2x}{P(x)} \pmod 4, \qquad B \equiv 1 + 2\left(\frac{1}{P(x)} - 1\right) \pmod 4. \tag{R8}$$

In particular $[x^n]A/2 \equiv p(n-1) \pmod 2$ *for* $n \geq 1$*, and* $[x^n]B/2 \equiv p(n) \pmod 2$ *for* $n \geq 1$.

Proof. All sums are well defined degree by degree. At index $j = 0$ the right sides are $1 - A^{-1}$ and $1 + B$, respectively; every other term has a positive power of x after the two summands in (R7) are separated. Consequently the equations determine unique integral coefficients, with leading coefficient 1 on each unknown coefficient. The constant series 1 solves both reduced equations modulo 2, proving that $A = 1 + 2H$ and $B = 1 + 2J$ with $H, J \in x\mathbb{Z}[[x]]$.

For every integer exponent r, $(1 + 2H)^r \equiv 1 + 2rH \pmod 4$. Thus (R6) reduces to $2x \equiv 2H\sum_j x^{g_j} \pmod 4$. After dividing this integer coefficient congruence by 2 and reducing modulo 2, the signs in P disappear and $H = x/P$.

In (R7), replace j by $-j$ in its first summand to write the entire sum as

$$\sum_{j\in\mathbb{Z}}(-1)^j x^{g_j}(B^{-3j} + B^{3j+1}).$$

The parenthesis is $2 + 2J$ modulo 4. Thus $1 = (1 + J)P$ modulo 2, giving $J = 1/P - 1$. This proves (R8) and both parity claims. □

The cross-entry relation $[x^n]A \equiv [x^{n-1}]B \pmod 4$ follows for $n > 1$ and adds no count. The reverse comparison printed in A359914 requires $n \geq 1$: its stated $n = 0$ endpoint compares 1 with 2. This endpoint correction is recorded but is not the assertion selected for counting at A359914.

4.2.8 An elementary exponential congruence

Theorem 4.2.12 (Both selected congruences in A380056)**.** *If*

$$\frac{e^t - 1}{\cos(2t)} = \sum_{n\geq 1} a_n \frac{t^n}{n!},$$

then $a_n \equiv 1 \pmod 4$ *for all* $n \geq 1$*, and* $5 \mid a_{4r}$ *for all* $r \geq 1$*.*

Proof. Let h_j be the jth derivative of $\sec(2t)$ at zero. Then $h_0 = 1$, all odd h_j vanish, and $h_{2k} = (-4)^k E_{2k}$. Consequently

$$a_n = \sum_{j=0}^{n-1} \binom{n}{j} h_j. \tag{R9}$$

All terms of this sum except $j = 0$ vanish modulo 4. Modulo 5, $h_j \equiv E_j$, because $-4 \equiv 1$ and the odd derivatives vanish. Lemma 4.2.1 with $M = 5$ changes (R9) into

$$a_n \equiv \sum_{j=0}^{4} (-1)^j \left((2j+2)^n - (2j+1)^n\right) \pmod 5.$$

For positive n divisible by 4, Fermat's theorem makes each nonzero base contribute 1. Only the bases $2j + 1 \equiv 0, -1$ matter: they occur at $j = 2, 4$, both with positive sign, and contribute 1 and -1. Their sum is zero. □

4.2.9 Cubing an integral exponential series

Lemma 4.2.13. *Let* $B = \sum_{n\geq 0} b_n x^n/n!$ *have integral derivatives and* $b_0 = 1$*. For a prime* p*, every positive-degree derivative of* B^p *is divisible by* p*. Consequently every derivative of* $\exp(xB^p)$ *is* 1 *modulo* p*.*

Proof. The nth derivative of B^p is the multinomial sum over ordered p-tuples $(n_1, \ldots, n_p)$ with total n. Rotation permutes these tuples. Nonconstant tuples have orbits of size p and equal summands. A fixed tuple has $n_1 = \cdots = n_p = k$; if $n > 0$, then $k > 0$ and its multinomial factor $(pk)!/(k!)^p$ has valuation $s_p(k) \geq 1$, by Legendre's formula. This proves the first assertion. The derivatives c_n of xB^p satisfy $c_1 = 1$ and $p \mid c_n$ for $n > 1$. The derivatives of its exponential are the complete Bell polynomials, which have integer coefficients. Their reduction is $B_n(1, 0, 0, \ldots) = 1$, proving the second assertion. □

Theorem 4.2.14 (The selected mod-3 claims in A368630, A368631)**.** *The derivatives* a_n *of either of the series*

$$\begin{aligned} &\textit{A368630:} \quad A(x/A(x)^2) = \exp(xA(x)), \\ &\textit{A368631:} \quad A(x/A(x)) = \exp(xA(x)^2) \end{aligned}$$

are 1 *modulo* 3 *for every* $n \geq 0$*.*

Proof. In the two cases set $r = 2$ and $r = 1$, respectively. Reversion of $x/A(x)^r$ gives the equivalent pair of formal identities

$$A = \exp(xB^3), \qquad B = A(xB^r), \qquad A(0) = B(0) = 1. \tag{R10}$$

For example, putting $z = xB(x)^r$ makes $z/A(z)^r = x$ and transforms the original equation into the first identity; the converse substitution is identical. The equations determine the derivatives recursively: a_n in the first identity uses only $b_0, \ldots, b_{n-1}$, and b_n in the second uses $a_0, \ldots, a_n$ and preceding b's. Products, integral powers and composition of integral exponential series with zero constant term in the inner series preserve integral derivatives, by the integer Bell polynomial formulas. Thus (R10) proves integrality inductively. Apply Lemma 4.2.13 with $p = 3$. The two separate parity conjectures on each page have not been selected. □

4.3 Elementary arithmetic, exact counterexamples, and a square-sum graph

This section addresses nine explicitly selected assertions on nine distinct OEIS entries. A result about a comment is counted only at the entry containing that comment. In particular, the result for A320192 concerns its reciprocal-square approximation comment, and the result for A228884 concerns part (ii) of its conjecture. We do not count already established claims at those entries as new results. Computations accompanying this section are exact checks of the arguments; they are not substitutes for the proofs.

4.3.1 A counterexample to the proposed recurrence in A190363

The definition and its equivalent formula on A190363 give

$$a(n) = 2n + \left\lfloor \frac{n\sqrt{5}}{2} \right\rfloor + \left\lfloor \frac{n}{4} \right\rfloor, \qquad n \geq 1.$$

The recurrence proposed by Harvey P. Dale on January 28, 2025 has its only nonzero coefficients at positions $4, 17, 21$, with values $1, 1, -1$.

Theorem 4.3.1. *The proposed recurrence*

$$a(n) = a(n-4) + a(n-17) - a(n-21)$$

fails at $n = 161$. It cannot hold for all sufficiently large n either.

Proof. For each of the following pairs (n, b) the inequalities $4b^2 \leq 5n^2 < 4(b+1)^2$ hold:

$$(161, 180), \quad (157, 175), \quad (144, 160), \quad (140, 156).$$

These are integer inequalities and certify all four irrational floors exactly. Consequently

$$a(161) = 542, \quad a(157) = 528, \quad a(144) = 484, \quad a(140) = 471.$$

The recurrence predicts $528 + 484 - 471 = 541$, not 542.

To rule out an eventual version, suppose the recurrence holds after some index and put $b_n = a(n) - a(n-4)$. Then $b_n = b_{n-17}$ eventually. Since 4 and 17 are coprime, the seventeen indices $n - 4j$ $(0 \leq j \leq 16)$ run through all residue classes modulo 17. Their b-values therefore have a constant integer sum B for all sufficiently large n. Telescoping gives

$$a(n) - a(n-68) = B.$$

Consequently $\lim_{n\to\infty} a(n)/n = B/68$ is rational. The defining floor formula instead gives the irrational limit $2 + \sqrt{5}/2 + 1/4$, a contradiction. □

This is a counterexample to the asserted universal recurrence. It is not a claim based on a numerical fit. The accompanying verifier additionally checks that 161 is the first failure among the permissible indices $n \geq 22$.

4.3.2 Cancellation in a divisor mean: A336838

Let p be prime and let q be the next prime after p. The prime shift defining A003961 sends p^e to q^e. Thus A336838, the reduced numerator of the mean of the divisors of that shifted number, satisfies

$$a(p^e) = \frac{S}{\gcd(S, e+1)}, \qquad S = 1 + q + \cdots + q^e.$$

Theorem 4.3.2 (The selected conjecture in A336838)**.** *If $e \geq 1$ is odd, then $(q+1)/2$ divides $a(p^e)$.*

Proof. Write $e+1 = 2h$. Pairing successive terms gives

$$S = (q+1)\sum_{j=0}^{h-1}(q^2)^j.$$

We use the elementary valuation fact

$$v_\ell(1+u+\cdots+u^{h-1}) = v_\ell(h) \tag{A1}$$

when ℓ is odd and $u \equiv 1 \pmod{\ell}$, or when $\ell = 2$ and $u \equiv 1 \pmod 4$. For completeness, factor $h = \ell^t r$ with $\ell \nmid r$. The geometric factor of length r is a unit modulo ℓ. In successively enlarging a length by a factor ℓ, the new factor is $1+v+\cdots+v^{\ell-1}$ with v satisfying the same congruence. The binomial expansion at $v=1$ shows this factor has valuation exactly one: for odd ℓ the constant term is ℓ and all remaining terms are divisible by ℓ^2; for $\ell = 2$ it is $1+v \equiv 2 \pmod 4$. This proves (A1).

If $\ell \mid q+1$ is odd, (A1) with $u = q^2$ gives

$$v_\ell(S) = v_\ell(q+1) + v_\ell(e+1).$$

Since q is odd, $q^2 \equiv 1 \pmod 8$, so it also gives

$$v_2(S) = v_2(q+1) + v_2(e+1) - 1.$$

Cancelling $\gcd(S, e+1)$ can remove at most $v_\ell(e+1)$ factors of each prime ℓ. The reduced numerator therefore retains at least $v_\ell(q+1)$ for odd $\ell \mid q+1$, and at least $v_2(q+1)-1$ for $\ell = 2$. These are exactly the exponents required by $(q+1)/2$. □

4.3.3 The closest reciprocal-square partial sum: A320192

Write $H_m^{(2)} = \sum_{k=1}^{m} k^{-2}$, with $H_0^{(2)} = 0$. The assertion selected here is Andrew Howroyd's November 24, 2018 *comment* in A320192; the separate formula for its main sequence, which involves finite Euler products, is outside this result's scope.

Theorem 4.3.3. *For every integer $n \geq 1$, the unique partial sum $H_m^{(2)}$ closest to $\zeta(2) - 1/n$ is $H_n^{(2)}$.*

Proof. Put $R_n = \sum_{k=n+1}^{\infty} k^{-2}$. The strict integral comparison for the decreasing function x^{-2} gives $R_n < \int_n^\infty x^{-2}\,dx = 1/n$. On each unit interval this function is strictly convex, so its integral is strictly smaller than the trapezoidal area. Summing these inequalities gives

$$\frac{1}{n} = \int_n^\infty x^{-2}\,dx < \frac{1}{2n^2} + R_n.$$

Hence

$$0 < H_n^{(2)} - \left(\zeta(2) - \frac{1}{n}\right) = \frac{1}{n} - R_n < \frac{1}{2n^2}. \tag{A2}$$

Since $H_n^{(2)} - H_{n-1}^{(2)} = 1/n^2$, the target lies strictly between the midpoint of these two sums and $H_n^{(2)}$. Thus $H_n^{(2)}$ is closer than $H_{n-1}^{(2)}$. Every earlier sum is still farther below, and every later sum is still farther above, proving uniqueness. □

4.3.4 An infinite explicit family for A111205

The entries of A111205 are nonsquare composite integers k for which $\gcd(k, 2\lfloor\sqrt{k}\rfloor + 1) > 1$. The second argument is the difference between the two consecutive squares surrounding k.

Theorem 4.3.4. *There are infinitely many such integers. In fact every*

$$k_t = (3t+1)^2 + 2 = 3(3t^2 + 2t + 1), \qquad t \geq 1,$$

belongs to A111205.

Proof. Set $r = 3t + 1 \geq 4$. Then $r^2 < r^2 + 2 < (r+1)^2$, so k_t is nonsquare and its integer square root is r. Its displayed factorization has both factors greater than one, so it is composite. Both k_t and $2r + 1 = 6t + 3$ are divisible by 3. Finally, k_t is strictly increasing with t. □

4.3.5 The signed representation conjecture in A025060

The original sequence uses positive $i < j < k$. We address Jon Perry's April 21, 2013 conjecture in A025060 concerning the variant that allows negative integers while retaining distinctness and excluding zero.

Theorem 4.3.5. *Every integer N has a representation $N = ij + jk + ki$ by three distinct nonzero integers $i < j < k$.*

Proof. Choose $a = |N| + 2$ and put

$$i = -a, \qquad j = a + 1, \qquad k = N + a(a+1).$$

Then $i + j = 1$ and $ij = -a(a+1)$, giving $ij + (i+j)k = N$. Also $i < 0 < j$, and

$$k - j = N + a^2 - 1 \geq -|N| + (|N| + 2)^2 - 1 > 0.$$

The three integers have all the required properties. □

4.3.6 Prime divisors of a gcd determinant: A228884(ii)

Let $D_n = \det(\gcd(i-j, n))_{0 \leq i,j < n}$. We prove only conjecture (ii) in A228884. Fedor Petrov's February 2021 answer to Zhi-Wei Sun's MathOverflow question proves positivity and gives an eigenvalue formula that also implies the divisibility assertion in part (i); these are not counted here. We recall that Fourier formula with attribution and give its short derivation so that this proof is complete. The prior answer is at https://mathoverflow.net/questions/384719.

Theorem 4.3.6. *If n is composite, every prime divisor of D_n is smaller than n.*

Proof. The identity $\gcd(t, n) = \sum_{d \mid \gcd(t,n)} \varphi(d)$ and the finite geometric sum show that the eigenvalue at the character indexed by $k \in \{0, \ldots, n-1\}$ is

$$\lambda_n(k) = \sum_{d \mid \gcd(k,n)} d\varphi(n/d). \tag{A3}$$

Indeed, the sum of the character over multiples of a divisor e of n is n/e if $(n/e) \mid k$, and zero otherwise; substitute $d = n/e$. The Fourier characters are a basis, so $D_n = \prod_{k=0}^{n-1} \lambda_n(k)$. All these eigenvalues are positive integers.

For $n = \prod p^a$, the divisor sum in (A3) factors over prime powers. At one such power, put $j = \min(v_p(k), a)$, interpreting $j = a$ when $k \equiv 0 \pmod{p^a}$. Direct summation gives

$$\lambda_{p^a}(k) = \begin{cases} (j+1)p^{a-1}(p-1), & j < a, \\ p^{a-1}((a+1)p - a), & j = a. \end{cases} \tag{A4}$$

Every prime factor of a value in the first line divides one of p, $p-1$, or $j+1 \le a$; each is smaller than the composite number n.

It remains to consider $B = (a+1)p - a$. If $a = 1$, the fact that n is composite and p occurs to exponent one implies $n \ge 2p > 2p - 1 = B$. If $a \ge 2$, then $B \le p^a$. At $a = 2$ the difference is $p^2 - (3p-2) = (p-1)(p-2) \ge 0$; increasing a increases p^a by $(p-1)p^a$ and increases B by only $p - 1$. Equality is possible only for $p = 2, a = 2$. Thus $B < n$ except possibly when $n = 4$ and $B = 4$, whose only prime factor is $2 < 4$. All prime factors of all eigenvalues in (A3) are therefore smaller than n, and the same holds for their product. □

4.3.7 A coprimality determinant: A228885

Let $M_n(i, j)$ be 1 if $\gcd(i + j, n) = 1$ and 0 otherwise, with rows and columns indexed by residues modulo n. Let $a(n) = \det M_n$. This is the definition of A228885.

Theorem 4.3.7. *For every $n \ge 1$,*

$$a(n) = (-1)^{\lfloor n/2 \rfloor} \mu(n)^2 \prod_{k=1}^{n} \varphi(\gcd(n, k)) = (-1)^{\lfloor n/2 \rfloor} \mu(n)^2 \prod_{d \mid n} \varphi(d)^{\varphi(n/d)}. \tag{A5}$$

In particular, the conjectured sign $(-1)^{n(n-1)/2} a(n) > 0$ holds whenever n is squarefree.

Proof. If $p^2 \mid n$, adding n/p to a row index does not change whether $i + j$ is divisible by any prime divisor of n. Two different rows are equal and the determinant is zero. This observation was already recorded in the entry; it agrees with the factor $\mu(n)^2$ in (A5).

Now assume n is squarefree and set $C_n(i, j) = 1_{\gcd(i-j,n)=1}$. The Chinese remainder theorem simultaneously reindexes rows and columns and identifies

$$C_n \cong \bigotimes_{p \mid n} (J_p - I_p).$$

The $p \times p$ matrix $J_p - I_p$ has eigenvalue $p - 1$ once and eigenvalue -1 with multiplicity $p - 1$. Consequently

$$|\det C_n| = \prod_{p \mid n} (p-1)^{n/p}. \tag{A6}$$

The sign is positive for odd n. For even squarefree n, the factor with $p = 2$ contributes sign $(-1)^{n/2} = -1$, and all odd-prime factors contribute positive sign.

The column permutation $j \mapsto -j$ changes C_n into M_n. It has $\lfloor (n-1)/2 \rfloor$ transpositions. Combining its sign with the preceding sign of C_n gives $(-1)^{\lfloor n/2 \rfloor}$ in both the odd and the even squarefree cases. For squarefree n, the factor $p - 1$ occurs in $\varphi(\gcd(n, k))$ precisely when $p \mid k$, which happens n/p times for $1 \le k \le n$. Thus its product over k is exactly (A6). Grouping the same product according to $d = \gcd(n, k)$ gives its divisor form in (A5), because there are $\varphi(n/d)$ such values of k. The case $n = 1$ is immediate.

Finally $n(n-1)/2$ and $\lfloor n/2 \rfloor$ have the same parity, and the product in (A6) is positive. This proves the sign assertion. □

The two product formulas in (A5), listed as conjectures by Ridouane Oudra on March 13, 2025, and the older sign assertion are treated as a single entry-level result. They do not contribute three counts.

4.3.8 Consecutive divisor gaps: A258409

For $n \geq 2$, let $1 = d_1 < d_2 < \cdots < d_t = n$ be its positive divisors. The definition of A258409 is $a(n) = \gcd(d_1 - 1, \ldots, d_t - 1)$.

Theorem 4.3.8 (The October 16, 2019 conjecture in A258409)**.**

$$a(n) = \gcd(d_2 - d_1, d_3 - d_2, \ldots, d_t - d_{t-1}).$$

Proof. Any integer dividing every $d_i - 1$ also divides their successive differences $d_{i+1} - d_i$. Conversely, if it divides all those differences, it divides every telescoping sum

$$d_j - 1 = d_j - d_1 = \sum_{i=1}^{j-1} (d_{i+1} - d_i).$$

The two lists have exactly the same common divisors, hence the same positive greatest common divisor. □

The earlier prime-divisor characterization in this entry was proved by Robert Israel on September 14, 2016. It is not part of the present count. The displayed gap statement is the numerical content of the later composition-of-sequences conjecture; no new count is assigned to its auxiliary sequence identifiers.

4.3.9 A square-root error term for A176615

Let $a(n)$ count pairs $1 \leq i < j \leq n$ for which $i + j$ is a square. The entry A176615 gives the leading asymptotic and describes an $O(\sqrt{n})$ error as probable. We prove that stronger error estimate.

Theorem 4.3.9. *For every integer $n \geq 1$,*

$$\left| a(n) - \frac{2\sqrt{2} - 2}{3} n^{3/2} \right| < 3\sqrt{n}.$$

In particular the error term is $O(\sqrt{n})$.

Proof. For a fixed upper endpoint j, the possible squares lie strictly above j and at most $2j - 1$. Therefore

$$a(n) = T(n) - U(n), \quad T(n) = \sum_{j=1}^{n} \lfloor \sqrt{2j-1} \rfloor, \quad U(n) = \sum_{j=1}^{n} \lfloor \sqrt{j} \rfloor.$$

Put $L = \lfloor \sqrt{n} \rfloor$ and $M = \lfloor \sqrt{2n} \rfloor$. Counting each threshold r^2 gives the exact identities

$$U(n) = L(n+1) - \frac{L(L+1)(2L+1)}{6} = Ln - \frac{L^3}{3} - \frac{L^2}{2} + \frac{5L}{6},$$

$$\begin{aligned} T(n) &= Mn - \sum_{r=1}^{M} \left\lfloor \frac{r^2}{2} \right\rfloor \\ &= Mn - \frac{M^3}{6} - \frac{M^2}{4} + \frac{M}{6} + \varepsilon_M, \qquad \varepsilon_M = \frac{1 - (-1)^M}{8} \in \{0, 1/4\}. \end{aligned} \tag{A7}$$

If $M^2 = 2n$, its threshold contributes zero, so the formula remains valid with this choice of M.

Write $x = \sqrt{n}$, $y = \sqrt{2n}$, $L = x - \alpha$ and $M = y - \beta$, where $0 \leq \alpha, \beta < 1$. Expanding (A7) gives

$$U(n) = \frac{2}{3} n^{3/2} - \frac{n}{2} + x\left(\frac{5}{6} + \alpha - \alpha^2\right) + u(\alpha), \quad u(t) = \frac{t^3}{3} - \frac{t^2}{2} - \frac{5t}{6},$$

$$T(n) = \frac{2\sqrt{2}}{3} n^{3/2} - \frac{n}{2} + y\left(\frac{1}{6} + \frac{\beta - \beta^2}{2}\right) + v(\beta) + \varepsilon_M, \quad v(t) = \frac{t^3}{6} - \frac{t^2}{4} - \frac{t}{6}.$$

On $0 \le t \le 1$, we have $0 \le t - t^2 \le 1/4$, $-1 \le u(t) \le 0$ and $-1/4 \le v(t) \le 0$. The latter two bounds follow, for example, by differentiating the two polynomials, which are decreasing on this interval. The two terms $-n/2$ cancel. The triangle inequality now bounds the remaining error by

$$\left(\frac{13}{12} + \frac{7\sqrt{2}}{24}\right)\sqrt{n} + \frac{5}{4} \le \left(\frac{13}{12} + \frac{7\sqrt{2}}{24} + \frac{5}{4}\right)\sqrt{n} < 3\sqrt{n}.$$

This proves the bound uniformly for all $n \ge 1$. □

4.4 Thirty-five generating-function and recurrence packages

This section treats 35 distinct OEIS entries absent from the 308 selected entries in rounds 1–3. The selected assertions were explicitly labeled *Conjecture* in the primary pages consulted on 20 October 2026. Thirty-four packages prove the selected assertion with an explicit index range. A243642 additionally requires a boundary correction: the source does not specify a starting index; the first natural index gives residual 24, while the recurrence holds at every subsequent index. That correction and its proof count as one package.

The defining formulas, established generating functions and existing shorter recurrences are prior input. In particular, the existing Kotesovec recurrences for A114464, A100095 and A000986 and the other Mathar recurrence for A128057 are not presented as newly discovered results. A102318 contributes only its first conjectured recurrence. A002026 is used as a supporting reference, not a further counted problem. These are entry-level selected assertions, often sharing the same differentiation method; no claim of worldwide priority or external acceptance is made.

4.4.1 Exact coefficient-extraction theorem

For an OGF $F(x) = \sum_{n\ge 0} a_n x^n$, extending undefined initial indices by zero only for this formal-series notation, put $\theta = x\,d/dx$. For polynomials $p_0(n), \dots, p_r(n)$ define

$$\mathcal{L}_p = \sum_{j=0}^{r} x^j p_j(\theta + j).$$

Directly on monomials,

$$[x^n]\mathcal{L}_p F = \sum_{j=0}^{r} p_j(n) a_{n-j}. \tag{4.1}$$

Consequently the exact identity $\mathcal{L}_p F = P(x)$, with P a polynomial, proves the displayed homogeneous recurrence at every $n > \deg P$ for which all source indices are defined. No agreement of a finite numerical prefix is used to infer the identity.

All 31 algebraic cases below are certified over a rational function field. Write $x = \xi(t)$ and

$$F(\xi(t)) = \sum_{i=0}^{s} R_i(t) Y_i(t), \qquad Y_i(t)^2 = D_i(t), \quad D_0 = Y_0 = 1.$$

For cases without a parameter, $t = x$ and $\xi(t) = t$. All fractions are interpreted in the formal Laurent field; any apparent poles cancel in the stated analytic branch of F. Define

$$\mathcal{T}_i R = \frac{\xi(t)}{\xi'(t)} \left(R'(t) + \frac{D_i'(t)}{2D_i(t)} R(t) \right), \qquad c_m(t) = \sum_{j=0}^{r} \xi(t)^j [n^m] p_j(n+j).$$

The chain rule gives $\theta(RY_i) = (\mathcal{T}_i R) Y_i$, and hence the finite certificate consists of the rational identities

$$\sum_m c_m \mathcal{T}_0^m R_0 = P(\xi(t)), \qquad \sum_m c_m \mathcal{T}_i^m R_i = 0 \quad (1 \le i \le s). \tag{4.2}$$

Every coefficient p_j, every R_i, D_i, ξ and the complete polynomial P are supplied below. Substitution in (4.2) and clearing denominators yields zero polynomials over $\mathbb{Q}$. Thus this is an exact finite certificate with a specified verification rule, not a numerical guess. The accompanying Python verifier repeats these rational operations, compares all recurrence coefficients with the retained source wording, and independently checks initial coefficients and branch choices.

For clarity, A162481 gives a small fully expanded instance. Its Euler operator is $c_1\theta + c_0$, where

$$c_1 = 1 - 8x + 10x^2 - 4x^3 + x^4, \quad c_0 = 1 - 6x + 14x^2 - 4x^3 + x^4.$$

Here $R_0 = 1/(2x)$, $R_1 = -1/(2x)$ and $D_1 = 1 - 4x/(1-x)^3$. The two residuals reduce respectively to

$$\frac{c_0 - c_1}{2x} = 1 + 2x, \qquad -\frac{1}{2x}\left(c_0 - c_1 + \frac{c_1 x D_1'}{2D_1}\right) = 0.$$

The second equality follows on inserting D_1, or from $c_1 x D_1' + 2D_1(c_0 - c_1) = 0$. The remaining algebraic certificates use exactly the same finite rule, with the supplied data.

4.4.2 Generating functions derived from the definitions

The entries defined by a Catalan transform use $C(z) = (1 - \sqrt{1-4z})/(2z)$. For example, $(1-x)^{-3}C(x/(1-x)^3)$ immediately gives the A162481 seed. The three peakless-Motzkin statistics A171851, A171847 and A171849 use the established seed $g = 1 + xg + x^2 g(g-1)$, so

$$g = \frac{1 - x + x^2 - \sqrt{1 - 2x - x^2 - 2x^3 + x^4}}{2x^2}.$$

Substitution in their respective established statistic OGFs gives the R_i, D_i below. A191792 uses the branch $C(0) = 1$ of $x^2(1+x^2)C^2 - (1 + x^2 + x^4)C + 1 + x^2 = 0$, followed by $F = C/(1 - xC)$. A182899 similarly uses $c = 1 + xc + x^2 c + x^3 c^2$.

For A176610, A176611, A176612, A176605 and A176607 a direct derivation from the defining convolution avoids any inference from data. If $a_0 = 1$, $a_1 = m$ and

$$a_{n+1} = \sum_{p=0}^{n} a_p a_{n-p} + k(n+1) + \ell \qquad (n \geq 1),$$

then summing after multiplication by x^{n+1} gives

$$F = xF^2 + 1 + (m-1)x + \frac{(k+\ell)x^2}{1-x} + \frac{kx^2}{(1-x)^2}.$$

The parameters (m, k, ℓ) are respectively $(0,1,1), (1,1,1), (2,1,1), (1,0,1), (4,0,1)$. The solution with constant term one is $(1-\sqrt{D})/(2x)$, with $D = 1 - 4x\{1 + (m-1)x + (k+\ell)x^2/(1-x) + kx^2/(1-x)^2\}$.

For A072100 let $B(x) = \sum_{k \geq 0} \binom{k}{\lfloor k/2 \rfloor} x^k$. Splitting even and odd indices in the central-binomial expansion gives

$$B(x) = \frac{(1+2x)/\sqrt{1-4x^2} - 1}{2x}, \qquad F(x) = \frac{x}{1-x} + \frac{x^2 B(x)}{1-x}.$$

The second identity is exactly its defining partial sum, with offset one. For A050168 its displayed binomial formula similarly gives $F = (1+x)(\sqrt{(1+2x)/(1-2x)} - 1)/(2x)$, with $a_0 = 1$. For A186940 the entry's printed "offset one" OGF must be divided by x to obtain the zero-offset OGF used here. This adjustment is part of the proof, and is checked independently against the source terms.

For A045743 use $t = T(x)$, $T(0) = 1$, and $xT^4 = (1-T)(1-3T+T^2)$; thus $\xi(t) = (1-t)(1-3t+t^2)/t^4$ and $F = x(T-1)$. For A308726, A285195 and A243642 take $t = \sqrt{1-4x}$, so $t(0) = 1$ and $\xi(t) = (1-t^2)/4$. The second radical has constant term one in the first two cases and constant term two in A243642. For A243816 set $u = \sqrt{1+4x^2}$, $t = (u-1)/(2x)$; then $t(0) = 0$, $x = t/(1-t^2)$, $u = (1+t^2)/(1-t^2)$ and the outer radical is $\sqrt{1-4t}$ with constant term one. These local branches have nonzero $\xi'(t_0)$, so the formal chain rule is legitimate.

4.4.3 The thirty-one algebraic certificates

In each item $D_0 = 1$ is implicit. Unless a parameter and base point are stated, the variable is x, $\xi = x$, and each radical has constant term one. The selected recurrence is $\sum_{j=0}^{r} p_j(n) a_{n-j} = 0$ in the stated range.

4.4.3.1 A210496: order 8, Oct 08 2016

Primary source: https://oeis.org/A210496. The generating function is established input, credited to the entry and its link to Jelinek, Mansour and Shattuck (2013), Theorem 4.27. It is not claimed as a new generating function.

$$\begin{aligned}
p_0(n) &= n+2\\
p_1(n) &= -2\,(3n+4)\\
p_2(n) &= 4\,(2n+3)\\
p_3(n) &= 13n-40\\
p_4(n) &= -5\,(7n-18)\\
p_5(n) &= 6\,(2n-1)\\
p_6(n) &= 22\,(n-7)\\
p_7(n) &= 134-19n\\
p_8(n) &= 2\,(2n-15)\,.
\end{aligned}$$

The exact generating-function representation is $F=\sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned}
R_0 &= -\frac{3x^5-15x^4+14x^3-2x^2-3x+1}{2x^2\,(x-1)^2\,(x^2-3x+1)},\\
R_1 &= -\frac{(x-1)\,(x+1)}{2x^2\,(x^2-3x+1)},\\
D_1 &= (x-1)\left(4x^2+x-1\right).
\end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x)=3x^7-30x^6+16x^5-32x^4+8x^3+16x^2-11x+2.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n\geq 8$. The source offset is 0.

4.4.3.2 A102318: order 5, Jun 08 2016

Primary source: https://oeis.org/A102318. The source contains two conjectured recurrences. Only the first, of order five, is selected here; the second is outside this package.

$$\begin{aligned}
p_0(n) &= -\,(n-1)\,(n+1)\\
p_1(n) &= 2\left(5n^2-9n+1\right)\\
p_2(n) &= -2\left(15n^2-58n+49\right)\\
p_3(n) &= 2\left(10n^2-76n+123\right)\\
p_4(n) &= (n-3)\,(31n-55)\\
p_5(n) &= -30\,(n-4)\,(n-3)\,.
\end{aligned}$$

The exact generating-function representation is $F=\sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned}
R_0 &= \frac{1}{2x},\\
R_1 &= -\frac{1}{4x},\\
R_2 &= -\frac{1}{4x},
\end{aligned}$$

$$D_1 = -\frac{3x-1}{x+1}$$
$$D_2 = \frac{5x-1}{x-1}.$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 11x^2 - 6x + 1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 5$. The source offset is 0.

4.4.3.3 A162481: order 4, Dec 11 2011

Primary source: https://oeis.org/A162481. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$p_0(n) = n + 1$$
$$p_1(n) = -2\,(4n - 1)$$
$$p_2(n) = 2\,(5n - 3)$$
$$p_3(n) = -4\,(n - 2)$$
$$p_4(n) = n - 3.$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$R_0 = \frac{1}{2x},$$
$$R_1 = -\frac{1}{2x},$$
$$D_1 = \frac{x^3 - 3x^2 + 7x - 1}{(x-1)^3}.$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 2x + 1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 4$. The source offset is 0.

4.4.3.4 A116391: order 5, Jan 23 2020

Primary source: https://oeis.org/A116391. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$p_0(n) = n$$
$$p_1(n) = n$$
$$p_2(n) = -3\,(3n - 4)$$
$$p_3(n) = -3\,(3n - 4)$$
$$p_4(n) = 20\,(n - 3)$$
$$p_5(n) = 20\,(n - 3)\,.$$

The exact generating-function representation is $F = \sum_i R_i \sqrt{D_i}$, with

$$R_0 = -\frac{x}{(x+1)\,(5x^2-1)},$$
$$R_1 = -\frac{1}{(x+1)\,(5x^2-1)},$$
$$D_1 = -\,(2x-1)\,(2x+1)\,.$$

For these data the complete rational certificate (4.2) gives

$$P(x) = x.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 5$. The source offset is 0.

4.4.3.5 A114464: order 5, Nov 26 2012

Primary source: https://oeis.org/A114464. Kotesovec already recorded a different, shorter recurrence on 13 February 2014. That existing result and the established generating function are credited. The selected target is Mathar's still-labeled order-five recurrence of 26 November 2012.

$$p_0(n) = n+1$$
$$p_1(n) = 3-5n$$
$$p_2(n) = 2\,(3n-7)$$
$$p_3(n) = -2\,(3n-11)$$
$$p_4(n) = 5n-27$$
$$p_5(n) = 7-n.$$

The exact generating-function representation is $F = \sum_i R_i \sqrt{D_i}$, with

$$R_0 = -\frac{x^3-3x^2+x-1}{2x},$$
$$R_1 = \frac{x-1}{2x},$$
$$D_1 = \left(x^2+1\right)\left(x^2-4x+1\right).$$

For these data the complete rational certificate (4.2) gives

$$P(x) = -3x^4+4x^3-6x^2+1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 5$. The source offset is 0.

4.4.3.6 A097332: order 3, Nov 26 2012

Primary source: https://oeis.org/A097332. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$p_0(n) = n+1$$
$$p_1(n) = -3n$$
$$p_2(n) = 5-n$$
$$p_3(n) = 3\,(n-2)\,.$$

The exact generating-function representation is $F = \sum_i R_i \sqrt{D_i}$, with

$$R_0 = -\frac{x+1}{2x\,(x-1)},$$
$$R_1 = \frac{1}{2x\,(x-1)},$$
$$D_1 = -\,(x+1)\,(3x-1)\,.$$

For these data the complete rational certificate (4.2) gives

$$P(x) = x + 1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 3$. The source offset is 0.

4.4.3.7 A128057: order 2, Dec 02 2014

Primary source: https://oeis.org/A128057. A different order-three D-finite recurrence is already recorded by Mathar on the same date. The selected target is only the still-labeled order-two recurrence.

$$p_0(n) = n\,(5n-9)$$
$$p_1(n) = 4$$
$$p_2(n) = 4\,(n-2)\,(5n-4)\,.$$

The exact generating-function representation is $F = \sum_i R_i \sqrt{D_i}$, with

$$R_0 = 0,$$
$$R_1 = \frac{x+1}{4x^2+1},$$
$$D_1 = 4x^2 + 1.$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 0.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 2$. The source offset is 0.

4.4.3.8 A100095: order 5, Nov 24 2012

Primary source: https://oeis.org/A100095. Kotesovec already recorded a shorter recurrence on 12 February 2014. The selected target is only Mathar's still-labeled order-five recurrence of 24 November 2012.

$$p_0(n) = 2 - n$$
$$p_1(n) = 3 - n$$
$$p_2(n) = 9n - 22$$
$$p_3(n) = 9n - 31$$
$$p_4(n) = -20\,(n-3)$$
$$p_5(n) = -20\,(n-4)\,.$$

The exact generating-function representation is $F = \sum_i R_i \sqrt{D_i}$, with

$$R_0 = -\frac{x}{5x^2-1},$$
$$R_1 = \frac{x^2}{(2x-1)(2x+1)(5x^2-1)},$$
$$D_1 = -(2x-1)(2x+1).$$

For these data the complete rational certificate (4.2) gives

$$P(x) = x(x+1).$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 5$. The source offset is 0.

4.4.3.9 A171851: order 9, Jul 22 2022

Primary source: https://oeis.org/A171851. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$p_0(n) = -2(n+2)(1088n-6417)$$
$$p_1(n) = 10968n^2 - 55789n - 59252$$
$$p_2(n) = -2\left(6704n^2 - 45808n + 27647\right)$$
$$p_3(n) = 2\left(2264n^2 - 18729n + 42256\right)$$
$$p_4(n) = -2\left(11968n^2 - 82555n + 88362\right)$$
$$p_5(n) = 24904n^2 - 243375n + 511724$$
$$p_6(n) = -4\left(1088n^2 - 9137n + 25680\right)$$
$$p_7(n) = 2\left(6616n^2 - 65333n + 164938\right)$$
$$p_8(n) = -2(n-7)(5616n-25895)$$
$$p_9(n) = (n-8)(2264n-10805).$$

The exact generating-function representation is $F = \sum_i R_i \sqrt{D_i}$, with

$$R_0 = -\frac{x^5 - 2x^4 + 3x^3 - 2x^2 + 5x - 2}{2x^2(x^2-3x+1)(x^2+x+1)},$$
$$R_1 = \frac{x^3 - x^2 + 3x - 2}{2x^2(x^2-3x+1)(x^2+x+1)},$$
$$D_1 = \left(x^2-3x+1\right)\left(x^2+x+1\right).$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 30x^3\left(103x^2 - 260x + 1051\right).$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 9$. The source offset is 0.

4.4.3.10 A171847: order 10, Jul 22 2022

Primary source: https://oeis.org/A171847. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned}
p_0(n) &= n+2\\
p_1(n) &= -5\,(n+1)\\
p_2(n) &= 7n-4\\
p_3(n) &= -2\,(3n-7)\\
p_4(n) &= 13n-18\\
p_5(n) &= -11\,(n-3)\\
p_6(n) &= 13n-60\\
p_7(n) &= -2\,(3n-11)\\
p_8(n) &= 7n-38\\
p_9(n) &= -5\,(n-7)\\
p_{10}(n) &= n-8.
\end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned}
R_0 &= -\frac{x^4+2x^3-x^2+2x-1}{2x^2\,(x^2-3x+1)\,(x^2+x+1)},\\
R_1 &= -\frac{x^2-x+1}{2x^2\,(x^2-3x+1)\,(x^2+x+1)},\\
D_1 &= \left(x^2-3x+1\right)\left(x^2+x+1\right).
\end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 2x^3\left(x^4-2x^3+3x^2-4x+5\right).$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 10$. The source offset is 0.

4.4.3.11 A171849: order 10, Jul 22 2022

Primary source: https://oeis.org/A171849. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned}
p_0(n) &= n+1\\
p_1(n) &= -5n\\
p_2(n) &= 7n-11\\
p_3(n) &= -2\,(3n-10)\\
p_4(n) &= 13n-31\\
p_5(n) &= -11\,(n-4)\\
p_6(n) &= 13n-73\\
p_7(n) &= -2\,(3n-14)\\
p_8(n) &= 7n-45\\
p_9(n) &= -5\,(n-8)
\end{aligned}$$

$$p_{10}(n) = n - 9.$$

The exact generating-function representation is $F = \sum_i R_i \sqrt{D_i}$, with

$$R_0 = \frac{x^4 - 2x^3 + x^2 - 2x + 1}{2x\,(x^2 - 3x + 1)\,(x^2 + x + 1)},$$
$$R_1 = -\frac{x^2 - x + 1}{2x\,(x^2 - 3x + 1)\,(x^2 + x + 1)},$$
$$D_1 = \left(x^2 - 3x + 1\right)\left(x^2 + x + 1\right).$$

For these data the complete rational certificate (4.2) gives

$$P(x) = -4x^3\,(x - 1)\,(x + 1).$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 10$. The source offset is 0.

4.4.3.12 A191792: order 8, Jul 22 2022

Primary source: https://oeis.org/A191792. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$p_0(n) = n + 1$$
$$p_1(n) = -2$$
$$p_2(n) = -2\,(n - 1)$$
$$p_3(n) = -4$$
$$p_4(n) = -5\,(n - 3)$$
$$p_5(n) = 4$$
$$p_6(n) = -2\,(n - 5)$$
$$p_7(n) = 2$$
$$p_8(n) = n - 7.$$

The exact generating-function representation is $F = \sum_i R_i \sqrt{D_i}$, with

$$R_0 = -\frac{1}{2x},$$
$$R_1 = \frac{1}{2x\,(x^4 - 2x^3 + x^2 - 2x + 1)},$$
$$D_1 = \left(x^4 - 2x^3 + x^2 - 2x + 1\right)\left(x^4 + 2x^3 + x^2 + 2x + 1\right).$$

For these data the complete rational certificate (4.2) gives

$$P(x) = -\,(x - 1)\,(x + 1)\left(x^4 + 3x^2 + 1\right).$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 8$. The source offset is 0.

4.4.3.13 A186338: order 4, Nov 17 2011

Primary source: https://oeis.org/A186338. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$p_0(n) = n + 1$$

$$
\begin{aligned}
p_1(n) &= -3\,(4n-1)\\
p_2(n) &= 15\,(3n-4)\\
p_3(n) &= -6\,(11n-26)\\
p_4(n) &= 16\,(2n-7)\,.
\end{aligned}
$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$
\begin{aligned}
R_0 &= \frac{3x-1}{2x\,(2x-1)},\\
R_1 &= \frac{1}{2x\,(2x-1)},\\
D_1 &= -\,(x-1)\left(16x^2-9x+1\right).
\end{aligned}
$$

For these data the complete rational certificate (4.2) gives

$$
P(x) = -\,(2x-1)\left(6x^2-3x+1\right).
$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 4$. The source offset is 0.

4.4.3.14 A182899: order 8, Jul 22 2022

Primary source: https://oeis.org/A182899. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$
\begin{aligned}
p_0(n) &= n\\
p_1(n) &= 3-4n\\
p_2(n) &= 2n-3\\
p_3(n) &= 0\\
p_4(n) &= 11\,(n-3)\\
p_5(n) &= 0\\
p_6(n) &= 2n-9\\
p_7(n) &= 21-4n\\
p_8(n) &= n-6.
\end{aligned}
$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$
\begin{aligned}
R_0 &= -\frac{x^2+x-1}{(x^2-3x+1)\,(x^2+x+1)},\\
R_1 &= -\frac{1}{(x^2-3x+1)\,(x^2+x+1)},\\
D_1 &= \left(x^2-3x+1\right)\left(x^2+x+1\right).
\end{aligned}
$$

For these data the complete rational certificate (4.2) gives

$$
P(x) = 2x^3\left(x^2-x+3\right).
$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 8$. The source offset is 0.

4.4.3.15 A187256: order 4, Nov 16 2011

Primary source: https://oeis.org/A187256. The entry links Barry's 2019 and 2021 papers. Their relevant discussions give existing generating-function, convolution and Hankel-transform results. These are credited as prior work; the selected linear recurrence is the separately labeled conjecture.

$$\begin{aligned} p_0(n) &= n+2 \\ p_1(n) &= -2\,(2n+1) \\ p_2(n) &= 2\,(n-1) \\ p_3(n) &= -2\,(2n-5) \\ p_4(n) &= n-4. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= \frac{(x-1)^2}{2x^2}, \\ R_1 &= -\frac{1}{2x^2}, \\ D_1 &= \left(x^2+1\right)\left(x^2-4x+1\right). \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = -2\,(x-1)\,(x+1)\,.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 4$. The source offset is 0.

4.4.3.16 A184120: order 4, Nov 17 2011

Primary source: https://oeis.org/A184120. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= 2\,(2n+1) \\ p_2(n) &= 4\,(n-1) \\ p_3(n) &= 4\,(2n-5) \\ p_4(n) &= 4\,(n-3)\,. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= \frac{1}{2x}, \\ R_1 &= -\frac{1}{2x}, \\ D_1 &= \frac{2x^2+1}{2x^2+4x+1}. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 1-2x^2.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 4$. The source offset is 0.

4.4.3.17 A186940: order 5, Nov 17 2011

Primary source: https://oeis.org/A186940. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= 3\,(n+1) \\ p_1(n) &= 10 - 17n \\ p_2(n) &= -6\,(n+1) \\ p_3(n) &= 10\,(2n-1) \\ p_4(n) &= -9\,(n-1) \\ p_5(n) &= n-2. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= -\frac{2x^2-3x-1}{2x\,(x^3-3x^2+x+3)}, \\ R_1 &= -\frac{1}{2x\,(x^3-3x^2+x+3)}, \\ D_1 &= x^2-6x+1. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = -x^3+9x^2-7x+3.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 5$. The source offset is 0.

4.4.3.18 A176610: order 5, Nov 17 2011

Primary source: https://oeis.org/A176610. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= 2-7n \\ p_2(n) &= 19n-29 \\ p_3(n) &= 110-37n \\ p_4(n) &= 36\,(n-4) \\ p_5(n) &= -12\,(n-5)\,. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= \frac{1}{2x}, \\ R_1 &= -\frac{1}{2x}, \\ D_1 &= \frac{12x^4-24x^3+13x^2-6x+1}{(x-1)^2}. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 6x^4-18x^3+18x^2-5x+1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 5$. The source offset is 0.

4.4.3.19 A179648: order 4, Nov 17 2011

Primary source: https://oeis.org/A179648. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= 2\,(2n+1) \\ p_2(n) &= -4\,(n-1) \\ p_3(n) &= -4\,(2n-5) \\ p_4(n) &= 4\,(n-3)\,. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= \frac{1}{2x}, \\ R_1 &= -\frac{1}{2x}, \\ D_1 &= \frac{2x^2-1}{2x^2-4x-1}. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 2x^2+1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 4$. The source offset is 0.

4.4.3.20 A166228: order 3, Nov 17 2011

Primary source: https://oeis.org/A166228. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= 4-5n \\ p_2(n) &= 1-5n \\ p_3(n) &= n-2. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= -\frac{x-1}{2x\,(x+1)}, \\ R_1 &= -\frac{1}{2x\,(x+1)}, \\ D_1 &= x^2-6x+1. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = x+1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 3$. The source offset is 0.

4.4.3.21 A050168: order 3, Nov 26 2012

Primary source: https://oeis.org/A050168. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= n-3 \\ p_2(n) &= -2\,(2n-1) \\ p_3(n) &= -4\,(n-3)\,. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= -\frac{x+1}{2x}, \\ R_1 &= \frac{x+1}{2x}, \\ D_1 &= -\frac{2x+1}{2x-1}. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = (x+1)^2\,.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 3$. The source offset is 0.

4.4.3.22 A072100: order 3, Nov 26 2012

Primary source: https://oeis.org/A072100. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= 1-n \\ p_1(n) &= n+1 \\ p_2(n) &= 2\,(2n-7) \\ p_3(n) &= -4\,(n-3)\,. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= -\frac{x}{2\,(x-1)}, \\ R_1 &= \frac{x}{2\,(x-1)\,(2x-1)}, \\ D_1 &= -\,(2x-1)\,(2x+1)\,. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = x^2.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 4$. The source offset is 1.

4.4.3.23 A176611: order 5, Mar 01 2016

Primary source: https://oeis.org/A176611. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= 2-7n \\ p_2(n) &= 3\,(5n-7) \\ p_3(n) &= 74-25n \\ p_4(n) &= 24\,(n-4) \\ p_5(n) &= -8\,(n-5)\,. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= \frac{1}{2x}, \\ R_1 &= -\frac{1}{2x}, \\ D_1 &= \frac{8x^4-16x^3+9x^2-6x+1}{(x-1)^2}. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 4x^4-12x^3+12x^2-3x+1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 5$. The source offset is 0.

4.4.3.24 A176612: order 5, Feb 29 2016

Primary source: https://oeis.org/A176612. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= 2-7n \\ p_2(n) &= 11n-13 \\ p_3(n) &= 38-13n \\ p_4(n) &= 12\,(n-4) \\ p_5(n) &= -4\,(n-5)\,. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= \frac{1}{2x}, \\ R_1 &= -\frac{1}{2x}, \\ D_1 &= \frac{4x^4-8x^3+5x^2-6x+1}{(x-1)^2}. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 2x^4-6x^3+6x^2-x+1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 5$. The source offset is 0.

4.4.3.25 A176605: order 4, Feb 29 2016

Primary source: https://oeis.org/A176605. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= -2\,(3n-1) \\ p_2(n) &= 9n-13 \\ p_3(n) &= -8\,(n-3) \\ p_4(n) &= 4\,(n-4)\,. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= \frac{1}{2x}, \\ R_1 &= -\frac{1}{2x}, \\ D_1 &= \frac{4x^3-4x^2+5x-1}{x-1}. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = -2x^3+4x^2-2x+1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 4$. The source offset is 0.

4.4.3.26 A176607: order 4, Feb 29 2016

Primary source: https://oeis.org/A176607. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= -2\,(3n-1) \\ p_2(n) &= 11-3n \\ p_3(n) &= 16\,(n-3) \\ p_4(n) &= -8\,(n-4)\,. \end{aligned}$$

The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned} R_0 &= \frac{1}{2x}, \\ R_1 &= -\frac{1}{2x}, \\ D_1 &= -\frac{8x^3-8x^2-5x+1}{x-1}. \end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = 4x^3-8x^2+4x+1.$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 4$. The source offset is 0.

4.4.3.27 A045743: order 4, Jul 26 2022

Primary source: https://oeis.org/A045743. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$
\begin{aligned}
p_0(n) &= 5n\,(n-2)\,(n-1)\,(1353818n-3651663)\\
p_1(n) &= 16\,(n-2)\,(n-1)\left(2236406n^2-6864025n+2014500\right)\\
p_2(n) &= -16\,(n-2)\left(46338048n^3-326486432n^2+734160494n-506411475\right)\\
p_3(n) &= 8\,(2n-7)\left(26592864n^3-436441232n^2+2071128458n-3030735075\right)\\
p_4(n) &= 128\,(2n-9)\,(4n-19)\,(4n-17)\,(94246n-334675)\,.
\end{aligned}
$$

Use $x=\xi(t)=-\frac{(t-1)\left(t^2-3t+1\right)}{t^4}$, $t(0)=1$. The exact generating-function representation is $F=\sum_i R_i\sqrt{D_i}$, with

$$
\begin{aligned}
R_0 &= -\frac{(t-1)^2\,(t^2-3t+1)}{t^4},\\
R_1 &= 0,\\
D_1 &= 1.
\end{aligned}
$$

For these data the complete rational certificate (4.2) gives

$$
P(x) = -6x^3\left(8739520x^2-37127156x-14413953\right).
$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n\geq 6$. The source offset is 2.

4.4.3.28 A308726: order 5, Jan 27 2020

Primary source: https://oeis.org/A308726. The stack-sorting interpretation and established generating function are prior input, with the entry citing Mansour, Skogman and Smith (2017–2019). The selected target is the 2020 polynomial recurrence.

$$
\begin{aligned}
p_0(n) &= 3n\,(n-1)\,(n+1)\\
p_1(n) &= -n\,(n-1)\,(67n-101)\\
p_2(n) &= 2\,(n-1)\left(286n^2-1112n+1089\right)\\
p_3(n) &= -4\left(580n^3-4200n^2+10106n-8049\right)\\
p_4(n) &= 24\left(184n^3-1784n^2+5770n-6221\right)\\
p_5(n) &= -96\,(2n-9)\,(4n-17)\,(4n-15)\,.
\end{aligned}
$$

Use $x=\xi(t)=-\frac{(t-1)(t+1)}{4}$, $t(0)=1$. The exact generating-function representation is $F=\sum_i R_i\sqrt{D_i}$, with

$$
\begin{aligned}
R_0 &= \frac{t^2-4t+1}{t\,(t-1)\,(t+1)},\\
R_1 &= \frac{2}{(t-1)\,(t+1)},\\
D_1 &= 2t-1.
\end{aligned}
$$

For these data the complete rational certificate (4.2) gives

$$P(x) = -12x^2\left(8x^2 - 2x + 1\right).$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 5$. The source offset is 0.

4.4.3.29 A285195: order 6, Jan 25 2020

Primary source: https://oeis.org/A285195. The established generating function and the He–Shapiro Riordan-array reference are prior input. Only the displayed conjectural recurrence is selected.

$$\begin{aligned}
p_0(n) &= 126n\,(n-1)\,(n+1)\\
p_1(n) &= -3n\,(n-1)\,(691n - 1037)\\
p_2(n) &= (n-1)\left(12023n^2 - 48049n + 47736\right)\\
p_3(n) &= -2\left(12889n^3 - 93970n^2 + 221427n - 169506\right)\\
p_4(n) &= -4\left(248n^3 - 12718n^2 + 88433n - 160113\right)\\
p_5(n) &= 16\left(2918n^3 - 41574n^2 + 198532n - 317121\right)\\
p_6(n) &= 240\,(2n-9)\,(4n-21)\,(4n-19).
\end{aligned}$$

Use $x = \xi(t) = -\frac{(t-1)(t+1)}{4}$, $t(0) = 1$. The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$\begin{aligned}
R_0 &= -\frac{6}{(t-5)\,(t+1)},\\
R_1 &= -\frac{2}{(t-5)\,(t+1)},\\
D_1 &= 2t - 1.
\end{aligned}$$

For these data the complete rational certificate (4.2) gives

$$P(x) = -90x^2\left(4x^3 + 80x^2 - 32x + 3\right).$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 6$. The source offset is 0.

4.4.3.30 A243642: order 6, Jan 25 2020

Primary source: https://oeis.org/A243642. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted. **Boundary correction.** The source gives no starting index. At its first natural admissible index $n = 6$, the displayed recurrence has residual 24, not zero. Thus its interpretation as holding at all admissible indices is refuted. The corrected tail assertion is proved for every $n \geq 7$. We do not attribute an explicit $n = 6$ claim to the source, and the counterexample and corrected theorem count together once.

$$\begin{aligned}
p_0(n) &= 16n\,(n-2)\,(n-1)\\
p_1(n) &= -4\,(n-2)\,(n-1)\,(43n - 54)\\
p_2(n) &= 2\,(n-2)\left(52n^2 + 695n - 2439\right)
\end{aligned}$$

$$
\begin{aligned}
p_3(n) &= 3479n^3 - 42714n^2 + 169789n - 219234 \\
p_4(n) &= -2\left(4633n^3 - 64020n^2 + 293825n - 447594\right) \\
p_5(n) &= 4\,(2n-11)\left(296n^2 - 3643n + 11187\right) \\
p_6(n) &= 24\,(n-7)\,(2n-13)\,(2n-11)\,.
\end{aligned}
$$

Use $x = \xi(t) = -\frac{(t-1)(t+1)}{4}$, $t(0) = 1$. The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$
\begin{aligned}
R_0 &= \frac{(t-3)\,(t+1)}{8}, \\
R_1 &= -\frac{t+1}{8}, \\
D_1 &= t^2 + 10t - 7.
\end{aligned}
$$

The chosen square root of D_1 has value 2 at $t = 1$. For these data the complete rational certificate (4.2) gives

$$
P(x) = 12x^3\left(2x^3 + 44x^2 + 31x - 2\right).
$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 7$. The source offset is 0.

4.4.3.31 A243816: order 6, Jan 25 2020

Primary source: https://oeis.org/A243816. The defining formula or generating function is established input from the cited primary entry. Only the selected, explicitly conjectural identity is counted.

$$
\begin{aligned}
p_0(n) &= 2n\,(n-1)\,(2n-3) \\
p_1(n) &= -\,(n-1)\left(47n^2 - 221n + 260\right) \\
p_2(n) &= 4\left(38n^3 - 300n^2 + 777n - 660\right) \\
p_3(n) &= -4\left(94n^3 - 1006n^2 + 3617n - 4355\right) \\
p_4(n) &= 16\,(n-5)\left(64n^2 - 510n + 1041\right) \\
p_5(n) &= -16\,(n-6)\,(n-5)\,(47n - 221) \\
p_6(n) &= 1920\,(n-7)\,(n-6)\,(n-5)\,.
\end{aligned}
$$

Use $x = \xi(t) = -\frac{t}{(t-1)(t+1)}$, $t(0) = 0$. The exact generating-function representation is $F = \sum_i R_i\sqrt{D_i}$, with

$$
\begin{aligned}
R_0 &= -\frac{2t-1}{2\,(t-1)\,(t+1)}, \\
R_1 &= \frac{1}{2\,(t-1)\,(t+1)}, \\
D_1 &= 1 - 4t.
\end{aligned}
$$

For these data the complete rational certificate (4.2) gives

$$
P(x) = -4x^2\left(8x^2 + 8x + 1\right).
$$

Every nonconstant radical component is identically zero. By (4.1), the recurrence therefore holds for every $n \geq 6$. The source offset is 0.

4.4.4 A000986: the exponential generating function

Primary source: https://oeis.org/A000986. The selected conjecture is Mathar's order-five relation dated 4 August 2013. Kotesovec's different order-four recurrence of 13 February 2014 is already known and is explicitly credited as such. The established EGF is

$$E(x) = (1-x)^{-1/2} \exp\left(-x - \frac{x^2}{4} + \frac{x}{2(1-x)}\right).$$

Its logarithmic derivative is

$$L(x) = \frac{E'}{E} = \frac{1}{2(1-x)} - 1 - \frac{x}{2} + \frac{1}{2(1-x)^2}.$$

The conjectured recurrence uses

$$\begin{aligned} p_0(n) &= 2 \\ p_1(n) &= -2\,(2n-1) \\ p_2(n) &= 2\left(n^2 - 2n - 1\right) \\ p_3(n) &= -2\,(n-4)\,(n-2) \\ p_4(n) &= (n-3)\,(n-2)\,(n-1) \\ p_5(n) &= -\,(n-4)\,(n-3)\,(n-2)\,. \end{aligned}$$

Put $b_n = a_n/n!$ and $r = 5$. For $n \geq r$, multiplication of the recurrence by $1/(n-r)!$ yields

$$\sum_{j=0}^{r} q_j(n) b_{n-j} = 0, \qquad q_j(n) = p_j(n)(n-j)_{r-j},$$

where $(z)_h = z(z-1)\cdots(z-h+1)$. Let $U_0 = 1$, $U_{m+1} = x(U_m' + LU_m)$, and $c_m(x) = \sum_j x^j [n^m] q_j(n+j)$. Then $\theta^m E = U_m E$ by induction. Exact rational simplification gives

$$\sum_m c_m(x) U_m(x) = 0.$$

All its inputs have been specified explicitly above, and the verifier reconstructs every q_j, c_m, U_m and clears denominators to obtain the zero polynomial. Thus $\mathcal{L}_q E = 0$, proving the selected recurrence for every $n \geq 5$. The factorial conversion is essential: no OGF operator is incorrectly applied directly to the integer sequence.

4.4.5 A121754: a harmonic-number certificate

Primary source: https://oeis.org/A121754; selected conjecture dated 26 July 2022. The established recurrence immediately preceding it in the entry is

$$a_n - (2n-3)a_{n-1} + (n-1)(n-3)a_{n-2} = (n-2)!\left(n - 2 + \frac{n-1}{2}(1 + (-1)^{n-1})\right),$$
$$a_1 = 0, \qquad a_2 = 1.$$

It gives $a_3 = 6$. Define

$$R(n) = \frac{8n^3 - 2n^2 - 14n + 7}{16}, \quad V(n) = \frac{2n-1}{16}, \quad W(n) = -\frac{n(n-1)}{2}.$$

Then the exact closed form is

$$a_n = (n-2)!\{R(n) + V(n)(-1)^{n-1} + W(n)H_{n-1}\} \qquad (n \geq 2). \tag{4.3}$$

One derivation sets $b_n = a_n/(n-2)!$ for $n \geq 2$, defines the auxiliary value $b_1 = 0$, and puts $d_n = b_n - b_{n-1}$ for $n \geq 2$. The given recurrence becomes, for $n \geq 4$,

$$(n-2)d_n - (n-1)d_{n-1} = (n-2)\left\{n - 2 + \frac{n-1}{2}(1 + (-1)^{n-1})\right\}.$$

Consequently $e_n = d_n/(n-1)$ satisfies

$$e_n - e_{n-1} = \frac{3}{2} - \frac{1}{n-1} + \frac{(-1)^{n-1}}{2}, \qquad e_2 = 1.$$

The same difference equation at $n = 3$ follows directly from $b_2 = 1$ and $b_3 = 6$, so summation may start with e_2. Summation gives $e_n = (6n - 3 + (-1)^{n-1})/4 - H_{n-1}$. A second summation uses

$$\sum_{j=1}^{N} jH_j = \frac{N(N+1)}{2}H_N - \frac{N(N-1)}{4}, \quad \sum_{j=1}^{N} j(-1)^j = \frac{(2N+1)(-1)^N - 1}{4},$$

and yields (4.3). Equivalently, substitute that formula into the known recurrence, divide by $(n-4)!$ for $n \geq 4$, and use $H_{n-2} = H_{n-1} - 1/(n-1)$: the rational residual is identically zero. The explicit values $a_2 = 1, a_3 = 6$ cover the indices omitted by that division, so induction proves the formula for every $n \geq 2$.

The conjectured homogeneous recurrence has the following coefficients.

$$\begin{aligned}
p_0(n) &= 16\,(n+1)\\
p_1(n) &= -16n^2 - 178n + 531\\
p_2(n) &= -16n^3 + 178n^2 - 393n - 510\\
p_3(n) &= 16n^4 + 98n^3 - 1439n^2 + 4222n - 3623\\
p_4(n) &= -146n^4 + 1479n^3 - 4483n^2 + 3054n + 2841\\
p_5(n) &= (n-6)\,(n-4)^2\,(130n - 311)\,.
\end{aligned}$$

For independent symbols H, ε, put

$$B_j = R(n-j) + (-1)^j V(n-j)\varepsilon + W(n-j)\left(H - \sum_{h=1}^{j}\frac{1}{n-h}\right).$$

The complete rational identity certificate is

$$\sum_{j=0}^{5} p_j(n)\left(\prod_{h=1}^{5-j}(n-7+h)\right)B_j = 0.$$

Its coefficients of $1, H, \varepsilon$ all vanish identically in $\mathbb{Q}(n)$; this is directly checked by clearing denominators. Set $H = H_{n-1}$, $\varepsilon = (-1)^{n-1}$ and multiply by $(n-7)!$. The result is the desired recurrence for every $n \geq 7$. At $n = 6$, the required values $(a_1, \ldots, a_6) = (0, 1, 6, 31, 211, 1530)$ give residual zero by direct substitution. Hence the selected recurrence holds for every $n \geq 6$.

4.4.6 A004395: three polynomial certificates

Primary source: https://oeis.org/A004395; selected conjecture dated 14 September 2025. The definition by successive ratios is

$$a_0 = 1, \qquad a_n = \left\lfloor \frac{2n+1}{3} \right\rfloor a_{n-1}.$$

The conjectured recurrence has coefficients

$$\begin{aligned} p_0(n) &= 243 \\ p_1(n) &= -216 \\ p_2(n) &= -54\,(2n-3) \\ p_3(n) &= -6\left(12n^3 - 60n^2 + 87n - 28\right) \\ p_4(n) &= 4\,(n-3)\,(2n-7)\,(2n-5)\,. \end{aligned}$$

For $n = 3k+s$, $s \in \{0,1,2\}$, divide the recurrence residual by the positive integer a_{n-4}. The result is

$$\sum_{j=0}^{4} p_j(3k+s) \prod_{h=j}^{3} \left(2k + \left\lfloor \frac{2(s-h)+1}{3} \right\rfloor\right).$$

For each of the three stated values of s, this is a polynomial in k and its complete expansion is zero. These three finite identities are exactly the retained certificates; they prove the recurrence for all $n \geq 4$. The proof does not assume convergence of the source's formal hypergeometric generating function.

4.4.7 A052731: the factorial-weighted A002026 identity

Primary sources: https://oeis.org/A052731 and https://oeis.org/A002026. The selected conjecture, dated 16 October 2013, is $a_n = n!\,\mathrm{A002026}(n-2)$. Let $D = 1 - 2x - 3x^2$ and let

$$M = \frac{1 - x - \sqrt{D}}{2x^2}.$$

The quadratic identity $M = 1 + xM + x^2M^2$ follows either by squaring or by substitution. The established OGF of A002026 is $B = xM^2$; the established EGF of A052731 is

$$E = \frac{1 - x - \sqrt{D}}{2x} - \frac{1 + x - \sqrt{D}}{2}.$$

Since $x^2M = (1 - x - \sqrt{D})/2$, we have

$$E = xM - x - x^2M = x(M - 1 - xM) = x^3M^2 = x^2B.$$

Equating coefficients proves $a_n = n!\,\mathrm{A002026}(n-2)$ for every $n \geq 2$. The source has offset zero, with $a_0 = a_1 = a_2 = 0$. The supporting A002026 entry is not counted separately. This is an exact formal-series identity.

4.4.8 Reproducibility and prior-work scope

The files `r4_gf_certificates.json`, `r4_gf_engine.py` and `r4_gf_verify.py` retain the complete exact inputs and the finite rational checks. The verifier uses only rational arithmetic, polynomial identities and elementary formal-series coefficient arithmetic. The compact primary-source records retain the selected mathematical statements, attributions, offsets, numeric prefixes and source URLs; raw webpage text is not required.

The Ekhad–Yang 2017 output proving 33 Mathar recurrence conjectures was checked by A-number; none of these 35 retained targets occurs in that output. A185010, which does occur there, was excluded. A032123 and A045406 were excluded after their 2026 proof references were identified, and A162477 was excluded because its entry explicitly credits a prior proof. Established generating functions, shorter recurrences and linked literature are credited above. This is a targeted source and prior-work audit, not an exhaustive worldwide priority search. No OEIS entry was edited or externally declared closed.

4.5 Forty-two empirical array recurrences

The claims in this section are constant-coefficient recurrences marked *Empirical* in their OEIS entries, consulted on 20 October 2026. They concern different widths, color requirements, or local constraints. All 42 A-numbers are absent from the 308 selected entries in rounds 1–3. They are 42 entry-level assertions proved by one general transfer method; they are not presented as 42 unrelated methods. The cited entries link numerical data tables and record no proof of the selected recurrences. This source check is not an exhaustive claim of priority over all literature.

4.5.1 The arrays and their exact finite state spaces

For integers $q \geq 1$, $w \geq 2$ and a set $S \subseteq \{1,2,3,4\}$, consider arrays with $n+1$ rows and w columns. Every adjacent 2×2 block must contain a number of distinct colors belonging to S. Colors are renamed $0, 1, \ldots$ in their order of first appearance when the array is read row by row. The superscript $=$ requires all q colors to occur; $\leq$ permits any positive number at most q. These are the precise conventions of the entries below.

For a row r, write $\mathrm{can}(r)$ for its renaming in order of first appearance. Let $\mathcal{R}_{j,w}$ contain all such canonical rows with at most j colors. For $r \in \mathcal{R}_{j,w}$ let $k(r)$ be its number of colors and put $m_j(r) = (j)_{k(r)}$. Define the integer matrix

$$T_{j,w,S}(r,s) = \#\left\{ v \in \{0,\ldots,j-1\}^w : \begin{array}{l} \mathrm{can}(v) = s, \\ \#\{r_i, r_{i+1}, v_i, v_{i+1}\} \in S \quad (1 \leq i < w) \end{array} \right\}. \tag{4.4}$$

All possible successor rows are included, including rows using colors absent from the preceding row. Simultaneous color permutations preserve the local condition, so the number of legal completions of a row depends only on its canonical form. Consequently the labeled count is

$$L_j(n) = m_j^{\mathsf{T}} T_{j,w,S}^{\,n} \mathbf{1}. \tag{4.5}$$

The factor $m_j(r)$ counts the actual labelings of the first row, and each subsequent multiplication adds one row. This proves (4.5) by induction on the number of rows.

An array that uses exactly q colors has exactly $q!$ labelings on a fixed q-element alphabet. Inclusion–exclusion for omitted colors gives

$$U_q^{=}(n) = \sum_{j=1}^{q} \frac{(-1)^{q-j}}{j!(q-j)!} L_j(n), \tag{4.6}$$

$$U_q^{\leq}(n) = \sum_{j=1}^{q} \left(\frac{1}{j!} \sum_{h=0}^{q-j} \frac{(-1)^h}{h!} \right) L_j(n). \tag{4.7}$$

The second identity follows by summing the first over the actual number of colors. Thus there is no assumption that every allowed color occurs unless the superscript is $=$. Empty-alphabet counts vanish because every array has at least one cell.

4.5.2 An all-index certificate theorem

For a displayed recurrence coefficient list $c = (c_1, \ldots, c_d)$ set

$$R(z) = z^d - \sum_{i=1}^{d} c_i z^{d-i}, \qquad Q(x) = 1 - \sum_{i=1}^{d} c_i x^i.$$

For every component with nonzero coefficient in (4.6) or (4.7), the retained exact integer certificate establishes

$$\boxed{TR(T)\mathbf{1} = \mathbf{0}.} \tag{4.8}$$

For $n \geq d+1$, multiply this identity on the left by $m_j^{\mathsf{T}} T^{n-d-1}$, and then take the linear combination in (4.6) or (4.7). This proves

$$\boxed{a(n) = \sum_{i=1}^{d} c_i a(n-i) \quad (n \geq d+1).} \tag{4.9}$$

Every term on the right is defined, since all selected OEIS entries start at $n = 1$. This is a universal proof from a finite matrix identity; it does not infer a recurrence from agreement with a finite sequence prefix.

The complete certificate contains 90 distinct matrices, each with at most 122 states; the largest sum of component dimensions is 155. The matrices are in `r4_arrays_matrices.json`. The verifier rebuilds every state and edge directly from (4.4), compares the result with the stored matrix, and checks every coordinate in (4.8) by integer arithmetic. No recurrence-fitting step or numerical approximation is used. The following pseudocode specifies the finite identity check completely:

```
v[0] = [1 for each canonical row]
for k = 1,...,d+1:
    for each canonical row r:
        v[k][r] = sum(T[r,s]*v[k-1][s] for all s)
for each canonical row r:
    assert v[d+1][r] == sum(c[i]*v[d+1-i][r] for i=1,...,d)
```

The coefficient signatures below, the finite definition (4.4), and this algorithm specify every certificate without relying on empirical data. Stored matrices make the arithmetic easy to inspect and reproduce. The source prefixes are compared only as a separate check of indexing and interpretation; all 42 computed prefixes are different, so there are no aliases within this section.

For $F(x) = \sum_{n \geq 1} a(n) x^n$, the proved recurrence gives $Q(x)F(x) = P(x)$, where $\deg P \leq d$ and

$$[x^k]P(x) = \sum_{\substack{0 \leq i \leq \min(k,d) \\ k-i \geq 1}} q_i a(k-i), \quad q_0 = 1, \quad q_i = -c_i \ (i \geq 1).$$

The values used here are calculated from the defining transfer matrices. All 16 generating functions explicitly given as empirical or conjectural on the retained source pages agree with this rational function exactly. Their numerator and denominator coefficient lists are included in `r4_arrays_certificates.json`. They add no extra entry count. Other assertions on an entry, such as a separately printed closed form, are outside the selected claim unless explicitly addressed here.

4.5.3 A fully displayed small example: A210100

For A210100, $(q, w, S) = (3, 2, \{1, 3\})$ and at most three colors are allowed. The only nonzero weights in (4.7) are $1/2$ for $j = 1$ and $1/6$ for $j = 3$. For $j = 1$, $T = (1)$ and $m = (1)$. For $j = 3$ the canonical rows are $00, 01$, and direct enumeration gives

$$T = \begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix}, \qquad m = \begin{pmatrix} 3 \\ 6 \end{pmatrix}.$$

For example, after 00 there is one constant successor (the same color) and two nonconstant successors using the other two colors; after 01 there is one constant successor using the third color and four nonconstant successors containing that color and one of $0, 1$. The characteristic polynomial of the latter matrix is $z^2 - 5z + 2$. Hence both components are annihilated by

$$(z-1)(z^2 - 5z + 2) = z^3 - 6z^2 + 7z - 2.$$

Therefore $a(n) = 6a(n-1) - 7a(n-2) + 2a(n-3)$ for $n \geq 4$. The transfer formula gives $a(1), a(2), a(3) = 7, 30, 135$, yielding

$$F(x) = \frac{x(7 - 12x + 4x^2)}{(1-x)(1-5x+2x^2)}.$$

This proves both the empirical recurrence and the conjectural generating function recorded at https://oeis.org/A210100. It illustrates the same exact certificate mechanism used for the larger state spaces.

4.5.4 Individual recurrence signatures and parameters

For each entry below, D is the total dimension of the components with nonzero weight. The list gives $c_1, \ldots, c_d$ in order, including zero coefficients. Each signature satisfies (4.8) in every component and therefore has the all-index range stated in (4.9).

4.5.4.1 A210005

The arrays have $n+1$ rows, width $w = 2$, exactly 3 colors up to order of first appearance, and $S = \{2, 3\}$. The recurrence (4.9) holds for $n \geq 5$ with order $d = 4$ and component dimension $D = 5$. Source: https://oeis.org/A210005.

$$(c_1, \ldots, c_d) = (11, -16, -34, -12).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.2 A210006

The arrays have $n+1$ rows, width $w = 3$, exactly 3 colors up to order of first appearance, and $S = \{2, 3\}$. The recurrence (4.9) holds for $n \geq 7$ with order $d = 6$ and component dimension $D = 10$. Source: https://oeis.org/A210006.

$$(c_1, \ldots, c_d) = (30, -109, -290, 96, 136, -48).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.3 A210007

The arrays have $n+1$ rows, width $w = 4$, exactly 3 colors up to order of first appearance, and $S = \{2, 3\}$. The recurrence (4.9) holds for $n \geq 11$ with order $d = 10$ and component dimension $D = 23$. Source: https://oeis.org/A210007.

```
75, -30, -7001, -18089, 52126, 91400, -118464, -92992, 85632, -13824
```

4.5.4.4 A210008

The arrays have $n+1$ rows, width $w = 5$, exactly 3 colors up to order of first appearance, and $S = \{2, 3\}$. The recurrence (4.9) holds for $n \geq 20$ with order $d = 19$ and component dimension $D = 58$. Source: https://oeis.org/A210008.

```
218, -672, -102513, 336965, 9441327, -64650448, 29276745, 733248704, -1733314135,
-814273096, 6961302814, -7307156292, -92994368, 3740262880, -1180127616, -466935296,
173254656, 18587648, -5406720
```

4.5.4.5 A210054

The arrays have $n+1$ rows, width $w=2$, at most 4 colors up to order of first appearance, and $S=\{1,2,4\}$. The recurrence (4.9) holds for $n \geq 5$ with order $d=4$ and component dimension $D=5$. Source: https://oeis.org/A210054.

$$(c_1,\ldots,c_d) = (13,-48,52,-16).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.6 A210055

The arrays have $n+1$ rows, width $w=3$, at most 4 colors up to order of first appearance, and $S=\{1,2,4\}$. The recurrence (4.9) holds for $n \geq 7$ with order $d=6$ and component dimension $D=10$. Source: https://oeis.org/A210055.

$$(c_1,\ldots,c_d) = (29,-266,902,-968,176,128).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.7 A210056

The arrays have $n+1$ rows, width $w=4$, at most 4 colors up to order of first appearance, and $S=\{1,2,4\}$. The recurrence (4.9) holds for $n \geq 12$ with order $d=11$ and component dimension $D=24$. Source: https://oeis.org/A210056.

```
66, -1511, 14904, -58950, 21360, 330508, -503712, -93160, 369120, -9504, -69120
```

4.5.4.8 A210057

The arrays have $n+1$ rows, width $w=5$, at most 4 colors up to order of first appearance, and $S=\{1,2,4\}$. The recurrence (4.9) holds for $n \geq 29$ with order $d=28$ and component dimension $D=68$. Source: https://oeis.org/A210057.

```
176, -12383, 457521, -9583668, 109074306, -432638439, -4419650349, 64230480793,
-252540170918, -595255299234, 8433913817808, -21427208559976, -38069222537892,
289460472761408, -336643109299632, -883702002797616, 2460214504431920, -361513807179872,
-4377499678974656, 3253938371960768, 2790969020407040, -3141539845016064,
-669537234834176, 1122282161785600, 46793169024000, -143518636118016, -1001036759040,
3148485230592
```

4.5.4.9 A210079

The arrays have $n+1$ rows, width $w=2$, exactly 4 colors up to order of first appearance, and $S=\{1,3,4\}$. The recurrence (4.9) holds for $n \geq 6$ with order $d=5$ and component dimension $D=7$. Source: https://oeis.org/A210079.

$$(c_1,\ldots,c_d) = (17,-71,67,-8,-4).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.10 A210080

The arrays have $n+1$ rows, width $w=3$, exactly 4 colors up to order of first appearance, and $S=\{1,3,4\}$. The recurrence (4.9) holds for $n \geq 8$ with order $d=7$ and component dimension $D=15$. Source: https://oeis.org/A210080.

$$(c_1,\ldots,c_d) = (42,-322,274,1887,-3124,76,1168).$$

4.5.4.11 A210081

The arrays have $n+1$ rows, width $w=4$, exactly 4 colors up to order of first appearance, and $S=\{1,3,4\}$. The recurrence (4.9) holds for $n \geq 20$ with order $d=19$ and component dimension $D=38$. Source: https://oeis.org/A210081.

```
111, -1551, -11620, 269796, -346276, -10657824, 37514734, 105399645, -595931403,
25518343, 3128667030, -3607436824, -4027147296, 9243102400, -2810113344, -3425981120,
1936238208, 61521408, -60604416
```

4.5.4.12 A210092

The arrays have $n+1$ rows, width $w=6$, exactly 3 colors up to order of first appearance, and $S=\{1,2\}$. The recurrence (4.9) holds for $n \geq 53$ with order $d=52$ and component dimension $D=155$. Source: https://oeis.org/A210092.

```
202, -14123, 380572, -1438616, -99274709, 1155724795, 8643300216, -166589849398,
-274497031426, 11515136131550, -1832292455645, -457188662253503, 336001687563753,
11327503957451248, -9113460943291491, -183665203305943843, 123571425473428963,
2003525661152721986, -990601469663807369, -14981189128638351022, 5154521066328382780,
78057653461845959313, -19250224843040180931, -287532733033011400278,
584172177021953987815, 757126134053256036905, -155693459113620706806,
-1433722830312214278540, 348602065891707730366, 1950547820706182800257,
-595414185159687071974, -188730074530425227 5773, 725686784875169133118,
1269686648151447607271, -608745622735310244088, -568946968704783843248,
341945871757434843792, 155435605971618737334, -124110444593762302930,
-19948113914003483822, 27435918496367785268, -714926210446370056, -3322485605775126312,
498971087766745072, 180395158795278272, -44929945871997536, -3093399295952128,
1451280469739136, -24128771207680, -15867004438528, 437469315072, 60523806720
```

4.5.4.13 A210100

The arrays have $n+1$ rows, width $w=2$, at most 3 colors up to order of first appearance, and $S=\{1,3\}$. The recurrence (4.9) holds for $n \geq 4$ with order $d=3$ and component dimension $D=3$. Source: https://oeis.org/A210100.

$$(c_1,\ldots,c_d)=(6,-7,2).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.14 A210103

The arrays have $n+1$ rows, width $w=5$, at most 3 colors up to order of first appearance, and $S=\{1,3\}$. The recurrence (4.9) holds for $n \geq 23$ with order $d=22$ and component dimension $D=42$. Source: https://oeis.org/A210103.

```
39, -424, -100, 25342, -98910, -353224, 2683456, -679949, -24899773, 39419608, 84219756,
-230024712, -56603460, 492890768, -143861456, -452015392, 208017984, 175455296,
-65216128, -32317440, 2433024, 925696
```

4.5.4.15 A210118

The arrays have $n+1$ rows, width $w=2$, at most 4 colors up to order of first appearance, and $S=\{2,4\}$. The recurrence (4.9) holds for $n \geq 4$ with order $d=3$ and component dimension $D=5$. Source: https://oeis.org/A210118.

$$(c_1,\ldots,c_d)=(10,-19,-14).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.16 A210119

The arrays have $n+1$ rows, width $w=3$, at most 4 colors up to order of first appearance, and $S=\{2,4\}$. The recurrence (4.9) holds for $n \geq 8$ with order $d=7$ and component dimension $D=10$. Source: https://oeis.org/A210119.

$$(c_1,\ldots,c_d) = (24,-174,368,303,-888,-340,240).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.17 A210129

The arrays have $n+1$ rows, width $w=4$, at most 4 colors up to order of first appearance, and $S=\{1,3,4\}$. The recurrence (4.9) holds for $n \geq 12$ with order $d=11$ and component dimension $D=24$. Source: https://oeis.org/A210129.

```
95, -1, -14266, 20367, 435937, -1093869, -1923590, 8479272, -8472184, 1936944, 631296
```

4.5.4.18 A210130

The arrays have $n+1$ rows, width $w=5$, at most 4 colors up to order of first appearance, and $S=\{1,3,4\}$. The recurrence (4.9) holds for $n \geq 33$ with order $d=32$ and component dimension $D=68$. Source: https://oeis.org/A210130.

```
319, -12448, -307770, 19003857, -47940781, -8062797435, 95630724963, 949633978540,
-23587913597875, 72618564306781, 1459146419334837, -13658086185029599,
17603864407218421, 283502331705919989, -1303129591973246857, -279014374825669271,
13815024039103815024, -22992880145677918265, -44182446460842415541,
152128697073903743642, -145056068503134502 36, -344933214811274872678,
261141694209971459720, 296779187097877685464, -389773691540956876144,
-446328883124471550 24, 204261362669091285120, -466391791007793221 76,
-307537102358789990 40, 113576547054029414 40, 637302851625194496, -414118546177001472
```

4.5.4.19 A210136

The arrays have $n+1$ rows, width $w=2$, exactly 4 colors up to order of first appearance, and $S=\{1,2,4\}$. The recurrence (4.9) holds for $n \geq 7$ with order $d=6$ and component dimension $D=7$. Source: https://oeis.org/A210136.

$$(c_1,\ldots,c_d) = (18,-111,266,-180,-24,32).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.20 A210137

The arrays have $n+1$ rows, width $w=3$, exactly 4 colors up to order of first appearance, and $S=\{1,2,4\}$. The recurrence (4.9) holds for $n \geq 11$ with order $d=10$ and component dimension $D=15$. Source: https://oeis.org/A210137.

```
40, -590, 3962, -11899, 12350, 3806, -12980, 4512, 1056, -256
```

4.5.4.21 A210138

The arrays have $n+1$ rows, width $w=4$, exactly 4 colors up to order of first appearance, and $S=\{1,2,4\}$. The recurrence (4.9) holds for $n \geq 20$ with order $d=19$ and component dimension $D=38$. Source: https://oeis.org/A210138.

```
89, -3068, 51966, -443001, 1546826, 1718332, -24576952, 30663994, 107350909, -226371596,
-140409398, 501187032, -47667796, -343494224, 74492360, 92532416, -19372128, -8588160,
1382400
```

4.5.4.22 A210139

The arrays have $n+1$ rows, width $w=5$, exactly 4 colors up to order of first appearance, and $S=\{1,2,4\}$. The recurrence (4.9) holds for $n \geq 49$ with order $d=48$ and component dimension $D=109$. Source: https://oeis.org/A210139.

```
235, -23547, 1326017, -46312598, 1033233394, -14208694709, 96208712962, 267300062626,
-11424901483364, 86498377558175, -31767158001113, -3617589382708047, 20693266346928210,
10888192487943923, -546652409630123972, 1681515865137829063, 4013358879516874873,
-33983772523000820266, 33111863198202404690, 257960652876911394845,
-727947325958421224552, -568221315698344240576, 480706018502676717071 2,
-313371955477289779 2592, -151729221642158600437 16, 2361397351961380714 0264,
219415716535722754129 52, -664625308165785126626 88, -242306463283566097048 0,
101519954129757123683008, -384825629736996712104 64, -911741779127357926285 44,
594933746796313036052 48, 484859593632688554752 0, -443322907473639315942 40,
-142020079292513484472 32, 189654023496745257108 48, 154065627907635924326 4,
-475311734799900737433 6, 269751803727187415040, 663177631600048187392,
-96002320931723030528, -44409102700361863168, 9151174318820646912, 918429987777970176,
-240590747390312448, -2948910065123328, 1511272910684160
```

4.5.4.23 A210158

The arrays have $n+1$ rows, width $w=2$, at most 4 colors up to order of first appearance, and $S=\{2,3\}$. The recurrence (4.9) holds for $n \geq 5$ with order $d=4$ and component dimension $D=5$. Source: https://oeis.org/A210158.

$$(c_1,\ldots,c_d) = (16,-19,-80,-36).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.24 A210159

The arrays have $n+1$ rows, width $w=3$, at most 4 colors up to order of first appearance, and $S=\{2,3\}$. The recurrence (4.9) holds for $n \geq 8$ with order $d=7$ and component dimension $D=10$. Source: https://oeis.org/A210159.

$$(c_1,\ldots,c_d) = (55,-186,-1230,2723,159,-1604,468).$$

4.5.4.25 A210160

The arrays have $n+1$ rows, width $w=4$, at most 4 colors up to order of first appearance, and $S=\{2,3\}$. The recurrence (4.9) holds for $n \geq 17$ with order $d=16$ and component dimension $D=24$. Source: https://oeis.org/A210160.

```
172, 2086, -61767, -221678, 6709264, -21628195, -28904236, 171575402, 51603, -460002202,
137911776, 512783040, -179902696, -200966160, 54584192, -885504
```

4.5.4.26 A210161

The arrays have $n+1$ rows, width $w=5$, at most 4 colors up to order of first appearance, and $S=\{2,3\}$. The recurrence (4.9) holds for $n \geq 41$ with order $d=40$ and component dimension $D=68$. Source: https://oeis.org/A210161.

```
651, 2491, -3033274, 75213828, 2871779009, -168547329017, 3025868330721,
-14913244152146, -227860409342090, 3284704577622354, -4647468768394855,
-151912308978989829, 746051218249104195, 2759862594209092375, -24875375814765882871,
-12826708391148092367, 412811120432989292070, -271333439668914274397,
-3949195606845988432506, 4878572324314521633479, 22621015632021698182303,
-35706585467435371532954, -76058853813329983983298, 141387729457080507256936,
```

```
13724770815149784865608 9, -3125679646287147139855 54, -9966257328323107784733 1,
3673612729538448576937 72, -2166169625741510563148 9, -2099979039638257290249 70,
5667611177564270790774 6, 5360010889262361277776 4, -2215698675576252410682 6,
-5080148629481596023904, 319819748971199957301 2, 14089783129953149268,
-171981655847047928312, 15964369340316597080, 1991492927343812928, -247274191930612320
```

4.5.4.27 A210167

The arrays have $n+1$ rows, width $w=2$, exactly 4 colors up to order of first appearance, and $S=\{2,4\}$. The recurrence (4.9) holds for $n \geq 6$ with order $d=5$ and component dimension $D=7$. Source: https://oeis.org/A210167.

$$(c_1,\ldots,c_d) = (14,-55,22,132,56).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.28 A210168

The arrays have $n+1$ rows, width $w=3$, exactly 4 colors up to order of first appearance, and $S=\{2,4\}$. The recurrence (4.9) holds for $n \geq 11$ with order $d=10$ and component dimension $D=15$. Source: https://oeis.org/A210168.

```
32, -362, 1648, -1561, -7568, 11440, 11360, -14768, -6400, 3840
```

4.5.4.29 A210169

The arrays have $n+1$ rows, width $w=4$, exactly 4 colors up to order of first appearance, and $S=\{2,4\}$. The recurrence (4.9) holds for $n \geq 25$ with order $d=24$ and component dimension $D=38$. Source: https://oeis.org/A210169.

```
63, -1431, 12729, 3336, -726316, 2597781, 14573977, -80969613, -143169465, 1105985892,
755457488, -8134284268, -2199693420, 33945204080, 3581733232, -80566448928, -3198195264,
104642408704, 1573518080, -66575366656, -375523328, 15376932864, -161906688, -629932032
```

4.5.4.30 A210260

The arrays have $n+1$ rows, width $w=2$, exactly 4 colors up to order of first appearance, and $S=\{3\}$. The recurrence (4.9) holds for $n \geq 5$ with order $d=4$ and component dimension $D=7$. Source: https://oeis.org/A210260.

$$(c_1,\ldots,c_d) = (12,-18,-64,-24).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.31 A210261

The arrays have $n+1$ rows, width $w=3$, exactly 4 colors up to order of first appearance, and $S=\{3\}$. The recurrence (4.9) holds for $n \geq 9$ with order $d=8$ and component dimension $D=15$. Source: https://oeis.org/A210261.

```
32, -225, 0, 2856, -2952, -4100, 3664, -384
```

4.5.4.32 A210262

The arrays have $n+1$ rows, width $w=4$, exactly 4 colors up to order of first appearance, and $S=\{3\}$. The recurrence (4.9) holds for $n \geq 19$ with order $d=18$ and component dimension $D=38$. Source: https://oeis.org/A210262.

```
63, -318, -15234, 103150, 1267432, -7283556, -49152712, 160312712, 877545008,
-931302192, -5537877952, 2212548864, 14304785408, -4034892800, -15800012800, 5312643072,
6186336256, -2686451712
```

4.5.4.33 A210302

The arrays have $n+1$ rows, width $w=2$, exactly 4 colors up to order of first appearance, and $S=\{1,3\}$. The recurrence (4.9) holds for $n \geq 6$ with order $d=5$ and component dimension $D=7$. Source: https://oeis.org/A210302.

$$(c_1,\ldots,c_d) = (15,-57,41,10,-8).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.34 A210303

The arrays have $n+1$ rows, width $w=3$, exactly 4 colors up to order of first appearance, and $S=\{1,3\}$. The recurrence (4.9) holds for $n \geq 8$ with order $d=7$ and component dimension $D=15$. Source: https://oeis.org/A210303.

$$(c_1,\ldots,c_d) = (35,-315,729,1132,-3804,688,1536).$$

4.5.4.35 A210304

The arrays have $n+1$ rows, width $w=4$, exactly 4 colors up to order of first appearance, and $S=\{1,3\}$. The recurrence (4.9) holds for $n \geq 21$ with order $d=20$ and component dimension $D=38$. Source: https://oeis.org/A210304.

```
81, -1553, 2061, 172685, -899169, -6513363, 44110455, 105719944, -862490004, -709417008,
7693929440, 1058374912, -32657532864, 6969586944, 63976482560, -23949775872,
-51839258624, 18903613440, 14501412864, -3227516928
```

4.5.4.36 A210313

The arrays have $n+1$ rows, width $w=2$, exactly 4 colors up to order of first appearance, and $S=\{2,3\}$. The recurrence (4.9) holds for $n \geq 7$ with order $d=6$ and component dimension $D=7$. Source: https://oeis.org/A210313.

$$(c_1,\ldots,c_d) = (24,-141,-24,718,768,216).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.37 A210314

The arrays have $n+1$ rows, width $w=3$, exactly 4 colors up to order of first appearance, and $S=\{2,3\}$. The recurrence (4.9) holds for $n \geq 11$ with order $d=10$ and component dimension $D=15$. Source: https://oeis.org/A210314.

```
79, -1474, 1450, 39515, -30297, -122076, 99228, 43912, -53472, 11232
```

4.5.4.38 A210315

The arrays have $n+1$ rows, width $w=4$, exactly 4 colors up to order of first appearance, and $S=\{2,3\}$. The recurrence (4.9) holds for $n \geq 22$ with order $d=21$ and component dimension $D=38$. Source: https://oeis.org/A210315.

```
237, -8494, -300237, 2482313, 58239270, -296444067, -2842201417, 12050668766,
41834205193, -197110345949, -109615007438, 1042214343696, -265321484840, -2283447354472,
1334391576272, 2102283712768, -1614381867264, -554944840192, 578116512768, -98061852672,
1530150912
```

4.5.4.39 A210316

The arrays have $n+1$ rows, width $w=5$, exactly 4 colors up to order of first appearance, and $S=\{2,3\}$. The recurrence (4.9) holds for $n \geq 51$ with order $d=50$ and component dimension $D=109$. Source: https://oeis.org/A210316.

```
848, -122300, -5801743, 682041320, -1213046081, -1079782780850, 27573638241302,
79725630193847, -11873109205022518, 131395804871513578, 986201712895823243,
-29652374531996895310, 132452849847042036360, 1622629993578919995350,
-18253509106659216838968, 6849069634272644477350, 71458757171051339150 1643,
-27604443296949269321045 49, -10516539662994187322703315, 86912495830872109675164069,
-95635420176926026268291 78, -12483651236283456975756390 65, 22491818794628119876795573 46,
90864354083613914185011204 28, -30736207455851587697978075075,
-25923589316759701283979550795, 20250020141581297553646585 7367,
-72715906863721870790944600743, -72078151478931850264351479 7441,
80178692373174840109005442 2513, 12869318025883072904224434 08052,
-25220610328423479704414736777 32, -67411943864170492443478125 2930,
37336160492017060127142141010 06, -10137704202225182006061933 39716,
-27073806607850611030245981917 72, 15795088135141845541095841 02836,
94268928470817838772086178 2888, -85971897946299753602719956 1344,
-12544692995354611939560997 5056, 22954147423587386798253979 5760,
-80192985777994314589000265 60, -31686621384563077338679511 744,
38534947579165216719747883 52, 22044063062316787899205504 512,
-36767309209353860714536755 2, -67950978187980533942738944, 13061961692650600938110976,
61387882781529487874457 6, -13369423189950802427904 0
```

4.5.4.40 A210397

The arrays have $n+1$ rows, width $w=2$, at most 4 colors up to order of first appearance, and $S=\{1,3\}$. The recurrence (4.9) holds for $n \geq 4$ with order $d=3$ and component dimension $D=5$. Source: https://oeis.org/A210397.

$$(c_1,\ldots,c_d) = (10,-5,-4).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.41 A210398

The arrays have $n+1$ rows, width $w=3$, at most 4 colors up to order of first appearance, and $S=\{1,3\}$. The recurrence (4.9) holds for $n \geq 6$ with order $d=5$ and component dimension $D=10$. Source: https://oeis.org/A210398.

$$(c_1,\ldots,c_d) = (28,-123,-20,500,-384).$$

The source generating function also agrees exactly with the proved rational function P/Q.

4.5.4.42 A210400

The arrays have $n+1$ rows, width $w=5$, at most 4 colors up to order of first appearance, and $S=\{1,3\}$. The recurrence (4.9) holds for $n \geq 33$ with order $d=32$ and component dimension $D=68$. Source: https://oeis.org/A210400.

```
222, -14466, 294924, 4656995, -258980402, 1519478244, 65765309856, -899946571108,
-6066410464024, 162105406266400, -88886529769536, -13947758522827904, 56368494845199616,
595808254992314112, -4168954910728206336, -10590192881040593920, 140772489264317126656,
-381656480867025182 72, -2466112169355411496960, 4370610701672862203904,
2213119315269638995 9680, -6652064103482542437 1712, -8413043192998315714 1504,
4442188918053522510 97088, -2452475326323893049 7536, -1357521916200660102021120,
9063833298423912637 39904, 1679955754896158917394432, -1768503113779591826636800,
-4690352669839256683 80672, 9709106370093691751 62880, -2553566680347570497 12640
```

4.6 Reproduction and verification

The accompanying archive is self-contained. It contains the complete standalone manuscript, the 100-entry ledger, four proof fragments, four main verifier programs, exact certificates, complete array matrices, and the count and prior-work audits. The proofs state their hypotheses, indexing conventions and selected scopes explicitly.

4.6.1 Running the verification

Use Python 3.10 or later with SymPy. The delivered run was checked with Python 3.12 and SymPy 1.14.0. From the extracted directory run:

```
python3 r4_run_all.py
```

This runs the component verifiers and independent array enumeration, checks all 100 unique A-numbers against the preceding 308, and writes a combined JSON summary. All mathematical checks use exact integers or rational expressions. No network connection is needed.

Program	What is checked
`r4_root_verify.py`	Fourteen congruence packages, including exact rational identities and supplementary sequence checks.
`r4_arith_verify.py`	Nine arithmetic packages, exact floor certificates, determinant formulas, tail enclosures and floor sums.
`r4_gf_verify.py`	Thirty-five symbolic certificates, recurrence coefficients, boundary ranges and independently expanded initial terms.
`r4_arrays_verify.py`	Forty-two recurrences, all 90 complete matrices reconstructed from their definitions, and 16 source generating functions.
`r4_gf_review_arrays.py`	A separate direct enumeration of small global color patterns, checking the transfer interpretation.

4.6.2 Why the finite certificates prove infinite statements

For an array recurrence, the defining count is a fixed linear combination of $m^{\mathsf{T}}T^n\mathbf{1}$. The complete identity $TR(T)\mathbf{1} = 0$ remains true after multiplication by every nonnegative power of T, proving the recurrence for all admissible indices. Checking a long initial prefix alone would not give this implication.

For the algebraic generating functions, the recurrence is translated to a differential operator. The verifier computes its exact action in a rational radical basis. All nonconstant basis components vanish and the remaining boundary polynomial is displayed. Its degree gives the precise range in which every recurrence coefficient is zero. This also detects the initial exception in A243642. The three special recurrence proofs and the factorial-shift identity have separate exact derivations in their sections.

For the remaining statements, the written arguments establish the quantified result. The accompanying finite checks guard against transcription and indexing errors and are not substituted for those arguments. For example, the persistent mod-4 obstruction in A126156 follows from an operator identity for every index, whereas the extra computed initial terms are a consistency check.

4.6.3 Rebuilding and integrity

The delivered `oeis_hunt_round4.tex` contains every proof directly and uses no external `\input` files. Compile it twice with pdfLaTeX and the packages listed in its preamble. To reconstruct the TEX and the ledger from the included component files, run:

```
python3 r4_build_dossier.py
```

The archive's README describes optional certificate rebuilds. Its `manifest.json` lists SHA-256 digests. Rerunning verifiers can regenerate reports and timing fields; those report digests may then change.

4.6.4 Source and priority records

Each entry in the catalogue links to its primary OEIS page. The portable records retain mathematical statements, source attribution and consulted URLs; full scraped webpages are omitted. The component audit files record known prior results, excluded candidates and limits of the source review. Three successful array candidates were held out to keep the selected catalogue at exactly 100; they contribute no count or unprovided proof to the final packet.

The count after this round is $107 + 101 + 100 + 100 = 408$ selected entry-level packages under the project's counting convention. This is a project ledger total, not a claim that 408 entries have been edited or formally closed by the OEIS editors.

Part V

Universal congruences, arithmetic, theta values, and priority corrections

Round overview

This fifth-round dossier treats 100 selected assertion packages on 100 distinct OEIS entries, disjoint from the 408 A-numbers previously selected by this project. The outcomes are 90 proofs of the original selected claims, two refutations, and eight corrected statements proved in full. The corrections comprise six recurrence starting-index qualifications, one malformed recurrence factor, and one theta-sum index normalization. Several assertions on one entry count only once.

The four parts contain 13 packages on formal series, periodicity and theta evaluations; 12 on arithmetic, averages and bounds; 30 on generating functions; and 45 on constrained arrays. The array results are proved by complete integer transfer matrices and vector identities. The generating-function recurrences are proved by exact differential identities, with every boundary term retained. The remaining results have written quantified proofs, accompanied by supplementary calculations.

The two refutations are explicit. For A051699, the nearest-prime distance sum at $N = 2^{20}$ is 5542944, exceeding the stated upper bound $\frac{3}{8}N \log N$. For A129366, the proposed recurrence has residual -73968 at $n = 6$ and cannot hold on any eventual tail; a corrected recurrence is proved. The eight correction cases are separated from these genuine counterexamples in the catalogue.

Scope and counting convention

Each catalogue row specifies one assertion package on one A-number. Different members of a parameter family can use a common theorem; the count does not mean 100 unrelated proof methods. The primary statements were labeled conjectural, empirical, or expressed as unproved bounds in the consulted sources. Established generating functions and classical identities are credited as inputs. A selected comment can be settled while other questions on the same entry remain outside scope. For example, this dossier does not settle the asymptotic constant in A051699, the mod-3 or asymptotic assertions in A397242, or every modular statement on A396804.

The six boundary cases have no starting index specified in the source. We give the precise valid tail and exhibit the earlier admissible exceptions; we do not attribute an explicit false starting-index claim to the author. For A279013, an undefined printed token is replaced by a specified factor and the repaired formula is proved. For A243763, both the original-index and zero-based evaluations are proved, and their factor e^{π} is explained.

Retrospective correction: five Round IV entries

The earlier new-resolution assessment is withdrawn for A176605, A176607, A176610, A176611, and A176612. Georg Fischer's general derivation dated 26 January 2020 already implies their selected recurrences. The Round IV proofs remain mathematically valid, but these five claims should not have been presented as newly resolved. The exact parameter substitutions and operator comparisons are documented in Section 5.4.5 and the accompanying audit certificate. None of these five entries contributes to the present 100. The preceding total of 408 is therefore used only as an exclusion ledger of selected A-numbers, not as a certified total of new discoveries.

Evidence and limits

Primary OEIS pages, relevant linked work and targeted literature searches were consulted on 20 October 2026; retrieved pages may be cached. The OEIS Open comparison covers only Appendix A: 100 costliest claims on 97 distinct A-numbers, with no present overlap. It does not cover the full project or constitute an exhaustive priority search.

The results are mathematical claims supported by the written arguments and certificates here. No worldwide-priority or OEIS editorial-acceptance claim is made, and no OEIS edit or submission was performed. The cross-reviews were performed by other agents in the same AI-assisted workflow; they are not external peer review or proof-assistant formalization.

5.1 Catalogue of the 100 selected entries

P: original selected claim proved. **R**: refuted. **B**: boundary corrected. **F**: malformed formula corrected. **I**: index normalization corrected. Every corrected statement is proved. Page references lead to the individual argument or recurrence signature; the JSON ledger retains exact scopes and attribution.

No.	Entry	Selected assertion package	Result	Page
Congruences, periodicity and theta values				
1	A338632	All three continued-fraction congruences, modulo 4 and 3	P	227
2	A338636	All three continued-fraction congruences, modulo 8 and 3	P	228
3	A397242	Odd precisely when the index plus one is a power of two	P	228
4	A396804	The coefficient at index n is congruent to n modulo 4	P	229
5	A383165	Eventual periodicity modulo every positive integer	P	229
6	A383166	Eventual periodicity modulo every positive integer	P	230
7	A179929	An eventual modular period dividing Euler phi	P	230
8	A210063	The empirical exponential-weighted theta sum	P	231
9	A243763	Two theta sums with original and shifted indices distinguished	I	231
10	A107635	The empirical exponential-weighted theta sum	P	231
11	A002512	The empirical exponential-weighted theta sum	P	231
12	A134416	The empirical exponential-weighted theta sum	P	232
13	A030204	The empirical exponential-weighted theta sum	P	232
Arithmetic, averages and exact bounds				
14	A010786	The floor-factorial logarithmic constant and a uniform error bound	P	232
15	A034444	The 2022 Hultman-polynomial Dirichlet-series family	P	233
16	A051699	Exact counterexample to the stated uniform nearest-prime bound	R	233
17	A129455	The cubic gcd triangle is a binomial coefficient times a sixth power	P	234
18	A161789	Existence of the empirical mean and a certified interval	P	235
19	A187011	Prime-distance determinant sign and nth-root divergence	P	235
20	A276646	The decimal-prefix average has sharp upper bound 0.55	P	236
21	A278083	Primitive-quadruple multiplicativity and the three-square limit	P	237
22	A280098	Complete classification of the divisor-sum moduli	P	238
23	A300984	Prime support and exponent restrictions for the divisor-sum condition	P	238
24	A339387	The remainder-sum inequality, strict at nonsquarefree indices	P	239
25	A394454	Every prime divides infinitely many central binomial coefficients	P	240
Generating-function certificates				
26	A025259	Conjectured polynomial recurrence of order 4; exact starting index supplied	B	242
27	A025272	Conjectured polynomial recurrence of order 5	P	243
28	A034267	Conjectured polynomial recurrence of order 2	P	243
29	A089324	Conjectured polynomial recurrence of order 6	P	244
30	A097331	Conjectured polynomial recurrence of order 3	P	244
31	A102319	Conjectured polynomial recurrence of order 6	P	245
32	A104624	Conjectured polynomial recurrence of order 3	P	246
33	A107587	Conjectured polynomial recurrence of order 5	P	246
34	A108600	Conjectured polynomial recurrence of order 6	P	247
35	A116388	Conjectured polynomial recurrence of order 7	P	248
36	A129366	The proposed recurrence fails on every tail; a replacement is proved	R	248
37	A134424	Conjectured polynomial recurrence of order 10	P	250
38	A162482	Conjectured polynomial recurrence of order 6	P	251
39	A164586	Conjectured polynomial recurrence of order 7	P	251
40	A174013	Conjectured polynomial recurrence of order 7	P	252
41	A182879	Conjectured polynomial recurrence of order 5	P	253
42	A182894	Conjectured polynomial recurrence of order 7	P	253

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44	A189176	Conjectured polynomial recurrence of order 2; exact starting index supplied	B	255
45	A190166	Conjectured polynomial recurrence of order 6	P	255
46	A218185	Conjectured polynomial recurrence of order 4; exact starting index supplied	B	256
47	A239201	Conjectured polynomial recurrence of order 3	P	257
48	A243764	Conjectured polynomial recurrence of order 6; exact starting index supplied	B	257
49	A262768	Conjectured polynomial recurrence of order 2; exact starting index supplied	B	258
50	A270490	Conjectured polynomial recurrence of order 4	P	259
51	A270530	Conjectured polynomial recurrence of order 4	P	259
52	A273351	Conjectured polynomial recurrence of order 5	P	260
53	A279013	Repair and proof of the malformed recurrence factor	F	260
54	A279014	Conjectured polynomial recurrence of order 4; exact starting index supplied	B	261
55	A387084	Conjectured polynomial recurrence of order 4	P	262
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56	A209505	Empirical array recurrence of order 2	P	266
57	A209506	Empirical array recurrence of order 3	P	266
58	A209507	Empirical array recurrence of order 5	P	267
59	A209508	Empirical array recurrence of order 7	P	267
60	A209509	Empirical array recurrence of order 13	P	267
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64	A209547	Empirical array recurrence of order 6	P	268
65	A209548	Empirical array recurrence of order 11	P	268
66	A209549	Empirical array recurrence of order 20	P	268
67	A209710	Empirical array recurrence of order 19	P	268
68	A209711	Empirical array recurrence of order 64	P	269
69	A209851	Empirical array recurrence of order 6	P	269
70	A209852	Empirical array recurrence of order 20	P	269
71	A209853	Empirical array recurrence of order 68	P	269
72	A209906	Empirical array recurrence of order 6	P	270
73	A209907	Empirical array recurrence of order 19	P	270
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76	A209955	Empirical array recurrence of order 14	P	271
77	A209956	Empirical array recurrence of order 36	P	271
78	A210070	Empirical array recurrence of order 6	P	271
79	A210071	Empirical array recurrence of order 20	P	271
80	A210072	Empirical array recurrence of order 71	P	272
81	A210082	Empirical array recurrence of order 53	P	272
82	A210128	Empirical array recurrence of order 5	P	272
83	A210149	Empirical array recurrence of order 5	P	273
84	A210150	Empirical array recurrence of order 13	P	273
85	A210151	Empirical array recurrence of order 48	P	273
86	A210263	Empirical array recurrence of order 50	P	273
87	A210269	Empirical array recurrence of order 3	P	274
88	A210270	Empirical array recurrence of order 9	P	274
89	A210271	Empirical array recurrence of order 25	P	274
90	A210272	Empirical array recurrence of order 69	P	274
91	A210305	Empirical array recurrence of order 53	P	275
92	A210328	Empirical array recurrence of order 6	P	275

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93	A210329	Empirical array recurrence of order 20	P	275
94	A210330	Empirical array recurrence of order 72	P	276
95	A210348	Empirical array recurrence of order 5	P	276
96	A210349	Empirical array recurrence of order 20	P	276
97	A210350	Empirical array recurrence of order 72	P	277
98	A210383	Empirical array recurrence of order 3	P	277
99	A210384	Empirical array recurrence of order 6	P	277
100	A210386	Empirical array recurrence of order 42	P	277

5.2 Formal-series congruences, periodicity, and theta values

The thirteen entry-level packages in this section use exact formal identities. The classical theta evaluations and the finite-ring argument are credited tools, not claims of new general theorems. Every selected assertion was still marked conjectural or empirical in the retrieved OEIS entry. Other assertions in the same entry are outside scope unless explicitly included below.

5.2.1 Two implicit continued fractions

Let $A_r(x)$, $r = 1, 2$, be the series with constant term 1 determined by

$$1 = A_r - \cfrac{x}{A_r - \cfrac{3^r x}{A_r - \cfrac{5^r x}{A_r - \cfrac{7^r x}{\ddots}}}}.$$

The cases $r = 1, 2$ define A338632 and A338636, respectively. Every denominator has constant term 1. Changing a tail beyond depth N changes the fraction only in degrees at least N. Thus the fraction exists formally. The equation is $A_r = 1 + xH(A_r, x)$, so its coefficient of degree n determines $[x^n]A_r$ from earlier coefficients, over any commutative ring. This proves existence, uniqueness, and integrality without division by n.

Theorem 5.2.1 (A338632)**.** *For $n > 1$, $a(n) \equiv 2 \pmod 4$. For $n > 0$, its residue modulo 3 is 1 exactly when the ternary expansion of n ends in 1 and all other digits are 0 or 1. The residue is 2 exactly when the final digit is 0 or 2 and all other digits are 0 or 1; otherwise it is 0. These are all three congruence conjectures selected from the entry.*

Proof. Modulo 4, the successive numerators have weights $1, 3, 1, 3, \ldots$. Let the two tail values be U, V. Then $U = A_1 - x/V$, $V = A_1 - 3x/U$, and the defining equation says $U = 1$. Therefore

$$A_1^2 - (1 + 3x)A_1 + 2x = 0 \pmod 4.$$

This polynomial equation has a unique root with constant term 1: its derivative with respect to A_1 has constant term 1 modulo 4. For $R = 1 + x + 2x^2/(1 - x)$, direct multiplication gives

$$R^2 - (1 + 3x)R + 2x = \frac{4x^4}{(1 - x)^2}.$$

Hence $A_1 \equiv R \pmod 4$, proving the first statement.

Modulo 3, the second numerator vanishes, so $1 = A_1 - x/A_1$ and $A_1^2 - A_1 - x = 0$. Put

$$H(x) = (1+x)^2 \prod_{j\geq 1}(1+x^{3^j}) \quad \text{in } \mathbb{F}_3[[x]].$$

For the product ending at $j = M$, squaring and using $(1+z)^3 = 1+z^3$ gives $(1+x)(1+x^{3^{M+1}})$. Consequently $H^2 = 1+x$. The unique root is $A_1 = (1+H)/2$. In positive degree, the coefficient of H is 2 for final ternary digit 1, is 1 for final digit 0 or 2, and is 0 if a higher digit is 2. Multiplication by $1/2 = 2$ gives the stated residues. The digit descriptions are the established descriptions of A191107 and A186776; their own assertions add no count here. □

Theorem 5.2.2 (A338636)**.** *For $n > 1$, $a(n) \equiv 0 \pmod 8$. The two mod-3 support descriptions in the preceding theorem hold for this sequence as well.*

Proof. Every odd square is 1 modulo 8. Thus the whole continued fraction has one repeated tail $U = A_2 - x/U$. Since $U = 1$, it follows that $A_2 = 1 + x \pmod 8$. Modulo 3, the numerator $3^2 x$ vanishes. The same equation $A_2^2 - A_2 - x = 0$ and the same uniqueness and digit argument apply. This proves all three displayed congruence conjectures. □

5.2.2 A397242: parity from diagonal powers

Theorem 5.2.3. *The coefficient $a(n)$ of A397242, including $a(0) = 1$, is odd if and only if $n+1$ is a power of 2.*

Proof. We first reproduce, with attribution to Round III, the integer-series lemma used there for A397241; that earlier entry is not counted again. Let $B(t) = 1 + t + \cdots$ satisfy

$$n[t^n]B(t)^n = (n-1)[t^n]B(t)^{n+1} \qquad (n > 1).$$

The coefficient of $b(n)$ in the difference is $n^2 - (n-1)(n+1) = 1$. Therefore this defines a unique integer series. Over $\mathbb{F}_2$, $B = (1-t)^{-1}$ satisfies it. Indeed the required vanishing is $\binom{2n}{n}$ when n is even and $\binom{2n-1}{n}$ when $n > 1$ is odd. Their 2-adic valuations are $s_2(n)$ and $s_2(n) - 1$, respectively; an odd $n > 1$ has at least two binary 1's. Thus every coefficient of B is odd.

For completeness we derive the change of variables relating B to the present entry. Define the unique integer series $A = 1 + x + \cdots$ by $A(x) = B(xA(x))$. Write $t = xA(x)$, so its inverse is $x = t/B(t)$. Lagrange inversion gives, for $n > 1$,

$$[t^n]B(t)^n = n[x^n]\log A(x), \qquad [t^n]B(t)^{n+1} = (n+1)[x^n]A(x).$$

The diagonal relation therefore gives $[x^n]\log A = (n^2-1)a(n)/n^2$. Together with $a(0) = a(1) = 1$, this is exactly the defining equation

$$A(x) = \exp\left(x + \sum_{n\geq 2} \frac{n^2-1}{n^2} a(n) x^n\right).$$

It determines $a(n)$ uniquely over $\mathbb{Q}$, since the coefficient left after moving its two occurrences to one side is $1/n^2$. Hence the constructed integral series is the source sequence.

Reducing $A = B(xA)$ modulo 2 gives $A = (1-xA)^{-1}$, or $A = 1 + xA^2$. The unique solution is

$$A(x) = \sum_{j\geq 0} x^{2^j - 1} \pmod 2,$$

as follows immediately by squaring. The other mod-3 assertions and the asymptotic observation in the entry are not selected. □

5.2.3 A396804: a Hurwitz-series congruence

An integer Hurwitz series means $F(x) = \sum f_n x^n/n!$ with $f_n \in \mathbb{Z}$. Composition of zero-constant such series, and the exponential of one, are given by integer Bell polynomials. Congruence therefore respects these operations; no reduction of $1/n!$ in a finite field is intended.

Lemma 5.2.4. *Let $F(x) = xe^x$. Then $F^{\circ 2} \equiv x \pmod 2$ and $F^{\circ 4} \equiv x \pmod 4$, coefficientwise in the Hurwitz sense.*

Proof. Lagrange inversion gives $n![x^n]F^{-1}(x) = (-n)^{n-1}$. For every $n \geq 1$ this is congruent to n, the coefficient of F, modulo 2. Thus $F^{-1} \equiv F$ and $F^{\circ 2} \equiv x$. Write $F^{\circ 2} = x + 2G$, with integral Hurwitz coefficients in G. Since $G \circ (x + 2G) \equiv G \pmod 2$,

$$(x + 2G) \circ (x + 2G) = x + 2G + 2G \circ (x + 2G) \equiv x \pmod 4.$$

□

Theorem 5.2.5. *For the exponential series $A = x \exp(A^{\circ 4})$ defining A396804, one has $a(n) \equiv n \pmod 4$ for every $n \geq 1$. This proves both its mod-4 and its parity conjectures as one package.*

Proof. The equation determines its Hurwitz coefficients by induction: its degree-n coefficient depends only on those of A of degree below n, and all defining coefficient polynomials are integral. Hence existence, integrality, and uniqueness hold modulo 4 as well as over $\mathbb{Z}$. By the lemma, $F = xe^x$ satisfies $x \exp(F^{\circ 4}) \equiv xe^x = F \pmod 4$. Uniqueness gives $A \equiv F$. The source's obsolete mod-3 claim already has a published counterexample in the entry; it is expressly excluded. Neither the replacement mod-3 claim nor the mod-5 claim is being asserted here. □

5.2.4 A383165 and A383166: eventual periodicity

Lemma 5.2.6 (Finite differential algebra)**.** *Suppose $y_i'(x) = P_i(y_1(x), \ldots, y_d(x))$, where each P_i is an integer polynomial and each $y_i(0)$ is an integer. For any integer polynomial Q, the derivatives $Q(y(x))^{(n)}|_{x=0}$ are integers and are eventually periodic modulo every positive integer m.*

Proof. Translate the initial values to zero, and set $D = \sum_i P_i \partial/\partial y_i$ in the translated coordinates. The n-th derivative is the constant term of $D^n Q$, proving integrality. For $m \geq 2$, consider the finite ring

$$R_m = (\mathbb{Z}/m\mathbb{Z})[y_1, \ldots, y_d]/(y_1^m, \ldots, y_d^m).$$

The derivation descends to this ring since $D(y_i^m) = m y_i^{m-1} P_i = 0$. Evaluation at zero also descends. The orbit $Q, DQ, D^2Q, \ldots$ in this finite ring eventually repeats, and iteration of the same map then repeats forever. Taking its constant term proves the assertion. For $m = 1$ the assertion is immediate. This argument gives existence of a period; it does not claim a bound involving Euler's totient. □

Set

$$L(x) = \log \frac{e^{2x} + 1}{2}, \quad U(x) = e^{-L(x)}, \quad H_2(x) = L(x)^2/2, \quad H_3(x) = L(x)^3/6.$$

These functions satisfy the integer polynomial system

$$L' = 2 - U, \qquad U' = U^2 - 2U, \qquad H_2' = L(2 - U), \qquad H_3' = H_2(2 - U),$$

with initial values $(L, U, H_2, H_3) = (0, 1, 0, 0)$.

Theorem 5.2.7 (A383165)**.** *The exponential coefficients of H_2 are eventually periodic modulo every positive integer, as conjectured in A383165.*

Proof. Apply Lemma 5.2.6 with $Q = H_2$ to the displayed system. The polynomial system also supplies integrality at primes dividing 2; there is no division by 2 in the finite ring. □

Theorem 5.2.8 (A383166)**.** *The exponential coefficients of H_3 are eventually periodic modulo every positive integer, as conjectured in A383166.*

Proof. Apply the same lemma with $Q = H_3$. This handles primes 2 and 3 without inverting 6. The source's sample modulus-5 period is a supplementary regression in the code; the proof is valid for all moduli. The linked Graves-method note supplies differentiation identities, not a proof of these modular-periodicity claims. □

5.2.5 A179929: a totient period with an exceptional prime

Theorem 5.2.9. *For the exponential coefficients $a(n)$ of $3/(1+2e^{-3x})$, the sequence modulo any $m \geq 1$ is eventually periodic with period dividing $\varphi(m)$, proving the selected conjecture in A179929.*

Proof. First take a prime power p^r with $p \neq 3$. Put $t = e^{3x} - 1$. The established exponential generating function gives

$$\frac{3}{1+2e^{-3x}} = \frac{1+t}{1+t/3} = 1 + \sum_{k\geq 1} 2(-1)^{k-1}3^{-k}t^k.$$

Consequently, for $n > 0$,

$$a(n) = \sum_{k\geq 1} 2(-1)^{k-1}3^{-k} \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} (3j)^n.$$

This is interpreted via $k!S(n,k)$, so terms with $k > n$ vanish. Modulo p^r, terms with $k \geq pr$ vanish as well, since $p^r \mid k!$. The remaining finite sum has fixed coefficients and bases $3j$. For $n \geq r$, nonunit bases vanish modulo p^r; unit bases have a period dividing $\lambda(p^r)$ by Euler's theorem (or the exponent of the unit group). Thus this part has that eventual period.

For $p = 3$, use $u' = 3u - u^2$, $u(0) = 1$. Its derivatives are $a(n) = (D^n u)(1)$ for $D = (3u - u^2)d/du$. Modulo 3, $D = -u^2 d/du$ and

$$D^3 u^j = -j(j+1)(j+2)u^{j+3} = 0 \pmod 3.$$

Therefore D^3 sends every integer polynomial into $3\mathbb{Z}[u]$, and iteration gives $D^{3r}\mathbb{Z}[u] \subseteq 3^r\mathbb{Z}[u]$. So $a(n) = 0 \pmod{3^r}$ for $n \geq 3r$. Write $m = 3^r M$ with $3 \nmid M$. The Chinese remainder theorem gives an eventual period $\lambda(M)$, with $\lambda(1) = 1$, starting no later than $\max(\{1, 3r\} \cup \{v_p(M) : p \mid M\})$. Since $\lambda(M) \mid \varphi(M) \mid \varphi(m)$, the desired bound follows. □

5.2.6 Classical theta evaluations used below

Here the nome is $q = e^{-\pi}$, and

$$T = \frac{\pi^{1/4}}{\Gamma(3/4)}, \quad \phi(z) = \sum_{n\in\mathbb{Z}} z^{n^2}, \quad \psi(z) = \sum_{n\geq 0} z^{n(n+1)/2}, \quad \chi(z) = \prod_{n\geq 0}(1+z^{2n+1}).$$

All these products and series converge in $|z| < 1$; the nonzero theta denominators used below follow from their convergent product formulas. We use the following classical special values, giving their derivation to make the substitutions independently reviewable:

$$\phi(q) = T, \qquad \phi(-q) = 2^{-1/4}T, \tag{5.1}$$

$$\psi(q) = 2^{-5/8}q^{-1/8}T, \qquad \psi(q^2) = 2^{-5/4}q^{-1/4}T, \tag{5.2}$$

$$\psi(q^4) = \tfrac{1}{4}q^{-1/2}T\sqrt{2-\sqrt{2}}, \qquad \chi(q) = 2^{1/4}q^{1/24}. \tag{5.3}$$

At lattice parameter i, the theta transformation gives $\theta_2(q) = \theta_4(q)$, and the quartic identity gives $\theta_2(q) = \theta_4(q) = 2^{-1/4}\theta_3(q)$. The elliptic-integral relation $K(1/\sqrt{2}) = \pi\theta_3(q)^2/2$, together with $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$ and the gamma reflection formula, gives $\theta_3(q) = T$. These are standard identities; see NIST DLMF §20.7, §20.9, and §19.20.

The elementary series relation $\theta_2(z) = 2z^{1/4}\psi(z^2)$ gives $\psi(q^2)$. The duplication formulas

$$\theta_2(z)^2 = 2\theta_2(z^2)\theta_3(z^2), \qquad \theta_2(z^2)^2 = \tfrac{1}{2}(\theta_3(z)^2 - \theta_4(z)^2)$$

give $\psi(q)$ and $\psi(q^4)$, respectively. Finally let $E(q) = \prod_{n\geq 1}(1-q^n)$ and $\eta(i) = q^{1/12}E(q^2)$. The product identities give $\phi(q) = E(q^2)\chi(q)^2$ and $2\eta(i)^3 = \theta_2(q)\theta_3(q)\theta_4(q)$. Hence $E(q^2) = q^{-1/12}T/\sqrt{2}$ and $\chi(q)^2 = \sqrt{2}q^{1/12}$, proving the last value. All roots are the positive real roots.

Source and priority scope. The six following statements were labelled empirical by Simon Plouffe in September 2025. His *Numbers in the base e^π* describes the numerical search and explicitly discusses shifting the initial index. The general theta identities above are classical; these packages are exact, source-specific deductions, not claims of six new special-function identities or of worldwide priority. Decimal-constant entries cross-referenced by the source are not separately counted.

Theorem 5.2.10 (A210063)**.** *For the coefficients of $\psi(x^4)/\phi(x)$,*

$$\sum_{n\geq 0} a(n)e^{-\pi n} = \frac{1}{4}e^{\pi/2}\sqrt{2-\sqrt{2}}.$$

Proof. The source's established generating function is analytic at q. By (5.1) and (5.3), its value is $(q^{-1/2}T\sqrt{2-\sqrt{2}}/4)/T$, exactly the asserted expression. □

Theorem 5.2.11 (A243763: index correction)**.** *The sequence has offset* 1*, and its generating function is $\sum_{n\geq 1} a(n)x^n = x\phi(x)^3\psi(x^2)^4$. Its exact evaluations are*

$$\sum_{n\geq 1} a(n)e^{-\pi n} = \frac{T^7}{32}, \qquad \sum_{j\geq 0} a(j+1)e^{-\pi j} = \frac{e^\pi T^7}{32}.$$

Thus the empirical right-hand side is correct for the zero-based list of terms; the displayed sum must use $a(j+1)$, not the original $a(j)$.

Proof. At $x = q$, substitute $\phi(q) = T$ and $\psi(q^2) = 2^{-5/4}q^{-1/4}T$ to get $qT^3(2^{-5/4}q^{-1/4}T)^4 = T^7/32$. Shifting the exponent down by one multiplies this by $q^{-1} = e^\pi$. This is classified as an index correction with a proof of the corrected formula, not as a counterexample to the intended normalized value. □

Theorem 5.2.12 (A107635)**.** *For the coefficients of $\chi(x)^3$,*

$$\sum_{n\geq 0} a(n)e^{-\pi n} = 2^{3/4}e^{-\pi/8}.$$

Proof. The generating function $\chi(x)^3$ is an established formula in the entry. Cubing $\chi(q) = 2^{1/4}q^{1/24}$ gives the result. □

Theorem 5.2.13 (A002512)**.** *For the coefficients of $\chi(x)^{10}/\phi(x)^4$,*

$$\sum_{n\geq 0} a(n)e^{-\pi n} = \frac{4\sqrt{2}\,e^{-5\pi/12}\Gamma(3/4)^4}{\pi}.$$

Proof. The defining generating function at q is $(2^{1/4}q^{1/24})^{10}/T^4 = 2^{5/2}q^{5/12}T^{-4}$. Replacing T and q by their definitions gives precisely the formula. □

Theorem 5.2.14 (A134416)**.** *For the source generating function* $1/(\phi(x)\phi(-x)^4)$*,*

$$\sum_{n\geq 0} a(n)e^{-\pi n} = \frac{2\Gamma(3/4)^5}{\pi^{5/4}}.$$

Proof. Michael Somos's established generating-function identity in the entry gives $1/(T(2^{-1/4}T)^4) = 2T^{-5}$ at q, as required. □

Theorem 5.2.15 (A030204)**.** *For the coefficients of* $\psi(x)\phi(-x)$*,*

$$\sum_{n\geq 0} a(n)e^{-\pi n} = \frac{2^{-7/8}e^{\pi/8}\sqrt{\pi}}{\Gamma(3/4)^2}.$$

Proof. The source's established generating function evaluates to $(2^{-5/8}q^{-1/8}T)(2^{-1/4}T) = 2^{-7/8}q^{-1/8}T^2$. This is the empirical expression, since $2^{-7/8} = 2^{1/8}/2$. □

5.3 Arithmetic, averages, and exact counterexamples

This section resolves twelve selected packages at twelve different OEIS entries. Nine packages prove conjectures, two prove explicitly empirical observations, and one refutes a stated uniform bound. The last classification matters: the counterexample in A051699 does not resolve the separate conjecture about the asymptotic constant. Known theorems used below are credited explicitly. The accompanying standard-library verifier `r5_arith_verify.py` produces `r5_arith_certificates.json`; finite checks supplement the universal arguments and do not establish any of their limits by extrapolation.

5.3.1 The floor-factorial constant in A010786

For $n \geq 1$, put $F(n) = \prod_{j=1}^{n} \lfloor n/j \rfloor$. The October 3, 2018 comment in A010786 conjectures that the constant in its leading logarithmic asymptotic is A085361.

Theorem 5.3.1. *With*

$$C = \sum_{r\geq 1} \frac{\zeta(r+1)-1}{r} = 0.7885305659\ldots,$$

one has, for every $n \geq 1$*,*

$$0 \leq Cn - \log F(n) \leq \log n + 1.$$

In particular $\log F(n) \sim Cn$ *and also* $\log F(n) \sim C(n - \frac{1}{2}\log(2\pi n))$*.*

Proof. For any positive integer h, $\log h = \sum_{m=2}^{h} \log(m/(m-1))$. Interchanging two finite sums gives

$$\log F(n) = \sum_{m=2}^{n} \left\lfloor \frac{n}{m} \right\rfloor \log \frac{m}{m-1}.$$

The convergent positive series $C = \sum_{m\geq 2} m^{-1}\log(m/(m-1))$ has the displayed zeta form because $-\log(1-1/m) = \sum_{r\geq 1}(rm^r)^{-1}$ and all terms are nonnegative. Subtracting the preceding finite identity from Cn yields

$$Cn - \log F(n) = \sum_{m=2}^{n} \left\{ \frac{n}{m} \right\} \log \frac{m}{m-1} + n \sum_{m>n} \frac{1}{m} \log \frac{m}{m-1}.$$

Both terms are nonnegative. The first is at most $\log n$, by telescoping. The inequality $\log(1+u) \leq u$ bounds the second by $n\sum_{m>n}(m(m-1))^{-1} = 1$. The conclusions follow. □

Scope and audit. The constant and its alternative series are already known; the selected claim identifies this constant in the floor-factorial asymptotic. The linked Alladi–Grinstead discussion was inspected and supplies the known constant, rather than a new claim of its discovery. Here $\sim$ means ratio asymptotics. The theorem does not assert a second-order expansion with coefficient $-C/2$ and bounded remainder.

5.3.2 The Hultman Dirichlet-series family in A034444

The selected target is Werner Schulte's February 17, 2022 conjecture, not the 2018 unitary-divisor divisibility conjecture, which the current entry already marks proved. Write $H(i,k) = \texttt{A164652}(i,k)$ for the Hultman numbers, and define the multiplicative function b_i by $b_i(1) = 1$ and

$$b_i(p^e) = \frac{i+2}{i!} \sum_{k=1}^{i+1} H(i,k) e^{k-1} \qquad (i \geq 0,\ e \geq 1).$$

Theorem 5.3.2. *For every integer $i \geq 0$ and every complex s with $\Re(s) > 1$,*

$$\sum_{n \geq 1} \frac{b_i(n)}{n^s} = \frac{\zeta(s)^{i+2}}{\zeta((i+2)s)}.$$

Proof. We use the established Hultman formula

$$H(i,k) = \begin{cases} c(i+2,k)/\binom{i+2}{2}, & i-k \text{ odd}, \\ 0, & i-k \text{ even}, \end{cases}$$

where c denotes unsigned Stirling numbers of the first kind. This is the known input recorded in A164652, and appears through the cycle polynomial in Section 3 of Bóna–Flynn, arXiv:0811.0740. Let $P_i(x) = \prod_{j=1}^{i+1}(x+j)$. Separating the two parities in $xP_i(x) = \sum_k c(i+2,k)x^k$ gives

$$\sum_k H(i,k)x^k = \frac{x\{P_i(x) + (-1)^i P_i(-x)\}}{(i+1)(i+2)}.$$

Consequently, for $e \geq 1$,

$$b_i(p^e) = \frac{\prod_{j=1}^{i+1}(e+j) - \prod_{j=1}^{i+1}(e-j)}{(i+1)!} = \binom{e+i+1}{i+1} - \binom{e-1}{i+1},$$

where the second binomial is zero when $1 \leq e \leq i+1$. With $m = i+2$, the local power series therefore is

$$1 + \sum_{e \geq 1} b_i(p^e) t^e = \frac{1-t^m}{(1-t)^m}.$$

Euler multiplication proves the assertion. Absolute convergence in $\Re(s) > 1$ follows, for example, by bounding each nonnegative local coefficient by $\binom{e+m-1}{m-1}$, the coefficient of $(1-t)^{-m}$. □

The known Hultman formula is not counted again. The special cases A226602 and A286779, and the parameter choices i, do not add to the one-entry count.

5.3.3 An exact counterexample to the uniform bound in A051699

Let $a(k)$ be the distance from k to its nearest prime and $S(N) = \sum_{k=1}^{N} a(k)$. The selected September 18, 2005 assertion in A051699 includes $S(N) < \frac{3}{8} N \log N$ for every $N > 130$.

Theorem 5.3.3. *This upper bound is false. At* $N = 2^{20} = 1048576$,

$$S(N) = 5542944 > 5505024 > \frac{3}{8} N \log N.$$

Proof. The exact certificate uses the sieve of Eratosthenes through $N + 100$. It gives $\pi(N) = 82025$, last prime 1048573, and next prime 1048583. For consecutive primes $p < q$, the total nearest-prime distance in the completed interval from p to q is $\lfloor (q-p)^2/4 \rfloor$. Summing this expression over all completed prime gaps up to 1048573 gives 5542937. The three remaining distances are $1, 2, 3$; the distance at 1 is 1. Thus $S(N) = 5542937 + 6 + 1 = 5542944$. The certificate includes the entire gap histogram and the verifier also recomputes the sum directly, one integer at a time.

The logarithm comparison is certified without floating point:

$$e^{7/10} > 1 + \frac{7}{10} + \frac{(7/10)^2}{2} + \frac{(7/10)^3}{6} = \frac{12013}{6000} > 2.$$

Hence $\log 2 < 7/10$, and $\frac{3}{8} N \log N < \frac{3}{8} N \cdot 20 \cdot \frac{7}{10} = 5505024$. The difference between the exact sum and this larger rational upper bound is $37920 > 0$. □

Scope and audit. This is a complete refutation of the explicit uniform bound. It does not prove or refute the separate proposed asymptotic $S(N) \sim CN \log N$, nor determine its constant. The current entry contains later numerical estimates inconsistent with the old bound, but no exact counterexample to that bound is supplied there.

5.3.4 Sixth powers in the cubic gcd triangle A129455

Put $G(0) = 1$ and $G(n) = \prod_{1 \le r,s,t \le n} \gcd(r, s, t)$ for $n \ge 1$. Thus $G(n) =$ A129454$(n+1)$. The triangle in A129455 is

$$T(n, k) = \frac{G(n)}{G(k)G(n-k)}, \qquad 0 \le k \le n.$$

Peter Bala's April 16, 2007 observations propose integrality, divisibility by $\binom{n}{k}$, and a perfect-sixth-power quotient.

Theorem 5.3.4. *For all integers* $n \ge 0$ *and* $0 \le k \le n$, *there is a positive integer* $Y(n, k)$ *such that* $T(n, k) = \binom{n}{k} Y(n, k)^6$.

Proof. For every prime p, threshold counting gives

$$v_p(G(n)) = \sum_{r \ge 1} \left\lfloor \frac{n}{p^r} \right\rfloor^3.$$

Indeed $v_p(\gcd(u, v, w)) = \sum_{r \ge 1} \mathbf{1}_{p^r \mid u} \mathbf{1}_{p^r \mid v} \mathbf{1}_{p^r \mid w}$. This valuation identity is already a known result: Ma's 2025 generalized-factorials paper, Section 2.2, includes the general power-q version. It is used here as an input, and is not counted as a new solution for A129454.

For a fixed p^r, set $u = \lfloor k/p^r \rfloor$, $v = \lfloor (n-k)/p^r \rfloor$, and $\epsilon = \lfloor n/p^r \rfloor - u - v \in \{0, 1\}$. Legendre's formula shows that the contribution to $v_p(T(n, k)/\binom{n}{k})$ is

$$(u + v + \epsilon)^3 - u^3 - v^3 - \epsilon = \begin{cases} 3uv(u+v), & \epsilon = 0, \\ 3(u+v)(u+1)(v+1), & \epsilon = 1. \end{cases}$$

Both expressions are nonnegative multiples of six. In the first case, $uv(u+v)$ is even; in the second, either $u+v$ is even or one of $u+1, v+1$ is even. Every prime valuation of the positive rational quotient is therefore a nonnegative multiple of six. Unique factorization proves the assertion, including the empty-product boundary cases. □

Only the sixth-power package is selected. The incorrectly shifted $\lfloor n/p^r \rfloor^3$ statement attached to the $n - 1$ convention in A129454 is not used to inflate the count.

5.3.5 The empirical mean in A161789

For $n \geq 1$, let $a(n) = \max\{j \geq 1 : 2^j - 1 \mid n\}$. The selected target is Amiram Eldar's June 30, 2025 empirical asymptotic mean in A161789. Its separate gcd conjecture was proved by Robert Israel on August 30, 2026; that one-page proof was inspected and is not claimed again here.

Theorem 5.3.5. *The mean* $\lim_{N\to\infty} N^{-1}\sum_{n=1}^{N} a(n) = C$ *exists, and*

$$1.9249135089 < C < 1.9249135166.$$

More generally it has the convergent rational approximation scheme below, with an explicit error bound for every truncation.

Proof. Put $M_j = 2^j - 1$ and $a_L(n) = \max\{1 \leq j \leq L : M_j \mid n\}$. This function is periodic with period $P_L = \operatorname{lcm}(M_1, \ldots, M_L)$, so it has a rational mean C_L. Pointwise,

$$0 \leq a(n) - a_L(n) \leq \sum_{j>L} (j-1)\mathbf{1}_{M_j \mid n}.$$

Averaging over any finite initial interval gives the uniform bound

$$0 \leq \frac{1}{N}\sum_{n\leq N}(a(n) - a_L(n)) \leq \sum_{j>L}\frac{j-1}{2^j-1} \leq E_L := \frac{2(L+1)}{2^{L+1}-1}.$$

For the last inequality, use $(2^j-1)^{-1} \leq 2^{-j}/(1-2^{-L-1})$ and $\sum_{j>L}(j-1)2^{-j} = (L+1)2^{-L}$. The C_L are increasing and bounded; their limit C satisfies $0 \leq C - C_L \leq E_L$. Taking the lower and upper limits of the finite means, then letting $L \to \infty$, proves their convergence to C.

For reproducible exact evaluation, represent $a_L(n) = \sum_{d\mid n} c_{L,d}$, with finite support. Start with $c_{1,1} = 1$. The identity

$$a_j(n) = a_{j-1}(n) + (j - a_{j-1}(n))\mathbf{1}_{M_j \mid n}$$

gives an integer update: retain the old coefficients, add j at M_j, and subtract each old $c_{j-1,d}$ at $\operatorname{lcm}(d, M_j)$. Then $C_L = \sum_d c_{L,d}/d = P_L^{-1}\sum_d c_{L,d}(P_L/d)$. The exact $L = 32$ certificate has 350208 nonzero coefficients and gives

$$C_{32} = \frac{20717720272364106658413433139135890106303010474495845226068481223389981 4}{10762935672948377150884689732602367548665840372701254861003025125458204 5},$$
$$E_{32} = \frac{66}{8589934591}.$$

Exact rational comparisons yield $1.9249135089 < C_{32} \leq C \leq C_{32} + E_{32} < 1.9249135166$. The displayed enclosure therefore certifies the initially observed digits; it is not a large-N numerical fit. □

5.3.6 Prime distance determinants in A187011

Let $p_1 < p_2 < \cdots$ be the primes and $D_n = \det(|p_i - p_j|)_{1\leq i,j\leq n}$, with $D_0 = 1$. Both parts of Zhi-Wei Sun's August 30, 2013 conjecture in A187011 are selected.

Theorem 5.3.6. *For every* $n \geq 2$,

$$D_n = (-1)^{n-1}2^{n-2}(p_n - 2)\prod_{j=1}^{n-1}(p_{j+1} - p_j).$$

In particular $(-1)^{n-1}D_n > 0$ *and* $|D_n|^{1/n} \to \infty$.

Proof. First let $x_1 < \cdots < x_n$ be arbitrary real numbers and set $d_j = x_{j+1} - x_j > 0$. Apply successive-difference operations to both rows and columns of $A = (|x_i - x_j|)$. Equivalently, apply the congruence transformation BAB^{T}, where B is the lower bidiagonal matrix with diagonal entries 1 and subdiagonal entries -1, whose determinant is 1. The resulting matrix is

$$\begin{pmatrix} 0 & d^{\mathsf{T}} \\ d & -2\,\mathrm{diag}(d_1, \ldots, d_{n-1}) \end{pmatrix}.$$

The Schur complement gives its determinant as

$$(-2)^{n-1}\Big(\prod_j d_j\Big)\frac{1}{2}\sum_j d_j,$$

which proves the formula and its sign.

For the growth claim we use the classical, unconditional prime number theorem in each fixed arithmetic progression: $\pi(X; q, a)/\pi(X) \to 1/\varphi(q)$ when $(a, q) = 1$. For a primary modern reference stating and refining this theorem, see Thorner–Zaman, arXiv:2108.10878, Section 1.1. No uniformity in growing moduli and no conjecture on prime tuples is used.

Fix $h \geq 1$. For any fixed product Q of odd primes ℓ not dividing h, a pair of primes $p, p + h$ larger than the prime factors of Q must lie in one of $\prod_{\ell|Q}(\ell - 2)$ admissible classes modulo Q. The fixed-modulus theorem therefore gives

$$\limsup_{X\to\infty} \frac{\#\{p \leq X : p, p + h \text{ prime}\}}{\pi(X)} \leq \prod_{\ell|Q} \frac{\ell - 2}{\ell - 1}.$$

Let Q include successively all the eligible primes. The product tends to zero, since the sum of prime reciprocals diverges and only finitely many primes were omitted. Thus each fixed difference h occurs in $o(\pi(X))$ prime pairs up to X.

For every fixed integer $K \geq 1$, it follows that among the first $n - 1$ consecutive prime gaps only $o(n)$ have size at most K. All gaps are at least 1, so

$$\prod_{j=1}^{n-1}(p_{j+1} - p_j) \geq (K + 1)^{n-1-o(n)}.$$

The determinant formula implies $\liminf_{n\to\infty} |D_n|^{1/n} \geq 2(K + 1)$. Since K is arbitrary, the desired divergence follows. □

The general distance-matrix calculation and the classical prime-distribution input are not presented as new theorems. Their application settles the two specific statements still labelled conjectural in this entry.

5.3.7 The sharp decimal-average bound in A276646

Write $d(k)$ for the number of decimal digits in k and $S(n) = \sum_{k=1}^{n} k/10^{d(k)}$. Thus the notation $0.k$ in A276646 means $k/10^{d(k)}$.

Theorem 5.3.7. *For all $n \geq 1$, $S(n)/n < 11/20 = 0.55$, and $\sup_{n\geq 1} S(n)/n = 11/20$.*

Proof. The sum of the d-digit block is

$$\sum_{k=10^{d-1}}^{10^d-1} \frac{k}{10^d} = \frac{99 \cdot 10^{d-1} - 9}{20}.$$

Summing the blocks gives the exact endpoint formula

$$S(10^d - 1) = \frac{11(10^d - 1) - 9d}{20}.$$

Let $A = 10^{d-1} - 1$ and $B = 10^d - 1$. Within the block, $20S(n)$ is a quadratic polynomial in n with positive leading coefficient, because

$$S(n) = S(A) + \frac{n(n+1) - A(A+1)}{2 \cdot 10^d}.$$

Hence $D(n) = 11n - 20S(n)$ is concave on the real interval $[A, B]$. Its endpoint values are $9(d-1)$ and $9d$, so it is nonnegative throughout. If $d \geq 2$ it is strictly positive. For $d = 1$, the left endpoint is $n = 0$ and strict concavity (or the formula) gives positivity at each integer $1 \leq n \leq 9$ as well. Finally the endpoint ratios tend to $11/20$, proving sharpness. □

This proves the explicit bound, and does not count the numerator and denominator companion entries separately.

5.3.8 Primitive quadruples in A278083

For odd $m \geq 1$, let $6b(m)$ count primitive ordered integral quadruples (x_1, x_2, x_3, x_4) with

$$\sum x_i = 2m, \qquad \sum x_i^2 = 2m^2.$$

Set $b(m) = 0$ for even m. Then $a(n) = b(2n-1)$ is A278083.

Theorem 5.3.8. *The function b is multiplicative, with $b(1) = 1$, $b(2^e) = 0$ for $e \geq 1$, and, for every odd prime p,*

$$b(p^e) = p^{e-1}(p - \chi(p)), \qquad \chi(p) = (-1)^{(p-1)/2}.$$

For odd $m \geq 1$ the additional October 14, 2025 limit conjecture holds:

$$b(m) = \frac{\varphi(m)}{6} \lim_{k \to \infty} \frac{r_3(m^{2k})}{m^k},$$

where $r_3(t)$ counts all ordered integral representations by three squares.

Proof. Use the change of variables

$$a = x_1 + x_2 - m, \quad b = x_1 + x_3 - m, \quad c = x_1 + x_4 - m.$$

The two constraints are equivalent to $a^2 + b^2 + c^2 = m^2$, with inverse

$$(x_1, x_2, x_3, x_4) = \tfrac{1}{2}(m + a + b + c,\ m + a - b - c,\ m - a + b - c,\ m - a - b + c).$$

For odd m, the three-square equation forces $a + b + c$ odd, so all four inverse coordinates are integers. The transformation is a bijection. The four coordinates cannot all be even, since their square sum is $2m^2 \equiv 2 \pmod 4$. For an odd prime p, divisibility of all four coordinates implies $p \mid m, a, b, c$; conversely these divisibilities imply $p \mid x_i$ for every i. Also any prime dividing a, b, c divides m, by the three-square equation. Thus the bijection preserves primitivity.

The required known input is Gauss's primitive three-square count

$$R_3(m^2) = 6 \prod_{p^e \| m} p^{e-1}(p - \chi(p)) \qquad (m \text{ odd}).$$

For a primary research source giving precisely this formula, see Hürlimann, *Exact and Asymptotic Evaluation of the Number of Distinct Primitive Cuboids*, J. Integer Sequences 18 (2015), Article

15.2.5, equation (2); it credits Gauss and Cooper–Hirschhorn. The bijection proves $6b(m) = R_3(m^2)$ and therefore the first assertion, including multiplicativity after adjoining the zero value on even integers.

The same paper's Theorem 2 records the known total-count formula

$$r_3(m^2) = 6 \prod_{p^e \| m} \left(\sum_{j=0}^{e} p^j - \chi(p) \sum_{j=0}^{e-1} p^j \right).$$

Apply it to m^k, divide by m^k, and let $k \to \infty$. Each factor tends to $(p - \chi(p))/(p-1)$, so multiplication by $\varphi(m)/6$ gives $m \prod_{p|m}(1 - \chi(p)/p) = b(m)$. For $m = 1$ the same assertion follows from $r_3(1) = 6$. □

The classical representation formulas are credited inputs. The selected OEIS resolution is the explicit bijection and application to both stated conjectures, counted together as one package.

5.3.9 All admissible divisor-sum moduli in A280098

The selected conjecture is the classification, proposed on December 25, 2016, of positive k for which $\sigma(kn-1)/k$ is always integral. The identical assertion at A295012 is an uncounted duplicate. Since σ here is defined for positive arguments, the domain is $n \geq 1$ with $kn - 1 \geq 1$; thus $k = 1$ starts at $n = 2$.

Theorem 5.3.9. *The admissible moduli are exactly* $1, 3, 4, 6, 8, 12, 24$.

Proof. Suppose $k > 1$ is admissible. Choose any unit r modulo k. Dirichlet's theorem provides distinct primes $p \equiv r \pmod{k}$ and $q \equiv -r^{-1} \pmod{k}$. Their product is $kn - 1$ for a positive integer n. Admissibility gives

$$0 \equiv \sigma(pq) = (p+1)(q+1) \equiv (r+1)(1 - r^{-1}) = r - r^{-1} \pmod{k}.$$

Thus every unit modulo k satisfies $r^2 \equiv 1 \pmod{k}$. Dirichlet's theorem is a classical input; for a formalized primary account see Carneiro, arXiv:1608.02029.

If an odd prime ℓ divides k, CRT lifts each nonzero residue modulo ℓ to a unit modulo k. The polynomial $X^2 - 1$ has at most two roots over $\mathbb{F}_\ell$, forcing $\ell = 3$. The factor 9 is impossible because the unit 4 is not a square root of 1 modulo 9; the factor 16 is impossible because $3^2 \not\equiv 1$ modulo 16. CRT again lifts these local obstructions. Therefore $k \mid 24$. The candidate $k = 2$ fails at $n = 1$, since $\sigma(1) = 1$.

Conversely, let $k \in \{3, 4, 6, 8, 12, 24\}$ and $m = kn - 1 > 0$. Every unit modulo such k has square 1. A divisor d of m is a unit, and its complementary divisor satisfies $m/d \equiv -d^{-1} \equiv -d \pmod{k}$. Their sum is divisible by k. There is no unpaired divisor: m cannot be a square because $m \equiv -1$ modulo 3 or modulo 4, and at least one of these moduli divides k. Pairing all divisors proves $k \mid \sigma(m)$. The case $k = 1$ is immediate. □

5.3.10 The prime support restriction in A300984

An integer N belongs to A300984 when the sum $U(N)$ of its squarefree divisors and the sum $V(N)$ of its nonsquarefree divisors are both positive squarefree integers.

Theorem 5.3.10. *Every such N has the conjectured form $2^i p^j$ for an odd prime p. In fact $i, j \geq 2$ and $p \equiv 1 \pmod{12}$ are necessary.*

Proof. Unique factorization gives $U(N) = \prod_{p|N}(1+p)$. Two distinct odd primes dividing N would make $4 \mid U(N)$, contradicting squarefreeness. Thus N has at most one odd prime factor, which proves the proposed form (allowing initially zero exponents).

For the stronger necessary conditions, a pure power of 2 has either $V = 0$ or $4 \mid V$, and an odd prime power has either $V = 0$ or $p^2 \mid V$. The value $N = 1$ also has $V = 0$. Hence both primes occur. If $N = 2p^j$, then $V(N) = 3p^2\sigma(p^{j-2})$ for $j \geq 2$, and $V = 0$ for $j = 1$. If $N = 2^i p$ with $i \geq 2$, then $V(N) = (2^{i+1} - 4)(p+1)$ is divisible by 4. This proves $i, j \geq 2$. Now $U(N) = 3(p+1)$ squarefree implies $4 \nmid p+1$ and $3 \nmid p+1$. It also excludes $p = 3$. Thus $p \equiv 1 \pmod 4$ and $p \equiv 1 \pmod 3$, giving $p \equiv 1 \pmod{12}$. □

These are necessary conditions. No converse or density claim about the allowed primes is asserted; the entry's separate majority-of-primes observation is not counted as resolved.

5.3.11 Remainder pairing in A339387

For $n \geq 1$ write

$$A(n) = \sum_{k=1}^{n} \left(\frac{\operatorname{lcm}(n,k)}{\gcd(n,k)} \bmod k \right), \qquad B(n) = \sum_{k=1}^{n} \left(\frac{\operatorname{lcm}(n,k)}{\gcd(n,k)} \bmod n \right).$$

These are respectively A339387 and A339384.

Theorem 5.3.11. *$B(n) \geq A(n)$ always. If n is not squarefree then $B(n) > A(n)$. Consequently the conjecture that $A(n) = B(n)$ implies n squarefree is true.*

Proof. The case $n = 1$ is immediate. For $n > 1$, group the terms by $d = \gcd(n,k)$, and put $g = n/d$, $k = dt$. For $g > 1$ the variable t runs over $1 \leq t < g$ coprime to g, and $\operatorname{lcm}(n,k)/\gcd(n,k) = gt$. The respective remainders are

$$t(g \bmod d), \qquad g(t \bmod d).$$

Set $r = g \bmod d$ and let $N_d = \#\{1 \leq t < g : (t,g) = 1,\ t \bmod d > r\}$. The involution $t \mapsto g - t$ preserves this set of totatives, and

$$(t \bmod d) + ((g-t) \bmod d) = r + d\,\mathbf{1}_{t \bmod d > r}.$$

Also $\sum_{(t,g)=1,\ 1 \leq t < g} t = g\varphi(g)/2$. It follows that the difference of the two grouped contributions is

$$g\sum (t \bmod d) - r\sum t = \frac{gd}{2} N_d = \frac{n}{2} N_d \geq 0.$$

The remaining group $g = 1$ consists of $k = n$ and contributes 1 to both sums. Therefore $B(n) - A(n) = \frac{n}{2} \sum_{d|n,\ n/d>1} N_d$.

If $p^2 \mid n$, choose $d = p$. Then $g = n/p$ is divisible by p, so $r = 0$, while the totative $t = 1$ has $t \bmod p = 1 > 0$. Thus $N_p \geq 1$, making the difference strictly positive. □

The existing prime-power evaluations and divisibility equivalences in the two entries were inspected; they are not counted again. Only the displayed implication and its strengthening by a nonnegative difference are selected.

5.3.12 Every prime occurs infinitely often in A394454's divisor sums

The entry A394454 sums the distinct prime divisors of $\binom{2n}{n}$. Its selected conjecture states that, for each prime p, divisibility by p occurs at infinitely many indices.

Theorem 5.3.12. *For every prime p and integer $r \geq 1$,*

$$v_p\binom{2(p^r-1)}{p^r-1} = r.$$

Thus each prime contributes to the defining sum infinitely often, and the sequence has infinite upper limit.

Proof. Put $n = p^r - 1$. Legendre's formula gives

$$v_p\binom{2n}{n} = \sum_{j\geq 1}\left(\left\lfloor\frac{2n}{p^j}\right\rfloor - 2\left\lfloor\frac{n}{p^j}\right\rfloor\right).$$

For $1 \leq j \leq r$ the two floors are respectively $2p^{r-j}-1$ and $p^{r-j}-1$, so each summand is 1. For $j > r$ both floors vanish, since $2(p^r-1) < p^{r+1}$. This proves the exact valuation, including $p = 2$. The indices $p^r - 1$ are distinct as r grows. Taking $r = 1$ and arbitrarily large primes gives $a(p-1) \geq p$, hence the final conclusion. □

The conclusion concerns occurrence as a prime divisor of the central binomial coefficient, hence as a summand in $a(n)$. It does not assert that each prime divides the integer $a(n)$ itself. This is an elementary application of classical factorial valuations, not a new theorem about the distribution of primes.

5.4 Thirty generating-function and recurrence packages

This section treats thirty distinct OEIS entries absent from the 408 previously counted entries. The count is 22 direct proofs, six boundary qualifications, one repair of a malformed printed factor, and one refutation with a proved replacement. Each entry counts once. The six boundary cases have no starting index specified in their source; we supply that index and retain all earlier admissible exceptions. The separate A129366 refutation is stronger: its original recurrence cannot become valid after any finite index. The source wording of A279013 contains an undefined token, which is repaired explicitly.

All selected statements were labeled *Conjecture* on their consulted primary OEIS entries. Known generating functions, combinatorial enumerations, closed forms and other recurrences are credited inputs. This is a selected-statement proof collection, with no assertion that the underlying generating functions or proof methods are new. The limited prior-work audit and the historical correction at the end of this section prevent source labels from being treated as guarantees of worldwide novelty. No external submission or acceptance is claimed.

5.4.1 A finite exact certificate proves an infinite recurrence

Let $F(x) = \sum_{n\geq 0} a_n x^n$, filling indices before the source offset with zero solely for formal-series notation. Put $\theta = x\,d/dx$ and

$$\mathcal{L}_p = \sum_{j=0}^{r} x^j p_j(\theta + j).$$

Since $\theta x^k = kx^k$, coefficient extraction gives

$$[x^n]\mathcal{L}_p F = \sum_{j=0}^{r} p_j(n) a_{n-j}. \tag{5.4}$$

An exact identity $\mathcal{L}_p F = P(x)$ with polynomial P therefore proves the recurrence for every $n > \deg P$ for which the source indices are defined. Coefficients of P give every exceptional initial residual.

For the 29 algebraic cases, write $x = \xi(t)$ and

$$F(\xi(t)) = \sum_{i=0}^{s} R_i(t)\sqrt{D_i(t)}, \qquad D_0 = 1.$$

The case without a change of variable means $t = x$ and $\xi = t$. All radicals below have constant term one at their indicated base point. Define rational operators and coefficients by

$$T_i R = \frac{\xi}{\xi'}\left(R' + \frac{D_i'}{2D_i}R\right), \qquad c_m(t) = \sum_{j=0}^{r} \xi(t)^j [n^m] p_j(n+j).$$

The chain rule proves $\theta(R\sqrt{D_i}) = (T_i R)\sqrt{D_i}$. Thus each supplied certificate consists of the exact rational identities

$$\sum_m c_m T_0^m R_0 = P(\xi(t)), \qquad \sum_m c_m T_i^m R_i = 0 \quad (i \geq 1). \tag{5.5}$$

Every p_j, R_i, D_i, ξ and the complete boundary polynomial P is printed below. Inserting these finite expressions and clearing denominators gives zero polynomials over $\mathbb{Q}$; this is a checkable algebraic certificate, not inference from a numerical prefix. Equation (5.4) then completes each all-index proof. The separate A034267 case uses a rational identity between successive hypergeometric terms.

For a fully expanded small illustration, A104624 has $F = \sqrt{1+4x^2} - 2x$ and

$$\mathcal{L}_p = (1+x)\big((1+4x^2)\theta - 4x^2\big).$$

Its action on $\sqrt{1+4x^2}$ is zero, because $\theta\sqrt{1+4x^2} = 4x^2/\sqrt{1+4x^2}$. Its action on $-2x$ is $-2x(1+x)$. This gives the complete boundary polynomial and proves the recurrence at every $n \geq 3$. The other algebraic entries apply precisely the same finite rule, with possibly more derivatives and radicals.

5.4.2 Indexing, seeds and parameter choices

The Catalan and Motzkin generating functions are

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}, \qquad M(x) = \frac{1-x-\sqrt{1-2x-3x^2}}{2x^2}.$$

For A162482, substituting $F = (1-x)^{-3}M(x/(1-x)^3)$ into $M = 1 + xM + x^2M^2$ gives $x^2F^2 - ((1-x)^3 - x)F + 1 = 0$, selecting the branch $F(0) = 1$. The A129366 definition gives $F = (C(x) - xC(x^2))/(1-x)$ by summing its Catalan partial sums. A134424 uses the existing statistic formula with $g = 1 + xg + x^2g + x^2g^2$, $g(0) = 1$; solving this quadratic gives the displayed radical data. The other ordinary algebraic seeds are the established generating functions identified in their individual source notes. A025259 requires multiplication of the source's offset-zero displayed GF by x, because its sequence begins at index 1.

For A243764, A270490, A270530, A279013 and A279014, set

$$t = xC(x) = \frac{1-\sqrt{1-4x}}{2}, \qquad x = \xi(t) = t(1-t), \qquad t(0) = 0.$$

Then $C(x) = 1/(1-t)$, so the nested square roots reduce to the rational radicands below. For example, A243764 becomes $F = -(1-t)(1+\sqrt{1-8t^2})/2$. The usual Catalan binomial kernel is

$$\sum_{n\geq 0} \binom{2n-1}{n-k} x^n = \frac{1-t}{1-2t}\left(\frac{t}{1-t}\right)^k \quad (k \geq 0),$$

using $\binom{-1}{0} = 1$ at the single initial case. Substituting the Catalan and Fibonacci generating functions for the k-weights gives the A279013 and A279014 representations. For A270530 the analogous kernel is

$$\sum_{n\geq 0} \binom{2n+2}{n-k} x^n = \frac{1}{(1-t)^2(1-2t)}\left(\frac{t}{1-t}\right)^k.$$

Both identities follow by the binomial expansion after $x = t(1-t)$ or by formal coefficient extraction; thus the sum definitions select the same analytic branches as the listed generating functions.

For A387084 let $g = 1 + xg^5$, $g(0) = 1$, and $t = g - 1$. Then $\xi(t) = t/(1+t)^5$, with $\xi'(0) = 1$. The central-binomial series $B = \sum_n \binom{5n}{n} x^n$ is $B = g/(5-4g)$, by Lagrange inversion. The defining expression $B/\sqrt{1+4(B-1)/5}$ therefore equals $g/\sqrt{5-4g}$. This proves the parameterized seed $F = (1+t)/\sqrt{1-4t}$. For every parameter used here $\xi'(0) = 1$, so the formal inverse exists. Apparent poles in individual R_i are interpreted in the Laurent field and cancel in F.

5.4.3 The individual statements and complete certificates

Each displayed recurrence means $\sum_j p_j(n) a_{n-j} = 0$ in the stated range. In the one refuted case the original coefficients are denoted q_j and the proved replacement coefficients are denoted p_j.

5.4.3.1 A025259: Sep 29 2012

Source: https://oeis.org/A025259. Selected conjecture: R. J. Mathar, Sep 29 2012. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

Boundary qualification. The source does not give a starting index. $n = 5$ gives residual 1. The exact valid tail is $n \geq 6$. No explicit smaller starting index is attributed to the source.

$$\begin{aligned}
p_0(n) &= n \\
p_1(n) &= -2\,(2n-3) \\
p_2(n) &= 6\,(n-3) \\
p_3(n) &= -4\,(2n-9) \\
p_4(n) &= n-6.
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned}
R_0 &= -\frac{x^2-2x-1}{2}, \\
R_1 &= -\frac{1}{2}, \\
D_1 &= x^4 - 8x^3 + 6x^2 - 4x + 1.
\end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = x\left(x^4 + 9x^2 - 6x + 2\right).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 6$. The source offset is 1.

5.4.3.2 A025272: Nov 21 2014

Source: https://oeis.org/A025272. Selected conjecture: R. J. Mathar, Nov 21 2014. Kotesovec already supplied the displayed generating function and a different order-four recurrence in 2015. Only the separately labeled Mathar order-five recurrence is selected.

$$\begin{aligned}
p_0(n) &= n \\
p_1(n) &= n + 1 \\
p_2(n) &= -10\,(2n - 5) \\
p_3(n) &= 2\,(2n - 9) \\
p_4(n) &= 2\,(12n - 67) \\
p_5(n) &= 20\,(n - 7)\,.
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned}
R_0 &= \frac{1}{2}, \\
R_1 &= -\frac{1}{2}, \\
D_1 &= 4x^4 + 4x^3 - 4x + 1.
\end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -x\,(5x + 1)\left(4x^3 + 3x^2 - 1\right).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 6$. The source offset is 1.

5.4.3.3 A034267: Feb 10 2025

Source: https://oeis.org/A034267. Selected conjecture: R. J. Mathar, Feb 10 2025. The diagonal formula follows directly from A034261, whose formula is credited to Michael Somos. The same diagonal formula was added by Amiram Eldar in 2025. These formulas are prior input.

$$\begin{aligned}
p_0(n) &= -\,(n + 2)\,(11n - 7) \\
p_1(n) &= 2\left(23n^2 + 44n + 30\right) \\
p_2(n) &= -4\,(n + 5)\,(2n - 3)\,.
\end{aligned}$$

Proof. For $n \geq 1$ the known formula is $a_n = C_n n(n^2 + n + 1)/(n + 2)$, where C_n is the Catalan number. Cancellation of factorials gives, for $n \geq 2$,

$$\frac{a_{n-1}}{a_n} = r(n) = \frac{(n-1)(n+2)(n^2-n+1)}{2n(2n-1)(n^2+n+1)}.$$

The same quotient gives $a_0/a_1 = r(1) = 0$ directly. The finite rational identity

$$p_0(n) + p_1(n)r(n) + p_2(n)r(n)r(n-1) = 0$$

follows by inserting the displayed polynomials and clearing denominators. Its denominators are nonzero for integer $n \geq 2$. Multiplication by a_n proves the recurrence for every $n \geq 2$. The source values $a_0 = 0, a_1 = 1, a_2 = 7$ also verify the first case.

5.4.3.4 A089324: Nov 24 2012

Source: https://oeis.org/A089324. Selected conjecture: R. J. Mathar, Nov 24 2012. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$\begin{aligned}
p_0(n) &= n+1\\
p_1(n) &= -3(n+1)\\
p_2(n) &= 13-5n\\
p_3(n) &= 18(n-2)\\
p_4(n) &= 7-5n\\
p_5(n) &= -3(n-5)\\
p_6(n) &= n-5.
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned}
R_0 &= -\frac{(x-1)^2}{2x(x^2-3x+1)},\\
R_1 &= \frac{1}{2x(x^2-3x+1)},\\
D_1 &= \left(x^2-2x-1\right)\left(x^2+2x-1\right).
\end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -(x-1)(x+1)\left(x^2-4x+1\right).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 6$. The source offset is 0.

5.4.3.5 A097331: Dec 02 2012

Source: https://oeis.org/A097331. Selected conjecture: R. J. Mathar, Dec 02 2012. Fung Lam already supplied the shorter step-two recurrence in 2014. We prove only the separately conjectured Mathar order-three recurrence from the defining generating function.

$$
\begin{aligned}
p_0(n) &= n+1\\
p_1(n) &= n\\
p_2(n) &= -4\,(n-2)\\
p_3(n) &= -4\,(n-3)\,.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$
\begin{aligned}
R_0 &= \frac{2x+1}{2x},\\
R_1 &= -\frac{1}{2x},\\
D_1 &= -\,(2x-1)\,(2x+1)\,.
\end{aligned}
$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = (x+1)\,(2x+1)\,.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 3$. The source offset is 0.

5.4.3.6 A102319: Feb 20 2015

Source: https://oeis.org/A102319. Selected conjecture: R. J. Mathar, Feb 20 2015. Only the first of the two listed conjectured recurrences, of order six, is counted. The second is outside the selected claim.

$$
\begin{aligned}
p_0(n) &= n\,(n-1)\\
p_1(n) &= -4\,(n-1)\,(3n-4)\\
p_2(n) &= 53n^2 - 221n + 232\\
p_3(n) &= -8\left(13n^2 - 85n + 134\right)\\
p_4(n) &= 51n^2 - 563n + 1308\\
p_5(n) &= 4\,(n-4)\,(29n-93)\\
p_6(n) &= -105\,(n-5)\,(n-4)\,.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$
\begin{aligned}
R_0 &= 0,\\
R_1 &= \frac{1}{2\,(x-1)\,(5x-1)},\\
R_2 &= -\frac{1}{2\,(x+1)\,(3x-1)},\\
D_1 &= (x-1)\,(5x-1)\,,
\end{aligned}
$$

$$D_2 = -(x+1)(3x-1).$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 0.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 6$. The source offset is 0.

5.4.3.7 A104624: Nov 15 2012

Source: https://oeis.org/A104624. Selected conjecture: R. J. Mathar, Nov 15 2012. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$\begin{aligned} p_0(n) &= n \\ p_1(n) &= n-1 \\ p_2(n) &= 4(n-3) \\ p_3(n) &= 4(n-4). \end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i \sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned} R_0 &= -2x, \\ R_1 &= 1, \\ D_1 &= 4x^2 + 1. \end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -2x(x+1).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 3$. The source offset is 0.

5.4.3.8 A107587: Feb 20 2015

Source: https://oeis.org/A107587. Selected conjecture: R. J. Mathar, Feb 20 2015. Eremin 2021, equation 2.10, proves the generating function used here. His OEIS check of the selected recurrence through 800 is finite verification, not used as an all-index proof. Only the first listed recurrence is selected.

$$\begin{aligned} p_0(n) &= n(n+2) \\ p_1(n) &= -5n^2 - n + 3 \\ p_2(n) &= 10n^2 - 16n + 3 \\ p_3(n) &= -10n^2 + 34n - 27 \end{aligned}$$

$$p_4(n) = -(n-3)(11n-5)$$
$$p_5(n) = 15(n-4)(n-3).$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned} R_0 &= 0, \\ R_1 &= \frac{1}{4x^2}, \\ R_2 &= -\frac{1}{4x^2}, \\ D_1 &= 5x^2 - 2x + 1, \\ D_2 &= -(x+1)(3x-1). \end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 0.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 5$. The source offset is 0.

5.4.3.9 A108600: Aug 24 2013

Source: https://oeis.org/A108600. Selected conjecture: R. J. Mathar, Aug 24 2013. The freely braided permutation enumeration and generating function are prior work of Green–Losonczy and Mansour. The current entry includes the initial value at zero, added by Heinz in January 2025. Only the later Mathar recurrence is targeted.

$$\begin{aligned} p_0(n) &= 1 - n \\ p_1(n) &= 7n - 10 \\ p_2(n) &= -2(4n-1) \\ p_3(n) &= -8(2n-11) \\ p_4(n) &= n - 1 \\ p_5(n) &= -3(n-2) \\ p_6(n) &= -2(2n-11). \end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned} R_0 &= -\frac{x^3 - 4x^2 - 7x + 2}{2(x^4 + 4x - 1)}, \\ R_1 &= -\frac{x(x+1)^2}{2(x^4 + 4x - 1)}, \\ D_1 &= 1 - 4x. \end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -x^5 + 15x^4 + 28x^3 - 12x^2 - 3x + 1.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 6$. The source offset is 0.

5.4.3.10 A116388: Jun 22 2016

Source: https://oeis.org/A116388. Selected conjecture: R. J. Mathar, Jun 22 2016. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$\begin{aligned}
p_0(n) &= n \\
p_1(n) &= -7\,(n-1) \\
p_2(n) &= 2\,(4n-9) \\
p_3(n) &= 25n - 58 \\
p_4(n) &= 65 - 18n \\
p_5(n) &= 199 - 52n \\
p_6(n) &= 135 - 31n \\
p_7(n) &= -6\,(n-5)\,.
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i \sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned}
R_0 &= -\frac{1}{2\,(x^3 + 3x^2 + 2x - 1)}, \\
R_1 &= -\frac{2x^2 + 3x - 1}{2\,(x+1)\,(3x-1)\,(x^3 + 3x^2 + 2x - 1)}, \\
D_1 &= -\,(x+1)\,(3x-1)\,.
\end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -x\left(3x^3 + 3x^2 + x - 1\right).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 7$. The source offset is 0.

5.4.3.11 A129366: Feb 05 2015

Source: https://oeis.org/A129366. Selected conjecture: R. J. Mathar, Feb 05 2015. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

Original conjecture refuted. Its original coefficients are:

$$\begin{aligned}
q_0(n) &= n\,(n-1)\,(12n + 35) \\
q_1(n) &= (n-1)\left(12n^2 - 701n + 1236\right)
\end{aligned}$$

$$
\begin{aligned}
q_2(n) &= 2\left(6n^3 - 385n^2 + 2285n - 3432\right)\\
q_3(n) &= -4\left(405n^3 - 5313n^2 + 19970n - 23175\right)\\
q_4(n) &= 8\left(156n^3 - 1724n^2 + 5498n - 5175\right)\\
q_5(n) &= 16\left(393n^3 - 4981n^2 + 20393n - 26820\right)\\
q_6(n) &= -32\,(n-5)\,(2n-9)\,(93n-268)\,.
\end{aligned}
$$

The defining Catalan sum gives $(a_0, \ldots, a_6) = (1, 1, 3, 7, 21, 61, 193)$. Exact substitution gives

$$\sum_{j=0}^{6} q_j(6)a_{6-j} = -73968 \neq 0.$$

This cannot be repaired by choosing a later starting index. Applying $\mathcal{L}_q$ to the same generating function gives

$$Q_1(x) = \frac{N(x)}{x(4x-1)^2}, \qquad Q_2(x) \in \mathbb{Q}(x),$$

$$
\begin{aligned}
N(x) = {} & 11040x^6 - 3200x^5 + 26064x^4\\
& - 8152x^3 - 2320x^2 + 756x - 23.
\end{aligned}
$$

Here $\mathcal{L}_q F = Q_1\sqrt{1-4x} + Q_2\sqrt{1-4x^2}$. The square classes of $1-4x$ and $1-4x^2$ are independent in $\mathbb{Q}(x)$: each factorization has simple roots, and their product also has a simple root. The biquadratic field therefore has an automorphism changing the sign of $\sqrt{1-4x}$ while fixing the other radical. If this residual were rational, that automorphism would force $2Q_1\sqrt{1-4x} = 0$, contrary to $Q_1 \neq 0$. It is consequently not a polynomial, so infinitely many recurrence residuals are nonzero. The original formula is not eventually valid.

Proved replacement. Write $F = U + V$ with the two radical terms specified below, and let $\ell_1 = U'/U$, $\ell_2 = V'/V$. Both are rational functions. Defining

$$A = -\frac{\ell_1' + \ell_1^2 - \ell_2' - \ell_2^2}{\ell_1 - \ell_2}, \qquad B = -\ell_1' - \ell_1^2 - A\ell_1$$

gives $U'' + AU' + BU = V'' + AV' + BV = 0$ directly. Clearing denominators, writing derivatives in terms of θ and extracting coefficients yields the following replacement. Its exact certificate is also verified independently by (5.5).

$$
\begin{aligned}
p_0(n) &= n\,(n+1)\\
p_1(n) &= -n\,(7n-5)\\
p_2(n) &= 2\,(n-2)\,(7n-9)\\
p_3(n) &= 12\,(4n-9)\\
p_4(n) &= -8\left(7n^2 - 41n + 63\right)\\
p_5(n) &= 16\left(7n^2 - 59n + 126\right)\\
p_6(n) &= -32\,(n-5)\,(2n-9)\,.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$
\begin{aligned}
R_0 &= 0,\\
R_1 &= \frac{1}{2x\,(x-1)},\\
R_2 &= -\frac{1}{2x\,(x-1)},\\
D_1 &= 1-4x,\\
D_2 &= -\,(2x-1)\,(2x+1)\,.
\end{aligned}
$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 0.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 6$. The source offset is 0.

5.4.3.12 A134424: Jul 24 2022

Source: https://oeis.org/A134424. Selected conjecture: R. J. Mathar, Jul 24 2022. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$
\begin{aligned}
p_0(n) &= -\,(n+2)\,(5n-7)\\
p_1(n) &= -\,(n+1)\,(5n-127)\\
p_2(n) &= 135n^2-655n-42\\
p_3(n) &= 2\left(5n^2-275n-108\right)\\
p_4(n) &= -725n^2+4941n-5734\\
p_5(n) &= -235n^2+1880n-1173\\
p_6(n) &= 725n^2-6659n+12606\\
p_7(n) &= 2\left(5n^2+195n-1988\right)\\
p_8(n) &= -135n^2+1505n-3358\\
p_9(n) &= -\,(n-9)\,(5n+87)\\
p_{10}(n) &= (n-10)\,(5n-33)\,.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$
\begin{aligned}
R_0 &= \frac{(x^2+1)\,(x^4+2x^3-3x^2-2x+1)}{2x^2\,(x^2-x-1)\,(x^2+3x-1)},\\
R_1 &= \frac{(x^2+1)\,(x^2+x-1)}{2x^2\,(x^2-x-1)\,(x^2+3x-1)},\\
D_1 &= \left(x^2-x-1\right)\left(x^2+3x-1\right).
\end{aligned}
$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 12x^2\left(x^2+1\right)\left(x^2+24x-1\right).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 10$. The source offset is 0.

5.4.3.13 A162482: Feb 10 2015

Source: https://oeis.org/A162482. Selected conjecture: R. J. Mathar, Feb 10 2015. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$\begin{aligned}
p_0(n) &= n+2\\
p_1(n) &= -4\,(2n+1)\\
p_2(n) &= 18\,(n-1)\\
p_3(n) &= -13\,(2n-5)\\
p_4(n) &= 17\,(n-4)\\
p_5(n) &= -3\,(2n-11)\\
p_6(n) &= n-7.
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned}
R_0 &= -\frac{x^3-3x^2+4x-1}{2x^2},\\
R_1 &= -\frac{1}{2x^2},\\
D_1 &= \left(x^3-3x^2+2x-1\right)\left(x^3-3x^2+6x-1\right).
\end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 2\,(x-1)^2\,(2x+1).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 6$. The source offset is 0.

5.4.3.14 A164586: Feb 10 2015

Source: https://oeis.org/A164586. Selected conjecture: R. J. Mathar, Feb 10 2015. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$\begin{aligned}
p_0(n) &= n+1\\
p_1(n) &= n-1\\
p_2(n) &= -2\,(2n+1)\\
p_3(n) &= -2\,(7n-10)
\end{aligned}$$

$$
\begin{aligned}
p_4(n) &= -14\,(n-2)\\
p_5(n) &= -2\,(2n-3)\\
p_6(n) &= n-7\\
p_7(n) &= n-5.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$
\begin{aligned}
R_0 &= -\frac{1}{2x\,(x+1)},\\
R_1 &= -\frac{1}{2x\,(x+1)\,(x^3+2x^2+2x-1)},\\
D_1 &= \left(x^3-2x^2-2x-1\right)\left(x^3+2x^2+2x-1\right).
\end{aligned}
$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = x^4+2x^3+2x+1.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 7$. The source offset is 0.

5.4.3.15 A174013: Jan 12 2013

Source: https://oeis.org/A174013. Selected conjecture: R. J. Mathar, Jan 12 2013. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$
\begin{aligned}
p_0(n) &= n+1\\
p_1(n) &= n\\
p_2(n) &= -12\,(n-1)\\
p_3(n) &= 3\,(3n-8)\\
p_4(n) &= 6\,(6n-23)\\
p_5(n) &= -11\,(n-2)\\
p_6(n) &= 253-49n\\
p_7(n) &= -20\,(n-6)\,.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$
\begin{aligned}
R_0 &= -\frac{x^2+3x-1}{2x\,(x^2+x-1)},\\
R_1 &= -\frac{1}{2x\,(x^2+x-1)},\\
D_1 &= 5x^4+6x^3-x^2-2x+1.
\end{aligned}
$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -(4x+1)\left(6x^4 - 3x^2 + x - 1\right).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 7$. The source offset is 0.

5.4.3.16 A182879: Jun 14 2016

Source: https://oeis.org/A182879. Selected conjecture: R. J. Mathar, Jun 14 2016. Kotesovec 2017 already gives a different order-four recurrence. Its existence is credited; the selected target is Mathar's order-five formula.

$$\begin{aligned}
p_0(n) &= (n-1)(310n - 781)\\
p_1(n) &= -882n^2 + 3845n - 3679\\
p_2(n) &= 214n^2 - 2903n + 3751\\
p_3(n) &= -358n^2 - 459n + 1955\\
p_4(n) &= 834n^2 - 3631n + 4065\\
p_5(n) &= -(n-3)(262n - 663).
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i \sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned}
R_0 &= 0,\\
R_1 &= -\frac{x\left(x^3 - 2x^2 - 1\right)}{\left(x^2 - 3x + 1\right)^2 \left(x^2 + x + 1\right)^2},\\
D_1 &= \left(x^2 - 3x + 1\right)\left(x^2 + x + 1\right).
\end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 0.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 5$. The source offset is 0.

5.4.3.17 A182894: Jun 14 2016

Source: https://oeis.org/A182894. Selected conjecture: R. J. Mathar, Jun 14 2016. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$\begin{aligned}
p_0(n) &= n\\
p_1(n) &= 3 - 4n\\
p_2(n) &= n - 3\\
p_3(n) &= -3\\
p_4(n) &= 3(5n - 14)
\end{aligned}$$

$$p_5(n) = 6\,(n-3)$$
$$p_6(n) = 6\,(n-4)$$
$$p_7(n) = -4\,(n-6)\,.$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$R_0 = \frac{x\,(x+1)}{4x^3+2x^2+2x-1},$$
$$R_1 = -\frac{1}{4x^3+2x^2+2x-1},$$
$$D_1 = \left(x^2-3x+1\right)\left(x^2+x+1\right).$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = x\left(2x^4+3x^2-x-1\right).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 7$. The source offset is 0.

5.4.3.18 A182904: Jun 14 2016

Source: https://oeis.org/A182904. Selected conjecture: R. J. Mathar, Jun 14 2016. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$p_0(n) = n$$
$$p_1(n) = n-2$$
$$p_2(n) = 10-7n$$
$$p_3(n) = -3\,(n-2)$$
$$p_4(n) = 2-n$$
$$p_5(n) = 14-5n$$
$$p_6(n) = 18-5n$$
$$p_7(n) = 3\,(n-4)\,.$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$R_0 = 0,$$
$$R_1 = \frac{1}{x^6-2x^5-x^4-x^2-2x+1},$$
$$D_1 = x^6-2x^5-x^4-x^2-2x+1.$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 0.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 7$. The source offset is 0.

5.4.3.19 A189176: Nov 15 2011

Source: https://oeis.org/A189176. Selected conjecture: R. J. Mathar, Nov 15 2011. Eldar 2025 gives the Catalan closed form for positive indices. The exceptional initial value at zero is essential to the boundary qualification below.

Boundary qualification. The source does not give a starting index. $n = 2$ gives residual -5. The exact valid tail is $n \geq 3$. No explicit smaller starting index is attributed to the source.

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= -3\,(3n - 1) \\ p_2(n) &= 10\,(2n - 3)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i \sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned} R_0 &= -\frac{x - 1}{2x}, \\ R_1 &= -\frac{5x - 1}{2x\,(4x - 1)}, \\ D_1 &= 1 - 4x. \end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -5x^2 - 2x + 1.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 3$. The source offset is 0.

5.4.3.20 A190166: Apr 09 2019

Source: https://oeis.org/A190166. Selected conjecture: R. J. Mathar, Apr 09 2019. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$\begin{aligned} p_0(n) &= 1 - n \\ p_1(n) &= 3n - 4 \\ p_2(n) &= -2\,(n - 1) \\ p_3(n) &= 3\,(n - 2) \\ p_4(n) &= -2\,(n - 3) \\ p_5(n) &= 3n - 8 \\ p_6(n) &= 3 - n. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i \sqrt{D_i}$, with $D_0 = 1$ and

$$R_0 = 0,$$

$$R_1 = \frac{x}{(x^2-3x+1)(x^2-x+1)(x^2+x+1)},$$
$$D_1 = \left(x^2-3x+1\right)\left(x^2+x+1\right).$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 0.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 6$. The source offset is 0.

5.4.3.21 A218185: Oct 08 2016

Source: https://oeis.org/A218185. Selected conjecture: R. J. Mathar, Oct 08 2016. Yildiz 2012, Proposition 6.13, supplies the generating function. Mathar's2016 recurrence is the selected target, with its initial boundary stated explicitly.

Boundary qualification. The source does not give a starting index. $n = 4$ gives residual -16. The exact valid tail is $n \geq 5$. No explicit smaller starting index is attributed to the source.

$$\begin{aligned}
p_0(n) &= n(n-2)(n-1)\\
p_1(n) &= -12(n-2)(n-1)(2n-5)\\
p_2(n) &= 4(n-2)\left(52n^2-312n+465\right)\\
p_3(n) &= -48(2n-7)\left(8n^2-56n+95\right)\\
p_4(n) &= 256(n-5)(2n-9)(2n-7).
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned}
R_0 &= \frac{4x-1}{4},\\
R_1 &= -\frac{1}{4},\\
R_2 &= \frac{1}{4},\\
R_3 &= \frac{1}{4},\\
D_1 &= 1-8x,\\
D_2 &= (4x-1)(8x-1),\\
D_3 &= 1-4x.
\end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -16x^4.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 5$. The source offset is 0.

5.4.3.22 A239201: Oct 07 2016

Source: https://oeis.org/A239201. Selected conjecture: R. J. Mathar, Oct 07 2016. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$\begin{aligned} p_0(n) &= 1 - n \\ p_1(n) &= 7n - 11 \\ p_2(n) &= 25 - 11n \\ p_3(n) &= 5\,(n-3)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i \sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned} R_0 &= -\frac{x}{2\,(x-1)}, \\ R_1 &= \frac{3x}{2\,(x-1)\,(5x-1)}, \\ D_1 &= (x-1)\,(5x-1)\,. \end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = x^2.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 4$. The source offset is 1.

5.4.3.23 A243764: Jun 07 2016

Source: https://oeis.org/A243764. Selected conjecture: R. J. Mathar, Jun 07 2016. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

Boundary qualification. The source does not give a starting index. $n = 6$ gives residual 24576. The exact valid tail is $n \geq 7$. No explicit smaller starting index is attributed to the source.

$$\begin{aligned} p_0(n) &= 245n\,(n-2)\,(n-1) \\ p_1(n) &= -140\,(n-2)\,(n-1)\,(28n-81) \\ p_2(n) &= 4\,(n-2)\left(5812n^2 - 41488n + 72261\right) \\ p_3(n) &= -16\left(3368n^3 - 45144n^2 + 191968n - 260859\right) \\ p_4(n) &= -128\left(152n^3 + 768n^2 - 16748n + 45483\right) \\ p_5(n) &= 2048\,(2n-11)\left(74n^2 - 715n + 1629\right) \\ p_6(n) &= -98304\,(n-7)\,(2n-13)\,(2n-11)\,. \end{aligned}$$

Use $x = \xi(t) = -t\,(t-1)$, $t(0) = 0$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned} R_0 &= \frac{t-1}{2}, \\ R_1 &= \frac{t-1}{2}, \\ D_1 &= 1 - 8t^2. \end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 12x^3\left(2048x^3 - 4096x^2 + 1352x - 105\right).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 7$. The source offset is 0.

5.4.3.24 A262768: Jun 07 2016

Source: https://oeis.org/A262768. Selected conjecture: R. J. Mathar, Jun 07 2016. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

Boundary qualification. The source does not give a starting index. $n = 2$ gives residual -96. The exact valid tail is $n \geq 3$. No explicit smaller starting index is attributed to the source.

$$\begin{aligned} p_0(n) &= -2\,(n+2)\,(49n - 95) \\ p_1(n) &= 325\,(n-1)\,(n+1) \\ p_2(n) &= 2\,(2n-5)\,(67n - 38)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$\begin{aligned} R_0 &= \frac{(x-1)\,(x+1)}{2x^2}, \\ R_1 &= \frac{2x^3 + 3x^2 + 2x - 1}{2x^2\,(4x-1)}, \\ D_1 &= 1 - 4x. \end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -96x^2.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 3$. The source offset is 0.

5.4.3.25 A270490: Jun 07 2016

Source: https://oeis.org/A270490. Selected conjecture: R. J. Mathar, Jun 07 2016. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

$$\begin{aligned}
p_0(n) &= 3n\,(n-2)\,(n+2)\\
p_1(n) &= -4\,(n+1)\left(8n^2-23n+12\right)\\
p_2(n) &= 16n\,(2n-5)\,(3n-4)\\
p_3(n) &= 8\,(2n-3)\,(4n-7)\\
p_4(n) &= -64\,(n-3)\,(2n-5)\,(2n-3)\,.
\end{aligned}$$

Use $x=\xi(t)=-t\,(t-1)$, $t(0)=0$.

The exact representation is $F=\sum_i R_i\sqrt{D_i}$, with $D_0=1$ and

$$\begin{aligned}
R_0 &= -\frac{1}{2t^2\,(2t-1)},\\
R_1 &= \frac{1}{2t^2\,(2t-1)},\\
D_1 &= -\,(2t-1)\,(2t+1)\,.
\end{aligned}$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -2x\,(8x-3)\,.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n\geq 4$. The source offset is 0.

5.4.3.26 A270530: Jun 07 2016

Source: https://oeis.org/A270530. Selected conjecture: R. J. Mathar, Jun 07 2016. Only the first listed recurrence is selected. The known Catalan-binomial sum and radical generating function are inputs.

$$\begin{aligned}
p_0(n) &= 2n\,(n+1)\,(2n+3)\\
p_1(n) &= -n\left(77n^2+27n-4\right)\\
p_2(n) &= 549n^3-987n^2+686n-168\\
p_3(n) &= -20\,(2n-3)\left(43n^2-104n+70\right)\\
p_4(n) &= 500\,(n-2)\,(2n-5)\,(2n-3)\,.
\end{aligned}$$

Use $x=\xi(t)=-t\,(t-1)$, $t(0)=0$.

The exact representation is $F=\sum_i R_i\sqrt{D_i}$, with $D_0=1$ and

$$R_0 = \frac{1}{2t\,(t-1)\,(2t-1)},$$

$$R_1 = \frac{1}{2t\,(t-1)^2\,(2t-1)},$$
$$D_1 = (t-1)\,(5t-1)\,.$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 0.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 4$. The source offset is 0.

5.4.3.27 A273351: Jun 06 2016

Source: https://oeis.org/A273351. Selected conjecture: R. J. Mathar, Jun 06 2016. The known bargraph generating function is an input. Only the first of the two separately conjectured recurrences is selected.

$$\begin{aligned}
p_0(n) &= n\,(13n-40)\\
p_1(n) &= -55n^2+211n-129\\
p_2(n) &= 38n^2-212n+255\\
p_3(n) &= -6n^2+56n-75\\
p_4(n) &= 13n^2-66n+3\\
p_5(n) &= -\,(n-6)\,(3n-13)\,.
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The exact representation is $F = \sum_i R_i\sqrt{D_i}$, with $D_0 = 1$ and

$$R_0 = -\frac{x+1}{2},$$
$$R_1 = -\frac{x^3+x^2+x-1}{2\,(x-1)\,(x^3+x^2+3x-1)},$$
$$D_1 = (x-1)\left(x^3+x^2+3x-1\right).$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = -28x^2.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 7$. The source offset is 2.

5.4.3.28 A279013: Jan 27 2020

Source: https://oeis.org/A279013. Selected conjecture: R. J. Mathar, Jan 27 2020. The defining formula or established generating function is credited input from the primary entry; only the selected conjectural recurrence is treated.

Printed-formula repair. The source has the undefined token `(2*n7)`. The coefficient of $a(n-4)$ below uses `(2*n-7)`. This repaired, well-defined formula is what is proved; the printed token is not silently normalized.

$$
\begin{aligned}
p_0(n) &= 2n\,(n+1)\,(2n-3)\\
p_1(n) &= -n\left(101n^2-312n+203\right)\\
p_2(n) &= 995n^3-5570n^2+9567n-4984\\
p_3(n) &= -2\left(2393n^3-19300n^2+50494n-42835\right)\\
p_4(n) &= 4\,(2n-7)\left(1408n^2-9889n+16690\right)\\
p_5(n) &= -2600\,(n-5)\,(2n-9)\,(2n-7)\,.
\end{aligned}
$$

Use $x=\xi(t)=-t\,(t-1)$, $t(0)=0$.

The exact representation is $F=\sum_i R_i\sqrt{D_i}$, with $D_0=1$ and

$$
\begin{aligned}
R_0 &= -\frac{(t-1)^2}{2t\,(2t-1)},\\
R_1 &= -\frac{t-1}{2t\,(2t-1)},\\
D_1 &= (t-1)\,(5t-1)\,.
\end{aligned}
$$

Inserting these data in (5.5) gives exactly

$$
\mathcal{L}_pF=P(x)=-6x^2\left(52x^2-14x+1\right).
$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n\geq 5$. The source offset is 0.

5.4.3.29 A279014: Mar 12 2017

Source: https://oeis.org/A279014. Selected conjecture: R. J. Mathar, Mar 12 2017. Only the first listed conjecture is selected. The Fibonacci-binomial definition and listed generating function are prior input.

Boundary qualification. The source does not give a starting index. $n=4$ gives residual 49. The exact valid tail is $n\geq 5$. No explicit smaller starting index is attributed to the source.

$$
\begin{aligned}
p_0(n) &= n\,(3n-62)\\
p_1(n) &= 3n^2+362n-247\\
p_2(n) &= -171n^2+220n+162\\
p_3(n) &= 417n^2-2570n+3551\\
p_4(n) &= 2\,(2n-7)\,(27n-59)\,.
\end{aligned}
$$

Use $x=\xi(t)=-t\,(t-1)$, $t(0)=0$.

The exact representation is $F=\sum_i R_i\sqrt{D_i}$, with $D_0=1$ and

$$
R_0=\frac{(t-1)^3}{(2t-1)\,(t^2-3t+1)},
$$

$$R_1 = 0,$$
$$D_1 = 1.$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = x^3 \left(49x - 159\right).$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 5$. The source offset is 0.

5.4.3.30 A387084: Aug 19 2025

Source: https://oeis.org/A387084. Selected conjecture: R. J. Mathar, Aug 19 2025. The already listed generating functions involving B and the quintic Fuss–Catalan series g are credited input. The selected claim is Mathar's August 2025 order-four recurrence.

$$\begin{aligned}
p_0(n) &= 3902464n\,(8n-5)\,(8n-3)\,(8n-1)\,(8n+1)\\
p_1(n) &= -1005260800000n^5 + 5475840000000n^4\\
&\quad - 13076601280000n^3 + 16038028320000n^2\\
&\quad + 2384377435440 - 9827447160880n\\
p_2(n) &= 16756875000000n^5 - 138724375000000n^4\\
&\quad + 465871453125000n^3 - 788390515625000n^2\\
&\quad + 665727720750000n - 221182599750000\\
p_3(n) &= -76019287109375n^5 + 773712158203125n^4\\
&\quad - 2961907958984375n^3 + 5069256591796875n^2\\
&\quad + 67940625000000 - 3308797753906250n\\
p_4(n) &= -95367431640625n^5 + 1621246337890625n^4\\
&\quad - 11005401611328125n^3 + 37288665771484375n^2\\
&\quad + 42583007812500000 - 63060607910156250n.
\end{aligned}$$

Use $x = \xi(t) = \frac{t}{(t+1)^5}$, $t(0) = 0$.

The exact representation is $F = \sum_i R_i \sqrt{D_i}$, with $D_0 = 1$ and

$$R_0 = 0,$$
$$R_1 = -\frac{t+1}{4t-1},$$
$$D_1 = 1 - 4t.$$

Inserting these data in (5.5) gives exactly

$$\mathcal{L}_p F = P(x) = 0.$$

All radical residuals in (5.5) vanish as rational identities. Therefore (5.4) proves the displayed recurrence for every integer $n \geq 4$. The source offset is 0.

5.4.4 Verification and scope of prior-work audit

The portable verifier uses exact rational arithmetic to check all29 algebraic certificates and the one hypergeometric identity. It compares each selected polynomial with the retained primary-source mathematical statement, treating the two explicit formula changes separately. It independently expands the generating functions through sixteen source terms for each entry, for 480 matched source terms, and checks ten of the sum definitions independently as well. These finite checks guard indexing and branch choices; the rational identities provide the proofs for all indices.

The thirty retained IDs are disjoint from the 408 earlier counted IDs, from the 33 explicit Ekhad–Yang 2017 proof IDs checked in the preceding round, and from the retained OEIS Open Appendix A comparison list. The last comparison covers only that stated appendix baseline, not every problem or result in the OEIS Open project. Current primary-entry proof notices were screened, and known covered entries were removed. The audit does not assert absence of a proof anywhere in the literature.

5.4.5 Retrospective correction to five Round IV priority assessments

The current A176604 page links Georg Fischer's dated 26 January 2020 derivation for a parameterized convolution family: https://oeis.org/A176604/a176604_2.txt. Its general differential equation and explicit parameter table cover the previously counted A176605, A176607, A176610, A176611 and A176612. For the last three, substituting $(m,k,j) = (0,1,1),(1,1,1),(2,1,1)$ gives exactly the Round IV recurrence coefficients. For the first two, substituting $(1,0,1)$ and $(4,0,1)$ gives

$$\mathcal{L}_{\text{Fischer}} = (1-x)\mathcal{L}_{\text{RoundIV}}.$$

Cancelling the common factor in Fischer's homogeneous equation for the radical part gives the order-four equations used in Round IV. All coefficient identities are retained in the accompanying historical audit and checked exactly. Therefore those five Round IV proofs remain mathematically valid reproductions, but their assessment as previously unresolved claims must be withdrawn. They are not counted in Round V. The family is excluded from the current hunt; its source retaining a conjecture label on neighboring entries does not establish novelty.

5.5 Forty-five empirical array recurrences

This section proves 45 different entry-level recurrence assertions marked *Empirical* on the cited OEIS pages, consulted on 20 October 2026. The underlying arrays differ in width, alphabet, or local condition. All 45 A-numbers are absent from the 408 entries counted in rounds I–IV. The argument uses one general transfer principle with two precisely specified state constructions. The count is a count of selected OEIS assertions, not a count of unrelated mathematical techniques. The consulted pages contain data links and no proof of the selected recurrence. This limited source audit does not establish global priority.

5.5.1 Signed clockwise differences: the new transfer construction

Forty-one entries count, or take a stated scalar multiple of the count of, arrays on $\{0,\dots,q-1\}$ with $n+1$ rows and fixed width w. For a block $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, define

$$e(a,b,c,d) = \#\{b-a, d-b, c-d, a-c\}.$$

These are ordinary signed integer differences; no reduction modulo q or absolute-value operation is used. The local requirement is $e(a,b,c,d) \in S$, where $S \subseteq \{1,2,3,4\}$ is specified below. The source normalization λ is one of 1, 1/2, 1/4 and is part of the definition, not an assumption about color orbits.

Let $H(r_1,\dots,r_w)=(r_w,\dots,r_1)$ and $C(r_1,\dots,r_w)=(q-1-r_1,\dots,q-1-r_w)$. Let $\mathcal{O}_{q,w}$ be the lexicographically least representatives of orbits under $\{I,H,C,HC\}$, and write $\gamma(r)$ for a representative. Horizontal reflection permutes and negates the clockwise differences; simultaneous complementation negates them. Hence each operation preserves their number, so it preserves the local condition. No other color permutation is used in this construction. For $r,s\in\mathcal{O}_{q,w}$ define

$$m(r)=\#\{v\in\{0,\dots,q-1\}^w:\gamma(v)=r\}, \tag{5.6}$$

$$T(r,s)=\#\left\{v\in\{0,\dots,q-1\}^w:\begin{array}{l}\gamma(v)=s,\\ e(r_i,r_{i+1},v_i,v_{i+1})\in S\quad(1\le i<w)\end{array}\right\}. \tag{5.7}$$

Every row in an orbit has the same number of legal continuations to each orbit: apply the same symmetry to both rows. Thus quotienting loses no information needed for counting. Summing the actual first-row labelings and adding each subsequent row gives, by induction,

$$a(n)=\lambda m^{\mathsf{T}}T^n\mathbf{1}\qquad(n\ge 1). \tag{5.8}$$

The n transitions correspond to $n+1$ rows. The vector m is not replaced by a uniform orbit size; fixed rows and two-element orbits are counted with their true sizes.

5.5.2 Four entries with colors renamed in first-occurrence order

The remaining four entries restrict the number of distinct *cell values* in each block and identify global color renamings by reading the array row by row. Their state construction adapts the method developed for the Round IV array packet. For j available labeled colors let $\mathcal{R}_{j,w}$ be the row patterns obtained by renaming colors in order of first appearance, and write $\mathrm{can}(v)$ for this row pattern. For $r\in\mathcal{R}_{j,w}$ with $k(r)$ distinct colors set $m_j(r)=(j)_{k(r)}=j!/(j-k(r))!$, and define

$$T_{j,w,S}(r,s)=\#\left\{v\in\{0,\dots,j-1\}^w:\begin{array}{l}\mathrm{can}(v)=s,\\ \#\{r_i,r_{i+1},v_i,v_{i+1}\}\in S\quad(1\le i<w)\end{array}\right\}. \tag{5.9}$$

All successor rows are enumerated, including rows using colors absent from the previous row. Simultaneous color permutations preserve this local condition. The same induction therefore gives the labeled count $L_j(n)=m_j^{\mathsf{T}}T_{j,w,S}^n\mathbf{1}$. Dividing a surjective labeled count by $q!$, and applying inclusion–exclusion to omitted colors, gives

$$U_q^{=}(n)=\sum_{j=1}^{q}\frac{(-1)^{q-j}}{j!(q-j)!}L_j(n), \tag{5.10}$$

$$U_q^{\le}(n)=\sum_{j=1}^{q}\left(\frac{1}{j!}\sum_{h=0}^{q-j}\frac{(-1)^h}{h!}\right)L_j(n). \tag{5.11}$$

The second identity is the sum of the exact-color identity over the actual number of colors used. In particular, the requirement to use all q colors is imposed only when the source says so. The empty alphabet contributes zero because the arrays are nonempty.

5.5.3 The all-index proof from exact finite certificates

For the coefficient vector $c=(c_1,\dots,c_d)$ printed for an entry, set

$$R(z)=z^d-\sum_{i=1}^{d}c_iz^{d-i},\qquad Q(x)=1-\sum_{i=1}^{d}c_ix^i.$$

For its edge-difference matrix, or for every color matrix carrying a nonzero weight in (5.10)–(5.11), the complete exact integer certificate establishes

$$\boxed{TR(T)\mathbf{1}=\mathbf{0}.} \tag{5.12}$$

For $n \geq d+1$, multiply (5.12) by $m^{\mathsf{T}} T^{n-d-1}$, then by the appropriate scalar weight, and sum the color components when necessary. Equations (5.8)–(5.11) give

$$\boxed{a(n) = \sum_{i=1}^{d} c_i a(n-i) \quad (n \geq d+1).} \tag{5.13}$$

All indices on the right are at least 1, as required by every selected entry's offset. We make no use of a source value $a(0)$. The transfer formula can have an auxiliary nonintegral value at $n = 0$ for a source normalized by $1/2$; this does not affect any asserted sequence value.

The proof package contains all 56 integer matrices (42,001 nonzero weighted edges), all Krylov vectors $v_k = T^k \mathbf{1}$ for $0 \leq k \leq d+1$, and every residual coordinate in (5.12). The largest single matrix has 574 states and the largest recurrence has order 72. The complete matrices are in `r5_arrays_matrices.json`; the vectors, coefficient lists, weights and residuals are in `r5_arrays_certificates.json`. These are finite proof certificates, supplied as part of this proof. The verifier rebuilds every state and every transition directly from (5.7) or (5.9), then performs the following exact calculation:

```
v[0] = [1 for each state]
for k = 1,...,d+1:
    v[k][r] = sum(T[r,s] * v[k-1][s] for all states s)
for every state r:
    assert v[d+1][r] == sum(c[i] * v[d+1-i][r] for i=1,...,d)
```

The matrix definitions, complete coefficient signatures below and this finite algorithm specify the entire certificate check without any fitted terms or numerical approximations. Multiplication by further powers of T is the reason the conclusion holds for every indicated index. The separate comparison with all 687 printed source-prefix terms checks the source interpretation and indexing; it is not used to infer the recurrences. The 20-term computed prefixes are pairwise distinct, so none of these 45 selected sequences are identical to another in this section.

For $F(x) = \sum_{n \geq 1} a(n) x^n$, the proved recurrence yields $QF = P$, where $\deg P \leq d$ and

$$[x^k]P(x) = \sum_{\substack{0 \leq i \leq \min(k,d) \\ k-i \geq 1}} q_i a(k-i), \qquad q_0 = 1, \quad q_i = -c_i \ (i \geq 1).$$

The initial values in this formula are calculated from the defining matrices. All 17 rational generating functions marked empirical or conjectural on the retained pages agree exactly with this P/Q. The full numerator, denominator and initial-value lists are in the certificate file. These generating functions add no extra entry count. Other assertions on a page, such as a separately displayed closed form, are outside the selected claim unless expressly listed as proved here.

5.5.4 A fully displayed example: A209953

Here $q = 3$, $w = 2$, $S = \{1, 3, 4\}$ and $\lambda = 1$. The four representatives are $00, 01, 02, 11$, with orbit sizes $2, 4, 2, 1$. The exact construction gives

$$T = \begin{pmatrix} 2 & 4 & 2 & 1 \\ 2 & 2 & 2 & 1 \\ 2 & 4 & 1 & 1 \\ 2 & 4 & 2 & 1 \end{pmatrix}, \qquad m = \begin{pmatrix} 2 \\ 4 \\ 2 \\ 1 \end{pmatrix}.$$

Direct multiplication gives

$$v_1 = \begin{pmatrix} 9 \\ 7 \\ 8 \\ 9 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 71 \\ 57 \\ 63 \\ 71 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 567 \\ 453 \\ 504 \\ 567 \end{pmatrix} = 6v_2 + 15v_1 + 6v_0.$$

Therefore $R(T)\mathbf{1} = 0$ for $R(z) = z^3 - 6z^2 - 15z - 6$, which in particular implies (5.12). The initial sequence values are $71, 567, 4521$, and

$$F(x) = \frac{x(71 + 141x + 54x^2)}{1 - 6x - 15x^2 - 6x^3}.$$

This proves the source recurrence for $n \geq 4$ and its empirical generating function. The matrices for larger widths are checked by exactly the same coordinate identity.

5.5.5 Source audit, exclusions and earlier work

The source status is the explicit word *Empirical* attached to the particular recurrence, not the OEIS editorial status "approved." The audit compares all IDs against the 408 previously counted entries. It also finds no intersection with the stored list of 97 distinct A-numbers appearing in the 100 costly claims in Appendix A of *OEIS Open*; that is a limited appendix audit, not an audit of that project's entire repository or all mathematical literature. A209722 was excluded because its current page explicitly identifies it as an elementary shifted form of a named parent sequence.

A210082, A210263 and A210305 were investigated provisionally during Round IV but were not in its final counted 100. Their current source claims and complete certificates were rebuilt here. Their inclusion means first inclusion in our counted packets, not first observation within this project. The color-orbit method and part of its software are credited to the earlier Round IV packet. No OEIS entry was edited or submitted by this work, and no official OEIS closure is claimed.

5.5.6 Individual assertions and complete recurrence signatures

For each entry, the coefficient list is $c_1, \ldots, c_d$ in that order, including zero coefficients. The given matrix dimensions refer to all nonzero weighted components. Every displayed signature satisfies (5.12), and the source's $a(1), a(2), a(3)$ are also printed to fix the indexing unambiguously.

5.5.6.1 A209505

Arrays on $0, \ldots, 2$ have $n + 1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 4\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 3$, with order $d = 2$ and component dimensions (10) (total $D = 10$). Source: https://oeis.org/A209505. The first three values are $(27, 123, 561)$.

$$(c_1, \ldots, c_d) = (5, -2).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.2 A209506

Arrays on $0, \ldots, 2$ have $n + 1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 4\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 4$, with order $d = 3$ and component dimensions (25) (total $D = 25$). Source: https://oeis.org/A209506. The first three values are $(81, 561, 3906)$.

$$(c_1, \ldots, c_d) = (9, -15, 6).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.3 A209507

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 5$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 4\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 6$, with order $d = 5$ and component dimensions (70) (total $D = 70$). Source: https://oeis.org/A209507. The first three values are $(243, 2559, 27225)$.

$$(c_1, \ldots, c_d) = (16, -65, 92, -48, 8).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.4 A209508

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 6$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 4\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 8$, with order $d = 7$ and component dimensions (196) (total $D = 196$). Source: https://oeis.org/A209508. The first three values are $(729, 11673, 189801)$.

$$(c_1, \ldots, c_d) = (30, -291, 1278, -2901, 3519, -2152, 516).$$

5.5.6.5 A209509

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 7$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 4\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 14$, with order $d = 13$ and component dimensions (574) (total $D = 574$). Source: https://oeis.org/A209509. The first three values are $(2187, 53247, 1323270)$.

```
55, -1109, 11330, -67206, 247404, -582440, 881876, -846764, 499200, -172400, 33152,
-3264, 128
```

5.5.6.6 A209530

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 4$, with order $d = 3$ and component dimensions (10) (total $D = 10$). Source: https://oeis.org/A209530. The first three values are $(9, 25, 65)$.

$$(c_1, \ldots, c_d) = (1, 8, -8).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.7 A209531

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 5$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 4$, with order $d = 3$ and component dimensions (70) (total $D = 70$). Source: https://oeis.org/A209531. The first three values are $(33, 193, 1025)$.

$$(c_1, \ldots, c_d) = (1, 32, -32).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.8 A209546

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 2$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 5$, with order $d = 4$ and component dimensions (6) (total $D = 6$). Source: https://oeis.org/A209546. The first three values are $(7, 20, 61)$.

$$(c_1, \ldots, c_d) = (5, -6, 0, 2).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.9 A209547

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 7$, with order $d = 6$ and component dimensions (20) (total $D = 20$). Source: https://oeis.org/A209547. The first three values are $(20, 105, 562)$.

$$(c_1, \ldots, c_d) = (8, -14, -3, 20, -8, -2).$$

5.5.6.10 A209548

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 12$, with order $d = 11$ and component dimensions (72) (total $D = 72$). Source: https://oeis.org/A209548. The first three values are $(61, 562, 5075)$.

```
15, -59, 27, 267, -488, 111, 312, -174, -30, 18, 2
```

5.5.6.11 A209549

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 5$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 21$, with order $d = 20$ and component dimensions (272) (total $D = 272$). Source: https://oeis.org/A209549. The first three values are $(191, 3051, 46515)$.

```
26, -177, 43, 3477, -10179, -6160, 62922, -59442, -82557, 141262, 24872, -112175, 5660,
41275, -1936, -7049, -319, 400, 56, 2
```

5.5.6.12 A209710

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{4\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 20$, with order $d = 19$ and component dimensions (20) (total $D = 20$). Source: https://oeis.org/A209710. The first three values are $(175, 2710, 40600)$.

```
9, 155, -718, -5017, 21111, 51148, -251141, -85963, 1092718, -438881, -1923975, 1461077,
1300250, -1385402, -184926, 434126, -73408, -12888, 1440
```

5.5.6.13 A209711

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{4\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 65$, with order $d = 64$ and component dimensions (72) (total $D = 72$). Source: https://oeis.org/A209711. The first three values are $(1241, 40600, 1277340)$.

```
15, 1008, -7876, -328904, 1926537, 52684864, -273951748, -4866393539, 24240653273,
284467521073, -1401913368926, -11199860935130, 55383623152305, 310221810205502,
-1550263153851120, -6235493002893665, 31634306714134207, 92998581142818874,
-480833347610958673, -1045915789998724699, 5532194062321947358, 8975769923693909839,
-48756799709225176849, -59299615298869844599, 332041426928348290121,
3037006875654294880003, -175825632478749823 7829, -121281119089979387 80012,
727046541274161986 4800, 379756669211429077 5489, -235378145480927488 56931,
-937951158230750647 2416, 597266610846821970 89538, 184019732909767092 36198,
-118737454342588779 056334, -289149586872464611 24391, 184561198236942732 186820,
366992651249140243 12999, -223429523113044656 251384, -378618050986156031 12860,
209387912188279839 779389, 317518879561621173 88848, -150583340090115952 826545,
-21431798119336812426970, 82105995000110413699734, 11401867924374467330034,
-33394113408712074250948, -4638404024514496142694, 9913173513982954582346,
1389760315203109342656, -2086440149254353119856, -293279980685597445408,
299468489710238488568, 41250131799294016000, -27806351608888814224,
-3591128372719351552, 1556040144706074432, 173746695509329664, -48068935504144192,
-3984150632877312, 742241025193984, 34721251272704, -4259930669056, -57036898304
```

5.5.6.14 A209851

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 2$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 4\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 7$, with order $d = 6$ and component dimensions (6) (total $D = 6$). Source: https://oeis.org/A209851. The first three values are $(25, 182, 1308)$.

$$(c_1, \ldots, c_d) = (4, 29, -28, -110, 82, -8).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.15 A209852

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 4\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 21$, with order $d = 20$ and component dimensions (20) (total $D = 20$). Source: https://oeis.org/A209852. The first three values are $(182, 2829, 43091)$.

```
11, 139, -972, -4071, 28751, 27819, -332527, 170449, 1380897, -1533070, -2378942,
3871609, 1392648, -4192842, 445370, 1893206, -633950, -236300, 99664, 2096
```

5.5.6.16 A209853

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 4\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 69$, with order $d = 68$ and component dimensions (72) (total $D = 72$). Source: https://oeis.org/A209853. The first three values are $(1308, 43091, 1386131)$.

```
20, 963, -12793, -312156, 3571443, 49605457, -553758368, -4457972689, 52612881372,
243417930360, -3275949406092, -8240107440663, 140666492498369, 160116943902217,
-4326535316148209, -817021352845345, 98006065095893985, -49203318829660909,
-1668148030882549725, 1723999493198171149, 21637008392085651838, -31773088576337392703,
-215843167651091106385, 399719679271900244237, 1664018938648323383901,
-3697329744805424755681, -9917310558011937270066, 25982170466786070131093,
```

```
4544797834652933196509l, -1411176918800015053155 99, -1578565879896869778899 53,
5980696707160017211496 53, 4019687170890206372792 50, -1987776050324662023234413,
-6866605719169720083225 47, 5191091867684820767344742, 5227332626326359908208 08,
-1064606943226382371117 7799, 9158090681377004172077 61, 1709857159952926933868 6083,
-3943750897982196417975435, -2139868517510661426234 9856, 7368362612864864601239602,
2070987906896948147244 7106, -9015916934517168198203128, -1533524786730745550575 0474,
7824208837288323351242620, 8561662593050307213736807, -4920492104705864543228925,
-3532336472390984912564596, 2242642650064493959464073, 1047216070797663839803065,
-7317758981014001599822 55, -2141078734940350078635 03, 1671321113134579735605 40,
2823856479497440082797 5, -2582555400325895241566 6, -2098980376078516736024,
2571400424053876584470, 52829982196209769868, -153859888970921746848,
2954600261893327248, 4977837135363185688, -211003278535383648, -72160266595102240,
3672896320734208, 335148663461120, -16066999343104
```

5.5.6.17 A209906

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 2$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 4\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 7$, with order $d = 6$ and component dimensions (6) (total $D = 6$). Source: https://oeis.org/A209906. The first three values are $(62, 562, 5008)$.

$$(c_1, \ldots, c_d) = (7, 21, -30, -30, 12, 8).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.18 A209907

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 4\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 20$, with order $d = 19$ and component dimensions (20) (total $D = 20$). Source: https://oeis.org/A209907. The first three values are $(562, 13098, 295448)$.

```
16, 227, -1440, -8244, 46409, 53926, -406100, -117231, 1484559, 166049, -2587430,
-386782, 2096907, 472125, -660956, -166954, 44616, 4488, -192
```

5.5.6.19 A209908

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 4\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 66$, with order $d = 65$ and component dimensions (72) (total $D = 72$). Source: https://oeis.org/A209908. The first three values are $(5008, 295448, 16664508)$.

```
55, 923, -47071, -231407, 15527254, 7880250, -2573669497, 3681229288, 249190408738,
-579450712511, -15545777999632, 42210038348626, 666118006004289, -1886332941998678,
-20450809195283436, 57107262824986158, 462837772656461540, -1233589210852233412,
-787241057201699273 4, 196309908041632912 43, 101986218315556639540,
-235100512508203944296, -101568279339048812 5566, 215000278974668190 5996,
782572174495150649823 4, -151631644221392971 44181, -468416163433125217 46791,
829951110443600457347 10, 218301161066990302616716, -353790955288161781138915,
-792601249983629087376567, 117594614598568837864 4800, 223998234361986090002 2721,
-304544638900622243925 2829, -491670157736285857325 6478, 613058914066465741379 0376,
835391830945520080352 7969, -955613200893580312168 4739, -109390499159292905008 08672,
114746886824559750321576 76, 109781249271943048786 29570, -105438318867601131615 36049,
-838398870073183063885 9763, 735225336882648323497 8608, 482672512928405250139 7921,
-384946305692262370981 0553, -206785415067596249209 7465, 149295267044449963283 8573,
647468497254091793151276, -421469451029786639157580, -144455861921375175788532,
846613072117070067600 20, 221568251806546492392 87, -117404397821167850760 12,
-221966654157521624312 8, 1077870975515912576792, 134607253301714192944,
-61608386055612989120, -4358301391935808576, 1995845043302066048, 56288530795526656,
-31330477096792576, 102010241398784, 162473872068608, -2652426567680
```

5.5.6.20 A209953

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 2$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 3, 4\}$, and the count is multiplied by $\lambda = 1$. The recurrence (5.13) holds for every $n \geq 4$, with order $d = 3$ and component dimensions (4) (total $D = 4$). Source: https://oeis.org/A209953. The first three values are $(71, 567, 4521)$.

$$(c_1, \ldots, c_d) = (6, 15, 6).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.21 A209955

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 3, 4\}$, and the count is multiplied by $\lambda = 1$. The recurrence (5.13) holds for every $n \geq 15$, with order $d = 14$ and component dimensions (25) (total $D = 25$). Source: https://oeis.org/A209955. The first three values are $(4521, 260201, 14922177)$.

```
48, 720, -8224, -127687, -63722, 3510704, 8214978, -17816696, -46577704, 37523312,
70734912, -30102144, -26296320, 10088448
```

5.5.6.22 A209956

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 5$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 3, 4\}$, and the count is multiplied by $\lambda = 1$. The recurrence (5.13) holds for every $n \geq 37$, with order $d = 36$ and component dimensions (70) (total $D = 70$). Source: https://oeis.org/A209956. The first three values are $(36057, 5569307, 856216289)$.

```
188, -2606, -449432, 4278439, 344696470, -1695340318, -106831277262, 260315668742,
14665133182628, -32541316090784, -1037044125842224, 3010738321041520, 39283976226587808,
-157080544206380960, -734266820477791744, 4221150804291177792, 4462953823466462848,
-55438390445568782720, 33810272263704976640, 347725953909047112192,
-562768188874634440704, -933791709840933227520, 2653805933866673033216,
4187553956154561863 68, -552152700194524173926 4, 241840731062986833920 0,
527864509251331191603 2, -422682259285020114944 0, -208534895177273861734 4,
261579649087842969190 4, 190791165053952327680, -696828367550619320320,
49927158958133346304, 76557678286106263552, -6934923112613937152, -2692851001743900672
```

5.5.6.23 A210070

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 2$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 3\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 7$, with order $d = 6$ and component dimensions (6) (total $D = 6$). Source: https://oeis.org/A210070. The first three values are $(33, 313, 2908)$.

$$(c_1, \ldots, c_d) = (9, 9, -50, -54, 12, 16).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.24 A210071

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 3\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 21$, with order $d = 20$ and component dimensions (20) (total $D = 20$). Source: https://oeis.org/A210071. The first three values are $(313, 7653, 183628)$.

```
40, -410, 162, 13887, -37312, -114069, 441850, 251010, -1865698, 353991, 3283497,
-1672828, -2287997, 1529824, 444317, -334882, -35990, 21286, 648, -288
```

5.5.6.25 A210072

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 3\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 72$, with order $d = 71$ and component dimensions (72) (total $D = 72$). Source: https://oeis.org/A210072. The first three values are $(2908, 183628, 11333893)$.

```
105, -2719, -15451, 1390943, -7924538, -266634929, 2783228062, 26184293659,
-399963589595, -1390685894189, 33792993132607, 30902632289869, -1890381555307832,
822619990486604, 74406664819656249, -89835422690076184, -2138656368143245036,
3544878797500349305, 46008317343363322887, -87903800402488590784,
-753473801980270591181, 1535436320348069965003, 9505348644817473245840,
-19817177440215843837829, -9312684295494448197374 5, 1940084727816302322528 45,
7124176299517871896970 06, -1464351302320119255078001, -4269039597123343663401254,
8611866713944156678864641, 2006233736929729296377 37396, -3972187051477659376 3639623,
-7388361801328615545107 9859, 1441864628568934146137 30955, 2125945752854552730509 00740,
-4121628062442283230320 35765, -4753627275576717304535 08543, 9260675831987936388315 71069,
8189436698209925627980 06291, -1628407201463016755238157088,
-1073377817747961549292055653, 2225907870727063061101510830,
1050655346169817900693995129, -2343867586300105570589047976,
-7464465281114419949894 43963, 1879795853747926271266149480, 3664266352858165050702131 67,
-1132874535089578641767501432, -1113239557332405416951092 81,
5051946389088219971084693 49, 1292762638247773374024097 9, -1638877994752139234548103 67,
4251111577638305441346089, 3796668138792674473430690 2, -2249309141628534316063828,
-6153601732703873799844070, 4762547148283615579413 33, 6811774572102320998399 86,
-5689789194824920199322 2, -4991056781550675854156 0, 4016897111990219118375,
2316946754025069893095, -163269106694177521346, -63879794902532341859,
3536992747961553187, 943996081685851752, -35450092561754998, -6289331583619410,
135657995058528, 14840999586360, -115072392784
```

5.5.6.26 A210082

Arrays have $n+1$ rows, width $w = 5$, exactly 4 colors renamed in row-major first-occurrence order, and $S = \{1, 3, 4\}$ distinct cell values per block. The recurrence (5.13) holds for every $n \geq 54$, with order $d = 53$ and component dimensions $(1, 16, 41, 51)$ (total $D = 109$). Source: https://oeis.org/A210082. The first three values are $(9405, 2718363, 735082380)$.

```
357, -24956, 287902, 26074079, -902939413, 1277468261, 399614425317, -6321060969799,
-24184081408184, 1587694606642173, -12981021635371611, -72434732281423272,
1851336712434285028, -8298795143833957130, -55370910495748475918, 6998857607776991950 37,
-12715893804894068989 05, -16850531904836117165732, 9743752975716317505 0052,
4842157519209399203743 5, -19744877106666877857698347, 4494801106206276016732911,
1597252705216752258074 5893, -85997411174773544395221229, 8070664493704334395329420,
7101347578676764190601 79075, -11614298861841527182418295 91,
-2761558002058205806660337818, 960746470117128279638642 4036,
1672650823286895899011692366, -38888570744662356012685589228,
2959477081152164902005 2740416, 8595772260087950306008623 6632,
-1328350613195718275636427 60504, -9141545235335324770140525 4720,
2821715667064588150564685 21216, -6031025209778057302947810368,
-3464432812457907595312691 00096, 1391250287292439712144438 53952,
2524885785028032352933387 45600, -1723479740037403534842835 47648,
-1045287262638908308514234 99264, 1033927291490845015491312 03584,
2142615335027805158016221 3888, -3378666011966298222689246 8224,
-1224295150346030527568994304, 6066487242636218074798030848,
-2145849238672480840830812 16, -5690475857909174069860761 60, 3022267176640326394419 6096,
2464660038953011060408320 0, -8009595428775823398666 24, -3833478817218655546245 12
```

5.5.6.27 A210128

Arrays have $n+1$ rows, width $w = 3$, at most 4 colors renamed in row-major first-occurrence order, and $S = \{1, 3, 4\}$ distinct cell values per block. The recurrence (5.13) holds for every $n \geq 6$, with order $d = 5$ and component dimensions $(1, 4, 5)$ (total $D = 10$). Source: https://oeis.org/A210128. The first three values are $(81, 2609, 84381)$.

$$(c_1, \ldots, c_d) = (35, -81, -153, 492, -292).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.28 A210149

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 2$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 3, 4\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 6$, with order $d = 5$ and component dimensions (6) (total $D = 6$). Source: https://oeis.org/A210149. The first three values are $(57, 820, 11783)$.

$$(c_1, \ldots, c_d) = (11, 46, 36, -10, -8).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.29 A210150

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 3, 4\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 14$, with order $d = 13$ and component dimensions (20) (total $D = 20$). Source: https://oeis.org/A210150. The first three values are $(820, 42689, 2219034)$.

```
49, 200, -2256, -3750, 29143, -5802, -84658, 57585, 56482, -40584, -14146, 6952, 1280
```

5.5.6.30 A210151

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 3, 4\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 49$, with order $d = 48$ and component dimensions (72) (total $D = 72$). Source: https://oeis.org/A210151. The first three values are $(11783, 2219034, 417014920)$.

```
145, 8371, -28223, -5099034, -6848727, 1239650435, 930448375, -145737260866,
174912243963, 8264631635949, -22596313142751, -230989466281390, 920787848497955,
3145294614986535, -17388014234115316, -20446231184431219, 183336119039290408,
35747712256243592, -1196660533703535191, 358715517055014311, 5151447312682008123,
-2881286940145422064, -15254072639096425931, 10457263543037282604, 31951721178336409991,
-23277237121895822264, -48167665472082343099, 34598067257613352063,
52718006780308703710, -35619157696064520092, -41957052053167847864,
25829709988559823846, 24165370410041239916, -13259152815213783132, -9956951932632964712,
4797896668438751400, 2878768836324951744, -1208406171336576976, -567099023575525152,
207214883425141632, 72862743480025024, -23352420721108352, -5706693362437376,
1628318996755456, 241536697018368, -62481141760000, -4133487042560, 987100119040
```

5.5.6.31 A210263

Arrays have $n+1$ rows, width $w = 5$, exactly 4 colors renamed in row-major first-occurrence order, and $S = \{3\}$ distinct cell values per block. The recurrence (5.13) holds for every $n \geq 51$, with order $d = 50$ and component dimensions $(1, 16, 41, 51)$ (total $D = 109$). Source: https://oeis.org/A210263. The first three values are $(4298, 592969, 74289956)$.

```
208, -11777, 94953, 11282777, -300230858, -2030020415, 154714365717, -835409506883,
-32817201385328, 387387151685882, 3044940024025706, -64835114300366009,
-42786707245576098, 5751830532275106073, -16103656230733623821, -295460333548848141092,
1596987697112378159646, 8716400092711373757801, -76829522478341353618635,
-11702552736549564447848, 22553864342355039482115 82, -10905276409944798079797 72,
-42924560123670593643529800, 79873007729548579232819824, 53045274082537323745424 9528,
-16789960154164016234332153 28, -39745971598422659988074437 28,
20442608956556896189560069024, 12572243416557611196531929600,
```

```
-158760447373287871211600674496, 623717460188588188643273251840,
790240531776664873390830228736, -895213241848646797338489708544,
-2390400685449497424892238006272, 4529311964034293254493883850752,
3661856275780121034325608316928, -1243637943073077822928427352064O,
-157232863171166993110527508480, 19218418014847258443838877335552,
-8524407983882505977655434674176, -157504539968925253721146659962888,
126329381519490828018242960752648, 555513505840154056450141637836B,
-76280493418725819068055845601288, -898099068367670880182992896O,
193614656831452656346472015462A, -32764273258022894961440273203Z,
-172830348605676587877088100352, 30197447438230952553989799936,
573749360085840586006855680O
```

5.5.6.32 A210269

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 2$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 3\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 4$, with order $d = 3$ and component dimensions (4) (total $D = 4$). Source: https://oeis.org/A210269. The first three values are $(35, 273, 2130)$.

$$(c_1, \ldots, c_d) = (8, 0, -12).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.33 A210270

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 3\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 10$, with order $d = 9$ and component dimensions (10) (total $D = 10$). Source: https://oeis.org/A210270. The first three values are $(273, 5558, 113248)$.

```
20, 26, -364, -218, 1203, -418, -394, 216, -20
```

5.5.6.34 A210271

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 3\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 26$, with order $d = 25$ and component dimensions (25) (total $D = 25$). Source: https://oeis.org/A210271. The first three values are $(2130, 113248, 6028871)$.

```
48, 480, -9807, -58253, 657082, 2138431, -18778839, -19795420, 235514744, -62769938,
-1259927930, 1276469157, 2637995603, -4298139072, -1484220465, 5231313990, -949603112,
-2426962356, 931229528, 400609008, -188316850, -12522776, 5412968, -237208, -448
```

5.5.6.35 A210272

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 5$. The number of distinct signed clockwise edge differences in each block lies in $S = \{2, 3\}$, and the count is multiplied by $\lambda = 1/2$. The recurrence (5.13) holds for every $n \geq 70$, with order $d = 69$ and component dimensions (70) (total $D = 70$). Source: https://oeis.org/A210272. The first three values are $(16620, 2307956, 321050788)$.

```
142, 1390, -255743, -110496, 169565270, -665183207, -52878902879, 395903765675,
8133200677492, -92576230139190, -542663156454254, 10497292789475974, 2631885281757262,
-646106145808214387, 1826441384938919701, 22362684125873346283, -123476318341203413496,
-395337011643732047682, 4142092565872768583256, 605782653333712666179,
-835187877050850311956627, 13663488837653017895271B, 1040134849078032981074015,
-340207491496490496082235Z, -72394650593142476521445A9, 443777374153119767518385Z4,
984074193386080198197944Z, -361590798269938925507784670, 32181868400904699276253250I,
188357671714632216481929645I, -34798901387906322052988505Z8,
```

-587839336365400947013012483 8, 190786672206894498039118345 52,
73928543419216578544464187 42, -65742784478773744652167403362,
19627223671361316822942496555, 14983990332823919122494447217 3,
-11763062987043559469524517919 3, -22676772849810540544363144 4233,
28487521532741695803054243589 9, 21903225925161870569547810 4114,
-42465721170267223100086586 3606, -11429856815396448624052520 8337,
42646367413477201100840659122 0, -14651112962082630549586674 49,
-29836170210803453405579044673 9, 50042360589416264853656878 477,
14749955724419246358269799237 2, -39645514435604456524417564 586,
-51787606560690998053903082159, 16872384336675821704842993456,
12907214080006987000746559571, -44718941389389042802092501035,
-22706492437979471599810704 88, 75743325628800204885271254 7, 27789894035962642791469122 8,
-80741549607760164976554158, -22888243320669979845951670, 51814366573838468947907 30,
11850299407226738276377 93, -18525146857396236796355 7, -33904216065585670936623,
327503211596843716446 0, 413089312994017424066, -25348341919215966472,
-1274153174398735370, 12493913728073620, 443471547890804, -2001148574736

5.5.6.36 A210305

Arrays have $n+1$ rows, width $w=5$, exactly 4 colors renamed in row-major first-occurrence order, and $S=\{1,3\}$ distinct cell values per block. The recurrence (5.13) holds for every $n \geq 54$, with order $d=53$ and component dimensions $(1,16,41,51)$ (total $D=109$). Source: https://oeis.org/A210305. The first three values are $(4553, 655743, 85724987)$.

260, -23288, 929838, -12001245, -334728110, 13677246260, -97982783232, -3038968483483,
61195800410360, 21617308239752, -10672008917494402, 77526891307988953,
740183820173087714, -11993209513343105860, 1757788347100408132, 831215814617340342320,
-3626826186455581358000, -28284540559276107423616, 25559249549715819002 5072,
25147292168155865574737 6, -89532259980337691756950 72, 16464543868595186192082752,
17660428024576228800537 47200, -73411826628162501514104 9088,
-17325907768388014097252042 24, 14878214282327021473824928768,
-16320488768678751418485790 72, -17645631387627519113895152 8448,
27532591951591123233356236 3904, 12361509315175432493935234 90816,
-37187614899678205804133739 88864, -42194449883260813464627925 81120,
26532685864866441003684492673024, -40310280092684478703270551 87968,
-11339494223973333246488021 5654400, 10647166945035102180146177 6384000,
28820655125091741735576489 7415168, -48170944305287083691261336 4858880,
-38403904010232870618415740 4856320, 11422689075641098138605326 85373440,
91027236998423324239131310030848, -15804648123146411362892777 39532288,
47396932522793849630765854 2284800, 12689807155920621526447004 09954304,
-68971155062185928771020318 8871168, -54817219455526837933715 6183719936,
40399716782818564676966618 1300224, 10690821576279573955903838 2243840,
-10299298352870953722196771 2616448, -72806082745538193030103 90507520,
10221635153981636051140175 986688, 41096544975325809960739143 680,
-23638264617310246189079 1997440

5.5.6.37 A210328

Arrays on $0,\ldots,3$ have $n+1$ rows and width $w=2$. The number of distinct signed clockwise edge differences in each block lies in $S=\{2,3\}$, and the count is multiplied by $\lambda=1/4$. The recurrence (5.13) holds for every $n \geq 7$, with order $d=6$ and component dimensions (6) (total $D=6$). Source: https://oeis.org/A210328. The first three values are $(39, 406, 4200)$.

$$(c_1,\ldots,c_d) = (12,-7,-108,22,258,124).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF=P$.

5.5.6.38 A210329

Arrays on $0,\ldots,3$ have $n+1$ rows and width $w=3$. The number of distinct signed clockwise edge differences in each block lies in $S=\{2,3\}$, and the count is multiplied by $\lambda=1/4$. The recurrence (5.13) holds for every $n \geq 21$, with order $d=20$ and component dimensions (20) (total $D=20$). Source: https://oeis.org/A210329. The first three values are $(406, 11877, 344659)$.

```
47, -547, -255, 35216, -129394, -465060, 2920628, 320982, -20531472, 17232541, 55329403,
-75077657, -49764339, 111287472, -12747232, -52250238, 26800488, -2501204, -451078,
-5036
```

5.5.6.39 A210330

Arrays on $0,\dots,3$ have $n+1$ rows and width $w=4$. The number of distinct signed clockwise edge differences in each block lies in $S=\{2,3\}$, and the count is multiplied by $\lambda=1/4$. The recurrence (5.13) holds for every $n\geq 73$, with order $d=72$ and component dimensions (72) (total $D=72$). Source: https://oeis.org/A210330. The first three values are $(4200, 344659, 27838382)$.

```
140, -5048, -23365, 4584810, -58220704, -1208071283, 30838348016, 46781002450,
-6610272669799, 33388421310781, 724931448827996, -7330140469767465, -38529835529169361,
750770095780529564, 104547760010327276, -45822745228789445629, 124886909490991773867,
1772584705154196067963, -9188006643225676802179, -4248755111966368679024 0,
3694413965679300164479 63, 4930463413318843302423 25, -9840007964400110991140626,
4577246401662065605851681, 1841191397891217570950007 06, -3246978805029640509678777 01,
-2457999961706566468164419681, 7357259533993676894189678070,
2291391425658037568285641156 2, -1059382949409546234882021723 45,
-1337973981878615065347875125 18, 1088287205984201969751840912702,
2141491730096480384787199607 91, -8295244297697700252477102952711,
4551271399893045426167401874520, 4749214865386727792516606318822 2,
-5546732808644924819844555489541 8, -2030447278364392472581803797081 15,
3649531619648148798028093488455 13, 6290006000035940429092002451310 02,
-1656973614766749241452655210247331, -1286419122570055663502048783109264,
5521435181568923625231432851494067, 1085213406458212188709424799285359,
-1376133278895135251639489780210047, 2932860737587723748361214175457686,
2567276685638581240732533309561147 0, -1392835741251889489109707182799909 2,
-3534231104962677749266104208936934 1, 3014900644080958312246798558770193 5,
3473103286904944678815700629290218 8, -4240992161480650853970233365080734 7,
-2252308146268707883739007594664142 0, 4147488525875081625032468464417121 3,
7300575902610538123306992258616268, -2851890413376211165121391570867528 8,
1573935469653866497475056486719028, 1360971032540850435051423064088125 6,
-3000959034483091478767274705019179, -4366242148634467525612510604850896,
1533561432404703029191784717830830, 8918783866219064862081616575607 30,
-4216532906261872524109285041689 62, -1057763734394892659461086775165 38,
6606793414014261680544388563624 2, 6007572598581250574435738268266,
-5638660751066341761268282571708, -7156569122318465321816647403 2,
2312502608422733316953024579 80, -3237179075004335463427133468,
-3230599808041743511640862256, -4172213429687612280517579 2
```

5.5.6.40 A210348

Arrays on $0,\dots,3$ have $n+1$ rows and width $w=2$. The number of distinct signed clockwise edge differences in each block lies in $S=\{1,2,3\}$, and the count is multiplied by $\lambda=1/4$. The recurrence (5.13) holds for every $n\geq 6$, with order $d=5$ and component dimensions (6) (total $D=6$). Source: https://oeis.org/A210348. The first three values are $(40, 431, 4615)$.

$$(c_1,\dots,c_d)=(14,-30,-66,104,136).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF=P$.

5.5.6.41 A210349

Arrays on $0,\dots,3$ have $n+1$ rows and width $w=3$. The number of distinct signed clockwise edge differences in each block lies in $S=\{1,2,3\}$, and the count is multiplied by $\lambda=1/4$. The recurrence (5.13) holds for every $n\geq 21$, with order $d=20$ and component dimensions (20) (total $D=20$). Source: https://oeis.org/A210349. The first three values are $(431, 13566, 423694)$.

```
55, -890, 4374, 14813, -179200, 198626, 1718220, -3858631, -4143905, 14290154, 1274736,
-19132266, 4460692, 9184880, -3036040, -1419448, 517280, 1792, -4096, -128
```

5.5.6.42 A210350

Arrays on $0, \ldots, 3$ have $n+1$ rows and width $w = 4$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 2, 3\}$, and the count is multiplied by $\lambda = 1/4$. The recurrence (5.13) holds for every $n \geq 73$, with order $d = 72$ and component dimensions (72) (total $D = 72$). Source: https://oeis.org/A210350. The first three values are $(4615, 423694, 38415623)$.

```
191, -12666, 362418, -2471771, -110401217, 2621836067, -5950956298, -457658156996,
4931495900625, 22595991223312, -660371000230224, 1576600613573351, 40726756485462035,
-261581043509847118, -1180848375506028374, 15214038374148596375, 2740239137031201481,
-500647751869384591185, 1003334958924814164002, 10265837091234582479490,
-3888575600529776124046l, -13061230819125689382l321, 81204142275520779616567 9,
855729427631059422140114, -11175171560989023064509960, 18895810618667646059108 91,
10792839941554511139272 4496, -10778428931541002944885 5934, -74812870257171692596513 6011,
12704469990464123864301 83438, 37121446932387486871887 01435,
-91243811225108747866177 73257, -12697180476385310458871841925,
45745095998148294054172238533, 25879826918923108106955540193,
-16725638570484622814705 3848315, -55227344396913587256618 62744,
45257100388327263259844 6053394, -16267695947157999523407 1136543,
-90494827155921928042276 8304046, 62603529366971517560486 0994661,
13142782018874001077741 75032607, -13550200356978430806121 69545698,
-13258733111919461669668 33531231, 19733023575361655032780 14656818,
81919679882410856872770 1573827, -20145560979857637446451 41103153,
-14129282510501188958318 5641794, 14444881361705979937722 31897282,
-24765571415076567698144 2117027, -71250704252351220759209 0491261,
26120653342928440031940 1980540, 23092216567133727149513 0560074,
-13029661800908943385584 3407234, -44578783099713406658548 437066,
38556121946149222868083 061076, 36919911895100513916387 68180,
-69959910144424441223565 72748, 25778983453208297526307 1992, 77234963043865261015603 0230,
-86947033325326193209399 996, -50550945071670308168118 284, 78042339874147986802121 12,
18805367712533561751580 40, -32139329928291579760024 8, -37501945544789391775192,
62207428866078029570 56, 344840850367579506624, -53336079334554500352,
-987610693682276352, 164858395676078080, -1278984298302976
```

5.5.6.43 A210383

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 2$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 2, 3\}$, and the count is multiplied by $\lambda = 1$. The recurrence (5.13) holds for every $n \geq 4$, with order $d = 3$ and component dimensions (4) (total $D = 4$). Source: https://oeis.org/A210383. The first three values are $(73, 597, 4881)$.

$$(c_1, \ldots, c_d) = (9, -6, -6).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.44 A210384

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 3$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 2, 3\}$, and the count is multiplied by $\lambda = 1$. The recurrence (5.13) holds for every $n \geq 7$, with order $d = 6$ and component dimensions (10) (total $D = 10$). Source: https://oeis.org/A210384. The first three values are $(597, 13399, 300653)$.

$$(c_1, \ldots, c_d) = (26, -76, -93, 144, 66, -32).$$

The source's empirical/conjectural rational generating function is also proved by the exact identity $QF = P$.

5.5.6.45 A210386

Arrays on $0, \ldots, 2$ have $n+1$ rows and width $w = 5$. The number of distinct signed clockwise edge differences in each block lies in $S = \{1, 2, 3\}$, and the count is multiplied by $\lambda = 1$. The recurrence

(5.13) holds for every $n \geq 43$, with order $d = 42$ and component dimensions (70) (total $D = 70$). Source: https://oeis.org/A210386. The first three values are $(39909, 6747079, 1140445889)$.

```
223, -9461, 38458, 3549663, -32588685, -508261230, 5706283402, 35105920095,
-470735135093, -1154545872838, 21185965166884, 11225843446480, -544537281641561,
302744827612151, 8161346038514766, -9528641858555588, -73104114337370148,
109184395751663568, 406815111946067415, -670442664892782833, -1455165639539505801,
2462053942187126418, 3420041227155372378, -5685667280406325350, -5317279202551076247,
8464088502302311604, 5417981379313428776, -8206507857103512021, -3524350348222019545,
5179718539525222811, 1395392552338466122, -2099353043062207344, -304790403886902836,
529249988183090028, 26637678526607096, -78134322897382024, 1358756843570944,
6043523752505920, -329711911635072, -200633167870464, 9138104982528, 2395048366080
```

5.6 Reproduction and verification

The archive includes this standalone manuscript, the numbered ledger, four proof fragments, four main verifiers, an independent array checker, complete transfer matrices, exact algebraic certificates, source-scope audits and cross-review reports. Full scraped webpages are omitted.

5.6.1 Running the offline checks

Use Python 3.10 or later with SymPy and mpmath. From the extracted directory run:

```
python3 r5_run_all.py
```

The runner checks all 100 unique A-numbers against the preceding 408, checks the outcome counts and proof labels, then runs all five programs. It writes `oeis_hunt_round5_verification_summary.json`. No network connection is required.

Program	Verification scope
`r5_root_verify.py`	Thirteen formal-series and theta packages: exact identities and supplementary coefficient and numerical checks.
`r5_arith_verify.py`	Twelve arithmetic packages: exact integer counterexamples, rational enclosures and bounded consistency checks supporting the separate written proofs.
`r5_gf_verify.py`	Thirty exact all-index certificates, all recurrence boundary terms, 480 independently expanded initial terms, and the five retrospective Fischer operator comparisons.
`r5_arrays_verify.py`	Forty-five recurrences, all 56 complete matrices rebuilt from their definitions, and 17 source generating functions proved without additional counts.
`r5_arrays_root_verify.py`	Ninety independent initial values across all 45 arrays, using full rows and a separate column dynamic program without importing the quotient-matrix engine.

5.6.2 Finite proof certificates and bounded checks

For an array recurrence, the defining count is a linear combination of $m^{\mathsf{T}}T^n\mathbf{1}$. The complete identity $TR(T)\mathbf{1} = 0$ remains true after multiplication by every nonnegative power of T, proving the recurrence on the stated infinite range. The largest component has 574 states, and the largest recurrence order is 72. Every coordinate is checked with exact integer arithmetic. An initial prefix by itself would not imply the recurrence.

For the generating functions, the selected recurrence is translated to a differential operator. Its exact action in a rational radical basis gives the displayed polynomial boundary term, whose degree determines the valid starting index. A nonzero irrational component in the A129366 residual rules out every eventual tail of the original recurrence. The repaired recurrence has an exact certificate.

The remaining universal conclusions follow from the written arguments. Their bounded computations check transcription and indexing. In particular, the 80-digit theta checks are supplementary; the exact evaluations follow from the cited classical identities and stated generating functions. No numerical match is used as the proof of an infinite identity or limit.

5.6.3 Rebuilding, source records and integrity

The delivered `oeis_hunt_round5.tex` contains all arguments directly and needs no external `\input` files. Compile it twice with pdfLaTeX and the standard packages listed in its preamble. To rebuild the standalone TEX, catalogue and audit index, run:

```
python3 r5_build_dossier.py
```

Optional certificate reconstruction commands and dependencies are listed in the README. `manifest.json` gives SHA-256 digests of the delivered files. Running the verifiers regenerates reports and timing fields, so those report digests can then change.

The metadata retain source attribution, indexing, exclusions and audit limits. The five retrospective corrections remain explicit. Project selection counts do not certify first discovery or OEIS editorial closure.

Part VI

Formal series, Bernoulli arithmetic, valuation formulas, and finite states

Round overview

This sixth-round dossier treats 100 selected assertion packages on 100 distinct OEIS entries, disjoint from the 508 A-numbers selected in the preceding five rounds. It proves 94 original selected claims, supplies and proves the valid starting range of five conjectural recurrences, and corrects and proves one periodicity starting index. Multiple assertions or generating-function consequences at one entry count only once.

The four parts contain 18 packages on formal series and theta values, 11 on arithmetic and exact congruences, 30 on generating functions, and 41 on constrained arrays. The formal-series proofs work over integer coefficient rings, including divided-power series where appropriate. The arithmetic arguments retain their precise source scope. The generating-function proofs use exact differential identities with boundary polynomials. The array proofs use complete integer transfer matrices and vector annihilator certificates.

What the count means

Every catalogue row names one selected assertion package on one distinct A-number. Different members of a family may follow from one general lemma; the count does not mean 100 unrelated methods. The consulted primary statements were marked conjectural or empirical. Established generating functions, different previously known recurrences, and classical identities are credited inputs.

The selected scope can be narrower than all questions on an entry. For example, the packages for A396805 and A396806 prove parity only, and leave the other modular conjectures on their pages outside scope. The mod-4 package for A389537 does not claim its separate sign-run observation. Such limits are recorded in the individual arguments and the source ledger.

The number 508 is a ledger of previously selected A-numbers, not a certified count of worldwide-new discoveries. The five Round IV priority corrections disclosed in Round V remain in force and all those IDs remain excluded. Known prior results found during this round's audit were excluded or explicitly credited as inputs.

Six qualifications and corrections

For five recurrences, the source gives no starting index. We show an earlier admissible exception and prove the exact valid tail; we do not attribute an explicitly false starting-index assertion to the source author. The residual is the left side of the displayed source recurrence, in its stated normalization.

Entry	Exceptional index	Residual	Proved range
A102880	4	3	$n \geq 5$
A121320	3	1	$n \geq 4$
A141351	3	2	$n \geq 4$
A278023	4	156	$n \geq 5$
A385252	5	−6912	$n \geq 6$

For A396798, the stated mod-8 pattern $(1, 1, 5, 5)$ starts at $n = 2$, with $a(1) = 1$. Starting it at $n = 1$, as printed, would predict $a(3) \equiv 5$, whereas $a(3) = 9 \equiv 1 \pmod 8$. The corrected pattern and all seven other source iterate-congruence statements are proved together in its single package.

Evidence and limits

Primary OEIS entries, relevant linked work and targeted literature searches were consulted on 20 October 2026; retrieved pages may be cached. The retained OEIS Open comparison covers Appendix A only: 100

costliest claims on 97 distinct A-numbers, with no current overlap. It is not a comparison with every result in that project or a complete historical priority search.

The written arguments and exact certificates support the stated mathematical conclusions. No worldwide-first-proof or OEIS editorial acceptance claim is made, and no OEIS edit or submission was performed. Cross-reviews are by separate agents within the same AI-assisted workflow; they are not external peer review or proof-assistant formalization.

6.1 Catalogue of the 100 selected entries

P: original selected claim proved. **B**: recurrence boundary qualified and proved. **I**: period starting index corrected and proved. Page references lead to the individual argument or complete recurrence signature. The JSON ledger preserves exact scope, indexing and attribution.

No.	Entry	Selected assertion package	Result	Page
Formal series and theta values				
1	A393856	Every term is 1 modulo 5; odd iff the positive index is a power of 2	P	285
2	A393857	Every term is 1 modulo 6	P	285
3	A393858	Every term is 1 modulo 7; odd iff the positive index is a power of 2	P	285
4	A393859	Every term is 1 modulo 8	P	286
5	A396793	Every coefficient after the first is divisible by 3	P	286
6	A396794	Every coefficient after the first is divisible by 4	P	286
7	A396795	Every coefficient after the first is divisible by 5	P	286
8	A393170	Complete parity and residue-two classification modulo four	P	287
9	A393173	Complete parity and residue-two classification modulo four	P	287
10	A393171	Complete residue-four support modulo eight	P	287
11	A389537	Residue-two coefficients are exactly at three times powers of two	P	288
12	A396798	Correct the first period start and prove all eight iterate congruences	I	288
13	A396807	Every integer iterate has coefficients k^(n-1) modulo ten	P	289
14	A396805	The exponential coefficient is odd exactly at odd positive indices	P	290
15	A396806	The exponential coefficient is odd exactly at odd positive indices	P	290
16	A096920	The empirical exponential-weighted theta sum	P	290
17	A083365	The empirical exponential-weighted theta sum	P	291
18	A273228	The empirical exponential-weighted theta sum	P	291
Arithmetic and exact congruences				
19	A061002	Harmonic-number numerator and quotient-denominator identity	P	291
20	A063956	The conjectured product identity for simple prime factors	P	292
21	A068991	Squarefree members divide the indicated Bernoulli denominator	P	292
22	A069208	The general multiplicative-function convolution identity	P	294
23	A116184	The harmonic-numerator progression and prime-residue restriction	P	294
24	A133907	Equivalence of the binomial and power-sum least-prime rules	P	296
25	A262735	The selected trinomial-triangle diagonal-sum identity	P	297
26	A299406	The Liouville product identity for the Dirichlet-series coefficients	P	297
27	A376484	Integrality and divisibility of the trigonometric power sum	P	298
28	A377278	The reciprocal-row-sum determinant quotient	P	299
29	A381355	Divisibility by the prime at the coefficient index	P	300
Generating-function certificates				
30	A025268	Conjectured polynomial recurrence of order 5	P	303
31	A025269	Conjectured polynomial recurrence of order 4	P	303
32	A025270	Conjectured polynomial recurrence of order 4	P	304
33	A025275	Conjectured polynomial recurrence of order 4	P	305
34	A026030	Conjectured polynomial recurrence of order 4	P	305
35	A026571	Conjectured polynomial recurrence of order 5	P	306
36	A026743	Conjectured polynomial recurrence of order 8	P	307
37	A052183	Conjectured polynomial recurrence of order 2	P	308
38	A095981	Conjectured polynomial recurrence of order 7	P	308
39	A102880	Conjectured polynomial recurrence of order 4; exact starting index supplied	B	309
40	A105864	Conjectured polynomial recurrence of order 4	P	310
41	A114584	Conjectured polynomial recurrence of order 7	P	310
42	A118093	Conjectured polynomial recurrence of order 3	P	311
43	A121320	Conjectured polynomial recurrence of order 3; exact starting index supplied	B	312

Continued on the next page

No.	Entry	Selected assertion package	Result	Page
44	A141351	Conjectured polynomial recurrence of order 3; exact starting index supplied	B	312
45	A157003	Conjectured polynomial recurrence of order 6	P	313
46	A159772	Conjectured polynomial recurrence of order 6	P	314
47	A160823	Conjectured polynomial recurrence of order 5	P	314
48	A162533	Conjectured polynomial recurrence of order 5	P	315
49	A174107	Conjectured polynomial recurrence of order 4	P	316
50	A174808	Conjectured polynomial recurrence of order 5	P	316
51	A190169	Conjectured polynomial recurrence of order 8	P	317
52	A208355	Conjectured polynomial recurrence of order 2	P	318
53	A228178	Conjectured polynomial recurrence of order 4	P	318
54	A243022	Conjectured polynomial recurrence of order 5	P	319
55	A243255	Conjectured polynomial recurrence of order 4	P	320
56	A254314	Conjectured polynomial recurrence of order 5	P	321
57	A257388	Conjectured polynomial recurrence of order 3	P	321
58	A278023	Conjectured polynomial recurrence of order 4; exact starting index supplied	B	322
59	A385252	Conjectured polynomial recurrence of order 5; exact starting index supplied	B	323
Finite-state array certificates				
60	A183586	Empirical array recurrence of order 2	P	327
61	A183587	Empirical array recurrence of order 4	P	327
62	A183589	Empirical array recurrence of order 31	P	327
63	A183596	Empirical array recurrence of order 4	P	328
64	A210176	Empirical array recurrence of order 4	P	328
65	A210177	Empirical array recurrence of order 8	P	328
66	A210178	Empirical array recurrence of order 16	P	328
67	A210406	Empirical array recurrence of order 2	P	329
68	A210407	Empirical array recurrence of order 4	P	329
69	A210408	Empirical array recurrence of order 9	P	329
70	A210409	Empirical array recurrence of order 31	P	329
71	A233921	Empirical array recurrence of order 6	P	330
72	A233982	Empirical array recurrence of order 3	P	330
73	A233983	Empirical array recurrence of order 13	P	330
74	A233984	Empirical array recurrence of order 25	P	330
75	A234162	Empirical array recurrence of order 4	P	331
76	A234163	Empirical array recurrence of order 17	P	331
77	A234220	Empirical array recurrence of order 3	P	331
78	A234221	Empirical array recurrence of order 6	P	331
79	A234222	Empirical array recurrence of order 11	P	331
80	A234557	Empirical array recurrence of order 5	P	332
81	A234558	Empirical array recurrence of order 14	P	332
82	A234559	Empirical array recurrence of order 14	P	332
83	A234762	Empirical array recurrence of order 5	P	332
84	A234763	Empirical array recurrence of order 13	P	332
85	A234764	Empirical array recurrence of order 27	P	333
86	A234875	Empirical array recurrence of order 4	P	333
87	A234876	Empirical array recurrence of order 9	P	333
88	A234877	Empirical array recurrence of order 11	P	333
89	A235011	Empirical array recurrence of order 5	P	334
90	A235012	Empirical array recurrence of order 8	P	334
91	A235080	Empirical array recurrence of order 4	P	334
92	A235081	Empirical array recurrence of order 10	P	334
93	A235271	Empirical array recurrence of order 4	P	335
94	A235272	Empirical array recurrence of order 9	P	335
95	A235273	Empirical array recurrence of order 9	P	335

Continued on the next page

No.	Entry	Selected assertion package	Result	Page
96	A235303	Empirical array recurrence of order 5	P	335
97	A235304	Empirical array recurrence of order 14	P	336
98	A235305	Empirical array recurrence of order 14	P	336
99	A235312	Empirical array recurrence of order 6	P	336
100	A235313	Empirical array recurrence of order 18	P	336

6.2 Formal-series congruences and three theta values

This section treats eighteen distinct entries. Every counted target is an explicit conjecture or empirical equality in its cited OEIS entry. The two exponential-series entries below are selected for parity only. The other congruence packages specify exactly which assertions they settle. Classical theta identities, earlier project lemmas, and Sundaram's comparison method are credited inputs. They are not presented as new general mathematics.

6.2.1 Four nested-reversion sequences

For an integer $k \geq 1$, let $A_k = x + O(x^2)$ satisfy

$$A_k^{-1}(x) = x - x\frac{A_k(kx)}{k}.$$

The superscript -1 here means compositional inversion. The apparent division by k is harmless: $A_k(kx)/k = \sum_{n\geq 1} a_k(n)k^{n-1}x^n$. Since the coefficient of degree n of a compositional inverse is $-a_k(n)$ plus an integer polynomial in earlier coefficients, this equation determines a unique integer series, over any coefficient ring after reducing the displayed integer polynomials.

Lemma 6.2.1. *For all $n \geq 1$, $a_k(n) \equiv 1 \pmod{k+1}$. If k is even, $a_k(n)$ is odd exactly when n is a power of 2.*

Proof. Modulo $k+1$ we have $A_k(kx)/k \equiv -A_k(-x)$. Set $F = x/(1-x)$. Its inverse is $x/(1+x)$, and

$$x + xF(-x) = x - \frac{x^2}{1+x} = \frac{x}{1+x} = F^{-1}(x).$$

Thus F satisfies the reduced equation; triangular uniqueness proves $A_k \equiv F \pmod{k+1}$.

For even k, the integer series $A_k(kx)/k$ reduces to x modulo 2. Consequently $A_k^{-1} \equiv x + x^2$. The series $S(x) = \sum_{j\geq 0} x^{2^j}$ satisfies $S + S^2 = x$ in $\mathbb{F}_2[[x]]$, so it is the unique inverse of $x + x^2$. Hence $A_k \equiv S \pmod 2$. □

Theorem 6.2.2 (A393856)**.** *For $k = 4$, every coefficient is 1 modulo 5, and the odd coefficients occur exactly at positive indices that are powers of 2.*

Proof. Apply Lemma 6.2.1 with $k = 4$. These are both congruence conjectures in A393856 (Paul D. Hanna, March 11, 2026). □

Theorem 6.2.3 (A393857)**.** *For $k = 5$, every coefficient is 1 modulo 6.*

Proof. Apply Lemma 6.2.1 with $k = 5$ to the defining equation in A393857. This proves its displayed congruence conjecture. □

Theorem 6.2.4 (A393858)**.** *For $k = 6$, every coefficient is 1 modulo 7, and the odd coefficients occur exactly at positive indices that are powers of 2.*

Proof. Apply Lemma 6.2.1 with $k = 6$ to A393858; both displayed congruence conjectures follow. □

Theorem 6.2.5 (A393859)**.** *For $k = 7$, every coefficient is 1 modulo 8.*

Proof. Apply Lemma 6.2.1 with $k = 7$ to A393859. All four applications have the source range $n \geq 1$. □

6.2.2 Divisibility in three products of iterates

Here $A^{\circ j}$ always denotes composition, not an ordinary power.

Lemma 6.2.6. *Let r, s be positive integers and $m = r + s$. The unique rational series $A = x + O(x^2)$ satisfying*

$$A^{\circ r}(x)A^{\circ s}(x) = x^2 + m^2x^3$$

belongs to $x + mx^2\mathbb{Z}[[x]]$.

Proof. In the original equation, the coefficient of $a(n)$ at degree $n + 1$ is $r + s = m$; all remaining terms depend on earlier coefficients. This gives existence and uniqueness over $\mathbb{Q}$.

To prove the stronger integrality statement without dividing a congruence by m, put $A = x + mB$, with $B \in x^2\mathbb{Z}[[x]]$ to be constructed. For an indeterminate t, induction under composition gives

$$(x + tB)^{\circ j} = x + jtB + t^2H_j(t, B),$$

where the coefficients of H_j are integer polynomials in the coefficients of B. Every term of H_j involves at least two occurrences of B or its formal derivatives. More precisely, changing B by $cx^n + O(x^{n+1})$ changes H_j only in degrees at least $n+1$; this follows by expanding $B(x+tB) - B(x)$ with the binomial theorem, since $\operatorname{ord} B \geq 2$. After multiplication and substitution $t = m$, the equation becomes

$$B + H_r + H_s + rs\frac{B^2}{x} + m\frac{rBH_s + sBH_r}{x} + m^2\frac{H_rH_s}{x} = x^2,$$

where H_j now means $H_j(m, B)$. Each displayed quotient has integer coefficients. All terms other than the first B depend at degree n only on $b(2), \ldots, b(n-1)$. Thus this is a monic triangular recursion for integer $b(n)$, starting with $b(2) = 1$. The resulting $A = x + mB$ solves the original equation and hence is its unique rational solution. □

Theorem 6.2.7 (A396793)**.** *For the sequence determined by $A\,A^{\circ 2} = x^2 + 9x^3$, every $a(n)$ with $n > 1$ is divisible by 3.*

Proof. Use $(r, s, m) = (1, 2, 3)$ in Lemma 6.2.6. This is the selected conjecture in A396793. □

Theorem 6.2.8 (A396794)**.** *For $A\,A^{\circ 3} = x^2 + 16x^3$, every $a(n)$ with $n > 1$ is divisible by 4.*

Proof. Use $(r, s, m) = (1, 3, 4)$; this is the conjecture in A396794. □

Theorem 6.2.9 (A396795)**.** *For $A^{\circ 2}A^{\circ 3} = x^2 + 25x^3$, every $a(n)$ with $n > 1$ is divisible by 5.*

Proof. Use $(r, s, m) = (2, 3, 5)$; this is the conjecture in A396795. All three source assertions are due to Paul D. Hanna, June 2026. □

6.2.3 Two diagonal-zero series and Catalan valuations

For $d = 1, 2$, let $A_d(0) = 1$ be determined by

$$[x^n]\frac{1}{A_d(x) - n^d x} = 0 \qquad (n \geq 1).$$

The coefficient of $a_d(n)$ is -1, so these equations define unique integer series, with uniqueness also after reduction modulo 4. Let $C_j = \binom{2j}{j}/(j+1)$ and put

$$C(x) = \sum_{j\geq 0}(-1)^j C_j x^{2j}, \qquad U(x) = 1 + x + x^2 C(x).$$

The Catalan equation gives $C = 1 - x^2C^2$. Consequently the following are exact identities in integer formal series:

$$U - x = C^{-1}, \qquad U^{-1} = 1 - xC. \tag{6.1}$$

Lemma 6.2.10. *Both A_1 and A_2 are congruent to U modulo 4.*

Proof. For $d = 2$, an even n has $n^2 \equiv 0 \pmod 4$ and $[x^n]U^{-1} = 0$; an odd n has $n^2 \equiv 1 \pmod 4$ and $[x^n](U - x)^{-1} = [x^n]C = 0$. Thus U satisfies every required diagonal equation modulo 4.

For $d = 1$, the classes $n \equiv 0, 1 \pmod 4$ follow in the same way. For $n \equiv 2 \pmod 4$,

$$(U - 2x)^{-1} \equiv U^{-1} + 2xU^{-2} \pmod 4.$$

The first summand has no positive even coefficients. Modulo 2 the square U^{-2} has only even degrees, so the second summand also has zero coefficient at this even n. Finally, for $n \equiv 3 \pmod 4$,

$$(U - 3x)^{-1} \equiv C + 2xC^2 \pmod 4.$$

The first summand has only even degrees, and C^2 modulo 2 has degrees divisible by 4. Since $n - 1 \equiv 2 \pmod 4$, the required coefficient vanishes. Triangular uniqueness proves the lemma in both cases. □

Theorem 6.2.11 (A393170)**.** *For $d = 1$ and $n > 0$, $a_1(n)$ is odd exactly when n is a power of 2. Moreover, $a_1(n) \equiv 2 \pmod 4$ exactly when $n/2$ is the sum of two distinct powers of 2.*

Theorem 6.2.12 (A393173)**.** *The same two assertions hold for $d = 2$.*

Proof of both theorems. Lemma 6.2.10 gives $a_d(1) \equiv 1$, $a_d(n) \equiv 0$ at odd $n > 1$, and

$$a_d(2k) \equiv (-1)^{k-1}C_{k-1} \pmod 4 \qquad (k \geq 1).$$

Legendre's factorial formula gives $v_2(C_{k-1}) = s_2(k) - 1$, where $s_2(k)$ is the number of 1's in the binary expansion of k. Thus oddness is equivalent to $s_2(k) = 1$, and residue 2 modulo 4 is equivalent to $s_2(k) = 2$; a sign does not change the latter residue. Including $n = 1$ proves both statements. These are the two congruence conjectures in each of `A393170` and `A393173` (Paul D. Hanna, March 5, 2026). Their asymptotic assertions are outside this selection. □

6.2.4 A393171: a mod-eight consequence of a Catalan substitution

Theorem 6.2.13. *Let $A = x + O(x^2)$ satisfy $A(x) = x + A(C_*(x))^2$, where $C_*(x) = x + C_*(x)^2$. For $n > 2$, its coefficient is 4 modulo 8 exactly when $n = 2^j$ or $2^j + 1$ for some $j \geq 1$; all other such coefficients are divisible by 8.*

Proof. Put $f = x - x^2$, so $C_* = f^{-1}$, and set $B = A \circ C_*$. Then $B(f(x)) = A(x) = x + B(x)^2$. This is the defining series of A393172, used in Round III and not counted again here. We reproduce the required short lemma. The equation for B is monic triangular. Modulo 2 its unique solution is $B = x$, so write $B = x + 2H$. Reduction modulo 4 gives $H(x + x^2) = x^2$ in $\mathbb{F}_2[[x]]$. Since the inverse of $x + x^2$ is $S = \sum_{j\ge 0} x^{2^j}$,

$$H \equiv S^2 = \sum_{j\ge 1} x^{2^j} \pmod 2.$$

Now square B modulo 8:

$$A = x + B^2 \equiv x + x^2 + 4xH + 4H^2 \pmod 8.$$

The two supports in positive degrees are $2^j + 1$ for $j \ge 1$ and 2^j for $j \ge 2$, respectively, and do not overlap. This is exactly the asserted set after imposing $n > 2$. It proves the displayed conjecture in A393171. □

6.2.5 A389537: lifting the inverse of a dyadic series

Theorem 6.2.14. *Let $R(x) = \sum_{j\ge 0}(-1)^j x^{2^j}$ and $A = R^{-1}$, the series in A389537. Then*

$$A(x) \equiv x + x^2 + 2\sum_{j\ge 0} x^{3\cdot 2^j} \pmod 4.$$

Thus for $n > 2$ its coefficient is 2 modulo 4 exactly at $n = 3 \cdot 2^j$, and is otherwise divisible by 4.

Proof. Modulo 2, $R = \sum x^{2^j}$ has inverse $B = x + x^2$. For $j \ge 1$, the binomial theorem gives

$$B^{2^j} \equiv x^{2^j} + 2x^{3\cdot 2^{j-1}} + x^{2^{j+1}} \pmod 4.$$

Indeed, among the interior binomial coefficients in row 2^j, only the central one has 2-adic valuation one: for $0 < i < 2^j$ the valuation is $j - v_2(i)$. Substitution in $R(B)$ cancels consecutive endpoint terms and yields

$$R(B) \equiv x + 2\sum_{j\ge 0} x^{3\cdot 2^j} \pmod 4.$$

Also $R'(x) \equiv 1 \pmod 2$. Therefore for any integer series H, $R(B + 2H) \equiv R(B) + 2H \pmod 4$ by the binomial theorem. Choosing $H = \sum x^{3\cdot 2^j}$ makes the composition equal to x modulo 4, so uniqueness of the compositional inverse proves the result. The separate observation about runs of signs in the source is not selected. □

6.2.6 A396798: an index correction and all eight iterate formulas

Let $A = x + A^{\circ 4}A^{\circ 5}$. This equation has a unique integer solution, and a unique solution over any coefficient ring: a change in degree n in A changes the product of its zero-constant iterates only in degrees at least $n + 1$.

Theorem 6.2.15. *Put $F_j = x/(1 - jx)$ and $h = x^4/(1 - x)^3$. For every integer k,*

$$A^{\circ k}(x) \equiv \begin{cases} F_k(x), & k \text{ even}, \\ F_k(x) + 4h(x), & k \text{ odd} \end{cases} \pmod 8.$$

In particular $a(1) = 1$, and the coefficients for $n \ge 2$ repeat $[1, 1, 5, 5]$ modulo 8. The first conjecture on the consulted page incorrectly starts that repeating list at $n = 1$; already $a(3) = 9 \equiv 1$, whereas that start would predict 5. Its seven other listed iterate conjectures, with their stated starting indices, are true.

Proof. Set $R = F_1 + 4h$. Since $F_r \circ F_s = F_{r+s}$, the first-order composition expansion modulo 8 gives

$$R^{\circ 2} \equiv F_2 + 4\big(F_1'(F_1)h + h(F_1)\big) = F_2 + \frac{8x^4}{(1-2x)^3} \equiv F_2 \pmod 8.$$

It follows that $R^{\circ 2j} \equiv F_{2j}$. For odd iterates, $h(F_{2j}) \equiv h(x) \pmod 2$, so $R^{\circ(2j+1)} \equiv F_{2j+1} + 4h$. This holds for negative iterates as well, because composition and inversion preserve coefficientwise congruences for series with linear coefficient 1.

To check the defining equation for R, the same formulas give $R^{\circ 4} \equiv F_4$ and $R^{\circ 5} \equiv F_5 + 4h$. Direct rational simplification gives

$$R(x) - x - F_4(x)\cdot(F_5(x) + 4h(x)) = \frac{8x^3(1-3x)(5x^2-2x+1)}{(x-1)^3(4x-1)(5x-1)}.$$

Every denominator here has unit constant term, so this is zero modulo 8. By uniqueness, $A \equiv R$, and the iterate formulas follow.

For odd k, coefficient extraction gives $[x^n]A^{\circ k} \equiv k^{n-1} + 4\binom{n-2}{2}$ for $n \geq 2$, with the binomial term interpreted as zero at $n = 2, 3$. The second term repeats $0, 0, 4, 4$ from $n = 2$. The four odd iterates therefore have patterns $[1, 1, 5, 5]$, $[3, 1, 7, 5]$, $[5, 1, 1, 5]$, and $[7, 1, 3, 5]$, respectively. For even $k = 2, 4, 6, 8$, k^{n-1} vanishes modulo 8 after $n = 3, 2, 3, 1$, respectively. These are exactly the source ranges. We count the entire eight-assertion package once, as an index correction. □

6.2.7 A396807: comparison with a rational iterate

Theorem 6.2.16. *For $A = x + A^{\circ 5}A^{\circ 6}$ and every integer k,*

$$A^{\circ k}(x) \equiv \frac{x}{1-kx} \pmod{10}.$$

Thus $a(n) \equiv 1 \pmod{10}$ for $n \geq 1$, and more generally $[x^n]A^{\circ k} \equiv k^{n-1} \pmod{10}$, with $0^0 = 1$.

Proof. We use the rational comparison method in Shiva Priyan Shanmuga Sundaram's September 2026 proof for A396797; that already proved entry is excluded. His note treats iterates $3, 4$ modulo 6. For the present iterates $5, 6$ the required separate computation is

$$(1-5x)(1-6x) = 1 - 11x + 30x^2 \equiv 1 - x \pmod{10}.$$

Consequently $F = x/(1-x)$ satisfies $F \equiv x + F^{\circ 5}F^{\circ 6}$ modulo 10. The same degree argument as in the previous subsection proves uniqueness, so $A \equiv F$. Composition gives the assertion for positive k; inversion extends it to negative k, and $k = 0$ gives x. This proves both conjectures in A396807. The method is an explicit extension of a credited proof, not a claim of a newly invented technique. □

6.2.8 A396805 and A396806: parity of exponential coefficients

An integer Hurwitz series is written $G = \sum g_n x^n/n!$ with $g_n \in \mathbb{Z}$. Its coefficient congruences refer to g_n. Products, composition of zero-constant series, and exponentiation of a zero-constant series use integer binomial or Bell polynomials. We never reduce a factorial denominator by inverting it modulo a prime.

Lemma 6.2.17. *For every positive integer ℓ, the unique Hurwitz solution*

$$A_\ell = x\exp(A_\ell^{\circ \ell})$$

has $a_\ell(n) \equiv n \pmod 2$ for all $n \geq 1$.

Proof. The degree-n Hurwitz coefficient on the right is n times a Bell polynomial depending only on coefficients of A_ℓ below degree n. Thus the equation has a unique integral solution, and is uniquely solvable after reduction modulo 2.

Let $F = xe^x$, whose n-th Hurwitz coefficient is n. By Lagrange inversion its compositional inverse has coefficient $(-n)^{n-1}$. This is congruent to n modulo 2, including $n = 1$, so $F^{\circ 2} \equiv x \pmod 2$ (the lemma used in Round V is reproduced here). Also the tree series $T = xe^T$ has Hurwitz coefficient n^{n-1}, hence $T \equiv F \pmod 2$. Reduction of its defining equation gives $F \equiv xe^F \pmod 2$.

If ℓ is even then $F^{\circ \ell} \equiv x$, and $x \exp(F^{\circ \ell}) \equiv xe^x = F$. If ℓ is odd then $F^{\circ \ell} \equiv F$, and the tree-series identity gives the same conclusion. Uniqueness proves $A_\ell \equiv F$ in both cases. □

Theorem 6.2.18 (A396805)**.** *For $\ell = 5$, $a(n)$ is odd if and only if n is odd.*

Proof. Apply Lemma 6.2.17 with $\ell = 5$. Only this explicit parity conjecture is selected from A396805; its mod-3 and mod-5 conjectures are not claimed here. □

Theorem 6.2.19 (A396806)**.** *For $\ell = 6$, $a(n)$ is odd if and only if n is odd.*

Proof. Apply Lemma 6.2.17 with $\ell = 6$. Only this parity conjecture is selected from A396806; its mod-3 and mod-6 assertions are outside scope. These two entries are distinct sequences, each counted once, despite sharing a general proof. □

6.2.9 Three empirical special values

Put

$$q = e^{-\pi}, \qquad T = \frac{\pi^{1/4}}{\Gamma(3/4)}, \qquad \phi(z) = \sum_{j\in\mathbb{Z}} z^{j^2}, \quad \psi(z) = \sum_{j\geq 0} z^{j(j+1)/2}, \quad \chi(z) = \prod_{j\geq 0}(1+z^{2j+1}).$$

We use the following classical theta evaluations:

$$\phi(q) = T, \quad \phi(-q) = 2^{-1/4}T, \quad \psi(q) = 2^{-5/8}q^{-1/8}T, \quad \chi(q) = 2^{1/4}q^{1/24}, \quad \psi(-q)^4 = 2^{-3}q^{-1/2}T^4. \tag{6.2}$$

For clarity, these are established inputs, not numerical discoveries. Here is their derivation from standard identities. At nome $e^{-\pi}$, the elliptic modulus is $1/\sqrt{2}$; the theta–elliptic relation and $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$ give $\theta_3(0,q) = T$ by the gamma reflection formula. The theta quartic identity gives $\theta_4(0,q) = 2^{-1/4}T$. Duplication gives $\psi(q)^2 = \phi(q)\psi(q^2)$ and $\psi(q^2) = \theta_2(0,q)/(2q^{1/4}) = 2^{-5/4}q^{-1/4}T$, which proves the stated $\psi(q)$ value. For the remaining product, put $E(z) = \prod_{n\geq 1}(1-z^n)$ and $\eta(i) = q^{1/12}E(q^2)$. The Jacobi products give $\phi(q) = E(q^2)\chi(q)^2$ and $2\eta(i)^3 = \theta_2(0,q)\theta_3(0,q)\theta_4(0,q)$. Thus $E(q^2) = q^{-1/12}T/\sqrt{2}$ and $\chi(q)^2 = \sqrt{2}q^{1/12}$, proving its stated value. Finally, applying the same duplication identity $\psi(z)^2 = \phi(z)\psi(z^2)$ at $z = -q$ gives $\psi(-q)^2 = \phi(-q)\psi(q^2)$, and hence the fourth-power value in (6.2). The standard identities and elliptic special value are documented in DLMF §§19.20, 20.7, 20.9 and the theta reference linked in the entries. All signs are fixed by the positive real products at $0 < q < 1$.

Theorem 6.2.20 (A096920)**.** *For the sequence with generating function $\psi(x)\chi(x)$,*

$$\sum_{n\geq 0} a(n)e^{-n\pi} = 2^{-3/8}e^{\pi/12}\frac{\pi^{1/4}}{\Gamma(3/4)}.$$

Proof. The known theta-product generating function is recorded in A096920. Multiplying the $\psi(q)$ and $\chi(q)$ values gives $2^{-3/8}q^{-1/12}T$, exactly the empirical equality attributed there to Simon Plouffe, September 17, 2025. The factors are analytic in $|x| < 1$, so the absolutely convergent evaluation is legitimate. □

Theorem 6.2.21 (A083365)**.** *For the sequence with generating function $\psi(x)/\phi(x)$,*

$$\sum_{n\geq 0} a(n)e^{-n\pi} = 2^{-5/8}e^{\pi/8}.$$

Proof. Divide the corresponding values in (6.2). This is the empirical equality in A083365 (Simon Plouffe, September 17, 2025). The Jacobi product for ϕ has no zeros in $|x| < 1$, so the quotient's Taylor series converges at q. □

Theorem 6.2.22 (A273228)**.** *For the sequence with generating function $1/\psi(-x)^4$,*

$$\sum_{n\geq 0} a(n)e^{-n\pi} = 8e^{-\pi/2}\frac{\Gamma(3/4)^4}{\pi}.$$

Proof. The source generating function is a previously known partition product; its derivation is not counted as new. Substituting the last value in (6.2) gives $8q^{1/2}T^{-4}$, which is the empirical equality in A273228 (Simon Plouffe, September 21, 2025). Its product has no zeros in the unit disk, so inversion and evaluation are valid. The three special values use classical identities and share one method; they are three distinct source assertions, not three new theta theorems. The linked decimal-constant entries add no counts. □

6.3 Arithmetic identities and exact modular certificates

This section resolves eleven selected packages at eleven distinct OEIS entries. Each entry still explicitly labels the selected statement a conjecture in the primary page inspected on October 20, 2026. Two packages also settle an accompanying empirical observation. The count is one per entry, including when a theorem proves more than one statement there. The standard-library program `r6_arith_verify.py` recomputes `r6_arith_certificates.json` using exact integer and rational arithmetic. The modular coefficient table for A116184 is a finite part of its proof; the other finite checks supplement universal arguments. Known formulas used as inputs are credited below. The audit concerns the inspected entries, relevant cross-references and targeted searches; it does not establish worldwide priority.

6.3.1 The harmonic denominator ratio in A061002

Let $H_m = \sum_{j=1}^{m} 1/j$, and write num and den for numerator and positive denominator in lowest terms. The selected statement is Gary Detlefs's February 20, 2013 conjecture in A061002.

Theorem 6.3.1. *For every prime $p > 3$,*

$$\frac{\mathrm{den}(H_p/H_{p-1})}{\mathrm{num}(H_{p-1}/p^2)} = p^3.$$

Proof. Write $H_{p-1} = A/B$ in lowest terms. None of its summands has a denominator divisible by p, so $p \nmid B$. We first recall the elementary harmonic form of Wolstenholme's theorem, with a proof to specify the input precisely. In the localization of the integers at p,

$$\frac{1}{p-j} \equiv -\frac{1}{j} - \frac{p}{j^2} \pmod{p^2}.$$

Summing for $1 \leq j < p$ gives

$$2H_{p-1} \equiv -p\sum_{j=1}^{p-1} j^{-2} \pmod{p^2}.$$

Inversion permutes the nonzero residues modulo p, and hence

$$\sum_{j=1}^{p-1} j^{-2} \equiv \sum_{j=1}^{p-1} j^2 = \frac{p(p-1)(2p-1)}{6} \equiv 0 \pmod{p}.$$

Here 6 and 2 are invertible because $p > 3$. Thus $p^2 \mid A$.

Now

$$\frac{H_p}{H_{p-1}} = \frac{pA+B}{pA}, \qquad \gcd(pA+B, pA) = \gcd(B, pA) = 1.$$

Its denominator is therefore pA. Also

$$\frac{H_{p-1}}{p^2} = \frac{A/p^2}{B}$$

is already reduced, with numerator A/p^2. The asserted quotient is $pA/(A/p^2) = p^3$. This reasoning also covers primes for which p^3 or a higher power divides A. □

Scope and verification. The classical integrality underlying the definition of A061002 is a known input, not an additional resolution. The selected target is the denominator ratio. The verifier checks every prime $5 \le p \le 200$ using reduced rational harmonic numbers and matches the displayed prefix.

6.3.2 The two-variable gcd identity in A063956

Define $f(n) = \sum_{p:\, v_p(n)=1} p$, the sum of prime divisors occurring to exponent exactly one. Velin Yanev's February 17, 2019 formula in A063956 is marked conjectural. The additive prime-power description, also stated by Amiram Eldar in July 2024, is credited as a known description of the sequence.

Theorem 6.3.2. *For all positive integers n, m,*

$$f(nm) = f(n) + f(m) - f(\gcd(n^2, m)) - f(\gcd(n, m^2)).$$

Proof. Fix a prime p and put $a = v_p(n)$, $b = v_p(m)$. The coefficient of p on each side is governed by

$$[a+b=1] = [a=1] + [b=1] - [\min(2a, b) = 1] - [\min(a, 2b) = 1],$$

where brackets denote indicators. If $a = 0$ or $b = 0$, both subtracted indicators are zero and the identity follows immediately. If $a, b > 0$, then $a + b \neq 1$, while $\min(2a, b) = 1$ precisely when $b = 1$, and $\min(a, 2b) = 1$ precisely when $a = 1$. The two positive indicators cancel the two negative ones. Multiplying by p and summing over primes proves the result. □

Verification. Independent factorization checks cover all $1 \le n, m \le 200$. The proof's case distinction applies to every exponent; the finite table in the certificate is a supplementary check, not a bound on the exponents.

6.3.3 Squarefree Bernoulli-denominator divisibility in A068991

Let

$$S(k) = \sum_{d \mid k} \frac{\sigma(d)}{\varphi(d)}.$$

The selected conjecture on A068991 says that a squarefree integer k for which $S(k)$ is an integer divides the denominator of B_{2k}. Here B_j denotes the Bernoulli numbers.

Theorem 6.3.3. *If k is squarefree and $S(k) \in \mathbb{Z}$, then*

$$k \mid \operatorname{den}(B_{2k}).$$

Proof. Write $r = \omega(k)$. Multiplicativity and squarefreeness give

$$S(k) = \prod_{p \mid k} \left(1 + \frac{p+1}{p-1}\right) = \frac{2^r k}{\varphi(k)}.$$

Consequently $\varphi(k) \mid 2^r k$. For each odd prime $p \mid k$, write

$$p - 1 = 2^{\alpha_p} u_p, \qquad u_p \text{ odd}, \quad \alpha_p \geq 1.$$

The odd part of $\varphi(k)$ divides the odd part of the squarefree integer k. Thus every u_p is squarefree, its prime factors divide k, and each odd prime can occur among all the u_p at most once.

If k is odd, there are r values of α_p and $\sum_{p \mid k} \alpha_p \leq r$. Each is at least one, so all equal one. It follows that $p - 1 \mid 2k$ for every $p \mid k$.

Suppose next that k is even, and let $t = r - 1$ be the number of its odd prime divisors. We have

$$\sum_{p \mid k,\ p > 2} \alpha_p \leq t + 2, \qquad \sum_{p \mid k,\ p > 2} (\alpha_p - 1) \leq 2.$$

It suffices to show that all $\alpha_p \leq 2$. The only alternative allowed by the displayed inequality is one prime with $\alpha_p = 3$ and all the other odd prime divisors having $\alpha_p = 1$. We show this alternative is impossible.

On the set of the t odd prime divisors of k, draw an arrow $q \to p$ whenever $q \mid p - 1$. Every arrow goes from a smaller prime to a larger one. Every vertex has outdegree at most one, because an odd prime occurs in the product of all the u_p at most once. A vertex with no incoming arrow has $u_p = 1$ and hence $p = 2^{\alpha_p} + 1$. Under the alternative being considered, α_p is either 1 or 3; the only prime of those two forms is 3, since $2^3 + 1 = 9$ is composite. In particular 3 is present, is the unique vertex of indegree zero, and all other vertices have indegree at least one.

The largest vertex has outdegree zero. There are therefore at most $t - 1$ arrows and, by the preceding indegree count, at least $t - 1$ arrows. It follows that every vertex except 3 has indegree one and every vertex except the largest has outdegree one. Since arrows strictly increase, the graph is a single chain starting at 3. Along that chain, a successor is 2^{α_p} times its predecessor plus one, because u_p is squarefree and has exactly one prime divisor. Before the unique exponent 3, a successor uses exponent 1. The chain can then begin only $3, 7$: a further such successor would be 15, which is composite. If exponent 3 follows 3, its value would be 25; if it follows 7, its value would be 57. Both are composite. The alternative is impossible.

Thus, for even k, each $\alpha_p \leq 2$ and u_p divides the odd part of k, so again $p - 1 \mid 2k$ for every $p \mid k$. The prime 2 causes no restriction; $k = 1$ and $k = 2$ are also covered directly.

Finally, the classical von Staudt–Clausen theorem states

$$\operatorname{den}(B_{2k}) = \prod_{q \text{ prime},\ q-1 \mid 2k} q.$$

Every prime divisor of k occurs in this product. Since k is squarefree, the desired divisibility follows. □

Known input and verification. The Bernoulli denominator theorem is used in its classical form; see https://dlmf.nist.gov/24.10.i. It is not counted as a new result. The verifier checks the derived condition $p - 1 \mid 2k$ for every qualifying squarefree $k \leq 100000$. It also checks the initial cases against Bernoulli numbers computed by their defining rational recurrence, independently of the denominator theorem.

6.3.4 The powerful-divisor transform in A069208

Call d powerful if every nonzero prime exponent in d is at least two; 1 is powerful. Let $\chi(d)$ be its indicator, namely A112526. Werner Schulte's January 23, 2025 conjecture on A069208 concerns the transform

$$P_f(n) = \sum_{\substack{d \mid n \\ d \text{ powerful}}} f(d).$$

Theorem 6.3.4. *For every multiplicative arithmetic function f with $f(1) = 1$, the function P_f is multiplicative. For any arithmetic function f, P_f is the inverse Möbius transform of the pointwise product χf; equivalently,*

$$P_f = \mathbf{1} * (\chi f), \qquad \mu * P_f = \chi f.$$

Here $*$ denotes Dirichlet convolution and $\mathbf{1}(n) = 1$.

Proof. For coprime u, v, unique factorization shows $\chi(uv) = \chi(u)\chi(v)$: the prime exponents of uv are precisely those of u and v. Thus χf is multiplicative whenever f is. Every divisor of uv has a unique form $d_1 d_2$ with $d_1 \mid u$ and $d_2 \mid v$, so

$$P_f(uv) = \sum_{d_1 \mid u} \sum_{d_2 \mid v} \chi(d_1) f(d_1) \chi(d_2) f(d_2) = P_f(u) P_f(v).$$

Independently of multiplicativity, the defining sum is exactly $\mathbf{1} * (\chi f)$. Convolving with μ, and using $\mu * \mathbf{1} = \varepsilon$, gives the second formula. □

Scope and verification. This is one package for the explicitly conjectured general transform. The entry's older formula for its base sequence and the already known powerful-number indicator are not additional counts. The verifier tests $f(n) = 1, n, n^2, (-1)^{\Omega(n)}, \tau(n)$ through 300, checking both Möbius inversion and every coprime product that stays within the bound.

6.3.5 A complete harmonic congruence certificate for A116184

For $n \geq 1$, put

$$H(36, n) = \sum_{j=1}^{36} j^{-n}.$$

The entry A116184, authored by Alexander Adamchuk on April 8, 2007, conjectures that every $3+36k$ belongs to the sequence defined by $37^3 \mid \operatorname{num} H(36, n)$. It also records the empirical observation that any prime term greater than 37 is -1 modulo 37.

Theorem 6.3.5. *For every $k \geq 0$,*

$$37^3 \mid \operatorname{num} H(36, 3 + 36k).$$

If a prime $\ell > 37$ satisfies $37^3 \mid \operatorname{num} H(36, \ell)$, then $\ell \equiv -1 \pmod{37}$.

Proof. All denominators are relatively prime to 37. Therefore the numerator condition is equivalent to $S(n) = 0$ modulo $M = 37^3 = 50653$, where

$$S(n) = \sum_{j=1}^{36} u_j^n \pmod{M}, \qquad u_j = j^{-1} \pmod{M}.$$

Choose least nonnegative residues for the u_j. Fermat's theorem gives $u_j^{36} \equiv 1 \pmod{37}$, so define h_j by

$$u_j^{36} \equiv 1 + 37 h_j \pmod{M}, \qquad 0 \leq h_j < 37^2.$$

For $0 \le r < 36$ define the following least nonnegative residues:

$$C_0(r) = \sum_j u_j^r \pmod{37^3},$$
$$C_1(r) = \sum_j u_j^r h_j \pmod{37^2},$$
$$C_2(r) = \sum_j u_j^r h_j^2 \pmod{37}.$$

The binomial theorem proves, for every $k \ge 0$,

$$S(r+36k) \equiv C_0(r) + 37kC_1(r) + 37^2\binom{k}{2}C_2(r) \pmod{37^3}. \tag{6.3}$$

Terms of binomial degree at least three vanish modulo 37^3. Direct exact evaluation of these 36-term sums gives

$$(C_0(3), C_1(3), C_2(3)) = (0, 0, 0).$$

Equation (6.3) proves the first assertion for all k, without a bound on k.

For the prime assertion, $\ell > 37$ has $\gcd(\ell, 6) = 1$. Its residue r modulo 36 therefore belongs to the twelve units modulo 36. For the eleven residues other than 35, the exact coefficient values are given below, with $C_0 = 37^2 A$, $C_1 = 37B$, and $C_2 = D$. All entries and polynomial equalities in the table are independently recomputed by the accompanying verifier.

r	A	B	D	c
1	24	13	24	12
5	1	12	5	21
7	29	2	5	21
11	19	3	30	15
13	20	13	36	18
17	23	18	12	6
19	30	34	6	3
23	21	26	23	30
25	19	27	30	15
29	33	20	21	29
31	23	24	4	2

For every row,

$$A + Bk + D\binom{k}{2} \equiv c(k-r)(k-r-1) \pmod{37}, \qquad c \not\equiv 0.$$

Thus $S(r+36k) = 0$ modulo 37^3 forces $k \equiv r$ or $r+1$ modulo 37. Since $36 \equiv -1 \pmod{37}$, this means $\ell = r + 36k \equiv 0$ or $-1 \pmod{37}$. The zero residue is impossible for a prime $\ell > 37$.

For the remaining residue $r = 35$, exact evaluation gives

$$(C_0(35), C_1(35), C_2(35)) = (17760, 203, 30).$$

Already modulo 37^2, the necessary condition $S(35+36k) = 0$ reduces, after division by 37, to

$$36 + 18k = 18(k-35) \equiv 0 \pmod{37}.$$

This forces $k \equiv 35 \pmod{37}$ and hence $\ell \equiv 35 - k \equiv 0 \pmod{37}$, again impossible. The second assertion follows. □

Reproducibility of the finite part. The following standard-library calculation specifies the coefficient certificate completely; it uses neither approximate harmonic sums nor large reduced rational powers.

```
p = 37
M = p**3
u = [pow(j, -1, M) for j in range(1, p)]
h = [(pow(v, 36, M) - 1)//p for v in u]
for r in range(36):
    a = [pow(v, r, M) for v in u]
    C0 = sum(a) % M
    C1 = sum(v*w for v, w in zip(a, h)) % (p*p)
    C2 = sum(v*w*w for v, w in zip(a, h)) % p
    print(r, C0, C1, C2)
```

The included verifier also proves the table's polynomial factorizations by comparing all three coefficients. As an independent check, $S(n)$ is periodic with period $36 \cdot 37^2 = 49284$ by Euler's theorem, and the verifier enumerates every residue in this full period directly. There are 2644 zero residues, of which 407 are relatively prime to 222; all of those 407 are 36 modulo 37. It also matches all 53 terms displayed on the source page. The theorem is therefore supported both by the small quadratic certificate and by a complete-period check, not a sampled prefix.

6.3.6 The power-sum criterion in A133907

Let $a(n)$ be the least prime p for which $\binom{n+p}{p} \equiv 1 \pmod{p}$. The selected conjecture in A133907, due to Thomas Ordowski on July 29, 2018, replaces the binomial test by a power-sum test. Robert Israel's May 11, 2017 floor criterion on the same entry is a known input, and is also derived below.

Theorem 6.3.6. *For every integer $n \geq 1$ and prime p,*

$$\binom{n+p}{p} \equiv 1 \pmod{p} \quad \Longleftrightarrow \quad \sum_{k=1}^{n} k^{p-1} \equiv n \pmod{p}.$$

Consequently the least primes satisfying the two tests are equal.

Proof. Write $n = qp + r$ with $0 \leq r < p$. In $\mathbb{F}_p[x]$,

$$(1+x)^{n+p} \equiv (1+x^p)^{q+1}(1+x)^r.$$

Extracting the coefficient of x^p gives $\binom{n+p}{p} \equiv q+1 \pmod{p}$. On the other hand, Fermat's theorem makes k^{p-1} equal to 1 modulo p for nonmultiples of p and to 0 for multiples. There are exactly q multiples among $1, \ldots, n$, and hence

$$\sum_{k=1}^{n} k^{p-1} \equiv n - q \pmod{p}.$$

Each test is therefore equivalent to $q \equiv 0 \pmod{p}$. The set of qualifying primes is nonempty because every prime $p > n$ has $q = 0$, so equality of the least primes follows as well. □

Verification. The verifier compares the actual binomial residue and the independently summed powers for all $1 \leq n \leq 200$ and all primes $p \leq 223$. The chosen prime bound also guarantees a qualifying prime for each of these n.

6.3.7 The trinomial diagonal sum in A262735

Let $a(n)$ be defined by

$$A(x) = \sum_{n\geq 0} a(n)x^n = \frac{x(1-x-x^2)}{(1-x)(1-x-2x^2-x^3)}.$$

Write $T(m,j) = [y^j](1+y+y^2)^m$, the ordinary trinomial coefficients of A027907. The selected statement in A262735 is the conjectural diagonal sum on the entry authored by Werner Schulte on September 29, 2015; the formula line itself is undated.

Theorem 6.3.7. *For every $n \geq 0$,*

$$a(n+1) = \sum_{j=0}^{\lfloor n/2 \rfloor} T(n-j, 2j).$$

Proof. Let $b(n)$ denote the sum on the right. Put $m = n - j$, so that $0 \leq j \leq m$. All subsequent operations are valid as formal power series: each coefficient receives only finitely many contributions. Taking the even part of the trinomial polynomial gives

$$\begin{aligned}
\sum_{n\geq 0} b(n)x^n &= \sum_{m\geq 0} x^m \sum_{j=0}^{m} T(m,2j)x^j \\
&= \frac{1}{2}\sum_{m\geq 0} x^m \left((1+\sqrt{x}+x)^m + (1-\sqrt{x}+x)^m\right) \\
&= \frac{1}{2}\left(\frac{1}{1-x-x^2-x\sqrt{x}} + \frac{1}{1-x-x^2+x\sqrt{x}}\right) \\
&= \frac{1-x-x^2}{(1-x-x^2)^2-x^3} \\
&= \frac{1-x-x^2}{(1-x)(1-x-2x^2-x^3)} = \frac{A(x)}{x}.
\end{aligned}$$

The square-root notation denotes an even part in the formal variable $\sqrt{x}$; the resulting expression contains only integer powers of x. Coefficient comparison proves the formula, including $n = 0$. □

Verification. The verifier builds the trinomial rows by polynomial multiplication and compares their diagonal sums with the independently supplied fourth-order recurrence for $a(n)$ through $n = 101$. It also checks the polynomial denominator factorization exactly. The rational generating function and the known trinomial definition are inputs, not extra resolutions.

6.3.8 The Liouville twist in A299406

The defining Dirichlet series of A299406 is

$$\sum_{n\geq 1} \frac{a(n)}{n^s} = \frac{\zeta(s)\zeta(6s)}{\zeta(2s)\zeta(3s)}.$$

Let $b(n) = $ A210826(n) be defined by its Lambert series

$$\sum_{n\geq 1} \frac{b(n)x^n}{1-x^n} = \sum_{m\geq 1} x^{m^3},$$

and let $\lambda(n) = (-1)^{\Omega(n)}$. The conjecture on Werner Schulte's 2018 entry is the following identity; its formula line is undated.

Theorem 6.3.8. *For all $n \geq 1$, $a(n) = \lambda(n)b(n)$.*

Proof. Extracting coefficients from the Lambert series gives

$$\sum_{d|n} b(d) = [n \text{ is a cube}].$$

Möbius inversion yields $b = \mu * \chi_3$, where χ_3 is the indicator of cubes. In particular b is multiplicative and its prime-power series is

$$B_p(t) = \sum_{e\geq 0} b(p^e)t^e = \frac{1-t}{1-t^3}.$$

These standard properties of b already appear on A210826 in comments by R. J. Mathar and Benoit Cloitre from 2012. Since λ is completely multiplicative, its pointwise twist replaces t by $-t$ in each prime-power series:

$$\sum_{e\geq 0} \lambda(p^e)b(p^e)t^e = B_p(-t) = \frac{1+t}{1+t^3}.$$

The defining zeta quotient for a has Euler factor

$$\frac{(1-t^2)(1-t^3)}{(1-t)(1-t^6)} = \frac{1+t}{1+t^3}.$$

Equality of every local coefficient, followed by unique factorization, proves the result. This can be read entirely as an identity of formal Dirichlet series, so no analytic convergence assumption is needed. □

Verification. For $n \leq 1000$, the verifier constructs a by Dirichlet convolution from its zeta quotient and b independently from its Lambert divisor recursion, then compares $a(n)$ with $\lambda(n)b(n)$. The six-periodic local coefficient formula is checked as a third representation.

6.3.9 Integrality of the sine-power array A376484

For $n, k \geq 0$, define

$$A(n,k) = 4^k \sum_{j=1}^{n} \sin^{2k}\left(\frac{2\pi j}{2n+1}\right).$$

The source A376484, by Cheng-Jun Li, September 24, 2024, conjectures integrality and observes divisibility by $2n+1$ when $n, k > 0$. The empty row has value zero; at $k = 0$ the sum has n terms of value one.

Theorem 6.3.9. *For $k \geq 1$ and $m = 2n+1$,*

$$A(n,k) = m\left(\frac{1}{2}\binom{2k}{k} + \sum_{\ell=1}^{\lfloor k/m \rfloor} (-1)^\ell \binom{2k}{k+\ell m}\right).$$

Thus all entries are integers, and $m \mid A(n,k)$ for $n, k > 0$.

Proof. Let $\zeta = e^{2\pi i/m}$. The elementary Laurent identity

$$(2 - z - z^{-1})^k = \sum_{r=-k}^{k} (-1)^r \binom{2k}{k+r} z^r$$

follows by expanding $(-1)^k z^{-k}(1-z)^{2k}$. Since m is odd, multiplication by 2 permutes residues modulo m. The terms indexed by j and $m-j$ in a full sine-power sum are equal, and the term at $j=0$ vanishes because $k \geq 1$. Therefore

$$\begin{aligned} 2A(n,k) &= \sum_{j=0}^{m-1} (2-\zeta^{2j}-\zeta^{-2j})^k \\ &= \sum_{j=0}^{m-1} (2-\zeta^{j}-\zeta^{-j})^k \\ &= m \sum_{\substack{-k\leq r\leq k \\ m\mid r}} (-1)^r \binom{2k}{k+r}. \end{aligned}$$

The last step is the root-of-unity filter $\sum_{j=0}^{m-1} \zeta^{jr} = m[m \mid r]$. Pair $r = \ell m$ with $r = -\ell m$. Their binomial coefficients agree, and $(-1)^{\ell m} = (-1)^{\ell}$ because m is odd. Dividing by two gives the displayed formula. Finally, $\binom{2k}{k} = 2\binom{2k-1}{k-1}$ is even for $k \geq 1$, so the expression in parentheses is an integer. The boundary values $A(0,k) = 0$ and $A(n,0) = n$ settle the remaining cases. □

Scope and verification. Integrality and the accompanying divisibility observation form one package. The quotient array A376499 is not an additional count. The source's other row and diagonal identifications are not needed here. The verifier compares the formula against exact cyclic convolution by $2 - z - z^{-1}$ for $0 \leq n \leq 12$, $0 \leq k \leq 30$, and matches all 66 displayed antidiagonal terms. No floating-point sine evaluations are used.

6.3.10 The harmonic row sums in A377278

Put $m_j = 2^j - 1$ and define the denominator triangle

$$T(n,k) = m_k m_{k+1} \quad (1 \leq k < n), \qquad T(n,n) = m_n.$$

The conjecture in A377278, an entry authored by Werner Schulte on October 22, 2024, identifies the sum of reciprocals in a row with a determinant ratio. The conjecture comment itself is undated. Let $P_n = \prod_{j=1}^{n} m_j =$ A005329(n) and let $D_n =$ A204243(n), the determinant of the $n \times n$ matrix whose first row and first column are all 1, whose remaining diagonal entries are $m_2, \ldots, m_n$, and whose other entries are zero.

Theorem 6.3.10. *For every $n \geq 1$,*

$$\sum_{k=1}^{n} \frac{1}{T(n,k)} = \frac{D_n}{P_n}.$$

Proof. Since $m_{k+1} = 2m_k + 1$,

$$\frac{1}{m_k m_{k+1}} = \frac{1}{m_k} - \frac{2}{m_{k+1}}.$$

Summing for $1 \leq k < n$ and adding $1/m_n$ gives

$$\sum_{k=1}^{n} \frac{1}{T(n,k)} = 1 - \sum_{j=2}^{n} \frac{1}{m_j}.$$

For the determinant, subtract $1/m_j$ times row j from row 1 for each $j = 2, \ldots, n$. This leaves all first-row entries except the first equal to zero, and the first entry becomes $1 - \sum_{j=2}^{n} 1/m_j$. Expansion along that row yields

$$D_n = \left(1 - \sum_{j=2}^{n} \frac{1}{m_j}\right) \prod_{j=2}^{n} m_j.$$

Because $m_1 = 1$, the product is P_n. The empty sums and products give the same conclusion for $n = 1$. □

Known input and verification. The determinant formula itself was already given by Robert Israel on A204243 on November 30, 2015; the proof above recalls it for completeness. The selected conjecture is its connection to the later harmonic triangle, not discovery of the determinant evaluation. For $1 \le n \le 20$, the verifier compares reduced rational row sums with determinants obtained by integer Bareiss elimination.

6.3.11 Prime divisibility of the logarithmic derivative in A381355

Let p_n be the nth prime. Define the formal series $F(x) = 1 + x + \sum_{n\ge2} f_n x^n$ by

$$[x^n]F(x)^{p_n} = 0 \qquad (n > 1),$$

and set

$$\frac{xF'(x)}{F(x)} = \sum_{n\ge1} a_n x^n.$$

The conjecture in A381355, an entry authored by Paul D. Hanna on March 11, 2025, is $p_n \mid a_n$ for $n > 1$. The comment itself is undated. The related entry A381353 contains the defining coefficients f_n and repeats the divisibility observation; it is not counted separately.

Theorem 6.3.11. *The defining equations uniquely determine $F \in \mathbb{Z}[[x]]$. All a_n are integers, and $p_n \mid a_n$ for every $n > 1$.*

Proof. We first prove integrality of F by induction. Suppose that the coefficients below degree n are integers, put $p = p_n$, and write $F_{<n} = \sum_{j=0}^{n-1} f_j x^j$. The defining equation is

$$pf_n + [x^n]F_{<n}^p = 0.$$

The freshman's dream gives $F_{<n}(x)^p \equiv F_{<n}(x^p) \pmod p$. Since $p_n > n$, its coefficient of x^n is zero modulo p. Thus $f_n = -[x^n]F_{<n}^p/p$ is an integer, uniquely determined. Starting from $f_0 = f_1 = 1$ proves the assertion about F. As a formal series with constant term 1, F has an inverse in $\mathbb{Z}[[x]]$, so all coefficients of xF'/F are integers.

Fix $n > 1$ again and put $p = p_n$. Differentiating F^p and using the defining equation gives the exact integer identity

$$p[x^{n-1}]F^{p-1}F' = n[x^n]F^p = 0,$$

so $[x^{n-1}]F^{p-1}F' = 0$. On the other hand, $F(x)^p \equiv F(x^p) \equiv 1 \pmod{(p, x^p)}$, and hence

$$F^{p-1} \equiv F^{-1} \pmod{(p, x^p)}.$$

The degree $n - 1$ is less than p. Multiplication by F' and extraction of that coefficient therefore yields

$$a_n = [x^{n-1}]\frac{F'}{F} \equiv [x^{n-1}]F^{p-1}F' = 0 \pmod p.$$

This proves the conjectured divisibility for every $n > 1$. □

Verification. The verifier constructs F through degree 50 by exact truncated polynomial powers, checks all defining equations, and computes xF'/F both by its coefficient recurrence and by an independent reciprocal series product. It checks every prime divisibility in that range and matches the 25 displayed source terms. The general proof does not depend on a bound for the coefficients or on numerical testing.

Excluded stale statements and counting boundary

Two initially promising entries were removed from the counted set after following their cross-references. The limit conjecture in A048651 follows from the known cumulative-sum relation in A177510 and Vaclav Kotesovec's 2014 asymptotic already stated on A008930; its separate Schulte series was also explicitly derived by Peter Bala in 2020. Neither is counted. The closure conjecture in A348713 already has a posted 2026 proof attributed to Bernard Schott and is excluded as well. Easy secondary recurrences in A349890 and A385590 are left uncounted while their principal permutation conjectures remain outside this dossier's conclusions. The detailed rejection ledger is `r6_arith_audit_notes.json`.

6.4 Thirty generating-function and recurrence packages

This section treats thirty distinct OEIS entries absent from the 508 previously counted A-IDs. The count is twenty-five direct proofs and five boundary qualifications. Every selected formula is explicitly labeled *Conjecture* in the consulted primary entry. In each boundary case the source does not specify a starting index; the exact valid tail and all earlier admissible exceptions are stated below. These qualifications do not attribute an unstated starting index to the conjecture's author.

Known generating functions, enumerations, transforms, closed forms and other recurrences are credited inputs. Thus these are proofs of selected source assertions, not claims that the sequences, generating functions, methods, or mathematical consequences are newly discovered. In particular three entries already contain different polynomial recurrences; those are identified individually. The prior-work audit has a stated finite scope and does not establish worldwide priority.

6.4.1 Exact finite certificates and all-index consequences

Let $F(x) = \sum_{n\ge 0} a_n x^n$, filling indices before the source offset with zero only to write a formal power series. Put $\theta = x\,d/dx$ and

$$\mathcal{L}_p = \sum_{j=0}^{r} x^j p_j(\theta + j).$$

Since $\theta x^k = kx^k$, for n such that all source indices are defined,

$$[x^n]\mathcal{L}_p F = \sum_{j=0}^{r} p_j(n) a_{n-j}. \tag{6.4}$$

Consequently an exact identity $\mathcal{L}_p F = P(x)$, with P a polynomial, proves the homogeneous recurrence for every $n > \deg P$. The coefficient $[x^n]P$ gives the exact initial residual. Taking $n \ge r + o$, where o is the original offset, ensures that no extension of the source's sequence is used in a claimed recurrence instance.

For the twenty-nine algebraic cases below, write $x = \xi(t)$ and

$$F(\xi(t)) = R_0(t) + R_1(t)\sqrt{D(t)}, \qquad D_0 = 1, \quad D_1 = D.$$

Without a change of variable this means $t = x$ and $\xi = x$. Every radical takes the formal branch with constant term one. For rational functions define

$$T_i R = \frac{\xi}{\xi'}\left(R' + \frac{D_i'}{2D_i}R\right), \qquad c_m(t) = \sum_{j=0}^{r} \xi(t)^j [n^m] p_j(n+j).$$

The chain rule gives $\theta(R\sqrt{D_i}) = (T_i R)\sqrt{D_i}$. Thus the exact certificate for each entry is the following pair of rational identities:

$$\sum_m c_m T_0^m R_0 = P(\xi(t)), \qquad \sum_m c_m T_1^m R_1 = 0. \tag{6.5}$$

All inputs p_j, R_0, R_1, D, ξ and the full polynomial P are printed below. Substitution followed by clearing denominators gives two zero polynomials over $\mathbb{Q}$. This is a finite algebraic certificate of an infinite identity. No linear independence of radical terms is needed: vanishing of each displayed residual suffices. Equation (6.4) completes the proof for every stated index.

As an expanded illustration, for A025269 let $D = 1 - 8x + 12x^2 + 12x^3 + 16x^4$ and $F = (1 - \sqrt{D})/2$. Its displayed coefficients give

$$\mathcal{L}_p = D\theta - \frac{xD'}{2}.$$

Using $\theta\sqrt{D} = xD'/(2\sqrt{D})$, one obtains

$$\mathcal{L}_p F = -\frac{xD'\sqrt{D}}{4} - \frac{xD'}{4} + \frac{xD'\sqrt{D}}{4} = 2x - 6x^2 - 9x^3 - 16x^4.$$

Hence the recurrence holds for every $n \geq 5$. All other algebraic certificates use the same chain rule, with at most four derivatives. The remaining A052183 certificate is a rational identity between successive factorial terms, proved separately below.

6.4.2 Definitions, indexing and the three rational parameters

Write

$$C(x) = \frac{1 - \sqrt{1 - 4x}}{2x}, \qquad M(x) = \frac{1 - x - \sqrt{1 - 2x - 3x^2}}{2x^2}.$$

These satisfy $C = 1 + xC^2$ and $M = 1 + xM + x^2M^2$, with constant term one. The large Schroeder series S satisfies $S = 1 + xS + xS^2$. For the four convolution entries A025268, A025269, A025270 and A025275, the definition is $a_n = \sum_{k=1}^{n-1} a_k a_{n-k}$ for $n \geq 5$. It makes $F - F^2$ a polynomial of degree at most four, fixed by their four initial values. Solving the quadratic with $F(0) = 0$ yields exactly the radicands listed below. This also derives their credited generating functions directly from the definitions.

For A026030 the reflection principle gives $a_n = \binom{2n}{n-1} - \binom{2n}{n-5}$ for $n \geq 1$. The Somos generating function used below starts with x and preserves this offset. A026571 starts at index two, and its given generating function already has valuation two. In A228178, the source displays $G = (xC + 2x^2C^4)/(1 - x)$, while its data and stated offset index the coefficient of x as a_0. Its supplied coefficient-list program uses this same convention. The indexed series used here is therefore $F = G/x = (C + 2xC^4)/(1 - x)$. This shift is explicit and is not a separately counted result. All other ordinary seeds are the established formulas identified in the individual source notes.

For A159772 let t count an ordered rooted tree whose nodes all have zero through three children. The ordered list of children gives $t = x(1 + t + t^2 + t^3)$. Allowing unrestricted root degree and retaining the root factor gives $F = x/(1 - t)$, hence

$$\xi(t) = \frac{t}{1 + t + t^2 + t^3}, \qquad F(\xi(t)) = \frac{t}{1 - t^4}.$$

This is Howroyd's already listed interpretation, used as input; the original binary-tree enumeration is not asserted anew.

For A243022 put $t = xC(x^2)$. The Catalan equation gives $t = x(1 + t^2)$ and $C(x^2) = 1 + t^2$. Thus its defining nested series is

$$\xi(t) = \frac{t}{1 + t^2}, \qquad -\frac{1}{C(x^2)C(xC(x^2))} = -\frac{1 + \sqrt{1 - 4t}}{2(1 + t^2)}.$$

For A243255 put $t = xM(x)$. Then $t = x(1 + t + t^2)$ and

$$\xi(t) = \frac{t}{1 + t + t^2}, \qquad F = xM(x)C(xM(x)) = tC(t) = \frac{1 - \sqrt{1 - 4t}}{2}.$$

In all three cases $\xi(0) = 0$ and $\xi'(0) = 1$, so the formal inverse $t(x)$ exists uniquely. Apparent poles in individual R_i are interpreted in the Laurent field; their sum is the power series specified by the input definition.

6.4.3 The thirty statements and their complete certificates

Each recurrence below means $\sum_j p_j(n)a_{n-j} = 0$ in its stated range. Sources were consulted on 20 October 2026. This date records retrieval, not a claim that every search cache was refreshed that day.

6.4.3.1 A025268: Jan 25 2015

Source: https://oeis.org/A025268. Selected conjecture: R. J. Mathar, Jan 25 2015. Kotesovec supplied the generating function and a different order-four recurrence on January 25, 2015. The selected target is the separately conjectured Mathar order-five relation. Both are consequences of the known convolution definition; no novelty is claimed for that definition or generating function.

Statement. With the source offset $o = 1$, the following recurrence holds for every integer $n \geq 6$.

$$\begin{aligned} p_0(n) &= n \\ p_1(n) &= n + 1 \\ p_2(n) &= -10\,(2n - 5) \\ p_3(n) &= 2\,(2n - 9) \\ p_4(n) &= 2\,(14n - 79) \\ p_5(n) &= 40\,(n - 7)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= \frac{1}{2}, \\ R_1 &= -\frac{1}{2}, \\ D &= (2x - 1)\left(4x^3 + 4x^2 + 2x - 1\right). \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -x\,(5x + 1)\left(8x^3 + 3x^2 - 1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 6$, with the initial residuals exactly as stated.

6.4.3.2 A025269: Nov 21 2014

Source: https://oeis.org/A025269. Selected conjecture: R. J. Mathar, Nov 21 2014. Kotesovec's January 25, 2015 generating function is credited input. The selected assertion is Mathar's separately labeled recurrence.

Statement. With the source offset $o = 1$, the following recurrence holds for every integer $n \geq 5$.

$$\begin{aligned} p_0(n) &= n \\ p_1(n) &= -4\,(2n-3) \\ p_2(n) &= 12\,(n-3) \\ p_3(n) &= 6\,(2n-9) \\ p_4(n) &= 16\,(n-6)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= \frac{1}{2}, \\ R_1 &= -\frac{1}{2}, \\ D &= (4x-1)\left(4x^3+4x^2+4x-1\right). \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -x\left(16x^3+9x^2+6x-2\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.3 A025270: Nov 21 2014

Source: https://oeis.org/A025270. Selected conjecture: R. J. Mathar, Nov 21 2014. Kotesovec's January 25, 2015 generating function is credited input. The selected assertion is Mathar's separately labeled recurrence.

Statement. With the source offset $o = 1$, the following recurrence holds for every integer $n \geq 5$.

$$\begin{aligned} p_0(n) &= n \\ p_1(n) &= -2\,(2n-3) \\ p_2(n) &= -4\,(n-3) \\ p_3(n) &= 6\,(2n-9) \\ p_4(n) &= 20\,(n-6)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= \frac{1}{2}, \\ R_1 &= -\frac{1}{2}, \end{aligned}$$

$$D = 20x^4 + 12x^3 - 4x^2 - 4x + 1.$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -x\left(20x^3 + 9x^2 - 2x - 1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.4 A025275: May 30 2014

Source: https://oeis.org/A025275. Selected conjecture: R. J. Mathar, May 30 2014. Kotesovec's January 25, 2015 generating function is credited input. The selected assertion is Mathar's separately labeled recurrence.

Statement. With the source offset $o = 1$, the following recurrence holds for every integer $n \geq 5$.

$$\begin{aligned} p_0(n) &= n \\ p_1(n) &= -2\,(2n - 3) \\ p_2(n) &= 4\,(n - 3) \\ p_3(n) &= -2\,(2n - 9) \\ p_4(n) &= 8\,(n - 6)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= \frac{1}{2}, \\ R_1 &= -\frac{1}{2}, \\ D &= (2x - 1)\left(4x^3 + 2x - 1\right). \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -x\left(8x^3 - 3x^2 + 2x - 1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.5 A026030: Jun 15 2014

Source: https://oeis.org/A026030. Selected conjecture: R. J. Mathar, Jun 15 2014. Stephan's 2005 binomial formula and Somos's 2012 generating function are prior input. Drube's 2022 raised-path enumeration is also credited. The displayed first generating function uses a shifted indexing convention; the Somos formula used here starts at the source index one.

Statement. With the source offset $o = 1$, the following recurrence holds for every integer $n \geq 5$.

$$
\begin{aligned}
p_0(n) &= n + 5 \\
p_1(n) &= -2\,(5n + 16) \\
p_2(n) &= 35n + 47 \\
p_3(n) &= -2\,(25n - 14) \\
p_4(n) &= 12\,(2n - 5)\,.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$
\begin{aligned}
R_0 &= \frac{(2x-1)^2\,(x^2 - 4x + 1)}{2x^5}, \\
R_1 &= \frac{(x-1)\,(2x-1)\,(3x-1)}{2x^5}, \\
D &= 1 - 4x.
\end{aligned}
$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$
\mathcal{L}_p F = P(x) = 6x\,(2x-1)^2\,.
$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.6 A026571: Jun 23 2013

Source: https://oeis.org/A026571. Selected conjecture: R. J. Mathar, Jun 23 2013. Gomez Aiza's 2024 Trees with flowers paper, pp. 18–19, supplies the generating function and related asymptotics. The selected target is the 2013 recurrence. The generating function already starts at degree two; no extra shift is made.

Statement. With the source offset $o = 2$, the following recurrence holds for every integer $n \geq 7$.

$$
\begin{aligned}
p_0(n) &= n + 2 \\
p_1(n) &= -3\,(n + 1) \\
p_2(n) &= -3n \\
p_3(n) &= 11\,(n - 1) \\
p_4(n) &= 2\,(n - 6) \\
p_5(n) &= -4\,(2n - 7)\,.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$
\begin{aligned}
R_0 &= -\frac{1}{2x^2}, \\
R_1 &= \frac{(x+1)\,(2x-1)}{2x^2\,(4x^2 + x - 1)},
\end{aligned}
$$

$$D = \frac{4x^2 + x - 1}{x - 1}.$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -2x^2 (x - 2).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 7$, with the initial residuals exactly as stated.

6.4.3.7 A026743: Jan 13 2023

Source: https://oeis.org/A026743. Selected conjecture: R. J. Mathar, Jan 13 2023. Callan's January 17, 2016 generating function is the prior input. The selected target is the 2023 recurrence.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 8$.

$$\begin{aligned}
p_0(n) &= n \\
p_1(n) &= -2 \\
p_2(n) &= 20 - 11n \\
p_3(n) &= 14 \\
p_4(n) &= 39n - 152 \\
p_5(n) &= -22 \\
p_6(n) &= 268 - 41n \\
p_7(n) &= -6 \\
p_8(n) &= -12 (n - 6).
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned}
R_0 &= -\frac{(2x + 1)(x^2 + 1)}{2(x^4 + 4x^2 - 1)}, \\
R_1 &= \frac{3x^2 - 1}{2(x^4 + 4x^2 - 1)}, \\
D &= -\frac{2x + 1}{2x - 1}.
\end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = 2x^2 (2x + 1)\left(3x^4 - 6x^2 + 1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 8$, with the initial residuals exactly as stated.

6.4.3.8 A052183: Sep 27 2012

Source: https://oeis.org/A052183. Selected conjecture: R. J. Mathar, Sep 27 2012. The factorial formula in the sequence definition is the prior input; it is related to Carlitz's two-line array enumeration and the Fuss–Catalan numbers. Only the explicitly conjectured three-term formula is selected.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 2$.

$$\begin{aligned} p_0(n) &= 2n\,(2n+1) \\ p_1(n) &= -47n^2 + 50n - 12 \\ p_2(n) &= 15\,(3n-5)\,(3n-4)\,. \end{aligned}$$

Proof. The defining factorial formula is

$$a_n = \frac{n+2}{2n+1}\binom{3n}{n}, \qquad n \geq 0.$$

Cancellation of factorials gives

$$r(n) = \frac{a_{n-1}}{a_n} = \frac{2n(n+1)(2n+1)}{3(n+2)(3n-1)(3n-2)}, \qquad n \geq 1.$$

The identity $p_0 + p_1 r(n) + p_2 r(n) r(n-1) = 0$ reduces, after division by $2n(2n+1)$ and clearing denominators, to

$$\begin{aligned} &3(n+2)(3n-1)(3n-2) \\ &\quad + (n+1)(-47n^2 + 50n - 12) + 10n(n-1)(2n-1) = 0. \end{aligned}$$

Expansion verifies this polynomial identity term by term. The cleared denominators are nonzero for every integer $n \geq 2$. Multiplication by a_n proves the recurrence throughout that range. Its first case also reads $20 \cdot 12 - 100 \cdot 3 + 30 \cdot 2 = 0$.

6.4.3.9 A095981: Nov 23 2014

Source: https://oeis.org/A095981. Selected conjecture: R. J. Mathar, Nov 23 2014. Callan's known first-return decomposition and its generating function are the prior input. The empty path is excluded by this entry's plateau convention, so its initial values are zero, zero, one.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 7$.

$$\begin{aligned} p_0(n) &= n + 2 \\ p_1(n) &= -5n - 4 \\ p_2(n) &= 6n \\ p_3(n) &= 4n - 13 \\ p_4(n) &= 43 - 13n \\ p_5(n) &= 3\,(3n - 13) \\ p_6(n) &= 4 - n \\ p_7(n) &= 7 - n. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= -\frac{x^3 - x^2 - 2x + 1}{2x^2\,(x-1)}, \\ R_1 &= \frac{1}{2x^2\,(x-1)}, \\ D &= \left(x^3 - x^2 + 1\right)\left(x^3 + 3x^2 - 4x + 1\right). \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = x^2\left(x^4 - x^3 + 6x^2 - 9x + 4\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 7$, with the initial residuals exactly as stated.

6.4.3.10 A102880: Nov 09 2012

Source: https://oeis.org/A102880. Selected conjecture: R. J. Mathar, Nov 09 2012. The defining Chebyshev transform of the Catalan series is prior input. The initial residual below is retained explicitly.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 5$.

Boundary qualification. The source specifies no starting index. $n = 4$ gives residual 3. These are all failures between the first index with all source terms defined and the proved tail. The coefficient formula itself is unchanged.

$$\begin{aligned} p_0(n) &= (n-3)\,(n+1) \\ p_1(n) &= -2\,(n-3)\,(2n-1) \\ p_2(n) &= 2\left(n^2 - 4n + 1\right) \\ p_3(n) &= -2\,(n-1)\,(2n-7) \\ p_4(n) &= (n-5)\,(n-1)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= -\frac{(x-1)\,(x+1)}{2x}, \\ R_1 &= \frac{(x-1)\,(x+1)}{2x}, \\ D &= \frac{x^2 - 4x + 1}{x^2 + 1}. \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = 3\,(x-1)\,(x+1)\left(x^2 + 1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.11 A105864: Nov 15 2011

Source: https://oeis.org/A105864. Selected conjecture: R. J. Mathar, Nov 15 2011. Barry's 2009 transformation paper and the already listed Catalan transform are prior input. A 2017 FindSequenceFunction program on the entry is computational guessing; it is not used as an all-index proof.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 4$.

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= -2\,(2n - 1) \\ p_2(n) &= -2\,(n - 1) \\ p_3(n) &= 2\,(2n - 3) \\ p_4(n) &= n - 3. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= \frac{1}{2x}, \\ R_1 &= -\frac{1}{2x}, \\ D &= \frac{x^2 + 4x - 1}{(x - 1)\,(x + 1)}. \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = x^2 + 1.$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 4$, with the initial residuals exactly as stated.

6.4.3.12 A114584: Feb 16 2018

Source: https://oeis.org/A114584. Selected conjecture: R. J. Mathar, Feb 16 2018. The established Motzkin-path generating function is prior input; the general forbidden-pattern treatment of Asinowski, Bacher, Banderier and Gittenberger (2019) is credited. Only the separately labeled recurrence is selected.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 7$.

$$\begin{aligned} p_0(n) &= n + 2 \\ p_1(n) &= n + 2 \\ p_2(n) &= -3\,(3n - 2) \\ p_3(n) &= 13 - 7n \\ p_4(n) &= 4n - 13 \\ p_5(n) &= -6\,(n - 5) \end{aligned}$$

$$p_6(n) = n - 7$$
$$p_7(n) = 3\,(n - 8)\,.$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$R_0 = \frac{x^3 - x + 1}{2x^2},$$
$$R_1 = -\frac{1}{2x^2},$$
$$D = \left(x^3 - 3x + 1\right)\left(x^3 + x + 1\right).$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -2\,(3x + 1)\left(2x^3 - 1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 7$, with the initial residuals exactly as stated.

6.4.3.13 A118093: Apr 05 2018

Source: https://oeis.org/A118093. Selected conjecture: R. J. Mathar, Apr 05 2018. The rooted toroidal hypermap enumeration is prior work. Coserea's November 6, 2018 generating function and the already listed finite sum are credited input.

Statement. With the source offset $o = 3$, the following recurrence holds for every integer $n \geq 6$.

$$p_0(n) = n\,(5n - 17)$$
$$p_1(n) = -15\,(n - 1)\,(5n - 16)$$
$$p_2(n) = 12\left(20n^2 - 103n + 140\right)$$
$$p_3(n) = 32\,(2n - 5)\,(5n - 12)\,.$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$R_0 = -\frac{4x^2 - 7x + 1}{8\,(x + 1)\,(8x - 1)},$$
$$R_1 = -\frac{3x - 1}{8\,(x + 1)\,(8x - 1)},$$
$$D = 1 - 8x.$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -6x^3.$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 6$, with the initial residuals exactly as stated.

6.4.3.14 A121320: Jun 22 2016

Source: https://oeis.org/A121320/internal. Selected conjecture: R. J. Mathar, Jun 22 2016. The plane-tree statistic generating function, reformulated by Georg Fischer on April 6, 2020, is prior input. This does not belong to Fischer's excluded A176604 parameter family.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 4$.

Boundary qualification. The source specifies no starting index. $n = 3$ gives residual 1. These are all failures between the first index with all source terms defined and the proved tail. The coefficient formula itself is unchanged.

$$\begin{aligned} p_0(n) &= 2 - n \\ p_1(n) &= 5n - 12 \\ p_2(n) &= 8 - 3n \\ p_3(n) &= -2\,(2n - 5)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= -\frac{x^2}{2\,(x^2 + x - 1)}, \\ R_1 &= \frac{x^2}{2\,(4x - 1)\,(x^2 + x - 1)}, \\ D &= 1 - 4x. \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = x^3.$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 4$, with the initial residuals exactly as stated.

6.4.3.15 A141351: Oct 15 2014

Source: https://oeis.org/A141351. Selected conjecture: R. J. Mathar, Oct 15 2014. The Catalan-plus-one definition and its other combinatorial interpretations are prior input. Only the separately conjectured recurrence is selected.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 4$.

Boundary qualification. The source specifies no starting index. $n = 3$ gives residual 2. These are all failures between the first index with all source terms defined and the proved tail. The coefficient formula itself is unchanged.

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= -2\,(3n - 1) \\ p_2(n) &= 9n - 13 \end{aligned}$$

$$p_3(n) = -2\,(2n-5)\,.$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= -\frac{2x^2 - x + 1}{2x\,(x-1)}, \\ R_1 &= -\frac{1}{2x}, \\ D &= 1 - 4x. \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = 2x^3 - 6x^2 + 1.$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 4$, with the initial residuals exactly as stated.

6.4.3.16 A157003: Nov 19 2014

Source: https://oeis.org/A157003. Selected conjecture: R. J. Mathar, Nov 19 2014. Kotesovec supplied a different order-five recurrence on February 1, 2015. Barry's 2023 elliptic-curve paper also gives the known generating function. The selected target is the separately labeled Mathar order-six relation; no Hankel-transform or pattern-avoidance conjecture is claimed here.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 6$.

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= n + 2 \\ p_2(n) &= 29 - 21n \\ p_3(n) &= 3n - 16 \\ p_4(n) &= 40\,(n-3) \\ p_5(n) &= -2\,(2n-7) \\ p_6(n) &= -10\,(2n-9)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= -\frac{1}{2x\,(x-1)\,(x+1)}, \\ R_1 &= \frac{1}{2x\,(x-1)\,(x+1)}, \\ D &= 4x^3 - 4x + 1. \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -(5x+1)\left(3x^2-1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 6$, with the initial residuals exactly as stated.

6.4.3.17 A159772: May 30 2014

Source: https://oeis.org/A159772. Selected conjecture: R. J. Mathar, May 30 2014. The binary-tree enumeration and Howroyd's 2017 rooted-tree generating function are prior input. Kotesovec's December 5, 2017 order-four recurrence is explicitly credited. The selected target is Mathar's separately conjectured order-six relation.

Statement. With the source offset $o = 1$, the following recurrence holds for every integer $n \geq 7$.

$$\begin{aligned}
p_0(n) &= 2(n-1)(2n-3)\\
p_1(n) &= -43n^2+172n-177\\
p_2(n) &= 4\left(44n^2-266n+411\right)\\
p_3(n) &= -8\left(43n^2-358n+741\right)\\
p_4(n) &= 96\left(3n^2-29n+69\right)\\
p_5(n) &= -128(n-6)(n-4)\\
p_6(n) &= 512(n-7)(n-6).
\end{aligned}$$

Use $x = \xi(t) = \frac{t}{(t+1)(t^2+1)}$, with $t(0) = 0$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned}
R_0 &= -\frac{t}{(t-1)(t+1)(t^2+1)},\\
R_1 &= 0,\\
D &= 1.
\end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -x^2\left(32x^3-24x^2-12x+3\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 7$, with the initial residuals exactly as stated.

6.4.3.18 A160823: Nov 16 2011

Source: https://oeis.org/A160823. Selected conjecture: R. J. Mathar, Nov 16 2011. Barry's established large-Schroeder transform and its finite sum are prior input.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 5$.

$$
\begin{aligned}
p_0(n) &= n+2\\
p_1(n) &= -3\,(n+1)\\
p_2(n) &= -3\,(n-2)\\
p_3(n) &= 11n-20\\
p_4(n) &= 11-5n\\
p_5(n) &= 4-n.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$
\begin{aligned}
R_0 &= \frac{x^2+x-1}{2x^2\,(x-1)},\\
R_1 &= \frac{1}{2x^2\,(x-1)},\\
D &= x^4+6x^3-5x^2-2x+1.
\end{aligned}
$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$
\mathcal{L}_p F = P(x) = -\,(x-2)\left(x^2-x+1\right).
$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.19 A162533: Nov 16 2011

Source: https://oeis.org/A162533. Selected conjecture: R. J. Mathar, Nov 16 2011. Barry's October 13, 2009 radical generating function is prior input. Its first two printed forms agree. The final factorization on the entry has a separate typo: the factors must contain 1-2x, rather than 1-x. That typo is not silently used and is not counted as a second result.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 5$.

$$
\begin{aligned}
p_0(n) &= n\\
p_1(n) &= 4-5n\\
p_2(n) &= 2\,(4n-7)\\
p_3(n) &= -4\,(n-3)\\
p_4(n) &= -4\,(n-2)\\
p_5(n) &= 4\,(n-4)\,.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$
R_0 = 0,
$$

$$R_1 = \frac{x-1}{(2x^2-2x+1)(2x^2+2x-1)},$$
$$D = -\left(2x^2-2x+1\right)\left(2x^2+2x-1\right).$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = 0.$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.20 A174107: Nov 15 2011

Source: https://oeis.org/A174107. Selected conjecture: R. J. Mathar, Nov 15 2011. The defining Catalan transform and Gladkovskii's June 27, 2013 radical form are prior input.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 4$.

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= -2(3n-1) \\ p_2(n) &= 7(n-1) \\ p_3(n) &= -2(3n-5) \\ p_4(n) &= n-3. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= \frac{1}{2x}, \\ R_1 &= -\frac{1}{2x}, \\ D &= \frac{x^2-5x+1}{x^2-x+1}. \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -(x-1)(x+1).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 4$, with the initial residuals exactly as stated.

6.4.3.21 A174808: Feb 10 2015

Source: https://oeis.org/A174808. Selected conjecture: R. J. Mathar, Feb 10 2015. The established large-Schroeder transform, including Somos's 2014 continued fraction, is prior input.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 5$.

$$
\begin{aligned}
p_0(n) &= n+1\\
p_1(n) &= 4-5n\\
p_2(n) &= 13-11n\\
p_3(n) &= -3\,(n-1)\\
p_4(n) &= 3\,(n-4)\\
p_5(n) &= n-5.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$
\begin{aligned}
R_0 &= -\frac{x^2+x-1}{2x\,(x+1)},\\
R_1 &= -\frac{1}{2x\,(x+1)},\\
D &= x^4+2x^3-5x^2-6x+1.
\end{aligned}
$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$
\mathcal{L}_p F = P(x) = (2x+1)\left(x^2+x+1\right).
$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.22 A190169: Apr 09 2019

Source: https://oeis.org/A190169. Selected conjecture: R. J. Mathar, Apr 09 2019. Deutsch's odd-level horizontal-step statistic generating function is prior input. This is a distinct statistic and A-ID from the previously treated even-level entry A190166.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 8$.

$$
\begin{aligned}
p_0(n) &= -\,(n-1)\,(n+1)\\
p_1(n) &= -n\,(n-19)\\
p_2(n) &= 2\,(n-1)\,(7n-40)\\
p_3(n) &= -\,(n-2)\,(17n-97)\\
p_4(n) &= 2\left(9n^2-64n+119\right)\\
p_5(n) &= -17\,(n-5)\,(n-4)\\
p_6(n) &= (n-5)\,(19n-59)\\
p_7(n) &= -2\,(n-6)\,(8n-21)\\
p_8(n) &= 2\,(n-7)\,(2n-5)\,.
\end{aligned}
$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$R_0 = -\frac{1}{2x},$$
$$R_1 = \frac{x^4 - 2x^3 + x^2 - 2x + 1}{2x\left(x^2 - 3x + 1\right)\left(x^2 - x + 1\right)\left(x^2 + x + 1\right)},$$
$$D = \left(x^2 - 3x + 1\right)\left(x^2 + x + 1\right).$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -8x^3.$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 8$, with the initial residuals exactly as stated.

6.4.3.23 A208355: Aug 04 2015

Source: https://oeis.org/A208355. Selected conjecture: R. J. Mathar, Aug 04 2015. The known duplicated-Catalan formula and Russell's January 2024 generating function are prior input.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 2$.

$$p_0(n) = -(n-2)(n+3)$$
$$p_1(n) = -4$$
$$p_2(n) = 4(n-1)^2.$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$R_0 = -\frac{(x-1)(2x+1)}{2x^3},$$
$$R_1 = -\frac{x+1}{2x^3},$$
$$D = -(2x-1)(2x+1).$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = 6.$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 2$, with the initial residuals exactly as stated.

6.4.3.24 A228178: Aug 25 2013

Source: https://oeis.org/A228178. Selected conjecture: R. J. Mathar, Aug 25 2013. The source offset is zero, but its displayed generating function begins with x. Its supplied PARI Vec program extracts the list beginning with that coefficient. Accordingly the proof identifies a(n) with the

coefficient of degree n+1 in G, and divides G by x. This explicitly resolves the source's indexing convention; the separate 'Apparently partial sums' claim is not counted.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 4$.

$$\begin{aligned} p_0(n) &= 2\,(n+3) \\ p_1(n) &= -2\,(7n+11) \\ p_2(n) &= 29n+7 \\ p_3(n) &= 19-21n \\ p_4(n) &= 2\,(2n-5)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= -\frac{5x^2-8x+2}{2x^3\,(x-1)}, \\ R_1 &= \frac{x^2-4x+2}{2x^3\,(x-1)}, \\ D &= 1-4x. \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = 5x^2-4x+6.$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 4$, with the initial residuals exactly as stated.

6.4.3.25 A243022: Jul 15 2017

Source: https://oeis.org/A243022. Selected conjecture: R. J. Mathar, Jul 15 2017. The nested Catalan generating function in the definition is prior input. Rational parametrization below is used solely to certify the selected polynomial recurrence.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 5$.

$$\begin{aligned} p_0(n) &= 2n\,(n-1)\,(2n-3)\,(5n-12) \\ p_1(n) &= -\,(n-1)\left(85n^3-459n^2+776n-400\right) \\ p_2(n) &= -4\left(40n^4-396n^3+1455n^2-2324n+1360\right) \\ p_3(n) &= 4\,(n-4)\left(170n^3-1258n^2+3003n-2185\right) \\ p_4(n) &= 16\,(n-5)\,(n-4)\left(20n^2-118n+143\right) \\ p_5(n) &= -272\,(n-6)\,(n-5)\,(n-4)\,(5n-7)\,. \end{aligned}$$

Use $x = \xi(t) = \frac{t}{t^2+1}$, with $t(0) = 0$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$R_0 = -\frac{1}{2\left(t^2+1\right)},$$
$$R_1 = -\frac{1}{2\left(t^2+1\right)},$$
$$D = 1-4t.$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = 4x^2\left(23x+1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.26 A243255: Jul 15 2017

Source: https://oeis.org/A243255. Selected conjecture: R. J. Mathar, Jul 15 2017. The nested Motzkin–Catalan generating function in the definition is prior input.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 4$.

$$p_0(n) = 2n\left(2n+1\right)$$
$$p_1(n) = -65n^2+119n-42$$
$$p_2(n) = 3\left(97n^2-373n+342\right)$$
$$p_3(n) = -9\left(n-3\right)\left(23n-68\right)$$
$$p_4(n) = -567\left(n-4\right)\left(n-3\right).$$

Use $x = \xi(t) = \frac{t}{t^2+t+1}$, with $t(0) = 0$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$R_0 = \frac{1}{2},$$
$$R_1 = -\frac{1}{2},$$
$$D = 1-4t.$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -6x\left(4x-1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 4$, with the initial residuals exactly as stated.

6.4.3.27 A254314: Jun 07 2016

Source: https://oeis.org/A254314. Selected conjecture: R. J. Mathar, Jun 07 2016. The established quadratic generating function, tree interpretation and Gomez Aiza's 2024 asymptotic analysis are prior input. Only the separately conjectured recurrence is selected.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 5$.

$$\begin{aligned} p_0(n) &= n \\ p_1(n) &= -2\,(5n-6) \\ p_2(n) &= 2\,(17n-39) \\ p_3(n) &= -6\,(8n-27) \\ p_4(n) &= 25n-114 \\ p_5(n) &= -2\,(n-6)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} R_0 &= -\frac{x^2-6x+3}{2\,(2x-1)}, \\ R_1 &= \frac{1}{2\,(2x-1)}, \\ D &= (x-1)\left(x^3-11x^2+7x-1\right). \end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -x\left(12x^3-21x^2+14x-3\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.28 A257388: Sep 24 2016

Source: https://oeis.org/A257388. Selected conjecture: R. J. Mathar, Sep 24 2016. The known colored Motzkin-path generating function and its quadratic equation, also listed by Gutkovskiy in 2020, are prior input.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 3$.

$$\begin{aligned} p_0(n) &= n+2 \\ p_1(n) &= -8\,(n+1) \\ p_2(n) &= 4\,(3n+1) \\ p_3(n) &= 8\,(2n-3)\,. \end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$R_0 = \frac{1}{2x^2},$$
$$R_1 = \frac{1}{2x^2\,(4x-1)},$$
$$D = (4x-1)\left(4x^2+4x-1\right).$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -2\,(2x-1)\,.$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 3$, with the initial residuals exactly as stated.

6.4.3.29 A278023: Jun 24 2018

Source: https://oeis.org/A278023. Selected conjecture: R. J. Mathar, Jun 24 2018. The generating function printed in the sequence definition is prior input. Baril's irreducible-permutation interpretation is credited; the initial residual below qualifies the selected recurrence.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 5$.

Boundary qualification. The source specifies no starting index. $n = 4$ gives residual 156. These are all failures between the first index with all source terms defined and the proved tail. The coefficient formula itself is unchanged.

$$p_0(n) = n\left(3n^2-12n+11\right)$$
$$p_1(n) = -\,(3n-5)\left(3n^2-9n+4\right)$$
$$p_2(n) = -2\,(2n-5)\left(3n^2-6n+2\right)$$
$$p_3(n) = n\left(3n^2-12n+11\right)$$
$$p_4(n) = -2\,(2n-5)\left(3n^2-6n+2\right).$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$R_0 = -\frac{2x^3+x+1}{2\,(x^3+x+1)},$$
$$R_1 = \frac{(x-1)\,(2x+1)}{2\,(4x-1)\,(x^3+x+1)},$$
$$D = 1-4x.$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = 2x\left(78x^3-3x^2-x+1\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 5$, with the initial residuals exactly as stated.

6.4.3.30 A385252: Jul 31 2025

Source: https://oeis.org/A385252. Selected conjecture: R. J. Mathar, Jul 31 2025. The known ternary-string generating function and finite-sum formula are prior input. The definition's exceptional value a(0)=0 is retained and is essential to the boundary residual.

Statement. With the source offset $o = 0$, the following recurrence holds for every integer $n \geq 6$.

Boundary qualification. The source specifies no starting index. $n = 5$ gives residual -6912. These are all failures between the first index with all source terms defined and the proved tail. The coefficient formula itself is unchanged.

$$\begin{aligned}
p_0(n) &= n \\
p_1(n) &= 36 - 37n \\
p_2(n) &= 4\,(131n - 245) \\
p_3(n) &= -16\,(221n - 605) \\
p_4(n) &= 192\,(59n - 213) \\
p_5(n) &= -6912\,(2n - 9)\,.
\end{aligned}$$

Use $t = x$ and $\xi = x$.

The credited input has the exact form $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned}
R_0 &= \frac{72x^2 - 12x + 1}{2\,(4x - 1)\,(9x - 1)}, \\
R_1 &= \frac{12x - 1}{2\,(8x - 1)\,(9x - 1)}, \\
D &= 1 - 8x.
\end{aligned}$$

Proof certificate. Inserting the displayed data into (6.5) gives exactly

$$\mathcal{L}_p F = P(x) = -16x^2\left(432x^3 - 360x^2 + 69x - 4\right).$$

The radical residual is identically zero as a rational function. The rational residual equals the displayed P; both assertions are verified by clearing denominators in (6.5). Taking coefficients in (6.4) proves the recurrence for every $n \geq 6$, with the initial residuals exactly as stated.

6.4.4 Verification and the precise prior-work audit

The portable verifier checks twenty-nine exact differential identities and one exact hypergeometric identity. It parses the retained primary-source mathematical statements and verifies every recurrence coefficient. It independently expands the input generating functions without using the target recurrence, matching twenty source terms per entry, or 600 terms in total. Additional finite-sum and convolution checks guard several defining inputs. For the three parameters it uses Lagrange inversion, $[x^n]t(x) = \frac{1}{n}[u^{n-1}]\phi(u)^n$ when $t = x\phi(t)$. These numerical checks guard branch and indexing choices; the rational identities are the all-index proofs.

All thirty IDs are disjoint from the previous 508 selected IDs, from the 85-entry parameter table in Georg Fischer's January 26, 2020 general proof for A176604, from the 33 explicit Ekhad–Yang 2017 proof IDs, and from the retained 97-ID OEIS Open Appendix A baseline. The latter represents the 100 costliest Appendix A claims among 153 combined-model resolutions, not all results of the OEIS

Open project. The same finite baseline limitation applies to the accompanying JSON audit. No completeness of a global literature search is claimed.

Primary-entry proof notices were inspected. A396722 was excluded because its entry contains Rodriguez's September 28, 2026 proof; A000781 was excluded because its entry contains Blanco Gomez's October 1, 2026 proof. A217333 was rejected as a shifted alias of the already proved A025273 recurrence, and A080243 was conservatively excluded as a signed little-Schroeder alias. None contributes to the thirty. The previously disclosed Round IV Fischer-family priority correction remains in force; no member of that 85-entry family is counted here.

The sources linked from the selected entries were checked selectively: Gomez Aiza's 2024 *Trees with flowers*, Drube's 2022 *Raised k-Dyck paths*, Barry's 2009 transformation paper and 2023 elliptic-curve paper, the 2019 forbidden-pattern paper of Asinowski and coauthors, Rowland's binary-tree paper, Baril's irreducible permutation paper, and the 2023 parking-function paper of Adeniran and Pudwell. Their known enumerations or generating functions are credited above. Targeted searches for the selected IDs or Mathar's name in those documents did not identify an explicit proof of these selected recurrence formulas. This restricted observation is not an absence-of-prior-proof theorem. Several broad web queries returned irrelevant results and are not treated as evidence of novelty. In particular A025268, A157003 and A159772 already have different polynomial recurrences on their OEIS pages. This collection proves the separately labeled formulas from their established inputs; it does not portray those sequences as previously unknown.

6.5 Forty-one empirical array recurrences

This section proves forty-one different entry-level recurrence assertions explicitly marked *Empirical* on their primary OEIS pages, consulted on 20 October 2026. Twenty-five displayed empirical or conjectural rational generating functions are proved within the same packages. The source definitions and conjectures are due to R. H. Hardin; the additional generating-function conjectures are credited on the relevant pages to Colin Barker. The forty-one A-numbers are absent from the 508 entries selected in rounds I–V. Several entries share a transfer principle: forty-one is the number of selected OEIS assertions, not the number of unrelated proof techniques.

6.5.1 Local conditions and the counting model

All arrays have $n+1$ rows, fixed width $w \geq 2$, and entries in $\{0, \ldots, q-1\}$. For a block $B = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ put

$$\Delta(B) = (b-a, d-b, c-d, a-c).$$

These differences are ordinary signed integers. The selected local conditions are the following; S and s are specified for each entry.

Cell values (C). $\#\{a, b, c, d\} \in S$.

Stress (D). $|a+d-b-c| = s$. When the source prohibits equal adjacent elements, additionally every entry of $\Delta(B)$ is nonzero.

Absolute edge sum (U). $\sum_{e \in \Delta(B)} |e| = s$ and every edge difference is nonzero. The sum includes all four edges with multiplicity.

Squared edge sum (V). $\sum_{e \in \Delta(B)} e^2 = s$ and every edge difference is nonzero, again including all four edges.

One permitted orientation (J). Exactly one of $\max \Delta(B) \leq 1$ and $\min \Delta(B) \geq -1$ holds. These are the clockwise and counterclockwise bounds in the source definition.

Every horizontal or vertical neighboring pair of a selected array belongs to a 2×2 block, since $n+1 \geq 2$ and $w \geq 2$. Thus the blockwise nonzero condition enforces precisely the source's prohibition of equal adjacent elements, including at the outer boundary. Diagonal neighbors are not subject to that prohibition.

6.5.2 Definition-derived transfer matrices

For C, canonicalize a row by renaming its colors in order of first appearance. For D, U, V and J, canonicalize under the four operations consisting of horizontal reflection, simultaneous complementation $x \mapsto q-1-x$, their product, and the identity. Write γ for the appropriate canonicalization and $\mathcal{O}_{q,w}$ for the lexicographically ordered representative rows. For each representative r define its actual orbit mass

$$m(r) = \#\{v \in \{0,\dots,q-1\}^w : \gamma(v) = r\}. \tag{6.6}$$

For the appropriate local predicate P, define the nonnegative integer matrix

$$T(r,s) = \#\left\{ v \in \{0,\dots,q-1\}^w : \begin{array}{l} \gamma(v) = s, \\ P(r_i, r_{i+1}, v_i, v_{i+1}) \quad (1 \le i < w) \end{array} \right\}. \tag{6.7}$$

For C, any simultaneous color permutation preserves P. For the other predicates, horizontal reflection and complementation permute and negate the edge differences and preserve the absolute stress. They preserve U and V and interchange the two bounds in J, preserving their exclusive disjunction. Consequently every row in an orbit has the same number of successors in each orbit. No arbitrary color permutation is used for D, U, V or J.

Adding rows, starting with the actual $m(r)$ labeled first rows, therefore gives the exact labeled count

$$L_q(n) = m^{\mathsf{T}} T^n \mathbf{1} \qquad (n \ge 1). \tag{6.8}$$

The exponent is n because an array has $n+1$ rows. Fixed representatives and smaller orbits retain their correct masses. In particular, a row that cannot extend to a legal array simply has no outgoing edges. For a scalar-normalized source definition, $a(n) = \lambda L_q(n)$, with precisely the printed scalar λ; no orbit interpretation of that scalar is assumed.

Seven C entries additionally identify global color renamings by row-major first-occurrence order. Let $U_q^{=}$ count such arrays using exactly q colors, and $U_q^{\le}$ those using at most q colors. Surjective labelings have $q!$ different color renamings. Inclusion–exclusion on omitted colors and then summation over the number actually used give

$$U_q^{=}(n) = \sum_{j=1}^{q} \frac{(-1)^{q-j}}{j!(q-j)!} L_j(n), \tag{6.9}$$

$$U_q^{\le}(n) = \sum_{j=1}^{q} \left(\frac{1}{j!} \sum_{h=0}^{q-j} \frac{(-1)^h}{h!} \right) L_j(n). \tag{6.10}$$

The empty alphabet contributes zero. A requirement to use every color is imposed only when the source explicitly includes it. In the other four C entries, the source normalization is simply $L_4/24$ or $L_3/6$. These constructions adapt the project's earlier color-orbit and geometric-orbit transfer methods; the present local predicates and selected source assertions are specified afresh here.

6.5.3 Exact certificates and the all-index conclusion

For each printed coefficient signature $c = (c_1,\dots,c_d)$ set

$$R(z) = z^d - \sum_{i=1}^{d} c_i z^{d-i}, \qquad Q(x) = 1 - \sum_{i=1}^{d} c_i x^i.$$

For every matrix in that entry with a nonzero scalar weight, the complete exact integer certificate establishes

$$\boxed{TR(T)\mathbf{1} = \mathbf{0}.} \tag{6.11}$$

For $n \geq d+1$, multiply this identity by $m^{\mathsf{T}} T^{n-d-1}$ and by the scalar weight, then add the weighted components. Equations (6.8)–(6.10) prove

$$\boxed{a(n) = \sum_{i=1}^{d} c_i a(n-i) \qquad (n \geq d+1).} \tag{6.12}$$

All source indices on the right are at least 1. The auxiliary value obtained by setting $n = 0$ in the transfer formula is not an asserted OEIS term and may be nonintegral under a scalar normalization.

The certificate file contains the forty-one signatures, every matrix component's exact weight, all vectors $v_k = T^k \mathbf{1}$ for $0 \leq k \leq d+1$, and every residual coordinate in (6.11). There are 49 distinct complete matrices and 6,105 nonzero weighted edges; the largest matrix has 342 states and the largest selected recurrence has order 31. The reproducible exact check is

```
v[0] = [1 for every state]
for k = 1,...,d+1:
    v[k][r] = sum(T[r,s] * v[k-1][s] for every state s)
for every state r:
    assert v[d+1][r] == sum(c[i] * v[d+1-i][r] for i=1,...,d)
```

The verifier rebuilds every state and transition directly from the finite definition (6.7), compares the stored matrices and all certificate coordinates, and uses only exact integers and fractions. This is a computer-assisted finite proof: further powers of T, not agreement with many numerical terms, establish the universal conclusion. The matrix definitions and all coefficient signatures below specify the entire finite calculation. The complete expanded certificates are supplied in `r6_arrays_matrices.json` and `r6_arrays_certificates.json`.

For $F(x) = \sum_{n \geq 1} a(n) x^n$ the proved recurrence yields $QF = P$, where $\deg P \leq d$ and

$$[x^k]P = \sum_{\substack{0 \leq i \leq \min(k,d) \\ k-i \geq 1}} q_i a(k-i), \qquad q_0 = 1, \quad q_i = -c_i \ (i \geq 1). \tag{6.13}$$

Every initial value in this expression is calculated from the defining transfer matrix. For the 25 source-stated rational generating functions, exact polynomial cross-multiplication proves equality with this P/Q; the relevant numerator P is also displayed below. The source's original rational expression is retained in the statement and certificate files. No generating function adds a second count to its A-number. Separately stated radical closed forms are not additional selected claims. All 832 printed source-prefix terms agree with the independent matrix counts; that comparison checks indexing and interpretation, not the universal recurrence. The forty-one computed 20-term prefixes are distinct.

6.5.4 A short hand-checkable example: A210406

Here $q = 4$, $w = 2$ and $S = \{3, 4\}$. A legal array uses at least three colors, so each orbit under four-color permutations has size 24. For the two row representatives $00, 01$, direct enumeration gives

$$T = \begin{pmatrix} 0 & 6 \\ 2 & 10 \end{pmatrix}, \qquad m = \begin{pmatrix} 4 \\ 12 \end{pmatrix}, \qquad a(n) = \frac{1}{24} m^{\mathsf{T}} T^n \mathbf{1} \quad (n \geq 1).$$

The identity $T^2 = 10T + 12I$ proves $a(n) = 10a(n-1) + 12a(n-2)$ for $n \geq 3$. Also $a(1) = 7$ and $a(2) = 78$, so

$$F(x) = \frac{x(7 + 8x)}{1 - 10x - 12x^2}.$$

This proves precisely the empirical recurrence and the conjectural rational generating function on https://oeis.org/A210406. The larger selected matrices use the same exact finite argument.

6.5.5 Source audit and counting limits

The status relevant to selection is the word *Empirical* attached to each recurrence, not OEIS's editorial label "approved." The consulted primary pages link numerical data tables and do not record a proof of their selected recurrence. Representative exact-A-number and family proof searches found no such proof among the inspected results. This is a limited primary-source audit, not a global-priority claim or a proof that no prior proof exists.

All forty-one IDs are disjoint from the 508 previously selected project entries, and from the retained 97 distinct A-numbers in the 100 costly claims in Appendix A of *OEIS Open*. That appendix check does not cover all repository results or all mathematical literature. Four larger certificate-verified candidates and two further sourced candidates were left outside this forty-one-entry selection and add no count. No OEIS entry has been edited or submitted by this work.

6.5.6 Individual definitions and complete recurrence signatures

Every list below is $c_1, \ldots, c_d$ in order, including zeros, in (6.12). The displayed component dimensions include all nonzero weighted components. Initial values begin at $n = 1$.

6.5.6.1 A183586

Arrays have $n{+}1$ rows and width $w = 2$ on the alphabet $0, \ldots, 3$. Predicate C requires $S = \{3\}$ distinct values per block. The labeled count is multiplied by $\lambda = 1/24$. The recurrence holds for every $n \geq 3$, with order $d = 2$ and component dimensions (2) (total $D = 2$). Source: https://oeis.org/A183586. The first three values are $(6, 56, 520)$.

$$(c_1, \ldots, c_d) = (8, 12).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 6x + 8x^2.$$

6.5.6.2 A183587

Arrays have $n{+}1$ rows and width $w = 3$ on the alphabet $0, \ldots, 3$. Predicate C requires $S = \{3\}$ distinct values per block. The labeled count is multiplied by $\lambda = 1/24$. The recurrence holds for every $n \geq 5$, with order $d = 4$ and component dimensions (5) (total $D = 5$). Source: https://oeis.org/A183587. The first three values are $(56, 1243, 27380)$.

$$(c_1, \ldots, c_d) = (24, -30, -296, 192).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 56x - 101x^2 - 772x^3 + 512x^4.$$

6.5.6.3 A183589

Arrays have $n + 1$ rows and width $w = 5$ on the alphabet $0, \ldots, 3$. Predicate C requires $S = \{3\}$ distinct values per block. The labeled count is multiplied by $\lambda = 1/24$. The recurrence holds for every $n \geq 32$, with order $d = 31$ and component dimensions (51) (total $D = 51$). Source: https://oeis.org/A183589. The first three values are $(4832, 603766, 74491726)$.

```
179, -6470, -111366, 8421556, -27797881, -3641734984, 33282463706, 695892591667,
-8771266871058, -59332612133336, 1069043025480468, 1401139251128208,
-67977787226742392, 103924226454772488, 2296282350133287328, -7835601773053827936,
-39951799309240069504, 217732921395304638464, 281480768985183547392,
-308871046940060599296O, 1060118307218454011904, 2323510682177754552729 6,
-298135028709049118228 48, -8620591085615221781 2992, 1763404553969134070 66112,
1203524695183646334648 32, -4130358369613784494899 20, 2341275689883680532 0704,
3365800174144712941240 32, -8748212139556775500 1856, -5261052670974917345 2800
```

6.5.6.4 A183596

Arrays have $n+1$ rows and width $w = 3$ on the alphabet $0, \ldots, 2$. Predicate C requires $S = \{3\}$ distinct values per block. The labeled count is multiplied by $\lambda = 1/6$. The recurrence holds for every $n \geq 5$, with order $d = 4$ and component dimensions (5) (total $D = 5$). Source: https://oeis.org/A183596. The first three values are $(27, 204, 1479)$.

$$(c_1, \ldots, c_d) = (8, -3, -16, 2).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 27x - 12x^2 - 72x^3 + 9x^4.$$

6.5.6.5 A210176

Arrays have $n + 1$ rows and width $w = 2$ on the alphabet $0, \ldots, 3$. Predicate C requires $S = \{3, 4\}$ distinct values per block. The array uses all 4 colors and is renamed in row-major first-occurrence order. The recurrence holds for every $n \geq 5$, with order $d = 4$ and component dimensions $(1, 2, 2, 2)$ (total $D = 7$). Source: https://oeis.org/A210176. The first three values are $(1, 51, 744)$.

$$(c_1, \ldots, c_d) = (14, -26, -68, -24).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = x + 37x^2 + 56x^3 + 20x^4.$$

6.5.6.6 A210177

Arrays have $n + 1$ rows and width $w = 3$ on the alphabet $0, \ldots, 3$. Predicate C requires $S = \{3, 4\}$ distinct values per block. The array uses all 4 colors and is renamed in row-major first-occurrence order. The recurrence holds for every $n \geq 9$, with order $d = 8$ and component dimensions $(1, 4, 5, 5)$ (total $D = 15$). Source: https://oeis.org/A210177. The first three values are $(51, 2316, 79017)$.

```
39, -211, -523, 2954, -1030, -4272, 2096, -192
```

6.5.6.7 A210178

Arrays have $n + 1$ rows and width $w = 4$ on the alphabet $0, \ldots, 3$. Predicate C requires $S = \{3, 4\}$ distinct values per block. The array uses all 4 colors and is renamed in row-major first-occurrence order. The recurrence holds for every $n \geq 17$, with order $d = 16$ and component dimensions $(1, 8, 14, 15)$ (total $D = 38$). Source: https://oeis.org/A210178. The first three values are $(744, 79017, 7363642)$.

```
93, 199, -28711, 35176, 2453690, -5602318, -51561340, 96505796, 378763976, -653899024,
-988302720, 1830649984, 381532160, -1548998656, 621346816, -44040192
```

6.5.6.8 A210406

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,3$. Predicate C requires $S=\{3,4\}$ distinct values per block. The array uses at most 4 colors and is renamed in row-major first-occurrence order. The recurrence holds for every $n \geq 3$, with order $d=2$ and component dimensions $(1,2,2)$ (total $D=5$). Source: https://oeis.org/A210406. The first three values are $(7,78,864)$.

$$(c_1,\dots,c_d)=(10,12).$$

The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x)=7x+8x^2.$$

6.5.6.9 A210407

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,3$. Predicate C requires $S=\{3,4\}$ distinct values per block. The array uses at most 4 colors and is renamed in row-major first-occurrence order. The recurrence holds for every $n \geq 5$, with order $d=4$ and component dimensions $(1,4,5)$ (total $D=10$). Source: https://oeis.org/A210407. The first three values are $(78,2520,80496)$.

$$(c_1,\dots,c_d)=(31,40,-280,96).$$

The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x)=78x+102x^2-744x^3+256x^4.$$

6.5.6.10 A210408

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,3$. Predicate C requires $S=\{3,4\}$ distinct values per block. The array uses at most 4 colors and is renamed in row-major first-occurrence order. The recurrence holds for every $n \geq 10$, with order $d=9$ and component dimensions $(1,8,15)$ (total $D=24$). Source: https://oeis.org/A210408. The first three values are $(864,80496,7380816)$.

```
81, 1151, -13041, -117154, 478740, 445664, -2873152, 2422784, -196608
```

6.5.6.11 A210409

Arrays have $n+1$ rows and width $w=5$ on the alphabet $0,\dots,3$. Predicate C requires $S=\{3,4\}$ distinct values per block. The array uses at most 4 colors and is renamed in row-major first-occurrence order. The recurrence holds for every $n \geq 32$, with order $d=31$ and component dimensions $(1,16,51)$ (total $D=68$). Source: https://oeis.org/A210409. The first three values are $(9576,2574592,677993552)$.

```
276, -844, -647137, 5856940, 402216721, -5862695617, -54643074114, 1382343039112,
-3102790608576, -89911455972644, 660007977690996, 263588914646272, -18055429128783448,
51374341139205968, 113931146918069408, -752286799961769696, 423125539061919552,
3650072001026978048, -6040761986311450112, -5385080244386082816, 17461611564608397312,
-24571305242592051 20, -16604319853477298176, 6207057768744222720, 6168586199070932992,
-2183260302370406400, -667625816443584512, 191550881093648384, 17950332629811200,
-3348386568732672, 49529562857472
```

6.5.6.12 A233921

Arrays have $n+1$ rows and width $w = 2$ on the alphabet $0, \ldots, 4$. Predicate U has target $s = 8$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 7$, with order $d = 6$ and component dimensions (9) (total $D = 9$). Source: https://oeis.org/A233921. The first three values are $(98, 518, 2792)$.

$$(c_1, \ldots, c_d) = (6, 5, -46, 5, 60, 20).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 98x - 70x^2 - 806x^3 + 396x^4 + 1288x^5 + 400x^6.$$

6.5.6.13 A233982

Arrays have $n + 1$ rows and width $w = 2$ on the alphabet $0, \ldots, 4$. Predicate V has target $s = 10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 4$, with order $d = 3$ and component dimensions (9) (total $D = 9$). Source: https://oeis.org/A233982. The first three values are $(56, 236, 976)$.

$$(c_1, \ldots, c_d) = (2, 8, 4).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 56x + 124x^2 + 56x^3.$$

6.5.6.14 A233983

Arrays have $n+1$ rows and width $w = 3$ on the alphabet $0, \ldots, 4$. Predicate V has target $s = 10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 14$, with order $d = 13$ and component dimensions (39) (total $D = 39$). Source: https://oeis.org/A233983. The first three values are $(236, 1540, 9632)$.

```
3, 53, -116, -892, 1620, 5804, -9766, -13812, 23148, 8224, -16848, 2560, 256
```

6.5.6.15 A233984

Arrays have $n+1$ rows and width $w = 4$ on the alphabet $0, \ldots, 4$. Predicate V has target $s = 10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 26$, with order $d = 25$ and component dimensions (169) (total $D = 169$). Source: https://oeis.org/A233984. The first three values are $(976, 9632, 92320)$.

```
0, 152, 150, -8071, -8102, 220582, 154134, -3549162, -957558, 35319910, -6392950,
-218035822, 131758738, 804014815, -803217908, -1602211348, 2304441408, 1290558096,
-3076728768, 201005888, 1628959232, -560044032, -245846016, 128188416, -7864320
```

6.5.6.16 A234162

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,4$. Predicate D has stress $s=2$. The recurrence holds for every $n \geq 5$, with order $d=4$ and component dimensions (9) (total $D=9$). Source: https://oeis.org/A234162. The first three values are $(136, 766, 4296)$.

$$(c_1,\dots,c_d) = (4, 41, -132, -264).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 136x + 222x^2 - 4344x^3 - 6480x^4.$$

6.5.6.17 A234163

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,4$. Predicate D has stress $s=2$. The recurrence holds for every $n \geq 18$, with order $d=17$ and component dimensions (39) (total $D=39$). Source: https://oeis.org/A234163. The first three values are $(766, 4970, 32070)$.

```
17, 16, -1651, 5750, 45373, -252572, -278797, 2737923, 518738, -11300724, 169584,
18786600, -2246544, -13289184, 2960064, 3359232, -1119744
```

6.5.6.18 A234220

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,4$. Predicate D has stress $s=3$. The recurrence holds for every $n \geq 4$, with order $d=3$ and component dimensions (9) (total $D=9$). Source: https://oeis.org/A234220. The first three values are $(104, 436, 1824)$.

$$(c_1,\dots,c_d) = (4, 20, -80).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 104x + 20x^2 - 2000x^3.$$

6.5.6.19 A234221

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,4$. Predicate D has stress $s=3$. The recurrence holds for every $n \geq 7$, with order $d=6$ and component dimensions (39) (total $D=39$). Source: https://oeis.org/A234221. The first three values are $(436, 1560, 5612)$.

$$(c_1,\dots,c_d) = (7, 20, -224, 144, 1680, -2880).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 436x - 1492x^2 - 14028x^3 + 47904x^4 + 106320x^5 - 360000x^6.$$

6.5.6.20 A234222

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,4$. Predicate D has stress $s=3$. The recurrence holds for every $n \geq 12$, with order $d=11$ and component dimensions (169) (total $D=169$). Source: https://oeis.org/A234222. The first three values are $(1824, 5612, 17448)$.

```
9, 20, -390, 460, 5520, -14240, -26400, 115776, -13824, -299520, 276480
```

6.5.6.21 A234557

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,4$. Predicate D has stress $s=4$. The recurrence holds for every $n \geq 6$, with order $d=5$ and component dimensions (9) (total $D=9$). Source: https://oeis.org/A234557. The first three values are $(70, 220, 618)$.

$$(c_1,\dots,c_d) = (3, 18, -54, -80, 240).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 70x + 10x^2 - 1302x^3 - 80x^4 + 6000x^5.$$

6.5.6.22 A234558

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,4$. Predicate D has stress $s=4$. The recurrence holds for every $n \geq 15$, with order $d=14$ and component dimensions (39) (total $D=39$). Source: https://oeis.org/A234558. The first three values are $(220, 546, 1240)$.

```
4, 30, -132, -331, 1720, 1530, -11280, -1324, 39136, -12360, -67968, 39456, 46080,
-34560
```

6.5.6.23 A234559

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,4$. Predicate D has stress $s=4$. The recurrence holds for every $n \geq 15$, with order $d=14$ and component dimensions (169) (total $D=169$). Source: https://oeis.org/A234559. The first three values are $(618, 1240, 2390)$.

```
4, 30, -132, -331, 1720, 1530, -11280, -1324, 39136, -12360, -67968, 39456, 46080,
-34560
```

6.5.6.24 A234762

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,3$. Predicate J permits exactly one orientation whose four edge jumps are all at most $+1$. The recurrence holds for every $n \geq 6$, with order $d=5$ and component dimensions (6) (total $D=6$). Source: https://oeis.org/A234762. The first three values are $(56, 240, 916)$.

$$(c_1,\dots,c_d) = (1, 14, -1, -25, -10).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 56x + 184x^2 - 108x^3 - 420x^4 - 156x^5.$$

6.5.6.25 A234763

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,3$. Predicate J permits exactly one orientation whose four edge jumps are all at most $+1$. The recurrence holds for every $n \geq 14$, with order $d=13$ and component dimensions (20) (total $D=20$). Source: https://oeis.org/A234763. The first three values are $(240, 1624, 9404)$.

```
8, 21, -252, 125, 1818, -2050, -4364, 6237, 2860, -5787, 336, 1311, -262
```

6.5.6.26 A234764

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,3$. Predicate J permits exactly one orientation whose four edge jumps are all at most $+1$. The recurrence holds for every $n \geq 28$, with order $d=27$ and component dimensions (72) (total $D=72$). Source: https://oeis.org/A234764. The first three values are $(916, 9404, 71280)$.

```
3, 137, -284, -6386, 9604, 145993, -160399, -1876827, 1488326, 14432970, -8040903,
-68366990, 25519880, 201876831, -47552365, -371183764, 52834992, 418743519, -37830792,
-280238238, 20541150, 104060106, -8730936, -18816300, 2167560, 1263600, -194400
```

6.5.6.27 A234875

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,4$. Predicate D has stress $s=1$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 5$, with order $d=4$ and component dimensions (9) (total $D=9$). Source: https://oeis.org/A234875. The first three values are $(32, 68, 128)$.

$$(c_1,\dots,c_d) = (0,5,0,-4).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 32x + 68x^2 - 32x^3 - 72x^4.$$

6.5.6.28 A234876

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,4$. Predicate D has stress $s=1$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 10$, with order $d=9$ and component dimensions (39) (total $D=39$). Source: https://oeis.org/A234876. The first three values are $(68, 148, 268)$.

```
2, 10, -21, -29, 68, 22, -73, 0, 22
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 68x + 12x^2 - 708x^3 + 40x^4 + 2332x^5 - 416x^6 - 2684x^7 + 440x^8 + 924x^9.$$

6.5.6.29 A234877

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,4$. Predicate D has stress $s=1$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 12$, with order $d=11$ and component dimensions (169) (total $D=169$). Source: https://oeis.org/A234877. The first three values are $(128, 268, 472)$.

```
2, 11, -23, -38, 87, 46, -130, -16, 78, 0, -16
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$\begin{aligned} P(x) = {} & 128x + 12x^2 - 1472x^3 + 168x^4 + 5700x^5 - 1248x^6 - 9048x^7 + 1968x^8 \\ & + 5984x^9 - 768x^{10} - 1376x^{11}. \end{aligned}$$

6.5.6.30 A235011

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\ldots,4$. Predicate D has stress $s=3$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 6$, with order $d=5$ and component dimensions (39) (total $D=39$). Source: https://oeis.org/A235011. The first three values are $(104, 224, 488)$.

$$(c_1,\ldots,c_d) = (3, 4, -14, -4, 16).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 104x - 88x^2 - 600x^3 + 208x^4 + 832x^5.$$

6.5.6.31 A235012

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\ldots,4$. Predicate D has stress $s=3$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 9$, with order $d=8$ and component dimensions (169) (total $D=169$). Source: https://oeis.org/A235012. The first three values are $(264, 488, 936)$.

```
4, 4, -30, 7, 74, -46, -60, 48
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 264x - 568x^2 - 2072x^3 + 4136x^4 + 5408x^5 - 9792x^6 - 4672x^7 + 7552x^8.$$

6.5.6.32 A235080

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\ldots,4$. Predicate D has stress $s=4$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 5$, with order $d=4$ and component dimensions (9) (total $D=9$). Source: https://oeis.org/A235080. The first three values are $(38, 90, 202)$.

$$(c_1,\ldots,c_d) = (2, 6, -10, -5).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 38x + 14x^2 - 206x^3 - 86x^4.$$

6.5.6.33 A235081

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\ldots,4$. Predicate D has stress $s=4$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 11$, with order $d=10$ and component dimensions (39) (total $D=39$). Source: https://oeis.org/A235081. The first three values are $(90, 198, 418)$.

```
3, 10, -33, -28, 111, 20, -111, -27, 30, 10
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$\begin{aligned} P(x) = {} & 90x - 72x^2 - 1076x^3 + 666x^4 + 4086x^5 - 1444x^6 - 5108x^7 - 514x^8 \\ & + 1544x^9 + 452x^{10}. \end{aligned}$$

6.5.6.34 A235271

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,4$. Predicate D has stress $s=5$. The recurrence holds for every $n \geq 5$, with order $d=4$ and component dimensions (9) (total $D=9$). Source: https://oeis.org/A235271. The first three values are $(40, 100, 208)$.

$$(c_1,\dots,c_d) = (0, 10, 0, -24).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 40x + 100x^2 - 192x^3 - 480x^4.$$

6.5.6.35 A235272

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,4$. Predicate D has stress $s=5$. The recurrence holds for every $n \geq 10$, with order $d=9$ and component dimensions (39) (total $D=39$). Source: https://oeis.org/A235272. The first three values are $(100, 208, 380)$.

```
1, 15, -15, -80, 80, 180, -180, -144, 144
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 100x + 108x^2 - 1328x^3 - 1156x^4 + 6236x^5 + 3864x^6 - 12312x^7 - 4032x^8 + 8640x^9.$$

6.5.6.36 A235273

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,4$. Predicate D has stress $s=5$. The recurrence holds for every $n \geq 10$, with order $d=9$ and component dimensions (169) (total $D=169$). Source: https://oeis.org/A235273. The first three values are $(208, 380, 632)$.

```
1, 15, -15, -80, 80, 180, -180, -144, 144
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 208x + 172x^2 - 2868x^3 - 1932x^4 + 13924x^5 + 6792x^6 - 28248x^7 - 7488x^8 + 20160x^9.$$

6.5.6.37 A235303

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,5$. Predicate D has stress $s=6$. The recurrence holds for every $n \geq 6$, with order $d=5$ and component dimensions (12) (total $D=12$). Source: https://oeis.org/A235303. The first three values are $(70, 210, 518)$.

$$(c_1,\dots,c_d) = (3, 13, -39, -40, 120).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 70x - 1022x^3 + 3600x^5.$$

6.5.6.38 A235304

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,5$. Predicate D has stress $s=6$. The recurrence holds for every $n \geq 15$, with order $d=14$ and component dimensions (63) (total $D=63$). Source: https://oeis.org/A235304. The first three values are $(210, 518, 1106)$.

```
4, 25, -112, -231, 1260, 875, -7280, -224, 22736, -7980, -36288, 21456, 23040, -17280
```

6.5.6.39 A235305

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,5$. Predicate D has stress $s=6$. The recurrence holds for every $n \geq 15$, with order $d=14$ and component dimensions (342) (total $D=342$). Source: https://oeis.org/A235305. The first three values are $(518, 1106, 2118)$.

```
4, 25, -112, -231, 1260, 875, -7280, -224, 22736, -7980, -36288, 21456, 23040, -17280
```

6.5.6.40 A235312

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,6$. Predicate D has stress $s=7$. The recurrence holds for every $n \geq 7$, with order $d=6$ and component dimensions (16) (total $D=16$). Source: https://oeis.org/A235312. The first three values are $(112, 392, 1120)$.

$$(c_1,\dots,c_d) = (0, 28, 0, -252, 0, 720).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 112x + 392x^2 - 2016x^3 - 7056x^4 + 8640x^5 + 30240x^6.$$

6.5.6.41 A235313

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,6$. Predicate D has stress $s=7$. The recurrence holds for every $n \geq 19$, with order $d=18$ and component dimensions (100) (total $D=100$). Source: https://oeis.org/A235313. The first three values are $(392, 1120, 2744)$.

```
4, 47, -200, -901, 4204, 8957, -48440, -47194, 334096, 101948, -1410080, 170136,
3549696, -1446912, -4861440, 2954880, 2764800, -2073600
```

6.6 Reproduction and verification

The archive includes this standalone manuscript, the numbered ledger, four proof fragments, four main verifiers, an independent array checker, complete transfer matrices, exact algebraic certificates, source-scope audits and four cross-review reports. Full scraped webpages are omitted.

6.6.1 Running the offline checks

Use Python 3.10 or later with SymPy and mpmath. From the extracted directory run:

```
python3 r6_run_all.py
```

The runner checks the 100 distinct IDs, the 508-ID exclusion ledger, the proof labels and the reviewed proof/code hashes. It executes the following independent components without network access.

Script	Purpose
`r6_root_verify.py`	Exact algebraic and modular certificates, with bounded series and numerical checks supplementing 18 written proofs.
`r6_arith_verify.py`	Exact arithmetic checks for 11 packages, including the complete modular certificate for A116184.
`r6_arith_independent_check.py`	Independent common-denominator enumeration of its complete 49,284-residue period.
`r6_gf_verify.py`	30 exact all-index differential certificates and 600 source-prefix checks, retaining all boundary terms.
`r6_arrays_verify.py`	41 recurrences from 49 complete rebuilt integer matrices and their vector identities; 25 source GFs.
`r6_arrays_root_verify.py`	82 independent initial counts using columns and global used-color sets, without importing the row engine or using its inclusion–exclusion normalization.

6.6.2 Why the finite certificates prove every index

For an array recurrence the defining count is a linear combination of $m^{\mathsf{T}}T^n\mathbf{1}$. The complete identity $TR(T)\mathbf{1} = 0$ remains true after multiplication by every nonnegative power of T, proving the recurrence on the stated infinite range. All 6,105 nonzero weighted edges are rebuilt from the local definitions; the largest matrix has 342 states and the largest recurrence has order 31. Every coordinate is checked with integers. Agreement of 832 source-prefix values is an interpretation check, not the proof of the universal recurrence.

For a generating-function recurrence, the exact action of its differential operator is evaluated in a rational algebraic basis. The residual is the displayed boundary polynomial; coefficients beyond its degree vanish identically. This proves the stated infinite tail, including each qualified starting range.

The other universal conclusions follow from the written algebraic, valuation, modular, or analytic arguments. Their bounded calculations check transcription and indexing. In particular, the theta evaluations follow from exact classical identities, with analytic convergence justified; their high-precision numerical checks are supplementary.

6.6.3 Rebuilding and integrity

The delivered `oeis_hunt_round6.tex` contains all proof text directly and needs no external `\input` files. Compile it twice with pdfLaTeX and the standard packages in its preamble. To rebuild the manuscript, catalogue and combined audit index, run:

```
python3 r6_build_dossier.py
```

The README records optional certificate reconstruction commands. `manifest.json` gives SHA-256 digests of the delivered files. Running the verifiers regenerates reports and timing fields, so those report digests can change after a successful rerun. Review bindings for immutable proof and code files are checked separately.

Source records retain authorship, the selected assertion, indexing, excluded candidates and search limits. The 100-entry selection count does not certify historical first discovery or official OEIS closure.

Part VII

Divisibility, arithmetic asymptotics, and exact array recurrences

Round overview

This seventh-round dossier resolves 100 selected assertion packages on 100 distinct OEIS entries, disjoint from the 608 A-numbers selected in Rounds I–VI. It proves 91 selected claims, specifies and proves the valid tail of four conjectural recurrences, corrects and proves four formulas or indexing/support statements, and refutes one asymptotic conjecture. Multiple assertions on one entry count once.

The four parts contain 14 packages on divisibility and formal congruences (including one theta-index correction), 10 on arithmetic and asymptotics, 31 on generating functions, and 45 on constrained arrays. The written arguments use integer coefficient rings, exact divisor and lattice formulas, differential identities with boundary polynomials, and finite transfer operators with integer vector certificates.

What the count means

Each catalogue row is one precisely scoped assertion package attached to one distinct A-number. Several rows can share a general lemma. The count does not assert 100 unrelated methods or 100 new general theorems. The consulted source statements explicitly mark the selected claims as conjectural or empirical. Established formulas, classical identities and previous project lemmas are credited inputs.

Some entries contain further questions outside the selected scope. In particular, the exponential-series packages do not claim their separate asymptotic observations. The chosen scope and every correction are recorded in the proofs and the machine-readable ledger. Two additional certified array candidates were held out when the total reached 100; they are not counted in this dossier.

The previous 608 is an exclusion ledger, not a certified discovery total. The five Round IV priority withdrawals disclosed in Round V remain in force; all those A-IDs remain excluded. New consequences at previously selected A397242 and equivalent quotient/decimal entries add no counts.

Qualifications and corrections

Four recurrence sources omit a starting index. The following earlier admissible residuals are nonzero; the proved tails start one index later. We do not attribute an unstated false starting index to their authors.

Entry	Exceptional index	Residual	Proved range
A163824	3	-6	$n \geq 4$
A165201	3	2	$n \geq 4$
A190788	7	16	$n \geq 8$
A228180	3	12	$n \geq 4$

The four corrected packages are A030212 (the special-value sum requires the actual offset), A108630 (replace coefficient arguments n by $n+1$), A168494 (one duplicated $a(n-2)$ should be $a(n-1)$), and A366561 (the support rule requires the correct 2-adic restriction). The proposed bound in A133205 is refuted by a divergent normalized summatory lower bound.

Two further input issues are disclosed within otherwise proved packages: the defining range in the title of A300732 contradicts its initial value, and a displayed generating function in A026270 is inconsistent with its defining triangle. Each proof explicitly derives and uses the consistent definition; these do not add extra counts.

Evidence and limits

Primary entries, relevant linked material and targeted searches were consulted on 20 October 2026; retrieval may use cached pages. The retained OEIS Open comparison covers the 100 costliest claims in Appendix A on 97 distinct A-numbers, with no current overlap. It does not cover all results in that project. This is not an exhaustive historical priority audit and no worldwide-first-proof claim is made.

No OEIS edit or submission was performed. Mathematical resolution of a selected assertion here is distinct from editorial acceptance by OEIS. Cross-review is by separate agents within the same AI-assisted workflow, not external peer review or proof-assistant formalization.

7.1 Catalogue of the 100 selected entries

P: selected claim proved. **B**: recurrence boundary qualified and proved. **C**: formula or support rule corrected and proved. **I**: indexing corrected and proved. **R**: selected asymptotic conjecture refuted. Page references lead to the argument or full recurrence signature.

No.	Entry	Selected assertion package	Result	Page
Divisibility and formal congruences				
1	A266489	Index divisibility and even coefficients after index 1	P	343
2	A300732	Index divisibility, with the defining range clarified	P	344
3	A292394	Square of the index divides every positive-index term	P	344
4	A300734	Square-index divisibility with exponent 2n^2	P	344
5	A395833	Divisibility by 2n-1 and by 3	P	344
6	A397348	Odd coefficients exactly at n=2^j-1	P	346
7	A397345	Odd coefficients exactly at n=2^j-1	P	346
8	A397245	Complete ternary description of coefficients modulo 3	P	346
9	A397349	Powers-of-two parity and alternating residues modulo 3	P	346
10	A397346	Period-four residue pattern modulo 8	P	347
11	A397347	Period-four logarithmic-derivative pattern modulo 4	P	347
12	A396846	Powers of three modulo 3 and period four modulo 8	P	347
13	A397244	Fibbinary characterization of odd coefficients	P	347
14	A030212	Correct the one-index shift in the empirical gamma value	I	348
Arithmetic and asymptotic assertions				
15	A092843	The shifted-totient summatory asymptotic	P	348
16	A108625	The general alternating binomial identity	P	349
17	A133205	Refute the proposed summatory big-O bound	R	349
18	A140110	The gcd-sum characterization	P	350
19	A179873	Monotonicity of the odd antiharmonic means	P	350
20	A309263	Membership of 4 times every nonnegative power of 5	P	351
21	A356459	The exact leading constant in the asymptotic	P	351
22	A366561	Local formulas and the corrected zero-support condition	C	352
23	A375838	The conjectured triangle recurrence	P	353
24	A385001	The fixed-parameter partition summatory asymptotics	P	353
Generating-function certificates				
25	A026026	Conjectured recurrence of order 2	P	355
26	A026270	Conjectured recurrence of order 3	P	356
27	A059279	Conjectured recurrence of order 3	P	357
28	A096471	Conjectured recurrence of order 5	P	357
29	A106391	Conjectured recurrence of order 6	P	358
30	A107026	Conjectured recurrence of order 2	P	359
31	A108630	Conjectured recurrence of order 6; coefficient argument corrected	I	360
32	A110198	Conjectured recurrence of order 5	P	361
33	A119975	Conjectured recurrence of order 3	P	361
34	A134389	Conjectured recurrence of order 3	P	362
35	A143020	Conjectured recurrence of order 2	P	363
36	A155051	Conjectured recurrence of order 4	P	363
37	A157418	Conjectured recurrence of order 3	P	364
38	A163824	Conjectured recurrence of order 3; starting range specified	B	365
39	A165201	Conjectured recurrence of order 3; starting range specified	B	366
40	A168494	Conjectured recurrence of order 4; term index corrected	C	366
41	A168503	Conjectured recurrence of order 4	P	367
42	A174016	Conjectured recurrence of order 4	P	368
43	A184018	Conjectured recurrence of order 4	P	368
44	A185020	Conjectured recurrence of order 2	P	369
45	A190725	Conjectured recurrence of order 4	P	370
46	A190736	Conjectured recurrence of order 5	P	370
47	A190788	Conjectured recurrence of order 7; starting range specified	B	371

Continued on the next page

No.	Entry	Selected assertion package	Result	Page
48	A191319	Conjectured recurrence of order 4	P	372
49	A191398	Conjectured recurrence of order 6	P	372
50	A220899	Conjectured recurrence of order 5	P	373
51	A228180	Conjectured recurrence of order 3; starting range specified	B	374
52	A278472	Conjectured recurrence of order 3	P	375
53	A372991	Conjectured recurrence of order 3	P	375
54	A383702	Conjectured recurrence of order 3	P	376
55	A386830	Conjectured recurrence of order 2	P	376
Finite-state array certificates				
56	A209550	Empirical array recurrence of order 37	P	380
57	A210179	Empirical array recurrence of order 50	P	381
58	A233637	Empirical array recurrence of order 2	P	381
59	A233638	Empirical array recurrence of order 7	P	381
60	A233639	Empirical array recurrence of order 13	P	381
61	A233640	Empirical array recurrence of order 31	P	382
62	A233684	Empirical array recurrence of order 3	P	382
63	A233685	Empirical array recurrence of order 9	P	382
64	A233686	Empirical array recurrence of order 15	P	382
65	A233710	Empirical array recurrence of order 8	P	382
66	A233711	Empirical array recurrence of order 27	P	383
67	A233749	Empirical array recurrence of order 9	P	383
68	A233750	Empirical array recurrence of order 44	P	383
69	A233885	Empirical array recurrence of order 10	P	383
70	A233886	Empirical array recurrence of order 32	P	384
71	A233910	Empirical array recurrence of order 11	P	384
72	A233911	Empirical array recurrence of order 38	P	384
73	A233951	Empirical array recurrence of order 4	P	384
74	A233952	Empirical array recurrence of order 19	P	385
75	A234146	Empirical array recurrence of order 8	P	385
76	A234147	Empirical array recurrence of order 37	P	385
77	A234651	Empirical array recurrence of order 5	P	385
78	A234652	Empirical array recurrence of order 15	P	386
79	A234653	Empirical array recurrence of order 21	P	386
80	A234731	Empirical array recurrence of order 7	P	386
81	A234732	Empirical array recurrence of order 23	P	386
82	A234733	Empirical array recurrence of order 23	P	386
83	A234753	Empirical array recurrence of order 6	P	387
84	A234754	Empirical array recurrence of order 17	P	387
85	A234770	Empirical array recurrence of order 6	P	387
86	A234771	Empirical array recurrence of order 23	P	387
87	A235091	Empirical array recurrence of order 5	P	388
88	A235092	Empirical array recurrence of order 16	P	388
89	A235093	Empirical array recurrence of order 22	P	388
90	A235216	Empirical array recurrence of order 4	P	388
91	A235217	Empirical array recurrence of order 8	P	389
92	A235218	Empirical array recurrence of order 21	P	389
93	A235282	Empirical array recurrence of order 3	P	389
94	A235283	Empirical array recurrence of order 6	P	389
95	A235284	Empirical array recurrence of order 6	P	389
96	A235291	Empirical array recurrence of order 4	P	390
97	A235292	Empirical array recurrence of order 10	P	390
98	A235293	Empirical array recurrence of order 10	P	390
99	A235294	Empirical array recurrence of order 10	P	390
100	A235314	Empirical array recurrence of order 18	P	391

7.2 Divisibility, formal congruences, and an indexing correction

This section contains fourteen selected A-IDs. Five concern zero diagonals under substitution, eight concern implicitly defined power series, and one corrects the index in an empirical theta evaluation. Different parameter specializations share lemmas; the count is a count of source assertion packages, not of independent general theorems. Asymptotic observations in the same entries are outside the selected scope.

7.2.1 Divisibility forced by a zero diagonal

Throughout this subsection $A(x) = 1 + x + \sum_{n\geq 2} a(n)x^n$. For positive integers m_n, impose

$$[x^n]A\left(\frac{x}{A(x)^{m_n}}\right) = 0 \qquad (n > 1). \tag{7.1}$$

The coefficient of $a(n)$ in this equation is 1; all other terms depend only on $a(1), \ldots, a(n-1)$. Negative integer powers of a series with constant term 1 have integer coefficients. Thus the defining equations give a unique integer series, and the same triangular uniqueness holds over every quotient of $\mathbb{Z}$.

Lemma 7.2.1. *If $d \geq 1$ and $n^d \mid m_n$ for every $n > 1$, the solution of (7.1) satisfies $n^d \mid a(n)$ for $n \geq 1$. If instead $m_n = 2n - 1$, then $(2n-1) \mid a(n)$ for $n \geq 1$.*

Proof. For $1 \leq k < n$ put $r = n - k$ and $c_{n,k} = [x^r]A(x)^{-m_n k}$. The defining recursion is

$$a(n) = -\sum_{k=1}^{n-1} a(k)c_{n,k}.$$

Formal differentiation gives the integer identity

$$rc_{n,k} = -m_n k[x^{r-1}]A'(x)A(x)^{-m_n k-1}. \tag{7.2}$$

Fix a prime p with $e = v_p(n) > 0$. Assume inductively that $k^d \mid a(k)$ for all $k < n$. If $v_p(k) \geq e$, the factor $a(k)$ is divisible by p^{de}. Otherwise $v_p(n-k) = v_p(k)$, and (7.2) shows $p^{de} \mid c_{n,k}$. Each summand is therefore divisible by p^{de}, proving the first claim for every prime dividing n. The initial value $a(1) = 1$ starts the induction.

For the second claim put $m = 2n - 1$. Equation (7.2) gives $m \mid rc_{n,k}$. Since $2k - 1 = m - 2r$, it also gives $m \mid (2k-1)c_{n,k}$. The induction hypothesis $(2k-1) \mid a(k)$ now makes every summand a multiple of m. Again the base case is $a(1) = 1$. □

We will also use the elementary fact

$$p \mid \binom{pr}{r} \qquad (r \geq 1, p \text{ prime}). \tag{7.3}$$

Indeed, write the binomial expansion in base p using $(1+t)^p = 1 + t^p$ over $\mathbb{F}_p$. At the least position with a nonzero digit of r, the corresponding digit of pr is zero. The coefficient in question therefore vanishes. This is the needed special case of Lucas's congruence.

Theorem 7.2.2 (A266489)**.** *For $m_n = n$, one has $n \mid a(n)$ for every $n \geq 1$, and $a(n)$ is even for every $n \geq 2$.*

Proof. The divisibility is Lemma 7.2.1, with $d = 1$. For parity, test $U(x) = 1 + x$ in the defining equation modulo 2:

$$[x^n]U\left(\frac{x}{U(x)^n}\right) = [x^{n-1}](1+x)^{-n} = (-1)^{n-1}\binom{2n-2}{n-1} \equiv 0 \pmod 2 \quad (n > 1).$$

Equation (7.3) and triangular uniqueness give $A \equiv 1 + x \pmod 2$. These are both conjectures C1 and C2 in A266489 (Paul D. Hanna, February 7, 2016). The equivalent integrality assertion in A268293 is a consequence and is not counted a second time. □

Theorem 7.2.3 (A300732)**.** *For $m_n = 2n$, the sequence with initial values $a(0) = a(1) = 1$ and (7.1) for $n > 1$ satisfies $n \mid a(n)$ for $n \geq 1$.*

Proof. Apply Lemma 7.2.1 with $d = 1$. The conjectured divisibility in A300732 is thereby proved. There is a separate defining-range typo in its title: the title says $n \geq 1$, although its listed $a(1) = 1$, example, and program use $n > 1$. At $n = 1$ the displayed coefficient is $a(1) = 1$, not zero. Our theorem uses the explicit initial values and the consistent range $n > 1$; it does not assert the existence of a series satisfying that contradictory literal title and $a(1) = 1$ simultaneously. □

Theorem 7.2.4 (A292394)**.** *For $m_n = n^2$, one has $n^2 \mid a(n)$ for every $n \geq 1$.*

Proof. Apply Lemma 7.2.1 with $d = 2$ to the conjecture in A292394 (Paul D. Hanna, September 15, 2017). The integrality of A292395 is the same assertion after division by n^2 and adds no count. □

Theorem 7.2.5 (A300734)**.** *For $m_n = 2n^2$, one has $n^2 \mid a(n)$ for every $n \geq 1$.*

Proof. Use $d = 2$ in Lemma 7.2.1. This proves the displayed divisibility conjecture in A300734 (March 11, 2018). □

Theorem 7.2.6 (A395833)**.** *For $m_n = 2n - 1$, one has $(2n - 1) \mid a(n)$ for every $n \geq 1$, and $3 \mid a(n)$ for every $n \geq 2$.*

Proof. The first assertion is the second part of Lemma 7.2.1. To prove the second, the test series $U = 1 + x$ gives, for $n > 1$,

$$[x^n]U\left(\frac{x}{U(x)^{2n-1}}\right) = (-1)^{n-1}\binom{3n-3}{n-1} \equiv 0 \pmod 3.$$

Use (7.3) with $p = 3$ and triangular uniqueness. This settles both displayed congruence conjectures in A395833 (Paul D. Hanna, May 7, 2026). □

7.2.2 A uniform implicit exponential family

For a positive integer c, define $F_c(x) = \sum_{n\geq 0} f_c(n)x^n$ by

$$F_c = \exp\left(x + \sum_{n\geq 2} \frac{cn^2 - 1}{cn^2} f_c(n)x^n\right), \qquad f_c(0) = f_c(1) = 1. \tag{7.4}$$

Also write $B_c = xF_c'/F_c = \sum_{n\geq 1} b_c(n)x^n$. The selected entries are F_2 (A397346), F_3 (A397348), F_4 (A397245), F_5 (A397345), and B_2, B_3, B_4 (A397347, A397349, A396846). The $c = 1$ parity case was already selected as A397242 in Round V; it is credited background and is not counted again.

Lemma 7.2.7. *Equation (7.4) defines a unique integer series. It is equivalent over $\mathbb{Q}[[x]]$ to*

$$F = 1 + x + c\left\{x(F-1)\frac{F' + xF''}{F} + x^2\frac{(F')^2}{F^2}\right\}. \tag{7.5}$$

With $F(0) = 1, F'(0) = 1$, the latter equation has a unique solution over every commutative coefficient ring. In particular reductions modulo integers can be checked in (7.5) without dividing by n.

Proof. Let $\theta = x\,d/dx$. Applying θ^2 to the logarithm of (7.4) gives

$$\theta^2 \log F = \theta^2 F - \frac{F - 1 - x}{c}.$$

Using $\theta^2 \log F = \theta^2 F/F - (\theta F)^2/F^2$ yields (7.5). Conversely the constant and linear coefficients of the logarithmic equation are fixed, so reversing this calculation over $\mathbb{Q}$ recovers (7.4). At degree $n \geq 2$ the right side of (7.5) involves only earlier coefficients: its two nonlinear terms have an extra factor of x beyond the possible appearance of $f_c(n)$. Inversion of a unit series and formal differentiation use integer coefficient polynomials. The recursion is therefore monic, proving all the assertions.

For reproducibility, an alternative exact recursion follows by comparing coefficients in $F' = F(\log F)'$:

$$f_c(n) = cnf_c(n-1) + n\sum_{j=2}^{n-1}(cj^2 - 1)\frac{f_c(j)}{j}f_c(n-j), \qquad n \geq 2. \tag{7.6}$$

Induction proves $n \mid f_c(n)$, so all its divisions are exact. The series B_c is integral because F_c is an integer unit series. □

Lemma 7.2.8. *Put $S_p(x) = \sum_{j\geq 0} x^{p^j}$ in $\mathbb{F}_p[[x]]$. For odd c,*

$$F_c \equiv \frac{S_2(x)}{x} \pmod 2, \qquad B_c \equiv S_2(x) \pmod 2.$$

For $c \equiv 1 \pmod 3$,

$$F_c \equiv \frac{S_3 + S_3^2 - x}{x^2} \pmod 3, \qquad B_c \equiv S_3 \pmod 3. \tag{7.7}$$

For $c \equiv 0 \pmod 3$, $F_c \equiv 1 + x$ and $B_c \equiv x/(1+x) \pmod 3$.

Proof. The last claim follows immediately from (7.5). For the first two, $S_p' = 1$ and $S_p = x + S_p^p$. In characteristic 2 take $R = S_2/x = 1/(1 - S_2)$. In characteristic 3 take

$$R = \frac{1}{1 - x - S_3^2} = \frac{S_3 + S_3^2 - x}{x^2}.$$

The second equality follows on substituting $x = S_3 - S_3^3$; its right side is a genuine power series with constant term 1. In either characteristic direct differentiation gives $\theta \log R = S_p$ and hence $\theta^2 \log R = x$. For characteristic 3, for example,

$$\frac{xR'}{R} = \frac{x(1 - S_3)}{1 - x - S_3^2} = S_3,$$

because $1 - x - S_3^2 = (1 - S_3)^2(1 + S_3)$. It follows that $\theta^2 R = R(x + S_p^2)$. In both cases $R(1 - x - S_p^2) = 1$, so

$$R - 1 - x = \theta^2 R - \theta^2 \log R.$$

This is (7.5) with $c = 1$. Its uniqueness proves the claims for the specified residues of c. □

Theorem 7.2.9 (A397348)**.** *For F_3, $f_3(n)$ is odd if and only if $n+1$ is a power of 2.*

Proof. Apply Lemma 7.2.8 with $c=3$. It proves the parity conjecture in A397348 for every $n \geq 0$. The asymptotic observation in that entry is not selected. □

Theorem 7.2.10 (A397345)**.** *For F_5, $f_5(n)$ is odd if and only if $n+1$ is a power of 2.*

Proof. Use the same lemma with $c=5$. This is the parity conjecture in A397345. These two applications extend the earlier $c=1$ project result; the common family argument is not counted separately. □

Theorem 7.2.11 (A397245)**.** *For F_4 and every $n \geq 0$, $f_4(n) \equiv 1 \pmod 3$ precisely when $n+2 = 3^j$ or $2 \cdot 3^j$ for a nonnegative integer j. One has $f_4(n) \equiv 2 \pmod 3$ precisely when $n+2 = 3^i + 3^j$ with $0 \leq i < j$. All other residues are zero.*

Proof. Use (7.7). The three types of exponents in $S_3 + S_3^2$ have, respectively, one ternary digit 1, one ternary digit 2, or two ternary digits 1. They are disjoint. The corresponding coefficients are $1, 1, 2$. Subtracting x removes the sole negative-power term before division by x^2. This proves both congruence conjectures in A397245. The same lemma additionally settles the previously unselected mod-3 assertions for A397242; that already selected A-ID receives no new count here. □

Theorem 7.2.12 (A397349)**.** *For B_3, $b_3(n)$ is odd exactly when n is a power of 2. For $n \geq 1$, $b_3(n) \equiv (-1)^{n-1} \pmod 3$.*

Proof. The identity $B_3 = xF_3'/F_3$ follows by differentiating the defining logarithmic series in A397349; the coefficient relation is $b_3(n) = (3n^2-1)f_3(n)/(3n)$ for $n > 1$, with $b_3(1) = 1$. The two reductions in Lemma 7.2.8 give $B_3 \equiv S_2 \pmod 2$ and $B_3 \equiv x/(1+x) \pmod 3$. These settle both displayed congruence conjectures. □

7.2.3 Exact rational certificates modulo eight

For a unit series R set

$$H_c(R) = R - 1 - x - cx(R-1)(R' + xR'')/R - cx^2(R')^2/R^2.$$

Define the rational functions

$$R_2 = 1 + x + \frac{4x^2 + 2x^3 + 2x^5}{1 - x^4},$$
$$R_4 = 1 + x + \frac{4x^3}{(1+x)^2}, \qquad J = x^4 - 2x^3 + 4x^2 + 1, \quad K = 5x^3 + 3x^2 + 3x + 1.$$

The following finite polynomial identity is an all-coefficient certificate:

$$H_2(R_2) = \frac{8x^3 N(x)}{(x-1)^3(x^2+1)^3 J^2}, \tag{7.8}$$

where

$$N(x) = 5x^{12} - 18x^{11} + 54x^{10} - 95x^9 + 118x^8 - 28x^7 - 68x^6 + 174x^5 - 83x^4 + 62x^3 - 2x^2 + x + 8.$$

It is verified by ordinary rational differentiation and clearing denominators; each denominator has constant term 1 or -1. Consequently $F_2 \equiv R_2 \pmod 8$ by monic uniqueness.

For $c = 4$, equation (7.5) first gives $F_4 \equiv 1 + x \pmod 4$. Substitute $1 + x$ into the expression in braces and reduce it modulo 2:

$$\frac{x^2}{1+x} + \frac{x^2}{(1+x)^2} = \frac{x^2(x+2)}{(1+x)^2} \equiv \frac{x^3}{(1+x)^2} \pmod 2.$$

Thus $F_4 \equiv R_4 \pmod 8$. Two further exact identities certify the logarithmic derivatives:

$$\frac{xR_2'}{R_2} - \frac{x + 3x^2 + 3x^3 + 3x^4}{1 - x^4} = \frac{4x^2(x^6 - x^5 + 2x^4 + 3x^2 + 2x - 1)}{(x-1)(x+1)(x^2+1)J}, \tag{7.9}$$

$$\frac{xR_4'}{R_4} - \frac{x + 7x^2 + 5x^3 + 7x^4}{1 - x^4} = \frac{8x^2(5x^4 + 2x^3 + 6x^2 + x + 1)}{(x-1)(x^2+1)K}. \tag{7.10}$$

Theorem 7.2.13 (A397346)**.** *Starting at $n = 2$, the residues of $f_2(n)$ modulo 8 repeat $4, 2, 0, 2$.*

Proof. The coefficients of $R_2 - 1 - x$ have precisely this pattern. The identity (7.8) therefore proves the conjecture in A397346 for its stated range. □

Theorem 7.2.14 (A397347)**.** *Starting at $n = 1$, the residues of $b_2(n)$ modulo 4 repeat $1, 3, 3, 3$.*

Proof. The definition in A397347 gives $B_2 = xF_2'/F_2$. Formal differentiation and inversion preserve coefficientwise congruences of integer unit series. Reduce (7.9) modulo 4 and use $F_2 \equiv R_2 \pmod 8$. This proves the selected residue pattern. □

Theorem 7.2.15 (A396846)**.** *For $n \geq 1$, $b_4(n) \equiv 1 \pmod 3$ if n is a power of 3, and $b_4(n) \equiv 0 \pmod 3$ otherwise. Its residues modulo 8, starting at $n = 1$, repeat $1, 7, 5, 7$.*

Proof. The logarithmic definition in A396846 gives $B_4 = xF_4'/F_4$; the correct coefficient relation is $b_4(n) = (4n^2 - 1)f_4(n)/(4n)$ for $n > 1$. Lemma 7.2.8 gives $B_4 \equiv S_3 \pmod 3$. For the second assertion use $F_4 \equiv R_4 \pmod 8$ and (7.10). Both source congruence conjectures are settled. □

7.2.4 A binary restriction from a cubic equation

Theorem 7.2.16 (A397244)**.** *Let $V(x) = \sum_{n\geq 0} v(n)x^n$, $v(0) = v(1) = 1$, be defined by*

$$(2n)[x^n]V^{2n} = (2n-1)[x^n]V^{2n+1} \qquad (n > 1).$$

For $n > 0$, $v(n)$ is odd exactly when $n = 2k + 1$ and the binary expansion of k has no adjacent 1 digits. Equivalently k is a Fibbinary number.

Proof. The coefficient of $v(n)$ in the difference of the two sides is $(2n)^2 - (2n-1)(2n+1) = 1$. Hence this is again an integer monic triangular definition, valid after reduction modulo 2. Let $U = 1 + x/U^2$, an integer unit series. Lagrange inversion, with $W = U - 1 = x(1 + W)^{-2}$, gives for $n > 1$

$$[x^n]U^{2n+1} = \frac{2n+1}{n}[t^{n-1}](1+t)^{2n}(1+t)^{-2n} = 0.$$

Thus U satisfies the defining relation for V modulo 2, and $V \equiv U \pmod 2$ by uniqueness. In characteristic 2, $D = U - 1$ satisfies $D = x + D^3$. The integer solution $E = x + E^3$ has

$$[x^{2k+1}]E = \frac{1}{2k+1}\binom{3k}{k}, \qquad [x^{2k}]E = 0.$$

These follow from Lagrange inversion applied to $E = x/(1 - E^2)$. Its reduction is D. Since $2k + 1$ is odd, the displayed odd-index coefficient has the parity of $\binom{3k}{k}$. The base-2 binomial rule says this coefficient is odd precisely when adding k and $2k$ produces no carry, equivalently $k \,\&\, (2k) = 0$. That is exactly the absence of adjacent binary 1 digits. This proves the parity conjecture in A397244. Its asymptotic assertion is not selected, and no potentially misindexed transform formula from that entry is needed. □

7.2.5 An offset correction in an empirical theta value

Theorem 7.2.17 (A030212: corrected index)**.** *Use the actual OEIS indexing* $a(1) = 1, a(2) = -4, a(3) = 0, \ldots$ *for Glaisher's sequence A030212. Then*

$$\sum_{n\geq 1} a(n)e^{-n\pi} = \frac{\pi^{5/2}}{64\Gamma(3/4)^{10}},$$

$$\sum_{n\geq 1} a(n)e^{-(n-1)\pi} = \frac{e^{\pi}\pi^{5/2}}{64\Gamma(3/4)^{10}}.$$

The second value is the constant printed in the empirical formula; it requires a shift of the exponent by one relative to the actual index.

Proof. Put $q = e^{-\pi}$, $T = \pi^{1/4}/\Gamma(3/4)$, $\phi(z) = \sum_{j\in\mathbb{Z}} z^{j^2}$, and $\psi(z) = \sum_{j\geq 0} z^{j(j+1)/2}$. The previously recorded generating function in A030212 is

$$\sum_{n\geq 1} a(n)z^n = z\prod_{j\geq 1}(1-z^j)^4(1-z^{2j})^2(1-z^{4j})^4 = z\phi(z)^2\psi(-z)^8.$$

The product/theta identity is also an immediate application of the Jacobi products; it is credited classical input, not a new identity. For completeness the required special values are

$$\phi(q) = T, \qquad \phi(-q) = 2^{-1/4}T, \qquad \psi(q^2) = 2^{-5/4}q^{-1/4}T.$$

At nome $e^{-\pi}$ the elliptic modulus is $1/\sqrt{2}$. The theta–elliptic relation, the classical value $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$, and the gamma reflection formula yield $\theta_3(0,q) = T$. Symmetry gives $\theta_2(0,q) = \theta_4(0,q)$, and $\theta_3^4 = \theta_2^4 + \theta_4^4$ gives the factor $2^{-1/4}$. Finally $\theta_2(0,q) = 2q^{1/4}\psi(q^2)$. The standard duplication identity $\psi(-q)^2 = \phi(-q)\psi(q^2)$ therefore implies $\psi(-q)^4 = 2^{-3}q^{-1/2}T^4$. Substitution in $q\phi(q)^2\psi(-q)^8$ gives $T^{10}/64$, proving both equalities. All products are analytic for $|z| < 1$, so these sums converge absolutely at q.

The consulted source's empirical expression starts at $n = 0$ although the entry has offset 1. Even if one naturally extends $a(0) = 0$, its unshifted left side is smaller than the printed constant by the factor $e^{-\pi}$. Reindexing $a(n)$ as $a(n+1)$, or replacing the exponent n by $n-1$ and starting at 1, repairs the formula. The linked decimal entry A388390 is not counted separately. The theta evaluations used here were already reproduced in Round VI and are independently included above; the selected contribution is this source-index correction. □

The classical identities in the last proof are documented in DLMF §20.7, §20.9, and §19.20. Source records, boundary notes, and exact rational certificates accompany the reproducibility packet. Finite coefficient checks supplement the universal arguments; they are not their logical basis.

7.3 Arithmetic identities, asymptotics, and two corrections

The ten packages in this section concern explicitly uncertain statements on the cited OEIS entries, inspected on 20 October 2026. Eight selected statements are proved. The proposed bound in A133205 is false, and the zero-support assertion in A366561 needs a correction; its accompanying general formulas are proved. Known identities used as inputs are credited. Finite computations in the accompanying script check transcription and independent definitions; the all-index arguments are the proofs below.

7.3.1 A092843: a shifted-totient divisor sum

Source and target. A092843 defines $a(n) = \sum_{d|n,\ d>1} \varphi(d-1)$. Kotesovec's 25 June 2024 conjecture is $\sum_{n\leq N} a(n) \sim N^2/2$.

Proof. Put $\Phi(y) = \sum_{1 \le m \le y} \varphi(m)$, with $\Phi(0) = 0$. The classical elementary estimate

$$\Phi(y) = \frac{y^2}{2\zeta(2)} + O(y \log(2y))$$

follows directly from $2\Phi(y) - 1 = \sum_{d \le y} \mu(d) \lfloor y/d \rfloor^2$: replace each square by $(y/d)^2 + O(y/d)$, and bound the tail of $\sum \mu(d)/d^2 = 1/\zeta(2)$. Reversing two finite sums gives

$$\sum_{n \le N} a(n) = \sum_{j \le N} \Phi(\lfloor N/j \rfloor - 1) = \frac{N^2}{2\zeta(2)} \sum_{j \le N} \frac{1}{j^2} + O\left(N \sum_{j \le N} \frac{\log(2N/j)}{j} \right).$$

The error is $O(N \log^2(2N))$, and the omitted tail of the zeta sum contributes only $O(N)$. Consequently

$$\boxed{\sum_{n \le N} a(n) = \frac{N^2}{2} + O(N \log^2(2N)).}$$

This proves the conjecture with an error term.

7.3.2 A108625: an arbitrary-parameter binomial identity

Source and scope. The 12 October 2024 conjecture in A108625 asks, for every positive integer m and every $n \ge 0$, whether

$$\sum_{k=0}^{n} (-1)^{n+k} \binom{n}{k} T(mn, k) = \frac{((m+1)n)!}{((m-1)n)!(n!)^2}.$$

The entry reports verification for $m \le 10$. We use and credit Peter Bala's 26 February 2024 identity $T(r, k) = \sum_i \binom{r}{i} \binom{r+i}{i} \binom{k}{i}$; the selected claim is the universal m assertion, not a new formula for the crystal-ball array or any of its unrelated congruences.

Proof. For $0 \le i \le n$, the finite binomial theorem gives

$$\sum_{k=i}^{n} (-1)^{n-k} \binom{n}{k} \binom{k}{i} = \binom{n}{i} \sum_{\ell=0}^{n-i} (-1)^{n-i-\ell} \binom{n-i}{\ell} = \begin{cases} 1 & i = n, \\ 0 & i < n. \end{cases}$$

Substitute the credited formula for $T(r, k)$ and interchange finite sums. The result is $\binom{r}{n} \binom{r+n}{n}$ for every $r \ge n$. Putting $r = mn$ gives the stated factorial quotient. This also includes $n = 0$, when both sides are 1.

7.3.3 A133205: the proposed logarithmic bound is false

Source and result. A133205 is completely multiplicative with $a(p) = p(p+1)/2$. Kotesovec's 28 January 2025 conjecture asserts $S(N) := \sum_{n \le N} a(n) = O(N^3/\log N)$. In fact,

$$\boxed{\frac{S(N) \log N}{N^3} \longrightarrow +\infty.}$$

Proof. We use the classical prime number theorem, as recorded in DLMF 27.12. Partial summation gives

$$\sum_{\substack{q\le X\\ q \text{ prime}}} a(q) = \frac{1}{2}\sum_{q\le X}(q^2+q) \sim \frac{X^3}{6\log X}.$$

Fix $M \ge 2$. The integers pq, with $p \le M < q$ both prime, are distinct for different pairs. Positivity and multiplicativity therefore give

$$S(N) \ge \sum_{\substack{p\le M\\ p \text{ prime}}} a(p) \sum_{\substack{M<q\le N/p\\ q \text{ prime}}} a(q).$$

Here M is fixed before N tends to infinity. Hence

$$\liminf_{N\to\infty} \frac{S(N)\log N}{N^3} \ge \frac{1}{12}\sum_{p\le M}\frac{p+1}{p^2}.$$

The right side tends to infinity with M, because $\sum_p 1/p$ diverges. For completeness, convergence of that sum would make $\prod_p(1-1/p)^{-1}$ bounded, whereas its finite products dominate arbitrarily long harmonic sums. For any prescribed real C, choose M so the displayed finite lower bound exceeds C; the definition of the lower limit then proves the boxed assertion. Thus no constant in the proposed big-O bound can exist. No numerical threshold or unproved assertion about prime pairs is needed.

7.3.4 A140110: the gcd-sum characterization

Source and target. A140110 contains those positive integers whose successive divisor ratios reduce to ratios of consecutive integers. Ratajczak's 9 January 2017 empirical assertion identifies precisely these integers by

$$n = 1 + \sum_{i=1}^{\tau(n)-1} \gcd(d_i, d_{i+1}), \qquad 1 = d_1 < \cdots < d_{\tau(n)} = n.$$

The entry's separate, already proved divisibility-by-4-or-6 statement is not part of this selection.

Proof. Let $g_i = \gcd(d_i, d_{i+1})$. Since g_i divides the positive difference $d_{i+1}-d_i$, one has $g_i \le d_{i+1}-d_i$. Summing gives $\sum_i g_i \le n-1$. Equality holds exactly when every individual inequality is equality. Write $d_i = g_i u$ and $d_{i+1} = g_i v$ with $\gcd(u,v)=1$. The equality $g_i = d_{i+1}-d_i$ is exactly $v-u=1$. This is the required reduced-ratio condition, proving both directions. For $n=1$ the empty sum also gives the correct result.

7.3.5 A179873: monotonicity and oddness of integral means

Source and target. A179873 lists the integral values, in increasing order of h, of

$$B(h) = \frac{\sum_{1\le k<h,\ (k,h)=1} k^2}{\sum_{1\le k<h,\ (k,h)=1} k}, \qquad B(1)=1.$$

Its comment conjectures that the list is nondecreasing and consists of odd integers. The general arithmetic formula for $B(h)$ already appears in Hasler's program on A175505 and in Eldar's later programs; we derive it here and use it as a credited input.

Proof. For $h > 1$, pairing k with $h - k$ gives $\sum_{(k,h)=1} k = h\varphi(h)/2$. Inclusion-exclusion gives

$$\sum_{(k,h)=1} k^2 = \sum_{d|h} \mu(d) d^2 \sum_{j=1}^{h/d} j^2 = \frac{h^2\varphi(h)}{3} + \frac{h}{6} \prod_{p|h} (1-p).$$

Put $t = h/\operatorname{rad}(h)$ and $\varepsilon = (-1)^{\omega(h)}$. Then

$$B(h) = \frac{2h + \varepsilon/t}{3}.$$

If $B(h)$ is integral, $3tB(h) = 2ht + \varepsilon$ forces $t = 1$. Thus every selected $h > 1$ is squarefree and $B(h) = (2h + \mu(h))/3$. Its numerator is odd, so an integral $B(h)$ is odd. If $h' < h$ are two selected indices greater than 1, then

$$B(h) - B(h') = \frac{2(h - h') + \mu(h) - \mu(h')}{3} \geq 0.$$

The separately prescribed $B(1) = 1$ is no larger than any selected value with $h > 1$. This proves the complete conjecture, including its initial repeated value 1.

7.3.6 A309263: an infinite family of admissible divisors

Source and target. A309263 selects the members of A140110 that are not divisible by 6. Its conjecture says that $4 \cdot 5^r$ is present for every $r \geq 0$.

Proof. The complete ordered divisor list of $4 \cdot 5^r$ consists of the blocks

$$(5^j,\ 2 \cdot 5^j,\ 4 \cdot 5^j), \qquad 0 \leq j \leq r.$$

They occur in precisely this order since $4 \cdot 5^j < 5^{j+1}$. Within a block each successive ratio is $2/1$; between blocks it is $5/4$. Both are ratios of consecutive coprime integers. Finally $3 \nmid 4 \cdot 5^r$, so the exclusion of multiples of 6 never applies.

7.3.7 A356459: the exact asymptotic constant

Source and target. A356459 defines $a(n) = n! \sum_{k \leq n} \sum_{d|k} d/(k/d)!$. Kotesovec's 12 August 2025 conjecture has the form $a(n) \sim c\, n! n^2$ with $c = 0.5732\ldots$.

Proof. Writing $k = rd$ gives the exact identity

$$\frac{a(n)}{n!} = \frac{1}{2} \sum_{r=1}^{n} \frac{\lfloor n/r \rfloor (\lfloor n/r \rfloor + 1)}{r!}.$$

For $u \geq 1$, $|\lfloor u \rfloor (\lfloor u \rfloor + 1) - u^2| \leq u$. The convergent sums of the resulting errors give

$$\boxed{\frac{a(n)}{n!} = cn^2 + O(n), \qquad c = \frac{1}{2} \sum_{r=1}^{\infty} \frac{1}{r! r^2}.}$$

More explicitly, the absolute error is at most

$$\frac{n}{2} \sum_{r \leq n} \frac{1}{r! r} + \frac{n^2}{2} \sum_{r > n} \frac{1}{r! r^2},$$

which is $O(n)$. The verification file encloses c in an exact rational interval: after M terms the omitted tail is at most $\frac{1}{2(M+1)!(M+1)^2} \frac{M+2}{M+1}$. This identifies the conjectured decimal constant without treating a decimal fit as a proof.

7.3.8 A366561: general counting formulas and the missing zeros

Source and selected scope. Let

$$T(n,k) = \#\{(x,y) \in (\mathbb{Z}/n\mathbb{Z})^2 : \gcd(x^2 - y^2, n) = k\}.$$

The 17 June 2025 conjectures of Oudra on A366561 propose a support criterion and several equivalent divisor-sum formulas. The formulas are correct. The proposed criterion $T(n,k) = 0$ exactly when $k \nmid n$ is false:

$$\boxed{T(4,2) = 0 \quad \text{although } 2 \mid 4.}$$

Squares modulo 4 are 0 and 1, so their difference cannot be 2. The corrected criterion, for positive n, k, is

$$\boxed{T(n,k) > 0 \iff k \mid n \text{ and } \neg(v_2(n) \geq 2 \text{ and } v_2(k) = 1).}$$

The diagonal and first-column identifications with A062803 and A082953 were already recorded on those entries; they are credited consequences, not additional selected problems.

Local enumeration. The Chinese remainder theorem makes the count a product over prime powers. Let $C_p(a,b) = T(p^a, p^b)$ for $a \geq 1$ and $0 \leq b \leq a$. For odd p, the substitution $(u,v) = (x-y, x+y)$ is bijective. Counting the possible valuations of u and v gives

$$C_p(a,b) = \begin{cases} (b+1)(p-1)^2 p^{2a-b-2}, & b < a, \\ ((a+1)p - a)p^{a-1}, & b = a. \end{cases}$$

For $b < a$, there are $b+1$ possible pairs of valuations, each contributing $(p-1)^2 p^{2a-b-2}$. For $b = a$, sum over $v_p(u) = i < a$ and include $u = 0$; the result is $a(p-1)p^{a-1} + p^a$.

For $p = 2$, the same map has a kernel of size 2 and image the pairs of equal parity. Odd u, v give 2^{2a-1} original pairs. Even u, v give $uv = 4u'v'$, with u', v' modulo 2^{a-1}. Thus

$$C_2(a,b) = \begin{cases} 2^{2a-1}, & b = 0, \\ 0, & b = 1 < a, \\ (b-1)2^{2a-b-1}, & 2 \leq b < a, \\ a2^a, & b = a. \end{cases}$$

For the last case and $a \geq 2$, reduce u', v' modulo 2^{a-2}. There are four lifts, and the original map contributes another factor 2; the product-zero count modulo 2^e is $(e+2)2^{e-1}$ for $e \geq 1$. The cases $a = 1, 2$ are respectively 2 and 8 by direct parity counting. This proves the local table, including all boundary cases. Every odd prime local count is positive; the displayed 2-adic zero is the only possible additional obstruction. The corrected support follows.

The source formulas. For $k \mid n$ define

$$F(n,k) = \gcd(n,2)(-1)^k \varphi(n/k) \sum_{d \mid k} (-1)^{k/d} \frac{k}{d} \varphi(dn/k).$$

For odd n the two signs cancel, giving the source's unsigned formula. The expression factors into local factors: signs depend only on the 2-part of k/d, and every other factor is multiplicative. For odd p, its local sum gives the first table. At 2, if $1 \leq b < a$, the inner signed sum is $(b-1)2^{a-1}$; if $b = a$, it is $a2^{a-1}$. For $b = 0$ it is $-\varphi(2^a)$ and the outer sign is negative. Multiplication by the outer factors gives exactly the second table. Therefore $T(n,k) = F(n,k)$ for every $k \mid n$.

Finally let $H(n,k)=\varphi(n/k)$ if $k\mid n$, and 0 otherwise, as in A054523. Extending the count by zero for $k>n$, the source's all-positive-integer version is also proved:

$$T(n,k)=\gcd(n,2)(-1)^k H(n,k)\sum_{d\mid k}(-1)^{k/d}\frac{k}{d}H(dn,k).$$

When $k\nmid n$ both sides vanish; when $k\mid n$ it is the identity already proved. The empirical row-sum assertion also follows, since the gcd values partition the n^2 ordered pairs.

7.3.9 A375838: the triangular recursion for rooted chains

Source and target. Kurkov's 21 June 2025 conjecture in A375838 defines

$$R(n,n)=1,\qquad R(n,k)=(k+1)\left(\sum_{i=0}^{n-1}R(n-1,i)+\sum_{j=0}^{k-1}R(n-1,j)\right),\quad k<n,$$

and asserts that $R(n,0)=a(n)$. The known rooted-chain recurrence, credited to Mohapatra on 1 July 2025, is

$$a(0)=1,\qquad a(n)=1+\sum_{r=1}^{n-1}c(n,r)a(r)\quad(n\ge 1),$$

where $c(n,r)$ denotes the unsigned Stirling number of the first kind.

Proof. Write $b(n)=R(n,0)$, and for $k<n$ put $Q(n,k)=R(n,k)/(k+1)$. Subtracting consecutive defining equations gives

$$Q(n,k)=Q(n,k-1)+kQ(n-1,k-1),\qquad Q(n,0)=b(n).$$

Induction on k, using the Stirling recurrence, yields

$$Q(n,k)=\sum_{j=0}^{k}c(k+1,k+1-j)b(n-j),\qquad k<n.$$

On the other hand, the definition and the boundary value $R(n-1,n-1)=1$ give $Q(n,n-1)=2b(n)-1$. Substitution of $k=n-1$ in the previous identity therefore gives

$$2b(n)-1=\sum_{r=1}^{n}c(n,r)b(r)=b(n)+\sum_{r=1}^{n-1}c(n,r)b(r).$$

Thus b satisfies exactly the credited recurrence for a, including $b(0)=1$. Inductive uniqueness proves $b(n)=a(n)$ for all $n\ge 0$. The related MathOverflow question was also inspected; its retrieved page contains the conjecture and comments but no proof of it.

7.3.10 A385001: every fixed column has the proposed mean

Source and target. In A385001, a partition with distinct part sizes $m_1<\cdots<m_k$ and positive multiplicities $r_1,\dots,r_k$ has $\prod_i r_i$ choices of designated summands. Kotesovec's 1 August 2025 conjecture states, for each fixed $k\ge 1$,

$$\sum_{j=1}^{N}T(j,k)\sim\frac{\pi^{2k}}{(2k)!(2k+1)!}\,N^{2k}.$$

The underlying MacMahon generating functions are classical and appear in A060043 and Andrews–Rose (2010). Only the marked summatory limit is selected here.

Proof. For a fixed tuple $\mathbf{m} = (m_1, \dots, m_k)$, let

$$S_{\mathbf{m}}(N) = \sum_{\substack{r_1,\dots,r_k \ge 1 \\ \sum_i m_i r_i \le N}} \prod_i r_i.$$

The Riemann sum with coordinates $u_i = m_i r_i / N$ gives

$$\lim_{N\to\infty} \frac{S_{\mathbf{m}}(N)}{N^{2k}} = \frac{1}{\prod_i m_i^2} \int_{u_i \ge 0,\ \sum u_i \le 1} \prod_i u_i \, d\mathbf{u} = \frac{1}{(2k)! \prod_i m_i^2}.$$

The last simplex integral follows by iterated beta integration. There is a summable bound independent of N: if every $m_i \le N$,

$$S_{\mathbf{m}}(N) \le \prod_i \sum_{r \le N/m_i} r \le \frac{N^{2k}}{\prod_i m_i^2};$$

otherwise the sum is zero. Dominated convergence for the countable sum over increasing tuples is therefore applicable. It gives

$$\lim_{N\to\infty} N^{-2k} \sum_{j \le N} T(j,k) = \frac{1}{(2k)!} \sum_{m_1 < \cdots < m_k} \frac{1}{m_1^2 \cdots m_k^2}.$$

The classical product

$$\prod_{m \ge 1} \left(1 + \frac{z^2}{m^2}\right) = \frac{\sinh(\pi z)}{\pi z}$$

(DLMF 4.36.1) identifies the final sum as $\pi^{2k}/(2k+1)!$. This proves the conjecture for every fixed k, with no Tauberian assumption and no numerical extrapolation.

7.4 Generating functions and exact recurrence proofs

This section treats 31 distinct selected assertions. Twenty-five recurrences are proved as stated, four require an explicit starting-index qualification, and two printed recurrences require a correction. Each target was explicitly marked conjectural in the consulted OEIS entry. Existing generating functions, finite sums, and product definitions are credited inputs. The selected scope does not claim that the sequence, its generating function, or the method of holonomic elimination is new.

The retained list excludes all 608 previously selected IDs and the inspected Fischer 2020 and Ekhad–Yang 2017 families. It also avoids the 97 IDs in the inspected OEIS Open Appendix A baseline; that appendix is only a bounded part of that project. In particular, A065942 and A226302 were removed because their marked recurrences are exact left multiples of already recorded stronger recurrences. A359643 was removed after the official Alkahest guide identified an earlier lower-order recurrence result. The audit file records these decisions and their limited scope. A remaining conjecture label does not by itself establish historical novelty or official closure.

The coefficient certificate

Write a_n for the sequence with its actual OEIS offset s and set $F(x) = \sum_{n \ge s} a_n x^n$, with zero coefficients below s. Every ordinary recurrence below uses the convention

$$R_n = \sum_{j=0}^{r} p_j(n) a_{n-j}, \qquad L = \sum_{j=0}^{r} x^j p_j(\theta + j), \qquad \theta = x\frac{d}{dx}. \tag{7.11}$$

Indeed, $\theta x^{n-j} = (n-j)x^{n-j}$, so $[x^n]LF = R_n$. Consequently the exact identity $LF = P(x)$ proves $R_n = 0$ for all $n > \deg P$. We additionally require $n \ge s + r$, so every sequence index in the displayed

recurrence belongs to its stated domain. The coefficients of P give the exceptional initial residuals exactly. No inference from agreement of finitely many terms is involved.

Here is a finite rational-arithmetic recipe for checking every displayed identity. If $x = \xi(t)$, put $M = \xi/\xi'$, and write $F = \sum_i R_i(t)\sqrt{D_i(t)}$, where $D_0 = 1$ and every square root is the formal branch with constant term one. Define

$$H_{i,0} = R_i, \qquad H_{i,m+1} = M\left(H'_{i,m} + \frac{D'_i}{2D_i}H_{i,m}\right), \qquad c_m = \sum_{j=0}^{r} \xi^j [z^m] p_j(z+j). \tag{7.12}$$

Then $LF = \sum_i \sqrt{D_i} \sum_m c_m H_{i,m}$. Thus it suffices to check the rational identities $\sum_m c_m H_{0,m} = P(\xi)$ and $\sum_m c_m H_{i,m} = 0$ for $i > 0$. For an unparametrized series take $t = x$, $\xi = x$, $M = x$. These identities use only differentiation, multiplication, and cancellation of rational functions; their exact numerators vanish. The manuscript gives all p_j, all generating functions, and every boundary polynomial, so this recipe is an independently reproducible finite certificate. The portable verification script carries it out over exact rational numbers and also checks offsets and independently generated initial coefficients.

Two formal-series facts used below

Let $t = x\phi(t)$ with $\phi(0) = 1$. The unique solution has $t(0) = 0$. For every formal series B,

$$\sum_{n\ge 0} x^n [u^n] B(u)\phi(u)^n = \frac{B(t)}{1 - t\phi'(t)/\phi(t)}. \tag{7.13}$$

To prove this, extract $[x^n]$ on the right by a formal residue and substitute $x = u/\phi(u)$. The differential $dx = (1 - u\phi'(u)/\phi(u))\, du/\phi(u)$ cancels the displayed denominator, leaving $\mathrm{Res}_u\, B(u)\phi(u)^n u^{-n-1} du$. This is exactly the coefficient on the left, including $n = 0$.

For an exponential generating function $F(x) = \sum a_n x^n/n!$ and a recurrence of order r, replace $p_j(n)$ in (7.11) by

$$q_j(n) = p_j(n) \prod_{h=0}^{r-j-1} (n - j - h). \tag{7.14}$$

The resulting coefficient of x^n is $R_n/(n-r)!$ for $n \ge r$, because $q_j(n)/(n-j)! = p_j(n)/(n-r)!$. An empty product is one. This explains the conversion used for A096471.

7.4.1 A026026: the selected recurrence

Source and selected assertion. A026026, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jun 22 2013.

The source offset is $s = 2$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \ge 4$, with $r = 2$ and

$$\begin{aligned} p_0(n) &= -(n+3)(11n - 38) \\ p_1(n) &= 2\left(35n^2 - 86n - 102\right) \\ p_2(n) &= -4(2n-5)(13n - 30) \end{aligned}$$

Known inputs and scope. The Catalan generating function is an existing input. Krause (May 9, 2026) also records a binomial difference, and Drube (2022) gives the path setting. Only Mathar's marked order-two recurrence is selected; neither the closed form nor the path interpretation is claimed as new.

Proof. With $C = (1-\sqrt{1-4x})/(2x)$, use the recorded series $x^2(1+x^2C^4)C^3$, whose lowest degree is the source offset two.

$$y^2 = 1 - 4x, \qquad y(0) = 1.$$

$$C = \frac{1-y}{2x}, \qquad F = x^2(1+x^2C^4)C^3.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 80x^2.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s + r = 4$, so no term below the source offset is used.

7.4.2 A026270: the selected recurrence

Source and selected assertion. A026270, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jun 23 2013.

The source offset is $s = 2$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 5$, with $r = 3$ and

$$\begin{aligned} p_0(n) &= n + 3 \\ p_1(n) &= -3\,(n+1) \\ p_2(n) &= -n - 1 \\ p_3(n) &= 3\,(n-5) \end{aligned}$$

Known inputs and scope. The generating function printed on A026270 has inconsistent constant and initial terms. We derive the needed generating function from the defining boundary of A026268 and the credited diagonal generating function on A026269. The selected Mathar recurrence is true from n=5; correction of the auxiliary input is disclosed separately and is not an extra count.

Proof. Let $d_n = T(n,n)$ and $a_n = T(n,n-1)$ for the defining triangle A026268. Its boundary rule gives $d_n = d_{n-1} + a_{n-1}$ for $n \geq 3$, with $d_2 = a_2 = 1$. The credited diagonal series in A026269 is

$$D(x) = \sum_{n\geq 2} d_n x^n = \frac{4x^2(1-x^2)}{(1-x+\sqrt{1-2x-3x^2})^2}.$$

Summing the boundary rule gives $D - x^2 = xD + xF$, hence

$$F(x) = \frac{(1-x)D(x) - x^2}{x}.$$

The expression printed directly as a GF on A026270 is $-1 + (1-x)D(x)$; it has constant term -1 and fails to reproduce the listed offset-two sequence. It is not a premise of this proof. The boundary-derived series below begins $x^2 + 2x^3 + 6x^4 + 15x^5$. The selected recurrence itself needs no coefficient correction.

$$y^2 = -3x^2 - 2x + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{4x\,(1-x)\,(1-x^2)}{(-x+y+1)^2} - x.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 7x^4 + 5x^2.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 5$.

The stated lower bound also ensures $n \geq s + r = 5$, so no term below the source offset is used.

7.4.3 A059279: the selected recurrence

Source and selected assertion. A059279, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Nov 15 2011.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 3$, with $r = 3$ and

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= -2\,(4n - 1) \\ p_2(n) &= 4\,(4n - 5) \\ p_3(n) &= -4\,(2n - 5) \end{aligned}$$

Known inputs and scope. The Catalan substitution is the definition. Greubel (January 4, 2017) and Gutkovskiy (June 30, 2020) record equivalent algebraic formulas. The linked Gomez Aiza material is background on the same enumerative family; the selected assertion is only the still-marked recurrence.

Proof. The defining Catalan substitution is $((1-x)/(1-2x))C(x(1-x)/(1-2x))$. Its radical form is the following expression.

$$y^2 = -\frac{4x\,(1-x)}{1-2x} + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{1-y}{2x}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 2x^2 - 2x + 1.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 3$.

The stated lower bound also ensures $n \geq s + r = 3$, so no term below the source offset is used.

7.4.4 A096471: the selected recurrence

Source and selected assertion. A096471, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jul 06 2023.

The source offset is $s = 2$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 7$, with $r = 5$ and

$$
\begin{aligned}
p_0(n) &= 1 \\
p_1(n) &= -3 \\
p_2(n) &= -2n^2 + 8n - 3 \\
p_3(n) &= (n-2)(4n-15) \\
p_4(n) &= (n-3)\left(n^3 - 9n^2 + 22n - 7\right) \\
p_5(n) &= -(n-5)^2(n-4)(n-3)
\end{aligned}
$$

Known inputs and scope. The cycle decomposition and its logarithmic EGF are known inputs recorded in the entry. The selected assertion is Mathar's July 2023 recurrence, with the ordinary-to-exponential coefficient conversion and source offset made explicit.

Proof. A permutation with exactly one even cycle consists of one even cycle and an arbitrary set of odd cycles. The cycle EGF therefore is

$$F(x) = \left(\sum_{m\geq 1} \frac{x^{2m}}{2m}\right) \exp\left(\sum_{m\geq 0} \frac{x^{2m+1}}{2m+1}\right) = -\frac{1}{2} H(x)\ell(x),$$

where $H = \sqrt{(1+x)/(1-x)}$ and $\ell = \log(1-x^2)$. This also derives the known EGF recorded in the entry. Its derivatives satisfy

$$\frac{H'}{H} = \frac{1}{1-x^2}, \qquad \ell' = -\frac{2x}{1-x^2}.$$

Use (7.14) with $r = 5$. On the basis $(H, H\ell)$, the Euler derivative acts on the rational coefficients by

$$(R_0, R_1) \longmapsto \left(x\left(R_0' + \frac{R_0 - 2xR_1}{1-x^2}\right), x\left(R_1' + \frac{R_1}{1-x^2}\right)\right).$$

Starting from $(R_0, R_1) = (0, -1/2)$ and applying $\sum_{j=0}^{5} x^j q_j(\theta + j)$ gives exactly $(0, 0)$. This is a rational identity, checked by the same finite derivative recipe as (7.12). Thus $R_n = 0$ for every $n \geq 7$, the first index for which all six source terms have index at least two.

7.4.5 A106391: the selected recurrence

Source and selected assertion. A106391, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Feb 20 2015.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 6$, with $r = 6$ and

$$
\begin{aligned}
p_0(n) &= -(n-2)(n+1) \\
p_1(n) &= 4n(3n-7) \\
p_2(n) &= -49n^2 + 193n - 148 \\
p_3(n) &= 8(n-4)(9n-19) \\
p_4(n) &= 5n^2 + 115n - 418 \\
p_5(n) &= -12(n-4)(7n-19) \\
p_6(n) &= 45(n-5)(n-4)
\end{aligned}
$$

Known inputs and scope. The finite binomial sum and Bessel EGF are existing inputs. The order-six recurrence is the selected source-marked assertion. The entry's second, order-four conjecture is not counted separately. The ordinary GF used here is derived directly by two binomial transforms.

Proof. Put $b_n = \binom{2n}{n-1}$, with $b_0 = 0$. The recorded defining sum is $a_n = \sum_k \binom{n}{2k} b_{n-2k}$. Since $b_n = \binom{2n}{n} - C_n$, its generating function is

$$B(x) = \frac{1-2x}{2x\sqrt{1-4x}} - \frac{1}{2x}.$$

The defining even-binomial transform equals the mean of the transforms with weights 1 and -1. Therefore

$$F(x) = \frac{1}{2}\left(\frac{B(x/(1-x))}{1-x} + \frac{B(x/(1+x))}{1+x}\right),$$

which is the three-component radical expression displayed below. This derives the needed ordinary GF from the existing finite sum.

$$F(x) = -\frac{1}{2x} + \frac{1-3x}{4x\sqrt{1-6x+5x^2}} + \frac{1-x}{4x\sqrt{1-2x-3x^2}}.$$

For (7.12), take

$$(D_0, D_1, D_2) = (1, 1-6x+5x^2, 1-2x-3x^2),$$
$$(R_0, R_1, R_2) = \left(-\frac{1}{2x}, \frac{1-3x}{4xD_1}, \frac{1-x}{4xD_2}\right).$$

The apparent negative powers cancel.

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 14x^3 - 8x^2 + 2x.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 6$.

The stated lower bound also ensures $n \geq s + r = 6$, so no term below the source offset is used.

7.4.6 A107026: the selected recurrence

Source and selected assertion. A107026, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Feb 20 2015.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 2$, with $r = 2$ and

$$p_0(n) = 3n\,(3n-2)\,(3n-1)\left(22n^2 - 62n + 43\right)$$
$$p_1(n) = -8\left(1892n^5 - 8280n^4 + 13330n^3 - 9660n^2 + 3048n - 315\right)$$
$$p_2(n) = 128\,(2n-3)\,(4n-7)\,(4n-5)\left(22n^2 - 18n + 3\right)$$

Known inputs and scope. The sequence is a Riordan inverse row sum. Manyama (November 8, 2025) records the quartic-tree generating function used as input. Only the second marked recurrence (order two) is selected; the first, order-four conjecture is outside the count.

Proof. The inverse of the defining Riordan array $(1/(1+x), x/(1+x)^4)$ has row-sum series $(1+t)/(1-t)$, where $t = x(1+t)^4$: inversion gives first component $1+t$ and second component t, and summing its columns divides by $1-t$. Equivalently use Manyama's November 8, 2025 formula $F = g/(2-g)$ with $g = 1 + xg^4$, and put $g = 1 + t$. This proves the rational parametrization used below.

$$x = \xi(t) = \frac{t}{(t+1)^4}, \qquad t(0) = 0, \quad \xi'(0) = 1.$$

$$F = -\frac{t+1}{t-1}.$$

Use $D_0 = 1$, $R_0 = F(\xi(t))$, and $M = \xi/\xi'$ in (7.12). All expressions are rational in t.

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = -84x.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 2$.

The stated lower bound also ensures $n \geq s + r = 2$, so no term below the source offset is used.

7.4.7 A108630: correction of the polynomial index

Source and selected assertion. A108630, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Sep 27 2014.

The source offset is $s = 1$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 7$, with $r = 6$ and

$$\begin{aligned}
p_0(n) &= n \\
p_1(n) &= 3 - n \\
p_2(n) &= -8\,(n-3) \\
p_3(n) &= 60 - 17n \\
p_4(n) &= 81 - 19n \\
p_5(n) &= -12\,(n-5) \\
p_6(n) &= -4\,(n-6)
\end{aligned}$$

Known inputs and scope. The generating function is the sequence definition at offset one. The printed recurrence fails at n=7 with residual 168. Replacing each polynomial coefficient p_j(n) by p_j(n+1), with sequence indices unchanged, yields the proved recurrence for n>=7. This is a correction, not a proof of the literal printed statement.

The literal source statement and its correction. The relevant source values are

$$(a_1, \ldots, a_7) = (1, 1, 2, 6, 17, 54, 178).$$

Substituting them into the printed recurrence gives residual $168 \neq 0$ at $n = 7$.

The corrected coefficients above are $p_j^{\text{source}}(n+1)$, while the sequence terms remain a_{n-j}. The coefficient shift is essential; the literal recurrence is refuted.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = -4x^4 - 8x^3 - 7x^2 - 2x + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{x - y + 1}{2x^2 + 2x + 2}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 2x^5 + 6x^4 + 6x^3 + 3x^2 + x.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 7$.

The stated lower bound also ensures $n \geq s + r = 7$, so no term below the source offset is used.

7.4.8 A110198: the selected recurrence

Source and selected assertion. A110198, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Nov 15 2012.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 5$, with $r = 5$ and

$$\begin{aligned} p_0(n) &= n \\ p_1(n) &= 1 - 3n \\ p_2(n) &= n \\ p_3(n) &= 2 - n \\ p_4(n) &= 3n - 5 \\ p_5(n) &= 2 - n \end{aligned}$$

Known inputs and scope. The antidiagonal generating function and a double binomial sum are already recorded. They are input formulas, not discoveries. Only the marked order-five recurrence is selected.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = \left(x^2 - 3x + 1\right)\left(x^2 + x + 1\right), \qquad y(0) = 1.$$

$$F(x) = \frac{1}{y\,(1 - x)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 0.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 5$.

The stated lower bound also ensures $n \geq s + r = 5$, so no term below the source offset is used.

7.4.9 A119975: the selected recurrence

Source and selected assertion. A119975, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Dec 10 2011.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 3$, with $r = 3$ and

$$
\begin{aligned}
p_0(n) &= n + 1 \\
p_1(n) &= n - 4 \\
p_2(n) &= 9 - 13n \\
p_3(n) &= -21\,(n - 2)
\end{aligned}
$$

Known inputs and scope. The ordinary GF is explicitly credited to Kotesovec's June 26, 2013 correction. The Bessel EGF is also known. Only Mathar's December 2011 recurrence is selected.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = -7x^2 - 2x + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{-1 + \frac{3x+1}{y}}{4x}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = x + 1.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 3$.

The stated lower bound also ensures $n \geq s + r = 3$, so no term below the source offset is used.

7.4.10 A134389: the selected recurrence

Source and selected assertion. A134389, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Oct 25 2012.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 3$, with $r = 3$ and

$$
\begin{aligned}
p_0(n) &= n - 1 \\
p_1(n) &= -2n \\
p_2(n) &= -4\,(n - 4) \\
p_3(n) &= 8\,(n - 3)
\end{aligned}
$$

Known inputs and scope. The quadratic GF is already recorded for the transform defining this sequence. Only the conjectured polynomial recurrence is selected.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = 1 - 4x^2, \qquad y(0) = 1.$$

$$F(x) = \frac{xy}{2\,(1 - 2x)^2} + \frac{2 - x}{2 - 4x}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 5x^2 - 2x - 1.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 3$.

The stated lower bound also ensures $n \geq s + r = 3$, so no term below the source offset is used.

7.4.11 A143020: the selected recurrence

Source and selected assertion. A143020, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, May 10 2018.

The source offset is $s = 2$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 2$ and

$$\begin{aligned} p_0(n) &= -(n-1)(n+1)\left(39n^2 - 136n + 60\right) \\ p_1(n) &= -60\left(18n^2 - 57n + 7\right) \\ p_2(n) &= 12(3n-8)(3n-7)\left(39n^2 - 58n - 37\right) \end{aligned}$$

Known inputs and scope. Deutsch's enumerative formula, the cubic GF for connected noncrossing graphs, and Howroyd's 2017 reversion program are known inputs. Only Mathar's May 2018 order-two recurrence is selected; the Flajolet–Noy setting is credited background.

Proof. The credited GF is $F = g(g-x)/x$, where $g^3 + g^2 - 3xg + 2x^2 = 0$ is the connected-noncrossing-graph series. Its branch starts $g = x + O(x^2)$. Substitute $g = x(1+t)$. After cancellation the cubic becomes $x = t(1-t)/(1+t)^3$, and $F = xt(1+t)$. Since $\xi'(0) = 1$, this specifies the branch uniquely.

$$x = \xi(t) = -\frac{t(t-1)}{(t+1)^3}, \qquad t(0) = 0, \quad \xi'(0) = 1.$$

$$F = xt(1+t).$$

Use $D_0 = 1$, $R_0 = F(\xi(t))$, and $M = \xi/\xi'$ in (7.12). All expressions are rational in t.

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 168x^2.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s + r = 4$, so no term below the source offset is used.

7.4.12 A155051: the selected recurrence

Source and selected assertion. A155051, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Feb 05 2015.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 4$ and

$$
\begin{aligned}
p_0(n) &= n + 2 \\
p_1(n) &= -2 \\
p_2(n) &= 4 - 5n \\
p_3(n) &= 8 \\
p_4(n) &= 4\,(n - 3)
\end{aligned}
$$

Known inputs and scope. The Catalan-substitution GF is the definition. Select the first Mathar recurrence, of order four. The second Mathar recurrence and Zumkeller's unrelated composition conjecture are outside the selected assertion.

Proof. The defining series is $C(x^2)(1 + x)/(1 - x)$. Substituting the Catalan radical gives the following expression.

$$y^2 = 1 - 4x^2, \qquad y(0) = 1.$$

$$F(x) = \frac{(1 - y)\,(x + 1)}{2x^2\,(1 - x)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 2x^2 + 4x + 2.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s + r = 4$, so no term below the source offset is used.

7.4.13 A157418: the selected recurrence

Source and selected assertion. A157418, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jul 24 2012.

The source offset is $s = 1$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 3$ and

$$
\begin{aligned}
p_0(n) &= n\,(7n - 29) \\
p_1(n) &= -21n^2 + 136n - 192 \\
p_2(n) &= -2\left(28n^2 - 165n + 210\right) \\
p_3(n) &= -4\,(n - 3)\,(7n - 22)
\end{aligned}
$$

Known inputs and scope. The GF belongs to the Guo–Sit Rota–Baxter enumeration (2006/2010). Spiegel (June 5, 2026) records another Catalan expression. These are credited inputs; only Mathar's marked recurrence is selected.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = -4x^2 - 4x + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{-2x^2 - x - y\,(x+2) + 2}{2\,(x+1)^2}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = -64x^2 - 22x.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s + r = 4$, so no term below the source offset is used.

7.4.14 A163824: the recurrence with its initial boundary

Source and selected assertion. A163824, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jul 24 2012.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 3$ and

$$\begin{aligned}
p_0(n) &= n\left(n^2 - 6n + 11\right) \\
p_1(n) &= -9n^3 + 56n^2 - 119n + 60 \\
p_2(n) &= 2\left(13n^3 - 83n^2 + 193n - 150\right) \\
p_3(n) &= -12\,(2n-5)\left(n^2 - 4n + 6\right)
\end{aligned}$$

Known inputs and scope. Bevan's recorded central-binomial GF with a rational correction is used as an input. The recurrence needs n>=4; its first fully indexed value n=3 has residual -6. No all-n version is asserted.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = 1 - 4x, \qquad y(0) = 1.$$

$$F(x) = -\frac{x\,(1-x)}{(1-3x)\,(1-2x)} + \frac{1}{y}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = -6x^3 + 2x^2 - 6x.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The first fully indexed value is $n = 3$, where $R_3 = -6$. This is exactly the coefficient of x^3 in P and explains the stated starting index.

7.4.15 A165201: the recurrence with its initial boundary

Source and selected assertion. A165201, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Nov 15 2011.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 3$ and

$$\begin{aligned} p_0(n) &= n\,(n-3) \\ p_1(n) &= -2\left(4n^2 - 15n + 10\right) \\ p_2(n) &= 15n^2 - 69n + 80 \\ p_3(n) &= 2\,(n-2)\,(2n-5) \end{aligned}$$

Known inputs and scope. The Catalan composition is the definition. The source recurrence needs n>=4; n=3 has residual 2. This is a boundary qualification of the selected recurrence.

Proof. Let $C = (1 - \sqrt{1-4x})/(2x)$ be the Catalan series. The defining expression $1/(1 - xC^3)$ rationalizes to the following F.

$$y^2 = 1 - 4x, \qquad y(0) = 1.$$

$$F(x) = \frac{-2x^2 - 3x + y\,(1-x) + 1}{-2x^2 - 8x + 2}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 2x^3 + 2x^2.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The first fully indexed value is $n = 3$, where $R_3 = 2$. This is exactly the coefficient of x^3 in P and explains the stated starting index.

7.4.16 A168494: correction of a duplicated shift

Source and selected assertion. A168494, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Sep 30 2012.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 4$ and

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= 4 - 11n \\ p_2(n) &= 5\,(7n - 11) \\ p_3(n) &= 92 - 37n \\ p_4(n) &= 6\,(2n - 7) \end{aligned}$$

Known inputs and scope. The period-three continued fraction and its algebraic GF are recorded inputs. The printed recurrence uses a(n-2) twice and fails at n=4 with residual 200. Replacing its first a(n-2), the term with coefficient 4-11n, by a(n-1) proves the corrected relation.

The literal source statement and its correction. The relevant source values are

$$(a_0, \ldots, a_4) = (1, 1, 2, 7, 32).$$

Substituting them into the printed recurrence gives residual $200 \neq 0$ at $n = 4$.

The printed term $(4 - 11n)a_{n-2}$ must instead be $(4 - 11n)a_{n-1}$. The separate term $5(7n - 11)a_{n-2}$ is retained. The literal recurrence is refuted.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = -12x^3 + 25x^2 - 10x + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{x - y + 1}{6x\,(1 - x)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = x^3 + 3x^2 - 5x + 1.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s + r = 4$, so no term below the source offset is used.

7.4.17 A168503: the selected recurrence

Source and selected assertion. A168503, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Feb 10 2015.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 4$ and

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= 5 - 13n \\ p_2(n) &= 6\,(8n - 13) \\ p_3(n) &= -12\,(5n - 13) \\ p_4(n) &= 12\,(2n - 7) \end{aligned}$$

Known inputs and scope. Barry's period-three continued fraction and its quadratic GF are known inputs. The selected claim is the marked order-four recurrence. This is not an alias of A168494: its defining GF and recurrence coefficients differ.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = -24x^3 + 36x^2 - 12x + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{1 - y}{6x\,(1 - x)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 6x^2 - 6x + 1.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s + r = 4$, so no term below the source offset is used.

7.4.18 A174016: the selected recurrence

Source and selected assertion. A174016, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Nov 13 2012.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 4$ and

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= -6n \\ p_2(n) &= 12\,(n - 1) \\ p_3(n) &= -12 \\ p_4(n) &= -8\,(2n - 7) \end{aligned}$$

Known inputs and scope. The recorded algebraic GF for the row sums of A174014 is an input. Only Mathar's marked order-four recurrence is selected.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = 8x^3 + 4x^2 - 4x + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{4x + y - 1}{2x\,(1 - 2x)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = -8x^3 + 6x^2 - 2x + 1.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s + r = 4$, so no term below the source offset is used.

7.4.19 A184018: the selected recurrence

Source and selected assertion. A184018, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Nov 15 2011.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 4$ and

$$\begin{aligned} p_0(n) &= n + 1 \\ p_1(n) &= -2\,(3n - 1) \\ p_2(n) &= 3\,(n - 1) \\ p_3(n) &= 2\,(3n - 5) \\ p_4(n) &= n - 3 \end{aligned}$$

Known inputs and scope. The Catalan substitution defines the sequence; Hanna's September 6, 2011 algebraic identity is credited. Only the marked recurrence is selected, not its known generating function or growth constant.

Proof. The defining Catalan substitution is $C(x/(1-x-x^2))/(1-x-x^2)$. Substituting the elementary expression for C gives the following F.

$$y^2 = \left(-x^2-5x+1\right)\left(-x^2-x+1\right), \qquad y(0)=1.$$

$$F(x) = \frac{-x^2-x-y+1}{2x\left(-x^2-x+1\right)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = x^2+1.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s+r = 4$, so no term below the source offset is used.

7.4.20 A185020: the selected recurrence

Source and selected assertion. A185020, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Oct 08 2016.

The source offset is $s=0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 2$, with $r=2$ and

$$\begin{aligned} p_0(n) &= n\,(n+1) \\ p_1(n) &= -4n\,(2n-1) \\ p_2(n) &= -8\,(2n-3)\,(2n-1) \end{aligned}$$

Known inputs and scope. The product C_n*A002605(n+1) is the definition. The second-order constant-coefficient recurrence of A002605 is known and credited. Dividing out Catalan factors proves the selected polynomial recurrence. The generic gauge method is elementary and not claimed as new.

Proof. The defining product is $a_n = C_n S_n$, where $S_n = A002605(n+1)$, $S_0 = 1$, $S_1 = 2$, and $S_n = 2S_{n-1} + 2S_{n-2}$. This constant-coefficient recurrence is a known input, recorded in the source programs as well as the defining sequence. The Catalan backward ratio is

$$\frac{C_{n-1}}{C_n} = \frac{n+1}{2(2n-1)}.$$

Dividing the displayed R_n by C_n and using this ratio twice gives

$$\frac{R_n}{C_n} = n(n+1)(S_n - 2S_{n-1} - 2S_{n-2}) = 0.$$

All divisions are legitimate for $n \geq 2$. This proves the selected recurrence for every such n.

7.4.21 A190725: the selected recurrence

Source and selected assertion. A190725, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jul 24 2012.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 4$ and

$$\begin{aligned} p_0(n) &= n\,(2n-3) \\ p_1(n) &= -6n^2 + 15n - 8 \\ p_2(n) &= -2\left(16n^2 - 32n + 11\right) \\ p_3(n) &= -6n^2 + 9n - 2 \\ p_4(n) &= (n-2)\,(2n-1) \end{aligned}$$

Known inputs and scope. Munarini's entry already records a nonconjectural recurrence of order six. The selected Mathar assertion is a shorter order-four recurrence, not a reindexing or scalar multiple of that known formula. Its lower order and boundary are proved from the credited GF; the existence of a recurrence is not claimed as new.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = x^2 - 6x + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{3x - y + 3}{y\,(2x^2 + 6x + 2)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = -2x.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s + r = 4$, so no term below the source offset is used.

7.4.22 A190736: the selected recurrence

Source and selected assertion. A190736, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jul 24 2012.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 5$, with $r = 5$ and

$$\begin{aligned} p_0(n) &= n\,(11n - 35) \\ p_1(n) &= -3\left(33n^2 - 149n + 136\right) \\ p_2(n) &= 2\left(77n^2 - 377n + 396\right) \\ p_3(n) &= -209n^2 + 1061n - 1200 \\ p_4(n) &= -12\,(11n - 30) \\ p_5(n) &= 4\,(n-4)\,(11n - 24) \end{aligned}$$

Known inputs and scope. Munarini's entry already records a nonconjectural order-nine recurrence. The selected Mathar assertion is a shorter order-five recurrence, not a reindexing or scalar multiple of that known formula. It is proved from the credited GF; the already-known holonomicity is not counted as a result.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = 4x^2 - 8x + 1, \qquad y(0) = 1.$$

$$F(x) = \frac{-2x^2 - 5x - y\,(x+2) + 4}{2x^3 + 4x^2 - 2x + 2}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 54x^3 - 102x^2 - 108x.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 5$.

The stated lower bound also ensures $n \geq s + r = 5$, so no term below the source offset is used.

7.4.23 A190788: the recurrence with its initial boundary

Source and selected assertion. A190788, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Nov 28 2013.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 8$, with $r = 7$ and

$$\begin{aligned}
p_0(n) &= 1 - n \\
p_1(n) &= 3n - 5 \\
p_2(n) &= 2\,(3n - 7) \\
p_3(n) &= -4\,(6n - 19) \\
p_4(n) &= 4\,(n - 3) \\
p_5(n) &= 4\,(11n - 53) \\
p_6(n) &= -16\,(3n - 16) \\
p_7(n) &= 16\,(n - 6)
\end{aligned}$$

Known inputs and scope. The algebraic GF is the definition. The selected recurrence is valid from n=8; its first fully indexed value n=7 has residual 16. No claim at that exceptional index is retained.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = 1 - 4x^2, \qquad y(0) = 1.$$

$$F(x) = \frac{y\,(x-1)}{x + y\,(x-1)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 16x^7 - 32x^6 + 12x^5 + 4x^4 + 4x^3 - 3x^2 - 2x + 1.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 8$.

The first fully indexed value is $n = 7$, where $R_7 = 16$. This is exactly the coefficient of x^7 in P and explains the stated starting index.

7.4.24 A191319: the selected recurrence

Source and selected assertion. A191319, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Dec 07 2017.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 4$, with $r = 4$ and

$$\begin{aligned} p_0(n) &= 2 - n \\ p_1(n) &= 2 \\ p_2(n) &= 5\,(n-2) \\ p_3(n) &= -2 \\ p_4(n) &= -4\,(n-2) \end{aligned}$$

Known inputs and scope. The recorded GF for pyramid weights in dispersed Dyck paths is an input. Only the marked order-four recurrence is selected.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = 1 - 4x^2, \qquad y(0) = 1.$$

$$F(x) = \frac{x^2}{y\,(1-2x)\,(1-x^2)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 0.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The stated lower bound also ensures $n \geq s + r = 4$, so no term below the source offset is used.

7.4.25 A191398: the selected recurrence

Source and selected assertion. A191398, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jun 14 2016.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 6$, with $r = 6$ and

$$
\begin{aligned}
p_0(n) &= -(n-3)(n+2)\\
p_1(n) &= 3n^2-3n-14\\
p_2(n) &= 2\left(n^2-7n+8\right)\\
p_3(n) &= -4\left(3n^2-12n+10\right)\\
p_4(n) &= 7n^2-31n+38\\
p_5(n) &= 4\\
p_6(n) &= 4(n-2)^2
\end{aligned}
$$

Known inputs and scope. The recorded dispersed-Dyck-path GF is an input. Only the marked order-six recurrence is selected.

Proof. Use the credited generating function displayed below, with the formal branch $y(0)=1$.

$$y^2 = 1-4x^2, \qquad y(0)=1.$$

$$F(x) = \frac{2}{-2x^3-x+y(1-x)+1}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = -4x^3-4x^2-8x+6.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 6$.

The stated lower bound also ensures $n \geq s+r = 6$, so no term below the source offset is used.

7.4.26 A220899: the selected recurrence

Source and selected assertion. A220899, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jul 17 2014.

The source offset is $s=2$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 7$, with $r=5$ and

$$
\begin{aligned}
p_0(n) &= 2(n+1)(2n+1)\\
p_1(n) &= -23n^2+31n-6\\
p_2(n) &= -2\left(4n^2+9n-25\right)\\
p_3(n) &= 2\left(23n^2-142n+174\right)\\
p_4(n) &= 4(n-5)(n+8)\\
p_5(n) &= -23(n-6)(n-5)
\end{aligned}
$$

Known inputs and scope. The map enumeration of Kitaev, Salimov, Severs and Ulfarsson and the composition using A000139 and A220898 are credited inputs. The offset-two GF removes the extension 2-x before the exact certificate is applied. Only the marked order-five recurrence is selected.

Proof. The credited map composition is $(1-x)G(x/(1+x))$, where A000139 has coefficients $2(3n)!/((2n+1)!(n+1)!)$. Its parameterization is $z=t(1-t)^2$ and $G(z)=(2-3t)/(1-t)^2$. It can also be checked directly by Lagrange inversion: for $n\geq 1$ the coefficient is

$$\frac{1}{n}\left(\binom{3n+1}{n-1}-3\binom{3n}{n-2}\right)=\frac{2(3n)!}{(2n+1)!(n+1)!},$$

and its constant term is 2. Thus, with $u=t(1-t)^2$, set $x=u/(1-u)$. The composition has initial extension $2-x$; subtracting it gives the actual offset-two series F. This boundary adjustment is included before applying the recurrence operator.

$$x=\xi(t)=-\frac{t\,(t-1)^2}{t^3-2t^2+t-1},\qquad t(0)=0,\quad \xi'(0)=1.$$

$$F=(1-x)\frac{2-3t}{(1-t)^2}-2+x.$$

Use $D_0=1$, $R_0=F(\xi(t))$, and $M=\xi/\xi'$ in (7.12). All expressions are rational in t.

Exact rational differentiation and cancellation in (7.12) give

$$LF=P(x),\qquad P(x)=40x^4+112x^3.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n\geq 7$.

The stated lower bound also ensures $n\geq s+r=7$, so no term below the source offset is used.

7.4.27 A228180: the recurrence with its initial boundary

Source and selected assertion. A228180, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Aug 25 2013.

The source offset is $s=0$. In the convention of (7.11), the selected recurrence is $R_n=0$ for every integer $n\geq 4$, with $r=3$ and

$$\begin{aligned}p_0(n)&=2\,(n+1)\\ p_1(n)&=5-13n\\ p_2(n)&=23n-37\\ p_3(n)&=-6\,(2n-5)\end{aligned}$$

Known inputs and scope. The recorded Catalan tree-boundary GF is an input. The selected recurrence requires n>=4; n=3 has residual 12. This boundary condition is part of the result.

Proof. With $C=(1-\sqrt{1-4x})/(2x)$, use the recorded series $(xC+2x^3C^4)/(1-x)$.

$$y^2=1-4x,\qquad y(0)=1.$$

$$C=\frac{1-y}{2x},\qquad F=\frac{xC+2x^3C^4}{1-x}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 12x^3 - 9x^2 + 4x.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 4$.

The first fully indexed value is $n = 3$, where $R_3 = 12$. This is exactly the coefficient of x^3 in P and explains the stated starting index.

7.4.28 A278472: the selected recurrence

Source and selected assertion. A278472, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Mar 06 2017.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 3$, with $r = 3$ and

$$\begin{aligned} p_0(n) &= (n+2)\left(n^2 - 4n + 1\right) \\ p_1(n) &= -2\left(4n^3 - 9n^2 - 16n + 6\right) \\ p_2(n) &= 15n^3 - 38n^2 - 9n + 14 \\ p_3(n) &= 2\,(2n-1)\left(n^2 - 2n - 2\right) \end{aligned}$$

Known inputs and scope. The Fibonacci-binomial sum and its GF are recorded inputs. The first marked recurrence, of order three, is selected. The second conjectured recurrence is not another count.

Proof. Use the credited generating function displayed below, with the formal branch $y(0) = 1$.

$$y^2 = 1 - 4x, \qquad y(0) = 1.$$

$$F(x) = \frac{-2x - y + 1}{x^2\,(2xy - 8x + 2)}.$$

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = 2.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 3$.

The stated lower bound also ensures $n \geq s + r = 3$, so no term below the source offset is used.

7.4.29 A372991: the selected recurrence

Source and selected assertion. A372991, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Jul 16 2024.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 3$, with $r = 3$ and

$$\begin{aligned} p_0(n) &= 1 \\ p_1(n) &= 0 \\ p_2(n) &= 0 \\ p_3(n) &= -2n\,(2n-1) \end{aligned}$$

Known inputs and scope. The nonlinear factorial quotient is the definition. Its positivity permits cancellation at consecutive indices. The selected assertion is exactly Mathar's marked step-three linear recurrence.

Proof. The definition gives $a_n a_{n-1} a_{n-2} = (2n)!$ for $n \geq 2$, with $a_0 = a_1 = 1$. All terms are positive by induction. Divide the identity at n by the identity at $n-1$ (valid when $n \geq 3$) to obtain

$$\frac{a_n}{a_{n-3}} = \frac{(2n)!}{(2n-2)!} = 2n(2n-1).$$

This is exactly the asserted recurrence. In particular the cancellation does not assume integrality or divide by a potentially zero term.

7.4.30 A383702: the selected recurrence

Source and selected assertion. A383702, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, May 07 2025.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 3$, with $r = 3$ and

$$\begin{aligned} p_0(n) &= 1 \\ p_1(n) &= -3\,(4n-5) \\ p_2(n) &= 48n^2 - 168n + 151 \\ p_3(n) &= -\,(4n-9)^3 \end{aligned}$$

Known inputs and scope. The coefficient -of-product definition is an input. Israel's March 12, 2026 proofs on neighboring A383700/A383701 are acknowledged and those IDs are excluded. The present step 4, constant 3 coefficient problem is distinct; its own exact polynomial identity is supplied, without claiming novelty of the general coefficient -elimination method.

Proof. Let $P_n(z) = \prod_{h=1}^{n}(z + 4h - 1)$, so the definition is $a_n = [z^2]P_n(z)$. With the displayed p_j, direct multiplication gives the polynomial identity in the two indeterminates n, z,

$$\sum_{j=0}^{3} p_j(n) \prod_{h=j}^{2} (z + 4(n-h) - 1) = z^3.$$

Multiplication by $P_{n-3}(z)$ yields $\sum_{j=0}^{3} p_j(n) P_{n-j}(z) = z^3 P_{n-3}(z)$ for $n \geq 3$. Taking the coefficient of z^2 proves the recurrence, since the right side has no such coefficient. The empty product is one. This argument needs no guessed initial values and no numerical fit.

7.4.31 A386830: the selected recurrence

Source and selected assertion. A386830, consulted on 20 October 2026, records the conjecture attributed to R. J. Mathar, Aug 19 2025.

The source offset is $s = 0$. In the convention of (7.11), the selected recurrence is $R_n = 0$ for every integer $n \geq 2$, with $r = 2$ and

$$\begin{aligned} p_0(n) &= 8n\,(2n-3) \\ p_1(n) &= -6\left(108n^2 - 207n + 80\right) \\ p_2(n) &= 405\,(3n-4)\,(3n-2) \end{aligned}$$

Known inputs and scope. Manyama's finite binomial sum and coefficient -extraction formula are existing inputs. The diagonal-series identity converts that definition to a rational parametrization. Only Mathar's marked order-two recurrence is selected.

Proof. The defining binomial sum is the coefficient

$$a_n = [u^n]\frac{(1+3u)^{3n+1}}{(1-2u)^{2n+1}}.$$

Expansion of the numerator and denominator gives the defining summand

$$3^k 2^{n-k}\binom{3n+1}{k}\binom{3n-k}{n-k}.$$

Apply (7.13) with $B(t) = (1+3t)/(1-2t)$ and $\phi(t) = (1+3t)^3/(1-2t)^2$. This proves the rational parametrization used below.

$$x = \xi(t) = \frac{t\,(2t-1)^2}{(3t+1)^3}, \qquad t(0) = 0, \quad \xi'(0) = 1.$$

$$\phi(t) = \frac{(1+3t)^3}{(1-2t)^2}, \qquad F = \frac{(1+3t)/(1-2t)}{1 - t\phi'(t)/\phi(t)}.$$

Use $D_0 = 1$, $R_0 = F(\xi(t))$, and $M = \xi/\xi'$ in (7.12). All expressions are rational in t.

Exact rational differentiation and cancellation in (7.12) give

$$LF = P(x), \qquad P(x) = -30x.$$

Coefficient extraction in (7.11) therefore proves the displayed recurrence for all $n \geq 2$.

The stated lower bound also ensures $n \geq s + r = 2$, so no term below the source offset is used.

7.5 Forty-five empirical array recurrence packages

This section proves forty-five entry-level recurrence assertions marked *Empirical* on the primary OEIS pages consulted on 20 October 2026. Nineteen source-stated empirical or conjectural rational generating functions are proved in the same packages. The original arrays and empirical recurrences are credited to R. H. Hardin; the additional generating-function conjectures are credited on their pages to Colin Barker. The counting unit is one selected A-number, even when several formulas on its page follow from the same proof. The forty-five entries are absent from the 608 A-numbers selected in rounds I–VI.

7.5.1 The local definitions

An array has $n+1$ rows, width $w \geq 2$, and entries in $\{0, \ldots, q-1\}$. For a block $B = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ let

$$\Delta(B) = (b-a, d-b, c-d, a-c).$$

All differences here are signed integers. The entry-specific parameters below select one of the following predicates.

Cell values (C). $\#\{a, b, c, d\} \in S$.

Distinct edge differences (E). The four entries of $\Delta(B)$ have a number of distinct values belonging to S.
Stress (D). $|a + d - b - c| = s$.
Absolute edge sum (U). $\sum_{e\in\Delta(B)} |e| = s$.
Squared edge sum (V). $\sum_{e\in\Delta(B)} e^2 = s$.
Exactly one orientation (J). Exactly one of $\max \Delta(B) \le 1$ and $\min \Delta(B) \ge -1$ holds.
At least one orientation (K). At least one of the same two inequalities holds, allowing both to hold.

The sums in U and V include all four edges with multiplicity. When the particular source additionally prohibits equal adjacent elements, every entry of $\Delta(B)$ is required to be nonzero. This extra condition is not imposed on sources that allow equality. Every horizontal and vertical edge of an array with at least two rows and columns lies in a 2×2 block. Thus this local nonzero condition is exactly the source's global adjacency prohibition, including boundary edges; diagonal neighbors are not covered by that prohibition.

7.5.2 Transfers derived from the array definition

For C, let $\gamma(r)$ rename a row's colors in first-occurrence order. For the numeric predicates E, D, U, V, J and K, let $\gamma(r)$ be the lexicographically least of r, its horizontal reversal, its simultaneous complement $q - 1 - r$, and the reversal of that complement. Let $\mathcal{O}_{q,w}$ be the sorted set of representatives and define

$$m(r) = \#\{v \in \{0, \dots, q-1\}^w : \gamma(v) = r\}, \tag{7.15}$$

$$T(r,s) = \#\left\{ v \in \{0,\dots,q-1\}^w : \begin{array}{l} \gamma(v) = s, \\ P(r_i, r_{i+1}, v_i, v_{i+1}) \quad (1 \le i < w) \end{array} \right\}. \tag{7.16}$$

In C, simultaneous color permutations preserve the local predicate. For all numeric predicates, reversal and complementation permute or negate the edge differences and preserve absolute stress. They also interchange the two orientation inequalities, preserving both exclusive and inclusive disjunction. Consequently each actual row in an orbit has the same number of successors in every target orbit. Arbitrary color permutations are not used for numeric predicates.

It follows by adding rows that the exact labeled count is

$$L_q(n) = m^{\mathsf{T}} T^n \mathbf{1} \qquad (n \ge 1). \tag{7.17}$$

The exponent is n because an array has $n + 1$ rows. All orbit masses are retained, including fixed representatives and rows with no successors. For a source with scalar normalization λ, the required count is $a(n) = \lambda L_q(n)$, with exactly the stated scalar. No interpretation of that scalar as a free symmetry quotient is needed.

The C entry normalizes global color names by row-major first-occurrence order and requires all four colors. The general counting formulas below also record the at-most-color variant used in the earlier rounds. Let $U_q^{=}$ and $U_q^{\le}$ count normalized arrays using exactly q and at most q colors, respectively. Since a surjective labeling has $q!$ distinct color renamings, inclusion–exclusion gives

$$U_q^{=}(n) = \sum_{j=1}^{q} \frac{(-1)^{q-j}}{j!(q-j)!} L_j(n), \tag{7.18}$$

$$U_q^{\le}(n) = \sum_{j=1}^{q} \left(\frac{1}{j!} \sum_{h=0}^{q-j} \frac{(-1)^h}{h!} \right) L_j(n). \tag{7.19}$$

The second formula sums the exact-color formula over the number of colors actually used. The empty alphabet contributes zero. All colors are required only when the entry explicitly says so.

7.5.3 An exact finite certificate proves every index

For the source signature $c = (c_1, \ldots, c_d)$ put

$$R(z) = z^d - \sum_{i=1}^{d} c_i z^{d-i}, \qquad Q(x) = 1 - \sum_{i=1}^{d} c_i x^i.$$

For each component with nonzero weight in the preceding counting formula, the complete certificate establishes the integer vector identity

$$\boxed{TR(T)\mathbf{1} = \mathbf{0}.} \tag{7.20}$$

Multiplying by $m^{\mathsf{T}} T^{n-d-1}$ for $n \geq d+1$, and then summing the weighted components, proves

$$\boxed{a(n) = \sum_{i=1}^{d} c_i a(n-i) \qquad (n \geq d+1).} \tag{7.21}$$

Every source term on the right has index at least 1. The auxiliary transfer value at $n = 0$ is not asserted to be an OEIS term and can be nonintegral for a scalar-normalized definition.

The 48 distinct complete matrices have 8,784 nonzero weighted edges; the largest has 1,056 states, and the largest recurrence order is 50. All 890 source-prefix values agree with the defining transfers. These comparisons check interpretation and indexing. The certificates contain all component weights, matrices, vectors $v_k = T^k\mathbf{1}$ for $0 \leq k \leq d+1$, and residual coordinates. The computation is exactly

```
v[0] = [1 for every state]
for k = 1,...,d+1:
    v[k][r] = sum(T[r,s]*v[k-1][s] for every state s)
for every state r:
    assert v[d+1][r] == sum(c[i]*v[d+1-i][r] for i=1,...,d)
```

The verifier regenerates all states and transitions directly from (7.16) and checks every stored vector entry with exact integer and rational arithmetic. Thus this is a finite computer-assisted proof: the vector identity implies every later index. Initial-value agreement alone is not used as the proof. Complete expanded data are supplied in `r7_arrays_matrices.json` and `r7_arrays_certificates.json`; the definitions and coefficient signatures in this manuscript determine the calculation without those files.

For $F(x) = \sum_{n\geq 1} a(n)x^n$, the proved recurrence gives $QF = P$, where $\deg P \leq d$ and

$$[x^k]P = \sum_{\substack{0 \leq i \leq \min(k,d) \\ k-i \geq 1}} q_i a(k-i), \qquad q_0 = 1, \quad q_i = -c_i \ (i \geq 1). \tag{7.22}$$

The initial values are computed from the defining transfers. Exact polynomial cross-multiplication identifies this P/Q with all nineteen selected source generating functions. Their numerators are displayed below and their original expressions are in the source audit. Separately stated radical closed forms are not extra selected assertions.

7.5.4 A hand-checkable example: A233637

For squared edge sum 2, $q = 3$, and $w = 2$, the representatives are $00, 01, 02, 11$. Direct enumeration gives

$$T = \begin{pmatrix} 0 & 2 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 \end{pmatrix}, \qquad m = \begin{pmatrix} 2 \\ 4 \\ 2 \\ 1 \end{pmatrix}.$$

The row 02 cannot extend to a legal block and has no outgoing edges. The first four vectors $T^k\mathbf{1}$ are

$$(1,1,1,1)^{\mathsf{T}},\quad (3,3,0,6)^{\mathsf{T}},\quad (12,12,0,18)^{\mathsf{T}},\quad (42,42,0,72)^{\mathsf{T}}.$$

The last is twice the preceding vector plus six times the second one. This is $T(T^2-2T-6I)\mathbf{1}=0$, proving $a(n)=2a(n-1)+6a(n-2)$ for $n\geq 3$. Since $a(1)=24$ and $a(2)=90$,

$$F(x)=\frac{x(24+42x)}{1-2x-6x^2}.$$

This proves the empirical recurrence and conjectural rational generating function on https://oeis.org/A233637.

7.5.5 Source status, provenance and the selection unit

The status used here is the explicit word *Empirical* attached to each recurrence, not OEIS's editorial label "approved." The consulted primary pages record no proof or proof link for these selected assertions. Representative searches for the exact A-numbers and related families did not identify a prior proof among the inspected results; many search hits were irrelevant. This limited audit does not establish worldwide priority.

All forty-five IDs are disjoint from the prior 608 selected project IDs and from the retained 97 distinct A-numbers in the 100 costly claims in Appendix A of *OEIS Open*. That check does not cover the entire repository. Five-value diagnostics under shifts $-1,0,1$ and rational rescaling found no aliases against 169 prior array certificates or among these forty-five entries; this is not a complete mathematical alias test. The transfer framework adapts rounds IV–VI. Two previously certified, uncounted Round VI reserves and one previously sourced, uncounted candidate have freshly checked sources here; their earlier research is credited in the ledger. Two further valid certificates, A210104 and A210170, are held outside this selection to keep the full dossier at 100 entries; this is count curation, not a failed proof. The additional certified reserve A235306 is also uncounted. The freshly sourced A234653, A235218 and A235284 reserves are included in the final selection. No OEIS editing or submission has been performed.

7.5.6 Individual definitions and full recurrence signatures

Every coefficient list below gives $c_1,\dots,c_d$ in order, including zeros, in (7.21). Component dimensions include every nonzero weighted component. Initial values start at $n=1$.

7.5.6.1 A209550

Arrays have $n+1$ rows and width $w=6$ on the alphabet $0,\dots,3$. Predicate E requires $S=\{2\}$ distinct signed clockwise edge differences; the labeled count is multiplied by $\lambda=1/4$. The recurrence holds for every $n\geq 38$, with order $d=37$ and component dimensions (1056) (total $D=1056$). Source: https://oeis.org/A209550. The first three values are $(603, 16582, 425803)$.

```
52, -856, 4014, 34244, -455004, 1232892, 6401520, -47507560, 64609279, 283015278,
-991731422, -25138511, 4334710815, -4388934859, -8391654500, 15874524375, 6183586137,
-26768058473, 3068141187, 25533122171, -9123208293, -14680808558, 7130958556,
5229264986, -2809641978, -1165852380, 598144917, 161163696, -66836588, -13174080,
3588829, 586993, -75654, -11622, 320, 76, 2
```

7.5.6.2 A210179

Arrays have $n+1$ rows and width $w=5$ on the alphabet $0,\dots,3$. Predicate C requires $S=\{3,4\}$ distinct values per block. The array uses all 4 colors and is renamed in row-major first-occurrence order. The recurrence holds for every $n \geq 51$, with order $d=50$ and component dimensions $(1,16,41,51)$ (total $D=109$). Source: https://oeis.org/A210179. The first three values are $(9042, 2563795, 677791782)$.

```
305, -8964, -592720, 25109411, 152644028, -18195830806, 181226187451, 3092095684717,
-66471049453238, 90264180231796, 6772584563156660, -49066540883042923,
-192090370791348766, 3446057586143788976, -5299468411446658785,
-101206410270018640248, 453991188789119119647, 1192283901355404923002,
-11763947474565853612062, 4976777711797107269820, 154357702742341519669776,
-3203575523053729414673 72, -10383718555956803136741 68, 4129078215906298103085296,
211912314624423565486 8816, -27118380193817405050087984, 17929710091623807641222208,
97507644778196665361489152, -14929763923888745332857 1904,
-16919129508197051070667 6864, 490220164048331643495809024, 23379223105512261258727168,
-83616707738599583632406 2720, 40626606709018058035447 1936,
74981900954321788163870 7200, -67669387912281854202681 7536,
-31044312397280558816903 1680, 49820394753650158822791 5776, 97447481920813614377205 76,
-18742099679295056537714 6880, 36325389074681359144845312, 34537447300013200634806272,
-11215019873068960788250624, -25804066939695119193866 24, 11326827385715769296814 08,
60903702041543212793856, -38872759178345280700416, -48239092475718166 1184,
345714358079352471552, -5401496006984466432
```

7.5.6.3 A233637

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,2$. Predicate V has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 3$, with order $d=2$ and component dimensions (4) (total $D=4$). Source: https://oeis.org/A233637. The first three values are $(24, 90, 324)$.

$$(c_1,\dots,c_d) = (2, 6).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 24x + 42x^2.$$

7.5.6.4 A233638

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,2$. Predicate V has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 8$, with order $d=7$ and component dimensions (10) (total $D=10$). Source: https://oeis.org/A233638. The first three values are $(90, 534, 3004)$.

$$(c_1,\dots,c_d) = (6, 9, -67, 26, 81, -48, 2).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 90x - 6x^2 - 1010x^3 + 624x^4 + 1238x^5 - 810x^6 + 34x^7.$$

7.5.6.5 A233639

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,2$. Predicate V has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 14$, with order $d=13$ and component dimensions (25) (total $D=25$). Source: https://oeis.org/A233639. The first three values are $(324, 3004, 26192)$.

```
2, 76, -5, -1165, 564, 5703, -4723, -9217, 10950, 1714, -5684, 1880, -96
```

7.5.6.6 A233640

Arrays have $n+1$ rows and width $w=5$ on the alphabet $0,\dots,2$. Predicate V has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 32$, with order $d=31$ and component dimensions (70) (total $D=70$). Source: https://oeis.org/A233640. The first three values are $(1188, 17424, 238570)$.

```
18, 104, -2900, 1703, 142773, -323741, -3138407, 9628188, 36089489, -132426888,
-233184244, 1009363199, 867273484, -4570942635, -1819188109, 12605037643, 1987026933,
-21096877188, -968267399, 20856565656, 317665620, -11488323206, -375223522,
3165049833, 183421296, -361542217, -20354018, 12534108, 274216, -114944, -832
```

7.5.6.7 A233684

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,3$. Predicate V has target $s=10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 4$, with order $d=3$ and component dimensions (6) (total $D=6$). Source: https://oeis.org/A233684. The first three values are $(32, 108, 356)$.

$$(c_1,\dots,c_d) = (2, 5, -2).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 32x + 44x^2 - 20x^3.$$

7.5.6.8 A233685

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,3$. Predicate V has target $s=10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 10$, with order $d=9$ and component dimensions (20) (total $D=20$). Source: https://oeis.org/A233685. The first three values are $(108, 548, 2628)$.

```
1, 28, -25, -103, 97, 68, -47, -12, 2
```

7.5.6.9 A233686

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,3$. Predicate V has target $s=10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 16$, with order $d=15$ and component dimensions (72) (total $D=72$). Source: https://oeis.org/A233686. The first three values are $(356, 2628, 19248)$.

```
7, 30, -198, -190, 1708, 65, -6015, 1916, 9134, -3808, -5930, 2155, 1461, -312, -112
```

7.5.6.10 A233710

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,4$. Predicate V has target $s=14$, allowing equal adjacent values. The recurrence holds for every $n \geq 9$, with order $d=8$ and component dimensions (9) (total $D=9$). Source: https://oeis.org/A233710. The first three values are $(96, 444, 1992)$.

```
0, 24, 14, -129, -68, 228, 64, -128
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 96x + 444x^2 - 312x^3 - 2900x^4 - 384x^5 + 5296x^6 + 832x^7 - 2944x^8.$$

7.5.6.11 A233711

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,4$. Predicate V has target $s=14$, allowing equal adjacent values. The recurrence holds for every $n \geq 28$, with order $d=27$ and component dimensions (39) (total $D=39$). Source: https://oeis.org/A233711. The first three values are $(444, 2624, 16004)$.

```
8, 56, -565, -766, 15139, -7664, -200684, 302993, 1413766, -3297757, -5115754,
18205660, 6468818, -56612688, 15267568, 100195210, -68399612, -95039845, 102899681,
36863396, -73128319, 5984214, 22126722, -7453432, -1251600, 860624, -95168
```

7.5.6.12 A233749

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,5$. Predicate U has target $s=10$, allowing equal adjacent values. The recurrence holds for every $n \geq 10$, with order $d=9$ and component dimensions (12) (total $D=12$). Source: https://oeis.org/A233749. The first three values are $(292, 3244, 33372)$.

```
8, 50, -191, -670, 1425, 2761, -3160, -644, 358
```

7.5.6.13 A233750

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,5$. Predicate U has target $s=10$, allowing equal adjacent values. The recurrence holds for every $n \geq 45$, with order $d=44$ and component dimensions (63) (total $D=63$). Source: https://oeis.org/A233750. The first three values are $(3244, 91116, 2317964)$.

```
28, 258, -8153, -21818, 884110, 641196, -49138214, 7881936, 1607832444, -1043204114,
-33437651311, 31731911114, 463391143344, -530799098345, -4407915878767, 5632602673516,
29289730685819, -40018456626533, -137180728733978, 194997456311814, 454206116121400,
-657831115817189, -1062348157268958, 1537508763715626, 1751866868878969,
-2476743191321207, -2036444198056430, 2723651246669803, 1674200095938075,
-2016791049187732, -976960955082435, 985471325349785, 401735680540938,
-307547802798483, -112638329429263, 58076876665437, 20060500861421, -6092017479043,
-2023405058657, 313961613786, 99780981302, -5783024558, -1748862224, -15322920
```

7.5.6.14 A233885

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,6$. Predicate V has target $s=30$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 11$, with order $d=10$ and component dimensions (16) (total $D=16$). Source: https://oeis.org/A233885. The first three values are $(112, 384, 1264)$.

```
0, 18, 4, -95, -30, 165, 36, -105, -8, 20
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 112x + 384x^2 - 752x^3 - 3068x^4 + 624x^5 + 5828x^6 + 224x^7 - 3760x^8 - 88x^9 + 720x^{10}.$$

7.5.6.15 A233886

Arrays have $n+1$ rows and width $w = 3$ on the alphabet $0, \ldots, 6$. Predicate V has target $s = 30$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 33$, with order $d = 32$ and component dimensions (100) (total $D = 100$). Source: https://oeis.org/A233886. The first three values are $(384, 1960, 9376)$.

```
4, 72, -278, -2093, 7846, 31943, -118256, -280293, 1059474, 1464690, -5968574,
-4541666, 21819710, 7753185, -52635660, -4645134, 84280382, -7066733, -89291380,
16609381, 61880902, -14399047, -27456604, 6388152, 7484866, -1438992, -1148164,
135572, 79124, -1776, -736, 32
```

7.5.6.16 A233910

Arrays have $n+1$ rows and width $w = 2$ on the alphabet $0, \ldots, 7$. Predicate V has target $s = 30$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 12$, with order $d = 11$ and component dimensions (20) (total $D = 20$). Source: https://oeis.org/A233910. The first three values are $(160, 640, 2440)$.

```
0, 24, 6, -177, -52, 516, 80, -646, 42, 284, -80
```

7.5.6.17 A233911

Arrays have $n+1$ rows and width $w = 3$ on the alphabet $0, \ldots, 7$. Predicate V has target $s = 30$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 39$, with order $d = 38$ and component dimensions (144) (total $D = 144$). Source: https://oeis.org/A233911. The first three values are $(640, 3944, 22632)$.

```
4, 122, -461, -6376, 22883, 188152, -648596, -3478827, 11715412, 42308852, -142371381,
-345931578, 1197824103, 1908669954, -7078856586, -6995027392, 29551303874,
16143386244, -87052258170, -19433628990, 179619345422, -2128109482, -255775086626,
46189589396, 245226941694, -73044737386, -152167442028, 56827325500, 57294922304,
-23672423108, -11706249412, 4940867468, 1040756704, -439680112, -20118176, 11155520,
-250624, -14592
```

7.5.6.18 A233951

Arrays have $n+1$ rows and width $w = 2$ on the alphabet $0, \ldots, 5$. Predicate V has target $s = 10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 5$, with order $d = 4$ and component dimensions (12) (total $D = 12$). Source: https://oeis.org/A233951. The first three values are $(80, 380, 1776)$.

$$(c_1, \ldots, c_d) = (4, 5, -7, -6).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 80x + 60x^2 - 144x^3 - 108x^4.$$

7.5.6.19 A233952

Arrays have $n+1$ rows and width $w = 3$ on the alphabet $0, \ldots, 5$. Predicate V has target $s = 10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 20$, with order $d = 19$ and component dimensions (63) (total $D = 63$). Source: https://oeis.org/A233952. The first three values are $(380, 2820, 20616)$.

```
3, 76, -161, -1932, 3277, 22377, -32880, -128727, 173771, 368053, -472908, -488012,
600604, 270148, -316676, -46496, 44736, 6656, -256
```

7.5.6.20 A234146

Arrays have $n+1$ rows and width $w = 2$ on the alphabet $0, \ldots, 5$. Predicate U has target $s = 10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 9$, with order $d = 8$ and component dimensions (12) (total $D = 12$). Source: https://oeis.org/A234146. The first three values are $(184, 1264, 8752)$.

```
6, 25, -105, -192, 494, 478, -322, -204
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 184x + 160x^2 - 3432x^3 - 2788x^4 + 16796x^5 + 12460x^6 - 10692x^7 - 6120x^8.$$

7.5.6.21 A234147

Arrays have $n+1$ rows and width $w = 3$ on the alphabet $0, \ldots, 5$. Predicate U has target $s = 10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 38$, with order $d = 37$ and component dimensions (63) (total $D = 63$). Source: https://oeis.org/A234147. The first three values are $(1264, 13936, 158812)$.

```
10, 188, -1705, -13231, 115068, 493976, -4229448, -11190946, 96334439, 164224925,
-1453766839, -1614673985, 15099553644, 10806854528, -110319519409, -49482794946,
573293794815, 155510560962, -2125916423322, -344648421788, 5607869207405,
602950452380, -10426809489560, -1017179390188, 13440589196562, 1660271297890,
-11680287015490, -2067440715994, 6524300789888, 1631686761052, -2149016060048,
-727542691768, 347865584016, 160185863728, -13689852000, -13226096640, -1457602560
```

7.5.6.22 A234651

Arrays have $n+1$ rows and width $w = 2$ on the alphabet $0, \ldots, 5$. Predicate D has stress $s = 4$. The recurrence holds for every $n \geq 6$, with order $d = 5$ and component dimensions (12) (total $D = 12$). Source: https://oeis.org/A234651. The first three values are $(160, 728, 3264)$.

$$(c_1, \ldots, c_d) = (4, 44, -176, -480, 1920).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 160x + 88x^2 - 6688x^3 - 1920x^4 + 69120x^5.$$

7.5.6.23 A234652

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,5$. Predicate D has stress $s=4$. The recurrence holds for every $n \geq 16$, with order $d=15$ and component dimensions (63) (total $D=63$). Source: https://oeis.org/A234652. The first three values are $(728, 2648, 9616)$.

```
9, 59, -741, -680, 23970, -24980, -382080, 836736, 3002976, -9941184, -8688384,
53360640, -15759360, -107827200, 99532800
```

7.5.6.24 A234653

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,5$. Predicate D has stress $s=4$. The recurrence holds for every $n \geq 22$, with order $d=21$ and component dimensions (342) (total $D=342$). Source: https://oeis.org/A234653. The first three values are $(3264, 9616, 28496)$.

```
8, 83, -804, -2479, 34112, 22457, -798036, 453896, 11262032, -15161188, -98135856,
191644944, 513355968, -1327868352, -1430555904, 5245827840, 1148359680, -11006668800,
3284582400, 9455616000, -5971968000
```

7.5.6.25 A234731

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,7$. Predicate D has stress $s=8$. The recurrence holds for every $n \geq 8$, with order $d=7$ and component dimensions (20) (total $D=20$). Source: https://oeis.org/A234731. The first three values are $(168, 672, 2184)$.

$$(c_1,\dots,c_d) = (4, 34, -136, -369, 1476, 1260, -5040).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 168x - 6216x^3 + 74160x^5 - 282240x^7.$$

7.5.6.26 A234732

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,7$. Predicate D has stress $s=8$. The recurrence holds for every $n \geq 24$, with order $d=23$ and component dimensions (144) (total $D=144$). Source: https://oeis.org/A234732. The first three values are $(672, 2184, 6048)$.

```
8, 53, -564, -888, 17184, -2382, -296784, 308859, 3196608, -5284083, -22169196,
48357478, 97763896, -273091508, -253071696, 976703832, 271829664, -2148744960,
304750080, 2640988800, -1133222400, -1378944000, 870912000
```

7.5.6.27 A234733

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,7$. Predicate D has stress $s=8$. The recurrence holds for every $n \geq 24$, with order $d=23$ and component dimensions (1056) (total $D=1056$). Source: https://oeis.org/A234733. The first three values are $(2184, 6048, 14760)$.

```
8, 53, -564, -888, 17184, -2382, -296784, 308859, 3196608, -5284083, -22169196,
48357478, 97763896, -273091508, -253071696, 976703832, 271829664, -2148744960,
304750080, 2640988800, -1133222400, -1378944000, 870912000
```

7.5.6.28 A234753

Arrays have $n+1$ rows and width $w = 2$ on the alphabet $0, \ldots, 3$. Predicate K permits at least one orientation whose four edge jumps are all at most $+1$, allowing both. The recurrence holds for every $n \geq 7$, with order $d = 6$ and component dimensions (6) (total $D = 6$). Source: https://oeis.org/A234753. The first three values are $(110, 1014, 8968)$.

$$(c_1, \ldots, c_d) = (8, 10, -16, -8, 4, 1).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 110x + 134x^2 - 244x^3 - 114x^4 + 66x^5 + 16x^6.$$

7.5.6.29 A234754

Arrays have $n+1$ rows and width $w = 3$ on the alphabet $0, \ldots, 3$. Predicate K permits at least one orientation whose four edge jumps are all at most $+1$, allowing both. The recurrence holds for every $n \geq 18$, with order $d = 17$ and component dimensions (20) (total $D = 20$). Source: https://oeis.org/A234754. The first three values are $(1014, 26642, 646036)$.

```
27, -26, -884, 1597, 7345, -14758, -23881, 52625, 29984, -81577, -7015, 51495, -7850,
-9139, 1340, 518, -36
```

7.5.6.30 A234770

Arrays have $n+1$ rows and width $w = 2$ on the alphabet $0, \ldots, 4$. Predicate J permits exactly one orientation whose four edge jumps are all at most $+1$. The recurrence holds for every $n \geq 7$, with order $d = 6$ and component dimensions (9) (total $D = 9$). Source: https://oeis.org/A234770. The first three values are $(88, 454, 2056)$.

$$(c_1, \ldots, c_d) = (1, 21, -1, -67, -10, 46).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 88x + 366x^2 - 246x^3 - 1426x^4 - 98x^5 + 1026x^6.$$

7.5.6.31 A234771

Arrays have $n+1$ rows and width $w = 3$ on the alphabet $0, \ldots, 4$. Predicate J permits exactly one orientation whose four edge jumps are all at most $+1$. The recurrence holds for every $n \geq 24$, with order $d = 23$ and component dimensions (39) (total $D = 39$). Source: https://oeis.org/A234771. The first three values are $(454, 3980, 28840)$.

```
9, 60, -657, -510, 14451, -9878, -143349, 195967, 711924, -1328829, -1738223, 4362370,
1713737, -7406892, 154943, 6781020, -1542017, -3326149, 1109694, 807890, -310656,
-75048, 30144
```

7.5.6.32 A235091

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,5$. Predicate D has stress $s=4$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 6$, with order $d=5$ and component dimensions (12) (total $D=12$). Source: https://oeis.org/A235091. The first three values are $(80, 256, 808)$.

$$(c_1,\dots,c_d) = (3, 13, -39, -26, 78).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 80x + 16x^2 - 1000x^3 - 52x^4 + 2004x^5.$$

7.5.6.33 A235092

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,5$. Predicate D has stress $s=4$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 17$, with order $d=16$ and component dimensions (63) (total $D=63$). Source: https://oeis.org/A235092. The first three values are $(256, 656, 1728)$.

```
6, 20, -178, -45, 2028, -1568, -11316, 15017, 32278, -57324, -42818, 104201, 14456,
-82056, 10920, 16380
```

7.5.6.34 A235093

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,5$. Predicate D has stress $s=4$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 23$, with order $d=22$ and component dimensions (342) (total $D=342$). Source: https://oeis.org/A235093. The first three values are $(808, 1728, 3832)$.

```
7, 20, -239, 36, 3285, -4086, -23439, 45882, 91959, -251226, -184209, 783240, 99087,
-1431666, 277611, 1499767, -513754, -855386, 293540, 256932, -54600, -32760
```

7.5.6.35 A235216

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,2$. Predicate K permits at least one orientation whose four edge jumps are all at most $+1$, allowing both. The recurrence holds for every $n \geq 5$, with order $d=4$ and component dimensions (4) (total $D=4$). Source: https://oeis.org/A235216. The first three values are $(59, 427, 3037)$.

$$(c_1,\dots,c_d) = (6, 9, -6, -2).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 59x + 73x^2 - 56x^3 - 18x^4.$$

7.5.6.36 A235217

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,2$. Predicate K permits at least one orientation whose four edge jumps are all at most $+1$, allowing both. The recurrence holds for every $n \geq 9$, with order $d=8$ and component dimensions (10) (total $D=10$). Source: https://oeis.org/A235217. The first three values are $(427, 8029, 145669)$.

```
16, 52, -188, -10, 188, -29, -40, 10
```

7.5.6.37 A235218

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,2$. Predicate K permits at least one orientation whose four edge jumps are all at most $+1$, allowing both. The recurrence holds for every $n \geq 22$, with order $d=21$ and component dimensions (25) (total $D=25$). Source: https://oeis.org/A235218. The first three values are $(3037, 145669, 6630699)$.

```
39, 410, -3737, -13178, 103275, 89722, -961878, -181366, 4083283, -48872, -8784678,
444362, 9715586, -378256, -5306393, 93253, 1319247, 1378, -124156, -3168, 2456
```

7.5.6.38 A235282

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,3$. Predicate D has stress $s=4$. The recurrence holds for every $n \geq 4$, with order $d=3$ and component dimensions (6) (total $D=6$). Source: https://oeis.org/A235282. The first three values are $(20, 40, 68)$.

$$(c_1,\dots,c_d) = (2,3,-6).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 20x - 72x^3.$$

7.5.6.39 A235283

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,3$. Predicate D has stress $s=4$. The recurrence holds for every $n \geq 7$, with order $d=6$ and component dimensions (20) (total $D=20$). Source: https://oeis.org/A235283. The first three values are $(40, 68, 104)$.

$$(c_1,\dots,c_d) = (3,3,-15,4,18,-12).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 40x - 52x^2 - 220x^3 + 272x^4 + 288x^5 - 336x^6.$$

7.5.6.40 A235284

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,3$. Predicate D has stress $s=4$. The recurrence holds for every $n \geq 7$, with order $d=6$ and component dimensions (72) (total $D=72$). Source: https://oeis.org/A235284. The first three values are $(68, 104, 148)$.

$$(c_1,\dots,c_d) = (3,3,-15,4,18,-12).$$

7.5.6.41 A235291

Arrays have $n+1$ rows and width $w=2$ on the alphabet $0,\dots,3$. Predicate D has stress $s=3$. The recurrence holds for every $n \geq 5$, with order $d=4$ and component dimensions (6) (total $D=6$). Source: https://oeis.org/A235291. The first three values are $(40, 108, 272)$.

$$(c_1,\dots,c_d) = (0, 14, 0, -48).$$

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 40x + 108x^2 - 288x^3 - 768x^4.$$

7.5.6.42 A235292

Arrays have $n+1$ rows and width $w=3$ on the alphabet $0,\dots,3$. Predicate D has stress $s=3$. The recurrence holds for every $n \geq 11$, with order $d=10$ and component dimensions (20) (total $D=20$). Source: https://oeis.org/A235292. The first three values are $(108, 232, 476)$.

```
3, 17, -57, -86, 372, 76, -972, 360, 864, -576
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 108x - 92x^2 - 2056x^3 + 1916x^4 + 13580x^5 - 13472x^6 - 36408x^7 + 37296x^8 + 33408x^9 - 34560x^{10}.$$

7.5.6.43 A235293

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,3$. Predicate D has stress $s=3$. The recurrence holds for every $n \geq 11$, with order $d=10$ and component dimensions (72) (total $D=72$). Source: https://oeis.org/A235293. The first three values are $(272, 476, 832)$.

```
3, 17, -57, -86, 372, 76, -972, 360, 864, -576
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 272x - 340x^2 - 5220x^3 + 6652x^4 + 34708x^5 - 44608x^6 - 93528x^7 + 119472x^8 + 86400x^9 - 108288x^{10}.$$

7.5.6.44 A235294

Arrays have $n+1$ rows and width $w=5$ on the alphabet $0,\dots,3$. Predicate D has stress $s=3$. The recurrence holds for every $n \geq 11$, with order $d=10$ and component dimensions (272) (total $D=272$). Source: https://oeis.org/A235294. The first three values are $(744, 1132, 1736)$.

```
3, 17, -57, -86, 372, 76, -972, 360, 864, -576
```

The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 744x - 1100x^2 - 14308x^3 + 21140x^4 + 95228x^5 - 139616x^6 - 256584x^7 + 369360x^8 + 237312x^9 - 331776x^{10}.$$

7.5.6.45 A235314

Arrays have $n+1$ rows and width $w=4$ on the alphabet $0,\dots,6$. Predicate D has stress $s=7$. The recurrence holds for every $n \geq 19$, with order $d=18$ and component dimensions (625) (total $D=625$). Source: https://oeis.org/A235314. The first three values are $(1120, 2744, 5968)$.

```
4, 47, -200, -901, 4204, 8957, -48440, -47194, 334096, 101948, -1410080, 170136,
3549696, -1446912, -4861440, 2954880, 2764800, -2073600
```

7.6 Verification, provenance and reuse

The accompanying archive contains the complete manuscript, numbered ledger, defining data, exact certificates, verifiers, source audits and cross-review reports. With Python, SymPy and mpmath installed, run

```
python3 r7_run_all.py
```

from its extracted directory. No network access is needed. The runner checks the distinct-A-ID count, all previous selections, the retained prior-work baseline, immutable review bindings, and each verification program. It records the actual versions and results in `oeis_hunt_round7_verification_summary.json`.

The array certificates use complete integer matrices derived from the definitions. An identity $TR(T)\mathbf{1}=0$ implies the recurrence for all later indices after multiplication by an arbitrary power of T. The generating-function certificates prove exact differential identities whose residuals are explicitly displayed polynomials; coefficients beyond the polynomial degree vanish for every index. These are finite exact certificates for infinite statements. Finite-prefix and numerical comparisons check interpretation and implementation but are not used as proofs of universal claims.

The arithmetic and formal-series parts give complete mathematical arguments. Their verification programs check exact identities and representative computations; they are not claimed to constitute proof-assistant formalizations. Separate review implementations and source-definition checks provide additional error detection.

7.6.1 Source and priority scope

Each selected theorem links to its OEIS source. The JSON records preserve the exact target, offset, correction, attribution, retrieval reference and the known formula used as an input. The audit distinguishes the source's conjectural wording from its editorial approval status. Source wording alone does not prove that a mathematical result was unknown elsewhere.

The combined audit compares the previous project ledgers and the retained OEIS Open Appendix A baseline. The generating-function audit additionally screens the relevant Fischer (2020) and Ekhad–Yang (2017) families. Other already proved or equivalent claims found during this round are excluded and explained in the component audits. The bibliography and links are not represented as an exhaustive historical search. In particular, the still-marked recurrences on A065942 and A226302 are exact consequences of already recorded shorter recurrences and were removed. A359643 was conservatively excluded after the official Alkahest guide identified its own earlier recurrence result; that project's complete result catalogue was not audited.

All previously selected IDs remain excluded, including the five Round IV entries whose priority claim was withdrawn in Round V: A176605, A176607, A176610, A176611 and A176612. Their valid formulas were already implied by Georg Fischer's 2020 general derivation. The current result count is a count of precisely scoped, source-marked assertions resolved in this packet, not a cumulative worldwide-discovery claim.

7.6.2 Standalone manuscript and reproducibility

The delivered `.tex` file contains all prose, formulas and individual recurrence signatures directly; it has no external `\input` or `\include` dependency. Compile it twice with pdfLaTeX and the standard packages in its preamble. `r7_build_dossier.py` reconstructs the manuscript and ledger from the retained component records. The README gives the certificate regeneration commands and dependencies. SHA-256 digests in the archive manifest bind the delivered files; review reports separately bind the proof and implementation versions they inspected.

7.6.3 Selected common references

1. The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/. Individual entry links accompany the statements throughout the dossier.
2. NIST Digital Library of Mathematical Functions, §19.20, §20.7 and §20.9: classical elliptic special values and theta identities used in the offset correction.
3. *OEIS Open*, retained Appendix A comparison, https://arxiv.org/html/2608.11941v1. The retained scope is 100 costliest claims on 97 distinct A-IDs among 153 combined-model resolutions, not the complete repository.
4. Georg Fischer, 2020 family derivation linked at https://oeis.org/A176604/a176604_2.txt; used for exclusion and priority auditing, not as a newly claimed result.
5. S. B. Ekhad and D. Yang, 2017 recurrence family and supporting output at https://sites.math.rutgers.edu/~zeilberg/tokhniot/oMathar1maple12.txt; used for exclusion and credit.
6. Alkahest, official user guide, https://alkahest-cas.github.io/alkahest/print.html, novelty-filtering discussion identifying A359643 as an earlier result; used as a conservative exclusion lead.

Part VIII

Denominators, exponent classification, arithmetic, and recurrence certificates

Round overview

This eighth-round dossier resolves 100 selected assertion packages on 100 distinct OEIS entries, disjoint from the 708 A-numbers selected in Rounds I–VII. It proves 98 selected claims, corrects and proves one modulo-four support rule, and refutes one empirical classification by an explicit counterexample and an infinite omitted family. The latter package also proves the lower bound in the same source comment.

The four parts contain 13 packages on formal series, denominators and congruences, 12 on arithmetic identities and a refutation, 26 on generating-function recurrences, and 49 on constrained arrays. They use coefficient-ring arguments, valuation and divisor identities, exact differential equations, and finite transfer operators.

What the count means

Each catalogue row selects one precisely scoped assertion package on one distinct A-number. All selected assertions are explicitly marked conjectural or empirical in the consulted sources. A recurrence and its generating function on the same entry count once. Several entries share a general lemma or method; in particular the 49 array packages are distinct fixed-parameter counting problems certified within one transfer framework. The count does not assert 100 unrelated methods.

The source wording is evidence of an unresolved label in the consulted entry. It does not certify worldwide historical priority. Established formulas, classical theorems and earlier project methods are credited inputs. Every selected scope is recorded in the numbered catalogue, the proofs and the machine-readable ledger.

Correction, refutation and auxiliary source issues

For A380680, the proposed square support modulo 4 is false: $a(1) = 4$ and $a(2) = 6$. The correct positive-index support is $n = 2j^2$, $j \geq 1$, where the residue is 2; elsewhere it is 0.

For A345993, the proposed classification of $a(n) = 3$ omits $n = 375$, where the least admissible m is 125. More generally, $n = 3p^e$ works whenever $p \equiv 2 \pmod 3$ is prime and e is odd. The statement $a(n) \geq 3$ for $n \geq 3$ is proved. A complete replacement classification is not asserted.

All 26 selected generating-function recurrence coefficient lists are proved unchanged, with explicit valid starting ranges. Auxiliary input issues are disclosed: A308815's title requires the alternating sign, an older displayed GF for A113409 conflicts with its defining sum, and older unmarked formulas for A174169 require care. The proofs derive or identify the consistent inputs explicitly; these repairs add no selected counts.

Prior work and uncounted material

The audit excludes A268149, already covered by a linked paper's general formula, and A101601, whose recurrence is an immediate consequence of the recorded hypergeometric closed form. Other stale labels and already proved claims found during the search are documented in the component audits. The previous 708 is an exclusion ledger, not a certified discovery total. The five Round IV priority withdrawals disclosed in Round V remain in force and remain excluded.

The integrality argument for A300617 is supplied as an uncounted supplement because its companion already states integrality without a conjecture qualifier. The stronger triangular-divisibility assertion is not proved here. The companion numerator consequence for A079269, the duplicate formulations A174490 and A177431, and the certified reserve A235252 add no counts.

Evidence and limits

Primary entries, relevant linked materials and targeted searches were consulted on 20 October 2026; retrieval may use cached pages. The retained OEIS Open comparison covers 100 costliest Appendix A claims on 97 distinct A-numbers, with no selected overlap. It does not cover that project's entire repository. The additional known-family and linked-paper checks are bounded, not an exhaustive priority search.

No OEIS edit or submission was performed. The results here are mathematical resolutions of selected assertions, separate from OEIS editorial acceptance. Internal cross-reviews were performed by separate agents in the same AI-assisted workflow; they are not external peer review or proof-assistant formalizations.

8.1 Catalogue of the 100 selected entries

P: selected claim proved. **C**: support rule corrected and proved. **R**: empirical classification refuted, with its lower bound proved. Page references lead to the full argument or recurrence signature.

No.	Entry	Selected assertion package	Result	Page
Formal series, denominators and congruences				
1	A381670	Dyadic denominators in the compositional equation	P	397
2	A352117	The convergent central-binomial moment formula	P	397
3	A174491	The exact power-of-two denominator	P	398
4	A079278	The denominator recurrence after reduction	P	399
5	A177430	The integral exponential transform: all real parameters	P	399
6	A383377	Parity of the absolute-value self-referential series	P	400
7	A382122	Powers-of-two parity support for every real solution	P	400
8	A365095	The full two-parameter square divisibility	P	401
9	A396102	All positive coefficients are 1 modulo 3	P	401
10	A378581	Square support modulo 4 in the theta composition	P	402
11	A380680	Correct the support to twice a square, modulo 4	C	402
12	A379764	Signed square support modulo 4	P	403
13	A380060	Square support modulo 4 in the bilateral equation	P	403
Arithmetic identities and one refutation				
14	A023887	The divisor-sum ratio for every positive exponent	P	404
15	A092287	The rectangular gcd-product valuation formula	P	405
16	A119689	Evenness and the extreme-prime-divisor criterion	P	405
17	A143991	Equality of reduced integral and series numerators	P	405
18	A179504	The doubled-argument divisor-power identity	P	407
19	A193679	The logarithmic Dirichlet series and convolution	P	407
20	A226787	The empirical inverse-nine rational generating function	P	408
21	A326691	The arithmetic-derivative gcd formula	P	408
22	A345993	Refute the value-three classification; prove its lower bound	R	409
23	A367865	The radical-average divisor identity	P	409
24	A373561	The fourfold gcd-indicator sum	P	410
25	A383044	The seven-ninths totient family	P	410
Generating-function recurrence certificates				
26	A014533	Conjectured polynomial recurrence of order 2	P	412
27	A024719	Conjectured polynomial recurrence of order 2	P	413
28	A026126	Conjectured polynomial recurrence of order 4	P	414
29	A098519	Conjectured polynomial recurrence of order 2	P	415
30	A103138	Conjectured polynomial recurrence of order 3	P	415
31	A112702	Conjectured polynomial recurrence of order 2	P	416
32	A113409	Conjectured polynomial recurrence of order 4	P	417
33	A114851	Conjectured polynomial recurrence of order 6	P	418
34	A162972	Conjectured polynomial recurrence of order 7	P	419
35	A168505	Conjectured polynomial recurrence of order 4	P	420
36	A174169	Conjectured polynomial recurrence of order 4	P	420
37	A182892	Conjectured polynomial recurrence of order 8	P	421
38	A220898	Conjectured polynomial recurrence of order 3	P	422
39	A225692	Conjectured polynomial recurrence of order 3	P	423
40	A240722	Conjectured polynomial recurrence of order 2	P	424
41	A246606	Conjectured polynomial recurrence of order 2	P	425
42	A268402	Conjectured polynomial recurrence of order 4	P	426
43	A278618	Conjectured polynomial recurrence of order 3	P	427
44	A302180	Conjectured polynomial recurrence of order 3	P	428
45	A302181	Conjectured polynomial recurrence of order 2	P	428
46	A302184	Conjectured polynomial recurrence of order 3	P	429
47	A302186	Conjectured polynomial recurrence of order 4	P	429
48	A308815	Conjectured polynomial recurrence of order 3	P	430

Continued on the next page

No.	Entry	Selected assertion package	Result	Page
49	A344502	Conjectured polynomial recurrence of order 4	P	431
50	A379463	Conjectured polynomial recurrence of order 5	P	432
51	A379464	Conjectured polynomial recurrence of order 7	P	433
Finite-state array recurrence certificates				
52	A233646	Empirical array recurrence of order 9	P	437
53	A233647	Empirical array recurrence of order 38	P	438
54	A233675	Empirical array recurrence of order 4	P	438
55	A233676	Empirical array recurrence of order 10	P	438
56	A233677	Empirical array recurrence of order 17	P	438
57	A233687	Empirical array recurrence of order 41	P	439
58	A233718	Empirical array recurrence of order 16	P	439
59	A233726	Empirical array recurrence of order 4	P	439
60	A233727	Empirical array recurrence of order 12	P	439
61	A233728	Empirical array recurrence of order 28	P	440
62	A233942	Empirical array recurrence of order 4	P	440
63	A233943	Empirical array recurrence of order 10	P	440
64	A233944	Empirical array recurrence of order 25	P	440
65	A234241	Empirical array recurrence of order 8	P	441
66	A234259	Empirical array recurrence of order 3	P	441
67	A234260	Empirical array recurrence of order 6	P	441
68	A234261	Empirical array recurrence of order 6	P	441
69	A234326	Empirical array recurrence of order 4	P	442
70	A234327	Empirical array recurrence of order 11	P	442
71	A234328	Empirical array recurrence of order 48	P	442
72	A234338	Empirical array recurrence of order 13	P	442
73	A234444	Empirical array recurrence of order 4	P	443
74	A234445	Empirical array recurrence of order 17	P	443
75	A234452	Empirical array recurrence of order 6	P	443
76	A234453	Empirical array recurrence of order 26	P	443
77	A234483	Empirical array recurrence of order 3	P	444
78	A234484	Empirical array recurrence of order 5	P	444
79	A234491	Empirical array recurrence of order 5	P	444
80	A234492	Empirical array recurrence of order 23	P	444
81	A234674	Empirical array recurrence of order 4	P	445
82	A234675	Empirical array recurrence of order 18	P	445
83	A234884	Empirical array recurrence of order 27	P	445
84	A234885	Empirical array recurrence of order 46	P	445
85	A234891	Empirical array recurrence of order 10	P	446
86	A234905	Empirical array recurrence of order 12	P	446
87	A234984	Empirical array recurrence of order 8	P	446
88	A234985	Empirical array recurrence of order 28	P	447
89	A235019	Empirical array recurrence of order 6	P	447
90	A235020	Empirical array recurrence of order 21	P	447
91	A235064	Empirical array recurrence of order 10	P	447
92	A235065	Empirical array recurrence of order 40	P	448
93	A235100	Empirical array recurrence of order 6	P	448
94	A235101	Empirical array recurrence of order 19	P	448
95	A235180	Empirical array recurrence of order 18	P	448
96	A235181	Empirical array recurrence of order 26	P	449
97	A235191	Empirical array recurrence of order 6	P	449
98	A235192	Empirical array recurrence of order 23	P	449
99	A235193	Empirical array recurrence of order 39	P	450
100	A235251	Empirical array recurrence of order 10	P	450

8.2 Formal series, divisibility, and corrected congruences

This section selects one assertion package for each of thirteen distinct A-numbers. The source statements are paraphrased, with their defining equations retained. The proofs concern all indices in the stated ranges. Calculations in `r8_root_verify.py` are supplementary checks, except where an exact symbolic identity is explicitly identified. Classical inversion, Euler products, and the six exponentials theorem are credited inputs, not new results.

8.2.1 A381670: dyadic denominators under self-composition

Source. A381670, Thomas Scheuerle, March 3, 2025, conjectures that the reduced denominators of the coefficients of

$$A(x) = x + \tfrac{1}{2}x^2 - \tfrac{1}{16}x^4 + \cdots, \qquad A(A(x)) = x(1 + A(x)),$$

are powers of two. We include $1 = 2^0$ and use denominator 1 for a zero coefficient.

Proposition 8.2.1. *There is a unique solution $A \in x + x^2\mathbb{Q}[[x]]$, and it belongs to $x + x^2\mathbb{Z}[1/2][[x]]$. Consequently every term of A381670 is a power of two.*

Proof. Write $A = x + \sum_{n\geq 2} c_n x^n$. For $n \geq 2$, the coefficient of c_n in $[x^n]A(A(x))$ is 2: one occurrence comes from the outer linear term, and one from $c_n A(x)^n$. All other contributions form an integer polynomial $P_n(c_2, \ldots, c_{n-1})$. Thus

$$2c_n = c_{n-1} - P_n(c_2, \ldots, c_{n-1}), \qquad c_1 = 1.$$

This determines the coefficients uniquely and inductively keeps them in $\mathbb{Z}[1/2]$. Reduction of a dyadic rational cannot introduce an odd prime in its denominator. □

8.2.2 A352117: a convergent central-binomial moment formula

Source. A352117 is defined by the exponential generating function $F(t) = (2 - e^{2t})^{-1/2}$. The infinite-sum formula below is labelled conjectural by Velin Yanev and Vaclav Kotesovec, July 21, 2026. The separate Stieltjes-fraction assertion is not selected here.

Proposition 8.2.2. *For every integer $n > 0$,*

$$a(n) = \sum_{k\geq 1} 2^{n-3k-1/2} \binom{2k}{k} k^n.$$

The series converges absolutely.

Proof. The binomial expansion, initially for $|e^{2t}| < 2$, gives

$$(2 - e^{2t})^{-1/2} = \frac{1}{\sqrt{2}} \sum_{k\geq 0} \binom{2k}{k} \frac{e^{2kt}}{8^k}.$$

For any $0 < \epsilon < \log(2)/2$ and $|t| \leq \epsilon$, the nth derivative of the kth summand has modulus at most a constant times $k^n(e^{2\epsilon}/2)^k$, using $\binom{2k}{k} \leq 4^k$. This is summable. Termwise differentiation is therefore valid near zero. Evaluating $F^{(n)}(0)$ proves the formula; the $k = 0$ summand vanishes when $n > 0$. The same estimate at $t = 0$ proves absolute convergence. For $n = 0$, one must retain $k = 0$ rather than apply the displayed $n > 0$ formula. □

8.2.3 A174491: the exact denominator, including odd-prime cancellation

Source. A174491, Paul D. Hanna, March 25, 2010, defines the denominator of $g_n = [x^n]G(x)$, where

$$G(x) = \exp\left(\sum_{j\geq 1} \frac{x^j}{j2^{j^2}}\right).$$

It conjectures the formula $2^{n^2+v_2(n)}$ for $n \geq 1$. The identical denominator assertion on A174490 is not counted again.

Lemma 8.2.3. *Put $b_n = 2^{-n^2}$ and $c_n = n^{-1}\sum_{d|n}\mu(d)b_{n/d}$. Then $c_n \in \mathbb{Z}[1/2]$, and*

$$G(x) = \prod_{n\geq 1}(1-x^n)^{-c_n} \in \mathbb{Z}[1/2][[x]].$$

Proof. Fix an odd prime p and let $n = p^s u$ with $s \geq 1$ and $p \nmid u$. For $p \nmid e$, Euler's theorem for units modulo p^s implies

$$b_{p^s e} - b_{p^{s-1}e} \in p^s\mathbb{Z}_{(p)}.$$

Indeed, the difference of the exponents of 2 is $(p^2-1)p^{2s-2}e^2$, divisible by $p^{s-1}(p-1)$. Pairing the divisors d and pd in the Mobius sum yields

$$\sum_{d|n}\mu(d)b_{n/d} = \sum_{d|u}\mu(d)(b_{p^s u/d} - b_{p^{s-1}u/d}) \in p^s\mathbb{Z}_{(p)}.$$

Division by n thus leaves no p in the denominator. This holds for every odd p, proving $c_n \in \mathbb{Z}[1/2]$. For $r \in \mathbb{Z}[1/2]$, all binomial coefficients $\binom{r}{j}$ also lie in that ring. To see this at an odd prime p, choose an integer M congruent to r modulo p^N with $N > v_p(j!)$; the falling products show $\binom{r}{j} \in \mathbb{Z}_{(p)}$. Each factor of the product is therefore dyadic coefficientwise. Finally, the logarithm of the product has coefficient $n^{-1}\sum_{d|n} dc_d = b_n/n$ by Mobius inversion. All products and logarithms are formal and coefficientwise finite. □

Proposition 8.2.4. *The reduced denominator of g_0 is 1. For every $n \geq 1$, that of g_n is exactly $2^{n^2+v_2(n)}$.*

Proof. The exponential formula expands g_n as

$$\sum_{\substack{m_1,m_2,\ldots\geq 0\\ \sum jm_j = n}} \prod_{j\geq 1} \frac{1}{m_j!\, j^{m_j}\, 2^{j^2 m_j}}.$$

The term $m_n = 1$ has 2-adic valuation $-n^2 - v_2(n)$. For every other partition of $n \geq 2$, convexity gives $\sum j^2 m_j \leq (n-1)^2 + 1 = n^2 - 2n + 2$. Write $r = \sum m_j$ and let t be the number of nonzero m_j. Since $v_2(j) \leq j-1$ and $v_2(m!) \leq m-1$ for $m \geq 1$,

$$\sum m_j v_2(j) + \sum v_2(m_j!) \leq (n-r) + (r-t) \leq n-1.$$

Thus every other term has denominator 2-adic exponent at most $n^2 - n + 1 < n^2 + v_2(n)$. A sum with exactly one term of least 2-adic valuation has that same valuation. Lemma 8.2.3 rules out all odd denominator primes. The cases $n = 0, 1$ are immediate. □

8.2.4 A079278: eliminate the numerators without losing reduction

Source. A079278 gives $b_1 = 1$, $b_{n+1} = b_n + b_n/(1 + b_n)$, and defines q_n as the reduced denominator of b_n. Leroy Quet's recurrence is explicitly conjectural.

Proposition 8.2.5. *For $n \geq 2$,*

$$q_{n+1} = q_n^2 + \frac{q_n^3}{q_{n-1}^2} - q_n q_{n-1}^2, \qquad q_1 = 1, \quad q_2 = 2.$$

Proof. Write $b_n = p_n/q_n$ with positive coprime integers. Direct substitution gives

$$p_{n+1} = p_n(p_n + 2q_n), \qquad q_{n+1} = q_n(p_n + q_n).$$

These are already reduced: each of p_n and $p_n + 2q_n$ is coprime to each of q_n and $p_n + q_n$. In particular no implicit cancellation changes the denominator recurrence. Now

$$p_{n-1} = q_n/q_{n-1} - q_{n-1}, \qquad p_n = p_{n-1}^2 + 2p_{n-1}q_{n-1} = q_n^2/q_{n-1}^2 - q_{n-1}^2.$$

Substitution into $q_{n+1} = q_n(p_n + q_n)$ gives the result. The quotients appearing here are integers by the first-order denominator recurrence. The associated numerator recurrence on A079269 is an uncounted corollary, obtained by the same elimination. □

8.2.5 A177430: which powers remain logarithmic derivatives?

Source. A177430, Paul D. Hanna, September 6, 2010, is the least strictly increasing sequence of squares a_n for which $\exp(\sum a_n x^n/n)$ has integer coefficients. Its selected conjecture says that replacing a_n by $a_n^{m/2}$ preserves this property precisely for nonnegative even integers m. The duplicate statement at A177431 is uncounted. We prove the assertion even if m is allowed to be real, using positive real powers. The integer-parameter proof is elementary.

Lemma 8.2.6. *For a sequence u_n of integers,*

$$\exp\left(\sum_{n\geq 1} u_n x^n/n\right) \in 1 + x\mathbb{Z}[[x]] \quad \Longleftrightarrow \quad u_n \equiv u_{n/p} \pmod{p^{v_p(n)}} \quad (p \mid n).$$

Sequences satisfying these congruences are closed under nonnegative integer powers taken term by term.

Proof. Every series in $1 + x\mathbb{Z}[[x]]$ has a unique formal factorization $\prod_{d\geq 1}(1 - x^d)^{-e_d}$ with $e_d \in \mathbb{Z}$: determine e_d successively from the coefficient of x^d. Its logarithmic derivative has $u_n = \sum_{d|n} d e_d$. Subtracting the expression for $u_{n/p}$ proves the congruences. Conversely $n e_n = \sum_{d|n} \mu(d) u_{n/d}$; pairing divisors with and without p shows this numerator divisible by every $p^{v_p(n)}$, hence by n. Integer powers preserve the congruences because $s^k - t^k$ is divisible by $s - t$; $k = 0$ gives the constant sequence 1. □

The defining greedy sequence exists at every step. If n is fixed and its earlier square roots are r_j, the Chinese remainder theorem supplies a positive integer r with $r \equiv r_{n/p} \pmod{p^{v_p(n)}}$ for every prime $p \mid n$. Taking r arbitrarily large makes r^2 an admissible next square. Lemma 8.2.6 gives consistency. Its first four terms are $1, 9, 16, 25$: at $n = 2, 3, 4$ the new square must be respectively $1 \pmod 2$, $1 \pmod 3$, and $9 \pmod 4$, as well as exceed the previous one.

Proposition 8.2.7. *For real m,*

$$\exp\left(\sum_{n\geq 1} a_n^{m/2} x^n/n\right) \in \mathbb{Z}[[x]] \quad \Longleftrightarrow \quad m \in \{0, 2, 4, 6, \ldots\}.$$

Proof. For $m = 2k \geq 0$, the assertion follows from Lemma 8.2.6. Conversely an integral unit series has an integral logarithmic derivative xF'/F, so integrality forces $3^m, 4^m, 5^m$ to be positive integers. If $m < 0$, already $0 < 3^m < 1$ is impossible. If m is rational, unique prime factorization applied to 3^m implies that m is a nonnegative integer.

For completeness, irrational real m is excluded by the classical six exponentials theorem. The numbers $1, m$ are linearly independent over $\mathbb{Q}$, as are $\log 3, \log 4, \log 5$. That theorem says that at least one of the six exponentials of their pairwise products is transcendental. Here those exponentials would be $3, 4, 5, 3^m, 4^m, 5^m$, all integers, a contradiction. We use the established theorem in the form stated and proved in Samit Dasgupta, *Ranks of matrices of logarithms of algebraic numbers I*, Theorem 4.2 and Section 5.1 (2023), https://sites.math.duke.edu/~dasgupta/papers/Baker.pdf.

Finally, for an odd integer m, the necessary ghost congruence at $n = 4$ would require $5^m \equiv 3^m \pmod 4$. Its two sides are 1 and -1, respectively. This excludes every odd m and completes the classification. □

8.2.6 A383377: absolute values disappear modulo two

Source. A383377, Paul D. Hanna, May 15, 2025, defines

$$A(x) = \sum_{k\geq 0} x^k \operatorname{abs}(A(x)^{-k}), \qquad A(0) = 1,$$

where abs acts on each integer coefficient. It conjectures that a_n is even for $n > 1$.

Proposition 8.2.8. *The equation defines a unique integral series, and* $A(x) \equiv 1 + x \pmod 2$.

Proof. For positive degree n, the term $k = 0$ contributes nothing, and each term $k \geq 1$ on the right uses coefficients of A only through degree $n - k < n$. This proves existence, uniqueness and integrality by recursion. For an integer c, $|c| \equiv c \pmod 2$. Reducing the equation in $\mathbb{F}_2[[x]]$ therefore gives

$$A = \sum_{k\geq 0} (x/A)^k = \frac{A}{A - x}.$$

Both A and $A - x$ are units. Cancellation yields $A - x = 1$, which is exactly the asserted parity pattern. □

8.2.7 A382122: a Catalan target forces powers-of-two parity

Source. A382122, Paul D. Hanna, March 16, 2025, uses

$$\sum_{k\geq 0} x^k \operatorname{abs}(A(x)^{-k}) = C(x), \qquad C(x) = 1 + xC(x)^2,$$

and conjectures that its positive-index odd terms occur precisely at powers of two. The conclusion below holds for every real-coefficient solution with $A(0) = 1$; it needs no uniqueness convention for the absolute-value branches.

Proposition 8.2.9. *Every such solution is integral and satisfies*

$$A(x) \equiv 1 + \sum_{j\geq 0} x^{2^j} \pmod 2.$$

In particular the conjectured parity of A382122 holds.

Proof. Assume $a_1, \ldots, a_{n-1}$ are integers. In degree $n+1$ the summand $x\,\mathrm{abs}(A^{-1})$ contributes $|I - a_n|$ for an integer I, and all other terms use earlier coefficients only. The defining equation therefore makes $|I - a_n|$ an integer, so a_n is an integer too. This starts with $|a_1| = 1$ and proves integrality for every solution.

Now reduce modulo two to obtain $A/(A - x) = C$. Thus $A(C - 1) = Cx$. Using $C - 1 = xC^2$ and cancelling x gives $A = C^{-1} = 1 + xC$ in characteristic two. The unique Catalan series there satisfies $C = 1 + xC^2$, hence by successive squaring $C = \sum_{j \geq 0} x^{2^j - 1}$. Substitution proves the assertion. This argument does not assert that the original absolute-value equation has only one real or positive branch; the parity is identical on every branch. □

8.2.8 A365095: a square-divisibility theorem for every integer parameter

Source. A365095, Paul D. Hanna, September 3, 2023, is defined by $A(0) = 1$ and

$$[x^{n-1}](A^{-1} + (n-1)xA)^n = 0 \qquad (n > 1).$$

Its selected conjecture replaces $n - 1$ by $kn - 1$ and asks for divisibility by n^2 for every integer k. Other observations about related series on the page are outside this selected assertion.

Proposition 8.2.10. *The defining series is integral, and for $n \geq 1$, $k \in \mathbb{Z}$,*

$$n^2 \mid [x^{n-1}]\frac{(1 + (kn-1)xA(x)^2)^n}{A(x)^n}.$$

Proof. For any $H \in \mathbb{Z}[[x]]$, differentiation gives

$$(n-1)[x^{n-1}]H^n = n[x^{n-2}]H'H^{n-1}.$$

Since n and $n - 1$ are coprime, $n \mid [x^{n-1}]H^n$. Suppose the coefficients of A below degree $n - 1$ are already integral, and temporarily put $a_{n-1} = 0$. Then $H = A^{-1} + (n-1)xA$ is integral. Restoring a_{n-1} changes the coefficient under consideration by $-na_{n-1}$. The preceding divisibility determines an integral a_{n-1}, proving existence and uniqueness inductively.

For $n > 1$, put $H = A^{-1} + (n-1)xA$. The target expression is $[x^{n-1}](H + n(k-1)xA)^n$. Subtract its known zero value $[x^{n-1}]H^n$. The term with one copy of $n(k-1)xA$ has the factor $\binom{n}{1}n = n^2$, and each term with $j \geq 2$ such copies has the factor n^j, hence n^2. All remaining coefficients are integers. For $n = 1$ the coefficient is 1, divisible by 1 as required. □

8.2.9 A396102: a triple-composition congruence

Source. A396102, Paul D. Hanna, June 6, 2026, has $A = x + O(x^2)$ and

$$A^{\langle 3 \rangle}(x) = (1 + x)A^{\langle 2 \rangle}(x),$$

where angle brackets denote composition, not multiplication. It conjectures $a_n \equiv 1 \pmod 3$ for $n \geq 1$.

Proposition 8.2.11. *The solution is integral and $A(x) \equiv x/(1 - x) \pmod 3$.*

Proof. For $A = x + \sum_{j \geq 2} a_j x^j$, the coefficient of a_n in the degree-n coefficient of the rth iterate is r. Thus the defining difference has coefficient $3 - 2 = 1$ on a_n; the extra factor x uses only earlier coefficients. This gives a unique solution by integer polynomial recurrences over any commutative coefficient ring.

Let $B = x/(1 - x)$. Direct composition gives $B^{\langle r \rangle} = x/(1 - rx)$. In characteristic three,

$$B^{\langle 3 \rangle} = x = (1 + x)\frac{x}{1 - 2x} = (1 + x)B^{\langle 2 \rangle}.$$

Uniqueness over $\mathbb{F}_3$ now gives $A = B$ there. The related but different iterate congruence in A196523 already has an explicit proof by Amol Bhave dated October 13, 2026, and is excluded. □

8.2.10 A378581: composition preserves the theta support modulo four

Source. A378581, Paul D. Hanna, January 8, 2025, defines

$$A(xA(x)) = 1 + 2\sum_{j\geq 1} x^{j^2}, \qquad A(0) = 1,$$

and conjectures $a_n \equiv 2 \pmod 4$ at positive squares and 0 elsewhere.

Proposition 8.2.12. *The defining equation has a unique integral solution and* $A(x) \equiv 1+2\sum_{j\geq 1} x^{j^2} \pmod 4$.

Proof. In degree n, the coefficient a_n occurs once, in $a_n x^n A(x)^n$. Every other contribution uses $a_1, \ldots, a_{n-1}$. This proves integral existence and uniqueness over every coefficient ring. Modulo two the unique solution is $A = 1$, since the right side is 1. Write $A = 1 + 2B$ with $B \in x\mathbb{Z}[[x]]$. Since $xA \equiv x \pmod 2$,

$$A(xA) = 1 + 2B(xA) \equiv 1 + 2B(x) = A(x) \pmod 4.$$

The defining equation gives the stated coefficientwise congruence. The transform differs from A378580, selected in an earlier round; no second count is assigned to that earlier A-number. □

8.2.11 A380680: the support is twice a square, not a square

Source. A380680, Paul D. Hanna, February 2025, defines

$$A(x) = \sum_{j\in\mathbb{Z}} x^{j^2} \left(\frac{2 - x^j}{1 - 2x^j}\right)^j$$

and labels the square-index residue pattern conjectural. Its own data already have $a_1 = 4$ and $a_2 = 6$, contradicting that pattern.

Proposition 8.2.13. *For every* $n > 0$,

$$a_n \equiv \begin{cases} 2 \pmod 4, & n = 2j^2 \text{ for an integer } j \geq 1, \\ 0 \pmod 4, & \text{otherwise.} \end{cases}$$

Proof. Replacing j by $-j$ leaves the summand invariant. Pairing the two signs therefore gives the well-defined integral formal series

$$A(x) = 1 + 2\sum_{j\geq 1} x^{j^2} \left(\frac{2 - x^j}{1 - 2x^j}\right)^j.$$

For positive j, the factor in parentheses is x^j modulo two, so its jth power is x^{j^2} modulo two. Hence

$$A(x) \equiv 1 + 2\sum_{j\geq 1} x^{2j^2} \pmod 4.$$

The positive exponents $2j^2$ are distinct. This proves the corrected classification and refutes the original one. We count this once as a corrected assertion, not once for the counterexample and again for its replacement. □

8.2.12 A379764: a bilateral series with an integral compositional inverse

Source. A379764, Paul D. Hanna, January 13, 2025, has $A = x + O(x^2)$ and $F(4A) = 2\theta_4(x)$, where

$$F(z) = \sum_{j\in\mathbb{Z}} (z^j - z)^{j+1}, \qquad \theta_4(x) = 1 + 2\sum_{m\geq 1} (-1)^m x^{m^2}.$$

The two selected conjectures are the square support of odd coefficients and its refinement $a_n \equiv (-1)^{n-1} \pmod 4$ on squares, zero otherwise.

Proposition 8.2.14. *Both conjectures hold for every positive index.*

Proof. The terms $j = -1, 0, 1$ of F are respectively $1, 1 - z, 0$. For $j \geq 2$, the term is an integral polynomial of order $j + 1$. For $j = -r \leq -2$ it is

$$z^{r(r-1)}(1 - z^{r+1})^{-(r-1)},$$

an integral power series of order $r(r - 1)$. Thus the bilateral sum is coefficientwise finite and $F(z) = 2 - z + \sum_{j\geq 2} f_j z^j$ with $f_j \in \mathbb{Z}$. Dividing $F(4A) = 2\theta_4(x)$ by 4 after subtracting 2 gives

$$A(x) = \sum_{m\geq 1} (-1)^{m-1} x^{m^2} + \sum_{j\geq 2} 4^{j-1} f_j A(x)^j.$$

This is a monic triangular equation, so it defines an integral solution. All terms in its second sum vanish modulo four. At a square $n = m^2$, the remaining coefficient is $(-1)^{m-1} = (-1)^{n-1}$, and at a nonsquare it is zero. The parity claim follows as well. □

8.2.13 A380060: the implicit bilateral theta-type series

Source. A380060, Paul D. Hanna, February 6, 2025, defines

$$A(x) = \sum_{j\in\mathbb{Z}} x^{j^2} \left(\frac{A(x) - x^j}{1 - x^j A(x)}\right)^j, \qquad A(0) = 1.$$

It conjectures that a_n is 2 modulo four at positive squares and 0 otherwise. Unlike A380680, the numerator here contains the unknown series $A(x)$ rather than the constant 2.

Proposition 8.2.15. *The equation has a unique integral formal solution, and*

$$A(x) \equiv 1 + 2\sum_{j\geq 1} x^{j^2} \pmod 4.$$

Proof. Pair the terms with indices j and $-j$. For $j > 0$ the two rational factors are reciprocals before the exponent is applied, so the summands are equal. The equation becomes

$$A(x) = 1 + 2\sum_{j\geq 1} x^{j^2} \left(\frac{A(x) - x^j}{1 - x^j A(x)}\right)^j.$$

All denominators are units in $\mathbb{Z}[[x]]$, and the factor x^{j^2} means the degree-n coefficient on the right uses only earlier coefficients of A. This proves unique integral existence. The same equation first gives $A \equiv 1 \pmod 2$. Consequently each quotient in parentheses is $(1 - x^j)/(1 - x^j) = 1$ modulo two. Multiplication by 2 then proves the claimed reduction modulo four at every index. □

8.2.14 Uncounted supplement: integrality for A300617

Source and scope. A300617, Paul D. Hanna, March 10, 2018, defines $A = \sum_{n\ge1} a_n x^n$ by

$$[x^n]e^{nA(x)} = n^2[x^{n-1}]e^{nA(x)} \qquad (n \ge 1),$$

and explicitly conjectures integrality. We give a proof here, but exclude this A-number from the thirteen selected packages: its companion A300616 states the same integrality without a conjecture qualifier, so the source status is inconsistent. No prior proof was located in the consulted entries. This conservative exclusion does not depend on a claim that the result was previously proved. The stronger, separate claim $n(n+1)/2 \mid a_n$ is *not* settled or counted by this package.

Proposition 8.2.16. *The defining series is the unique solution in $x\mathbb{Q}[[x]]$ of*

$$A(1-xA')^3 = x + x^3A'',$$

and all its coefficients are integers.

Proof. Let $W(t)$ be the inverse of $t = xe^{-A(x)}$, so $W = te^{A(W)}$. Lagrange inversion gives, for $n \ge 1$,

$$[t^n]W = \frac{1}{n}[x^{n-1}]e^{nA(x)}, \qquad [t^n]A(W) = \frac{1}{n}[x^{n-1}]A'e^{nA} = \frac{1}{n}[x^n]e^{nA}.$$

The source equations are consequently equivalent to $A(W) = \Theta_t^2 W$, where $\Theta_t = t\,d/dt$. Since $\Theta_t = \frac{x}{1-xA'}\frac{d}{dx}$ after setting $x = W(t)$,

$$\Theta_t W = \frac{x}{1-xA'}, \qquad \Theta_t^2 W = \frac{x(1+x^2A'')}{(1-xA')^3}.$$

This proves the differential equation, and every step is reversible for formal series with $A(0) = 0$.

In the equivalent form

$$A = x + x^3A'' + 3xAA' - 3x^2A(A')^2 + x^3A(A')^3,$$

the coefficient of degree n on the right uses only $a_1, \dots, a_{n-1}$, with integer coefficients. Thus $a_1 = 1$ and the remaining coefficients are uniquely determined integers by induction. This establishes precisely the stated integrality conjecture. □

8.3 Arithmetic identities, exact counts, and one refutation

The twelve packages in this section concern twelve previously uncounted OEIS identifiers. Each selected statement is explicitly conjectural or empirical in the inspected primary entry. Known formulas used below are credited as inputs; the section does not claim that its elementary methods or every consequence of those formulas are historically new. In particular, the square case of the gcd-product formula, and the $m = 2$ case of the divisor-sum ratio, already have published proofs. Neither known special case is counted separately here.

8.3.1 The divisor-sum ratio for arbitrary positive exponent: A023887

Write $\sigma_m(n) = \sum_{d\mid n} d^m$, $\sigma = \sigma_1$, and $R = \mathrm{rad}(n) = \prod_{p\mid n} p$. The primary entry labels Yanev's August 24, 2017 identity conjectural for positive m, n. The usual prime-power formula for σ_m is already recorded there. Fried's 2025 paper, Theorem 3, proves the special case $m = 2$; that case is credited.

Theorem 8.3.1. *For all integers $m, n \ge 1$,*

$$\sigma_m(n) = \frac{\sigma(n^m R^{m-1})}{\sigma(R^{m-1})}.$$

Proof. If $n = \prod_p p^{e_p}$, where the displayed exponents are positive, then $n^m R^{m-1} = \prod_p p^{m(e_p+1)-1}$. Consequently the quotient on the right is

$$\prod_{p|n} \frac{(p^{m(e_p+1)}-1)/(p-1)}{(p^m-1)/(p-1)} = \prod_{p|n}(1+p^m+\cdots+p^{me_p}) = \sum_{d|n} d^m.$$

For $n = 1$, all products are empty and both sides are 1. This proves the unrestricted positive-integer parameter statement. □

8.3.2 The rectangular gcd product: A092287

The entry records a proved formula for the square product, and separately retains Kaydalov's March 11, 2019 conjecture for a rectangle. Ma's 2025 paper, Section 2.2, proves the square and equal-side higher-dimensional versions by threshold counting. We explicitly credit that argument and apply it to the two independent side lengths in the selected statement.

Theorem 8.3.2. *For nonnegative integers n, m and every prime p, define*

$$F(n,m) = \prod_{j=1}^{n}\prod_{k=1}^{m} \gcd(j,k).$$

Then

$$v_p(F(n,m)) = \sum_{r\geq 1} \left\lfloor \frac{n}{p^r} \right\rfloor \left\lfloor \frac{m}{p^r} \right\rfloor.$$

Proof. For positive j, k,

$$v_p(\gcd(j,k)) = \min(v_p(j), v_p(k)) = \sum_{r\geq 1} \mathbf{1}_{p^r|j}\mathbf{1}_{p^r|k}.$$

Sum this equality over the rectangle. The number of pairs divisible coordinatewise by p^r is exactly $\lfloor n/p^r \rfloor \lfloor m/p^r \rfloor$. All sums are finite: the summands vanish once $p^r > \min(n,m)$. When a side length is zero, the product is 1 and both sides are zero. □

8.3.3 Evenness of the extreme-prime-divisor sequence: A119689

The members of A119689 are integers $n \geq 2$ whose smallest and largest prime divisors have prime sum. Dale's September 22, 2016 evenness statement remains explicitly empirical.

Theorem 8.3.3. *Every member is even. More precisely, n is a member if and only if n is even and its largest prime divisor P satisfies that $P+2$ is prime.*

Proof. Let p and P be the smallest and largest prime divisors of n. Their sum is at least 4, so a prime value of $p+P$ must be odd. Exactly one of p, P is therefore the even prime 2. Since $p \leq P$, this forces $p = 2$, whence n is even and $p+P = P+2$. Conversely, for even n the smallest prime divisor is 2, so primality of $P+2$ is precisely the defining condition. This also handles powers of two: they fail because $2+2 = 4$. □

8.3.4 Equality of the integral and generating-function numerators: A143991

Let

$$c_n = [x^n]\exp(1-\sqrt{1-x}), \qquad J_n = \frac{e}{\pi}\int_{-\infty}^{\infty} \frac{\cos t}{(1+t^2)^n}\,dt \quad (n \geq 1).$$

The numerator sequence of c_n is A143991. Spezia's March 12, 2025 conjecture says that c_n and J_n have the same reduced numerator. The companion A381845 actually lists the denominators of J_n; its cross-reference also marks the numerator identification conjectural. The Bessel numbers used in the proof are the classical sequence A001515.

Theorem 8.3.4. *For every integer $n \geq 1$,*

$$J_n = 2nc_n, \qquad 2n \mid \mathrm{den}(c_n).$$

In particular $\mathrm{num}(J_n) = \mathrm{num}(c_n)$.

Proof. Put

$$B_m = \sum_{k=0}^{m} \frac{(m+k)!}{(m-k)!\,k!\,2^k} \quad (m \geq 0).$$

Every summand is an integer, since it equals $\binom{m+k}{2k}(2k-1)!!$, with $(-1)!! = 1$. These are the Bessel polynomial values $y_m(1)$; their standard recurrence is

$$B_0 = 1, \quad B_1 = 2, \qquad B_m = (2m-1)B_{m-1} + B_{m-2} \quad (m \geq 2).$$

For completeness, define the polynomial

$$y_m(z) = \sum_{k=0}^{m} \frac{(m+k)!z^k}{(m-k)!\,k!\,2^k}.$$

The recurrence follows by comparing coefficients: the coefficient of z^k in $(2m-1)zy_{m-1}(z) + y_{m-2}(z)$ equals the coefficient in $y_m(z)$ after putting the two terms over the common factorial denominator. The boundary coefficients $k = 0, m-1, m$ obey the same identity directly.

First we identify the two rational numbers. Set $w = 1 - \sqrt{1-x}$, so $w = x/(2-w)$. Lagrange inversion gives, for $n \geq 1$,

$$\begin{aligned} c_n &= \frac{1}{n}[u^{n-1}]e^u(2-u)^{-n} \\ &= \frac{1}{n}\sum_{k=0}^{n-1} \frac{\binom{n+k-1}{k}}{2^{n+k}(n-1-k)!} = \frac{B_{n-1}}{2^n n!}. \end{aligned}$$

To evaluate the integral, integrate $e^{iz}/(1+z^2)^n$ over an upper semicircle of radius $R > 1$. On the arc its modulus is at most $(R^2-1)^{-n}$, so the arc integral is $O(R^{1-2n})$ and tends to zero. The only enclosed pole is $z = i$. Writing $z = i + u$, its residue is

$$e^{-1}[u^{n-1}]e^{iu}(2i+u)^{-n} = -ie^{-1}\sum_{k=0}^{n-1} \frac{\binom{n+k-1}{k}}{2^{n+k}(n-1-k)!}.$$

The residue theorem, and oddness of the imaginary part on the real axis, yield

$$J_n = 2\sum_{k=0}^{n-1} \frac{\binom{n+k-1}{k}}{2^{n+k}(n-1-k)!} = \frac{B_{n-1}}{2^{n-1}(n-1)!} = 2nc_n.$$

Absolute convergence on the real line holds for every $n \geq 1$.

It remains to control cancellation; the identity $J_n = 2nc_n$ alone would not prove equality of numerators. We prove two elementary congruence facts about B_m.

For an odd prime p, the finite factorial sum shows

$$B_{p-1} \equiv B_p \equiv 1 \pmod{p}.$$

For B_{p-1} every summand with $1 \leq k \leq p-1$ has a factor p in its numerator and none in its denominator. For B_p, the same is true when $1 \leq k < p$. Its final summand has valuation $v_p((2p)!/p!) = 1$, so it too vanishes modulo p. Define $B_{-1} = 1$; the recurrence is then valid also at

$m = 1$. The coefficient $2m - 1$ has period p modulo p, and the state (B_{p-1}, B_p) equals (B_{-1}, B_0) modulo p. Induction in the recurrence therefore proves

$$B_{m+p} \equiv B_m \pmod{p} \qquad (m \geq -1).$$

In particular, if $p \mid n$, then $B_{n-1} \equiv 1 \pmod{p}$.

Modulo 4, the recurrence has coefficient period 2. Its first twelve values, indexed 0 through 11, are

$$1, 2, 3, 1, 2, 3, 3, 2, 1, 3, 2, 1 \pmod{4}.$$

The next two values are again $1, 2$. Thus the initial pair and the coefficient phase recur after twelve steps, which proves periodicity for all subsequent indices by induction. None of these twelve residues is zero; hence $v_2(B_m) \leq 1$ for every m.

Let $v = \mathrm{den}(c_n)$. For every odd prime $p \mid n$, the preceding congruence shows that cancellation removes no factor of p from $2^n n!$, so $v_p(v) = v_p(n!) \geq v_p(n)$. For $n \geq 2$, the modulo-4 result gives

$$v_2(v) \geq n + v_2(n!) - 1 \geq 1 + v_2(n) = v_2(2n).$$

For $n = 1$, $c_1 = 1/2$ gives the same divisibility directly. Thus $2n \mid v$. If $c_n = u/v$ in lowest terms, then $J_n = u/(v/(2n))$ is also in lowest terms, since its denominator divides v. Its numerator is therefore exactly u. □

8.3.5 The doubled-argument divisor-power formula: A179504

For $n > 0$, A179504 is $\sigma_n(2n)$. Yanev's July 8, 2017 formula is still labelled conjectural. The diagonal divisor sum $\sigma_n(n)$ is A023887.

Theorem 8.3.5. *For every integer $n \geq 1$, with $e = v_2(n)$,*

$$\sigma_n(2n) = \sigma_n(n)\left(2^n + \frac{2^n - 1}{2^{n(e+1)} - 1}\right).$$

Proof. Write $n = 2^e u$ with u odd, and put $q = 2^n$ and $S_e = 1 + q + \cdots + q^e$. Multiplicativity in the argument, with the exponent n fixed, gives $\sigma_n(n) = \sigma_n(u)S_e$ and $\sigma_n(2n) = \sigma_n(u)S_{e+1}$. Since $S_{e+1} = qS_e + 1$ and $S_e = (q^{e+1} - 1)/(q - 1)$, their ratio is $q + (q - 1)/(q^{e+1} - 1)$, as claimed. The sequence's separately specified $n = 0$ term is outside the source conjecture. □

8.3.6 The logarithmic Dirichlet series of A193679

The known cyclotomic-discriminant formula in A193679 is

$$a(1) = 1, \qquad a(n) = \prod_{p \mid n} p^{\phi(n)/(p-1)}.$$

Irwin's July 14, 2018 conjecture concerns the Dirichlet series of its logarithm and the equivalent von Mangoldt convolution. We use the displayed classical product as an input, not as a new discriminant formula.

Theorem 8.3.6. *For every $n \geq 1$,*

$$\log a(n) = \sum_{d \mid n} \Lambda(d)\phi(n/d).$$

For $\Re s > 2$, both sides below converge absolutely and

$$\sum_{n \geq 1} \frac{\log a(n)}{n^s} = -\frac{\zeta(s-1)\zeta'(s)}{\zeta(s)^2}.$$

Proof. Fix a prime $p^e \| n$ and write $n = p^e u$, $(p, u) = 1$. The coefficient of $\log p$ in the asserted convolution is

$$\sum_{j=1}^{e} \phi(n/p^j) = \phi(u) \sum_{h=0}^{e-1} \phi(p^h) = \phi(u) p^{e-1} = \frac{\phi(n)}{p-1}.$$

The middle identity follows by telescoping $\phi(p^h) = p^h - p^{h-1}$ for $h \geq 1$, together with $\phi(1) = 1$. These are exactly the coefficients of $\log p$ in the known product for $a(n)$. For $n = 1$ the convolution is zero, as is $\log a(1)$.

For $\sigma = \Re s > 2$, the series $\sum \phi(m) m^{-s}$ and $\sum \Lambda(d) d^{-s}$ are absolutely convergent, since $\phi(m) \leq m$ and $\Lambda(d) \leq \log d$. The convolution may therefore be summed and factored absolutely. The classical Euler-product identities give

$$\sum_{m \geq 1} \frac{\phi(m)}{m^s} = \frac{\zeta(s-1)}{\zeta(s)}, \qquad \sum_{d \geq 1} \frac{\Lambda(d)}{d^s} = -\frac{\zeta'(s)}{\zeta(s)}.$$

For the first identity one may use $\phi = \mu * \mathrm{id}$; the second follows by logarithmically differentiating the absolutely convergent Euler product. Their product proves the stated formula on the specified half-plane. □

8.3.7 The empirical inverse-nine generating function: A226787

Let $a(1) = 0$, let $a(n) = 0$ when $3 \mid n$, and otherwise let $a(n)$ be the least nonnegative inverse of 9 modulo n. Barker's June 20, 2013 rational generating function in A226787 remains empirical.

Theorem 8.3.7. *With $A(x) = \sum_{n \geq 1} a(n) x^n$,*

$$A(x) = \frac{x^2 + x^4 + 4x^5 + 4x^7 + x^8 + 9x^{10} + 3x^{11} + x^{13} + 3x^{14} + x^{16} - x^{19}}{(1 - x^9)^2}.$$

This is exactly the posted rational function after expanding its numerator.

Proof. For $n = 9q + r$ with $r \in \{1, 2, 4, 5, 7, 8\}$, choose t_r with $1 \leq t_r \leq 8$ and $r t_r \equiv -1 \pmod 9$. The complete table is

r	1	2	4	5	7	8
t_r	8	4	2	7	5	1
$c_r = (r t_r + 1)/9$	1	1	1	4	4	1

For $n > 1$, the integer $(t_r n + 1)/9 = t_r q + c_r$ lies strictly between 0 and n, and multiplying it by 9 gives 1 modulo n. It is therefore $a(n)$. The formula at $q = 0, r = 1$ instead gives 1, whereas the prescribed value $a(1)$ is zero; this requires subtracting x. All three residue classes divisible by 3 contribute zero. Hence

$$A(x) = -x + \sum_{r \in \{1,2,4,5,7,8\}} x^r \left(\frac{c_r}{1 - x^9} + \frac{t_r x^9}{(1 - x^9)^2} \right).$$

Putting this expression over the common denominator and using the six displayed table entries gives the stated numerator identically. The calculation is an identity of formal power series, not a fit to a finite prefix. □

8.3.8 The arithmetic-derivative gcd conjecture: A326691

In A326691, define

$$a(n) = \frac{n}{\operatorname{den}\left(\sum_{p \mid n} \frac{1}{p} - \frac{1}{n} \right)},$$

where the denominator is positive and reduced. Karttunen's March 15, 2021 conjecture identifies this with a gcd involving the arithmetic derivative $D(n) = \sum_{p^e \| n} e n/p$, with $D(1) = 0$.

Theorem 8.3.8. *For every $n \geq 1$,*

$$a(n) = \gcd(n, 1 + (n-1)D(n)) = \gcd(n, D(n) - 1).$$

Proof. Set $S(n) = \sum_{p|n} n/p$. Reduction of $(S(n) - 1)/n$ gives $a(n) = \gcd(n, S(n) - 1)$, including $n = 1$. Fix $p^e \| n$. If $e \geq 2$, every summand of both $S(n)$ and $D(n)$ is divisible by p. Neither $S(n) - 1$ nor $D(n) - 1$ is then divisible by p, so their gcds with n both have p-valuation zero. If $e = 1$, the summand corresponding to p has coefficient 1 in both sums, while all summands belonging to other primes are divisible by p. Therefore $S(n) - 1 \equiv D(n) - 1 \pmod{p}$. Since the valuation in either gcd is at most $e = 1$, the two valuations again agree. Comparing every prime divisor of n proves the second expression for $a(n)$. Finally $1 + (n-1)D(n) \equiv 1 - D(n) \pmod{n}$, which gives the first expression. □

8.3.9 A lower bound survives, but the proposed value-three classification fails: A345993

Let $m(n)$ be the least positive integer m with $n \mid m(m+1)$, and put $a(n) = \gcd(n, m(n) + 1)$, as in A345993. The empirical comment proposes both $a(n) \geq 3$ for $n \geq 3$ and a classification of $a(n) = 3$ using only $n = 3 \cdot 2^{\text{odd}}$ and $n = 3p$ with primes $p \equiv -1 \pmod{6}$.

Theorem 8.3.9. *For all $n \geq 3$, $a(n) \geq 3$. However, the proposed classification is false: $m(375) = 125$ and $a(375) = 3$. More generally, if $p \equiv 2 \pmod{3}$ is prime and e is any positive odd integer, then*

$$m(3p^e) = p^e, \qquad a(3p^e) = 3.$$

Thus $3 \cdot 5^{2r+1}$, $r \geq 1$, is an infinite family omitted by the proposed classification.

Proof. For $n \geq 2$, the value $m = n - 1$ is admissible, so $m(n) < n$. Write $m = m(n)$ and $g = \gcd(n, m + 1)$. Since consecutive integers are coprime, every prime power dividing n divides exactly one of $m, m + 1$; in particular n/g divides m. If $g = 1$, this would force $n \mid m$, impossible for $0 < m < n$. If $g = 2$, then m is a positive multiple of $n/2$ smaller than n, so $m = n/2$. Because $m+1$ is even, m is odd. For $n \geq 3$ this odd m is at least 3, and $m - 1$ is a smaller positive admissible integer: $2m = n$ divides $(m-1)m$ because $m - 1$ is even. This contradicts minimality. Hence $g \geq 3$.

For the family, put $q = p^e$. Then $q \equiv 2 \pmod{3}$. A root of $u(u+1) \equiv 0 \pmod{q}$ is either $u \equiv 0$ or $u \equiv -1 \pmod{q}$, since q is a prime power. Combining either possibility with $u \equiv 0$ or $-1 \pmod{3}$ gives exactly four roots modulo $3q$, namely

$$0, \qquad q, \qquad 2q - 1, \qquad 3q - 1.$$

They are distinct, and the least positive one is q. Consequently $m(3q) = q$ and $a(3q) = \gcd(3q, q+1) = 3$. Taking $p = 5, e = 3$ proves the explicit counterexample. The integer 125 is neither a power of 2 nor a prime, so 375 belongs to neither proposed family. This resolves the two parts as a true lower bound and a false classification; it does not assert a complete replacement classification of all value-three inputs. □

8.3.10 The radical average identity: A367865

The definition of A367865 is $a(n) = \sum_{d|n} d\phi(d)\mu(d)^2$. Yanev's September 14, 2025 conjecture expresses it as a radical average.

Theorem 8.3.10. *For every integer $n \geq 1$,*

$$a(n) = \frac{\operatorname{rad}(n)}{n} \sum_{k=1}^{n} \frac{\operatorname{rad}(nk)}{\operatorname{rad}(k)}.$$

Proof. Let $R = \mathrm{rad}(n)$. Prime by prime,

$$\frac{\mathrm{rad}(nk)}{\mathrm{rad}(k)} = \frac{R}{\gcd(R,k)}.$$

This is periodic in k with period R, and $R \mid n$. The right side of the proposed identity therefore equals

$$\sum_{k=1}^{R} \frac{R}{\gcd(R,k)}.$$

For a divisor $d \mid R$, exactly $\phi(d)$ of these k satisfy $\gcd(R,k) = R/d$. Grouping them gives $\sum_{d\mid R} d\phi(d)$. The divisors of R are exactly the squarefree divisors of n, so this is $\sum_{d\mid n} d\phi(d)\mu(d)^2 = a(n)$. The same argument includes $n = 1$. □

8.3.11 The fourfold gcd-indicator sum: A373561

The polynomial sequence A373561 is $a(n) = n^3(n+1)(2n+1)/6$ for $n \geq 0$. The entry conjectures the following weighted fourfold expression, where $f(x,y,z) = x^2 + y^2 - z^2$.

Theorem 8.3.11. *For every integer $n \geq 0$,*

$$\sum_{k=1}^{n}\sum_{z=1}^{n}\sum_{y=1}^{n}\sum_{x=1}^{n} \mathbf{1}_{\gcd(f(x,y,z),n)=k}\, f(x,y,z) = \frac{n^3(n+1)(2n+1)}{6}.$$

Proof. For $n \geq 1$ and every integer f, including zero and negative values, $\gcd(f,n)$ is one positive integer in $\{1,\dots,n\}$. Thus the sum of its n equality indicators is exactly 1. Interchanging the finite sums removes the outer indicator sum and leaves

$$\sum_{x,y,z=1}^{n} (x^2+y^2-z^2) = n^2\left(\sum_{j=1}^{n} j^2 + \sum_{j=1}^{n} j^2 - \sum_{j=1}^{n} j^2\right) = \frac{n^3(n+1)(2n+1)}{6}.$$

For $n = 0$, the sums are empty and the polynomial is zero. □

8.3.12 The empirical seven-ninths totient family: A383044

The selected April 25, 2025 empirical statement in A383044 concerns the ratio 7/9, not the entry's main ratio-one classification. It proposes the family $m = 18\,\mathtt{A003586}(n)$, where A003586 consists of 2^u3^v with $u, v \geq 0$. Steinerberger's linked 2025 paper studies the ratio-one equation and its six known families; that analysis is credited and is outside this selected assertion.

Theorem 8.3.12. *For all nonnegative integers u, v, if $m = 18 \cdot 2^u3^v$, then*

$$\phi(m) + \phi(m + \phi(m)) = \frac{7}{9}m.$$

In particular this ratio has infinitely many positive integer solutions.

Proof. Here $m = 2^{u+1}3^{v+2}$, so the ordinary prime-power formula gives $\phi(m) = m(1-1/2)(1-1/3) = m/3$. Consequently $m + \phi(m) = 4m/3 = 2^{u+3}3^{v+1}$ still has exactly the prime divisors $2, 3$. Its totient is therefore $(4m/3)/3 = 4m/9$. Adding gives $7m/9$. Taking $v = 0$ and letting u increase provides infinitely many distinct solutions. □

8.4 Generating functions and exact recurrence certificates

This section proves 26 distinct source-marked polynomial recurrences. Each selected target is explicitly labelled a conjecture in its consulted OEIS entry. The generating functions, finite sums and combinatorial definitions already in those entries are credited inputs. Auxiliary source formulas also need correction: the missing alternating sign in the title of A308815, an incorrect generating function printed for A113409, and older substitution and finite-sum expressions for A174169. These corrections do not add any counted items.

The selection excludes the 708 previously selected IDs, the inspected Fischer 2020 and Ekhad–Yang 2017 proof families, and the limited 97-ID Appendix A baseline from OEIS Open. Same-entry established recurrences and linked proof papers were checked for stale labels. In particular, A101601 and A268149 are uncounted because of previously established hypergeometric evaluations; A278992 is uncounted because its linked paper proves a recurrence system for the same counts. The audit is bounded, and does not assert worldwide priority, official OEIS closure, or novelty of any credited input. A220898 and the previously treated A220899 are related by a finite difference, not a shift or scalar multiple; their selected recurrence assertions are distinct.

Exact coefficient identities

Set $a_n = 0$ below the actual source offset s, and let $F(x) = \sum_{n\ge s} a_n x^n$. In every ordinary recurrence below the convention is

$$R_n = \sum_{j=0}^{r} p_j(n) a_{n-j}, \qquad L = \sum_{j=0}^{r} x^j p_j(\theta + j), \qquad \theta = x\frac{d}{dx}. \tag{8.1}$$

Termwise differentiation gives $[x^n]LF = R_n$. Thus an exact identity $LF = P(x)$ proves the recurrence for $n > \deg P$. We also require $n \ge s + r$ so all displayed sequence indices belong to their source domain. No division by $p_0(n)$ is required, including at its integer zeros.

All the algebraic certificates below are finite rational identities. For $x = \xi(t)$, put $M = \xi/\xi'$. Reduce the displayed seed modulo $y^2 = D(t)$ to $F = R_0(t) + R_1(t)y$, where $y(0) = 1$. Set $D_0 = 1$, $D_1 = D$, and define

$$H_{i,0} = R_i, \quad H_{i,m+1} = M\left(H'_{i,m} + \frac{D'_i}{2D_i} H_{i,m}\right), \quad c_m = \sum_{j=0}^{r} \xi^j [z^m] p_j(z + j). \tag{8.2}$$

The certificate is exactly $\sum_m c_m H_{0,m} = P(\xi)$ and $\sum_m c_m H_{1,m} = 0$. Each entry supplies every polynomial p_j, the seed, its radicand and parameter, and P. Formula (8.2) therefore specifies a finite, independently reproducible proof using rational differentiation and polynomial cancellation. Its two residual numerators are zero identically, not merely on a finite set of coefficients. The accompanying verifier repeats these exact operations and checks the source transcription. For an ordinary unparametrized series use $t = x$, $\xi = x$ and $M = x$.

If $t = x\phi(t)$, $\phi(0) = 1$, formal inversion gives a unique $t(0) = 0$. For any formal series B we use the diagonal form of Lagrange inversion,

$$\sum_{n\ge 0} x^n [u^n] B(u)\phi(u)^n = \frac{B(t)}{1 - t\phi'(t)/\phi(t)}. \tag{8.3}$$

Indeed, substituting $x = u/\phi(u)$ in the coefficient residue on the right cancels its denominator by the Jacobian and leaves $\operatorname{Res}_u B(u)\phi(u)^n u^{-n-1} du$. The formula includes $n = 0$. We also use $[x^n]H(t) = n^{-1}[u^{n-1}]H'(u)\phi(u)^n$ for $n \ge 1$, which follows from the same substitution and formal integration by parts.

For an exponential or even exponential series $F(z) = \sum a_n z^{kn}/(kn)!$, where $k = 1$ or 2, set

$$q_j(n) = p_j(n) \prod_{h=kj}^{kr-1} (kn - h), \qquad \Theta = \frac{z}{k}\frac{d}{dz}, \qquad \mathcal{L} = \sum_{j=0}^{r} z^{kj} q_j(\Theta + j). \tag{8.4}$$

Then $[z^{kn}]\mathcal{L}F = R_n/(kn - kr)!$ for $n \geq r$. This follows by cancellation of the factorials term by term. A differential basis b_i with $b_j' = \sum_i A_{ji} b_i$ is handled by

$$(R_i)_i \longmapsto \left(\frac{z}{k} \left(R_i' + \sum_j R_j A_{ji} \right) \right)_i .$$

Repeated use of this rational matrix rule, with the polynomials q_j, is again a finite exact certificate.

Walk components for the Bessel certificates

Define the formal series

$$I_0(2z) = \sum_{m \geq 0} \frac{z^{2m}}{(m!)^2}, \qquad I_1(2z) = \sum_{m \geq 0} \frac{z^{2m+1}}{m!(m+1)!}.$$

Differentiating these sums gives $I_0' = 2I_1$ and $I_1' = 2I_0 - I_1/z$. In one dimension, excursions constrained to stay nonnegative have EGF $I_1(2z)/z$, unrestricted bridges have EGF $I_0(2z)$, and nonnegative meanders have EGF $I_0(2z) + I_1(2z)$. The coefficient formulas are respectively C_m at length $2m$, $\binom{2m}{m}$ at length $2m$, and $\binom{j}{\lfloor j/2 \rfloor}$ at length j. The last formula follows from reflection of paths first reaching -1; the first is the same reflection argument with final height zero. Interleaving independent components corresponds to multiplication of EGFs, since choosing their positions gives the multinomial coefficient. One or two free directions contribute e^z or e^{2z}. These are the components used in Dershowitz's *Touchard's Drunkard* (2017), Sections 4–5 and Table 3.

For a fixed degree d use the basis $b_j = e^{cz} I_0(2z)^{d-j} I_1(2z)^j$, $0 \leq j \leq d$. Its derivative is

$$b_j' = (c - j/z) b_j + 2(d - j) b_{j+1} + 2j b_{j-1}, \tag{8.5}$$

with out-of-range basis terms omitted. This explicitly supplies every matrix entry used by the four walk certificates below.

8.4.1 A014533: the selected recurrence

Source and assertion. A014533, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Feb 25 2015.

The source offset is 1. With $R_n = \sum_{j=0}^{2} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= -(n-1)(n+7) \\ p_1(n) &= (n+3)(2n+5) \\ p_2(n) &= 3(n+2)(n+3) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 3$.

Definition and generating-function derivation. The fourth trinomial column is $a_n = [u^{n-1}](1 + u + u^2)^{n+3}$ for $n \geq 1$. Equivalently it counts unrestricted Motzkin walks of length $n+3$ ending at height four. The bridge GF is $1/y$, $y = \sqrt{1 - 2x - 3x^2}$, and the GF for endpoint $h \geq 0$ is $(xM)^h/y$, obtained by solving the walk kernel recurrence with the branch $xM = O(x)$. At $h = 4$, shifting length down by three gives $F = xM^4/y$, with no artificial constant term. This agrees with the credited source GF and fixes the offset explicitly.

Exact certificate.

$$F(x) = \frac{(-x-y+1)^4}{16x^7y}.$$

Here $y^2 = -3x^2 - 2x + 1$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 0.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 3$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.2 A024719: the selected recurrence

Source and assertion. A024719, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Sep 29 2012.

The source offset is 0. With $R_n = \sum_{j=0}^{2} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= 2n\,(2n-1) \\ p_1(n) &= -31n^2 + 29n - 6 \\ p_2(n) &= 3\,(3n-2)\,(3n-1) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 2$.

Definition and generating-function derivation. The source defines

$$a_n = \frac{2 + \sum_{k=0}^{n} \binom{3k}{k}}{3}.$$

The identity $\binom{3n}{n} = [u^n](1-u)^{-2n-1}$ lets us apply the diagonal formula with $\phi = (1-t)^{-2}$, giving

$$\sum_{n\geq 0} \binom{3n}{n} x^n = \frac{1}{1-3t}, \qquad x = t(1-t)^2.$$

Partial summation therefore gives

$$F = \frac{2 + 1/(1-3t)}{3(1-x)} = \frac{1-2t}{(1-3t)(1-x)},$$

which is van Hoeij's November 9, 2011 parameterization. Greubel's later first-difference formula is the same defining partial sum in inhomogeneous form; no novelty is claimed for it.

Exact certificate.

$$F(\xi(t)) = \frac{1-2t}{(3t-1)\,(t^3-2t^2+t-1)}.$$

$$\xi(t) = t\,(1-t)^2\,, \qquad \phi(t) = \frac{1}{(1-t)^2}, \qquad y^2 = 1.$$

The formal parameter satisfies $t = x\phi(t)$; every square root uses its branch with constant term one. Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = -4x.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 2$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.3 A026126: the selected recurrence

Source and assertion. A026126, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jun 10 2013.

The source offset is 4. With $R_n = \sum_{j=0}^{4} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned}
p_0(n) &= -\,(n-4)\,(n+8)\\
p_1(n) &= 4n^2+5n-73\\
p_2(n) &= -2n^2+13n+38\\
p_3(n) &= -\,(n-3)\,(4n+5)\\
p_4(n) &= 3\,(n-4)\,(n-3)
\end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 8$.

Definition and generating-function derivation. Let $M = 1 + xM + x^2M^2$ be the Motzkin GF. A nonnegative Motzkin walk from 0 to height h has GF $x^h M^{h+1}$, by cutting at the last upward crossing of each level. Walks from 1 to 5 whose minimum is at least 1 contribute x^4M^5. The other walks first reach zero, with first-passage GF xM, then continue with GF x^5M^6. Thus their total GF is $x^4M^5(1+x^2M^2) = x^4(1-x)M^6$. Forbidding a horizontal first step multiplies by $1-x$, proving $F = x^4(1-x)^2M^6$ with the actual offset four.

Exact certificate.

$$F(x) = \frac{(1-x)^2\,(-x-y+1)^6}{64x^8}.$$

Here $y^2 = -3x^2-2x+1$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 0.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 8$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.4 A098519: the selected recurrence

Source and assertion. A098519, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Nov 23 2011.

The source offset is 0. With $R_n = \sum_{j=0}^{2} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= (n-1)\,(n+1) \\ p_1(n) &= -n\,(2n-1) \\ p_2(n) &= -11n\,(n-1) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 2$.

Definition and generating-function derivation. The defining EGF is $e^z I_1(2\sqrt{3}z)/\sqrt{3}$. Expanding its two factors gives

$$a_n = \sum_{k\geq 0} \binom{n}{k} \binom{n-k}{k+1} 3^k.$$

The untransformed sequence has OGF $2x/(1-12x^2+\sqrt{1-12x^2})$. The binomial-transform rule $F(x) = B(x/(1-x))/(1-x)$, proved by summing $\binom{n}{j}$, gives the source radical expression below. No property of special functions beyond their displayed power series is needed for this ordinary generating-function certificate.

Exact certificate.

$$F(x) = \frac{2x}{-11x^2 - 2x + y\,(1-x) + 1}.$$

Here $y^2 = -11x^2 - 2x + 1$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 0.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 2$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.5 A103138: the selected recurrence

Source and assertion. A103138, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Dec 14 2011.

The source offset is 0. With $R_n = \sum_{j=0}^{3} p_j(n) a_{n-j}$, the exact source coefficients are:

$$p_0(n) = 2n$$
$$p_1(n) = 13n - 20$$
$$p_2(n) = 8n - 27$$
$$p_3(n) = n - 5$$

We prove $R_n = 0$ for every integer $n \geq 3$.

Definition and generating-function derivation. The source defining GF is $F = (1 - xS(-x))xS(-x)$, where $S = 1 + xS + xS^2$ is the large-Schroeder GF. Put $t = xS(-x)$. Its quadratic equation gives $x = t(1 + t)/(1 - t)$, and $F = t - t^2$. This also proves Hanna's August 6, 2013 composition identity used in the certificate. The parameter has $\xi'(0) = 1$, selecting the formal inverse uniquely.

Exact certificate.

$$F(\xi(t)) = -t^2 + t.$$

$$\xi(t) = \frac{t\,(t+1)}{1-t}, \qquad \phi(t) = \frac{1-t}{t+1}, \qquad y^2 = 1.$$

The formal parameter satisfies $t = x\phi(t)$; every square root uses its branch with constant term one. Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = -6x^2 + 2x.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 3$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.6 A112702: the selected recurrence

Source and assertion. A112702, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jun 08 2016.

The source offset is 0. With $R_n = \sum_{j=0}^{2} p_j(n)a_{n-j}$, the exact source coefficients are:

$$p_0(n) = n + 1$$
$$p_1(n) = -3\,(11n - 5)$$
$$p_2(n) = 16\,(2n - 1)$$

We prove $R_n = 0$ for every integer $n \geq 2$.

Definition and generating-function derivation. The definition is $a_n = \sum_{k=0}^{n} 8^k C_k$. Summing first over n gives $F = C(8x)/(1 - x)$, where $C(x) = (1 - \sqrt{1 - 4x})/(2x)$. Thus the generating function used below follows directly from the definition.

Exact certificate.

$$F(x) = \frac{1-y}{16x\,(1-x)}.$$

Here $y^2 = 1 - 32x$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 1.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 2$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.7 A113409: the selected recurrence

Source and assertion. A113409, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Nov 07 2012.

The source offset is 0. With $R_n = \sum_{j=0}^{4} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= n + 2 \\ p_1(n) &= -2\,(n+1) \\ p_2(n) &= n - 4 \\ p_3(n) &= 2 \\ p_4(n) &= -4\,(n-2) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 4$.

Definition and generating-function derivation. The defining finite sum is

$$a_n = \sum_k \binom{n-k}{k} \binom{k}{\lfloor k/2 \rfloor}.$$

Put $B(z) = \sum_k \binom{k}{\lfloor k/2 \rfloor} z^k$. Separating even and odd k gives

$$B(z) = \frac{1 + 2z - \sqrt{1-4z^2}}{2z\sqrt{1-4z^2}}.$$

Since $\sum_{n \geq 2k} \binom{n-k}{k} x^n = x^{2k}/(1-x)^{k+1}$, we obtain $F = B(x^2/(1-x))/(1-x)$, which simplifies to the seed below. The source's older expression $(1 - xC(x^2))/(1 - x^2 - x^4 C(x^4))$ has coefficient -1 at x, instead of the correct $a_1 = 1$. It is explicitly rejected as a premise. The recurrence itself is proved without changing any coefficient.

Exact certificate.

$$F(x) = \frac{2x^2 - x - y + 1}{2x^2 y}.$$

Here $y^2 = -4x^4 + x^2 - 2x + 1$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 2 - x.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 4$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.8 A114851: the selected recurrence

Source and assertion. A114851, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Mar 04 2015.

The source offset is 0. With $R_n = \sum_{j=0}^{6} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned}
p_0(n) &= n + 2 \\
p_1(n) &= -2\,(n + 1) \\
p_2(n) &= 2 - n \\
p_3(n) &= 4\,(n - 2) \\
p_4(n) &= 18 - 5n \\
p_5(n) &= 2\,(n - 4) \\
p_6(n) &= n - 6
\end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 6$.

Definition and generating-function derivation. A binary lambda term is a de Bruijn index of length at least two, an abstraction of cost two followed by a term, or an application of cost two followed by two ordered terms. Thus

$$F = \frac{x^2}{1 - x} + x^2 F + x^2 F^2, \qquad a_0 = a_1 = 0.$$

This is the nonlinear recurrence proved by Grygiel–Lescanne, arXiv:1401.0379, Section 5. Solving the quadratic with zero constant term gives the seed below. Its radicand is $(1-x)^2(1-x^2)^2 - 4x^4(1-x)$. The selected finite-order linear recurrence is certified separately.

Exact certificate.

$$F(x) = \frac{-y + (1 - x)\,(1 - x^2)}{2x^2\,(1 - x)}.$$

Here $y^2 = -4x^4\,(1 - x) + (1 - x)^2\,(1 - x^2)^2$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = -x^5 - 3x^3 + 4x^2.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 6$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.9 A162972: the selected recurrence

Source and assertion. A162972, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jul 26 2022.

The source offset is 1. With $R_n = \sum_{j=0}^{7} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned}
p_0(n) &= 1 \\
p_1(n) &= 8 - 3n \\
p_2(n) &= (n-3)(3n-14) \\
p_3(n) &= -n^3 + 19n^2 - 98n + 156 \\
p_4(n) &= -4n^3 + 56n^2 - 258n + 395 \\
p_5(n) &= -6n^3 + 84n^2 - 395n + 624 \\
p_6(n) &= -(n-5)\left(4n^2 - 41n + 106\right) \\
p_7(n) &= -(n-6)^2(n-5)
\end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 8$.

Definition and generating-function derivation. Mark each cycle by u. The EGF for permutations is $(1-z)^{-u}$, and that for derangements is $(e^{-z}/(1-z))^u$, by the exponential formula for disjoint cycles. Differentiating their difference at $u = 1$ counts cycles in the non-derangement permutations and gives

$$F(z) = \frac{ze^{-z} + (e^{-z} - 1)\log(1-z)}{1-z}.$$

For $\ell = \log(1-z)$ and $e = e^{-z}$, the basis is $(1, \ell, e, e\ell)$, whose derivatives are $(0, -1/(1-z), -e, -e/(1-z) - e\ell)$. The seed coefficients are $(0, -1/(1-z), z/(1-z), 1/(1-z))$. Apply (8.4) with $k = 1$ and $r = 7$ using these derivatives. All four rational residuals vanish identically. This proves the recurrence from $n = 8$, where every source index is at least one.

Exact certificate. The seed and basis derivatives just given, together with the displayed p_j and (8.4), give an identically zero rational vector. The exact matrix and its zero residual are also recorded in the machine-readable certificate. Thus every required coefficient vanishes, proving the all-index assertion.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.10 A168505: the selected recurrence

Source and assertion. A168505, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Feb 10 2015.

The source offset is 0. With $R_n = \sum_{j=0}^{4} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= n+1 \\ p_1(n) &= -3n \\ p_2(n) &= 3\,(n-1) \\ p_3(n) &= 3\,(n-4) \\ p_4(n) &= -2\,(2n-7) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 4$.

Definition and generating-function derivation. The defining continued fraction repeats the three signs $-, +, -$. Writing one full period gives

$$F = \cfrac{1}{1 - \cfrac{x}{1 + \cfrac{x}{1 - xF}}}.$$

Clearing denominators yields $x(1-x)F^2 - (1+x)F + (1+x) = 0$. The branch $F(0) = 1$ is exactly the radical below. This also derives Somos's January 20, 2017 identity $(F-1)/F^2 = x(1-x)/(1+x)$. That known nonlinear identity is an input; it is not the selected linear polynomial recurrence.

Exact certificate.

$$F(x) = \frac{x - y + 1}{2x\,(1-x)}.$$

Here $y^2 = 4x^3 + x^2 - 2x + 1$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = -x^3 - 3x^2 - x + 1.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 4$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.11 A174169: the selected recurrence

Source and assertion. A174169, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Sep 30 2012.

The source offset is 0. With $R_n = \sum_{j=0}^{4} p_j(n) a_{n-j}$, the exact source coefficients are:

$$
\begin{aligned}
p_0(n) &= n + 2 \\
p_1(n) &= -2n - 1 \\
p_2(n) &= 3\,(n - 1) \\
p_3(n) &= -3\,(2n - 5) \\
p_4(n) &= 9\,(n - 4)
\end{aligned}
$$

We prove $R_n = 0$ for every integer $n \geq 4$.

Definition and generating-function derivation. Manyama recorded the correct quadratic identity on October 17, 2025:

$$(1 - x + 3x^2)F = 1 + x^2F^2, \qquad a_0 = a_1 = 1.$$

It also follows from the Chebyshev substitution

$$F = \frac{1}{1 + 3x^2} M\left(\frac{x}{1 + 3x^2}\right).$$

The branch with constant term one gives the displayed radical. For clarity, the older printed substitution denominator $1 + 3x$ is incorrect, and its older finite sum omits a factor: the corrected sum is

$$a_n = \sum_k (-3)^k \binom{n - k}{k} M_{n-2k}.$$

Neither erroneous expression is a premise, and these auxiliary fixes are not additional counted assertions.

Exact certificate.

$$F(x) = \frac{3x^2 - x - y + 1}{2x^2}.$$

Here $y^2 = 9x^4 - 6x^3 + 3x^2 - 2x + 1$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 2 - 6x^2.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 4$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.12 A182892: the selected recurrence

Source and assertion. A182892, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jun 14 2016.

The source offset is 0. With $R_n = \sum_{j=0}^{8} p_j(n) a_{n-j}$, the exact source coefficients are:

$$
\begin{aligned}
p_0(n) &= n \\
p_1(n) &= n-2 \\
p_2(n) &= -2\,(9n-16) \\
p_3(n) &= 5\,(2n-5) \\
p_4(n) &= 10n-33 \\
p_5(n) &= 2\,(26n-109) \\
p_6(n) &= 13n-37 \\
p_7(n) &= 13n-63 \\
p_8(n) &= -10\,(n-7)
\end{aligned}
$$

We prove $R_n = 0$ for every integer $n \geq 8$.

Definition and generating-function derivation. The source permits heights of either sign. An unrestricted weighted bridge with horizontal weights x, x^2, upward weight x^2 and downward weight x has GF

$$B(x) = \sum_{m\geq 0} \binom{2m}{m} \frac{x^{3m}}{(1-x-x^2)^{2m+1}} = \frac{1}{\sqrt{(1-x-x^2)^2-4x^3}}.$$

Decompose a bridge into its successive returns to height zero. Removing the weight-two horizontal component at zero increases the inverse GF by x^2, so the required GF is $F = 1/(B^{-1}+x^2)$. The radicand factors as $(1+x+x^2)(1-3x+x^2)$. This proves the GF recorded by Deutsch, independently of the background composition/fence-poset papers linked in the entry.

Exact certificate.

$$F(x) = \frac{1}{x^2+y}.$$

Here $y^2 = (x^2-3x+1)\,(x^2+x+1)$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 5x^6+6x^5+16x^4-7x^3-2x^2.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 8$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.13 A220898: the selected recurrence

Source and assertion. A220898, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, May 30 2014.

The source offset is 0. With $R_n = \sum_{j=0}^{3} p_j(n)\,a_{n-j}$, the exact source coefficients are:

$$
\begin{aligned}
p_0(n) &= 2\,(n+1)\,(2n+1) \\
p_1(n) &= -15n^2+19n-8 \\
p_2(n) &= -2\,(n-2)\,(21n-22) \\
p_3(n) &= -23\,(n-3)\,(n-2)
\end{aligned}
$$

We prove $R_n = 0$ for every integer $n \geq 3$.

Definition and generating-function derivation. Kitaev, Salimov, Severs and Ulfarsson (2012), Theorem 1, prove that the primitive-map GF is $F(x) = G(x/(1+x))$. The established map series has $[z^n]G = 2(3n)!/((2n+1)!(n+1)!)$, including constant term two. Its parameterization is $z = t(1-t)^2$, $G = (2-3t)/(1-t)^2$. For completeness Lagrange inversion gives, for $n \geq 1$,

$$[z^n]G = \frac{1}{n}\left(\binom{3n+1}{n-1} - 3\binom{3n}{n-2}\right) = \frac{2(3n)!}{(2n+1)!(n+1)!}.$$

Thus with $u = t(1-t)^2$, setting $x = u/(1-u)$ gives the seed below. The known map GF and composition are credited premises; the selected 2014 recurrence is the assertion certified here.

Exact certificate.

$$F(\xi(t)) = \frac{2-3t}{(1-t)^2}.$$

$$\xi(t) = \frac{t\,(1-t)^2}{-t\,(1-t)^2+1}, \qquad \phi(t) = \frac{-t\,(1-t)^2+1}{(1-t)^2}, \qquad y^2 = 1.$$

The formal parameter satisfies $t = x\phi(t)$; every square root uses its branch with constant term one. Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 4x + 4.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 3$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.14 A225692: the selected recurrence

Source and assertion. A225692, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jul 09 2013.

The source offset is 2. With $R_n = \sum_{j=0}^{3} p_j(n)a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= -10\,(n-4)\,(n+1) \\ p_1(n) &= 51n^2 - 196n + 80 \\ p_2(n) &= -45n^2 + 134n - 99 \\ p_3(n) &= 2\,(n-3)\,(2n-5) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 5$.

Definition and generating-function derivation. In the source's Dyck-pattern order, containment means a scattered subword. A Dyck path of semilength n contains $U^{n-1}D^{n-1}$ exactly when its first n steps have at most one down step. Indeed a cut separating the selected up and down steps must have at least $n-1$ ups and at most one down, and can be taken at the midpoint; conversely that midpoint condition supplies the selected subword. There is one path with no down step in its first half. For exactly one down step, the nonnegative first half has $n-1$ choices, and by reversal the second half has $n-1$ choices. Thus

$$a_n = C_n - 1 - (n-1)^2 \qquad (n \geq 2).$$

This also follows by subtracting the first two terms from the ballot-square sum in Bernini–Ferrari–Pinzani–West (2013), Section 3.3. Writing $G = \sum_{n\geq 0}(n^2-2n+2)x^n$, the actual offset-two series is $F = C - G + 1$; the extra 1 removes the spurious constant coefficient. The exact operator identity below includes that boundary adjustment.

Exact certificate.

$$F(x) = \frac{2x}{(1-x)^2} - \frac{x\,(x+1)}{(1-x)^3} + 1 - \frac{2}{1-x} + \frac{1-y}{2x}.$$

Here $y^2 = 1-4x$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 0.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 5$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.15 A240722: the selected recurrence

Source and assertion. A240722, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jun 14 2016.

The source offset is 0. With $R_n = \sum_{j=0}^{2} p_j(n)a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= -\,(n+1)\,(2n+1)\,(3n-1) \\ p_1(n) &= 2\left(21n^3 + 14n^2 - 2n - 1\right) \\ p_2(n) &= 8n\,(2n-1)\,(3n+2) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 2$.

Definition and generating-function derivation. The source definition is $F = B'/B$, where

$$B(x) = \frac{-1 - \sqrt{1-8x} + \sqrt{2 + 2\sqrt{1-8x} + 8x}}{4x}.$$

Set $x = t(1-2t)$ and $y^2 = 1-4t^2$, $y(0) = 1$. Then $\sqrt{1-8x} = 1-4t$ and

$$B = \frac{-1+2t+y}{2t(1-2t)}, \qquad F = \frac{1}{1-4t}\frac{dB/dt}{B}, \qquad y' = -4t/y.$$

These exact substitutions specify the seed used in the rational certificate. The removable singularities select $B(0) = 1$. The proof uses the original offset zero, not the shifted convention of the source's later congruence comments.

Exact certificate. Use the B, F, x, y expressions immediately above, differentiating with $y\prime = -4t/y$.

$$\xi(t) = t\,(1-2t)\,, \qquad \phi(t) = \frac{1}{1-2t}, \qquad y^2 = 1-4t^2.$$

The formal parameter satisfies $t = x\phi(t)$; every square root uses its branch with constant term one. Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 1-20x.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 2$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.16 A246606: the selected recurrence

Source and assertion. A246606, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jul 01 2022.

The source offset is 1. With $R_n = \sum_{j=0}^{2} p_j(n)a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= 3-2n \\ p_1(n) &= 4\,(n-1)\left(2n^2-4n+1\right) \\ p_2(n) &= (n-2)\,(n-1)\,(2n-1) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 3$.

Definition and generating-function derivation. The terminating hypergeometric expression recorded by Luschny on November 4, 2018 expands to

$$a_n = \sum_{j=0}^{n-1}(-1)^j\binom{n-1}{j}(2n-1-j)! = \int_0^\infty e^{-t}t^n(t-1)^{n-1}\,dt.$$

The integral equality follows by expanding $(t-1)^{n-1}$ and integrating monomials against e^{-t}. For $n \geq 3$, put $g = e^{-t}t^{n-2}(t-1)^{n-3}$ and $R = p_0t^2(t-1)^2 + p_1t(t-1) + p_2$. Direct polynomial multiplication gives

$$R = t(t-1)Q' + (-t^2 + (2n-2)t - (n-1))Q,$$

where

$$Q = (2n-3)t^2 + (4n^2 - 10n + 6)t - 2n^2 + 5n - 2.$$

Thus the recurrence residual is the integral of $\frac{d}{dt}\{e^{-t}t^{n-1}(t-1)^{n-2}Q\}$. The bracket vanishes at zero because $n-1 \geq 2$, and at infinity because of e^{-t}. There is no singularity at $t = 1$, since the exponents are nonnegative integers. The integral is therefore zero for every $n \geq 3$.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.17 A268402: the selected recurrence

Source and assertion. A268402, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Oct 07 2016.

The source offset is 5. With $R_n = \sum_{j=0}^{4} p_j(n)a_{n-j}$, the exact source coefficients are:

$$\begin{aligned}
p_0(n) &= -(n-5)(n-3)(55n-618)\\
p_1(n) &= -55n^3 - 840n^2 + 6969n - 5742\\
p_2(n) &= 3135n^3 - 37532n^2 + 138815n - 163614\\
p_3(n) &= -7645n^3 + 93072n^2 - 343985n + 391386\\
p_4(n) &= -18(n-3)(2n-5)(55n-126)
\end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 9$.

Definition and generating-function derivation. Cut a north-east path at its returns to the diagonal. Each nonempty piece is a prime excursion, on either side, with GF $E = xC(x)$. A right bounce occurs precisely between two consecutive pieces on the right. If u marks right bounces and U, V count nonempty words ending on the left or right, then $U = E(1+U+V)$ and $V = E(1+U+uV)$. Consequently the total GF is

$$H(u,x) = \frac{1 + (1-u)E}{1 - E - E^2 - uE(1-E)}.$$

For $r \geq 1$ its coefficient of u^r is $E^{r+1}(1-E)^{r-1}/(1-E-E^2)^{r+1}$. Taking $r = 4$ gives the required GF and, after substituting $E = (1-\sqrt{1-4x})/2$, simplifies to the source expression below. This also gives an independent derivation of the specialization of Pan–Remmel's proved bivariate GF (2016, Section 3.2).

Exact certificate.

$$F(x) = -\frac{x^3(y-1)^5(2x+y-1)}{2(x(y-5)-y+1)^5}.$$

Here $y^2 = 1 - 4x$, with $y(0) = 1$.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 0.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 9$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.18 A278618: the selected recurrence

Source and assertion. A278618, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Dec 02 2016.

The source offset is 0. With $R_n = \sum_{j=0}^{3} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= 2n\,(n+1)\,(2n-1)\,(5n-8) \\ p_1(n) &= -n\left(115n^3 - 344n^2 + 299n - 82\right) \\ p_2(n) &= -4\,(2n-1)\left(5n^3 + 27n^2 - 74n + 30\right) \\ p_3(n) &= 36\,(n-2)\,(2n-3)\,(2n-1)\,(5n-3) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 3$.

Definition and generating-function derivation. Set $B(u) = 1+u^2/(1-u) = (1-u+u^2)/(1-u)$. Expanding its $(2n+1)$-st power gives

$$[u^n]B(u)^{2n+1} = \sum_j \binom{2n+1}{j}\binom{n-j-1}{n-2j} = a_n.$$

The $j = 0$ term is interpreted as 1 for $n = 0$ and 0 otherwise. Formula (8.3), with $\phi = B^2$, gives the rational parameterization below, without using the longer nested radical printed in the source.

Exact certificate.

$$F(\xi(t)) = \frac{t^2 - t + 1}{(1-t)\left(-\dfrac{t(1-t)^2\left(\frac{(4t-2)(t^2-t+1)}{(1-t)^2} + \frac{2(t^2-t+1)^2}{(1-t)^3}\right)}{(t^2-t+1)^2} + 1\right)}.$$

$$\xi(t) = \frac{t\,(1-t)^2}{(t^2-t+1)^2}, \qquad \phi(t) = \frac{(t^2-t+1)^2}{(1-t)^2}, \qquad y^2 = 1.$$

The formal parameter satisfies $t = x\phi(t)$; every square root uses its branch with constant term one. Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 12x.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 3$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.19 A302180: the selected recurrence

Source and assertion. A302180, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Oct 29 2025.

The source offset is 0. With $R_n = \sum_{j=0}^{3} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= (n+2)\,(n+4) \\ p_1(n) &= -3\left(n^2+3n+1\right) \\ p_2(n) &= -13\,(n-1)\,(n+1) \\ p_3(n) &= 15\,(n-2)\,(n-1) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 3$.

Definition and generating-function derivation. These are Dershowitz's walks of type *aad*: two independent nonnegative excursions and one free direction. Choosing the positions of their steps proves $F(z) = e^z (I_1(2z)/z)^2$. Use the degree-two basis (8.5) with $c = 1$, seed $(0, 0, z^{-2})$, and $k = 1$ in (8.4). Exact application of $\mathcal{L}$ gives the zero vector $(0, 0, 0)$. All apparent Laurent poles cancel because $I_1(2z) = z + O(z^3)$.

Exact certificate. The seed and basis derivatives just given, together with the displayed p_j and (8.4), give an identically zero rational vector. The exact matrix and its zero residual are also recorded in the machine-readable certificate. Thus every required coefficient vanishes, proving the all-index assertion.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.20 A302181: the selected recurrence

Source and assertion. A302181, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Oct 29 2025.

The source offset is 0. With $R_n = \sum_{j=0}^{2} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= (n+1)^3 \\ p_1(n) &= -2\,(2n-1)\left(10n^2+10n+3\right) \\ p_2(n) &= 36n\,(2n-3)\,(2n-1) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 2$.

Definition and generating-function derivation. These are the even-length walks of type *abb*: a nonnegative excursion and two unrestricted bridges. Hence

$$\sum_{n\geq 0} \frac{a_n z^{2n}}{(2n)!} = I_0(2z)^2 I_1(2z)/z.$$

The even-index convolution recorded by Tebni on November 6, 2024 is another form of this decomposition. Use $d = 3$, $c = 0$, seed $(0, z^{-1}, 0, 0)$, and crucially $k = 2$ in (8.4). The basis derivative remains with respect to z, while the Euler operator is $(z/2)d/dz$. The resulting four rational residuals are identically zero, proving the asserted recurrence in the index n.

Exact certificate. The seed and basis derivatives just given, together with the displayed p_j and (8.4), give an identically zero rational vector. The exact matrix and its zero residual are also recorded in the machine-readable certificate. Thus every required coefficient vanishes, proving the all-index assertion.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.21 A302184: the selected recurrence

Source and assertion. A302184, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Oct 29 2025.

The source offset is 0. With $R_n = \sum_{j=0}^{3} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned} p_0(n) &= (n+2)^2 \\ p_1(n) &= -2\left(3n^2+5n+1\right) \\ p_2(n) &= -4\,(n-1)\,(n+3) \\ p_3(n) &= 24\,(n-2)\,(n-1) \end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 3$.

Definition and generating-function derivation. These are walks of type *abe*: one nonnegative excursion, one unrestricted bridge and two free directions. The multinomial interleaving gives $F(z) = e^{2z} I_0(2z) I_1(2z)/z$. Equivalently this is Tebni's November 30, 2024 source convolution of the aerated Catalan numbers with $\binom{2m}{m}$; the latter has EGF $e^{2z} I_0(2z)$. Use $d = 2$, $c = 2$, seed $(0, z^{-1}, 0)$ and $k = 1$. Formula (8.5) gives $\mathcal{L}F = (0, 0, 0)$ exactly.

Exact certificate. The seed and basis derivatives just given, together with the displayed p_j and (8.4), give an identically zero rational vector. The exact matrix and its zero residual are also recorded in the machine-readable certificate. Thus every required coefficient vanishes, proving the all-index assertion.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.22 A302186: the selected recurrence

Source and assertion. A302186, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Oct 29 2025.

The source offset is 0. With $R_n = \sum_{j=0}^{4} p_j(n) a_{n-j}$, the exact source coefficients are:

$$
\begin{aligned}
p_0(n) &= (n+2)\,(n+3)\\
p_1(n) &= -4\left(3n^2+8n+3\right)\\
p_2(n) &= 8\left(4n^2+3n-3\right)\\
p_3(n) &= 48\,(n-2)^2\\
p_4(n) &= -144\,(n-3)\,(n-2)
\end{aligned}
$$

We prove $R_n = 0$ for every integer $n \geq 4$.

Definition and generating-function derivation. These are walks of type *ace*: a nonnegative excursion, a nonnegative meander and two free directions. Interleaving gives

$$F(z) = e^{2z} I_1(2z)(I_0(2z) + I_1(2z))/z.$$

This also derives the source double-binomial sum from the component counts. Use $d = 2$, $c = 2$, seed $(0, z^{-1}, z^{-1})$ and $k = 1$. The rational matrix calculation in (8.5) gives the zero vector after applying $\mathcal{L}$.

Exact certificate. The seed and basis derivatives just given, together with the displayed p_j and (8.4), give an identically zero rational vector. The exact matrix and its zero residual are also recorded in the machine-readable certificate. Thus every required coefficient vanishes, proving the all-index assertion.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.23 A308815: the selected recurrence

Source and assertion. A308815, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jan 27 2020.

The source offset is 0. With $R_n = \sum_{j=0}^{3} p_j(n) a_{n-j}$, the exact source coefficients are:

$$
\begin{aligned}
p_0(n) &= n\,(2n+1)\\
p_1(n) &= -32\,(n-1)\,(9n-4)\\
p_2(n) &= 1024\left(3n^2-23n+29\right)\\
p_3(n) &= 65536\,(n-2)\,(2n-5)
\end{aligned}
$$

We prove $R_n = 0$ for every integer $n \geq 3$.

Definition and generating-function derivation. Farkas, Moschetti, Naranjo and Pirola, Theorems 1.1–1.2 in arXiv:1906.10406, prove the alternating sum and generating series. The intended sequence is

$$a_n = 16^n \sum_{i=0}^{n} (-2)^i \binom{n}{i} C_{2n-i}.$$

The OEIS title prints 2^i instead of $(-2)^i$. At $n = 1$ the printed title gives 64, while the data, PARI program and cited paper give 0. We explicitly correct that title; the selected recurrence is unchanged. Squaring and rationalizing the paper's displayed nested-radical GF gives

$$F(x) = \sqrt{\frac{2}{1 + 64x + \sqrt{1 - 128x}}}.$$

With $x = t(1 - 32t)$ the inner square root is $1 - 64t$ and hence $F = 1/\sqrt{1 - 1024t^2}$. Constant terms select the asserted branches.

Exact certificate.

$$F(\xi(t)) = \frac{1}{y}.$$

$$\xi(t) = t\,(1 - 32t)\,, \qquad \phi(t) = \frac{1}{1 - 32t}, \qquad y^2 = 1 - 1024t^2.$$

The formal parameter satisfies $t = x\phi(t)$; every square root uses its branch with constant term one. Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 0.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 3$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.24 A344502: the selected recurrence

Source and assertion. A344502, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Nov 02 2021.

The source offset is 0. With $R_n = \sum_{j=0}^{4} p_j(n) a_{n-j}$, the exact source coefficients are:

$$\begin{aligned}
p_0(n) &= -4\,(n+1)\,(2n+1)\,(3035n - 11997)\\
p_1(n) &= 2\left(153546n^3 - 490325n^2 - 62942n + 71982\right)\\
p_2(n) &= -2\left(452090n^3 - 1745622n^2 + 1405285n + 226200\right)\\
p_3(n) &= -2\,(n-2)\left(83741n^2 - 200458n - 19650\right)\\
p_4(n) &= -3\,(n-3)\,(n-2)\,(46423n + 6938)
\end{aligned}$$

We prove $R_n = 0$ for every integer $n \geq 4$.

Definition and generating-function derivation. Let $T(n, k)$ count nonnegative Motzkin prefixes of length n ending at height k. Reflection at the first visit to -1 gives

$$T(n, k) = [u^{n-k}](1 - u^2)(1 + u + u^2)^n.$$

Expanding by the number j of down steps gives

$$T(n,k)=\sum_{j\ge 0}\frac{n!(k+1)}{j!(j+k+1)!(n-k-2j)!}.$$

The terminating hypergeometric factor in the source, multiplied by $\binom{n}{k}$, equals this sum, since $4^j((k-n)/2)_j((k-n+1)/2)_j=(k-n)_{2j}$. Therefore the source definition is $\sum_k\binom{n}{k}T(n,k)$, and the binomial theorem proves

$$a_n=[u^n](1-u^2)\big((1+u)(1+u+u^2)\big)^n.$$

Apply (8.3) with $B=1-t^2$ and $\phi=(1+t)(1+t+t^2)$ to obtain the certificate seed.

Exact certificate.

$$F(\xi(t))=\frac{1-t^2}{-\frac{t(t^2+t+(t+1)(2t+1)+1)}{(t+1)(t^2+t+1)}+1}.$$

$$\xi(t)=\frac{t}{(t+1)\left(t^2+t+1\right)},\qquad \phi(t)=(t+1)\left(t^2+t+1\right),\qquad y^2=1.$$

The formal parameter satisfies $t=x\phi(t)$; every square root uses its branch with constant term one. Substitution in (8.2), followed by exact cancellation, gives

$$LF=P(x)=47988-225302x.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n\ge 4$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.25 A379463: the selected recurrence

Source and assertion. A379463, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jan 29 2025.

The source offset is 0. With $R_n=\sum_{j=0}^{5}p_j(n)a_{n-j}$, the exact source coefficients are:

$$\begin{aligned}
p_0(n)&=3n\,(n-3)\,(3n+4)\,(3n+8)\\
p_1(n)&=-3\left(45n^4-54n^3-192n^2+27n+20\right)\\
p_2(n)&=9\,(n-1)\left(30n^3-72n^2-7n+20\right)\\
p_3(n)&=-3\,(n-2)\,(n-1)\left(90n^2-234n+95\right)\\
p_4(n)&=-\,(n-3)\,(n-2)\,(n-1)\,(121n+499)\\
p_5(n)&=229\,(n-4)\,(n-3)\,(n-2)\,(n-1)
\end{aligned}$$

We prove $R_n=0$ for every integer $n\ge 5$.

Definition and generating-function derivation. First derive the finite sum from the path definition. For $r = 2$ or 3, remove the horizontal steps from a path with steps $1, 0, -3$, ending at zero and staying at least $-(r-1)$. Its remaining $4k$ steps consist of $3k$ upward and k downward steps. Reverse their order, negate them, and start at height $r-1$. This gives a nonnegative path with steps $-1, 3$, ending at height $r-1$. Append r downward steps. The resulting Lukasiewicz word encodes an ordered forest of r rooted four-ary trees with k internal vertices: each internal vertex contributes a step 3 and each leaf a step -1. Conversely, every nontrivial such forest ends with at least four leaf steps, because its last internal vertex has four leaf children. Deleting the final r steps and reversing the transformation recovers the original path. The forest of r isolated leaves gives the empty path. The horizontal steps can be inserted in $\binom{n}{4k}$ ways.

For this entry $r = 2$, so the defining path count has the finite sum

$$a_n = \sum_{k=0}^{\lfloor n/4 \rfloor} \binom{n}{4k} \frac{2}{4k+2} \binom{4k+2}{k}.$$

This is the same as Scheuerle's January 7, 2025 program, since $\binom{4k+2}{k}/(4k+2) = \binom{4k+1}{k}/(3k+2)$. Let $C = 1 + zC^4$. Lagrange inversion gives $[z^k]C^r = r\binom{4k+r}{k}/(4k+r)$. The binomial insertion identity $\sum_{n \geq m} \binom{n}{m} x^n = x^m/(1-x)^{m+1}$ therefore yields $F = C((x/(1-x))^4)^2/(1-x)$. Put $t = xC/(1-x)$. Then $C = 1 + t^4$, $x = t/(1+t+t^4)$ and $F = (1+t^4)(1+t+t^4)$, as used in the certificate.

Exact certificate.

$$F(\xi(t)) = \left(t^4 + 1\right)\left(t^4 + t + 1\right).$$

$$\xi(t) = \frac{t}{t^4 + t + 1}, \qquad \phi(t) = t^4 + t + 1, \qquad y^2 = 1.$$

The formal parameter satisfies $t = x\phi(t)$; every square root uses its branch with constant term one.

Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = 0.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 5$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.4.26 A379464: the selected recurrence

Source and assertion. A379464, consulted on 20 October 2026, explicitly marks the recurrence as a conjecture attributed to R. J. Mathar, Jan 29 2025.

The source offset is 0. With $R_n = \sum_{j=0}^{7} p_j(n) a_{n-j}$, the exact source coefficients are:

$$
\begin{aligned}
p_0(n) &= 3\,(n+4)\,(3n+4)\,(3n+8)\\
p_1(n) &= -3\left(63n^3+297n^2+349n+60\right)\\
p_2(n) &= 3\left(189n^3+270n^2-229n-140\right)\\
p_3(n) &= -15\left(63n^3-117n^2-44n+64\right)\\
p_4(n) &= 689n^3-5372n^2+6946n-1288\\
p_5(n) &= (n-4)\left(201n^2+2767n-3011\right)\\
p_6(n) &= -\,(n-5)\,(n-4)\,(579n+257)\\
p_7(n) &= 229\,(n-6)\,(n-5)\,(n-4)
\end{aligned}
$$

We prove $R_n = 0$ for every integer $n \geq 7$.

Definition and generating-function derivation. Apply the path-to-forest bijection just proved for A379463 with $r = 3$: the defining lower boundary is -2, so deleting horizontal steps, reversing and negating, shifting the starting height to two, and appending three downward steps gives a forest of three four-ary trees. Lagrange inversion and horizontal-step insertion give

$$a_n = \sum_{k=0}^{\lfloor n/4 \rfloor} \binom{n}{4k} \frac{3}{4k+3} \binom{4k+3}{k}.$$

For $C = 1 + zC^4$, Lagrange inversion gives the second factor as $[z^k]C^3$. Summing the binomial insertion factor gives $F = C((x/(1-x))^4)^3/(1-x)$. With $t = xC/(1-x)$ one obtains $x = t/(1+t+t^4)$ and $F = (1+t^4)^2(1+t+t^4)$. This differs from A379463 in its defining lower boundary and forest power; it is not a shift or scalar multiple.

Exact certificate.

$$F(\xi(t)) = \left(t^4+1\right)^2\left(t^4+t+1\right).$$

$$\xi(t) = \frac{t}{t^4+t+1}, \qquad \phi(t) = t^4+t+1, \qquad y^2 = 1.$$

The formal parameter satisfies $t = x\phi(t)$; every square root uses its branch with constant term one. Substitution in (8.2), followed by exact cancellation, gives

$$LF = P(x) = -384x^3+1152x^2-1152x+384.$$

The two rational residuals are $P(\xi)$ and 0. Their coefficient interpretation proves the recurrence for all $n \geq 7$; this is at least the source offset plus the recurrence order.

Independent consistency check. The companion validation computes the defining finite sum, combinatorial dynamic program, or nonlinear grammar through index 47, without using the conjectured recurrence. All listed source coefficients match. These checks audit the transcription and branch; the exact identities above supply the proof.

8.5 Forty-nine empirical array recurrence packages

This section proves forty-nine entry-level recurrence assertions that are explicitly marked *Empirical* on the primary OEIS pages consulted on 20 October 2026. Seventeen source-stated empirical or conjectural rational generating functions are proved in the same packages. The arrays and empirical recurrences are credited to R. H. Hardin; the supplementary generating functions are credited on their pages to Colin Barker. One selected A-number contributes one package, regardless of the number of formulas proved on that page. All forty-nine A-numbers are absent from the 708 entries selected in Rounds I–VII. These are freshly sourced and certified candidates; none of the earlier rounds' uncounted reserves is promoted here.

8.5.1 The local definitions and the counting convention

An array has $n+1$ rows, width $w \geq 2$, and entries in $\{0, \ldots, q-1\}$. For a block $B = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, write its four signed clockwise edge differences as

$$\Delta(B) = (b-a, d-b, c-d, a-c).$$

The entry-specific parameters select one of three predicates:

Stress (D). $|a+d-b-c| = s$.
Absolute edge sum (U). $\sum_{e \in \Delta(B)} |e| = s$.
Squared edge sum (V). $\sum_{e \in \Delta(B)} e^2 = s$.

Both edge sums include all four edges with multiplicity. In entries that additionally prohibit equal adjacent values, every component of $\Delta(B)$ must be nonzero. Adjacency here is horizontal or vertical, not diagonal. Every edge of an array with at least two rows and two columns belongs to a 2×2 block, so this local test enforces the source condition even on boundary edges.

All selected definitions count labeled arrays literally. Colors may be absent; there is no requirement to use every alphabet symbol, no first-occurrence renaming, and no division by an orbit size. The row orbits below compress a counting computation, while retaining the full labeled count through their multiplicities.

8.5.2 Transfers obtained directly from the array definition

Let $\gamma(r)$ be the lexicographically least of a row r, its horizontal reversal, its simultaneous complement $q-1-r$, and the reversal of that complement. Let $\mathcal{O}_{q,w}$ be the sorted set of these representatives. Define the mass vector and transition matrix by

$$m(r) = \#\{u \in \{0, \ldots, q-1\}^w : \gamma(u) = r\}, \tag{8.6}$$

$$T(r,t) = \#\left\{ u \in \{0, \ldots, q-1\}^w : \begin{array}{l} \gamma(u) = t, \\ P(r_i, r_{i+1}, u_i, u_{i+1}) \quad (1 \leq i < w) \end{array} \right\}. \tag{8.7}$$

Reversal and complementation preserve each of D, U and V, as well as the prohibition on zero edge differences. Applying either operation simultaneously to a pair of rows preserves compatibility and permutes each target orbit. Thus every row in a given orbit has the same number of successors in every target orbit. Arbitrary permutations of the numeric alphabet are not used.

Adding rows therefore gives the exact count

$$a(n) = m^{\mathsf{T}} T^n \mathbf{1} \qquad (n \geq 1). \tag{8.8}$$

The exponent is n because there are $n+1$ rows. In particular, $\sum_r m(r) = q^w$; fixed rows, shorter orbits, and rows with no successors retain their proper multiplicities. The auxiliary transfer value at $n = 0$ is not asserted to be a term of the source sequence, whose offset is 1.

8.5.3 A finite exact certificate proves every index

For the source coefficient signature $c = (c_1, \ldots, c_d)$, put

$$R(z) = z^d - \sum_{i=1}^{d} c_i z^{d-i}, \qquad Q(x) = 1 - \sum_{i=1}^{d} c_i x^i.$$

The complete certificate for each selected definition establishes the integer vector identity

$$\boxed{TR(T)\mathbf{1} = \mathbf{0}.} \tag{8.9}$$

For $n \geq d+1$, multiply this identity by $m^{\mathsf{T}} T^{n-d-1}$ and use (8.8). This proves

$$\boxed{a(n) = \sum_{i=1}^{d} c_i a(n-i) \qquad (n \geq d+1).} \tag{8.10}$$

Every index on the right is at least 1. The proof does not assume an empirical generating function or infer a recurrence from a numerical prefix.

The forty-nine complete matrices contain 3,067 states in total and 5,879 nonzero weighted edges. The largest matrix has 625 states; the largest recurrence order is 48. Every matrix, every orbit mass, and every coordinate of $v_k = T^k\mathbf{1}$ for $0 \leq k \leq d+1$ is retained in the certificate data. The exact calculation is

```
v[0] = [1 for every state]
for k = 1,...,d+1:
    v[k][r] = sum(T[r,s]*v[k-1][s] for every state s)
for every state r:
    assert v[d+1][r] == sum(c[i]*v[d+1-i][r] for i=1,...,d)
```

The verifier rebuilds all states and transitions from the predicates, then recomputes and compares every stored vector coordinate using integer arithmetic. This is a finite computer-assisted proof of an all-index assertion. The parameters and full coefficient lists below specify the calculation independently of the data files; the expanded matrices and vectors are supplied in `r8_arrays_matrices.json` and `r8_arrays_certificates.json`.

All 954 displayed source-prefix values agree with the defining transfers. An additional program, which does not import the primary engine, counts two-row and three-row arrays using unquotiented column-by-column dynamic programming; all 98 values agree. These checks validate the interpretation and offset. They are supplementary to the exact vector identities that establish every later index.

For $F(x) = \sum_{n\geq 1} a(n)x^n$, the proved recurrence implies $QF = P$, where $\deg P \leq d$ and

$$[x^k]P = \sum_{\substack{0 \leq i \leq \min(k,d) \\ k-i \geq 1}} q_i a(k-i), \qquad q_0 = 1, \quad q_i = -c_i \ (i \geq 1). \tag{8.11}$$

The initial values come from the defining transfer. Polynomial cross-multiplication identifies this rational function exactly with all seventeen selected source generating functions. Their numerators are displayed below. Other formulas on a source page, such as radical closed forms, are not additional selected assertions.

8.5.4 A hand-checkable example: A234259

For $q = 3$, $w = 2$, and stress $s = 2$, the row representatives are $00, 01, 02, 11$. Direct enumeration from (8.7) gives

$$T = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \\ 2 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \end{pmatrix}, \qquad m = \begin{pmatrix} 2 \\ 4 \\ 2 \\ 1 \end{pmatrix}.$$

The vectors $v_k = T^k\mathbf{1}$, for $k = 1, 2, 3, 4$, are

$$(2, 2, 3, 2)^{\mathsf{T}}, \quad (6, 4, 6, 6)^{\mathsf{T}}, \quad (12, 8, 18, 12)^{\mathsf{T}}, \quad (36, 16, 36, 36)^{\mathsf{T}}.$$

They satisfy $v_4 = 2v_3 + 6v_2 - 12v_1$. Consequently

$$a(n) = 2a(n-1) + 6a(n-2) - 12a(n-3) \qquad (n \geq 4).$$

The defining counts are $a(1) = 20$, $a(2) = 46$, $a(3) = 104$, so

$$F(x) = \frac{20x + 6x^2 - 108x^3}{1 - 2x - 6x^2 + 12x^3} = \frac{2x(10 + 3x - 54x^2)}{(1 - 2x)(1 - 6x^2)}.$$

This proves the empirical recurrence and conjectural rational generating function on https://oeis.org/A234259.

8.5.5 Source status, prior work, and the selection unit

Each selected primary page and each of the 25 associated parent-table pages was retrieved and inspected for this round. All selected recurrences are explicitly empirical; no proof link or established encompassing recurrence was identified on those consulted pages. Retrieval dates and the search tool's reported cache ages are recorded separately in the source audit. A retrieval on the audit date does not guarantee that an indexed page reflects every live edit.

Exact-ID and family searches did not identify a prior proof of a selected assertion among the inspected results. Several searches returned irrelevant matches, so this is a bounded audit, not a claim of worldwide priority. The inspected README of the additional https://github.com/Eppie/oeis-proofs lead lists no selected array entry; its entire repository was not audited.

The IDs are also disjoint from the retained 97 distinct A-numbers in the 100 costly claims of Appendix A of *OEIS Open*. This comparison does not cover every result in that project's repository. The method here adapts the geometric-orbit transfers used in Rounds IV–VII, with a new definition-derived matrix certificate for every selected entry.

A separate exact comparison with 236 previously certified array sequences from Rounds II–VII and between these forty-nine sequences tests twenty values for each shift from -3 through 3, allowing an arbitrary rational scale. Every tested proportional equality has a mismatch within the first three compared values. This rules out those specific shift-and-scale aliases; unbounded shifts, affine changes, and other transformations are not covered. A shared recurrence polynomial alone does not make two different counting sequences the same selected claim.

One further fresh certificate, A235252, is retained as an uncounted reserve. It is not a failed proof and does not contribute to the forty-nine selected packages. No OEIS editing, submission, official closure, or worldwide-first claim is made.

8.5.6 Individual definitions and complete coefficient signatures

Every list below gives $c_1, \ldots, c_d$ in order, including zeros, for (8.10). Initial values start at $n = 1$.

8.5.6.1 A233646

Arrays have $n + 1$ rows, width $w = 2$, and labeled alphabet $0, \ldots, 5$. Predicate V has target $s = 14$, allowing equal adjacent values. The recurrence holds for every $n \geq 10$, with order $d = 9$ and a complete $D = 12$ state matrix. The first three source values are $(144, 772, 3984)$.

```
2, 31, -36, -261, 206, 714, -356, -528, 256
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233646; parent column 1 of https://oeis.org/A233653. Both consulted pages mark the relevant recurrence empirical.

8.5.6.2 A233647

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,5$. Predicate V has target $s=14$, allowing equal adjacent values. The recurrence holds for every $n \geq 39$, with order $d=38$ and a complete $D=63$ state matrix. The first three source values are $(772, 5232, 36784)$.

```
13, 60, -1403, 915, 60037, -161542, -1305952, 5581957, 14886303, -99457303, -65691409,
1070532941, -439804539, -7350827624, 8431876793, 32422650909, -59093935245,
-87614113660, 242031773814, 116489014001, -628983746855, 50730140495, 1046367995789,
-482503791934, -1075303683128, 846072582698, 610370785910, -745472433880,
-117896421864, 344109162632, -46130175152, -71419513632, 22452177920, 3530443776,
-1777822976, 30745088, 22653952, -933888
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233647; parent column 2 of https://oeis.org/A233653. Both consulted pages mark the relevant recurrence empirical.

8.5.6.3 A233675

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,3$. Predicate V has target $s=6$, allowing equal adjacent values. The recurrence holds for every $n \geq 5$, with order $d=4$ and a complete $D=6$ state matrix. The first three source values are $(48, 184, 648)$.

$$(c_1,\dots,c_d) = (1, 12, -4, -16).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233675; parent column 1 of https://oeis.org/A233682. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 48x + 136x^2 - 112x^3 - 224x^4.$$

8.5.6.4 A233676

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,3$. Predicate V has target $s=6$, allowing equal adjacent values. The recurrence holds for every $n \geq 11$, with order $d=10$ and a complete $D=20$ state matrix. The first three source values are $(184, 928, 4448)$.

```
7, 11, -157, 202, 554, -1224, -112, 1584, -928, 64
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233676; parent column 2 of https://oeis.org/A233682. Both consulted pages mark the relevant recurrence empirical.

8.5.6.5 A233677

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,3$. Predicate V has target $s=6$, allowing equal adjacent values. The recurrence holds for every $n \geq 18$, with order $d=17$ and a complete $D=72$ state matrix. The first three source values are $(648, 4448, 28544)$.

```
5, 70, -332, -1384, 6780, 11520, -61568, -45216, 280096, 84864, -653696, -81664,
762880, 41984, -397312, -8192, 65536
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233677; parent column 3 of https://oeis.org/A233682. Both consulted pages mark the relevant recurrence empirical.

8.5.6.6 A233687

Arrays have $n+1$ rows, width $w=5$, and labeled alphabet $0,\dots,3$. Predicate V has target $s=10$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 42$, with order $d=41$ and a complete $D=272$ state matrix. The first three source values are $(1188, 13324, 145512)$.

```
4, 222, -895, -14806, 61290, 479263, -2067766, -8897249, 40799029, 102512949,
-513857547, -760171900, 4350919810, 3628256006, -25577829290, -10454924632,
106480049675, 12586615989, -317308345201, 27167936641, 679221981450, -157201990004,
-1040817140533, 351215509546, 1129423930543, -469396649218, -850891317254,
403441486513, 431368323585, -223830914820, -140453027219, 77851050483, 27476109266,
-16042803856, -2950519700, 1797075893, 152255054, -93497323, -3429989, 1518688, 74280
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233687; parent column 4 of https://oeis.org/A233691. Both consulted pages mark the relevant recurrence empirical.

8.5.6.7 A233718

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,7$. Predicate V has target $s=26$, allowing equal adjacent values. The recurrence holds for every $n \geq 17$, with order $d=16$ and a complete $D=20$ state matrix. The first three source values are $(280, 1820, 11016)$.

```
2, 58, -58, -1227, 461, 12296, 722, -62278, -22804, 155159, 90798, -162636, -121112,
35968, 28416, -1152
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233718; parent column 1 of https://oeis.org/A233724. Both consulted pages mark the relevant recurrence empirical.

8.5.6.8 A233726

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,3$. Predicate U has target $s=6$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 5$, with order $d=4$ and a complete $D=6$ state matrix. The first three source values are $(40, 148, 556)$.

$$(c_1,\dots,c_d) = (3, 5, -7, -2).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233726; parent column 1 of https://oeis.org/A233733. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 40x + 28x^2 - 88x^3 - 24x^4.$$

8.5.6.9 A233727

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,3$. Predicate U has target $s=6$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 13$, with order $d=12$ and a complete $D=20$ state matrix. The first three source values are $(148, 756, 3972)$.

```
3, 30, -71, -192, 415, 410, -847, -269, 544, 38, -42, -4
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233727; parent column 2 of https://oeis.org/A233733. Both consulted pages mark the relevant recurrence empirical.

8.5.6.10 A233728

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,3$. Predicate U has target $s=6$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 29$, with order $d=28$ and a complete $D=72$ state matrix. The first three source values are $(556, 3972, 29816)$.

```
9, 46, -508, -481, 10680, -3945, -112548, 115263, 657688, -958275, -2192200, 4058993,
4067658, -9776097, -3651898, 13759514, 407041, -11221093, 1862871, 5154086, -1437504,
-1269561, 444011, 147392, -60418, -4972, 2810, -123
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233728; parent column 3 of https://oeis.org/A233733. Both consulted pages mark the relevant recurrence empirical.

8.5.6.11 A233942

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,2$. Predicate U has target $s=4$, allowing equal adjacent values. The recurrence holds for every $n \geq 5$, with order $d=4$ and a complete $D=4$ state matrix. The first three source values are $(44, 226, 1150)$.

$$(c_1,\dots,c_d) = (5, 3, -13, 2).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233942; parent column 1 of https://oeis.org/A233949. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 44x + 6x^2 - 112x^3 + 18x^4.$$

8.5.6.12 A233943

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,2$. Predicate U has target $s=4$, allowing equal adjacent values. The recurrence holds for every $n \geq 11$, with order $d=10$ and a complete $D=10$ state matrix. The first three source values are $(226, 2108, 19484)$.

```
10, 9, -154, 24, 569, -305, -372, 163, 31, -6
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233943; parent column 2 of https://oeis.org/A233949. Both consulted pages mark the relevant recurrence empirical.

8.5.6.13 A233944

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,2$. Predicate U has target $s=4$, allowing equal adjacent values. The recurrence holds for every $n \geq 26$, with order $d=25$ and a complete $D=25$ state matrix. The first three source values are $(1150, 19484, 328508)$.

```
22, 20, -2402, 8315, 57619, -297894, -394397, 3586512, -630788, -18290282, 13131067,
44278092, -40653252, -53889205, 50657365, 32681066, -27199613, -9122229, 6034975,
846023, -477739, -19115, 11259, -26, -40
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A233944; parent column 3 of https://oeis.org/A233949. Both consulted pages mark the relevant recurrence empirical.

8.5.6.14 A234241

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 9$, with order $d=8$ and a complete $D=20$ state matrix. The first three source values are $(630, 6706, 72616)$.

```
7, 181, -1022, -7657, 21504, 54144, -96768, -80640
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234241; parent column 1 of https://oeis.org/A234245. Both consulted pages mark the relevant recurrence empirical.

8.5.6.15 A234259

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,2$. Predicate D has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 4$, with order $d=3$ and a complete $D=4$ state matrix. The first three source values are $(20, 46, 104)$.

$$(c_1,\dots,c_d) = (2, 6, -12).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234259; parent column 1 of https://oeis.org/A234266. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 20x + 6x^2 - 108x^3.$$

8.5.6.16 A234260

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,2$. Predicate D has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=10$ state matrix. The first three source values are $(46, 88, 170)$.

$$(c_1,\dots,c_d) = (3, 6, -24, 4, 36, -24).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234260; parent column 2 of https://oeis.org/A234266. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 46x - 50x^2 - 370x^3 + 424x^4 + 588x^5 - 648x^6.$$

8.5.6.17 A234261

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,2$. Predicate D has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=25$ state matrix. The first three source values are $(104, 170, 292)$.

$$(c_1,\dots,c_d) = (3, 6, -24, 4, 36, -24).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234261; parent column 3 of https://oeis.org/A234266. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 104x - 142x^2 - 842x^3 + 1160x^4 + 1332x^5 - 1752x^6.$$

8.5.6.18 A234326

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,3$. Predicate D has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 5$, with order $d=4$ and a complete $D=6$ state matrix. The first three source values are $(80, 432, 2368)$.

$$(c_1,\dots,c_d) = (0, 32, 0, -48).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234326; parent column 1 of https://oeis.org/A234333. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 80x + 432x^2 - 192x^3 - 768x^4.$$

8.5.6.19 A234327

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,3$. Predicate D has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 12$, with order $d=11$ and a complete $D=20$ state matrix. The first three source values are $(432, 3296, 25728)$.

```
10, 59, -768, 608, 7580, -11628, -18088, 32880, 3232, -17600, 4224
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234327; parent column 2 of https://oeis.org/A234333. Both consulted pages mark the relevant recurrence empirical.

8.5.6.20 A234328

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,3$. Predicate D has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 49$, with order $d=48$ and a complete $D=72$ state matrix. The first three source values are $(2368, 25728, 287888)$.

```
0, 507, 0, -102295, 0, 10646830, 0, -635915377, 0, 23832852746, 0, -596280829182, 0,
10371633339873, 0, -128744093807520, 0, 1159671655063068, 0, -7660667577596808, 0,
37368059037902308, 0, -135219186772494408, 0, 364034788822163472, 0,
-730004249849579520, 0, 1089413122161376064, 0, -1205481757762071168, 0,
981742220067000064, 0, -581000394777277440, 0, 244895745327540224, 0,
-71316845470162944, 0, 13706205520330752, 0, -1618885433491456, 0, 103796813856768, 0,
-2721872609280
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234328; parent column 3 of https://oeis.org/A234333. Both consulted pages mark the relevant recurrence empirical.

8.5.6.21 A234338

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,7$. Predicate U has target $s=14$, allowing equal adjacent values. The recurrence holds for every $n \geq 14$, with order $d=13$ and a complete $D=20$ state matrix. The first three source values are $(604, 9076, 120838)$.

```
11, 100, -657, -3362, 13855, 47579, -118877, -254743, 334913, 167982, -130132, -15912,
10008
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234338; parent column 1 of https://oeis.org/A234343. Both consulted pages mark the relevant recurrence empirical.

8.5.6.22 A234444

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,4$. Predicate D has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 5$, with order $d=4$ and a complete $D=9$ state matrix. The first three source values are $(160, 1120, 8064)$.

$$(c_1,\dots,c_d) = (0, 60, 0, -344).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234444; parent column 1 of https://oeis.org/A234450. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 160x + 1120x^2 - 1536x^3 - 8480x^4.$$

8.5.6.23 A234445

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,4$. Predicate D has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 18$, with order $d=17$ and a complete $D=39$ state matrix. The first three source values are $(1120, 11574, 125080)$.

```
16, 122, -2854, 3035, 114162, -324660, -1581792, 6105836, 6998392, -41646592, 4188416,
100180352, -55355328, -54337536, 19902720, 7110912, -1815552
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234445; parent column 2 of https://oeis.org/A234450. Both consulted pages mark the relevant recurrence empirical.

8.5.6.24 A234452

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=12$ state matrix. The first three source values are $(280, 2400, 21344)$.

$$(c_1,\dots,c_d) = (0, 100, 0, -1384, 0, 1440).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234452; parent column 1 of https://oeis.org/A234458. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 280x + 2400x^2 - 6656x^3 - 47040x^4 + 8640x^5 + 51840x^6.$$

8.5.6.25 A234453

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 27$, with order $d=26$ and a complete $D=63$ state matrix. The first three source values are $(2400, 31112, 427956)$.

```
24, 202, -8375, 18366, 832456, -4372543, -32845569, 252119346, 475222218, -6386106484,
1228195800, 78155524488, -93005666672, -459032184864, 814396694400, 1253941668864,
-2968493257344, -1338994735872, 5126543725056, -147546633728, -4053307347968,
1013644265472, 1200673271808, -333045153792, -113446158336, 22613852160
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234453; parent column 2 of https://oeis.org/A234458. Both consulted pages mark the relevant recurrence empirical.

8.5.6.26 A234483

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,3$. Predicate D has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 4$, with order $d=3$ and a complete $D=6$ state matrix. The first three source values are $(62, 242, 944)$.

$$(c_1,\dots,c_d) = (7,-9,-12).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234483; parent column 1 of https://oeis.org/A234490. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 62x - 192x^2 - 192x^3.$$

8.5.6.27 A234484

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,3$. Predicate D has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 6$, with order $d=5$ and a complete $D=20$ state matrix. The first three source values are $(242, 926, 3540)$.

$$(c_1,\dots,c_d) = (10,-28,1,54,24).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234484; parent column 2 of https://oeis.org/A234490. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 242x - 1494x^2 + 1056x^3 + 3840x^4 + 1536x^5.$$

8.5.6.28 A234491

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 6$, with order $d=5$ and a complete $D=12$ state matrix. The first three source values are $(250, 1834, 13464)$.

$$(c_1,\dots,c_d) = (5,74,-295,-1008,840).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234491; parent column 1 of https://oeis.org/A234497. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 250x + 584x^2 - 14206x^3 - 30144x^4 + 29520x^5.$$

8.5.6.29 A234492

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=2$, allowing equal adjacent values. The recurrence holds for every $n \geq 24$, with order $d=23$ and a complete $D=63$ state matrix. The first three source values are $(1834, 17022, 158288)$.

```
27, -7, -5642, 38876, 328643, -3991917, -1314423, 134764409, -274870060, -1754310602,
6188057089, 6383741530, -45869779722, 19646271368, 128555834088, -144208395792,
-115200081792, 233207698368, -26459811072, -102846620160, 35034937344, 13514784768,
-5918883840
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234492; parent column 2 of https://oeis.org/A234497. Both consulted pages mark the relevant recurrence empirical.

8.5.6.30 A234674

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=5$, allowing equal adjacent values. The recurrence holds for every $n \geq 5$, with order $d=4$ and a complete $D=12$ state matrix. The first three source values are $(112, 404, 1264)$.

$$(c_1,\dots,c_d) = (0, 22, 0, -120).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234674; parent column 1 of https://oeis.org/A234681. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 112x + 404x^2 - 1200x^3 - 4320x^4.$$

8.5.6.31 A234675

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=5$, allowing equal adjacent values. The recurrence holds for every $n \geq 19$, with order $d=18$ and a complete $D=63$ state matrix. The first three source values are $(404, 1152, 2924)$.

```
4, 47, -200, -901, 4204, 8957, -48440, -47194, 334096, 101948, -1410080, 170136,
3549696, -1446912, -4861440, 2954880, 2764800, -2073600
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234675; parent column 2 of https://oeis.org/A234681. Both consulted pages mark the relevant recurrence empirical.

8.5.6.32 A234884

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=1$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 28$, with order $d=27$ and a complete $D=63$ state matrix. The first three source values are $(272, 980, 3164)$.

```
2, 46, -91, -852, 1670, 8364, -16318, -48531, 94856, 177211, -349287, -422181, 844527,
666796, -1362305, -697540, 1464198, 475379, -1029412, -204338, 455683, 52461, -118912,
-7142, 16187, 376, -846
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234884; parent column 2 of https://oeis.org/A234890. Both consulted pages mark the relevant recurrence empirical.

8.5.6.33 A234885

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=1$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 47$, with order $d=46$ and a complete $D=342$ state matrix. The first three source values are $(856, 3164, 10640)$.

```
2, 86, -171, -3254, 6420, 72208, -140996, -1058026, 2037844, 10934873, -20686144,
-83167418, 153691479, 479339096, -859425200, -2137842471, 3687468872, 7489913552,
-12299566373, -20830586705, 32156843576, 46298748518, -66176437253, -82513440614,
107256565124, 117915942463, -136506202014, -134651989808, 135526272729, 121980787038,
-103852594014, -86649823814, 60484275256, 47465104086, -26205941074, -19592874823,
8200629244, 5905922924, -1777329681, -1244167978, 250186878, 171453636, -20378088,
-13746096, 723168, 482112
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234885; parent column 3 of https://oeis.org/A234890. Both consulted pages mark the relevant recurrence empirical.

8.5.6.34 A234891

Arrays have $n + 1$ rows, width $w = 2$, and labeled alphabet $0, \ldots, 6$. Predicate D has target $s = 1$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 11$, with order $d = 10$ and a complete $D = 16$ state matrix. The first three source values are $(160, 760, 3456)$.

```
0, 30, 0, -194, 0, 428, 0, -288, 0, 48
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234891; parent column 1 of https://oeis.org/A234898. Both consulted pages mark the relevant recurrence empirical.

8.5.6.35 A234905

Arrays have $n + 1$ rows, width $w = 2$, and labeled alphabet $0, \ldots, 7$. Predicate D has target $s = 1$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 13$, with order $d = 12$ and a complete $D = 20$ state matrix. The first three source values are $(280, 1720, 10344)$.

```
0, 55, 0, -725, 0, 3663, 0, -7525, 0, 6335, 0, -1805
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234905; parent column 1 of https://oeis.org/A234911. Both consulted pages mark the relevant recurrence empirical.

8.5.6.36 A234984

Arrays have $n + 1$ rows, width $w = 2$, and labeled alphabet $0, \ldots, 5$. Predicate D has target $s = 2$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 9$, with order $d = 8$ and a complete $D = 12$ state matrix. The first three source values are $(90, 370, 1640)$.

```
5, 6, -33, -24, 56, 34, -20, -4
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234984; parent column 1 of https://oeis.org/A234991. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 90x - 80x^2 - 750x^3 + 104x^4 + 1832x^5 + 692x^6 - 648x^7 - 120x^8.$$

8.5.6.37 A234985

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=2$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 29$, with order $d=28$ and a complete $D=63$ state matrix. The first three source values are $(370, 1490, 6908)$.

```
10, 9, -317, 216, 4392, -4481, -34461, 37067, 168161, -177688, -531245, 547375,
1099678, -1121930, -1478131, 1530027, 1252813, -1355278, -636426, 748324, 180286,
-245436, -25096, 44796, 1272, -4080, 0, 144
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A234985; parent column 2 of https://oeis.org/A234991. Both consulted pages mark the relevant recurrence empirical.

8.5.6.38 A235019

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=3$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=12$ state matrix. The first three source values are $(88, 332, 1248)$.

$$(c_1,\dots,c_d) = (0, 18, 0, -56, 0, 32).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235019; parent column 1 of https://oeis.org/A235026. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 88x + 332x^2 - 336x^3 - 1272x^4 + 192x^5 + 768x^6.$$

8.5.6.39 A235020

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=3$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 22$, with order $d=21$ and a complete $D=63$ state matrix. The first three source values are $(332, 1200, 4312)$.

```
5, 40, -213, -616, 3529, 4898, -29657, -24043, 140862, 80302, -394226, -183852,
650340, 268328, -614608, -222048, 318656, 91776, -82432, -14336, 8192
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235020; parent column 2 of https://oeis.org/A235026. Both consulted pages mark the relevant recurrence empirical.

8.5.6.40 A235064

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=3$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 11$, with order $d=10$ and a complete $D=16$ state matrix. The first three source values are $(168, 848, 4352)$.

```
0, 35, 0, -235, 0, 543, 0, -458, 0, 96
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235064; parent column 1 of https://oeis.org/A235071. Both consulted pages mark the relevant recurrence empirical.

8.5.6.41 A235065

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=3$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 41$, with order $d=40$ and a complete $D=100$ state matrix. The first three source values are $(848, 4348, 22936)$.

```
7, 114, -826, -5390, 40509, 141982, -1101107, -2376591, 18759386, 27110607,
-214422274, -218647229, 1714423248, 1266295692, -9847282701, -5264745392, 41312388336,
15449236953, -127755960707, -30541686755, 292034903938, 35307790177, -491841781764,
-7382786774, 604462209267, -48701691090, -533018275889, 88708374022, 328737830185,
-79723218859, -136723211260, 42223796030, 36429154260, -13346749528, -5768901376,
2417530544, 474735456, -225680256, -14561280, 8110080
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235065; parent column 2 of https://oeis.org/A235071. Both consulted pages mark the relevant recurrence empirical.

8.5.6.42 A235100

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=4$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=16$ state matrix. The first three source values are $(154, 670, 2900)$.

$$(c_1,\dots,c_d) = (7,-2,-47,3,84,36).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235100; parent column 1 of https://oeis.org/A235107. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 154x - 408x^2 - 1482x^3 + 856x^4 + 3312x^5 + 1296x^6.$$

8.5.6.43 A235101

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=4$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 20$, with order $d=19$ and a complete $D=100$ state matrix. The first three source values are $(670, 2498, 9328)$.

```
12, -14, -322, 949, 3384, -12901, -21740, 86236, 107744, -322761, -412538, 641088,
1018740, -478292, -1306432, -247248, 584736, 372096, 65664
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235101; parent column 2 of https://oeis.org/A235107. Both consulted pages mark the relevant recurrence empirical.

8.5.6.44 A235180

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,5$. Predicate D has target $s=5$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 19$, with order $d=18$ and a complete $D=63$ state matrix. The first three source values are $(212, 560, 1400)$.

```
3, 29, -88, -327, 1003, 1816, -5577, -5295, 15566, 8705, -20575, -9813, 12552, 5962,
-3436, -1676, 336, 168
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235180; parent column 2 of https://oeis.org/A235186. Both consulted pages mark the relevant recurrence empirical.

8.5.6.45 A235181

Arrays have $n + 1$ rows, width $w = 4$, and labeled alphabet $0, \ldots, 5$. Predicate D has target $s = 5$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 27$, with order $d = 26$ and a complete $D = 342$ state matrix. The first three source values are $(576, 1400, 3240)$.

```
3, 44, -133, -842, 2563, 9221, -28202, -64059, 195733, 296446, -893437, -937956,
2714749, 2064161, -5442684, -3205526, 7001040, 3522128, -5502080, -2622832, 2452848,
1173648, -555264, -271584, 48384, 24192
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235181; parent column 3 of https://oeis.org/A235186. Both consulted pages mark the relevant recurrence empirical.

8.5.6.46 A235191

Arrays have $n + 1$ rows, width $w = 2$, and labeled alphabet $0, \ldots, 6$. Predicate D has target $s = 5$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 7$, with order $d = 6$ and a complete $D = 16$ state matrix. The first three source values are $(136, 512, 1880)$.

$$(c_1, \ldots, c_d) = (0, 30, 0, -257, 0, 468).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235191; parent column 1 of https://oeis.org/A235198. Both consulted pages mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 136x + 512x^2 - 2200x^3 - 8288x^4 + 4592x^5 + 17504x^6.$$

8.5.6.47 A235192

Arrays have $n + 1$ rows, width $w = 3$, and labeled alphabet $0, \ldots, 6$. Predicate D has target $s = 5$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 24$, with order $d = 23$ and a complete $D = 100$ state matrix. The first three source values are $(512, 1544, 4724)$.

```
6, 52, -367, -995, 9350, 7034, -128869, 26637, 1045428, -801048, -5098437, 5754163,
14714982, -20381230, -23985995, 38063780, 20215810, -35199080, -8246780, 13170048,
2801112, -1572480, -393120
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235192; parent column 2 of https://oeis.org/A235198. Both consulted pages mark the relevant recurrence empirical.

8.5.6.48 A235193

Arrays have $n+1$ rows, width $w = 4$, and labeled alphabet $0, \ldots, 6$. Predicate D has target $s = 5$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 40$, with order $d = 39$ and a complete $D = 625$ state matrix. The first three source values are $(1880, 4724, 12176)$.

```
9, 57, -731, -878, 26304, -15900, -552844, 899723, 7521105, -18125633, -69122561,
220983779, 431291855, -1831340243, -1746606119, 10803108780, 3716404250, -46315168788,
2550194940, 145490417695, -46954531755, -335197745077, 166429786799, 564880422518,
-334400168840, -694120308016, 424547635156, 620744392928, -343613520080,
-401608244880, 170947437904, 182786646240, -46865973504, -54609655680, 4665398976,
9369063936, 551978496, -679311360, -113218560
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235193; parent column 3 of https://oeis.org/A235198. Both consulted pages mark the relevant recurrence empirical.

8.5.6.49 A235251

Arrays have $n+1$ rows, width $w = 2$, and labeled alphabet $0, \ldots, 7$. Predicate D has target $s = 7$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 11$, with order $d = 10$ and a complete $D = 20$ state matrix. The first three source values are $(184, 744, 2672)$.

```
0, 43, 0, -617, 0, 3073, 0, -1762, 0, 264
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (8.10) follows. Source: https://oeis.org/A235251; parent column 1 of https://oeis.org/A235258. Both consulted pages mark the relevant recurrence empirical.

8.6 Verification, provenance and reuse

The accompanying archive contains the standalone manuscript, numbered ledger, source definitions, complete exact certificates, verifiers, compact source audits and independent internal review reports. With Python, SymPy and mpmath installed, run

```
python3 r8_run_all.py
```

from its extracted directory. No network access is needed. The runner checks the 100 distinct selections, all 708 previous selections, the retained prior-work baseline, the immutable review bindings and every included verification program. It records actual versions and results in `oeis_hunt_round8_verification_summary.json`.

8.6.1 Why the finite certificates prove all indices

For each array problem, the complete transfer matrix is derived from the definition. The identity $TR(T)\mathbf{1} = 0$ implies every later recurrence instance after multiplication by an arbitrary power of T. An independently written implementation exhaustively constructs the uncompressed labeled-row relation: all 10,978 states and 24,364 directed edges across the 49 problems. It checks the annihilator on every full state and verifies every row of the proposed orbit quotient. Thus the symmetry reduction has a separate complete check.

The generating-function certificates establish exact differential identities $\mathcal{L}F = P$, with every nonzero boundary polynomial displayed. Coefficients beyond its degree vanish for all indices. Radical-field,

rational-parameter and differential-module calculations are exact. They do not infer an infinite recurrence from a fitted prefix. Independent coefficient calculations additionally check the intended sequence and offset.

The arithmetic and formal-series parts supply written universal proofs. Their finite rational, modular, divisor and numerical checks are supplementary. In particular, numerical integration is not the proof of reduced-numerator equality in A143991: the proof also controls every possible cancellation prime. The real-parameter result in A177430 explicitly invokes the classical six-exponentials theorem for irrational parameters.

8.6.2 Source and priority scope

Each selected proof links to its primary entry. The compact JSON records retain its precise target, uncertainty marker, offset, attribution, retrieval reference, relevant known input and bounded prior-work notes. Source hashes identify consulted snapshots without redistributing full raw web pages. An entry's editorial approved status is not interpreted as a mathematical proof of every formula on that page.

The audit compares all prior project selections and the retained OEIS Open Appendix A list. It also compares the Fischer (2020) and Ekhad–Yang (2017) proof-family lists. Relevant linked arithmetic and generating-function papers were inspected as described in the component audits. Finding a known input does not itself establish priority for the selected consequence; no worldwide-first-proof claim is made.

All previously selected IDs remain excluded, including A176605, A176607, A176610, A176611 and A176612, whose Round IV priority claims were withdrawn in Round V because Fischer's 2020 general derivation already covered them. The current and cumulative selection counts are not presented as counts of verified new discoveries.

8.6.3 Standalone manuscript and reproducibility

The delivered `.tex` contains all prose, formulas and individual recurrence signatures directly, with no external `\input` or `\include` fragments. Compile with pdfLaTeX until cross-references stabilize. `r8_build_dossier.py` reconstructs the manuscript, ledger and combined audit. The README gives all offline certificate-regeneration commands and dependencies. An isolated regeneration check compares the regenerated immutable outputs with the delivered versions.

The archive manifest records SHA-256 digests. The internal review reports separately bind the exact proof, code and certificate versions they inspected. The layout review binds the visually inspected PDF.

8.6.4 Selected common references

1. The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/. Individual primary-entry links accompany every selected assertion.
2. S. Dasgupta, *Ranks of matrices of logarithms of algebraic numbers I: the theorems of Baker and Waldschmidt–Masser*, 3 March 2023, Theorem 4.2 and Section 5.1, https://sites.math.duke.edu/~dasgupta/papers/Baker.pdf. Used for the six-exponentials theorem, not claimed as a new result.
3. *OEIS Open*, retained Appendix A comparison, https://arxiv.org/html/2608.11941v1. The retained scope is 100 costliest claims on 97 distinct A-IDs among 153 combined-model resolutions, not the complete repository.
4. Georg Fischer, 2020 family derivation linked at https://oeis.org/A176604/a176604_2.txt; used for exclusion and credit.

5. S. B. Ekhad and D. Yang, 2017 recurrence family and supporting output at https://sites.math.rutgers.edu/~zeilberg/tokhniot/oMathar1maple12.txt; used for the bounded known-family comparison.

Part IX

Gauss congruences, matrix symmetry, valuations, and exact algebra

Round overview

This ninth-round dossier resolves 100 precisely selected assertion packages on 100 distinct OEIS entries, disjoint from the 808 A-numbers selected in Rounds I–VIII. It proves 96 selected assertions and gives explicit corrections, with proofs over the full stated domains, for four others. The four parts contain ten formal-series, congruence and matrix packages, nine arithmetic packages, nineteen generating-function packages, and sixty-two constrained-array recurrence packages.

The methods include formal inversion in finite characteristic, integer exponential series, cyclic actions, an operator symmetry, elementary divisor and valuation arguments, exact generating-function identities, and finite transfer operators derived directly from the definitions.

What the count means

One catalogue row selects one assertion package on one distinct A-number. All selected statements are explicitly conjectural or empirical in the consulted primary entries. A recurrence and its generating function count once, and duplicate formulations of a general assertion are not promoted to separate results. The 62 array entries are distinct fixed-parameter counting problems within a shared transfer framework. Thus the count is not a claim of 100 independent techniques or 100 worldwide-first proofs.

The word "proved" refers to the exact target and domain stated in each package. Other conjectures on the same OEIS page may remain untouched. Established formulas, prior project methods and linked results are credited as inputs, including cases where the new package is an elementary consequence or an explicit repair of an old formula.

The four corrections

A202943. The parity-product sum must start at $j = 0$, not $j > 0$. The omitted constant already contradicts $a_0 = 1$. The corrected full product and integrality of the underlying cube root are proved.

A048896. The 2026 factorial sum ending at $n + 1$ has reduced numerator 13 when $n = 1$, whereas $a(1) = 1$. Its upper limit must be $n + 2$; the repaired numerator identity is proved for every $n \geq 0$.

A001784. The printed recurrence predates an offset change. Its coefficient index is shifted by one, giving the correct recurrence for $n \geq 2$. The previously recorded closed form is credited.

A052736. The stated identity fails at $n = 1$. Its corrected domain is $n \geq 2$, established from the defining exponential generating function.

No correction is counted twice. Additional notation and convention issues are disclosed within the relevant proofs; in particular A180654's integrality theorem covers both the printed extra-x^m convention and the program/example convention. The recurrences of A025256 and A025260 start at $n = 6$, as their exact boundary polynomials show.

Source and prior-work audit

The primary entries, relevant linked material and targeted searches were consulted on 20 October 2026. Cached retrievals can predate that date; source wording does not certify every live edit or worldwide priority. The earlier 808 is an exclusion ledger, not a cumulative discovery count. The five Round IV priority withdrawals disclosed in Round V remain in force and remain excluded. The retained OEIS Open comparison covers 100 Appendix A claims on 97 distinct A-numbers, with zero overlap; it does not cover that project's complete repository.

Several initially promising targets were removed after finding prior proofs, including A354766 and A278085 in an October 2026 paper, the linked A192069/A192248/A192249 family, and source entries that already contain

proofs or immediate duplicate recurrences. A234202 has a fresh exact certificate but remains an uncounted reserve. Previously certified reserves from earlier rounds are not counted as new work here.

All 62 selected array primary pages were retrieved. Of the 24 parent tables for the selected and reserve set, 21 were retrievable. The unretrieved parents A233883, A234114 and A234175 are disclosed in the affected records. The child definitions and certificates do not rely on those missing pages, but their historical proof status is not asserted to have been audited.

No OEIS submission or edit was performed. These are mathematical resolutions of selected assertions, separate from editorial acceptance. Independent internal reviews were performed by separate agents in the same AI-assisted workflow. They are not external peer review or proof-assistant formalization.

9.1 Catalogue of the 100 selected entries

P: selected assertion proved over its stated domain. **C**: source formula or domain corrected and proved. Page references lead to the complete argument or exact certificate.

No.	Entry	Selected assertion package	Result	Page
Formal series, congruences and matrix symmetry				
1	A158119	Powers-of-two parity from the continued fraction	P	457
2	A180654	Three exponential-integrality assertions	P	457
3	A384270	Dyadic parity of the composition equation	P	458
4	A376226	The complete ternary residue rule	P	458
5	A397588	Dyadic parity of the differential equation	P	459
6	A355414	The prime-index logarithmic congruence	P	459
7	A032312	All prime-power Gauss congruences	P	460
8	A167138	Integrality under every positive pointwise power	P	461
9	A091351	Symmetry between matrix-power parameters	P	461
10	A202943	Corrected parity-product lower limit	C	462
Arithmetic identities and a factorial-sum correction				
11	A026741	The general-prime gcd and lcm identities	P	463
12	A040003	A gcd formula for the partial quotients of sqrt(6)	P	464
13	A048896	Corrected upper limit in the factorial sum	C	464
14	A084068	The square-triangular Pell gcd	P	465
15	A096617	The reduced-numerator recurrence for n H(n)	P	466
16	A329354	The summatory asymptotic and its constant	P	466
17	A363409	The selected exact 7-adic valuation	P	468
18	A363411	The selected exact 5-adic valuation	P	468
19	A363412	The selected 3-adic assertions	P	469
Generating functions and polynomial recurrences				
20	A000483	Conjectured polynomial recurrence of order 3	P	470
21	A001784	Correction of the recurrence after the offset change	C	471
22	A025256	Conjectured polynomial recurrence of order 4	P	472
23	A025260	Conjectured polynomial recurrence of order 4	P	472
24	A026023	Conjectured polynomial recurrence of order 3	P	473
25	A026027	Conjectured polynomial recurrence of order 3	P	474
26	A026570	Conjectured polynomial recurrence of order 3	P	475
27	A026572	Conjectured polynomial recurrence of order 5	P	476
28	A026582	Conjectured polynomial recurrence of order 6	P	477
29	A026732	Conjectured polynomial recurrence of order 8	P	478
30	A026765	Conjectured polynomial recurrence of order 6	P	479
31	A052736	Correction of the boundary range in the identity	C	480
32	A071715	Conjectured polynomial recurrence of order 3	P	481
33	A098520	Conjectured polynomial recurrence of order 2	P	481
34	A123071	Conjectured polynomial recurrence of order 7	P	482
35	A143013	Conjectured polynomial recurrence of order 3	P	483
36	A372018	The identity for the odd-index subsequence	P	484
37	A212341	The empirical coefficient generating function	P	484
38	A110668	The empirical generating function of partial sums	P	485
Finite-state array recurrence certificates				
39	A233787	Empirical array recurrence of order 34	P	488
40	A233877	Empirical array recurrence of order 6	P	489
41	A233878	Empirical array recurrence of order 12	P	489
42	A233879	Empirical array recurrence of order 29	P	489
43	A233880	Empirical array recurrence of order 52	P	489
44	A233960	Empirical array recurrence of order 5	P	490
45	A233961	Empirical array recurrence of order 14	P	490
46	A233962	Empirical array recurrence of order 43	P	490

Continued on the next page

No.	Entry	Selected assertion package	Result	Page
47	A234029	Empirical array recurrence of order 4	P	490
48	A234030	Empirical array recurrence of order 9	P	491
49	A234031	Empirical array recurrence of order 16	P	491
50	A234032	Empirical array recurrence of order 44	P	491
51	A234107	Empirical array recurrence of order 6	P	491
52	A234108	Empirical array recurrence of order 18	P	492
53	A234109	Empirical array recurrence of order 40	P	492
54	A234117	Empirical array recurrence of order 5	P	492
55	A234118	Empirical array recurrence of order 11	P	492
56	A234119	Empirical array recurrence of order 17	P	493
57	A234124	Empirical array recurrence of order 6	P	493
58	A234125	Empirical array recurrence of order 14	P	493
59	A234126	Empirical array recurrence of order 27	P	493
60	A234133	Empirical array recurrence of order 3	P	494
61	A234135	Empirical array recurrence of order 6	P	494
62	A234154	Empirical array recurrence of order 3	P	494
63	A234155	Empirical array recurrence of order 6	P	494
64	A234156	Empirical array recurrence of order 11	P	495
65	A234157	Empirical array recurrence of order 26	P	495
66	A234158	Empirical array recurrence of order 53	P	495
67	A234170	Empirical array recurrence of order 6	P	496
68	A234171	Empirical array recurrence of order 35	P	496
69	A234177	Empirical array recurrence of order 5	P	496
70	A234178	Empirical array recurrence of order 11	P	497
71	A234179	Empirical array recurrence of order 22	P	497
72	A234201	Empirical array recurrence of order 10	P	497
73	A234210	Empirical array recurrence of order 4	P	497
74	A234211	Empirical array recurrence of order 15	P	498
75	A234212	Empirical array recurrence of order 56	P	498
76	A234659	Empirical array recurrence of order 5	P	498
77	A234660	Empirical array recurrence of order 16	P	498
78	A234661	Empirical array recurrence of order 39	P	499
79	A234667	Empirical array recurrence of order 7	P	499
80	A234668	Empirical array recurrence of order 31	P	499
81	A234683	Empirical array recurrence of order 6	P	500
82	A234684	Empirical array recurrence of order 24	P	500
83	A234697	Empirical array recurrence of order 5	P	500
84	A234698	Empirical array recurrence of order 15	P	500
85	A234699	Empirical array recurrence of order 31	P	501
86	A234706	Empirical array recurrence of order 31	P	501
87	A234707	Empirical array recurrence of order 36	P	501
88	A234721	Empirical array recurrence of order 8	P	502
89	A234722	Empirical array recurrence of order 30	P	502
90	A234816	Empirical array recurrence of order 7	P	502
91	A234817	Empirical array recurrence of order 23	P	503
92	A234818	Empirical array recurrence of order 23	P	503
93	A234992	Empirical array recurrence of order 10	P	503
94	A234993	Empirical array recurrence of order 49	P	503
95	A235169	Empirical array recurrence of order 9	P	504
96	A235170	Empirical array recurrence of order 46	P	504
97	A235205	Empirical array recurrence of order 6	P	504
98	A235206	Empirical array recurrence of order 29	P	505
99	A235233	Empirical array recurrence of order 28	P	505
100	A235234	Empirical array recurrence of order 40	P	505

9.2 Formal series, congruences, and matrix symmetry

The ten packages in this section concern precisely identified conjectures on ten primary OEIS entries. Nine are proved as stated; the parity formula at A202943 requires its summation to include the zero term. Known definitions, classical identities, and closely related earlier results are credited below. Finite calculations in `r9_root_verify.py` check transcription and boundary cases; the arguments in this section prove the universal statements.

9.2.1 A158119: parity from a continued fraction

Source. A158119 conjectures that $a(n)$ is odd exactly when $n = 0$ or n is a power of two. Its ordinary generating function, with $A(0) = 1$, is characterized by Hanna's displayed continued fraction

$$1 = A(x) - \cfrac{x}{A(x) - \cfrac{4x}{A(x) - \cfrac{9x}{A(x) - \cdots}}}.$$

The continued-fraction characterization is a known input from the entry.

Proposition 9.2.1. *The integral solution satisfies*

$$A(x) \equiv 1 + \sum_{j \geq 0} x^{2^j} \pmod 2.$$

Proof. All denominators of the fraction are units. Truncation at depth N fixes coefficients through a degree tending to infinity with N, because each additional level is multiplied by x. The fraction therefore has a precise coefficientwise meaning. In degree $n \geq 1$, the unknown coefficient of A occurs with coefficient 1 in the initial A; the rest uses only earlier coefficients. This proves existence and uniqueness over $\mathbb{Z}$.

Modulo two the numerator $4x$ vanishes, so $1 = A - x/A$, or $A^2 + A = x$. Set $B = A - 1$. In characteristic two this is $B + B(x^2) = x$. Comparing coefficients gives $[x]B = 1$, all other odd-index coefficients zero, and $[x^{2n}]B = [x^n]B$. Iteration gives exactly the stated support, including the constant coefficient of A. □

Prior-work scope. The linked Bozejko–Hasebe paper concerns free infinite divisibility; its inspected text supplied no parity theorem for this sequence. The related entries A157308, A157310, A155585, and A157309 describe the transforms underlying the continued fraction. They are not additional selected packages.

9.2.2 A180654: integrality of the logarithmic construction

Source. A180654 states three integrality conjectures: the binomial-square case, every fixed positive integer power, and the case in which the power equals the summation index. Its program and expanded example use the following series:

$$P_{m,p}(x) = \sum_{k=0}^{m} \binom{m}{k}^p x^k, \qquad F(x) = 1 + \sum_{m \geq 1} \frac{1}{m!} \left(\frac{\log P_{m,p_m}(x)}{m} \right)^m.$$

Here each p_m is a positive integer. The source title and two displayed generalizations contain an extra factor x^m absent from the program and coefficient list. The theorem covers both conventions; the numerical sequence is checked against the program convention.

Proposition 9.2.2. *For every choice of positive integers $(p_m)_{m\ge1}$, all coefficients $n![x^n]F$ are integers. The same assertion holds after multiplying the mth summand by x^m. Thus all three source integrality claims hold under either of the two displayed conventions.*

Proof. For $1 \le k \le m$,

$$\frac{k}{m}\binom{m}{k}^{p_m} = \binom{m-1}{k-1}\binom{m}{k}^{p_m-1} \in \mathbb{Z}.$$

Since $P_{m,p_m}(0) = 1$, this shows

$$\left(\frac{\log P_{m,p_m}}{m}\right)' = \frac{P'_{m,p_m}/m}{P_{m,p_m}} \in \mathbb{Z}[[x]].$$

Consequently $H_m = \log(P_{m,p_m})/m$ has zero constant term and integral exponential coefficients: if $H'_m = \sum b_j x^j$, then $r![x^r]H_m = (r-1)!b_{r-1}$.

For any such $H = \sum_{r\ge1} h_r x^r/r!$ with $h_r \in \mathbb{Z}$,

$$n![x^n]\frac{H^m}{m!} = \sum_{\substack{\sum j_r = m \\ \sum r j_r = n}} \frac{n!}{\prod_r (r!)^{j_r} j_r!} \prod_r h_r^{j_r} \in \mathbb{Z}.$$

The prefactor counts set partitions with prescribed block sizes. Each H_m^m starts in degree m, so the sum is locally finite. Finally, $n![x^n](x^m G) = n!/(n-m)! \cdot (n-m)![x^{n-m}]G$, preserving integrality. Taking $p_m = 2$, a fixed p, or $p_m = m$ gives the three cases. They form one package at this A-number, not three separate counts. □

9.2.3 A384270: a composition quotient modulo two

Source. A384270, Hanna, July 10, 2025, defines

$$A = x + O(x^2), \qquad A\,A(x^3 + 3xA^3) = A(x^4 + 4xA^4),$$

and conjectures odd coefficients precisely at powers of two.

Proposition 9.2.3. *There is a unique integral solution and $A \equiv \sum_{j\ge0} x^{2^j} \pmod 2$.*

Proof. For $n \ge 2$, the coefficient a_n appears only in the leading product Ax^3 when the defining equation is read in degree $n+3$; every other occurrence has larger degree. Its coefficient is 1. This gives a monic integer recursion, also unique over every commutative coefficient ring.

Over $\mathbb{F}_2$, the equation becomes $A\,A(x^3 + xA^3) = A(x^4) = A^4$. Let $S = \sum_{j\ge0} x^{2^j}$, so $x = S + S^2$. A direct identity gives

$$x^3 + xS^3 = S^3 + S^6.$$

The series $S(y)$ is the compositional inverse of $y + y^2$; hence $S(x^3 + xS^3) = S^3$. Thus S satisfies the reduced equation. The monic uniqueness just proved forces $A = S$ modulo two. □

9.2.4 A376226: the full ternary support rule

Source. A376226, Hanna, October 4, 2024, defines

$$A = x + O(x^2), \qquad A\,A(x^2 + 4xA^2) = A(x^3 + 6xA^3).$$

Its two conjectures describe the coefficients congruent to 1 and 2 modulo three.

Proposition 9.2.4. *For every $n \geq 1$,*

$$a_n \equiv \begin{cases} 1, & n = 3^i \quad (i \geq 0), \\ 2, & n = (3^i + 3^j)/2 \quad (0 \leq i < j), \\ 0, & \textit{otherwise} \end{cases} \pmod 3.$$

Proof. In degree $n+2$ the new coefficient a_n occurs only in Ax^2, with coefficient 1. All other contributions involve earlier coefficients. This establishes integral existence and uniqueness, also modulo three. The reduced equation, after cancellation of the nonzero series A, is

$$A(x^2 + xA^2) = A^2.$$

Let B be the inverse of $u + u^2 + u^3$. For $x = B + B^2 + B^3$ one has

$$x^2 + xB^2 = B^2 + B^4 + B^6.$$

The right side is $u + u^2 + u^3$ evaluated at $u = B^2$, so B satisfies the reduced equation. Uniqueness gives $A = B$ modulo three.

To read its coefficients, put $U(t) = \sum_{i \geq 0} t^{3^i}$. Then $U - U^3 = t$, and

$$t^2 = (U - U^3)^2 = U^2 + U^4 + U^6.$$

Thus $B(t^2) = U(t)^2$. Its diagonal terms are $t^{2 \cdot 3^i}$ and its off-diagonal terms are $2t^{3^i + 3^j}$. Base-three expansions show that these exponents are distinct and that no diagonal exponent is off-diagonal. Dividing all exponents by two proves the three cases. □

9.2.5 A397588: parity of a differential recursion

Source. A397588, Hanna, July 3, 2026, has $A = x + (xA^2)'$, $A \in x\mathbb{Z}[[x]]$. The selected conjecture says that a_n is odd exactly at powers of two, for $n \geq 1$.

Proposition 9.2.5. $A \equiv \sum_{j \geq 0} x^{2^j} \pmod 2$.

Proof. The defining equation gives $a_1 = 1$ and

$$a_n = (n+1) \sum_{k=1}^{n-1} a_k a_{n-k} \qquad (n \geq 2),$$

which establishes integrality and uniqueness. In characteristic two, $(xA^2)' = A^2 + 2xAA' = A^2$, so $A = x + A^2$. Coefficient comparison gives the required support as in A158119. □

9.2.6 A355414: the prime-index logarithmic congruence

Source. A355414, Melika Tebni, July 1, 2022, defines

$$\sum_{n \geq 0} a_n \frac{x^n}{n!} = \frac{\log((1-x)/(1-2x))}{(1-x)^5}$$

and conjectures $a_p \equiv -1 \pmod p$ for every prime p.

Proposition 9.2.6. *More generally, if $H \in \mathbb{Z}[[x]]$, $H(0) = 1$, and $b_n = n![x^n]H(x)\log((1-x)/(1-qx))$ with integer q, then $b_n \in \mathbb{Z}$ and*

$$b_p \equiv 1 - q \pmod p \qquad (p \textit{ prime}).$$

Proof. Write $H = \sum h_j x^j$. For $n \geq 1$,

$$b_n = \sum_{k=1}^{n} \frac{n!}{k} h_{n-k}(q^k - 1).$$

Each $n!/k$ is an integer. At $n = p$ every term with $k < p$ is divisible by p. The remaining term is $(p-1)!(q^p-1)$, congruent to $-(q-1)$ by Wilson's theorem and $q^p \equiv q \pmod{p}$. Taking $H = (1-x)^{-5}$ and $q = 2$ proves the selected assertion, including $p = 2$. □

Credit and counting. Fried's 2025 note and the 2026 compilation, Section 3.9, Theorem 52, already prove the power-three case A355372. The proof above is a routine extension of that type of argument, not an independent new method. A355372 is excluded; the power-four consequence A355407 is not counted separately. Only the still-marked power-five assertion is selected here.

9.2.7 A032312: all Gauss congruences by a cyclic action

Source. A032312 has exponential generating function $\prod_{r\geq 1}(1 + x^r/r!)^2$. Peter Bala, September 5, 2022, conjectures its Gauss congruences and the equivalent integrality of its logarithmic exponential transform.

Proposition 9.2.7. *For every prime p and all positive integers n, k,*

$$a(np^k) \equiv a(np^{k-1}) \pmod{p^k}.$$

Furthermore $\exp(\sum_{m\geq 1} a(m)x^m/m) \in \mathbb{Z}[[x]]$. *The congruence proof holds for any fixed nonnegative number of colors.*

Proof. The exponential product counts set partitions whose blocks have one of two colors, with at most one block of each given size and color. Let a cyclic group C_{p^k} act freely on np^k labels, as n disjoint regular orbits, and hence on these partitions. An orbit of partitions whose size is less than p^k is fixed by the unique subgroup H of order p; conversely, a partition not fixed by H has orbit size exactly p^k.

If a partition is fixed by H, each block is fixed individually: the action preserves its size and color, and a second block with the same pair is forbidden. Each block is therefore a union of H-orbits of labels. Quotienting by H gives exactly the same kind of colored partition on np^{k-1} labels, with block sizes divided by p. This is a bijection. All other partitions occur in orbits of size p^k, proving the congruence.

For completeness, put

$$c_m = \frac{1}{m} \sum_{d|m} \mu(d) a(m/d).$$

If $m = p^s u$, $p \nmid u$, pair divisors d and pd to see that the numerator equals $\sum_{d|u} \mu(d)(a(p^s u/d) - a(p^{s-1}u/d))$, divisible by p^s. Doing this for every prime factor gives $c_m \in \mathbb{Z}$. Formal logarithms then give

$$\exp\left(\sum_{m\geq 1} \frac{a(m)}{m} x^m\right) = \prod_{m\geq 1} (1 - x^m)^{-c_m} \in \mathbb{Z}[[x]].$$

All products and sums are locally finite coefficientwise. □

Prior-work scope. A007837 is the one-color case, and its entry credits Jovovic with a September 26, 2026 proof of the Gauss congruences. That case and its companion A168268 are excluded. The selected two-color case is proved directly above; the common Euler-product criterion is classical, not a new theorem.

9.2.8 A167138: closure under powers after logarithmic substitution

Source. A167138 asks whether integrality of $\sum_{n\ge 0} L(n)\log(1+x)^n/n!$ implies integrality of the same expression with $L(n)$ replaced by $L(n)^m$, for every positive integer m. The same general conjecture is repeated at A167139, A167141, and A167532; these repetitions contribute no additional count.

Proposition 9.2.8. *The asserted closure holds. In fact, the class is closed under pointwise products of any finite number of sequences.*

Proof. Define $s_j(n) = j!\left\{{n \atop j}\right\}$, the number of surjections from an n-set to a fixed j-set, with $s_0(0) = 1$ and $s_0(n) = 0$ for $n > 0$. If the original integral series is $F(x) = \sum c_j x^j$, substitution $x = e^t - 1$ shows

$$L(n) = \sum_{j=0}^{n} c_j s_j(n), \qquad c_j \in \mathbb{Z}.$$

For nonnegative $j_1, \dots, j_m$, let $C(j_1, \dots, j_m; r)$ be the number of r-element subsets of $[j_1] \times \cdots \times [j_m]$ whose projection onto each factor is surjective. A tuple of surjections has such a joint image; after choosing that image there are $s_r(n)$ surjections onto it. Hence

$$\prod_{i=1}^{m} s_{j_i}(n) = \sum_{r\ge 0} C(j_1, \dots, j_m; r)\, s_r(n).$$

This includes empty sets with the stated conventions. For a fixed r a nonzero coefficient forces every $j_i \le r$, except that the $r = 0$ case forces all $j_i = 0$. Thus collecting coefficients in a product of the expansions of $L(n)$ is finite at each r and yields integral coefficients d_r. Finally

$$\sum_{n\ge 0} s_r(n)\frac{\log(1+x)^n}{n!} = x^r.$$

The transformed product is $\sum d_r x^r \in \mathbb{Z}[[x]]$. For the entry itself, c_j is the partition number $p(j)$ and $m = 2$. □

9.2.9 A091351: symmetry of two matrix-power parameters

Source. A091351 is the lower triangular matrix T in which $T(n,k)$ equals the row sum in row $n-k$ of T^k. Hanna, July 16, 2026, conjectures

$$(T^p)_{n+q-1,q-1} = (T^q)_{n+p-1,p-1} \quad (n \ge 0,\ p, q \ge 1).$$

We prove it from a formal operator identity and also justify that identity from an equivalent recursive construction of this matrix.

Proposition 9.2.9. *The displayed symmetry holds for every stated n, p, q.*

Proof. Let S be the down-shift matrix, $S_{i,j} = 1$ if $i = j + 1$ and 0 otherwise, and let $e_0 = (1, 0, \dots)^t$. First construct a lower triangular matrix T by $T(n, 0) = 1$ and

$$TS = ST + (ST)^2. \tag{2.M1}$$

This determines column $k+1$ in row n using only rows smaller than n, so there is a unique integral matrix with diagonal entries 1. Put $D = ST$ and $f(z) = z + z^2$, and let g be the compositional inverse of f. Since D is strictly lower triangular, every power series in D is well-defined entrywise. The shift relation gives

$$TD = f(D)T, \qquad DS = Sf(D), \qquad SD = g(D)S.$$

The last identity follows by applying g to the middle intertwining identity. Thus, for any formal h,

$$Th(D) = h(f(D))T, \qquad Sh(D) = h(g(D))S.$$

Since Te_0 is the all-ones vector, $Te_0 = (1+D)e_0$. In particular, $De_0 = S(1+D)e_0 = (1+g(D))Se_0$. As $D = g(D)(1+g(D))$, cancellation of the unit $1+g(D)$ gives $Se_0 = g(D)e_0$. It follows that

$$Se_0 = g(D)e_0, \quad S^r e_0 = \prod_{i=1}^{r} g^{\circ i}(D)e_0, \quad T^r e_0 = \frac{f^{\circ r}(D)}{D}e_0.$$

The quotient in the last formula is a power series with constant term 1; no inverse of the shift matrix is being assumed. These identities follow by induction and $f(z)/z = 1+z$.

Define $H_1(z) = 1$ and $H_r(z) = \prod_{i=0}^{r-2} g^{\circ i}(z)$ for $r \geq 2$. Applying the preceding intertwining identities and cancelling equal formal factors yields

$$S^{p-1}T^pS^{q-1}e_0 = (1+D)H_p(D)H_q(D)e_0. \tag{2.M2}$$

One direct check of the cancellation starts from

$$T^pS^{q-1}e_0 = \frac{f^{\circ p}(D)}{D}\prod_{i=1}^{q-1} f^{\circ(p-i)}(D)e_0,$$

where negative iterates denote iterates of g. Apply S^{p-1}, which substitutes $g^{\circ(p-1)}(D)$ and contributes $\prod_{i=1}^{p-1} g^{\circ i}(D)$. The cancelled factors leave exactly (2.M2). Each quotient after cancellation is an ordinary power series, so this calculation is valid in the strictly triangular matrix algebra.

Putting $q = 1$ in (2.M2) proves $S^{p-1}T^pe_0 = TS^{p-1}e_0$. Therefore column k of T, shifted up by k, equals $T^{k+1}e_0 = T^k\mathbf{1}$. This is precisely the source definition. That definition also determines rows successively: for $k > 0$, the relevant power uses only rows through $n-k < n$. Hence the matrix just constructed is the source matrix.

Finally the right side of (2.M2) is symmetric in p, q. Comparing coordinate $n+p+q-2$ proves the desired equality. □

9.2.10 A202943: a corrected parity product and integral cube root

Source. A202943, Hanna, December 26, 2011, defines the cube root with constant term 1 by

$$A(x)^3 = R(x) := \sum_{n\geq 0}\binom{n+2}{2}2^{n(n-1)}x^n.$$

The conjectured parity product uses $\sum_{j>0} x^{32M(j)}$, where M =A000695 has offset 0 and $M(0) = 0$. That lower limit omits the constant term, contradicting $a_0 = 1$ (and $a_1 = 1$). The source's expanded example already includes the missing 1. We make the correction explicit.

Proposition 9.2.10. *$A \in \mathbb{Z}[[x]]$, and its corrected parity formula is*

$$A(x) \equiv (1+x)(1+x^2)(1+x^8)\sum_{j\geq 0} x^{32M(j)} \pmod 2.$$

In addition, $a_n \equiv 1 \pmod 3$ for all $n \geq 0$; this last assertion is already unmarked in the entry and is not an extra selected claim.

Proof. First, $R(x) \equiv (1-x)^{-3} \pmod 9$. Indeed, if $n \equiv 0$ or $1 \pmod 3$, then $n(n-1)$ is divisible by six and $2^{n(n-1)} \equiv 1 \pmod 9$. If $n \equiv 2 \pmod 3$, the binomial factor is divisible by three and $2^{n(n-1)} - 1$ is also divisible by three. Thus

$$(1-x)^3R(x) = 1 + 9K(x), \qquad K \in x\mathbb{Z}[[x]].$$

For every $j \geq 1$, the coefficient

$$9^j \binom{1/3}{j} = \frac{3^j \prod_{i=0}^{j-1}(1-3i)}{j!}$$

is an integer divisible by three. For a prime $\ell \neq 3$, among j successive terms of the numerator progression at least $\lfloor j/\ell^r \rfloor$ are divisible by ℓ^r, for every r. This supplies the ℓ-valuation of $j!$. At three the valuation is $j - v_3(j!) \geq 1$. Hence $A = (1-x)^{-1}(1+9K)^{1/3}$ is integral and congruent to $(1-x)^{-1}$ modulo three.

Modulo two the defining equation is $A^3 = 1 + x$. Its constant-term-one root is unique, since the new coefficient at each positive degree has multiplier $3 = 1$ in $\mathbb{F}_2$. Let

$$B = (1+x)\prod_{i\geq 0}(1+x^{2\cdot 4^i}).$$

Then, coefficientwise,

$$B^3 = (1+x)^3 \prod_{r\geq 1}(1+x^{2^r}) = \frac{(1+x)^3}{1+x^2} = 1+x.$$

Thus $A = B$. Separating the factors with exponents 2 and 8 leaves $\prod_{i\geq 0}(1+x^{32\cdot 4^i})$, the generating function for all finite sums of distinct powers of four scaled by 32, including the empty sum. This is the corrected formula. □

9.3 Arithmetic identities, valuations, and an exact correction

The nine packages below concern nine identifiers absent from the 808 previously selected identifiers. Eight selected assertions are proved; the ninth, a 2026 factorial-sum assertion on A048896, is refuted as written and repaired for every index. The primary entries were inspected on October 20, 2026. Each package states its selected scope explicitly. In particular, the valuation results do not purport to settle the wider prime-classification or asymptotic conjectures on the same pages. The accompanying verifier distinguishes exact finite proof certificates from numerical checks of the universal arguments.

9.3.1 A026741: the prime-parameter gcd and lcm formulas

Source. A026741, Timothy Hopper, February 26, 2017, labels the following pair of formulas "Conjecture". For a prime p, put $f_p(n) = n/p$ if $p \mid n$, and $f_p(n) = n$ otherwise. The original sequence is the case $p = 2$.

Proposition 9.3.1. *For every prime p and positive integer n,*

$$\operatorname{lcm}(n,p) = p f_p(n), \qquad \gcd(n,p) = \frac{n}{f_p(n)}.$$

Proof. If $p \mid n$, then $\gcd(n,p) = p$, $\operatorname{lcm}(n,p) = n$, and $f_p(n) = n/p$. If $p \nmid n$, primality gives $\gcd(n,p) = 1$, $\operatorname{lcm}(n,p) = np$, and $f_p(n) = n$. Substitution proves both identities in both cases. □

Scope and prior work. These are elementary consequences of the definitions. The $p = 2$ gcd and lcm formulas already occur elsewhere in the entry and are credited; the selected package is the stated general-prime assertion, counted once. The linked $p = 3, 5, 7$ entries A051176, A060791, and A106608 were inspected as companions and are not separate results. At $n = 0$ the lcm identity remains valid, but the printed gcd quotient is $0/0$; its meaningful domain is $n \geq 1$. Fermat's little theorem also immediately verifies Hopper's associated formula involving $n^{p-1} \bmod p$.

9.3.2 A040003: a gcd expression for the continued fraction of sqrt(6)

Source. Michel Marcus's September 3, 2020 formula on A040003 is marked "empirical". Let $a(j)$, $j \geq 0$, be the successive partial quotients of $\sqrt{6}$.

Proposition 9.3.2. *For every integer $n \geq 1$,*

$$a(n-1) = \gcd(2^n, 3^n + 1).$$

Proof. Writing $s = \sqrt{6}$, we have $2 < s < 3$ and

$$\frac{1}{s-2} = \frac{s+2}{2}, \qquad \frac{1}{(s+2)/2 - 2} = s + 2.$$

The latter two complete quotients have integer parts 2 and 4 and alternate. Thus $\sqrt{6} = [2; \overline{2, 4}]$. The proposed gcd is 2 when $n = 1$. For positive even n, $3^n + 1 \equiv 2 \pmod 8$, so the gcd is 2. For odd $n \geq 3$, $3^n + 1 \equiv 4 \pmod 8$, so the gcd is 4. These are exactly $a(n-1)$ in the three cases. □

Prior-work scope. The continued fraction is classical and is the definition of the entry. The inspected companion pages A040001 and A010694 describe the related periodic sequences; they supplied no proof of this gcd expression. No claim of novelty for the continued fraction itself is made.

9.3.3 A048896: a factorial-sum counterexample and its corrected limit

Source. The entry A048896 defines $a(n) = 2^{s_2(n+1)-1}$ for $n \geq 0$, where s_2 counts binary ones. Stefano Spezia's August 20, 2026 assertion, introduced by "It appears that", identifies $a(n)$ with the reduced numerator of

$$R_n = \sum_{k=1}^{n+1} \frac{k^2}{(n+2-k)!(n+2+k)!}.$$

Here and below the numerator of a rational number means the numerator after reduction to lowest terms with positive denominator.

Proposition 9.3.3. *The displayed assertion is false: $R_1 = 13/240$, while $a(1) = 1$. Changing the upper summation limit to $n+2$ gives a valid identity for every $n \geq 0$:*

$$a(n) = \operatorname{num}\left(\sum_{k=1}^{n+2} \frac{k^2}{(n+2-k)!(n+2+k)!} \right).$$

Proof. Direct exact arithmetic gives

$$R_1 = \frac{1}{2!4!} + \frac{4}{1!5!} = \frac{1}{48} + \frac{1}{30} = \frac{13}{240}.$$

Since $\gcd(13, 240) = 1$, this is a counterexample to the literal formula. For the correction put $m = n + 2$. The binomial second-moment identity, obtained by differentiating $(1+x)^{2m}$ twice and once and evaluating at $x = 1$, is

$$\sum_{j=0}^{2m} (j-m)^2 \binom{2m}{j} = m 2^{2m-1}.$$

Pairing $j = m - k$ and $j = m + k$ therefore yields

$$\sum_{k=1}^{m} \frac{k^2}{(m-k)!(m+k)!} = \frac{m 2^{2m-2}}{(2m)!} = \frac{4^{n+1}}{2(2n+3)!}.$$

Legendre's identity $v_2(N!) = N - s_2(N)$ follows by summing $\lfloor N/2^r \rfloor$ over $r \geq 1$. The exponent of two remaining in the reduced numerator of the last fraction is consequently

$$2n + 1 - v_2((2n+3)!) = s_2(2n+3) - 2 = s_2(n+1) - 1 \geq 0.$$

Its unreduced numerator is a power of two, so no odd prime can remain in the numerator. This proves the corrected identity. □

Prior-work scope. The older Lenard parity-sum observation is not the counted target. Its content already appears as an unmarked central-entry identity on A261363 (August 16, 2015): the middle partial sum in an even row of Pascal's triangle modulo two is A048896. The present package counts only the literal 2026 factorial-sum assertion and its exact correction, once.

9.3.4 A084068: the square-triangular Pell gcd

Source. Joseph A. Stocke's July 28, 2025 conjecture on A084068 states

$$a(n) = \gcd(\mathrm{A001108}(n), \mathrm{A001109}(n)).$$

The source defines $a(1) = 1$, $a(2) = 2$, $a(2k) = 2a(2k-1) - a(2k-2)$, and $a(2k+1) = 4a(2k) - a(2k-1)$.

Proposition 9.3.4. *For every $n \geq 1$, the displayed gcd equals $a(n)$.*

Proof. Define positive integers u_n, v_n by

$$(1+\sqrt{2})^n = u_n + v_n\sqrt{2}, \qquad u_n^2 - 2v_n^2 = (-1)^n.$$

Their norm shows $\gcd(u_n, v_n) = 1$. Also u_n is odd, since $u_{n+1} = u_n + 2v_n$ and $u_0 = 1$. The standard Pell formulas in A001108 and A001109 give

$$T_n := \mathrm{A001108}(n) = \frac{u_n^2 + 2v_n^2 - 1}{2}, \qquad S_n := \mathrm{A001109}(n) = u_n v_n.$$

These formulas also follow by squaring $u_n + v_n\sqrt{2}$; they have $T_0 = S_0 = 0$, $T_1 = S_1 = 1$ and the familiar recurrences $T_{n+2} = 6T_{n+1} - T_n + 2$ and $S_{n+2} = 6S_{n+1} - S_n$. For even n the norm identity gives $T_n = 2v_n^2$, whence

$$\gcd(T_n, S_n) = v_n \gcd(2v_n, u_n) = v_n.$$

For odd n it gives $T_n = u_n^2$, whence $\gcd(T_n, S_n) = u_n$. Finally set $c_n = v_n$ for even n and $c_n = u_n$ for odd n. The relations $u_{n+1} = u_n + 2v_n$ and $v_{n+1} = u_n + v_n$ imply

$$v_{2k} = 2u_{2k-1} - v_{2k-2}, \qquad u_{2k+1} = 4v_{2k} - u_{2k-1}.$$

Together with $c_1 = 1, c_2 = 2$, these are precisely the defining recurrences of a. Thus $c_n = a(n)$ and the gcd assertion follows. □

Prior-work scope. Mathar's October 15, 2021 even/odd Pell description of $a(n)$ is already on the primary entry and is credited. The companion square-triangular pages were inspected. This package concerns Stocke's gcd formula only; the separate conjecture about first occurrences in A348295 is not selected.

9.3.5 A096617: the reduced recurrence for harmonic numerators

Source. M. F. Hasler's July 4, 2019 observation on A096617 is introduced by "It appears that" and asks whether the recurrence is easy to prove. The sequence is the reduced numerator of nH_n, where $H_n = \sum_{j=1}^{n} 1/j$.

Proposition 9.3.5. *Set $b_1 = f_0 = 1$. For $n \geq 1$ define*

$$g_n = \gcd((n+1)b_n, nf_{n-1}), \quad f_n = \frac{nf_{n-1}}{g_n}, \quad b_{n+1} = \frac{(n+1)b_n}{g_n} + f_n.$$

Then b_n is the reduced numerator of nH_n for every $n \geq 1$. This initialization gives $g_1 = f_1 = 1$, as in the source.

Proof. We prove the stronger invariant

$$nH_n = \frac{b_n}{f_{n-1}}, \qquad \gcd(b_n, f_{n-1}) = 1.$$

It holds at $n = 1$. If it holds at n, then

$$(n+1)H_{n+1} = \frac{(n+1)b_n + nf_{n-1}}{nf_{n-1}}.$$

The gcd of this numerator and denominator is $\gcd((n+1)b_n, nf_{n-1}) = g_n$. Dividing by it gives exactly b_{n+1}/f_n in lowest terms. Induction proves the assertion. □

Scope and prior work. The auxiliary $f_0 = 1$ makes explicit the first denominator that is implicit in the printed initialization. The standard Stirling/harmonic identity and the linked A000254 recurrence are known inputs motivating the source. The inspected pages A000254 and A193758 do not supply this reduction invariant.

9.3.6 A329354: the summatory asymptotic, including its constant

Source. Vaclav Kotesovec's March 3, 2023 conjecture on A329354 concerns $a(n) = \sum_{d|n} d\omega(d)$, where ω counts distinct prime divisors. Let B_1 denote the prime Mertens constant, A077761.

Proposition 9.3.6. *As real $x \to \infty$,*

$$\sum_{n \leq x} a(n) = \frac{\pi^2 x^2}{12}(\log\log x + B_1) + o(x^2).$$

In particular the asymptotic equivalence in the source holds.

Proof. We use the classical prime-harmonic estimate

$$\sum_{p \leq y} \frac{1}{p} = \log\log y + B_1 + O(1/\log y)$$

and $\pi(y) = O(y/\log y)$. These are established inputs; see Axler, *New estimates for some functions defined over primes*, Integers 18 (2018), A52, equation (5.1) and the prime-counting estimates in its introduction. Put $W(y) = \sum_{d \leq y} d\omega(d)$. Counting each distinct prime divisor once gives

$$W(y) = \sum_{p \leq y} p \sum_{m \leq y/p} m = \frac{y^2}{2} \sum_{p \leq y} \frac{1}{p} + O(y\pi(y)) = \frac{y^2}{2}(\log\log y + B_1) + o(y^2).$$

The floor in the triangular sum introduces $O(y)$ per prime because $y/p \geq 1$. This argument also gives the uniform bound $W(y) = O(y^2 \log\log(3y))$ for $y \geq 1$.

Changing the order of the finite divisor sum yields

$$\sum_{n\leq x} a(n) = \sum_{m\leq x} W(x/m).$$

Let $M = \lfloor (\log x)^2 \rfloor$. For $m \leq M$, the error in the formula for $W(x/m)$ is uniformly $o(x^2/m^2)$ because $x/M \to \infty$. Moreover,

$$\log\log(x/m) = \log\log x + O(\log M/\log x) \quad (1 \leq m \leq M).$$

Since $\sum_{m\leq M} m^{-2} = \zeta(2) + O(M^{-1})$, the contribution from $m \leq M$ is $\frac{1}{2}\zeta(2)x^2(\log\log x + B_1) + o(x^2)$. The nonnegative tail is at most

$$O\left(x^2 \log\log(3x) \sum_{m>M} m^{-2}\right) = O\left(\frac{x^2 \log\log(3x)}{M}\right) = o(x^2).$$

Finally $\zeta(2) = \pi^2/6$ proves the result. □

Prior-work scope. The entry already gives its divisor and prime convolutions and its Dirichlet generating function. Those are credited, not counted as discoveries. The inspected companions A328260, A329375, and A323599 did not provide the selected summatory formula. Our stated $o(x^2)$ remainder is stronger than the displayed asymptotic equivalence. Numerical checks of finite sums are transcription checks, not evidence used to prove an asymptotic limit.

A finite-block lemma for the next two packages

Write

$$P_D(n) = \prod_{k=1}^{n} (1 + ks) = A_D(n) + B_D(n)s, \qquad s^2 = -D, \quad P_D(0) = 1.$$

All products are taken in the commutative ring $\mathbb{Z}[s]/(s^2 + D)$. Their coefficients are generated exactly by

$$A_D(n) = A_D(n-1) - DnB_D(n-1), \qquad B_D(n) = B_D(n-1) + nA_D(n-1).$$

If p is an odd prime and $-D$ is a quadratic nonresidue modulo p, then

$$P_D(n+p) \equiv 2P_D(n) \pmod{p}, \qquad P_D(n+p^2) \equiv 2^p P_D(n) \pmod{p^2}. \tag{9.1}$$

Indeed, over $\mathbb{F}_p$, $\prod_{a=0}^{p-1}(1 + as) = 1 - s^{p-1} = 2$. For a full block modulo p^2, group the factors as $k = a + pt$, $1 \leq a \leq p$, $0 \leq t < p$. Expansion gives

$$\prod_{t=0}^{p-1} (1 + (a + pt)s) \equiv (1 + as)^p \pmod{p^2} :$$

every term of order at least two in pt is divisible by p^2, and the linear coefficient is divisible by $p\sum_{t=0}^{p-1} t = p^2(p-1)/2$. Thus the full block is congruent to C^p, where $C = \prod_{a=1}^{p}(1 + as) \equiv 2 \pmod{p}$, and hence to $2^p \pmod{p^2}$. Any p or p^2 consecutive factors have the same residue multiset, proving both shift formulas for every $n \geq 0$.

9.3.7 A363409: the complete 7-adic valuation rule

Source. The entry A363409 (Peter Bala, June 1, 2023) defines $a(n) = A_2(n)$ and says "It appears that" before the rule $v_7(a(7n+3)) = 1$, with valuation zero at all other indices. The assertion has no separate signature or date within the entry.

Proposition 9.3.7. *For every $n \geq 0$,*

$$v_7(A_2(n)) = \begin{cases} 1, & n \equiv 3 \pmod 7, \\ 0, & n \not\equiv 3 \pmod 7. \end{cases}$$

Proof. Here -2 is a nonresidue modulo 7, so (9.1) gives the multipliers 2 modulo 7 and $2^7 \equiv 30 \pmod{49}$. The exact coefficient recurrence in the lemma gives the following small certificates:

r	0	1	2	3	4	5	6
$A_2(r) \bmod 7$	1	1	4	0	6	3	2
$(A_2(7r+3) \bmod 49)/7$	4	6	4	5	2	5	4

In the second row of values, the residue modulo 49 is chosen in $\{0, \ldots, 48\}$ before division. The first shift formula and first row show that divisibility by 7 occurs exactly at indices $3 \pmod 7$. Write any such index uniquely as $49q + 7r + 3$ with $0 \leq r < 7$. The second shift formula multiplies its residue modulo 49 by the unit 30^q. Every entry in the second row is nonzero modulo 7, so none of these values is divisible by 49. This proves the exact valuation at every index. □

Audit and verification. The displayed 14 residues are generated in integer arithmetic by `r9_arith_verify.py`; together with the proved shifts they are a finite certificate for the universal result. The linked Moll paper (2012) studies $D = 1$. Its statements and examples do not give the selected $D = 2$, $p = 7$ rule. Other prime classifications on the OEIS entry remain outside this package.

9.3.8 A363411: the complete 5-adic valuation rule

Source. A363411 (Peter Bala, June 1, 2023) defines $a(n) = A_3(n)$ and explicitly conjectures $v_5(a(n+2)) = \text{A079998}(n)$, reporting checks through 5000. The assertion has no separate signature or date within the entry. The companion A079998 is the characteristic function of multiples of five, with offset zero.

Proposition 9.3.8. *For every $n \geq 0$,*

$$v_5(A_3(n)) = \begin{cases} 1, & n \equiv 2 \pmod 5, \\ 0, & n \not\equiv 2 \pmod 5. \end{cases}$$

Consequently the source's shifted identity holds for every $n \geq 0$.

Proof. Here -3 is a nonresidue modulo 5, so (9.1) gives multipliers 2 modulo 5 and $2^5 \equiv 7 \pmod{25}$. Exact multiplication gives

r	0	1	2	3	4
$A_3(r) \bmod 5$	1	1	0	3	2
$(A_3(5r+2) \bmod 25)/5$	4	4	2	1	4

The first shift and first row give exactly the residue class $2 \pmod 5$ for divisibility by 5. Each such index is $25q + 5r + 2$ with $0 \leq r < 5$. The second shift multiplies the corresponding entry modulo 25 by the unit 7^q; all five quotients in the table are nonzero modulo 5. Thus the valuation is exactly one there. □

Audit and verification. The ten table entries and full-block multiplier are reproduced exactly in the verifier. This closes the unbounded shifted valuation assertion, rather than merely extending its 5000-term check. The linked Moll paper is credited as background for $D = 1$; the wider $D = 3$ prime-type conjectures are not selected.

9.3.9 A363412: a stronger exact 3-adic valuation formula

Source. A363412 (Peter Bala, June 1, 2023) defines $a(n) = B_3(n)$ and labels two statements "Conjectures": $3 \nmid a(3n+1)$, and $v_3(a(3n)) = v_3(a(3n-1))$. The second statement is meaningful for $n \geq 1$; the first includes $n = 0$. These assertions have no separate signature or date in the entry.

Proposition 9.3.9. *For every $n \geq 1$,*

$$v_3(B_3(n)) = v_3\left(\frac{n(n+1)}{2}\right).$$

In particular $B_3(n) \neq 0$ for $n \geq 1$, and both selected conjectures hold on their stated meaningful domains.

Proof. Let $e_r = e_r(1, 2, \ldots, n)$ denote an elementary symmetric function and put $T_n = n(n+1)/2$. We use the established Hultman-number formula

$$H(N,k) = \frac{c(N+2,k)}{\binom{N+2}{2}} \quad \text{if } N-k \text{ is odd},$$

where $H(N,k)$ counts permutations and c is an unsigned Stirling number of the first kind. See Bóna and Flynn, arXiv:0811.0740, Section 3, the final displayed formula; A164652 also records this known formula. Since

$$e_{2j+1} = c(n+1, n-2j), \qquad (n-1)-(n-2j) = 2j-1,$$

it follows that e_{2j+1}/T_n is an integer whenever $2j+1 \leq n$. Also $e_1/T_n = 1$.

Extracting the coefficient of s in $P_3(n)$ gives

$$\frac{B_3(n)}{T_n} = \sum_{j \geq 0} (-3)^j \frac{e_{2j+1}}{T_n} \equiv 1 \pmod 3.$$

The finite sum on the right is an integer not divisible by three. This proves the valuation formula and nonvanishing. At $n = 3m+1$, neither n nor $n+1$ is divisible by three. For $m \geq 1$ the two triangular numbers at $3m$ and $3m-1$ have the same 3-adic valuation $1 + v_3(m)$. The two source conjectures follow. □

Prior-work scope. Hultman integrality is a credited classical input, not a new theorem of this report. The linked Moll paper concerns the related $D = 1$ product. No assertion is made here about the source's global Type 2 or Type 3 prime classification. The empty-Type-1 statement is also not counted: its literal version includes the zero term $a(0) = 0$.

9.4 Generating functions and exact algebra: 19 selected packages

These nineteen packages concern primary OEIS assertions carrying an explicit conjectural or empirical marker on 20 October 2026. Seventeen packages prove the selected assertion with its index range stated precisely. A001784 and A052736 instead correct an index or boundary error, give a counterexample to the unqualified printed version, and prove the corrected statement. Existing generating functions, formulas and combinatorial interpretations are credited as inputs. One A-number contributes one package, even if both its generating function and recurrence are established. No claim of first publication, independent research novelty, or OEIS acceptance is made.

9.4.1 A finite exact certificate for every recurrence

Put $\theta = x\,d/dx$ and extend a_n by zero below its stated offset. If $F(x) = \sum a_n x^n$ and $p_0, \dots, p_r$ are polynomials, then

$$L = \sum_{j=0}^{r} x^j p_j(\theta + j), \qquad [x^n]LF = \sum_{j=0}^{r} p_j(n) a_{n-j}.$$

Consequently an exact identity $LF = P(x)$, with P polynomial, proves the recurrence for every $n > \deg P$ at which the sequence indices exist. This is an algebraic proof for all indices, not a finite numerical test.

For an exponential generating function $F = \sum a_n x^n/n!$, replace $p_j(n)$ by $q_j(n) = p_j(n)(n - j)_{r-j}$, where $(v)_h$ is the falling factorial. For $n \geq r$ the coefficient of the resulting operator is $\sum_j p_j(n) a_{n-j}/(n-r)!$. The same polynomial-residual argument therefore applies.

For a radical $F = R_0 + R_1\sqrt{D}$, use

$$\theta(R_0 + R_1\sqrt{D}) = xR_0' + x\left(R_1' + \frac{D'}{2D}R_1\right)\sqrt{D}.$$

For a rational parameter $x = X(t)$, replace $x\,d/dx$ by $X(t)/X'(t)\,d/dt$. Every table below gives the actual p_j, the actual series and branch, and the exact residual. All coefficients lie in a rational-function field; the scripts clear denominators and verify zero polynomials. A displayed radical is the branch with constant term $+1$. Removable negative powers in the rational components do not introduce negative coefficients in the complete formal series.

The independently calculated prefixes are checks on source indexing and transcription. They are not substitutes for the identities above.

9.4.2 A000483

Source and selected claim. A000483, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{3} p_j(n) a_{n-j} = 0 \qquad (n \geq 9).$$

$$\begin{aligned}
p_0(n) &= (n-4)\,(n-2)\,,\\
p_1(n) &= -\,(n-1)\left(3n^2 - 21n + 35\right),\\
p_2(n) &= (n-2)\,(n-1)\left(3n^2 - 24n + 47\right),\\
p_3(n) &= -\,(n-5)\,(n-4)\,(n-3)\,(n-2)\,(n-1)\,,
\end{aligned}$$

Offset and initial data. Offset 6.

$$(a_6, \dots, a_{13}) = (15, 210, 2380, 26432, 303660, 3678840, 47324376, 647536032)\,.$$

Proof. The defining objects are permutations with exactly three cycles, each of length at least two. A labelled cycle of length j contributes x^j/j; the unordered set of three cycles therefore has exponential generating function

$$F(x) = \frac{1}{6}\left(\sum_{j\geq 2} \frac{x^j}{j}\right)^3 = -\frac{(x + \log(1-x))^3}{6}.$$

This argument derives the series from the objects, without using either conjectural recurrence. Put $\ell = x + \log(1-x)$, so $\ell' = -x/(1-x)$. In the basis $1, \ell, \ell^2, \ell^3$ the coefficient vector of F is $(0, 0, 0, -1/6)$. For a vector R the Euler derivative has component $x(R_i' + (i+1)\ell' R_{i+1})$, with $R_4 = 0$. Applying the exponential version of the certificate lemma above gives the displayed exact residual, and hence the recurrence over the stated full range.

Exact certificate. Using $q_j(n) = p_j(n)(n-j)_{3-j}$, $LF = 20x^6$; all three positive log-power residuals are zero. **Verification.** The standalone verifier reproduces the exact identity, compares 20 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.3 A001784

Source and selected claim. A001784, checked 20 October 2026. Conjecture. The printed first-order recurrence requires the index correction proved below.

$$\sum_{j=0}^{1} p_j(n) a_{n-j} = 0 \qquad (n \geq 2).$$

$$\begin{aligned} p_0(n) &= (n-1)\left(20n^2 + 35n + 12\right), \\ p_1(n) &= -(n+1)(2n+3)\left(20n^2 + 75n + 67\right), \end{aligned}$$

Offset and initial data. Offset 0.

$$(a_0, \ldots, a_7) = (1, 24, 924, 26432, 705320, 18858840, 520059540, 14980405440).$$

Proof. Here $a_0 = 1$ is an added initial convention, and for $n \geq 1$ the sequence counts permutations of $2n+3$ elements having n cycles and no fixed points. The source's first-order conjecture was entered in 2015, before the offset was changed to zero in 2016. With the current indexing its residual at $n = 3$ is

$$162a_3 - 6237a_2 = 162 \cdot 26432 - 6237 \cdot 924 = -1481004.$$

Thus that printed formula is false with the current offset. The corrected coefficients displayed here result from replacing its coefficient variable n by $n+1$ while leaving a_n, a_{n-1} in place.

To prove the correction, subtract two from every cycle length. The excess is three, so the only possibilities are one 5-cycle; one 4-cycle and one 3-cycle; or three 3-cycles. All other cycles have length two. The standard labelled cycle count gives

$$a_n = (2n+3)!\left\{\frac{1}{5\,2^{n-1}(n-1)!} + \frac{1}{12\,2^{n-2}(n-2)!} + \frac{1}{162\,2^{n-3}(n-3)!}\right\},$$

where a term with a negative factorial is omitted. Factoring the first denominator and using the factors $n-1$ and $(n-1)(n-2)$ gives, for every $n \geq 1$,

$$a_n = \frac{(20n^2 + 75n + 67)(2n+3)!}{405\,2^n(n-1)!}.$$

This closed form is already credited to Kotesovec (2016) in the entry; the present package is an index correction, not a claim of a new closed form or a newly discovered recurrence. Dividing successive expressions, and clearing the nonzero denominators for $n \geq 2$, proves the correction.

Verification. The standalone verifier reproduces the exact identity, compares 16 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.4 A025256

Source and selected claim. A025256, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{4} p_j(n)a_{n-j} = 0 \qquad (n \geq 6).$$

$$\begin{aligned} p_0(n) &= n, \\ p_1(n) &= 3 - 2n, \\ p_2(n) &= 3 - n, \\ p_3(n) &= -5\,(2n - 9)\,, \\ p_4(n) &= n - 6, \end{aligned}$$

Offset and initial data. Offset 1.

$$(a_1, \ldots, a_8) = (1, 1, 3, 3, 6, 18, 42, 105)\,.$$

Proof. The source defines $a_1 = 1$, $a_2 = 1$, $a_3 = 3$ and $a_n = \sum_{j=1}^{n-3} a_j a_{n-j}$ for $n \geq 4$. Set $F = \sum_{n\geq 1} a_n x^n$. The omitted last two terms in the full convolution are $a_1 a_{n-1} + a_2 a_{n-2}$. Summing over $n \geq 4$ and retaining the three initial coefficients gives

$$F^2 - (1 + 1x + (1)x^2)F + 1x + (1)x^2 + 3x^3 = 0.$$

The branch with zero constant term is precisely the radical displayed below. Inserting it in the certificate lemma proves the selected linear recurrence, independent of the nonlinear recurrence used to define the terms. The other quadratic branch has constant term one and is excluded.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} D &= x^4 - 10x^3 - x^2 - 2x + 1, \\ R_0 &= \frac{x^2 + x + 1}{2}, \\ R_1 &= -\frac{1}{2}, \end{aligned}$$

Exact substitution gives $LF = P(x) = -3x^5 + x^4 + 6x^3 + x^2 + x$. The degree-five boundary term is why the recurrence starts at $n = 6$, not $n = 5$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 30 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.5 A025260

Source and selected claim. A025260, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{4} p_j(n)a_{n-j} = 0 \qquad (n \geq 6).$$

$$\begin{aligned} p_0(n) &= n, \\ p_1(n) &= -2\,(2n - 3)\,, \end{aligned}$$

$$p_2(n) = 8\,(n-3)\,,$$
$$p_3(n) = -6\,(2n-9)\,,$$
$$p_4(n) = 4\,(n-6)\,,$$

Offset and initial data. Offset 1.

$$(a_1, \ldots, a_8) = (2, -2, 1, 2, 2, 1, 2, 10)\,.$$

Proof. The source defines $a_1 = 2$, $a_2 = -2$, $a_3 = 1$ and $a_n = \sum_{j=1}^{n-3} a_j a_{n-j}$ for $n \geq 4$. Set $F = \sum_{n\geq 1} a_n x^n$. The omitted last two terms in the full convolution are $a_1 a_{n-1} + a_2 a_{n-2}$. Summing over $n \geq 4$ and retaining the three initial coefficients gives

$$F^2 - (1 + 2x + (-2)x^2)F + 2x + (-2)x^2 + 1x^3 = 0.$$

The branch with zero constant term is precisely the radical displayed below. Inserting it in the certificate lemma proves the selected linear recurrence, independent of the nonlinear recurrence used to define the terms. The other quadratic branch has constant term one and is excluded.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$D = 4x^4 - 12x^3 + 8x^2 - 4x + 1,$$
$$R_0 = -\frac{2x^2 - 2x - 1}{2},$$
$$R_1 = -\frac{1}{2},$$

Exact substitution gives $LF = P(x) = 2x^5 - 6x^4 + 15x^3 - 8x^2 + 2x$. The degree-five boundary term is why the recurrence starts at $n = 6$, not $n = 5$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 35 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.6 A026023

Source and selected claim. A026023, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{3} p_j(n) a_{n-j} = 0 \qquad (n \geq 3).$$

$$p_0(n) = (n-1)\,(n+4)\,,$$
$$p_1(n) = (n-1)\,(n+1)\,,$$
$$p_2(n) = -2\,(n+1)\,(2n+1)\,,$$
$$p_3(n) = -4\,(n-1)\,(n+1)\,,$$

Offset and initial data. Offset 0.

$$(a_0, \ldots, a_7) = (1, 2, 4, 8, 15, 30, 56, 112)\,.$$

Proof. For a nonnegative nearest-neighbor walk starting at height three, the reflection principle gives $T(n,k) = \binom{n}{k} - \binom{n}{k-4}$ when the terminal height is $n+3-2k \geq 0$. The reflected forbidden walks

start at -5, which explains the difference of four in the binomial indices. Summing in k telescopes to four neighboring binomial coefficients. Pascal's identity yields

$$a_{2m} = \binom{2m+2}{m}, \qquad a_{2m+1} = 2\binom{2m+2}{m}.$$

These bisections are known in the entry; the reflection argument also provides their justification from the definition. Writing $C(z) = (1-\sqrt{1-4z})/(2z)$, the binomial generating function is $C(z)^2/\sqrt{1-4z}$. Consequently $F(x) = (1+2x)C(x^2)^2/\sqrt{1-4x^2}$, which reduces to the displayed radical coefficients. The exact differential residual proves the conjectured mixed-parity recurrence.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} D &= -(2x-1)(2x+1), \\ R_0 &= -\frac{2x+1}{2x^4}, \\ R_1 &= \frac{2x^2-1}{2x^4(2x-1)}, \end{aligned}$$

Exact substitution gives $LF = P(x) = -4$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 34 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.7 A026027

Source and selected claim. A026027, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{3} p_j(n) a_{n-j} = 0 \qquad (n \geq 5).$$

$$\begin{aligned} p_0(n) &= (n+5)(13n+118), \\ p_1(n) &= 13n^2 - 969n - 3490, \\ p_2(n) &= -2\left(203n^2 - 1080n - 758\right), \\ p_3(n) &= 4(2n-5)(73n-182), \end{aligned}$$

Offset and initial data. Offset 2.

$$(a_2, \ldots, a_9) = (1, 5, 21, 84, 329, 1274, 4900, 18768).$$

Proof. Use the equivalent triangle definition in the source: the path has $2n-1$ nearest-neighbor steps from height three to height six and remains nonnegative. The title's step-condition range ending at n is a typo; the triangle and the binomial formula specify all $2n-1$ steps. Reflection gives

$$a_n = \binom{2n-1}{n-2} - \binom{2n-1}{n-6}, \quad n \geq 2,$$

where an out-of-range binomial coefficient is zero. For $q \geq 0$, $\sum_{m\geq 0}\binom{2m+q}{m}x^m = C(x)^q/\sqrt{1-4x}$. Apply this identity with $q = 3, 11$ to the two sums. It follows that

$$F(x) = \frac{x^2C(x)^3 - x^6C(x)^{11}}{\sqrt{1-4x}} = x^2\{C(x)^5 + x^2C(x)^9\}.$$

The source writes the latter expression without x^2, because that display starts its exponent at the sequence offset; here exponents are the actual indices. Put $t = C(x) - 1$, so $x = t/(1+t)^2$ and $F = t^2(1+t)(1+t^2)$. The rational parameter is locally invertible at zero. The parameter version of the certificate lemma now proves the recurrence for every admissible index.

Exact certificate. $x = \frac{t}{(t+1)^2}$, $F = t^2\,(t+1)\,(t^2+1)$. Exact substitution gives $LF = P(x) = 1008x^2$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 24 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.8 A026570

Source and selected claim. A026570, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{3} p_j(n) a_{n-j} = 0 \qquad (n \geq 4).$$

$$\begin{aligned} p_0(n) &= n+1, \\ p_1(n) &= -2n, \\ p_2(n) &= -3n-1, \\ p_3(n) &= 2\,(2n-3)\,, \end{aligned}$$

Offset and initial data. Offset 1.

$$(a_1, \ldots, a_8) = (1, 1, 4, 7, 20, 43, 111, 259)\,.$$

Proof. The parent triangle A026568 counts walks on the integers starting at zero, with steps $+1, -1$, and with a zero step allowed only at an even height. Let e_j, o_j be the generating functions for endpoint $2j, 2j+1$. Arrival at a vertex gives

$$(1-x)e_j = \delta_{j0} + x(o_{j-1} + o_j), \qquad o_j = x(e_j + e_{j+1}).$$

Set $d = (1-x)(1-x-4x^2)$, $h = 1-x-2x^2$, and $r = (h - \sqrt{d})/(2x^2) = O(x^2)$. Eliminating the odd vertices and using symmetry about zero gives

$$e_j = \frac{r^{|j|}}{\sqrt{d}}, \qquad o_j = x(e_j + e_{j+1}).$$

Indeed $x^2r^2 - hr + x^2 = 0$, and the equation at zero reduces to $(h - 2x^2r)/\sqrt{d} = 1$. This is the unique formal walk solution: the coefficient of degree N depends only on paths of length at most N, hence on finitely many vertices, and the arrival equations determine it inductively. Thus no analytic convergence or unproved recurrence is being assumed.

For endpoint one, $F = o_0 = x(1+r)/\sqrt{d} = ((1-x)/\sqrt{d} - 1)/(2x)$. This produces the displayed radical certificate. In particular it also proves the entry's conditional identity with the difference of A026569, but that identity is not counted as an additional package.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} D &= (x-1)\left(4x^2 + x - 1\right), \\ R_0 &= -\frac{1}{2x}, \end{aligned}$$

$$R_1 = -\frac{1}{2x\left(4x^2+x-1\right)},$$

Exact substitution gives $LF = P(x) = -x^2 + 2x$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 29 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.9 A026572

Source and selected claim. A026572, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{5} p_j(n) a_{n-j} = 0 \qquad (n \geq 8).$$

$$\begin{aligned}
p_0(n) &= n+3,\\
p_1(n) &= -3n-5,\\
p_2(n) &= -4\left(n+1\right),\\
p_3(n) &= 13n-9,\\
p_4(n) &= 5n-9,\\
p_5(n) &= -6\left(2n-7\right),
\end{aligned}$$

Offset and initial data. Offset 3.

$$(a_3, \ldots, a_{10}) = (1, 2, 8, 19, 56, 140, 376, 953)\,.$$

Proof. The parent triangle A026568 counts walks on the integers starting at zero, with steps $+1, -1$, and with a zero step allowed only at an even height. Let e_j, o_j be the generating functions for endpoint $2j, 2j+1$. Arrival at a vertex gives

$$(1-x)e_j = \delta_{j0} + x(o_{j-1} + o_j), \qquad o_j = x(e_j + e_{j+1}).$$

Set $d = (1-x)(1-x-4x^2)$, $h = 1-x-2x^2$, and $r = (h-\sqrt{d})/(2x^2) = O(x^2)$. Eliminating the odd vertices and using symmetry about zero gives

$$e_j = \frac{r^{|j|}}{\sqrt{d}}, \qquad o_j = x(e_j + e_{j+1}).$$

Indeed $x^2r^2 - hr + x^2 = 0$, and the equation at zero reduces to $(h - 2x^2r)/\sqrt{d} = 1$. This is the unique formal walk solution: the coefficient of degree N depends only on paths of length at most N, hence on finitely many vertices, and the arrival equations determine it inductively. Thus no analytic convergence or unproved recurrence is being assumed.

Endpoint three has $F = o_1 = x(r + r^2)/\sqrt{d}$. Reduction using $r = (1-x-2x^2-\sqrt{d})/(2x^2)$ gives

$$F = \frac{(1-x-3x^2)\sqrt{(1-x)/(1-x-4x^2)} - 1 + x + x^2}{2x^3},$$

which is exactly the conjectured generating function (its apparent pole is removable). The separate conjectural recurrence follows from the exact residual below. These two linked assertions constitute one package.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$D = (x-1)\left(4x^2+x-1\right),$$
$$R_0 = \frac{x^2+x-1}{2x^3},$$
$$R_1 = \frac{3x^2+x-1}{2x^3\left(4x^2+x-1\right)},$$

Exact substitution gives $LF = P(x) = -3x^4 + 6x^3$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 27 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.10 A026582

Source and selected claim. A026582, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{6} p_j(n)a_{n-j} = 0 \qquad (n \geq 6).$$

$$p_0(n) = n,$$
$$p_1(n) = -2\left(2n-1\right),$$
$$p_2(n) = -2\left(n-3\right),$$
$$p_3(n) = 20\left(n-2\right),$$
$$p_4(n) = 6-7n,$$
$$p_5(n) = -2\left(12n-41\right),$$
$$p_6(n) = 8\left(2n-7\right),$$

Offset and initial data. Offset 0.

$$(a_0,\ldots,a_7) = (1,3,8,20,50,125,314,792)\,.$$

Proof. The parent triangle A026568 counts walks on the integers starting at zero, with steps $+1,-1$, and with a zero step allowed only at an even height. Let e_j, o_j be the generating functions for endpoint $2j, 2j+1$. Arrival at a vertex gives

$$(1-x)e_j = \delta_{j0} + x(o_{j-1}+o_j), \qquad o_j = x(e_j+e_{j+1}).$$

Set $d = (1-x)(1-x-4x^2)$, $h = 1-x-2x^2$, and $r = (h-\sqrt{d})/(2x^2) = O(x^2)$. Eliminating the odd vertices and using symmetry about zero gives

$$e_j = \frac{r^{|j|}}{\sqrt{d}}, \qquad o_j = x(e_j+e_{j+1}).$$

Indeed $x^2r^2 - hr + x^2 = 0$, and the equation at zero reduces to $(h-2x^2r)/\sqrt{d} = 1$. This is the unique formal walk solution: the coefficient of degree N depends only on paths of length at most N, hence on finitely many vertices, and the arrival equations determine it inductively. Thus no analytic convergence or unproved recurrence is being assumed.

The definition sums, through row n, the half-row with nonnegative endpoint. Reflection in height zero shows that its row sum is half the sum of all walks plus the number ending at zero. If E, O sum over all even and odd endpoints, then $(1-x)E = 1+2xO$ and $O = 2xE$. Thus $E+O = (1+2x)/(1-x-4x^2)$ and $e_0 = 1/\sqrt{d}$. Taking cumulative sums gives

$$F = \frac{1}{2(1-x)}\left\{\frac{1+2x}{1-x-4x^2} + \frac{1}{\sqrt{d}}\right\}.$$

This derivation fixes both the half-row boundary and the initial term. The displayed certificate proves the selected recurrence.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} D &= (x-1)\left(4x^2+x-1\right), \\ R_0 &= \frac{2x+1}{2\,(x-1)\,(4x^2+x-1)}, \\ R_1 &= -\frac{1}{2\,(x-1)^2\,(4x^2+x-1)}, \end{aligned}$$

Exact substitution gives $LF = P(x) = 2x^4 + x$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 28 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.11 A026732

Source and selected claim. A026732, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{8} p_j(n)\,a_{n-j} = 0 \qquad (n \geq 8).$$

$$\begin{aligned} p_0(n) &= 1-n, \\ p_1(n) &= 2, \\ p_2(n) &= 3\,(3n-7), \\ p_3(n) &= -10, \\ p_4(n) &= 95-23n, \\ p_5(n) &= 6, \\ p_6(n) &= 11n-95, \\ p_7(n) &= 2, \\ p_8(n) &= 4\,(n-7), \end{aligned}$$

Offset and initial data. Offset 0.

$$(a_0, \ldots, a_7) = (1, 2, 4, 9, 18, 40, 80, 176).$$

Proof. In A026725 the Pascal rule has an extra $T(2m-1, m-1)$ only at position $(2m+1, m)$ for $m \geq 1$. Boundary entries remain one. Put $b_m = T(2m+1, m)$ and $P_n(u) = \sum_k T(n,k)u^k$. Propagation of each extra entry by Pascal's rule gives

$$P_n(u) = (1+u)^n + \sum_{\substack{j\geq 1 \\ 2j+1\leq n}} b_{j-1}u^j(1+u)^{n-2j-1}.$$

The corresponding row-polynomial identity also occurs in Fried's 2026 paper, equation (57), where it is used to study a different Riordan-array claim; it is not claimed here as new. Extracting u^m in P_{2m+1} gives

$$b_m = \binom{2m+1}{m} + \sum_{j=1}^{m} b_{j-1} \binom{2(m-j)}{m-j}.$$

Hence $B(z) = \sum b_m z^m = C(z)/(\sqrt{1-4z} - z)$. Putting $u = 1$ in the row identity gives the row-sum generating function

$$F(x) = \frac{1 + x^3 B(x^2)}{1 - 2x}.$$

This is the radical used below; the differential certificate proves the selected row-sum recurrence, which is distinct from the diagonal and Riordan-array assertions in the cited earlier work.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} D &= -(2x-1)(2x+1), \\ R_0 &= -\frac{x^3 - 2x^2 + 3x + 2}{2(x^4 + 4x^2 - 1)}, \\ R_1 &= -\frac{x(x-1)(x+1)}{2(2x-1)(x^4 + 4x^2 - 1)}, \end{aligned}$$

Exact substitution gives $LF = P(x) = -5x^6 + 18x^5 + 7x^4 - 8x^3 - 3x^2 + 2x + 1$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 31 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.12 A026765

Source and selected claim. A026765, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{6} p_j(n) a_{n-j} = 0 \qquad (n \geq 6).$$

$$\begin{aligned} p_0(n) &= n + 1, \\ p_1(n) &= -2, \\ p_2(n) &= -2(5n - 3), \\ p_3(n) &= 12, \\ p_4(n) &= 29n - 71, \\ p_5(n) &= -10, \\ p_6(n) &= -20(n - 5), \end{aligned}$$

Offset and initial data. Offset 0.

$$(a_0, \ldots, a_7) = (1, 2, 4, 9, 18, 41, 82, 188).$$

Proof. The parent array A026758 adds $T(n-2, k-1)$ to the Pascal rule exactly when n is odd and $1 \leq k \leq (n-1)/2$. At odd row $2m+1$ use the integer coordinate $j = m - k$. Two successive ordinary Pascal steps move j by $-1, 0, 1$ with weights $1, 2, 1$; the extra term raises the middle weight to three on $j \geq 0$. The initial odd row places one at each of $j = 0, -1$. These statements also hold

at the finite row boundaries after extending the rows by zero, because the extra predecessor there is zero.

Let $U_j(z)$ sum these odd-row entries over $m \geq 0$. Write

$$r_+ = \frac{1-3z-\sqrt{(1-z)(1-5z)}}{2z}, \qquad r_- = \frac{1-2z-\sqrt{1-4z}}{2z}.$$

Both roots have zero constant term. The unique solution with finite support at each coefficient is $U_j = Ar_+^j$ for $j \geq 0$ and $U_{-1-j} = Br_-^j$ for $j \geq 0$. The two boundary equations are $A/r_+ - B = 1/z$ and $B/r_- - A = 1/z$. Therefore

$$A = \frac{r_+(1+r_-)}{z(1-r_+r_-)}, \qquad B = \frac{r_-(1+r_+)}{z(1-r_+r_-)}.$$

Their sum over all endpoints simplifies exactly to

$$H(z) = \frac{A}{1-r_+} + \frac{B}{1-r_-} = \frac{\sqrt{(1-z)/((1-4z)(1-5z))}-1}{2z}.$$

The accompanying certificate reduces this equality modulo the two quadratic radical equations, so it is an algebraic identity, not a fit. Every positive even row has twice the preceding odd row sum, whence $F(x) = 1 + (x+2x^2)H(x^2)$. This is exactly the conjectured generating function printed below. Applying the differential certificate also proves the conjectural recurrence. These are one package; no asymptotic claim is needed for its count.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} D &= -(x-1)(x+1)(2x-1)(2x+1)\left(5x^2-1\right), \\ R_0 &= -\frac{1}{2x}, \\ R_1 &= \frac{1}{2x(2x-1)(5x^2-1)}, \end{aligned}$$

Exact substitution gives $LF = P(x) = 5x^4 - 8x^3 - 6x^2 + 2x + 1$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 30 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.13 A052736

Source and selected claim. A052736, checked 20 October 2026. Conjecture. The printed identity requires the lower-bound correction proved below. **Offset and initial data.** Offset 0.

$$(a_0, \ldots, a_7) = (0, 0, 2, 24, 384, 8160, 218880, 7116480).$$

Proof. The source defines a_n by its exponential generating function

$$F(x) = \frac{1-3x-\sqrt{1-6x+x^2}-x(1-x-\sqrt{1-6x+x^2})}{2}.$$

Let b_n denote A006319. Its known ordinary generating function, credited in that entry to Catalani (2003), is

$$B(x) = \frac{(1-x)^2-(1-x)\sqrt{1-6x+x^2}}{2x}.$$

For example it follows from $B = C(x/(1-x)^2)$, the standard Catalan composition for the royal-path sequence. Direct subtraction gives the exact identity $F(x) = x(B(x) - 1)$. Consequently

$$a_0 = a_1 = 0, \qquad a_n = n!\, b_{n-1} \quad (n \geq 2).$$

The entry's conjecture omits this lower-bound qualification: at $n = 1$, $a_1 = 0$ but $1!b_0 = 1$. The result is therefore recorded as a corrected boundary statement, with a counterexample and a full-range proof, rather than as a proof of the literally unqualified claim. The existing unmarked recurrence in A052736 is not counted.

Verification. The standalone verifier reproduces the exact identity, compares 20 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.14 A071715

Source and selected claim. A071715, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{3} p_j(n) a_{n-j} = 0 \qquad (n \geq 3).$$

$$\begin{aligned}
p_0(n) &= (n+7)\,(891n + 6824)\,, \\
p_1(n) &= 891n^2 - 38453n - 250970, \\
p_2(n) &= -4\left(5040n^2 + 4271n - 8095\right), \\
p_3(n) &= 4\,(2n-1)\,(1170n + 3847)\,,
\end{aligned}$$

Offset and initial data. Offset 0.

$$(a_0, \ldots, a_7) = (1, 5, 22, 92, 373, 1482, 5810, 22576)\,.$$

Proof. The defining generating function is $F = C(x)^4 + xC(x)^8$, where $C = 1 + xC^2$ and $C(0) = 1$. Put $t = C - 1$. Then $x = t/(1+t)^2$ and $F = (1+t)^4 + t(1+t)^6$. This parameter change is invertible as a formal power series since dx/dt is one at zero. Euler differentiation becomes $\theta = t(1+t)/(1-t)\, d/dt$. Applying the displayed conjectural operator to this rational expression gives the exact polynomial residual below and proves the recurrence for all indicated indices. The entry's 2025 binomial formula is a known alternative description, not an additional claim of this package.

Exact certificate. $x = \frac{t}{(t+1)^2}$, $F = t\,(t+1)^6 + (t+1)^4$. Exact substitution gives $LF = P(x) = 20068x + 47768$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 26 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.15 A098520

Source and selected claim. A098520, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{2} p_j(n) a_{n-j} = 0 \qquad (n \geq 2).$$

$$\begin{aligned}
p_0(n) &= (n-1)\,(n+1)\,, \\
p_1(n) &= -n\,(2n-1)\,,
\end{aligned}$$

$$p_2(n) = -15n\,(n-1)\,,$$

Offset and initial data. Offset 0.

$$(a_0, \ldots, a_7) = (0, 1, 2, 15, 52, 285, 1206, 6027)\,.$$

Proof. Expand the defining exponential generating function $e^x I_1(4x)/2$. Since $I_1(4x)/2 = \sum_{k\geq 0} 4^k x^{2k+1}/(k!(k+1)!)$, its coefficients are

$$a_n = \sum_{2k+1\leq n} \frac{n!\,4^k}{k!(k+1)!(n-2k-1)!}.$$

Summing first in n gives

$$F(x) = \frac{x}{(1-x)^2} \sum_{k\geq 0} \binom{2k+1}{k} \left(\frac{4x^2}{(1-x)^2}\right)^k.$$

Use $\sum_{k\geq 0} \binom{2k+1}{k} z^k = (1/\sqrt{1-4z} - 1)/(2z)$. After simplification, $F = ((1-x)/\sqrt{1-2x-15x^2} - 1)/(8x)$. The displayed radical differential certificate now proves the selected recurrence. This argument fixes $a_0 = 0$ and the offset without using the conjectural recurrence to generate data.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$\begin{aligned} D &= -\,(3x+1)\,(5x-1)\,,\\ R_0 &= -\frac{1}{8x},\\ R_1 &= \frac{x-1}{8x\,(3x+1)\,(5x-1)}, \end{aligned}$$

Exact substitution gives $LF = P(x) = 0$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 27 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.16 A123071

Source and selected claim. A123071, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{7} p_j(n) a_{n-j} = 0 \qquad (n \geq 7).$$

$$\begin{aligned} p_0(n) &= 2,\\ p_1(n) &= 1,\\ p_2(n) &= -2n,\\ p_3(n) &= -n - 10,\\ p_4(n) &= -2\,(n-2)\,(n+2)\,,\\ p_5(n) &= -n^2 - 2n + 23,\\ p_6(n) &= 2\,(n-5)\left(n^2 - 7n + 11\right),\\ p_7(n) &= (n-6)\,(n-5)^2\,, \end{aligned}$$

Offset and initial data. Offset 0.

$$(a_0, \ldots, a_7) = (1, 2, 4, 12, 36, 120, 400, 1520)\,.$$

Proof. Use the sequence specification $a_n = B(n)B(n+1)$ in A123071, with $B(2m) = B(2m+1) = u_m$ and

$$u_0 = 1, \quad u_1 = 2, \quad u_m = 2u_{m-1} + 2(m-1)u_{m-2} \quad (m \geq 2).$$

This is the established symmetric-involution sequence A000898; the source attributes the bishop interpretation to Robinson (1976), equation (24). Thus $a_{2m} = u_m^2$ and $a_{2m+1} = u_m u_{m+1}$. The elementary involution recurrence also follows by choosing whether a distinguished item is a singleton (two colors) or belongs to a pair (two colors and $m-1$ choices for its partner).

The proof of the selected order-seven recurrence is a finite polynomial identity in two independent indeterminates. For even $n = 2m$, $m \geq 4$, set $u_{m-4} = X, u_{m-3} = Y$ and use the displayed second-order recurrence to express u_{m-2}, u_{m-1}, u_m. Substituting the eight products into $\sum_{j=0}^{7} p_j(2m)a_{2m-j}$ gives the zero polynomial in m, X, Y. For odd $n = 2m+1$, $m \geq 3$, start with $u_{m-3} = X, u_{m-2} = Y$, extend through u_{m+1}, and the analogous residual is again identically zero. The machine-readable certificate records every product and the zero polynomial in each parity. These two identities cover every $n \geq 7$; they require no induction based on the target recurrence and no division by a possibly zero coefficient.

Verification. The standalone verifier reproduces the exact identity, compares 26 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.17 A143013

Source and selected claim. A143013, checked 20 October 2026. Conjecture. The selected polynomial recurrence is

$$\sum_{j=0}^{3} p_j(n)a_{n-j} = 0 \qquad (n \geq 3).$$

$$\begin{aligned} p_0(n) &= n + 2, \\ p_1(n) &= -2n - 1, \\ p_2(n) &= -3\,(n-1)\,, \\ p_3(n) &= -2\,(2n-5)\,, \end{aligned}$$

Offset and initial data. Offset 0.

$$(a_0, \ldots, a_7) = (1, 2, 3, 7, 17, 43, 114, 310)\,.$$

Proof. In the source's precise word specification the special level step F may be followed only by a down step or the end of the word. At the ground level a nonempty admissible Motzkin path is either the singleton F, an ordinary level step followed by a path, or a first-return $UADB$, with A, B admissible paths. Consequently its generating function obeys $F(x) = 1 + x + xF(x) + x^2F(x)^2$. The root with constant term one is $(1 - x - \sqrt{1 - 2x - 3x^2 - 4x^3})/(2x^2)$. The linked 2023 papers give this known algebraic generating function and related enumerations; the selected claim here is the particular polynomial recurrence still marked conjectural. Its exact differential residual is given below, completing its full-range proof.

Exact certificate. $F = R_0 + R_1\sqrt{D}$, with

$$D = -4x^3 - 3x^2 - 2x + 1,$$

$$R_0 = -\frac{x-1}{2x^2},$$
$$R_1 = -\frac{1}{2x^2},$$

Exact substitution gives $LF = P(x) = -x^2 + 3x + 2$. The coefficient of the radical in the residual is identically zero. **Verification.** The standalone verifier reproduces the exact identity, compares 28 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.18 A372018

Source and selected claim. A372018, checked 20 October 2026. Conjecture. The selected generating-function or coefficient identity is stated and proved below. **Offset and initial data.** Offset 0.

$$(a_0, \ldots, a_7) = (1, 2, 4, 10, 30, 98, 336, 1194)\,.$$

Proof. Write $A(x) = \sum a_n x^n$ for the defining series. Squaring its defining square-root equation and using the branch $A(0) = 1$ gives

$$xA^3 - A^2 + 3xA + 1 = 0.$$

Separate parity by $A(x) = E(z) + xO(z)$, $z = x^2$. The odd part of this equation is $E(E^2 + 3zO^2 - 2O + 3) = 0$. Since $E(0) = 1$, it is a unit, and hence $E^2 = 2O - 3 - 3zO^2$. The even part is $E^2(3zO - 1) + z^2O^3 - zO^2 + 3zO + 1 = 0$. Eliminating E^2 yields

$$4z^2O^3 - 4zO^2 + 3zO + O - 2 = 0.$$

For $B = O/2$ this becomes $B(1 - 4zB)^2 = 1 - 3zB$, with $B(0) = 1$. If $W = zB$, then W is the compositional inverse of $u(1-4u)^2/(1-3u)$, exactly the definition of A371364. The derivative of the last implicit equation with respect to B is one at $(z, B) = (0, 1)$, so that formal solution is unique. Therefore

$$a_{2n+1} = 2\,\mathrm{A371364}(n) \qquad (n \geq 0).$$

The source prints empty parentheses after A371364; the intended index n is made explicit here. Neither a separate unmarked recurrence in A372018 nor the partner sequence is counted as another package.

Verification. The standalone verifier reproduces the exact identity, compares 27 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.19 A212341

Source and selected claim. A212341, checked 20 October 2026. Empirical generating function. The selected generating-function or coefficient identity is stated and proved below. **Offset and initial data.** Offset 5.

$$(a_5, \ldots, a_{12}) = (14, 56, 188, 603, 1907, 5615, 15968, 44464)\,.$$

Proof. Let $Q(x, u)$ count 132-avoiding permutations, with u marking entries having at least four smaller entries to their left. This is the source's quadrant mesh statistic. Split a permutation at its maximum. All values on the left exceed those on the right, so the smaller-to-left statistic in either subpermutation is unchanged. The maximum is marked precisely when its left subtree has at least four entries. Subtrees of sizes zero through three have no marked entries and have Catalan counts $P(x) = 1 + x + 2x^2 + 5x^3$. Therefore

$$Q = 1 + xPQ + xu(Q - P)Q.$$

This is the bivariate equation of Kitaev–Remmel–Tiefenbruck (2012), Theorem 9; it is credited as a known starting point. Put $Q_0 = Q(x,0) = 1/(1-xP)$ and $Q_1 = [u]Q$. Taking the coefficient of u in the equation gives $(1-xP)Q_1 = x(Q_0 - P)Q_0$, and thus

$$Q_1 = \frac{x(1-P+xP^2)}{(1-xP)^3} = \frac{x^5(14+14x+20x^2+25x^3)}{(1-x-x^2-2x^3-5x^4)^3}.$$

This is the entry's empirical generating function, now proved for every coefficient. Section 4 of the cited paper gives the bivariate formula and initial terms and explicitly leaves this coefficient extraction without a simple formula; the present one-line extraction is the selected assertion. The initial zero coefficients through degree four explain the source offset five.

Verification. The standalone verifier reproduces the exact identity, compares 26 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.4.20 A110668

Source and selected claim. A110668, checked 20 October 2026. Conjecture. The selected generating-function or coefficient identity is stated and proved below. **Offset and initial data.** Offset 0.

$$(a_0, \dots, a_7) = (0, 1, 3, 3, -3, -15, -27, -32)\,.$$

Proof. Let $u_n = a(0,n)$ denote the base row in A110665 and let $U_m(x) = \sum_{n\ge 0} a(m,n)x^n$. The defining array recurrence and the zero first column give $(1-x)U_m = U_{m-1}$, hence $U_3 = U_0/(1-x)^3$. The base-row generating function $U_0 = x(1-2x)/(1-x+x^2)^2$ is already credited to Adams-Watters (2006) in A110665. For completeness it can be recovered from the diagonal condition alone: if $C = 1 + xC^2$ and $z = xC$, then

$$\sum_{n\ge 0} a(n,n)x^n = \frac{U_0(z)}{C(x)\sqrt{1-4x}}, \qquad x = z(1-z).$$

Indeed the coefficient of u_k on the diagonal is $\binom{2n-k-1}{n-1}$ for $n \ge k \ge 1$, whose generating function is $(xC)^k/(C(x)\sqrt{1-4x})$. Since the diagonal equals n, the left side is $x/(1-x)^2$. Substituting $x = z(1-z)$, $C(x) = 1/(1-z)$, and $\sqrt{1-4x} = 1-2z$ gives the stated U_0. Thus

$$U_3(x) = \frac{x(1-2x)}{(1-x+x^2)^2(1-x)^3} = \frac{x(-1+2x)}{(x^2-x+1)^2(x-1)^3},$$

which proves the selected conjectural formula. The base row and the other rows are credited or uncounted; this package selects only A110668.

Verification. The standalone verifier reproduces the exact identity, compares 54 published terms with an independent definition calculation, and checks all applicable indices through 100.

9.5 Empirical array recurrences: 62 selected packages

This section proves 62 entry-level recurrences explicitly marked *Empirical* on primary OEIS pages retrieved on 20 October 2026. The local array definitions and recurrence conjectures are due to R. H. Hardin. In 20 of the same packages, a source rational generating function, credited to Colin Barker, is also proved. One selected A-number contributes one package. No additional count is assigned to its generating function or to a common theorem.

9.5.1 Four local predicates and labeled counting

For a block $B = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, let

$$\mathcal{D}(B) = (b-a, d-b, c-d, a-c, d-a, c-b).$$

These are all six pairwise differences among its four positions, with signs chosen immaterially for squares or absolute values. Each selected sequence counts arrays with $n+1$ rows, width $w \geq 2$, and entries in $\{0, \ldots, q-1\}$, subject to one of the following requirements in every 2×2 block:

Stress (D). $|a + d - b - c| = s$.
Absolute six-difference sum (U). $\sum_{u \in \mathcal{D}(B)} |u| = s$.
Squared six-difference sum (V). $\sum_{u \in \mathcal{D}(B)} u^2 = s$.
Bounded six-difference range (B). $|u| \leq s$ for every $u \in \mathcal{D}(B)$, equivalently $\max(a,b,c,d) - \min(a,b,c,d) \leq s$.

An additional inequality condition is imposed only where the source requires every horizontally or vertically adjacent pair to be unequal. In a block this says $a \neq b$, $b \neq d$, $d \neq c$, and $c \neq a$. Every edge belongs to a block when both array dimensions are at least two, so this also treats boundary edges correctly.

All counts retain the given numeric labels. Symbols may be absent; there is no requirement to use every symbol and no quotient by color permutations or geometric symmetries. The row orbits used below are only a compression of the exact labeled count.

9.5.2 A transfer obtained directly from the definition

For a row $r \in \{0, \ldots, q-1\}^w$, let $\gamma(r)$ be the lexicographically least among r, its reversal, its simultaneous complement $q-1-r$, and the reversal of that complement. Let $\mathcal{O}_{q,w}$ be the sorted set of representatives. Define

$$m(r) = \#\{u \in \{0, \ldots, q-1\}^w : \gamma(u) = r\}, \tag{9.2}$$

$$T(r,t) = \#\left\{ u \in \{0, \ldots, q-1\}^w : \begin{array}{l} \gamma(u) = t, \\ P(r_i, r_{i+1}, u_i, u_{i+1}) \quad (1 \leq i < w) \end{array} \right\}. \tag{9.3}$$

Here P is the complete local predicate specified above, including the edge inequalities when required. Reversal and complementation preserve each predicate and permute every target orbit when applied to both rows. Consequently every row in a source orbit has the same number of successors in each target orbit. The weighted quotient is therefore equitable, and adding rows gives

$$a(n) = m^{\mathsf{T}} T^n \mathbf{1} \qquad (n \geq 1). \tag{9.4}$$

The exponent n counts the joins between $n+1$ rows. The mass sum is $\sum_r m(r) = q^w$; short orbits, fixed rows, and zero-successor rows retain their actual multiplicities. The auxiliary value at $n = 0$ is not asserted to be an OEIS term; all selected sources have offset 1.

9.5.3 An exact finite identity proves all later indices

For the source signature $c = (c_1, \ldots, c_d)$, define

$$R(z) = z^d - \sum_{i=1}^{d} c_i z^{d-i}, \qquad Q(x) = 1 - \sum_{i=1}^{d} c_i x^i.$$

The full integer certificate for every selected entry verifies

$$\boxed{TR(T)\mathbf{1} = \mathbf{0}.} \tag{9.5}$$

Multiplication by $m^{\mathsf{T}}T^{n-d-1}$ and (9.4) prove

$$\boxed{a(n)=\sum_{i=1}^{d} c_i a(n-i) \qquad (n \geq d+1).} \tag{9.6}$$

Thus every index on the right belongs to the original offset-one sequence. No recurrence is fitted to a numerical prefix, and no conjectural generating function is assumed.

The 62 complete matrices have 10537 states and 30926 nonzero weighted edges in total. The largest dimension is 1056; the largest source recurrence order is 56. For every matrix the data retain the representative rows, all orbit masses, every weighted transition, and every coordinate of $v_k = T^k\mathbf{1}$ for $0 \leq k \leq d+1$. The decisive calculation is

```
v[0] = [1 for every state]
for k = 1,...,d+1:
    v[k][r] = sum(T[r,s]*v[k-1][s] for every state s)
for every state r:
    assert v[d+1][r] == sum(c[i]*v[d+1-i][r] for i=1,...,d)
```

The verifier rebuilds the matrix from the definition and recomputes all coordinates with exact integer arithmetic. This is a finite computer-assisted proof of an all-index assertion. The full parameter and coefficient lists below specify the calculation without relying on opaque numerical output. Expanded data are in `r9_arrays_matrices.json` and `r9_arrays_certificates.json`.

All 1112 displayed source terms match the defining transfers. A separate program, which does not import this engine, uses labeled column-by-column dynamic programming to obtain two-row and three-row counts. All 124 values agree. These checks confirm interpretation and offset; the vector identity supplies the universal proof.

For $F(x)=\sum_{n\geq 1} a(n)x^n$, the recurrence gives $QF=P$ with $\deg P \leq d$, where

$$[x^k]P = \sum_{\substack{0\leq i\leq \min(k,d)\\ k-i\geq 1}} q_i a(k-i), \quad q_0=1, \quad q_i=-c_i \ (i\geq 1). \tag{9.7}$$

Exact polynomial cross-multiplication proves agreement with all 20 source rational generating functions. Their numerators are printed in the corresponding packages; other source formulas are not counted. For every order-d recurrence, the certificate file also retains all initial values $a(1),\ldots,a(d)$, computed from the defining transfer; the entry paragraphs print the first three values for orientation.

9.5.4 A hand-checkable example: A234154

For $q=4$, width $w=2$, and predicate B with $s=1$, the orbit representatives are $00, 01, 02, 03, 11, 12$. Direct enumeration gives

$$T=\begin{pmatrix} 1 & 2 & 0 & 0 & 1 & 0\\ 1 & 2 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0\\ 1 & 2 & 0 & 0 & 2 & 2\\ 0 & 0 & 0 & 0 & 2 & 2 \end{pmatrix}, \qquad m=(2,4,4,2,2,2)^{\mathsf{T}}.$$

The successive column vectors are

$$v_1=(4,4,0,0,7,4)^{\mathsf{T}}, \qquad v_2=(19,19,0,0,34,22)^{\mathsf{T}},$$
$$v_3=(91,91,0,0,169,112)^{\mathsf{T}}, \qquad v_4=(442,442,0,0,835,562)^{\mathsf{T}}.$$

Every coordinate satisfies $v_4 = 7v_3 - 9v_2 - 6v_1$. It follows that

$$a(n) = 7a(n-1) - 9a(n-2) - 6a(n-3) \qquad (n \geq 4).$$

The first values $46, 226, 1108$ come from $m^{\mathsf{T}} v_i$, so

$$F(x) = \frac{46x - 96x^2 - 60x^3}{1 - 7x + 9x^2 + 6x^3}.$$

This proves both empirical formulas on https://oeis.org/A234154.

9.5.5 Source status, prior work and audit limits

All selected primary entries were retrieved for this round and explicitly mark the selected recurrence empirical. No proof or established encompassing recurrence was identified in their consulted content. The 24 associated parent tables for the selected and reserve packages were sought separately: 21 were retrieved and their relevant statements are also empirical. The contents of A233883, A234114, and A234175 could not be retrieved after ordinary, internal-format and public text-endpoint attempts. Their proof status is not claimed to have been audited. The affected child entries remain directly sourced, and their mathematical certificates are independent of those inaccessible parent pages.

Retrieval dates and reported cache ages are retained separately. Some retrieved copies predate this round; no claim is made that they reflect every live edit. Exact-ID and family searches did not identify a prior proof among the inspected results, but several searches had poor recall or irrelevant results. This bounded audit does not establish worldwide priority.

The selected IDs are absent from all 808 earlier selected IDs and from the retained 97 distinct IDs in the 100 costly claims of Appendix A of *OEIS Open*. The latter comparison is not a full audit of that project's repository. Previously proved array reserves were also excluded from new selection.

The alias screen compares each new sequence with 289 retained array packages from Rounds II–VIII, including four earlier uncounted reserves, and with earlier entries in this new set. It compares 20 exact values at each shift $-3, \ldots, 3$, allowing an arbitrary rational scale. Every proposed equality has a mismatch within the first 3 compared values. This disproves those specific shift-and-scale identities; unbounded shifts and other transformations are not covered. Shared recurrence polynomials alone do not identify two different counting sequences.

One further fresh certificate is retained as an uncounted reserve. None was proved in an earlier round. No OEIS submission, edit, or message was performed, and no official closure is claimed.

9.5.6 Entry-level coefficients and certificates

In each package below, the displayed coefficient signature is ordered as $(c_1, \ldots, c_d)$ in (9.6). Every zero coefficient is retained. All arrays have offset 1.

9.5.6.1 A233787

Arrays have $n + 1$ rows, width $w = 4$, and labeled alphabet $0, \ldots, 3$. Predicate V has target $s = 11$, allowing equal adjacent values. The recurrence holds for every $n \geq 35$, with order $d = 34$ and a complete $D = 72$ state matrix. The first three source values are $(720, 3932, 20816)$.

```
0, 139, 0, -7929, 0, 248319, 0, -4798442, 0, 60800038, 0, -523147257, 0, 3118918441,
0, -13016374051, 0, 38139962547, 0, -78218266028, 0, 111158046394, 0, -107387771932,
0, 68279402362, 0, -27119287096, 0, 6185757504, 0, -702040320, 0, 29030400
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A233787; parent column 3 of https://oeis.org/A233792. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.2 A233877

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,2$. Predicate V has target $s=11$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=10$ state matrix. The first three source values are $(76, 300, 1224)$.

$$(c_1,\dots,c_d) = (3, 19, -60, 8, 36, -12).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A233877; parent column 2 of https://oeis.org/A233883. The primary entry marks the recurrence empirical; the parent content was not retrieved, as disclosed in the source audit. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 76x + 72x^2 - 1120x^3 + 344x^4 + 688x^5 - 256x^6.$$

9.5.6.3 A233878

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,2$. Predicate V has target $s=11$, allowing equal adjacent values. The recurrence holds for every $n \geq 13$, with order $d=12$ and a complete $D=25$ state matrix. The first three source values are $(240, 1224, 6200)$.

```
0, 56, 0, -819, 0, 2791, 0, -3096, 0, 961, 0, -84
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A233878; parent column 3 of https://oeis.org/A233883. The primary entry marks the recurrence empirical; the parent content was not retrieved, as disclosed in the source audit.

9.5.6.4 A233879

Arrays have $n+1$ rows, width $w=5$, and labeled alphabet $0,\dots,2$. Predicate V has target $s=11$, allowing equal adjacent values. The recurrence holds for every $n \geq 30$, with order $d=29$ and a complete $D=70$ state matrix. The first three source values are $(760, 5156, 33656)$.

```
10, 111, -1347, -2952, 61962, -23589, -1303743, 2061738, 13676265, -32627870,
-74205472, 248050297, 184715702, -1058894761, 18879451, 2642739884, -1291495298,
-3771977707, 3244720225, 2774675096, -3693043205, -677649206, 2114258648, -285661000,
-570117698, 190279800, 48545460, -28512000, 2851200
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A233879; parent column 4 of https://oeis.org/A233883. The primary entry marks the recurrence empirical; the parent content was not retrieved, as disclosed in the source audit.

9.5.6.5 A233880

Arrays have $n+1$ rows, width $w=6$, and labeled alphabet $0,\dots,2$. Predicate V has target $s=11$, allowing equal adjacent values. The recurrence holds for every $n \geq 53$, with order $d=52$ and a complete $D=196$ state matrix. The first three source values are $(2400, 22020, 178704)$.

```
0, 390, 0, -63115, 0, 5633508, 0, -311287762, 0, 11395143071, 0, -290608508897, 0,
5364876782159, 0, -73770016903887, 0, 771455610322102, 0, -6226548780129951, 0,
39172372910038001, 0, -193237451888156699, 0, 749448322438960607, 0,
-2284924382785662062, 0, 5461859355511846682, 0, -10185644885765121071, 0,
14709447103939892100, 0, -16281042118217455735, 0, 13617119182197701100, 0,
-8438611741506693900, 0, 3769223995920232000, 0, -1166605374110880000, 0,
236321446856400000, 0, -28818965889000000, 0, 1865383020000000, 0, -48478500000000
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A233880; parent column 5 of https://oeis.org/A233883. The primary entry marks the recurrence empirical; the parent content was not retrieved, as disclosed in the source audit.

9.5.6.6 A233960

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,3$. Predicate U has target $s=10$, allowing equal adjacent values. The recurrence holds for every $n \geq 6$, with order $d=5$ and a complete $D=6$ state matrix. The first three source values are $(48, 188, 728)$.

$$(c_1,\dots,c_d) = (4, 10, -36, -8, 16).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A233960; parent column 1 of https://oeis.org/A233967. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 48x - 4x^2 - 504x^3 - 48x^4 + 224x^5.$$

9.5.6.7 A233961

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,3$. Predicate U has target $s=10$, allowing equal adjacent values. The recurrence holds for every $n \geq 15$, with order $d=14$ and a complete $D=20$ state matrix. The first three source values are $(188, 864, 3976)$.

```
9, 6, -231, 402, 1164, -3026, -973, 5685, -1687, -2838, 1862, -188, -88, 16
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A233961; parent column 2 of https://oeis.org/A233967. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.8 A233962

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,3$. Predicate U has target $s=10$, allowing equal adjacent values. The recurrence holds for every $n \geq 44$, with order $d=43$ and a complete $D=72$ state matrix. The first three source values are $(728, 3976, 21272)$.

```
15, 31, -1396, 2653, 51714, -174135, -1048642, 4568556, 13206552, -69675134,
-110040243, 692530682, 626223287, -4730984068, -2448897493, 22858856946, 6357801357,
-79394291518, -9325291758, 199923021296, -23907331, -365908349411, 34128121167,
485101885188, -82901695912, -461141546148, 108852864503, 308761092196, -88956808820,
-141933147555, 45953728822, 43469138372, -14674989574, -8649583694, 2801908488,
1098165952, -309043744, -85338496, 18684416, 3638272, -557056, -64000, 6144
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A233962; parent column 3 of https://oeis.org/A233967. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.9 A234029

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,4$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 5$, with order $d=4$ and a complete $D=9$ state matrix. The first three source values are $(61, 307, 1543)$.

$$(c_1,\dots,c_d) = (7, -6, -18, -6).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234029; parent column 1 of https://oeis.org/A234036. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 61x - 120x^2 - 240x^3 - 78x^4.$$

9.5.6.10 A234030

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,4$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 10$, with order $d=9$ and a complete $D=39$ state matrix. The first three source values are $(307, 3059, 30263)$.

```
18, -79, -56, 459, 56, -597, -88, 90, -10
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234030; parent column 2 of https://oeis.org/A234036. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.11 A234031

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,4$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 17$, with order $d=16$ and a complete $D=169$ state matrix. The first three source values are $(1543, 30263, 591711)$.

```
41, -473, 378, 15090, -24456, -191256, 160111, 1124270, 180876, -2156866, -1359614,
738304, 501820, -106672, -46728, 7920
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234031; parent column 3 of https://oeis.org/A234036. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.12 A234032

Arrays have $n+1$ rows, width $w=5$, and labeled alphabet $0,\dots,4$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 45$, with order $d=44$ and a complete $D=819$ state matrix. The first three source values are $(7783, 300597, 11637549)$.

```
110, -4333, 67258, -49727, -9381974, 63529143, 464873722, -5105834204, -11512298940,
199030863273, 171749489662, -4640901058548, -2279932923269, 69427761078243,
37219734180920, -679717373578792, -517634627173445, 4315879589285745,
4575357081875100, -17189611279421840, -24043742578423549, 40454429041355663,
73834278266333876, -51489559946124003, -132873735034278651, 30290331332947967,
146882875991746168, -208125301638949, -104800983506559948, -9888619515464597,
50085499296310162, 5296975913473720, -16275620572050062, -948007832112512,
3507290470767932, -86886144072040, -454894672567616, 54652514222304, 27282841319424,
-5902990277920, -173722766592, 96048316160, -753748992, -473149440
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234032; parent column 4 of https://oeis.org/A234036. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.13 A234107

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,4$. Predicate U has target $s=12$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=9$ state matrix. The first three source values are $(56, 186, 604)$.

$$(c_1,\dots,c_d) = (5, 6, -50, 16, 80, -16).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234107; parent column 1 of https://oeis.org/A234114. The primary entry marks the recurrence empirical; the parent content was not retrieved, as disclosed in the source audit. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 56x - 94x^2 - 662x^3 + 728x^4 + 1424x^5 - 304x^6.$$

9.5.6.14 A234108

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,4$. Predicate U has target $s=12$, allowing equal adjacent values. The recurrence holds for every $n \geq 19$, with order $d=18$ and a complete $D=39$ state matrix. The first three source values are $(186, 608, 2086)$.

```
8, 6, -172, 144, 1438, -1805, -6160, 8467, 14464, -20123, -17822, 25464, 9216, -15948,
136, 3552, -928, 64
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234108; parent column 2 of https://oeis.org/A234114. The primary entry marks the recurrence empirical; the parent content was not retrieved, as disclosed in the source audit.

9.5.6.15 A234109

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,4$. Predicate U has target $s=12$, allowing equal adjacent values. The recurrence holds for every $n \geq 41$, with order $d=40$ and a complete $D=169$ state matrix. The first three source values are $(604, 2086, 7732)$.

```
7, 60, -438, -1782, 12360, 35330, -205978, -512290, 2229224, 5503649, -16243635,
-43372518, 79850408, 248228062, -254990128, -1024802383, 454865089, 3032850159,
-63270053, -6374308278, -1896654228, 9366697820, 5199067584, -9374661980, -7431981604,
6110120008, 6536677712, -2372079792, -3650095360, 416839296, 1285380768, 31536704,
-277159232, -25366784, 34816896, 3406336, -2369024, -107520, 69632, -4096
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234109; parent column 3 of https://oeis.org/A234114. The primary entry marks the recurrence empirical; the parent content was not retrieved, as disclosed in the source audit.

9.5.6.16 A234117

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,2$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 6$, with order $d=5$ and a complete $D=25$ state matrix. The first three source values are $(673, 12593, 231713)$.

$$(c_1,\dots,c_d) = (20, -10, -324, -277, 144).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234117; parent column 3 of https://oeis.org/A234122. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 673x - 867x^2 - 13417x^3 - 11213x^4 + 5904x^5.$$

9.5.6.17 A234118

Arrays have $n+1$ rows, width $w=5$, and labeled alphabet $0,\dots,2$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 12$, with order $d=11$ and a complete $D=70$ state matrix. The first three source values are $(3127, 116801, 4279065)$.

```
50, -427, -2748, 22158, 50476, -220478, -236864, 237555, 152850, -74775, -7956
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234118; parent column 4 of https://oeis.org/A234122. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.18 A234119

Arrays have $n+1$ rows, width $w=6$, and labeled alphabet $0,\dots,2$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 18$, with order $d=17$ and a complete $D=196$ state matrix. The first three source values are $(14527, 1082977, 79003521)$.

```
104, -1893, -33284, 535522, 4829920, -36672922, -308181352, 266551299, 4516883912,
2174714575, -20675388468, -11598332564, 40919536464, 9741869952, -32943946752,
6233518080, 2468413440
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234119; parent column 5 of https://oeis.org/A234122. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.19 A234124

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,6$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=16$ state matrix. The first three source values are $(91, 469, 2413)$.

$$(c_1,\dots,c_d) = (10,-24,-18,57,54,12).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234124; parent column 1 of https://oeis.org/A234131. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 91x - 441x^2 - 93x^3 + 1221x^4 + 1038x^5 + 228x^6.$$

9.5.6.20 A234125

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,6$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 15$, with order $d=14$ and a complete $D=100$ state matrix. The first three source values are $(469, 4757, 47933)$.

```
26, -218, 474, 1603, -5034, -4660, 14440, 7897, -13910, -6694, 2654, 581, -250, 20
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234125; parent column 2 of https://oeis.org/A234131. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.21 A234126

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,6$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 28$, with order $d=27$ and a complete $D=625$ state matrix. The first three source values are $(2413, 47933, 951709)$.

```
62, -1377, 11706, 3232, -539522, 1362867, 10121999, -31760118, -116725179, 307724737,
923620665, -1359253203, -4570974318, 1827227267, 11806964282, 3166332198,
-13354128233, -8963914586, 4581307330, 4739020728, -551989326, -1042026878, 26127344,
107962116, -4177212, -4538160, 475200
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234126; parent column 3 of https://oeis.org/A234131. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.22 A234133

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,3$. Predicate U has target $s=9$, allowing equal adjacent values. The recurrence holds for every $n \geq 4$, with order $d=3$ and a complete $D=6$ state matrix. The first three source values are $(32, 80, 192)$.

$$(c_1,\dots,c_d) = (2, 8, -16).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234133; parent column 1 of https://oeis.org/A234140. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 32x + 16x^2 - 224x^3.$$

9.5.6.23 A234135

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,3$. Predicate U has target $s=9$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=72$ state matrix. The first three source values are $(192, 296, 488)$.

$$(c_1,\dots,c_d) = (3, 8, -30, 4, 48, -32).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234135; parent column 3 of https://oeis.org/A234140. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 192x - 280x^2 - 1936x^3 + 2896x^4 + 3296x^5 - 4672x^6.$$

9.5.6.24 A234154

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,3$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 4$, with order $d=3$ and a complete $D=6$ state matrix. The first three source values are $(46, 226, 1108)$.

$$(c_1,\dots,c_d) = (7, -9, -6).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234154; parent column 1 of https://oeis.org/A234161. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 46x - 96x^2 - 60x^3.$$

9.5.6.25 A234155

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,3$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=20$ state matrix. The first three source values are $(226, 2210, 21428)$.

$$(c_1,\dots,c_d) = (16, -59, -31, 118, 35, -10).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234155; parent column 2 of https://oeis.org/A234161. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 226x - 1406x^2 - 598x^3 + 3078x^4 + 900x^5 - 260x^6.$$

9.5.6.26 A234156

Arrays have $n + 1$ rows, width $w = 4$, and labeled alphabet $0, \ldots, 3$. Predicate B has target $s = 1$, allowing equal adjacent values. The recurrence holds for every $n \geq 12$, with order $d = 11$ and a complete $D = 72$ state matrix. The first three source values are $(1108, 21428, 411712)$.

```
38, -412, 740, 5548, -11466, -28213, 27544, 43860, -11608, -8172, 1584
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234156; parent column 3 of https://oeis.org/A234161. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.27 A234157

Arrays have $n + 1$ rows, width $w = 5$, and labeled alphabet $0, \ldots, 3$. Predicate B has target $s = 1$, allowing equal adjacent values. The recurrence holds for every $n \geq 27$, with order $d = 26$ and a complete $D = 272$ state matrix. The first three source values are $(5446, 208530, 7952532)$.

```
90, -2653, 24943, 89825, -2218609, 1432019, 71490923, -78048525, -1164227315,
761044775, 9924372783, -667076603, -41948446347, -15809876567, 78900501505,
41327430666, -75139994803, -37071907734, 40119015622, 12686351262, -12371945864,
-754232304, 1729058752, -288379744, -6642240, 2957184
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234157; parent column 4 of https://oeis.org/A234161. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.28 A234158

Arrays have $n + 1$ rows, width $w = 6$, and labeled alphabet $0, \ldots, 3$. Predicate B has target $s = 1$, allowing equal adjacent values. The recurrence holds for every $n \geq 54$, with order $d = 53$ and a complete $D = 1056$ state matrix. The first three source values are $(26794, 2031286, 153793284)$.

```
217, -17184, 575274, -4536464, -187521807, 3793789039, 16856009949, -838186556820,
566323425072, 100894927353562, -209284767431095, -7820748769614354, 15947814680267338,
413533369489268493, -552537406964179485, -15031120049686806579, 6646617078976384635,
3706596658964532168 59, 135453696144913493680, -6128560315932176342756,
-6096479285302210528097, 67946244106778846892950, 98850271203664956510736,
-5128657589686915864130 28, -9127554113331561005068 67, 27149781387475527179864 92,
538736656210972055421854 2, -1049436135011112699045 6413, -2146660571844822748656 6037,
309831773884507238567300 13, 59383999283149719935113021, -7242133327338416709065 8391,
-1147674578600906243283174 90, 13507182656937204671095 4792,
151628599152427140815306228, -1955065830854474104842602076,
-1269093361497407457019196 08, 20802378914037392457956 8064, 50774598029282885589249600,
-1512060176309953249488648 64, 11766570188661225224139904, 67386656389693313023099968,
-2358794008976279537695 4112, -1452214043448586654064 1792, 10175560813561814524537344,
535272730559164256256, -1562960541170454941048832, 42577516405397968345 4976,
9006299347542687350784, -19411131518479690629120, 2198238267084808126464,
101315221269019361280, -21080039086463385600
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234158; parent column 5 of https://oeis.org/A234161. Both retrieved entries mark the relevant recurrence empirical. The empirical coefficient list is in the primary entry's linked attachment: https://oeis.org/A234158/a234158.txt.

9.5.6.29 A234170

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=16$ state matrix. The first three source values are $(448, 4536, 47936)$.

$$(c_1,\dots,c_d) = (0, 154, 0, -4144, 0, 16176).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234170; parent column 1 of https://oeis.org/A234175. The primary entry marks the recurrence empirical; the parent content was not retrieved, as disclosed in the source audit. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 448x + 4536x^2 - 21056x^3 - 181440x^4 + 105216x^5 + 787584x^6.$$

9.5.6.30 A234171

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 36$, with order $d=35$ and a complete $D=100$ state matrix. The first three source values are $(4536, 70486, 1173940)$.

```
33, 323, -20479, 66827, 4196961, -32408967, -381895309, 4395837462, 15113145790,
-293548976068, -51289577932, 10922776315912, -16766805352904, -235952101172000,
626750889441184, 2967198544937088, -10881957631245888, -20625630149777664,
106077230575696640, 62012982701743616, -603993107147691008, 80220923857108992,
1983943732124129280, -1120944653090611200, -3571475321182953472, 3062535903549603840,
3148640202567516160, -3550878140723822592, -1004211355174305792, 1711073233311105024,
-31999222443147264, -285978658647048192, 29150317966786560, 14870074192035840,
-1882287872409600
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234171; parent column 2 of https://oeis.org/A234175. The primary entry marks the recurrence empirical; the parent content was not retrieved, as disclosed in the source audit.

9.5.6.31 A234177

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,5$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 6$, with order $d=5$ and a complete $D=12$ state matrix. The first three source values are $(76, 388, 1978)$.

$$(c_1,\dots,c_d) = (10, -27, 3, 36, 12).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234177; parent column 1 of https://oeis.org/A234184. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 76x - 372x^2 + 150x^3 + 588x^4 + 192x^5.$$

9.5.6.32 A234178

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,5$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 12$, with order $d=11$ and a complete $D=63$ state matrix. The first three source values are $(388, 3908, 39098)$.

```
24, -182, 349, 938, -2021, -1914, 2375, 1747, -219, -140, 20
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234178; parent column 2 of https://oeis.org/A234184. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.33 A234179

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,5$. Predicate B has target $s=1$, allowing equal adjacent values. The recurrence holds for every $n \geq 23$, with order $d=22$ and a complete $D=342$ state matrix. The first three source values are $(1978, 39098, 771710)$.

```
59, -1253, 10658, -12393, -283822, 938714, 2905662, -12071951, -18482610, 67209819,
84336539, -168519304, -218405166, 167608180, 253438774, -36105806, -96411410, 1835064,
14870944, -529572, -839520, 95040
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234179; parent column 3 of https://oeis.org/A234184. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.34 A234201

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,5$. Predicate U has target $s=15$, allowing equal adjacent values. The recurrence holds for every $n \geq 11$, with order $d=10$ and a complete $D=12$ state matrix. The first three source values are $(104, 440, 1792)$.

```
0, 36, 0, -396, 0, 1392, 0, -1664, 0, 512
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234201; parent column 1 of https://oeis.org/A234208. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.35 A234210

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,4$. Predicate V has target $s=11$, allowing equal adjacent values. The recurrence holds for every $n \geq 5$, with order $d=4$ and a complete $D=9$ state matrix. The first three source values are $(72, 308, 1248)$.

$$(c_1,\dots,c_d) = (0, 24, 0, -112).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234210; parent column 1 of https://oeis.org/A234217. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 72x + 308x^2 - 480x^3 - 2048x^4.$$

9.5.6.36 A234211

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,4$. Predicate V has target $s=11$, allowing equal adjacent values. The recurrence holds for every $n \geq 16$, with order $d=15$ and a complete $D=39$ state matrix. The first three source values are $(308, 1540, 7348)$.

```
6, 51, -354, -656, 6790, -457, -52786, 47381, 158504, -209462, -148184, 260800, 12192,
-94128, 21312
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234211; parent column 2 of https://oeis.org/A234217. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.37 A234212

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,4$. Predicate V has target $s=11$, allowing equal adjacent values. The recurrence holds for every $n \geq 57$, with order $d=56$ and a complete $D=169$ state matrix. The first three source values are $(1248, 7348, 40384)$.

```
0, 223, 0, -22136, 0, 1303842, 0, -51198565, 0, 1426660738, 0, -29308622249, 0,
455180602960, 0, -5436994724661, 0, 50553279723332, 0, -368988227986457, 0,
2126351180635715, 0, -9709003146923883, 0, 35189269788128342, 0, -101259728165773203,
0, 231060390001397360, 0, -417017561764832798, 0, 592925182387254959, 0,
-660503855304387479, 0, 572258984227035715, 0, -381879096938313788, 0,
193753758860748742, 0, -73452183198353516, 0, 20317503519116744, 0, -3966504433028624,
0, 520682605459552, 0, -42615909301440, 0, 1905181236480, 0, -34505662464
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234212; parent column 3 of https://oeis.org/A234217. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.38 A234659

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=4$, allowing equal adjacent values. The recurrence holds for every $n \geq 6$, with order $d=5$ and a complete $D=16$ state matrix. The first three source values are $(298, 1820, 11112)$.

$$(c_1,\dots,c_d) = (11, 17, -492, 1050, 1260).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234659; parent column 1 of https://oeis.org/A234665. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 298x - 1458x^2 - 13974x^3 + 61530x^4 + 61740x^5.$$

9.5.6.39 A234660

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=4$, allowing equal adjacent values. The recurrence holds for every $n \geq 17$, with order $d=16$ and a complete $D=100$ state matrix. The first three source values are $(1820, 9774, 52646)$.

```
24, -116, -1573, 17655, -7512, -599532, 2201681, 4899218, -40315144, 38576320,
130316460, -132305400, -263116800, 8496000, 134568000, 45360000
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234660; parent column 2 of https://oeis.org/A234665. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.40 A234661

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=4$, allowing equal adjacent values. The recurrence holds for every $n \geq 40$, with order $d=39$ and a complete $D=625$ state matrix. The first three source values are $(11112, 52646, 250792)$.

```
37, -377, -2618, 77447, -294799, -4757619, 45540630, 61891376, -2437351502,
6395028748, 63315544498, -379499690085, -574635388877, 9755368316951, -11625591036256,
-133055653485659, 418266455931349, 823106610584361, -5467891684835430,
1031251784123218, 36193728208088096, -44705964178611952, -124471844677347304,
241456322415418264, 272081034415319536, -671531537886950352, -480780534963511680,
1128701330288740800, 770576980895712000, -1120320571251686400, -920761444521216000,
573508008695808000, 665567689121280000, -74650786306560000, -24754797711360000,
-49525866547200000, 32441969664000000, 15456946176000000, 1881169920000000
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234661; parent column 3 of https://oeis.org/A234665. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.41 A234667

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=4$, allowing equal adjacent values. The recurrence holds for every $n \geq 8$, with order $d=7$ and a complete $D=20$ state matrix. The first three source values are $(492, 3816, 29568)$.

$$(c_1,\dots,c_d) = (14, 25, -950, 2091, 7326, -3780, -10080).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234667; parent column 1 of https://oeis.org/A234672. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 492x - 3072x^2 - 36156x^3 + 187320x^4 + 425832x^5 - 322560x^6 - 645120x^7.$$

9.5.6.42 A234668

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=4$, allowing equal adjacent values. The recurrence holds for every $n \geq 32$, with order $d=31$ and a complete $D=144$ state matrix. The first three source values are $(3816, 28860, 217968)$.

```
35, -234, -5393, 83088, 72599, -7727281, 30601466, 288437560, -2223662740,
-2827663592, 61030210280, -68134000167, -723492961431, 1379135043174, 4956504550981,
-10336340466744, -22249573436775, 42072453243081, 68185197096618, -103181037099516,
-142550751258288, 157651414286112, 199205608411008, -149554834911360,
-179827747799040, 84774205497600, 99173800742400, -26026116480000, -29990476800000,
3292047360000, 3762339840000
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234668; parent column 2 of https://oeis.org/A234672. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.43 A234683

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=5$, allowing equal adjacent values. The recurrence holds for every $n \geq 7$, with order $d=6$ and a complete $D=16$ state matrix. The first three source values are $(232, 1148, 5472)$.

$$(c_1,\dots,c_d) = (0, 72, 0, -1712, 0, 13440).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234683; parent column 1 of https://oeis.org/A234690. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 232x + 1148x^2 - 11232x^3 - 55312x^4 + 134400x^5 + 658560x^6.$$

9.5.6.44 A234684

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=5$, allowing equal adjacent values. The recurrence holds for every $n \geq 25$, with order $d=24$ and a complete $D=100$ state matrix. The first three source values are $(1148, 4416, 16580)$.

```
7, 136, -1036, -7837, 67291, 245614, -2526790, -4355332, 60808804, 36086728,
-982312744, 135510432, 10839179904, -6945594624, -81450996480, 81465827328,
407151166464, -514409232384, -1284949370880, 1874710241280, 2296420761600,
-3685898649600, -1755758592000, 3009871872000
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234684; parent column 2 of https://oeis.org/A234690. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.45 A234697

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=5$, allowing equal adjacent values. The recurrence holds for every $n \geq 6$, with order $d=5$ and a complete $D=20$ state matrix. The first three source values are $(408, 2628, 16848)$.

$$(c_1,\dots,c_d) = (6, 90, -540, -2016, 12096).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234697; parent column 1 of https://oeis.org/A234702. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 408x + 180x^2 - 35640x^3 - 8064x^4 + 774144x^5.$$

9.5.6.46 A234698

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=5$, allowing equal adjacent values. The recurrence holds for every $n \geq 16$, with order $d=15$ and a complete $D=144$ state matrix. The first three source values are $(2628, 13952, 74308)$.

```
15, 125, -2865, -1490, 219360, -507740, -8437740, 34905816, 161734320, -1006299360,
-1062547200, 13887417600, -8926848000, -75188736000, 121927680000
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234698; parent column 2 of https://oeis.org/A234702. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.47 A234699

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=5$, allowing equal adjacent values. The recurrence holds for every $n \geq 32$, with order $d=31$ and a complete $D=1056$ state matrix. The first three source values are $(16848, 74308, 329984)$.

```
20, 141, -5340, 5451, 616740, -2678805, -39706740, 278440182, 1491646680,
-16092380232, -27470360160, 601403119168, -194332088960, -15257818574160,
26729605928640, 263906583339744, -801813357648000, -2982678610973184,
13948745101747200, 18588639920403456, -156794289434542080, -1681574850293760,
1140373919202508800, -1022866437252096000, -5015897891205120000, 8536532135362560000,
10656426280550400000, -31534442652303360000, 815374290124800000, 46191901645209600000,
-31856483893248000000
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234699; parent column 3 of https://oeis.org/A234702. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.48 A234706

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=6$, allowing equal adjacent values. The recurrence holds for every $n \geq 32$, with order $d=31$ and a complete $D=144$ state matrix. The first three source values are $(1730, 7154, 28170)$.

```
12, 159, -2412, -9452, 217248, 161262, -11578680, 10718427, 405901116, -824307633,
-9828766476, 28650887914, 167599827384, -632763934044, -2003077953360, 9664372451624,
16138030431072, -105143721192576, -76625123046912, 819575849887488, 75093856109568,
-4528143100320768, 1633754973450240, 17236678723430400, -11938042414080000,
-42782336704512000, 40275310510080000, 62010467205120000, -70124998164480000,
-39609913835520000, 50565847449600000
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234706; parent column 2 of https://oeis.org/A234712. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.49 A234707

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=6$, allowing equal adjacent values. The recurrence holds for every $n \geq 37$, with order $d=36$ and a complete $D=1056$ state matrix. The first three source values are $(8738, 28170, 87698)$.

```
13, 163, -2779, -9420, 268200, 60770, -15439130, 22995763, 589369361, -1588530069,
-15656100243, 58787313370, 294085396250, -1450484076640, -3845621898440,
25550700910752, 32289252593664, -330371143423376, -113428722066352, 3163397610252320,
-1019517140684000, -22338341568842880, 18872130722296320, 114348342295028736,
-148610586143628288, -409381709070163968, 722386331480346624, 952125730341396480,
-2277950012653363200, -1180887662923776000, 4529957329207296000, 69777190748160000,
-5142605210910720000, 1663475920404480000, 2524921315983360000, -1415843728588800000
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234707; parent column 3 of https://oeis.org/A234712. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.50 A234721

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=7$, allowing equal adjacent values. The recurrence holds for every $n \geq 9$, with order $d=8$ and a complete $D=20$ state matrix. The first three source values are $(240, 1096, 4192)$.

```
0, 68, 0, -1724, 0, 19312, 0, -80640
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234721; parent column 1 of https://oeis.org/A234728. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 240x + 1096x^2 - 12128x^3 - 55408x^4 + 203200x^5 + 928672x^6 - 1128960x^7 - 5160960x^8.$$

9.5.6.51 A234722

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=7$, allowing equal adjacent values. The recurrence holds for every $n \geq 31$, with order $d=30$ and a complete $D=144$ state matrix. The first three source values are $(1096, 3976, 12464)$.

```
4, 137, -560, -8456, 35504, 310702, -1349320, -7561533, 34294092, 128206281,
-615707400, -1548288698, 8040276992, 13333179376, -77453548480, -79946374816,
552146144704, 309222145232, -2893327015040, -567224779296, 10948880162304,
-974085452928, -28950298675200, 9103597862400, 50436504576000, -24908850432000,
-51674112000000, 32903055360000, 23410114560000, -17557585920000
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234722; parent column 2 of https://oeis.org/A234728. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.52 A234816

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=6$, allowing equal adjacent values. The recurrence holds for every $n \geq 8$, with order $d=7$ and a complete $D=16$ state matrix. The first three source values are $(168, 686, 2380)$.

$$(c_1,\dots,c_d) = (4, 41, -164, -558, 2232, 2520, -10080).$$

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234816; parent column 1 of https://oeis.org/A234823. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F=P/Q$, with $Q=1-\sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 168x + 14x^2 - 7252x^3 - 378x^4 + 103896x^5 + 2520x^6 - 493920x^7.$$

9.5.6.53 A234817

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=6$, allowing equal adjacent values. The recurrence holds for every $n \geq 24$, with order $d=23$ and a complete $D=100$ state matrix. The first three source values are $(686, 2220, 6282)$.

```
8, 60, -620, -1210, 20740, 1580, -392180, 353267, 4606324, -7135240, -34677440,
72387960, 165390560, -443380480, -464032320, 1696321584, 566607168, -3948347520,
465384960, 5084985600, -2142028800, -2757888000, 1741824000
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234817; parent column 2 of https://oeis.org/A234823. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.54 A234818

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=6$, allowing equal adjacent values. The recurrence holds for every $n \geq 24$, with order $d=23$ and a complete $D=625$ state matrix. The first three source values are $(2380, 6282, 15224)$.

```
8, 60, -620, -1210, 20740, 1580, -392180, 353267, 4606324, -7135240, -34677440,
72387960, 165390560, -443380480, -464032320, 1696321584, 566607168, -3948347520,
465384960, 5084985600, -2142028800, -2757888000, 1741824000
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234818; parent column 3 of https://oeis.org/A234823. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.55 A234992

Arrays have $n+1$ rows, width $w=2$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=2$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 11$, with order $d=10$ and a complete $D=16$ state matrix. The first three source values are $(172, 916, 5280)$.

```
5, 21, -70, -144, 326, 381, -549, -351, 198, 54
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234992; parent column 1 of https://oeis.org/A234998. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.56 A234993

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=2$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 50$, with order $d=49$ and a complete $D=100$ state matrix. The first three source values are $(916, 5112, 32256)$.

```
6, 145, -752, -9334, 41421, 350517, -1354277, -8617439, 29691561, 147908106,
-465456923, -1844354978, 5418592827, 17142962010, -47956757782, -120746606447,
327447020322, 650926343809, -1740121457481, -2698809487907, 7230222191327,
8610347154836, -23519481347436, -21060770608040, 59812533502435, 39165695095186,
-118451096631369, -54588838411578, 181491721699373, 55729099282996, -213160956817772,
-40145457720166, 189498178497244, 19155239362192, -125401972381171, -5433880826151,
60455696914684, 890128451025, -20656751347661, -330921215331, 4825908656200,
227383374666, -729157775008, -75605102388, 62860038344, 11433294720, -2022154176,
-653147712, -43972992
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A234993; parent column 2 of https://oeis.org/A234998. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.57 A235169

Arrays have $n + 1$ rows, width $w = 2$, and labeled alphabet $0, \ldots, 7$. Predicate D has target $s = 4$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 10$, with order $d = 9$ and a complete $D = 20$ state matrix. The first three source values are $(268, 1528, 8688)$.

```
6, 37, -234, -141, 1290, 108, -2340, 0, 1296
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A235169; parent column 1 of https://oeis.org/A235175. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.58 A235170

Arrays have $n + 1$ rows, width $w = 3$, and labeled alphabet $0, \ldots, 7$. Predicate D has target $s = 4$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 47$, with order $d = 46$ and a complete $D = 144$ state matrix. The first three source values are $(1528, 8380, 45776)$.

```
19, 34, -2719, 8988, 156665, -916999, -4550486, 40037121, 64400627, -1012400300,
-148677233, 16600041004, -9873660689, -189875213441, 187284339366, 1593590554819,
-1874014494735, -10158650544956, 12545534952885, 50324143130662, -60649685406985,
-196489095492359, 219272447037552, 609171221202828, -603073067708424,
-1502594545151892, 1270051970766120, 2940961682306376, -2045313082168560,
-4535601595588512, 2497426659701280, 5448277567918080, -2274890615189568,
-5012380431150912, 1506311626822272, 3449065504346496, -697075003450368,
-1716839475671808, 212360144961024, 588874155660288, -38772262462464,
-129032521869312, 3632324124672, 15786025500672, -130606940160, -783641640960
```

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A235170; parent column 2 of https://oeis.org/A235175. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.59 A235205

Arrays have $n + 1$ rows, width $w = 2$, and labeled alphabet $0, \ldots, 7$. Predicate D has target $s = 5$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 7$, with order $d = 6$ and a complete $D = 20$ state matrix. The first three source values are $(240, 1172, 5696)$.

$$(c_1, \ldots, c_d) = (3, 35, -81, -264, 144, 384).$$

The defining transfer satisfies $TR(T)\mathbf{1} = 0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A235205; parent column 1 of https://oeis.org/A235212. Both retrieved entries mark the relevant recurrence empirical. The source rational generating function is also proved: $F = P/Q$, with $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 240x + 452x^2 - 6220x^3 - 10856x^4 + 11016x^5 + 19136x^6.$$

9.5.6.60 A235206

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,7$. Predicate D has target $s=5$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 30$, with order $d=29$ and a complete $D=144$ state matrix. The first three source values are $(1172, 4616, 18444)$.

```
15, 23, -1373, 3547, 50847, -236177, -957405, 6739749, 8784137, -112122375, -8734283,
1205207433, -682793243, -8845512699, 8015164257, 45997463512, -47860413548,
-173398534628, 175140255652, 476432770592, -402722060848, -936013467040, 551524974720,
1245278007168, -366681768960, -999476834304, 6506053632, 362586931200, 89712230400
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A235206; parent column 2 of https://oeis.org/A235212. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.61 A235233

Arrays have $n+1$ rows, width $w=3$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=6$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 29$, with order $d=28$ and a complete $D=100$ state matrix. The first three source values are $(420, 1288, 3688)$.

```
6, 47, -331, -846, 7735, 6531, -99876, -4062, 780931, -294777, -3829232, 2187642,
11922251, -7106223, -23840076, 11482195, 31200479, -9197246, -26295345, 2967600,
13895070, 468804, -4403844, -634896, 759672, 171504, -54432, -15552
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A235233; parent column 2 of https://oeis.org/A235239. Both retrieved entries mark the relevant recurrence empirical.

9.5.6.62 A235234

Arrays have $n+1$ rows, width $w=4$, and labeled alphabet $0,\dots,6$. Predicate D has target $s=6$, with every horizontal and vertical neighboring pair unequal. The recurrence holds for every $n \geq 41$, with order $d=40$ and a complete $D=625$ state matrix. The first three source values are $(1328, 3688, 9656)$.

```
6, 75, -499, -2477, 18893, 46844, -431641, -544644, 6650508, 3692164, -73169776,
-7965360, 593911464, -108754278, -3624939268, 1348847209, 16812874388, -8151710835,
-59533275997, 31896084725, 160970283883, -85813326662, -331183400635, 159993289668,
514651400122, -202610873932, -596529887736, 165384664128, 505501523616, -76023552288,
-303948460656, 8914601088, 124404317856, 9463220928, -32567277312, -5301282816,
4869517824, 1128950784, -313528320, -89579520
```

The defining transfer satisfies $TR(T)\mathbf{1}=0$ in every coordinate, so (9.6) follows. Source: https://oeis.org/A235234; parent column 3 of https://oeis.org/A235239. Both retrieved entries mark the relevant recurrence empirical.

9.6 Verification, provenance and reuse

The accompanying archive contains this standalone manuscript, the numbered ledger, precise source statements, complete exact certificates, verifiers, compact source audits and independent internal review reports. With the Python dependencies in `requirements.txt` installed, run

```
python3 r9_run_all.py
```

from the extracted directory. No network access is needed. The runner checks all 100 selections, the 808-ID exclusion ledger, the retained prior-work baseline, review bindings and the included verification programs. Actual versions and results are recorded in `oeis_hunt_round9_verification_summary.json`.

9.6.1 Universal proofs and supplementary finite checks

The array definitions give complete finite transition matrices. The identity $TR(T)\mathbf{1} = 0$ proves the recurrence at every subsequent index by multiplication by arbitrary nonnegative powers of T. The 62 compressed matrices have 10,537 states and 30,926 nonzero weighted edges. A separately written verifier, importing none of the author's transfer engine, constructs all compatible pairs of labelled rows. It checks the annihilator on all 39,370 uncompressed states and every one of their 139,886 directed edges, then verifies every row of the proposed equitable quotient. Source generating functions in 20 of these packages also follow by exact polynomial cross-multiplication, without an additional count.

Generating-function certificates establish exact differential or rational identities. Every inhomogeneous boundary term is retained; vanishing beyond its degree proves the stated recurrence for every index in the specified range. The seed series is derived from the definition or explicitly credited established input, independently of the conjectural target. This avoids assuming the recurrence being proved.

The arithmetic and formal-series sections supply written universal arguments. Complete residue tables are finite proof certificates where the proof reduces all indices to those classes. Initial-term comparisons, additional modular tests and numerical asymptotic checks are labelled supplementary; a long finite prefix is never used as an infinite proof.

9.6.2 Counting and priority limits

The compact source records retain each selected target, its uncertainty marker, attribution, offset, date, known inputs and audit notes. Source hashes identify consulted snapshots without distributing entire raw web pages. An entry's editorial "approved" status is not a mathematical proof of every conjecture on that entry.

The audit excludes all prior project selections, including A176605, A176607, A176610, A176611 and A176612, whose Round IV priority claims were withdrawn in Round V. The present and cumulative counts refer to selection packages. They are not certified totals of new discoveries. Exact-ID and family searches, linked-paper inspections and bounded alias screens are recorded with their actual coverage. Unretrieved material and transformations outside the alias screen remain explicit limits.

9.6.3 Standalone manuscript and reproducibility

The delivered `.tex` includes every proof and recurrence signature directly, without external `\input` or `\include` fragments. Compile with pdfLaTeX until cross-references stabilize. `r9_build_dossier.py` reconstructs the manuscript, ledger and combined audit. The README lists all offline regeneration commands. An isolated regeneration check compares the reconstructed immutable mathematical outputs byte for byte.

The archive manifest records SHA-256 digests. Internal review reports bind the precise mathematical files reviewed. The manuscript consistency check compares printed coefficient lists, definitions, indices and catalogue statuses against the exact records. The layout review binds the PDF whose rendered pages were inspected.

9.6.4 Common references

1. The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/. Individual links and attributions accompany each selected statement; established input results are credited in the relevant proofs.
2. *OEIS Open*, retained Appendix A comparison, https://arxiv.org/html/2608.11941v1. The comparison covers the 100 costliest Appendix A claims on 97 distinct A-numbers, not the entire project repository.
3. Daniel Axler, *New estimates for some functions defined over primes*, Integers 18 (2018), A52, https://math.colgate.edu/~integers/s52/s52.pdf. Used as an established input for the prime-harmonic estimate in A329354.
4. Miklós Bóna and Ryan Flynn, *The average number of block interchanges needed to sort a permutation and a recent result of Stanley*, https://arxiv.org/abs/0811.0740. The Hultman-number identity is a credited input to A363412.

Part X

Universal periods, dimension formulas, tatami polynomials, and word arrays

Round overview and scope

This tenth-round dossier gives mathematical resolutions for 100 precisely specified assertion packages on 100 distinct OEIS entries, excluding the 908 A-numbers selected in Rounds I–IX. It proves 97 selected assertions or explicit questions and supplies corrected statements with full proofs for three others. The four parts contain ten formal-series congruence packages, five arithmetic packages, twenty generating-function or recurrence packages, and sixty-five constrained-word recurrence packages.

Counting unit and source status

One catalogue row corresponds to one selected package on one A-number. A recurrence and its generating function count once. The 65 word entries belong to twelve fixed-pattern families and share an automaton theorem; thirteen modular-form dimension packages specialize established general dimension formulas. These are not 100 independent proof techniques.

The selected source wording is conjectural or empirical in the consulted primary material, except for A243046, which explicitly asks for a direct formula. The exact result and domain are stated for each entry. A parity proof at A337758 or A338328 does not resolve its separate modulo-four claim. A339100 is treated only for the selected three-adic valuation. The no-three-in-line comparison mentioned at A326478 is not asserted. Known definitions, theorems and related formulas are credited as inputs.

Corrections and boundary conventions

A339100. The empirical three-adic valuation has its parity reversed. The correct valuation is one at positive even indices and zero at odd indices.

A094287 and A094288. The printed conjectural generating functions omit the constant coefficient added when the entries' offsets were changed to include the empty path. Adding one restores the complete present-index generating functions. Each correction is counted once; the underlying known bounded-path theory is credited.

Within the A326478 package, the auxiliary comparison with A103517 also requires the index $n-1$ in place of $n+1$. Its selected empirical even-index generating function is true as stated, so this auxiliary repair is not counted as another correction package.

Audit and priority limits

Primary entries, relevant linked material and targeted searches were consulted on 20 October 2026. Retrieval can be cached, so these source records do not certify the latest live edit state. The retained OEIS Open comparison covers 100 Appendix A claims on 97 unique A-numbers, with no overlap; it does not cover that project's entire repository.

Known proofs found during the audit were excluded. Examples include the general A272462/A272468 integrality assertion, A195986 after a recent explicit proof was found, nine fractional-factorial candidates resolved in an updated public repository, and the A242468–A242471 triple-equality avoidance family covered by an earlier generating-function theorem. Earlier priority withdrawals remain in force; 908 is a selection ledger, not a cumulative total of worldwide-new discoveries. Fresh reserve certificates are kept outside the 100 selected entries.

All selected word entries and their parent definitions were retrieved. An unsuccessful retrieval of root companion A373313 is disclosed in its audit; that companion supplies no needed proof input. Independent reviews are internal checks by separate agents within one AI-assisted workflow, not external peer review or proof-assistant formalization.

No OEIS submission or edit was performed. The dossier proves or repairs the selected assertions; editorial acceptance is a separate event.

10.1 Catalogue of the 100 selected entries

P: the selected assertion or question is proved over its stated domain. **C**: the source formula is corrected and the corrected form is proved. Page references lead to the complete argument or exact finite certificate.

No.	Entry	Selected assertion package	Result	Page
Formal series and universal congruences				
1	A373312	Binary-minus-one parity	P	512
2	A325286	Ternary digit residues	P	513
3	A320949	Divisibility by 3 of rationally summed moments	P	513
4	A205572	Full powers-of-two congruence	P	514
5	A227051	Last decimal digit 3	P	515
6	A292177	Prime-index vanishing in a bilateral limit	P	515
7	A382124	Cube-root Frobenius congruence	P	516
8	A338328	Parity of a diagonal coefficient definition	P	518
9	A337758	Parity with even diagonal exponents	P	518
10	A230218	Square-exponent parity	P	518
Arithmetic, a direct formula and a valuation correction				
11	A165886	Consecutive normalized Hirzebruch quotient	P	519
12	A243046	Direct formula for the simultaneous integrality count	P	520
13	A308833	Tetrahedral divisibility and odd composites	P	521
14	A326478	Even-index Bernoulli-denominator generating function	P	522
15	A339100	Corrected three-adic valuation	C	523
Generating functions, dimensions and polynomial recurrences				
16	A063081	Conjectural modular-form dimension formulas	P	525
17	A063087	Conjectural modular-form dimension formulas	P	526
18	A063098	Conjectural modular-form dimension formulas	P	526
19	A063105	Conjectural modular-form dimension formulas	P	527
20	A063110	Conjectural modular-form dimension formulas	P	528
21	A063116	Conjectural modular-form dimension formulas	P	528
22	A063118	Conjectural modular-form dimension formulas	P	529
23	A063126	Conjectural modular-form dimension formulas	P	530
24	A063140	Conjectural modular-form dimension formulas	P	530
25	A063147	Conjectural modular-form dimension formulas	P	531
26	A063148	Conjectural modular-form dimension formulas	P	532
27	A063168	Conjectural modular-form dimension formulas	P	533
28	A063247	Conjectural modular-form dimension formulas	P	533
29	A006502	The second Carlitz polynomial diagonal	P	535
30	A259709	The first Carlitz polynomial diagonal	P	535
31	A174015	Polynomial recurrence from the signed Catalan fraction	P	536
32	A226302	Tatami-polynomial specialization at minus one	P	536
33	A273905	Polynomial recurrence for symmetric bargraphs	P	538
34	A094287	Corrected constant term for bounded Motzkin paths	C	539
35	A094288	Corrected constant term for bounded Motzkin paths	C	540
Consecutive-pattern word recurrences				
36	A212824	Empirical word recurrence of order 4	P	545
37	A212825	Empirical word recurrence of order 6	P	546
38	A212826	Empirical word recurrence of order 8	P	546
39	A212827	Empirical word recurrence of order 10	P	546
40	A242543	Empirical word recurrence of order 4	P	547
41	A242544	Empirical word recurrence of order 8	P	547
42	A242546	Empirical word recurrence of order 16	P	547
43	A242547	Empirical word recurrence of order 20	P	547
44	A242548	Empirical word recurrence of order 24	P	548
45	A242578	Empirical word recurrence of order 4	P	548
46	A242579	Empirical word recurrence of order 7	P	548

Continued on the next page

No.	Entry	Selected assertion package	Result	Page
47	A242580	Empirical word recurrence of order 10	P	549
48	A242581	Empirical word recurrence of order 13	P	549
49	A242582	Empirical word recurrence of order 16	P	549
50	A243027	Empirical word recurrence of order 3	P	549
51	A243028	Empirical word recurrence of order 6	P	550
52	A243029	Empirical word recurrence of order 9	P	550
53	A243038	Empirical word recurrence of order 4	P	550
54	A243039	Empirical word recurrence of order 7	P	551
55	A243040	Empirical word recurrence of order 10	P	551
56	A243041	Empirical word recurrence of order 13	P	551
57	A243042	Empirical word recurrence of order 16	P	551
58	A243043	Empirical word recurrence of order 19	P	552
59	A243383	Empirical word recurrence of order 4	P	552
60	A243384	Empirical word recurrence of order 7	P	552
61	A243385	Empirical word recurrence of order 10	P	552
62	A243386	Empirical word recurrence of order 13	P	553
63	A243387	Empirical word recurrence of order 16	P	553
64	A243388	Empirical word recurrence of order 19	P	553
65	A243513	Empirical word recurrence of order 3	P	553
66	A243514	Empirical word recurrence of order 4	P	554
67	A243515	Empirical word recurrence of order 5	P	554
68	A243516	Empirical word recurrence of order 6	P	554
69	A243517	Empirical word recurrence of order 7	P	555
70	A243518	Empirical word recurrence of order 8	P	555
71	A243600	Empirical word recurrence of order 5	P	555
72	A243601	Empirical word recurrence of order 7	P	556
73	A243602	Empirical word recurrence of order 9	P	556
74	A243603	Empirical word recurrence of order 11	P	556
75	A243604	Empirical word recurrence of order 13	P	556
76	A243605	Empirical word recurrence of order 15	P	557
77	A243606	Empirical word recurrence of order 17	P	557
78	A243635	Empirical word recurrence of order 6	P	557
79	A243636	Empirical word recurrence of order 8	P	557
80	A243637	Empirical word recurrence of order 10	P	558
81	A243638	Empirical word recurrence of order 12	P	558
82	A243640	Empirical word recurrence of order 16	P	558
83	A243724	Empirical word recurrence of order 7	P	559
84	A243727	Empirical word recurrence of order 13	P	559
85	A244177	Empirical word recurrence of order 6	P	559
86	A244178	Empirical word recurrence of order 12	P	560
87	A244179	Empirical word recurrence of order 18	P	560
88	A244180	Empirical word recurrence of order 24	P	560
89	A244181	Empirical word recurrence of order 30	P	560
90	A244182	Empirical word recurrence of order 36	P	561
91	A244183	Empirical word recurrence of order 42	P	561
92	A244184	Empirical word recurrence of order 48	P	561
93	A244694	Empirical word recurrence of order 7	P	562
94	A244695	Empirical word recurrence of order 13	P	562
95	A244696	Empirical word recurrence of order 19	P	562
96	A244697	Empirical word recurrence of order 25	P	562
97	A244698	Empirical word recurrence of order 31	P	563
98	A244699	Empirical word recurrence of order 37	P	563
99	A244700	Empirical word recurrence of order 43	P	563
100	A244701	Empirical word recurrence of order 49	P	563

10.2 Formal series and universal congruences

This section proves ten precisely selected assertions marked conjectural in the consulted primary entries. Definitions and unmarked companion formulas are credited as existing inputs. The proofs are over the full stated domains; the coefficient checks in `r10_root_verify.py` are auxiliary checks of the definitions and their transcription. In particular, a proof of a parity clause is not a claim to settle every other conjecture on the same entry.

10.2.1 A373312: parity after inversion of a binary product

Source. A373312, Hanna, June 25, 2024, defines

$$A = x + O(x^2), \qquad A(x)^2 = A\left(\frac{xA(x)}{(1-A(x))^2}\right),$$

and conjectures that a_n is odd precisely when $n = 2^j - 1$, $j \geq 1$. The product and reversion characterizations are also displayed in the entry.

Proposition 10.2.1. *The defining equation has a unique solution in $x + x^2\mathbb{Z}[[x]]$, and*

$$A(x) \equiv \sum_{j\geq 1} x^{2^j-1} \pmod 2.$$

Proof. Let R be the compositional inverse of A. Substitution of $t = A(x)$ gives

$$R(t^2) = \frac{tR(t)}{(1-t)^2}.$$

Writing $R(t) = tP(t)$, with $P(0) = 1$, yields $P(t) = (1-t)^2P(t^2)$. Iterating this identity coefficientwise proves

$$R(t) = t\prod_{j\geq 0}(1-t^{2^j})^2.$$

Conversely, this product satisfies the equation for R and its inverse satisfies the original equation for A. All coefficients of R are integral and its linear coefficient is one, so coefficient recursion proves integral existence and uniqueness of its inverse.

Over $\mathbb{F}_2$, uniqueness of binary expansions gives $\prod_{j\geq 0}(1+t^{2^j}) = 1/(1+t)$. Hence $R(t) = t/(1+t)^2$ in this field. Its inverse obeys $A = x(1+A)^2 = x + xA^2$. Starting with its zero constant term and comparing successive coefficients gives exactly $A = \sum_{j\geq 1} x^{2^j-1}$. □

The companion entries A106407 and A372530 describe the binary product and a related inverse. They are known constructions, not additional selected packages. Direct retrieval of the further companion A373313 failed; no assertion about that entry is needed here.

10.2.2 A320949 and A325286: defining rationally summed moments

The two source sums below are not locally finite as ordinary formal sums in x: their summands already have nonzero constant terms. Their precise meaning is obtained by summing each polynomial moment as a rational function first. The following elementary construction also proves that reduction modulo three is legitimate.

For an integer-valued polynomial $b(r)$ of degree at most d on nonnegative integers, Newton expansion gives

$$b(r) = \sum_{j=0}^{d} c_j \binom{r}{j}, \qquad c_j = \Delta^j b(0) \in \mathbb{Z}.$$

Consequently, its ordinary moment series is the rational function

$$\sum_{r\geq 0} b(r)t^r = \sum_{j=0}^{d} c_j \frac{t^j}{(1-t)^{j+1}}. \tag{10.1}$$

Both identities follow by finite differences and $\sum_{r\geq 0}\binom{r}{j}t^r = t^j/(1-t)^{j+1}$.

Lemma 10.2.2 (Odd-period reflection)**.** *For $k \geq 1$, let $M_k(t)$ be the rational function associated by (10.1) with $b(r) = \binom{r^2}{k}$. Its reduction over $\mathbb{F}_3$ is defined at $t = -1$, and $M_k(-1) = 0$ there.*

Proof. Choose $L = 3^h > k$. The coefficient of z^k in $(1+z)^v$ depends, modulo three, only on v mod L, because $(1+z)^L = 1 + z^L$ in characteristic three. Thus $b(r)$ has period L modulo three and $b(L-r) = b(r)$ for $1 \leq r < L$. As rational functions over $\mathbb{F}_3$,

$$M_k(t) = \frac{\sum_{r=0}^{L-1} b(r)t^r}{1-t^L}.$$

At $t = -1$, the denominator is $2 \neq 0$. The term $r = 0$ vanishes for $k > 0$; the terms r and $L - r$ cancel because L is odd. This proves the lemma. The evaluation is also allowed in (10.1), whose denominators are powers of $1 - t$. □

10.2.2.1 A320949: every positive coefficient is a multiple of three

Source. A320949, Hanna, November 13, 2018, defines $A(0) = 1$ by

$$\sum_{r\geq 0} \frac{(1+x)^{r^2}}{(1+A(x))^{r+1}} = 1$$

and marks $a_n \equiv 0 \pmod 3$, $n > 0$, as a conjecture.

Proposition 10.2.3. *The rational-moment interpretation of the defining equation has a unique integral solution, and $A(x) \equiv 1 \pmod 3$.*

Proof. Let $c_{k,j} = \Delta^j \binom{r^2}{k}|_{r=0}$, with $0 \leq j \leq 2k$. Put $t = (1+A)^{-1}$ in (10.1), and multiply by t. The defining equation becomes

$$1 = \sum_{k\geq 0} x^k \sum_{j=0}^{2k} c_{k,j} A^{-j-1}, \qquad A = 1 + \sum_{k\geq 1} x^k \sum_{j=0}^{2k} c_{k,j} A^{-j}.$$

The second expression is now locally finite. In degree $n > 0$ its right side depends only on coefficients of A below degree n, because every summand has a factor x^k, $k \geq 1$. This is a monic recursion over $\mathbb{Z}$, and over any commutative coefficient ring, with constant term one.

In $\mathbb{F}_3$, try $A = 1$, for which $t = 1/2 = -1$. Every positive-degree moment is zero by Lemma 10.2.2. The constant moment is $t/(1-t) = 1$ at this value. Thus the constant series 1 satisfies the reduced defining equation. Uniqueness of the monic recursion proves the claim. □

10.2.2.2 A325286: the complete ternary-digit rule

Source. A325286, Hanna, April 18, 2019, defines $A(0) = 1$ by

$$\frac{1}{2}\sum_{r\geq 0} \frac{(1+x)^{r(r-1)/2}}{(2A(x))^r} = 1.$$

Its conjecture assigns residue one to indices containing only zeroes and ones in base three, and residue zero to every other index.

Proposition 10.2.4. *There is a unique integral solution and*

$$A(x) \equiv \prod_{j\geq 0}(1+x^{3^j}) \pmod 3.$$

Proof. Set $c_{k,j} = \Delta^j \binom{r(r-1)/2}{k}|_{r=0}$ for $0 \leq j \leq 2k$. Taking $t = (2A)^{-1}$ in (10.1) gives

$$1 = A\sum_{k\geq 0} x^k \sum_{j=0}^{2k} \frac{c_{k,j}}{(2A-1)^{j+1}}, \qquad A-1 = A\sum_{k\geq 1} x^k \sum_{j=0}^{2k} \frac{c_{k,j}}{(2A-1)^{j}}.$$

Since $2A-1$ has constant term one, the second equation gives a monic integral recursion exactly as before, including uniqueness after reduction modulo three.

Work in $\mathbb{F}_3[[u]]$ and substitute $x = 2u+u^2$ and $A = (1+u)^{-1}$. The change of variable has invertible linear coefficient. The original rational-moment expression becomes

$$\frac{1}{2}\sum_{r\geq 0} \frac{(1+u)^{r^2}}{2^r}.$$

This substitution is legitimate coefficientwise: before reduction every coefficient is a polynomial in r times 2^{-r}, summed by (10.1). Its positive-degree coefficients are zero modulo three by Lemma 10.2.2, and its constant term is one. Uniqueness therefore gives $A(x)^2(1+x) = 1$ over $\mathbb{F}_3$.

Let $D(x) = \prod_{j\geq 0}(1+x^{3^j})$. Frobenius gives $D = (1+x)D^3$, so $D^2(1+x) = 1$. A square root with constant term one is unique here, since two such roots have sum with invertible constant term two. Hence $A = D$. Unique ternary expansion of the selected exponents proves both residue cases, including $a_0 = 1$. □

The related moment equation at A303290 is an existing construction; no additional claim for that entry is counted.

10.2.3 A205572: the full family of powers-of-two periods

Source. A205572, Hanna, January 29, 2012, defines $F(x) = 1/(\cos x - \sinh x) = \sum a_n x^n/n!$ and conjectures the congruence below for every indicated pair of indices.

Proposition 10.2.5. *For all $r \geq 1$ and $k \geq 0$,*

$$a_{k+2^r} \equiv a_k \pmod{2^r}.$$

This proves that 2^r is a period; it does not assert that it is the least period.

Proof. Use the graded integral ring

$$R = \mathbb{Z}[C,S,H,K,(C-H)^{-1}], \qquad DC = -S, \quad DS = C, \quad DH = K, \quad DK = H.$$

Each of C, S, H, K has degree one and $(C-H)^{-1}$ has degree minus one. The derivation D preserves degree and extends to the localization by the quotient rule. Modulo two, D^2 is a derivation: the middle term in the second-derivative product formula vanishes. On the four generators it equals the Euler derivation $E = C\partial_C + S\partial_S + H\partial_H + K\partial_K$. These derivations therefore agree on the localization as well. On its homogeneous degree-minus-one component V, E is multiplication by -1, which is the identity modulo two. Thus $D^2 - I$ maps V into $2V$.

The ring is torsion-free and the kernel of reduction modulo two is exactly $2R$. Consequently $B = (D^2-I)/2$ is a well-defined integral linear endomorphism of V. For $r \geq 1$,

$$D^{2^r} = (I+2B)^{2^{r-1}} \equiv I \pmod{2^r} \quad \text{on } V.$$

Indeed, for $1 \le j \le 2^{r-1}$,

$$v_2\left(2^j\binom{2^{r-1}}{j}\right) \ge j + r - 1 - v_2(j) \ge r.$$

The first inequality follows from $j\binom{N}{j} = N\binom{N-1}{j-1}$.

Take $f = (C - H)^{-1} \in V$, apply D^k, and evaluate at $(C, S, H, K) = (1, 0, 0, 1)$. This evaluation is integral because the localized denominator is one. Under $(C, S, H, K) = (\cos x, \sin x, \sinh x, \cosh x)$, the derivation is ordinary differentiation, so the evaluated values are precisely a_{k+2^r} and a_k. This proves the assertion for every k, r. □

10.2.4 A227051: every positive term ends in three

Source. A227051, Hanna, June 29, 2013, defines

$$F(x) = \frac{1}{\cos x - \sin x \sinh x} = \sum_{m\ge 0} b_m \frac{x^m}{m!}, \qquad a_n = b_{2n},$$

and conjectures $a_n \equiv 3 \pmod{10}$ for $n > 0$.

Proposition 10.2.6. *For every $n \ge 1$, $a_n \equiv 3 \pmod{10}$.*

Proof. The denominator has integral exponential coefficients and constant term one. Its reciprocal has integral exponential coefficients by the recursion $b_m = -\sum_{j=1}^{m}\binom{m}{j} g_j b_{m-j}$, where g_j are the denominator's exponential coefficients.

First work modulo five in $\mathbb{F}_5[C, S, H, K, (C - SH)^{-1}]$ with the preceding derivation D. In characteristic five, D^5 is a derivation. On each of C, S, H, K, $D^5 = D$ because D^4 is the identity on these generators. The derivations therefore agree on the localized ring. For $f = (C - SH)^{-1}$ this gives $D^{m+4}f = D^m f$ whenever $m \ge 1$. Evaluation at $(1, 0, 0, 1)$ implies $b_{m+4} \equiv b_m \pmod 5$ for $m \ge 1$. The initial values $b_2 = 3$ and $b_4 = 53$ give $b_{2n} \equiv 3 \pmod 5$ for all $n > 0$.

For the modulo-two part, use the integral Hurwitz-series ring: the product of exponential coefficient sequences is $(uv)_m = \sum_{j=0}^{m}\binom{m}{j} u_j v_{m-j}$. Modulo two, the square of every zero-constant series is zero. Terms with $j \ne m - j$ cancel in pairs, and the middle coefficient $\binom{2j}{j}$ is even for $j > 0$. Since $\sin x \equiv \sinh x$ in this ring, their product is zero. Also $(\cos x)^2 = 1$ modulo two. Thus $F \equiv \cos x$ in the Hurwitz ring, and every even-index exponential coefficient is one modulo two. Combining residues modulo two and five proves the claim. □

The different square-root EGF at A185071 has its own unmarked congruence. That statement is not used to infer this result and is not counted.

10.2.5 A292177: a justified bilateral limit and prime-index zeroes

Source. A292177, Hanna, September 10, 2017, defines

$$S_K(x) = \sum_{n\in\mathbb{Z}} x^{n-K}\left(1 - x^n + \frac{n(n+1)}{6}x^{n+K}\right)^n, \qquad A(x) = \lim_{K\to\infty} S_K(x),$$

and conjectures $a_p = 0$ for odd primes p.

Proposition 10.2.7. *The limit exists coefficientwise in $\mathbb{Q}[[x]]$ and equals the integral series*

$$A(x) = \sum_{d\ge 1} \frac{d^2(d+1)}{6} x^{2d}(1 - x^d)^{d-1}$$

$$+\sum_{d\geq 2}\frac{(-1)^d d^2(d-1)}{6}\frac{x^{d(d-1)}}{(1-x^d)^{d+1}}.$$

In particular, $a_p = 0$ for every odd prime p.

Proof. First establish the bilateral base identity

$$\sum_{n\in\mathbb{Z}} x^n(1-x^n)^n = 0.$$

For $n = -m < 0$, rewrite the summand as $(-1)^m x^{m^2-m}(1-x^m)^{-m}$. These summands are now locally finite; their constant term at $m = 1$ cancels the $n = 0$ term. For a positive degree $N = dr$, the contribution of the positive index d is $(-1)^{r-1}\binom{d}{r-1}$, and that of the negative index $-r$ is $(-1)^r\binom{d}{r-1}$. Summing over divisors proves cancellation in every degree, with binomial coefficients outside their usual range interpreted as zero.

Expand S_K in powers j of the perturbation $n(n+1)x^{n+K}/6$. The $j = 0$ part is x^{-K} times the preceding zero series. The $j = 1$ part is independent of K and equals

$$\sum_{n\in\mathbb{Z}}\frac{n^2(n+1)}{6}x^{2n}(1-x^n)^{n-1}.$$

For $n \geq 1$, a nonzero $j \geq 2$ term has valuation at least $(j-1)K+(j+1)n \geq K+3$. For $n = -m < 0$, rewriting its negative powers gives valuation at least $m^2 - m + (j-1)K$. The case $m = 1$ has zero perturbation; for $m \geq 2$ and $j \geq 2$ this is at least $K + 2$. For each fixed $K \geq 1$ the rewritings define a locally finite Laurent series, and the bounds are uniform in the indices. The higher perturbations therefore disappear coefficientwise as K tends to infinity. Rewriting the negative indices in the linear term gives the displayed two unilateral sums.

For completeness their coefficients also have the paired-divisor formula

$$a_N = \sum_{dr=N}(-1)^r\frac{r(r-1)(d+r)}{6}\binom{d+1}{r} \quad (N \geq 1). \tag{10.2}$$

This follows by pairing the positive index d with the negative index $-r$ in the two unilateral sums. Every summand is integral. Divisibility by two follows from $r(r-1)$. For divisibility by three, only $r \equiv 2 \pmod 3$ needs attention. If $d \equiv 1 \pmod 3$, then $d + r$ is a multiple of three. Otherwise the units ternary digit of $d + 1$ is zero or one, less than the units digit two of r, so $\binom{d+1}{r} \equiv 0 \pmod 3$. This last fact follows directly by expanding $(1+z)^{d+1}$ using $(1+z)^3 = 1+z^3$ modulo three.

Finally, a coefficient in either unilateral sum can occur only at a multiple of d. For an odd prime degree p, the only possible indices are $d = 1, p$. The positive $d = 1$ summand is just $x^2/3$ and the negative $d = 1$ summand is absent. The $d = p$ summands start at $2p$ and $p(p-1)$, both greater than p. Hence $a_p = 0$. □

The base bilateral identity and several different weighted identities also appear at A291937; they are known context. Here the perturbation limit and the exact selected prime-index assertion are proved explicitly.

10.2.6 A382124: integrality before Frobenius

Source. A382124, Hanna, April 6, 2025, defines

$$L_n = \frac{\sigma(n)\sigma(2n)}{3}, \qquad A(x) = \exp\left(\sum_{n\geq 1}\frac{L_n}{n}x^n\right).$$

The companion A382125 is $B(x) = A(x)^3$. The selected conjecture is $[x^n]A \equiv [x^{3n}]B \pmod 3$ for all $n \geq 0$.

Proposition 10.2.8. *The coefficients of A are integers and $B(x) \equiv A(x^3) \pmod 3$.*

Proof. The prime-power formulas for L_n are known at A382123; they also follow immediately from multiplicativity of the divisor sum. For odd primes p and $r \geq 1$,

$$L_{p^r} = \sigma(p^r)^2, \qquad L_{2^r} = C(r) = \frac{(2^{r+1}-1)(2^{r+2}-1)}{3}, \qquad L_1 = C(0) = 1.$$

Define the Euler exponents $e_n = n^{-1}\sum_{d|n} \mu(d) L_{n/d}$. They are multiplicative. On odd prime powers,

$$e_{p^r} = \frac{\sigma(p^r)^2 - \sigma(p^{r-1})^2}{p^r} = \sigma(p^r) + \sigma(p^{r-1}) \in \mathbb{Z}.$$

For powers of two, $C(r) - C(r-1) = 2^r(2^{r+1}-1)$, so $e_{2^r} = 2^{r+1}-1$; also $e_1 = 1$. Thus every Euler exponent is a nonnegative integer. Taking formal logarithms and using $L_n = \sum_{d|n} d e_d$ proves

$$A(x) = \prod_{n \geq 1} (1-x^n)^{-e_n} \in \mathbb{Z}[[x]].$$

Frobenius may now be applied in $\mathbb{F}_3[[x]]$ to give $A(x)^3 = A(x^3)$. Extracting degree $3n$ proves the selected assertion, including $n = 0$. □

The zero residues of the other two index classes in A382125 follow at once but are uncounted consequences. Its duplicate statement of the same cube congruence does not supply a second package.

10.2.7 A338328, A337758 and A230218: integral diagonal recursions

The same standard cyclic-action argument justifies the divisions needed in three different source definitions. The entries and selected assertions are distinct; shared lemmas do not turn them into three independent new methods.

Lemma 10.2.9 (Coprime diagonal division)**.** *Let $n, q \geq 1$ with $\gcd(n,q) = 1$, and let $F = 1 + \sum_{i \geq 1} f_i x^i$. Then*

$$\frac{1}{q}[x^n]F(x)^q$$

is a polynomial with integer coefficients in $f_1, \ldots, f_n$. Its coefficient of f_n is one and it is affine in f_n.

Proof. Expand F^q using ordered q-tuples of nonnegative indices with total weight n. The cyclic group of order q acts by rotation. If a tuple were a repetition of $d > 1$ identical blocks, then d would divide both q and n, a contradiction. Every orbit therefore has q elements, all contributing the same monomial. Dividing orbit sums by q proves integrality as a polynomial identity, so it remains valid for arbitrary signed integer specializations. The terms involving f_n have exactly one nonzero index and together give qf_n before division. □

Write

$$S(x) = \sum_{j \geq 0} x^{2^j} \in \mathbb{F}_2[[x]], \qquad S + S^2 = x.$$

We will use the coefficient identity

$$[x^n]H(S(x)) = [t^n]H(t)(1+t)^{-n-1}. \tag{10.3}$$

Indeed, substitute $x = t + t^2$ in the formal residue giving the left side. The derivative is one in characteristic two, and $(t+t^2)^{-n-1} = t^{-n-1}(1+t)^{-n-1}$. This is a valid change of formal variable because its linear coefficient is one.

10.2.7.1 A338328: odd coefficients exactly at powers of two

Source. A338328, Hanna, October 24, 2020, defines $A(0) = 0$ by

$$[x^n](1 + nx - A(x))^{2n-1} = 0 \qquad (n > 0).$$

We select only its conjectural parity statement. Its separate modulo-four assertion and its asymptotic formula are outside this package.

Proposition 10.2.10. *The defining recursion is integral and* $A(x) \equiv S(x) \pmod 2$.

Proof. Since $\gcd(n, 2n-1) = 1$, Lemma 10.2.9 makes the divided coefficient equation monic with coefficient -1 in the new coefficient a_n. It gives a unique integral solution, and a unique solution of the divided equations modulo two.

Test $A = S$. If n is odd, the reduced base is $1 + x + S = 1 + S^2$, which contains only even degrees; hence its odd-degree coefficient n is zero. If n is even, $n \geq 2$, the reduced coefficient is

$$[x^n](1 + S)^{2n-1} = [t^n](1 + t)^{n-2} = 0$$

by (10.3). The exponent $2n - 1$ is odd, so the divided equation also vanishes modulo two. Monic uniqueness proves $A = S$. □

10.2.7.2 A337758: handling the even exponents correctly

Source. A337758, Hanna, October 24, 2020, defines $A(0) = 0$ by

$$[x^n](1 + nx + nx^2 - A(x))^{n+1} = 0 \qquad (n > 0).$$

Only the parity conjecture is selected; the separate modulo-four clause is not asserted here.

Proposition 10.2.11. *The coefficients are integral and* $A(x) \equiv S(x) \pmod 2$.

Proof. Here $\gcd(n, n+1) = 1$, so Lemma 10.2.9 again gives a monic integer recursion. Let $S_0 = \sum_{j \geq 0} x^{2^j}$ denote an integral lift of S, and put $F = 1 + nx + nx^2 - S_0$ and $q = n + 1$. It suffices to show that the integer $q^{-1}[x^n]F^q$ is even.

If n is even, q is odd and the reduced base is $1 + S$. Then

$$[x^n](1 + S)^{n+1} = [t^n]1 = 0.$$

If n is odd, division by q directly modulo two would be invalid. Instead use the exact derivative identity

$$n\left(\frac{1}{q}[x^n]F^q\right) = [x^{n-1}]F'F^{q-1}.$$

The quantity in parentheses is integral by the lemma. Also $F' = n + 2nx - S_0' \equiv 1 - 1 = 0 \pmod 2$. As n is odd, the quantity in parentheses must be even. Thus S satisfies all reduced divided equations, and monic uniqueness completes the proof. □

10.2.7.3 A230218: a missing binary digit in a square

Source. A230218, Hanna, October 11, 2013, has $a_0 = a_1 = 1$ and

$$[x^n]A(x)^{n^2+n+1} = 0 \qquad (n > 1).$$

The selected conjecture says that, for $n > 0$, a_n is odd precisely at powers of two.

Proposition 10.2.12. *The recursion is integral and* $A(x) \equiv 1 + S(x) \pmod 2$.

Proof. For $q = n^2 + n + 1$ one has $\gcd(n, q) = 1$ and q is odd. Lemma 10.2.9 gives integral monic uniqueness from the specified initial coefficients. By (10.3), the candidate $1 + S$ gives

$$[x^n](1 + S)^{n^2+n+1} = [t^n](1 + t)^{n^2} = \binom{n^2}{n} \pmod 2.$$

This binomial coefficient is even for every $n > 1$. If n is even and its least nonzero binary digit is at position $r \geq 1$, then $2^{2r} \mid n^2$, so that digit is missing from n^2. If $n > 1$ is odd, let $r \geq 1$ be its first nonzero binary digit above the units digit. Then $n \equiv 1 + 2^r \pmod{2^{r+1}}$ but $n^2 \equiv 1 \pmod{2^{r+1}}$, so again digit r is missing from the square. The binary factorization of $(1 + t)^{n^2}$ in characteristic two shows that its coefficient of t^n is zero whenever such a digit is missing. Thus all defining equations hold for $1 + S$, and uniqueness proves the result. □

The inspected companion entries A185072, A229041, A229044, A171791 and A292877 concern known reversion relations or other parity patterns. None is counted as a further result of this package.

Scope of the source and prior-work audit

All ten selected A-numbers are disjoint from the previous 908 selected A-numbers, the retained uncounted earlier reserves, and the 97 unique A-numbers in the inspected OEIS Open Appendix A. The audit includes the primary entries, the named companions above and targeted searches. It is a bounded source audit: cached retrievals and unsuccessful companion retrievals cannot certify worldwide historical priority or the latest live edit state. Known tools used here include finite differences, formal inversion, cyclic group actions, formal residues, Euler products, and elementary congruences. The count records ten source-marked assertion packages with written proofs, not ten new general methods or an official change to OEIS.

10.3 Arithmetic identities, a divisor formula, and a valuation correction

These five packages concern five distinct entries absent from the previous 908 selected identifiers and from the supplied 97-identifier OEIS Open Appendix A baseline. Four selected assertions or questions are resolved affirmatively; one empirical valuation clause is corrected. The primary entries and relevant companion records were inspected on October 20, 2026. The source audit is bounded: it does not establish historical priority. Every theorem below has a proof valid on its stated infinite domain. The finite computations in the accompanying standard-library verifier are independent checks of definitions and boundary cases.

10.3.1 A165886: the quotient of consecutive normalized Hirzebruch numbers

Source and selected assertion. The comment on A165886 marks the quotient identity with A141459 and A006519 as a conjecture. The entry was submitted by Paul Curtz on September 29, 2009; the comment is not separately dated. R. J. Mathar records a finite check through 2000 on July 4, 2011. Define

$$H(n) = \prod_{p \text{ prime}} p^{\lfloor n/(p-1) \rfloor}, \qquad F(n) = \frac{H(n)}{2^{s_2(n)}} \quad (n \geq 0),$$

where $s_2(n)$ counts the ones in the binary expansion of n. The product is finite, since primes greater than $n + 1$ have exponent zero. These are the definitions of A091137 and A165641; A165886 is $a(n) = F(n + 1)/F(n)$.

Proposition 10.3.1. *For every integer $n \geq 0$,*

$$a(n) = 2^{v_2(n+1)} \prod_{\substack{p \text{ odd prime} \\ p-1 \mid n+1}} p = \mathrm{A006519}(n+1)\,\mathrm{A141459}(n+1).$$

Proof. For any positive integer d,

$$\left\lfloor \frac{n+1}{d} \right\rfloor - \left\lfloor \frac{n}{d} \right\rfloor = \begin{cases} 1, & d \mid n+1, \\ 0, & d \nmid n+1. \end{cases}$$

Consequently

$$\frac{H(n+1)}{H(n)} = 2 \prod_{\substack{p \text{ odd prime} \\ p-1 \mid n+1}} p.$$

Put $t = v_2(n+1)$. Adding one to n erases exactly t trailing binary ones and inserts one binary one, including the case $n = 0$. Thus $s_2(n+1) - s_2(n) = 1 - t$. Dividing the preceding product by $2^{s_2(n+1)-s_2(n)}$ gives the assertion. The companion definitions are precisely the odd-prime product and the greatest power of two dividing the argument. □

Prior-work scope. The Hirzebruch prime product, its consecutive quotient, and Gould's binary-digit formula are established inputs, explicitly credited to A091137, A001316, A141459, and A006519. The selected result is their combination in the still-marked assertion on A165886. A search also located the same formula in the conjecture-list mirror and in Sequence Machine; neither inspected result supplied a proof. No novelty claim is made for the elementary floor or binary-carry identities.

10.3.2 A243046: a finite formula using paired divisors

Source and selected question. A243046 counts positive integers $k \neq n$ for which both $kn/(k+n)$ and $kn/(k-n)$ are integers. Antti Karttunen's February 18, 2023 comment explicitly asks for a direct formula. The following formula uses only the positive divisors of n^2; it does not search over the original k interval.

Proposition 10.3.2. *Let $n \geq 1$, let $D_n = \{d > 0 : d \mid n^2\}$, and let $[P]$ be one or zero according as the proposition P is true or false. Then*

$$\boxed{a(n) = \sum_{d \in D_n} [d + 2n \in D_n] + \sum_{\substack{d \in D_n \\ d < n}} [2n - d \in D_n].} \tag{A243046}$$

The full set of admissible values of k is the disjoint union

$$\{n + d : d \in D_n,\ d + 2n \in D_n\} \,\dot\cup\, \{n - d : d \in D_n,\ d < n,\ 2n - d \in D_n\}.$$

In particular, for a prime p and integer $e \geq 1$,

$$a(p^e) = \begin{cases} 1, & p = 3, \\ 1, & p = 2 \text{ and } e \geq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. For $k > 0$ and $k \neq n$, the exact identities

$$\frac{kn}{k+n} = n - \frac{n^2}{k+n}, \qquad \frac{kn}{k-n} = n + \frac{n^2}{k-n}$$

show that admissibility is equivalent to $(k+n) \mid n^2$ and $(k-n) \mid n^2$, with the second divisor allowed to be negative. If $k > n$, put $d = k - n > 0$. Then the two positive divisors are d and $d + 2n$, and $k = n + d$. This is a bijection with the first set. If $0 < k < n$, put $d = n - k$. Then $0 < d < n$ and the two positive divisors are d and $2n - d$, with $k = n - d$. This is a bijection with the second set. The ranges $k > n$ and $0 < k < n$ are disjoint, which proves both the formula and the stated solution set.

For the prime-power corollary, distinct divisors of p^{2e} have the form $p^b < p^c$. A pair contributing to the first sum must satisfy

$$p^b(p^{c-b} - 1) = 2p^e.$$

If p is odd, comparison of p-adic valuations gives $b = e$; then $p^{c-b} = 3$, forcing $p = 3$ and $c = e + 1$. This pair exists for every $e \geq 1$. If $p = 2$, the odd factor forces $b = e + 1$ and $c = b + 1 = e + 2$, which lies at most $2e$ exactly when $e \geq 2$. A pair contributing to the second sum would satisfy $p^b(p^{c-b} + 1) = 2p^e$ with $b < c$. For odd p this would give $b = e$ and $p^{c-b} + 1 = 2$, which is impossible. For $p = 2$, its odd factor $2^{c-b} + 1 > 1$ cannot divide a power of two. Thus the second sum has no contribution for a prime power. □

Effective range and scope. If $n = \prod_i p_i^{e_i}$, the set in the formula is explicitly $D_n = \{\prod_i p_i^{f_i} : 0 \leq f_i \leq 2e_i\}$, of size $\prod_i(2e_i + 1) = \tau(n^2)$. Once these divisors have been generated, the answer requires fewer than $2\tau(n^2)$ set-membership checks. Any admissible k also satisfies $k \leq n^2 - n$, because $k + n$ is a positive divisor of n^2. This proves finiteness even for the boundary $n = 1$, when no k exists. The inspected companions A063647 and A146564 count either individual integrality condition, while A243017 and A243045 record the minimum and maximum. Umberto Cerruti's linked note *Percorsi tra i numeri*, Theorems 1–2 and Corollary 4, proves the signed-divisor parametrization for the single minus condition and the associated companion counts. Those results are credited inputs, with the elementary equivalences reproved above. The simultaneous divisor-pair formula is the answer selected here; the companion counts are not additional packages.

10.3.3 A308833: tetrahedral divisibility is governed by an odd composite

Source and selected assertion. A308833, submitted by Lorenzo Sauras Altuzarra on June 28, 2019, consists of the positive integers r for which $r(r+1)(r+2)/6$ divides $r!$. Its conjectural comment, not separately dated, compares consecutive odd/even indices with the compositeness of the following odd integer.

Proposition 10.3.3. *For every odd integer $r > 1$, the following statements are equivalent:*

$$\frac{r(r+1)(r+2)}{6} \mid r!, \qquad \frac{(r+1)(r+2)(r+3)}{6} \mid (r+1)!, \qquad r+2 \text{ is composite.}$$

Proof. Write $m = r + 2$, so m is odd and $m \geq 5$. If m is prime, it is coprime to 6. Each of the two tetrahedral numbers contains a factor m, whereas neither $(m-2)!$ nor $(m-1)!$ does. Both divisibility assertions are therefore false.

Suppose instead that m is composite; then $m \geq 9$. We first prove $m \mid (m-3)!$. Choose a factorization $m = uv$ with $3 \leq u \leq v$. If $u < v$, both distinct factors lie at most $m - 3$ and their product occurs in the factorial. If $u = v$, the distinct numbers u and $2u$ lie at most $u^2 - 3 = m - 3$ for $u \geq 3$, and their product is $2m$. This proves the claim in both cases.

The even integer $m - 1 \geq 8$ is the product of the distinct numbers 2 and $(m-1)/2$, both at most $m - 3$. Hence $m - 1$ also divides $(m-3)!$. Since $\gcd(m, m-1) = 1$, we obtain

$$\frac{(m-2)!}{(m-2)(m-1)m/6} = \frac{6(m-3)!}{m(m-1)} \in \mathbb{Z}.$$

Likewise $m+1 \geq 10$ is the product of the distinct integers 2 and $(m+1)/2$, both at most $m-2$. Together with $m \mid (m-3)!$ and $\gcd(m, m+1) = 1$, this gives

$$\frac{(m-1)!}{(m-1)m(m+1)/6} = \frac{6(m-2)!}{m(m+1)} \in \mathbb{Z}.$$

Thus both divisibilities hold exactly in the composite case. □

Prior-work scope. The companion A153238 describes the associated odd composites. The inspected source and companion provide definitions and computational tables, but no proof of this selected three-way equivalence. The factorial arguments above are elementary; neither that technique nor the companion classification is claimed as new mathematics.

10.3.4 A326478: the even Bernoulli-denominator subsequence

Source and selected assertion. A326478, submitted by Peter Luschny on July 16, 2019, defines

$$a(m) = m\frac{\operatorname{den}(mB_{m-1})}{\operatorname{den}(B_{m-1})} \quad (m \geq 1),$$

where B_j is the jth Bernoulli number, $B_1 = -1/2$, and $\operatorname{den}(0) = 1$. The even-index generating-function comment is explicitly marked empirical; that comment is not separately dated. Only this even-index assertion is selected.

Proposition 10.3.4. *For every integer $n \geq 1$,*

$$a(2n) = \begin{cases} 1, & n = 1, \\ 2n, & n \geq 2, \end{cases} \qquad \sum_{n\geq 1} a(2n)x^n = x\left(\frac{2}{(1-x)^2} - 1\right).$$

With the currently stated offset of A103517, the correct companion relation is

$$a(2n) = \mathrm{A103517}(n-1).$$

Proof. The defining Bernoulli series satisfies

$$\frac{z}{e^z - 1} + \frac{z}{2} = \frac{z}{2}\frac{e^z + 1}{e^z - 1},$$

an even formal power series, since substitution of $-z$ leaves it unchanged. Consequently $B_j = 0$ for every odd $j > 1$. For $n \geq 2$, both B_{2n-1} and $2nB_{2n-1}$ are zero; their reduced denominators are 1, so $a(2n) = 2n$. For $n = 1$, the denominators of $B_1 = -1/2$ and $2B_1 = -1$ are 2 and 1, respectively, so $a(2) = 1$. Therefore

$$\sum_{n\geq 1} a(2n)x^n = \frac{2x}{(1-x)^2} - x.$$

The source A103517 has offset zero and generating function $2/(1-x)^2 - 1$, so shifting its coefficients by one yields the companion relation. □

Exact scope and an offset correction. The same source sentence prints the comparison with $\mathrm{A103517}(n+1)$. This index is incorrect with its present offset: at $n = 1$, $a(2) = 1$ but $\mathrm{A103517}(2) = 6$. The correction to $n-1$ is included in this one package and is not counted separately. The further comparison with A272651, a maximum in the no-three-in-line problem, does not follow from a Bernoulli calculation and is not asserted here. The odd-fixed-point question on A326478 is also outside the selected scope. The Bernoulli vanishing fact and the companion rational generating function are classical inputs, not novelty claims.

10.3.5 A339100: correcting the three-adic valuation clause

Source and selected clause. A339100, submitted by Luc Rousseau on November 23, 2020, uses the triangle A214406 with

$$T(0,0)=1, \qquad T(n,k)=(4n-2k-1)T(n-1,k-1)+(2k+1)T(n-1,k),$$

and $T(n,k)=0$ outside $0\le k\le n$. It defines

$$a(n)=\gcd_{0\le k\le n}\big((2n-k)T(n,k)+(k+1)T(n,k+1)\big), \quad n\ge 1.$$

The empirical prime-valuation comment includes the clause $v_3(a(n))=n \bmod 2$. This particular clause is reversed.

Proposition 10.3.5. *The literal three-adic clause is false already at $n=1$, where $a(1)=\gcd(3,1)=1$. Its corrected version, valid for every $n\ge 1$, is*

$$\boxed{v_3(a(n))=\begin{cases}1, & n \text{ even},\\ 0, & n \text{ odd}.\end{cases}}$$

Proof. Put $P_n(y)=\sum_{k=0}^{n}T(n,k)y^k$ and

$$R_n(x)=\frac{P_n(x^2)}{(1-x^2)^{2n}}, \qquad L(f)=\frac{d}{dx}\left(\frac{xf}{1-x^2}\right).$$

Summing the triangular recurrence gives

$$P_{n+1}(y)=2y(1-y)P_n'(y)+(1+(4n+1)y)P_n(y).$$

Differentiation then shows $R_0=1$ and $R_{n+1}=L(R_n)$. If $Q_n(y)$ denotes the polynomial whose coefficients occur in the defining gcd, a further exact differentiation gives

$$R_n'(x)=\frac{2x}{(1-x^2)^{2n+1}}Q_n(x^2), \qquad Q_n(y)=(1-y)P_n'(y)+2nP_n(y).$$

Reduction modulo 3 is legitimate in the rational-function field $\mathbb{F}_3(x)$, since all displayed denominators remain nonzero. In that field $3\mid a(n)$ is equivalent to $R_n'=0$.

The first two iterates are

$$R_1=\frac{1+x^2}{(1-x^2)^2}, \qquad R_2=\frac{1+8x^2+3x^4}{(1-x^2)^4}\equiv(1-x^2)^{-3}=:H \pmod 3.$$

Here $H'=0$ in characteristic 3. Hence $L(H^m f)=H^m L(f)$, and induction gives

$$R_{2m}=H^m, \qquad R_{2m+1}=H^m R_1 \quad \text{in } \mathbb{F}_3(x), \quad m\ge 0.$$

Thus every even iterate has zero derivative, whereas

$$R_{2m+1}'=H^m R_1', \qquad R_1'=\frac{2x(3+x^2)}{(1-x^2)^3}\equiv\frac{2x^3}{(1-x^2)^3}\neq 0 \pmod 3.$$

It follows that 3 divides the defining gcd exactly when n is even.

It remains to show that the even gcd has only one factor of 3. The triangular recurrence gives $T(n,0)=1$. Setting $c_n=T(n,1)$, it also gives $c_0=0$ and $c_n=3c_{n-1}+4n-3$; induction solves this as

$$c_n=\frac{3^{n+1}-4n-3}{2}.$$

Therefore the constant coefficient of Q_n is

$$Q_n(0) = 2n + c_n = \frac{3(3^n - 1)}{2},$$

whose three-adic valuation is exactly 1 for every $n \geq 1$. As $a(n)$ divides this coefficient, its valuation is at most 1. Combined with the preceding divisibility criterion, this proves the corrected formula. □

Scope and prior work. This is one correction package for the three-adic clause alone. It proves neither the two-adic clause nor the claims for primes at least 5 in the same empirical paragraph. The recurrence and rational-function construction are credited to A214406 and A339100. The related quotient sequence A165886 is not identified with this gcd sequence: $a(20) = 3300$, whereas A165886(19) = 660. The source's own displayed factorization table agrees with the corrected parity. The bounded search and inspected companion notes supplied no prior proof of the selected correction.

10.4 Generating functions, dimension formulas, and exact recurrence certificates

This section contains twenty selected A-number packages: eighteen proofs of source-marked assertions and two corrections to generating functions after an offset change. The thirteen modular-form packages are direct applications of classical dimension theorems. The continued-fraction and bargraph generating functions are also credited to earlier work. A226302 includes an independent proof of the generating-function conjecture in its defining paper. One A-number contributes one package, even when several displayed claims are addressed. These counts do not assert twenty independent research discoveries, first publication, or acceptance of an edit by OEIS.

10.4.1 Exact coefficient and operator certificates

For $F(x) = \sum a_n x^n$, extend a_n by zero below the stated offset and set $\theta = x\,d/dx$. For polynomial coefficients p_j,

$$L = \sum_{j=0}^{r} x^j p_j(\theta + j), \qquad [x^n]LF = \sum_{j=0}^{r} p_j(n) a_{n-j}. \tag{10.4}$$

This follows by applying $(\theta + j)$ to x^{n-j}. Therefore an exact identity $LF = P$, with P polynomial, proves the recurrence at every $n > \deg P$ for which all source indices exist. Constant p_j reduce this certificate to multiplication by $Q(x) = \sum_j p_j x^j$.

For a radical, always take $\sqrt{D(0)} = 1$. In the basis $1, \sqrt{D}$,

$$\theta(R_0 + R_1\sqrt{D}) = xR_0' + x\left(R_1' + \frac{D'}{2D}R_1\right)\sqrt{D}.$$

The displayed residuals below are exact rational identities, checked after clearing denominators. Each proof first obtains the series from its independent definition. The accompanying coefficient comparisons through index 100, and comparisons with the published source terms, are supplementary checks of indexing and transcription; they do not replace these identities.

10.4.2 A classical input for thirteen dimension assertions

For this subsection let $a_N(n) = \dim S_{2n}(\Gamma_0(N))$ for $n \geq 1$, and put $a_N(0) = 0$ only as a generating-function convention. We use the classical dimension theorem in William Stein, *Modular Forms, a Computational Approach*, Propositions 6.1 and 6.4 (Chapter 6). Define

$$\mu = N\prod_{p|N}(1 + 1/p), \quad c = \sum_{d|N}\varphi(\gcd(d, N/d)), \quad g = 1 + \frac{\mu}{12} - \frac{e_2}{4} - \frac{e_3}{3} - \frac{c}{2},$$

where e_2 and e_3 count roots modulo N of u^2+1 and u^2+u+1, respectively. The theorem gives

$$a_N(n) = (2n-1)(g-1) + (n-1)c + e_2\lfloor n/2\rfloor + e_3\lfloor 2n/3\rfloor + [n=1]. \tag{10.5}$$

For newforms it gives

$$a_N^{\text{new}}(n) = \sum_{M\mid N} \overline{\mu}(N/M)a_M(n), \tag{10.6}$$

where $\overline{\mu}$ is multiplicative with values $-2, 1, 0$ on p, p^2, p^e $(e \geq 3)$. These theorems are existing results.

We now sum (10.5). Splitting indices by their residues modulo 2 or 3 gives

$$\sum_{n\geq 1}\lfloor n/2\rfloor x^n = \frac{x^2}{(1-x)(1-x^2)}, \qquad \sum_{n\geq 1}\lfloor 2n/3\rfloor x^n = \frac{x^2(1+x)}{(1-x)(1-x^3)}.$$

Together with $\sum nx^n = x/(1-x)^2$, these prove

$$G_N(x) = x + \frac{(g-1)x(1+x)+cx^2}{(1-x)^2} + \frac{e_2x^2}{(1-x)(1-x^2)} + \frac{e_3x^2(1+x)}{(1-x)(1-x^3)}. \tag{10.7}$$

The root counts and divisor sums are finite, exact computations. Substitution in this identity proves each rational function below. Multiplication by its stated Q proves the recurrence on its full range. None of these series is defined by the recurrence being proved. Closed forms follow from (10.5); the verifier checks the six symbolic identities $n = 6m + r$, $r = 2, \ldots, 7$, $m \geq 0$, covering all $n \geq 2$.

10.4.3 A063081: cusp forms of level 13

Source and scope. A063081, checked 20 October 2026. Conjecture; Chai Wah Wu; Bill McEachen (2016-05-27; 2025-06-16).

Indexing. Offset 1, with

$$(a_1, \ldots, a_8) = (0, 3, 5, 7, 9, 13, 13, 17).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x^2\left(x^4+x^3+4x^2+5x+3\right)}{x^5-x^3-x^2+1}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{5} p_j(n)a_{n-j} = 0 \qquad (n \geq 7).$$

$$(p_0, \ldots, p_5) = (1, 0, -1, -1, 0, 1).$$

The further selected assertion is $a_n = a_{n-6} + 14$ for $n \geq 8$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (14, 2, 2, 2, 0).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^6 + x^5 + 4x^4 + 5x^3 + 3x^2.$$

The residual has degree below 7, so coefficient extraction proves the selected recurrence for every stated index.

In (10.5), increasing $n \geq 2$ by 6 increases the dimension by 14, proving the additional assertion.

10.4.4 A063087: cusp forms of level 19

Source and scope. A063087, checked 20 October 2026. Conjecture; Wesley Ivan Hurt; Colin Barker (2016-12-27; 2019-06-09).

Indexing. Offset 1, with

$$(a_1, \dots, a_8) = (1, 4, 8, 10, 14, 18, 20, 24).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^4 + x^3 + 4x^2 + 3x + 1\right)}{(1-x)^2\left(x^2 + x + 1\right)}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{4} p_j(n) a_{n-j} = 0 \qquad (n \geq 6).$$

$$(p_0, \dots, p_4) = (1, -1, 0, -1, 1).$$

The selected closed form is $a_n = 4n - 2\left\lfloor \frac{n}{3} - \frac{1}{3} \right\rfloor - 4$ for $n \geq 2$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (20, 0, 2, 2, 1).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^5 + x^4 + 4x^3 + 3x^2 + x.$$

The residual has degree below 6, so coefficient extraction proves the selected recurrence for every stated index.

The closed form follows by substituting the same invariants in (10.5); its exception at $n = 1$ is retained in the displayed initial data.

10.4.5 A063098: cusp forms of level 30

Source and scope. A063098, checked 20 October 2026. Conjecture; Colin Barker (2019-06-09).

Indexing. Offset 1, with

$$(a_1, \dots, a_8) = (3, 14, 26, 38, 50, 62, 74, 86).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^2 + 8x + 3\right)}{(1-x)^2}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{2} p_j(n) a_{n-j} = 0 \qquad (n \geq 4).$$

$$(p_0, \ldots, p_2) = (1, -2, 1).$$

The selected closed form is $a_n = 12n - 10$ for $n \geq 2$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (72, 0, 0, 8, 3).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^3 + 8x^2 + 3x.$$

The residual has degree below 4, so coefficient extraction proves the selected recurrence for every stated index.

The closed form follows by substituting the same invariants in (10.5); its exception at $n = 1$ is retained in the displayed initial data.

10.4.6 A063105: cusp forms of level 37

Source and scope. A063105, checked 20 October 2026. Conjecture; Colin Barker (2012-09-27).

Indexing. Offset 1, with

$$(a_1, \ldots, a_8) = (2, 9, 15, 21, 27, 35, 39, 47).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^5 + 3x^4 + 10x^3 + 13x^2 + 9x + 2\right)}{(1-x)^2\,(x+1)\,(x^2+x+1)}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{5} p_j(n)\, a_{n-j} = 0 \qquad (n \geq 7).$$

$$(p_0, \ldots, p_5) = (1, 0, -1, -1, 0, 1).$$

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (38, 2, 2, 2, 2).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^6 + 3x^5 + 10x^4 + 13x^3 + 9x^2 + 2x.$$

The residual has degree below 7, so coefficient extraction proves the selected recurrence for every stated index.

10.4.7 A063110: cusp forms of level 42

Source and scope. A063110, checked 20 October 2026. Conjecture; Colin Barker (2012-09-23).

Indexing. Offset 1, with

$$(a_1, \dots, a_8) = (5, 20, 36, 52, 68, 84, 100, 116).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^2 + 10x + 5\right)}{(1-x)^2}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{2} p_j(n) a_{n-j} = 0 \qquad (n \geq 4).$$

$$(p_0, \dots, p_2) = (1, -2, 1).$$

The selected closed form is $a_n = 16n - 12$ for $n \geq 2$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (96, 0, 0, 8, 5).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^3 + 10x^2 + 5x.$$

The residual has degree below 4, so coefficient extraction proves the selected recurrence for every stated index.

The closed form follows by substituting the same invariants in (10.5); its exception at $n = 1$ is retained in the displayed initial data.

10.4.8 A063116: cusp forms of level 48

Source and scope. A063116, checked 20 October 2026. Conjecture; Colin Barker (2019-06-09).

Indexing. Offset 1, with

$$(a_1, \dots, a_8) = (3, 18, 34, 50, 66, 82, 98, 114).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^2 + 12x + 3\right)}{(1-x)^2}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{2} p_j(n) a_{n-j} = 0 \qquad (n \geq 4).$$

$$(p_0, \ldots, p_2) = (1, -2, 1).$$

The selected closed form is $a_n = 16n - 14$ for $n \geq 2$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (96, 0, 0, 12, 3).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^3 + 12x^2 + 3x.$$

The residual has degree below 4, so coefficient extraction proves the selected recurrence for every stated index.

The closed form follows by substituting the same invariants in (10.5); its exception at $n = 1$ is retained in the displayed initial data.

10.4.9 A063118: cusp forms of level 50

Source and scope. A063118, checked 20 October 2026. Conjecture; Klaus Brockhaus; Bruno Berselli; Vincenzo Librandi (2011-03-10; 2011-03-24).

Indexing. Offset 1, with

$$(a_1, \ldots, a_8) = (2, 17, 31, 47, 61, 77, 91, 107).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^3 + 12x^2 + 15x + 2\right)}{(x-1)^2(x+1)}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{3} p_j(n) a_{n-j} = 0 \qquad (n \geq 5).$$

$$(p_0, \ldots, p_3) = (1, -1, -1, 1).$$

The selected closed form is $a_n = \frac{(-1)^n}{2} + 15n - \frac{27}{2}$ for $n \geq 2$.

Also selected are $a_n = a_{n-2} + 30$ and $a_n = 2a_{n-1} - a_{n-2} + 2(-1)^n$ for $n \geq 4$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (90, 2, 0, 12, 2).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^4 + 12x^3 + 15x^2 + 2x.$$

The residual has degree below 5, so coefficient extraction proves the selected recurrence for every stated index.

The closed form follows by substituting the same invariants in (10.5); its exception at $n = 1$ is retained in the displayed initial data.

The two further identities follow by subtracting the closed form at indices differing by 2, and by taking its second difference, respectively.

10.4.10 A063126: cusp forms of level 58

Source and scope. A063126, checked 20 October 2026. Conjecture; Colin Barker (2013-02-07).

Indexing. Offset 1, with

$$(a_1, \dots, a_8) = (6, 21, 35, 51, 65, 81, 95, 111).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\,(x+2)\,(x^2+6x+3)}{(x-1)^2\,(x+1)}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{3} p_j(n) a_{n-j} = 0 \qquad (n \geq 5).$$

$$(p_0, \dots, p_3) = (1, -1, -1, 1).$$

The selected closed form is $a_n = \frac{(-1)^n}{2} + 15n - \frac{19}{2}$ for $n \geq 2$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (90, 2, 0, 4, 6).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^4 + 8x^3 + 15x^2 + 6x.$$

The residual has degree below 5, so coefficient extraction proves the selected recurrence for every stated index.

The closed form follows by substituting the same invariants in (10.5); its exception at $n = 1$ is retained in the displayed initial data.

10.4.11 A063140: cusp forms of level 72

Source and scope. A063140, checked 20 October 2026. Conjecture; Colin Barker; Stefano Spezia (2012-09-23; 2025-03-02).

Indexing. Offset 1, with

$$(a_1, \dots, a_8) = (5, 28, 52, 76, 100, 124, 148, 172).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^2 + 18x + 5\right)}{(1-x)^2}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{2} p_j(n) a_{n-j} = 0 \qquad (n \geq 4).$$

$$(p_0, \ldots, p_2) = (1, -2, 1).$$

The selected closed form is $a_n = 24n - 20$ for $n \geq 2$.

The selected exponential generating function is $20 + x + 4e^x(6x - 5)$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (144, 0, 0, 16, 5).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^3 + 18x^2 + 5x.$$

The residual has degree below 4, so coefficient extraction proves the selected recurrence for every stated index.

The closed form follows by substituting the same invariants in (10.5); its exception at $n = 1$ is retained in the displayed initial data.

Since $a_0 = 0$, $a_1 = 5$ and $a_n = 24n - 20$ for $n \geq 2$, summing $(24n - 20)x^n/n!$ and correcting its first two coefficients gives $24xe^x - 20e^x + 20 + x$, exactly the selected exponential generating function.

10.4.12 A063147: cusp forms of level 79

Source and scope. A063147, checked 20 October 2026. Conjecture; Colin Barker (2019-06-10).

Indexing. Offset 1, with

$$(a_1, \ldots, a_8) = (6, 19, 33, 45, 59, 73, 85, 99).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^4 + 6x^3 + 14x^2 + 13x + 6\right)}{(1-x)^2\left(x^2 + x + 1\right)}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{4} p_j(n) a_{n-j} = 0 \qquad (n \geq 6).$$

$$(p_0, \ldots, p_4) = (1, -1, 0, -1, 1).$$

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (80, 0, 2, 2, 6).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^5 + 6x^4 + 14x^3 + 13x^2 + 6x.$$

The residual has degree below 6, so coefficient extraction proves the selected recurrence for every stated index.

The nearby newform sequence A063233 is not the same sequence: their values at $n = 6$ are 73 and 71.

10.4.13 A063148: cusp forms of level 80

Source and scope. A063148, checked 20 October 2026. Conjecture; R. J. Mathar; Colin Barker (2015-07-21; 2019-06-08).

Indexing. Offset 1, with

$$(a_1, \ldots, a_8) = (7, 30, 54, 78, 102, 126, 150, 174).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\,(x^2 + 16x + 7)}{(1 - x)^2}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{2} p_j(n) a_{n-j} = 0 \qquad (n \geq 4).$$

$$(p_0, \ldots, p_2) = (1, -2, 1).$$

The selected closed form is $a_n = 24n - 18$ for $n \geq 2$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (144, 0, 0, 12, 7).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^3 + 16x^2 + 7x.$$

The residual has degree below 4, so coefficient extraction proves the selected recurrence for every stated index.

The closed form follows by substituting the same invariants in (10.5); its exception at $n = 1$ is retained in the displayed initial data.

The level-60 companion A063128 has the same invariants, so it is equal termwise. It is credited as a companion and contributes no package.

10.4.14 A063168: cusp forms of level 100

Source and scope. A063168, checked 20 October 2026. Conjecture; Chai Wah Wu; Bill McEachen (2016-06-15; 2025-06-16).

Indexing. Offset 1, with

$$(a_1, \dots, a_8) = (7, 36, 66, 96, 126, 156, 186, 216).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^2 + 22x + 7\right)}{(x-1)^2}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{2} p_j(n) a_{n-j} = 0 \qquad (n \geq 4).$$

$$(p_0, \dots, p_2) = (1, -2, 1).$$

The further selected assertion is $a_n = a_{n-1} + 30$ for $n \geq 3$.

Proof. This is a specialization of the credited dimension theorem.

The exact invariants are

$$(\mu, e_2, e_3, c, g) = (180, 0, 0, 18, 7).$$

Substituting these values in (10.7) gives the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^3 + 22x^2 + 7x.$$

The residual has degree below 4, so coefficient extraction proves the selected recurrence for every stated index.

For $n \geq 2$, (10.5) reduces to $30n - 24$, proving the further first-difference assertion.

10.4.15 A063247: newforms of level 100

Source and scope. A063247, checked 20 October 2026. Conjecture; Colin Barker (2019-06-08).

Indexing. Offset 1, with

$$(a_1, \dots, a_8) = (1, 5, 8, 11, 14, 18, 20, 24).$$

The defining weight is $2n$. The selected ordinary generating function is

$$G(x) = \frac{x\left(x^4 + 5x^3 + 7x^2 + 5x + 1\right)}{(1-x)^2\,(x+1)\,(x^2+x+1)}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{5} p_j(n) a_{n-j} = 0 \qquad (n \geq 6).$$

$$(p_0, \dots, p_5) = (1, 0, -1, -1, 0, 1).$$

Proof. This is a specialization of the credited dimension theorem.

For (10.6), the divisor data are as follows; $b = \overline{\mu}(100/M)$.

M	μ	e_2	e_3	c	g	b
1	1	1	1	1	0	1
2	3	1	0	2	0	−2
4	6	0	0	3	0	1
5	6	2	0	2	0	−2
10	18	2	0	4	0	4
20	36	0	0	6	1	−2
25	30	2	0	6	0	1
50	90	2	0	12	2	−2
100	180	0	0	18	7	1

The sum $\sum_{M|100} bG_M$ from (10.7) simplifies exactly to the stated G.

With $Q(x) = \sum_j p_j x^j$, direct rational simplification gives

$$Q(x)G(x) = x^5 + 5x^4 + 7x^3 + 5x^2 + x.$$

The residual has degree below 6, so coefficient extraction proves the selected recurrence for every stated index.

10.4.16 Two diagonals of Carlitz's polynomial triangle

The independent definition is the polynomial recurrence in A259708, from L. Carlitz, *Some Polynomials Related to Fibonacci and Eulerian Numbers*, Fibonacci Quarterly 16 (1978), 216–226, equation (1.6):

$$r_0(z) = 1, \quad r_1(z) = z, \qquad r_{m+1}(z) = (m+1)zr_m(z) + z(1-z)r_m'(z) + (1-z)^2 r_{m-1}(z).$$

The initial value $r_1 = z$ corrects the printed $r_1 = 1$ typo, as explicitly recorded by Eric Schmidt in A259708; the displayed polynomial table also has $r_1 = z$. Let $T(m,k) = [z^k]r_m(z)$, zero outside $0 \le k \le m$. Coefficient extraction gives

$$\begin{aligned} T(m,k) = {} & (m-k+1)T(m-1,k-1) + kT(m-1,k) \\ & + T(m-2,k) - 2T(m-2,k-1) + T(m-2,k-2). \end{aligned}$$

Set $D_j(x) = \sum_{k\ge 0} T(k+j,k)x^k$, with $D_{-1} = D_{-2} = 0$. Multiplying this coefficient identity by x^k and summing proves

$$(1 - (j+1)x - x^2)D_j = xD_{j-1}' - 2xD_{j-1} + D_{j-2} + [j=0]. \tag{10.8}$$

Indeed, the five terms give $(j+1)xD_j$, xD_{j-1}', D_{j-2}, $-2xD_{j-1}$ and x^2D_j; the exceptional cell is $T(0,0) = 1$. Thus, entirely from the defining triangle,

$$D_0 = \frac{1}{1-x-x^2}, \quad D_1 = \frac{xD_0' - 2xD_0}{1-2x-x^2}, \quad D_2 = \frac{xD_1' - 2xD_1 + D_0}{1-3x-x^2}.$$

10.4.17 A006502: a Carlitz diagonal

Source and scope. A006502, checked 20 October 2026. Conjecture; R. J. Mathar (2019-06-18).

Indexing. Offset 0, with

$$(a_0, \dots, a_7) = (1, 3, 14, 60, 279, 1251, 5329, 21532).$$

The independent sequence is $a_n = T(n+2, n)$ for $n \geq 0$.

The source-marked rational function is $G = P/Q$, where

$$\begin{aligned} P &= -3x^8 - 20x^7 - 39x^6 - 15x^5 + 5x^4 - 10x^3 + 18x^2 - 7x + 1, \\ Q &= \left(x^2 + x - 1\right)^3 \left(x^2 + 2x - 1\right)^2 \left(x^2 + 3x - 1\right). \end{aligned}$$

Proof. Equation (10.8) gives $G = D_2$; substitution and rational simplification give exactly the stated numerator and denominator. This proves the generating function for all coefficients.

The two diagonals have different reduced denominators, so they are not scalar multiples or index shifts of each other.

10.4.18 A259709: a Carlitz diagonal

Source and scope. A259709, checked 20 October 2026. Empirical; Eric M. Schmidt; R. J. Mathar (2015-07-10; 2015-07-15).

Indexing. Offset 1, with

$$(a_1, \dots, a_8) = (0, -1, 0, 4, 22, 78, 240, 679).$$

The independent sequence is $a_n = T(n, n-1)$ for $n \geq 1$.

The source-marked rational function is $G = P/Q$, where

$$\begin{aligned} P &= 2x^4 + 4x^3 - x^2, \\ Q &= -\left(x^2 + x - 1\right)^2 \left(x^2 + 2x - 1\right). \end{aligned}$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{6} p_j(n) a_{n-j} = 0 \qquad (n \geq 7).$$

$$\begin{aligned} &p_0(n) = 1, \quad p_1(n) = -4, \quad p_2(n) = 2, \\ &p_3(n) = 6, \quad p_4(n) = -2, \quad p_5(n) = -4, \\ &p_6(n) = -1. \end{aligned}$$

Proof. Equation (10.8) gives $G = xD_1$; substitution and rational simplification give exactly the stated numerator and denominator. This proves the generating function for all coefficients.

For the recurrence polynomial $Q_0 = \sum_j p_j x^j$, the exact identity is

$$Q_0(x) G(x) = 2x^4 + 4x^3 - x^2.$$

Its residual has degree 4, and all source indices exist for $n \geq 7$, proving the selected recurrence.

The two diagonals have different reduced denominators, so they are not scalar multiples or index shifts of each other.

10.4.19 A174015: a period-three continued fraction

Source and scope. A174015, checked 20 October 2026. Conjecture; R. J. Mathar (2012-11-13).

Indexing. Offset 0, with

$$(a_0, \dots, a_7) = (1, 1, 2, 3, 4, 4, 2, -3).$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{4} p_j(n) a_{n-j} = 0 \qquad (n \geq 4).$$

$$p_0(n) = n + 1, \qquad p_1(n) = -3n, \qquad p_2(n) = 3n - 3,$$
$$p_3(n) = 3n - 12, \qquad p_4(n) = 14 - 4n.$$

Independent construction and prior credit. A174015 is the first column of A174014. The defining Delέham transform has period-three data $r = (1, 1, -1, \dots)$, $s = (1, 0, 0, \dots)$. The constant-column continued fraction is the case $y = 0$ of A084938. Its three tails obey

$$C_0 = \frac{1}{1 - xC_1}, \qquad C_1 = \frac{1}{1 - xC_2}, \qquad C_2 = \frac{1}{1 + xC_0}.$$

The generating function was already treated by Mathar, *Properties of Delέham's Delta Transformation* (2015), Theorem 4 and Examples 17, 20. That generating function is an existing input, not a new claim here.

Proof. Eliminating C_1, C_2 and setting $F = C_0$ gives

$$x(1 - x)F^2 + (1 - 3x)F + x - 1 = 0.$$

The derivative with respect to F at $(x, F) = (0, 1)$ is 1, so this has a unique formal solution with constant term 1. Solving the quadratic gives

$$F = \frac{3x - 1 + \sqrt{1 - 2x + x^2 + 4x^3}}{2x(1 - x)}.$$

Apply (10.4) with the displayed p_j. In the basis $1, \sqrt{1 - 2x + x^2 + 4x^3}$, the exact result is

$$LF = 1 - x + 3x^2 - 3x^3 + 0\sqrt{1 - 2x + x^2 + 4x^3}.$$

The radical coefficient cancels identically and the remaining polynomial has degree 3. Consequently the recurrence holds for all $n \geq 4$.

10.4.20 A226302: tatami polynomials at minus one

Source and scope. A226302, checked 20 October 2026. Conjecture; R. J. Mathar (2013-11-06).

Indexing. Offset 2, with

$$(a_2, \dots, a_9) = (1, -1, 2, -4, 6, -14, 20, -48).$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{5} p_j(n) a_{n-j} = 0 \qquad (n \geq 7).$$

$$p_0(n) = 2 - n, \qquad p_1(n) = 2 - n, \qquad p_2(n) = 6n - 22,$$
$$p_3(n) = 6n - 28, \qquad p_4(n) = 36 - 8n, \qquad p_5(n) = 48 - 8n.$$

Defining data and selected scope. The sequence is $a_n = P_n(-1)$, $n \geq 2$, for the polynomials of A. Erickson and F. Ruskey, *Enumerating Maximal Tatami Mat Coverings of Square Grids with v Vertical Dominoes*, arXiv:1304.0070. Their equations (1)–(2) define $P_n = V_n/D_n$ using $S_m(z) = \prod_{j=1}^{m}(1+z^j)$:

$$V_n(z) = 2\sum_{i=1}^{\lfloor (n-1)/2 \rfloor} S_{n-i-2}(z)S_{i-1}(z)z^{n-i-1} + S_{\lfloor (n-2)/2 \rfloor}(z)^2,$$
$$D_n(z) = \prod_{j \geq 1} S_{\lfloor (n-2)/2^j \rfloor}(z).$$

The product is finite because $S_0 = 1$. The paper proves the divisibility needed to make P_n a polynomial. Its Conjecture 2(e) states the generating function of $P_{n+2}(-1)$. We prove it below from the defining products; we do not assume the later unmarked generating-function entry as a theorem. It and the OEIS recurrence form one package.

Proof by leading terms at $z = -1$. Put $z = -1 + h$ and

$$v_m = \lceil m/2 \rceil, \qquad L_m = \frac{m!}{\lfloor m/2 \rfloor!}.$$

Each odd j contributes $jh + O(h^2)$ and each even j contributes $2 + O(h)$, hence

$$S_m(-1+h) = L_m h^{v_m} + O(h^{v_m+1}). \tag{10.9}$$

For $q = \lfloor (n-2)/2 \rfloor$, the denominator has valuation q and leading coefficient $q!$. Both statements follow inductively from $\lceil q/2 \rceil + \lfloor q/2 \rfloor = q$ and $L_q \lfloor q/2 \rfloor! = q!$.

First let $n = 2m + 2$, $m \geq 1$. Every summand in V_n has valuation at least m, and its coefficient at h^m is $2(-1)^{i+1}L_{2m-i}L_{i-1}$. The terms $i = 2r$ and $i = 2r + 1$ cancel, because $L_{2s} = 2L_{2s-1}$ for $s \geq 1$. If m is even, the last unpaired negative term cancels the central square's coefficient $L_m^2 = 2L_mL_{m-1}$. If m is odd, that square has larger valuation. Only the $i = 1$ term remains. Thus, also checking $m = 0$ directly,

$$P_{2m+2}(-1) = \frac{2L_{2m-1}}{m!} = \binom{2m}{m} \quad (m \geq 1), \qquad P_2(-1) = 1. \tag{10.10}$$

Next let $n = 2m + 3$. Only odd $i = 2r + 1$ contribute at order m; each contributes $-2L_{2(m-r)}L_{2r}$. The central square, when m is even, compensates exactly for the duplicated middle term. With $u_r = L_{2r} = (2r)!/r!$ and $T_m = \sum_{r=0}^{m} u_r u_{m-r}$, it follows that

$$P_{2m+3}(-1) = -\frac{T_m}{m!}.$$

The identity $u_r = 2(2r-1)u_{r-1}$ and symmetry of this convolution give, for $m \geq 1$,

$$T_m - u_m = 2\sum_{r=0}^{m-1}(2r+1)u_r u_{m-1-r} = 2mT_{m-1}.$$

Therefore $b_m = T_m/m!$ satisfies $b_0 = 1$ and $b_m = 2b_{m-1} + \binom{2m}{m}$. Induction proves

$$P_{2m+3}(-1) = -\sum_{j=0}^{m} 2^{m-j}\binom{2j}{j} \qquad (m \geq 0). \tag{10.11}$$

Using $\sum_{m\ge0}\binom{2m}{m}t^m=(1-4t)^{-1/2}$ now proves

$$F(x)=\sum_{n\ge2}P_n(-1)x^n=\frac{x^2}{\sqrt{1-4x^2}}\left(1-\frac{x}{1-2x^2}\right).$$

After dividing by x^2, this is precisely the paper's Conjecture 2(e). Finally write

$$F=\frac{x^2(1-x-2x^2)}{(1-2x^2)(1-4x^2)}\sqrt{1-4x^2}.$$

Substitution in (10.4) gives $LF=0$ exactly in both quadratic-field components. All source indices exist for $n\ge7$, so the selected OEIS recurrence follows for its entire stated range.

10.4.21 A273905: symmetric bargraphs

Source and scope. A273905, checked 20 October 2026. Conjecture; R. J. Mathar (2022-07-22).

Indexing. Offset 2, with

$$(a_2,\dots,a_9)=(1,2,3,5,9,15,27,46).$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{8}p_j(n)a_{n-j}=0\qquad(n\ge10).$$

$$\begin{aligned}&p_0(n)=n+1,\qquad p_1(n)=-2,\qquad p_2(n)=6-4n,\\&p_3(n)=0,\qquad p_4(n)=2n-12,\qquad p_5(n)=6,\\&p_6(n)=-2,\qquad p_7(n)=0,\qquad p_8(n)=n-9.\end{aligned}$$

Independent definition and prior credit. The objects are bargraphs invariant under vertical reflection, counted by semiperimeter. Deutsch and Elizalde, *Statistics on Bargraphs Viewed as Cornerless Motzkin Paths*, Section 4.1, prove the generating function. Heinz's 2016 OEIS program already contains the recurrence; its exact proof, rather than discovery priority, is at issue.

Their decomposition can be summarized as follows. Let $M(u,v)$ count cornerless Motzkin paths by horizontal and up steps. First return gives $M=(1+v(M-1))(1+uM)$. Prefixes ending at height h have series $P_h=M(v/K)^h$, where $K=1-u-uv(M-1)$, because $P_h=vP_{h-1}+(u+uv(M-1))P_h$. Every nonempty symmetric path is uniquely AHA' or $AHHA'$, with A' reflected. If A ends at height h, its reflection supplies h fewer up steps. The bargraph bijection adds one up step. Thus

$$F=x(x+x^2)M(x^2,x^2)\frac{K}{K-x},\qquad K=1-x^2-x^4(M(x^2,x^2)-1).\tag{10.12}$$

Solving the quadratic with constant term 1 gives

$$M(x^2,x^2)=\frac{(1-x^2)^2-\sqrt{D}}{2x^4},\qquad D=(1-x^2)(1-x-x^2-x^3)(1+x-x^2+x^3).$$

Substitution and exact reduction modulo $(\sqrt{D})^2-D$ yield

$$F=\frac{(1+x)(x^4+2x^3+2x^2-1+\sqrt{D})}{2x(1-x-x^2-x^3)}.$$

The verifier independently enumerates palindromic positive column heights.

Recurrence proof. The operator (10.4) with the displayed p_j gives

$$LF=3x^2+6x^3+x^4-4x^5-x^6+2x^7+x^8+0\sqrt{D}.$$

The residual has degree 8. The zero-extended identity starts at $n=9$; for the source sequence of offset 2, every listed term exists at $n\ge10$. This proves the selected recurrence on that range.

10.4.22 Two current-offset corrections for bounded Motzkin walks

The classical finite-path enumeration is credited in advance: Felsner and Heldt, *Lattice Path Enumeration and Toeplitz Matrices*, Journal of Integer Sequences 18 (2015), Fact 10 and Example 17, give the general solution and explicitly list both A-numbers below. Accordingly these packages correct a boundary mismatch in the currently printed generating functions, not an unsolved enumeration theorem.

Let G_m count closed walks beginning at the bottom of a path with m states, with a loop at each state. Decomposing at the first return gives $G_1 = 1/(1-x)$ and $G_m = 1/(1-x-x^2G_{m-1})$. Equivalently, with

$$D_0 = 1, \quad D_1 = 1-x, \quad D_m = (1-x)D_{m-1} - x^2 D_{m-2},$$

induction proves $G_m = D_{m-1}/D_m$. All denominators have constant term 1, so these are identities of formal power series with $G_m(0) = 1$.

10.4.23 A094287: a boundary correction after adding the empty walk

Source and scope. A094287, checked 20 October 2026. Conjecture; R. J. Mathar (2011-12-20).

Indexing. Offset 0, with

$$(a_0, \dots, a_7) = (1, 1, 2, 4, 9, 21, 51, 127).$$

The definition counts walks on states $1, \dots, 6$, starting and ending at 1, with increments in $\{-1, 0, 1\}$.

The source-marked 2011 generating function currently printed in the entry is

$$H(x) = -\frac{x\,(x^5 + 2x^4 - 5x^3 - 2x^2 + 4x - 1)}{(x^3 - x^2 - 2x + 1)\,(x^3 + 3x^2 - 4x + 1)}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{6} p_j(n) a_{n-j} = 0 \qquad (n \geq 6).$$

$$\begin{aligned} &p_0(n) = 1, && p_1(n) = -6, && p_2(n) = 10, \\ &p_3(n) = 0, && p_4(n) = -9, && p_5(n) = 2, \\ &p_6(n) = 1. \end{aligned}$$

Correction and proof. The empty walk gives $a_0 = 1$, whereas $H(0) = 0$. The term $a_0 = 1$ was prepended in 2023, after the 2011 statement. Thus this is a mismatch with the current offset; the old positive-index coefficients are unaffected.

The continuant calculation gives $G = P/Q$, where

$$\begin{aligned} P &= -4x^4 + 2x^3 + 6x^2 - 5x + 1, \\ Q &= x^6 + 2x^5 - 9x^4 + 10x^2 - 6x + 1. \end{aligned}$$

Exact rational simplification yields $G = 1 + H$. This proves the corrected generating function at every index and supplies the explicit counterexample to the unadjusted constant term. Since $Q = \sum_j p_j x^j$ and $\deg P < 6$, the selected recurrence also follows for all $n \geq 6$.

10.4.24 A094288: a boundary correction after adding the empty walk

Source and scope. A094288, checked 20 October 2026. Conjecture; R. J. Mathar (2011-12-20).

Indexing. Offset 0, with

$$(a_0, \dots, a_7) = (1, 1, 2, 4, 9, 21, 51, 127).$$

The definition counts walks on states $1, \dots, 7$, starting and ending at 1, with increments in $\{-1, 0, 1\}$.

The source-marked 2011 generating function currently printed in the entry is

$$H(x) = -\frac{x\left(x^6 - 2x^5 - 7x^4 + 6x^3 + 5x^2 - 5x + 1\right)}{(x-1)\left(x^2 + 2x - 1\right)\left(x^4 - 4x^3 - 2x^2 + 4x - 1\right)}.$$

The selected recurrence, in coefficient-vector notation, is

$$\sum_{j=0}^{7} p_j(n) a_{n-j} = 0 \qquad (n \geq 7).$$

$$\begin{aligned} &p_0(n) = 1, \qquad p_1(n) = -7, \qquad p_2(n) = 15, \\ &p_3(n) = -5, \qquad p_4(n) = -15, \qquad p_5(n) = 9, \\ &p_6(n) = 3, \qquad p_7(n) = -1. \end{aligned}$$

Correction and proof. The empty walk gives $a_0 = 1$, whereas $H(0) = 0$. The term $a_0 = 1$ was prepended in 2023, after the 2011 statement. Thus this is a mismatch with the current offset; the old positive-index coefficients are unaffected.

The continuant calculation gives $G = P/Q$, where

$$\begin{aligned} P &= x^6 + 2x^5 - 9x^4 + 10x^2 - 6x + 1, \\ Q &= -x^7 + 3x^6 + 9x^5 - 15x^4 - 5x^3 + 15x^2 - 7x + 1. \end{aligned}$$

Exact rational simplification yields $G = 1 + H$. This proves the corrected generating function at every index and supplies the explicit counterexample to the unadjusted constant term. Since $Q = \sum_j p_j x^j$ and $\deg P < 7$, the selected recurrence also follows for all $n \geq 7$.

10.4.25 Scope of the audit and reproducibility

The selected IDs were compared against the parent's 908 previously selected IDs, its prior uncounted reserve and known-result exclusions, and the 97-ID OEIS Open appendix baseline. The portable exclusion copy and the per-entry source statements accompany this fragment. Freshly retrieved OEIS pages and the defining references listed above delimit the prior-result audit; no exhaustive claim about all unpublished or unindexed proofs is made. The modular dimension formulas, the A174015 and A273905 generating functions, and the bounded-walk general theorem are expressly credited as known results. The two blocked candidates A099553 and A308620 were not selected.

The dimension packages are related specializations. In particular, for $n \geq 1$,

$$a_{063110}(n) = a_{063116}(n) + 2, \qquad a_{063148}(n) = a_{063140}(n) + 2.$$

Also $a_{063140}(n) = 2a_{063098}(n)$ and $2a_{063148}(n) = 3a_{063110}(n)$ for $n \geq 2$, with respective differences -1 at $n = 1$. These exact identities are disclosed in the companion audit; the count is of source-assertion packages, not unrelated mathematical methods. The exact companion A063128 is not counted.

Run `python r10_gf_verify.py` for the exact certificates, or `python r10_gf_build.py` to regenerate this fragment and those certificates. Python 3 and SymPy are sufficient; the scripts use no network and import no prior-round module. The JSON certificates contain the actual operator residuals, rational numerators and denominators, all dimension parameters, source-prefix comparisons, and independent coefficient checks through index 100. Their universal identities, together with the mathematical definition-to-series arguments above, prove the stated full ranges.

10.5 Restricted-growth word arrays: exact recurrence certificates

This section proves **65 distinct primary-entry assertion packages** and retains 3 further freshly certified packages outside that count. In each selected OEIS snapshot the recurrence is explicitly marked *Empirical*. The selection contains twelve closely related local-avoidance families. It is a collection of specific source assertions proved by a common finite-state argument, not a claim to 65 unrelated major open problems. The 31 listed source generating functions that are also verified add no further count.

The primary entries and all 12 named parent tables were retrieved for this round on October 20, 2026. Cache ages are retained in the source audit; the retrieval date does not establish that the snapshots include every later live edit. The primary recurrence of A212827 is empirical, although its column is absent from the parent table's displayed empirical formula list. The primary entries list numeric b-files rather than linked proofs; those b-files were not expanded, and the 1452 printed selected-entry terms were compared with the definition. The retained source authorship and formula attributions distinguish R. H. Hardin's entries from later generating-function contributions.

10.5.1 Definitions and the twelve local predicates

A word $w = w_1 \cdots w_\ell$ over $\{0, \dots, q-1\}$ is a *restricted-growth word* if $w_1 = 0$ and, for every $j > 1$,

$$0 \le w_j \le \min\bigl(q-1,\, 1 + \max(w_1, \dots, w_{j-1})\bigr).$$

Thus new labels appear in the order $0, 1, \dots$; a word need not use every available label. This is the precise meaning of the entries' introduction-order condition. Existing labels remain distinct and are never divided out by a color-permutation factor.

For any finite word v, let $\mathrm{can}(v)$ relabel its distinct letters by $0, 1, \dots$ in their order of first appearance. For example, $\mathrm{can}(7,2,7,2) = 0101$ and $\mathrm{can}(7,7,7,7) = 0000$. The forbidden patterns below concern *equality*, not the numerical order of distinct letters. Consequently a forbidden *aba* with $a \neq b$ excludes both numerical orders of its two letters. Forbidding *abc* with three unequal letters excludes all six numerical orders. Only consecutive windows are tested.

Each entry specifies an alphabet size q, a memory K, a length shift s, and a set $\mathcal{B}$ of canonical words of length $K+1$. Its source term $a(n)$ counts the permitted words of length $n+s$ for $n \ge 1$. The complete translation of the local conditions is as follows.

Source local condition	$\mathcal{B}$	(K, s)
Repeated adjacent pair, allowing a constant pair	$\{0000, 0101\}$	$(3, 0)$
aabb, possibly $a = b$	$\{0000, 0011\}$	$(3, 3)$
aabb, $a \neq b$	$\{0011\}$	$(3, 3)$
abba, possibly $a = b$	$\{0000, 0110\}$	$(3, 3)$
abab, $a \neq b$	$\{0101\}$	$(3, 3)$
abba, $a \neq b$	$\{0110\}$	$(3, 3)$
aba or *abb*, $a \neq b$	$\{010, 011\}$	$(2, 2)$
aba, $a \neq b$	$\{010\}$	$(2, 2)$
Three pairwise unequal letters	$\{012\}$	$(2, 2)$
Three equal or three pairwise unequal letters	$\{000, 012\}$	$(2, 2)$
ababa or *abbba*, possibly $a = b$	$\{00000, 01010, 01110\}$	$(4, 4)$
ababa or *abbba*, $a \neq b$	$\{01010, 01110\}$	$(4, 4)$

In the first row, two consecutive adjacent pairs are equal exactly when the four-letter window has its first and third letters equal and its second and fourth letters equal. The word length is n in that family; shifting it to $n+3$ would change both the source sequence and its recurrence threshold.

10.5.2 The complete state space and weighted transitions

Let $\mathcal{P}_{K,q}$ be all canonical restricted-growth words of length K using at most q letters. Define

$$\mathcal{S} = \{(m,p) : 1 \le m \le q,\ p \in \mathcal{P}_{K,q},\ r(p) \le m\}, \qquad r(p) = 1 + \max p.$$

The state records the number m of labels introduced anywhere in the prefix and the equality pattern p of its final K letters. The state order is increasing m, then lexicographic p. All these states are retained, including states unreachable from the initial words. In particular the matrix dimension is

$$D = \sum_{r=1}^{\min(K,q)} \left\{ {K \atop r} \right\} (q - r + 1),$$

where the braces count partitions of a K-element set into r blocks.

Let T be the row-to-column transition matrix. From (m,p), put $r = r(p)$. Every legal append belongs to exactly one of the following three cases.

1. For each letter $c \in \{0, \dots, r-1\}$ already present in the suffix, if $\mathrm{can}(pc) \notin \mathcal{B}$, add weight one to $(m, \mathrm{can}(p_2 \cdots p_K c))$.
2. There are $m - r$ previously introduced letters absent from the suffix. They all have the same equality pattern when appended, represented by $c = r$. If $\mathrm{can}(pr) \notin \mathcal{B}$, add weight $m - r$ to $(m, \mathrm{can}(p_2 \cdots p_K r))$.
3. If $m < q$, the single next new label may be introduced. With the same test $\mathrm{can}(pr) \notin \mathcal{B}$, add weight one to $(m+1, \mathrm{can}(p_2 \cdots p_K r))$.

Weights with the same destination are added; zero weights are omitted. These cases exhaust the actual labels available to an introduced-order word. The forbidden-window test and destination depend only on the stated equality pattern, so two actual prefixes with the same state have exactly these same numbers of extensions to each destination. This proves that the weighted compression preserves the number of words, rather than assuming a symmetry quotient for the counted objects.

Let u be the row vector with $u_{(m,p)} = 1$ when $m = r(p)$ and zero otherwise. Every length-K restricted-growth word is one of these p's, and cannot yet contain a forbidden window of length $K+1$. Hence u is the exact initial vector. With $\mathbf{1}$ the all-ones column, induction on the number of appended letters gives

$$a(n) = uT^{n+s-K}\mathbf{1} \qquad (n + s \ge K). \tag{10.13}$$

When $n + s < K$, the finitely many short words are counted directly. Only A212824–A212827 use this exception, and their first two source terms are $1, 2$.

10.5.3 A finite vector identity proves every later recurrence term

For an entry let the displayed source coefficients be $c_1, \dots, c_d$ and let

$$b = d + 1 + K - s, \qquad e = b + s - K = d + 1, \qquad v_j = T^j\mathbf{1} \quad (0 \le j \le e).$$

The exact integer certificate retains every coordinate of every v_j and of

$$z = v_e - \sum_{i=1}^{d} c_i v_{e-i}. \tag{10.14}$$

Let W be the set of states reachable by nonzero transitions from the support of u. It contains that support and is closed under transitions by definition; the verifiers also check both properties directly, and independently reconstruct the whole reachable set. Every certificate has $z_t = 0$ for every $t \in W$. In fact, in all 68 retained packages the stronger statement $z = 0$ on the full state space holds: all 2188 residual coordinates are zero. No unreachable coordinate has been removed from the certificate.

For $n \geq b$ all indices $n - i$ in the recurrence are source indices at least one, all powers in (10.13) are nonnegative, and

$$\begin{aligned} a(n) - \sum_{i=1}^{d} c_i a(n-i) &= uT^{n-b}\left(T^e\mathbf{1} - \sum_{i=1}^{d} c_i T^{e-i}\mathbf{1}\right) \\ &= uT^{n-b}z = 0. \end{aligned}$$

The last equality already follows from $z|_W = 0$ and the closure of W. We have therefore proved, universally,

$$a(n) = \sum_{i=1}^{d} c_i a(n-i) \qquad (n \geq b). \tag{10.15}$$

For length $n + K$ entries this means $b = d + 1$. For A212824–A212827, $s = 0, K = 3$ and therefore $b = d + 4$, namely $8, 10, 12, 14$ respectively, exactly the explicit thresholds in their source assertions.

This proof does not infer a recurrence from a long list of scalar values. The source supplies the candidate coefficients, the definition supplies T, u, and (10.14) is checked as an exact vector equality. The finite identity is sufficient for all subsequent n by the displayed algebra. No assumption about minimal recurrence order is needed.

For reproducibility, the central identity check is the following finite integer calculation. The full verifier additionally reconstructs T, u from the definition, checks the source coefficients and threshold, checks the reachability set, and checks every stored vector against this calculation.

```
# rows[t] contains every (destination, positive integer weight).
# initial[t] is u_t. c contains the d source coefficients.
v = [[1] * len(rows)]
for j in range(d + 1):
    v.append([sum(weight * v[-1][dest] for dest, weight in row)
              for row in rows])
z = [v[d+1][t] - sum(c[i-1] * v[d+1-i][t]
                     for i in range(1, d+1))
     for t in range(len(rows))]
W = {t for t, value in enumerate(initial) if value}
pending = list(W)
while pending:
    t = pending.pop()
    for dest, weight in rows[t]:
        if dest not in W:
            W.add(dest)
            pending.append(dest)
assert all(dest in W for t in W for dest, weight in rows[t])
assert all(z[t] == 0 for t in W)
# All coordinates were retained; the present 68 cases also satisfy:
assert z == [0] * len(rows)
```

10.5.4 Generating functions and an explicit small certificate

Put $F(x) = \sum_{n \geq 1} a(n)x^n$ and $Q(x) = 1 - \sum_{i=1}^{d} c_i x^i$. For this formal series let $A_0 = 0$ and $A_j = a(j)$ for $j \geq 1$; A_0 is a series convention, not an assertion about an auxiliary transfer value at source index zero. If $Q(x) = \sum_{i=0}^{d} q_i x^i$, define

$$P(x) = \sum_{j=0}^{b-1} \left(\sum_{i=0}^{\min(j,d)} q_i A_{j-i} \right) x^j.$$

Equation (10.15) makes every coefficient of QF of degree at least b zero; the lower coefficients are exactly those displayed in P. Thus $QF = P$ as formal power series. Since $Q(0) = 1$, division is legitimate, and $F = P/Q$. This proves each recorded source generating function by exact polynomial cross-multiplication, not by agreement through finitely many terms.

For example A243513 has $q = 5$, $K = s = 2$, and $\mathcal{B} = \{010, 011\}$. The full matrix has nine states. Its reachable states, in order, are

$$(1, 00),\ (2, 01),\ (3, 01),\ (4, 01),\ (5, 01).$$

Their closed transition block and initial row are

$$T_W = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}, \qquad u_W = (1, 1, 0, 0, 0).$$

The four successive vectors, restricted here to W for a readable worked example, are

j	$(T_W^j \mathbf{1})^{\mathsf{T}}$				
1	2	1	2	3	3
2	3	2	5	9	9
3	5	5	14	27	27
4	10	14	41	81	81

Every column satisfies $v_4 - 5v_3 + 7v_2 - 3v_1 = 0$. Therefore $a(n) = 5a(n-1) - 7a(n-2) + 3a(n-3)$ for all $n \geq 4$, with initial terms $3, 5, 10$, and

$$F(x) = \frac{3x - 10x^2 + 6x^3}{1 - 5x + 7x^2 - 3x^3}.$$

The machine certificate also retains all four other state coordinates and checks them; the small display is not a deletion from the full proof data. The explicitly empirical source and its parent are https://oeis.org/A243513 and https://oeis.org/A243519.

10.5.5 Independent checks, exclusions, and scope

The 65 selected matrices have 2098 full states and 6816 nonzero weighted edges. They retain 55400 Krylov-vector coordinates, with maximum dimension 128 and maximum recurrence order 49. The independent implementation enumerates every actual suffix over the introduced labels, tests forbidden windows by pairwise equality, and checks that every actual transition aggregates to the stated compressed row. It imports neither the compressed engine nor its verifier. For the selected set it checked 131159 labelled states, 1052986 legal labelled transitions, all 2098 compressed rows, and 406 direct short-word/source comparisons. It independently checks the exact vector identity and closed reachable set as well. This is an internal independent implementation check, not external peer review or proof-assistant formalization.

No selected ID occurs among the retained 908 earlier selected IDs, five earlier uncounted reserves, 22 retained known-result exclusions, or the 97 distinct IDs of the OEIS Open Appendix A baseline. The latter audit concerns that appendix's 100 costly claims, not every result in its repository. Identity and arbitrary rational-scale relations were screened at shifts $-3, \ldots, 3$ against all 352 retained earlier array packages, including their available reserves. Each of the 167552 prior comparisons failed within 20 terms. Each of the 15946 within-round comparisons failed within 60 terms. These mismatches exclude those particular relations; they do not exclude larger shifts or other transforms.

Four initially considered three-equal-letter avoidance entries, A242468–A242471, were removed after a direct prior result was located. Mansour and Munagi, *Enumeration of Partitions by Long Rises,*

Levels, and Descents, J. Integer Sequences 12 (2009), Article 09.1.8, Theorem 2.1 and Section 2.1, give for exactly k blocks

$$L_3(x,0;k) = \frac{(x+x^2)^k}{\prod_{j=0}^{k-1}(1-j(x+x^2))}.$$

Summing this expression over $1 \le k \le q$ and removing the length-one and length-two terms gives exactly the source series $(\sum_{k=1}^{q} L_3(x,0;k) - x - 2x^2)/x^2$. Those four direct specializations have count zero. The primary paper is https://cs.uwaterloo.ca/journals/JIS/VOL12/Munagi/munagi15.pdf.

The bounded literature review also read Mansour, Shattuck and Yan, *Counting Subwords in a Partition of a Set*, Electron. J. Combin. 17 (2010), R19, at https://www.combinatorics.org/ojs/index.php/eljc/article/download/v17i1r19/pdf/, and Mansour and Shattuck, *Counting Peaks and Valleys in a Partition of a Set*, J. Integer Sequences 13 (2010), Article 10.6.8, at https://emis.de/ft/3989. The former treats individual numerical-order subwords, so its separate 121 and 212 assertions are not themselves a joint equality-pattern avoidance formula. The latter's inequalities for peaks and valleys differ from the selected equal-endpoint conditions. No direct proof of a retained selected assertion was identified in the inspected portions. A potentially relevant 2020 IEEE paper by Sabri and Tran on restricted-growth functions avoiding vincular patterns was found, but its full text could not be retrieved; that is an explicit coverage gap. The compact literature audit preserves the primary URLs, search scope, and this limitation.

These facts establish the proofs and the source-marked status of the selected assertions in the retrieved material. They do not establish worldwide priority or the absence of every earlier proof. No OEIS edit, message, or submission was performed, and no official closure is claimed. Other formulas in the same entries and unselected columns of parent tables are not silently counted.

10.5.6 Entry-by-entry assertions and certificates

Each coefficient list below is in the order $(c_1, \dots, c_d)$ used in (10.15). Each package supplies its full D-state matrix, initial vector, all $e+1 = d+2$ full vectors, and every residual coordinate in the portable JSON certificate files. The transition definition above and the per-entry parameters completely specify the finite calculation; the listed assertion follows from (10.14) and the universal argument, not from the three displayed initial terms.

10.5.6.1 A212824

Parameters: $q = 4$, $K = 3$, $s = 0$, and $\mathcal{B} = \{0000, 0101\}$. The source counts words of length n for $n \ge 1$. Its empirical recurrence has order $d = 4$ and is proved for every $n \ge 8$, with coefficients

$$(c_1, \dots, c_d) = (4, 1, -6, -3).$$

The complete matrix has $D = 15$ states, with $|W| = 15$ reachable states. All 90 coordinates of $v_0, \dots, v_5$ are retained. The residual in (10.14) is zero in all 15 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(1, 2, 5)$. Source: https://oeis.org/A212824; column 3 of https://oeis.org/A212829. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = x - 2x^2 - 4x^3 - 3x^4 + x^5 + 3x^6 + x^7.$$

10.5.6.2 A212825

Parameters: $q = 5$, $K = 3$, $s = 0$, and $\mathcal{B} = \{0000, 0101\}$. The source counts words of length n for $n \geq 1$. Its empirical recurrence has order $d = 6$ and is proved for every $n \geq 10$, with coefficients

$$(c_1, \ldots, c_d) = (7, -7, -20, 10, 24, 8).$$

The complete matrix has $D = 20$ states, with $|W| = 20$ reachable states. All 160 coordinates of $v_0, \ldots, v_7$ are retained. The residual in (10.14) is zero in all 20 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(1, 2, 5)$. Source: https://oeis.org/A212825; column 4 of https://oeis.org/A212829. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = x - 5x^2 - 2x^3 + 12x^4 + 18x^5 + 6x^6 - 12x^7 - 12x^8 - 3x^9.$$

10.5.6.3 A212826

Parameters: $q = 6$, $K = 3$, $s = 0$, and $\mathcal{B} = \{0000, 0101\}$. The source counts words of length n for $n \geq 1$. Its empirical recurrence has order $d = 8$ and is proved for every $n \geq 12$, with coefficients

```
11, -30, -21, 112, 63, -119, -120, -30
```

The complete matrix has $D = 25$ states, with $|W| = 25$ reachable states. All 250 coordinates of $v_0, \ldots, v_9$ are retained. The residual in (10.14) is zero in all 25 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(1, 2, 5)$. Source: https://oeis.org/A212826; column 5 of https://oeis.org/A212829. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$\begin{aligned} P(x) = x &- 9x^2 + 13x^3 + 39x^4 - 19x^5 - 108x^6 - 106x^7 + 6x^8 + 84x^9 \\ &+ 55x^{10} + 11x^{11}. \end{aligned}$$

10.5.6.4 A212827

Parameters: $q = 7$, $K = 3$, $s = 0$, and $\mathcal{B} = \{0000, 0101\}$. The source counts words of length n for $n \geq 1$. Its empirical recurrence has order $d = 10$ and is proved for every $n \geq 14$, with coefficients

```
16, -79, 70, 361, -372, -964, 144, 1116, 720, 144
```

The complete matrix has $D = 30$ states, with $|W| = 30$ reachable states. All 360 coordinates of $v_0, \ldots, v_{11}$ are retained. The residual in (10.14) is zero in all 30 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(1, 2, 5)$. Source: https://oeis.org/A212827; column 6 of https://oeis.org/A212829. The primary recurrence is explicitly empirical. The parent definition was read, but its displayed empirical formulas do not include this column.

10.5.6.5 A242543

Parameters: $q = 3$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0000, 0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 4$ and is proved for every $n \geq 5$, with coefficients

$$(c_1, \ldots, c_d) = (2, 0, 4, 4).$$

The complete matrix has $D = 10$ states, with $|W| = 10$ reachable states. All 60 coordinates of $v_0, \ldots, v_5$ are retained. The residual in (10.14) is zero in all 10 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(12, 32, 88)$. Source: https://oeis.org/A242543; column 2 of https://oeis.org/A242549. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 12x + 8x^2 + 24x^3 + 18x^4.$$

10.5.6.6 A242544

Parameters: $q = 4$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0000, 0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 8$ and is proved for every $n \geq 9$, with coefficients

```
4, -3, 10, -2, -12, -9, -18, -9
```

The complete matrix has $D = 15$ states, with $|W| = 15$ reachable states. All 150 coordinates of $v_0, \ldots, v_9$ are retained. The residual in (10.14) is zero in all 15 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(13, 42, 150)$. Source: https://oeis.org/A242544; column 3 of https://oeis.org/A242549. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 13x - 10x^2 + 21x^3 - 50x^4 - 88x^5 - 78x^6 - 99x^7 - 42x^8.$$

10.5.6.7 A242546

Parameters: $q = 6$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0000, 0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 16$ and is proved for every $n \geq 17$, with coefficients

```
11, -41, 100, -259, 283, -178, 383, 1393, 312, 723, -1707, -5129, -5430, -5400, -3600,
-900
```

The complete matrix has $D = 25$ states, with $|W| = 25$ reachable states. All 450 coordinates of $v_0, \ldots, v_{17}$ are retained. The residual in (10.14) is zero in all 25 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(13, 43, 166)$. Source: https://oeis.org/A242546; column 5 of https://oeis.org/A242549. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.8 A242547

Parameters: $q = 7$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0000, 0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 20$ and is proved for every $n \geq 21$, with coefficients

```
16, -95, 326, -998, 2268, -2953, 5002, -877, -8932, -5460, -46536, -36728, -4608,
43488, 180288, 260208, 254016, 207360, 103680, 20736
```

The complete matrix has $D = 30$ states, with $|W| = 30$ reachable states. All 660 coordinates of $v_0, \ldots, v_{21}$ are retained. The residual in (10.14) is zero in all 30 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(13, 43, 166)$. Source: https://oeis.org/A242547; column 6 of https://oeis.org/A242549. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.9 A242548

Parameters: $q = 8$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0000, 0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 24$ and is proved for every $n \geq 25$, with coefficients

```
22, -190, 924, -3465, 10962, -24402, 45828, -74016, 19016, -23090, -159064, 571211,
704098, 1846562, 2774132, 616507, -2599138, -8906720, -16117180, -18096476, -15823920,
-10584000, -4233600, -705600
```

The complete matrix has $D = 35$ states, with $|W| = 35$ reachable states. All 910 coordinates of $v_0, \ldots, v_{25}$ are retained. The residual in (10.14) is zero in all 35 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(13, 43, 166)$. Source: https://oeis.org/A242548; column 7 of https://oeis.org/A242549. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.10 A242578

Parameters: $q = 3$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 4$ and is proved for every $n \geq 5$, with coefficients

$$(c_1, \ldots, c_d) = (4, -5, 6, -4).$$

The complete matrix has $D = 10$ states, with $|W| = 10$ reachable states. All 60 coordinates of $v_0, \ldots, v_5$ are retained. The residual in (10.14) is zero in all 10 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(13, 35, 97)$. Source: https://oeis.org/A242578; column 2 of https://oeis.org/A242584. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 13x - 17x^2 + 22x^3 - 20x^4.$$

10.5.6.11 A242579

Parameters: $q = 4$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 7$ and is proved for every $n \geq 8$, with coefficients

```
7, -18, 32, -45, 37, -21, 9
```

The complete matrix has $D = 15$ states, with $|W| = 15$ reachable states. All 135 coordinates of $v_0, \ldots, v_8$ are retained. The residual in (10.14) is zero in all 15 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 45, 159)$. Source: https://oeis.org/A242579; column 3 of https://oeis.org/A242584. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 14x - 53x^2 + 96x^3 - 164x^4 + 152x^5 - 87x^6 + 45x^7.$$

10.5.6.12 A242580

Parameters: $q = 5$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 10$ and is proved for every $n \geq 11$, with coefficients

```
11, -48, 131, -282, 452, -541, 530, -364, 176, -64
```

The complete matrix has $D = 20$ states, with $|W| = 20$ reachable states. All 240 coordinates of $v_0, \ldots, v_{11}$ are retained. The residual in (10.14) is zero in all 20 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 46, 174)$. Source: https://oeis.org/A242580; column 4 of https://oeis.org/A242584. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.13 A242581

Parameters: $q = 6$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 13$ and is proved for every $n \geq 14$, with coefficients

```
16, -106, 425, -1284, 3083, -5823, 9091, -11623, 11784, -9883, 6151, -2730, 900
```

The complete matrix has $D = 25$ states, with $|W| = 25$ reachable states. All 375 coordinates of $v_0, \ldots, v_{14}$ are retained. The residual in (10.14) is zero in all 25 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 46, 175)$. Source: https://oeis.org/A242581; column 5 of https://oeis.org/A242584. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.14 A242582

Parameters: $q = 7$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0011\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 16$ and is proved for every $n \geq 17$, with coefficients

```
22, -206, 1156, -4714, 15358, -40643, 89400, -167115, 262308, -349485, 393316,
-361252, 276048, -160848, 67392, -20736
```

The complete matrix has $D = 30$ states, with $|W| = 30$ reachable states. All 540 coordinates of $v_0, \ldots, v_{17}$ are retained. The residual in (10.14) is zero in all 30 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 46, 175)$. Source: https://oeis.org/A242582; column 6 of https://oeis.org/A242584. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.15 A243027

Parameters: $q = 3$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0000, 0110\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 3$ and is proved for every $n \geq 4$, with coefficients

$$(c_1, \ldots, c_d) = (2, 1, 2).$$

The complete matrix has $D = 10$ states, with $|W| = 10$ reachable states. All 50 coordinates of $v_0, \ldots, v_4$ are retained. The residual in (10.14) is zero in all 10 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(12, 32, 85)$. Source: https://oeis.org/A243027; column 2 of https://oeis.org/A243033. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 12x + 8x^2 + 9x^3.$$

10.5.6.16 A243028

Parameters: $q = 4$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0000, 0110\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 6$ and is proved for every $n \geq 7$, with coefficients

$$(c_1, \ldots, c_d) = (4, -1, 2, -6, -2, -3).$$

The complete matrix has $D = 15$ states, with $|W| = 15$ reachable states. All 120 coordinates of $v_0, \ldots, v_7$ are retained. The residual in (10.14) is zero in all 15 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(13, 42, 147)$. Source: https://oeis.org/A243028; column 3 of https://oeis.org/A243033. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 13x - 10x^2 - 8x^3 - 36x^4 - 16x^5 - 14x^6.$$

10.5.6.17 A243029

Parameters: $q = 5$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0000, 0110\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 9$ and is proved for every $n \geq 10$, with coefficients

```
7, -10, 1, -21, 13, 6, 27, 10, 8
```

The complete matrix has $D = 20$ states, with $|W| = 20$ reachable states. All 220 coordinates of $v_0, \ldots, v_{10}$ are retained. The residual in (10.14) is zero in all 20 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(13, 43, 162)$. Source: https://oeis.org/A243029; column 4 of https://oeis.org/A243033. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 13x - 48x^2 - 9x^3 - 45x^4 + 108x^5 + 98x^6 + 154x^7 + 63x^8 + 37x^9.$$

10.5.6.18 A243038

Parameters: $q = 3$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0101\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 4$ and is proved for every $n \geq 5$, with coefficients

$$(c_1, \ldots, c_d) = (3, 0, -1, -1).$$

The complete matrix has $D = 10$ states, with $|W| = 10$ reachable states. All 60 coordinates of $v_0, \ldots, v_5$ are retained. The residual in (10.14) is zero in all 10 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(13, 36, 101)$. Source: https://oeis.org/A243038; column 2 of https://oeis.org/A243044. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 13x - 3x^2 - 7x^3 - 5x^4.$$

10.5.6.19 A243039

Parameters: $q = 4$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0101\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 7$ and is proved for every $n \geq 8$, with coefficients

```
5, -3, -5, -3, 3, 3, 1
```

The complete matrix has $D = 15$ states, with $|W| = 15$ reachable states. All 135 coordinates of $v_0, \ldots, v_8$ are retained. The residual in (10.14) is zero in all 15 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 46, 163)$. Source: https://oeis.org/A243039; column 3 of https://oeis.org/A243044. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 14x - 24x^2 - 25x^3 - 5x^4 + 22x^5 + 17x^6 + 5x^7.$$

10.5.6.20 A243040

Parameters: $q = 5$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0101\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 10$ and is proved for every $n \geq 11$, with coefficients

```
8, -14, -10, 13, 28, 9, -12, -14, -6, -1
```

The complete matrix has $D = 20$ states, with $|W| = 20$ reachable states. All 240 coordinates of $v_0, \ldots, v_{11}$ are retained. The residual in (10.14) is zero in all 20 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 47, 178)$. Source: https://oeis.org/A243040; column 4 of https://oeis.org/A243044. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.21 A243041

Parameters: $q = 6$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0101\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 13$ and is proved for every $n \geq 14$, with coefficients

```
12, -41, 13, 96, 33, -134, -166, -33, 82, 87, 41, 10, 1
```

The complete matrix has $D = 25$ states, with $|W| = 25$ reachable states. All 375 coordinates of $v_0, \ldots, v_{14}$ are retained. The residual in (10.14) is zero in all 25 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 47, 179)$. Source: https://oeis.org/A243041; column 5 of https://oeis.org/A243044. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.22 A243042

Parameters: $q = 7$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0101\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 16$ and is proved for every $n \geq 17$, with coefficients

```
17, -95, 154, 222, -448, -773, 213, 1374, 1181, 27, -736, -690, -334, -95, -15, -1
```

The complete matrix has $D = 30$ states, with $|W| = 30$ reachable states. All 540 coordinates of $v_0, \ldots, v_{17}$ are retained. The residual in (10.14) is zero in all 30 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 47, 179)$. Source: https://oeis.org/A243042; column 6 of https://oeis.org/A243044. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.23 A243043

Parameters: $q = 8$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0101\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 19$ and is proved for every $n \geq 20$, with coefficients

```
23, -190, 614, -135, -2559, 369, 7121, 5143, -7257, -15107, -9219, 2219, 7955, 6635,
3195, 982, 190, 21, 1
```

The complete matrix has $D = 35$ states, with $|W| = 35$ reachable states. All 735 coordinates of $v_0, \ldots, v_{20}$ are retained. The residual in (10.14) is zero in all 35 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 47, 179)$. Source: https://oeis.org/A243043; column 7 of https://oeis.org/A243044. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.24 A243383

Parameters: $q = 3$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0110\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 4$ and is proved for every $n \geq 5$, with coefficients

$$(c_1, \ldots, c_d) = (4, -4, 2, -1).$$

The complete matrix has $D = 10$ states, with $|W| = 10$ reachable states. All 60 coordinates of $v_0, \ldots, v_5$ are retained. The residual in (10.14) is zero in all 10 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(13, 35, 96)$. Source: https://oeis.org/A243383; column 2 of https://oeis.org/A243389. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 13x - 17x^2 + 8x^3 - 5x^4.$$

10.5.6.25 A243384

Parameters: $q = 4$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0110\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 7$ and is proved for every $n \geq 8$, with coefficients

```
7, -16, 18, -15, 9, -3, 1
```

The complete matrix has $D = 15$ states, with $|W| = 15$ reachable states. All 135 coordinates of $v_0, \ldots, v_8$ are retained. The residual in (10.14) is zero in all 15 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 45, 158)$. Source: https://oeis.org/A243384; column 3 of https://oeis.org/A243389. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 14x - 53x^2 + 67x^3 - 59x^4 + 40x^5 - 13x^6 + 5x^7.$$

10.5.6.26 A243385

Parameters: $q = 5$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0110\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 10$ and is proved for every $n \geq 11$, with coefficients

```
11, -44, 87, -107, 101, -71, 37, -16, 4, -1
```

The complete matrix has $D = 20$ states, with $|W| = 20$ reachable states. All 240 coordinates of $v_0, \ldots, v_{11}$ are retained. The residual in (10.14) is zero in all 20 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 46, 173)$. Source: https://oeis.org/A243385; column 4 of https://oeis.org/A243389. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.27 A243386

Parameters: $q = 6$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0110\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 13$ and is proved for every $n \geq 14$, with coefficients

```
16, -99, 313, -584, 752, -757, 592, -370, 191, -74, 25, -5, 1
```

The complete matrix has $D = 25$ states, with $|W| = 25$ reachable states. All 375 coordinates of $v_0, \ldots, v_{14}$ are retained. The residual in (10.14) is zero in all 25 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 46, 174)$. Source: https://oeis.org/A243386; column 5 of https://oeis.org/A243389. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.28 A243387

Parameters: $q = 7$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0110\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 16$ and is proved for every $n \geq 17$, with coefficients

```
22, -195, 914, -2539, 4594, -6074, 6382, -5368, 3726, -2167, 1026, -419, 130, -36, 6,
-1
```

The complete matrix has $D = 30$ states, with $|W| = 30$ reachable states. All 540 coordinates of $v_0, \ldots, v_{17}$ are retained. The residual in (10.14) is zero in all 30 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 46, 174)$. Source: https://oeis.org/A243387; column 6 of https://oeis.org/A243389. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.29 A243388

Parameters: $q = 8$, $K = 3$, $s = 3$, and $\mathcal{B} = \{0110\}$. The source counts words of length $n + 3$ for $n \geq 1$. Its empirical recurrence has order $d = 19$ and is proved for every $n \geq 20$, with coefficients

```
29, -349, 2287, -9065, 23201, -41175, 55373, -60041, 53229, -39663, 25113, -13424,
6232, -2386, 806, -209, 49, -7, 1
```

The complete matrix has $D = 35$ states, with $|W| = 35$ reachable states. All 735 coordinates of $v_0, \ldots, v_{20}$ are retained. The residual in (10.14) is zero in all 35 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(14, 46, 174)$. Source: https://oeis.org/A243388; column 7 of https://oeis.org/A243389. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.30 A243513

Parameters: $q = 5$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010, 011\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 3$ and is proved for every $n \geq 4$, with coefficients

$$(c_1, \ldots, c_d) = (5, -7, 3).$$

The complete matrix has $D = 9$ states, with $|W| = 5$ reachable states. All 45 coordinates of $v_0, \ldots, v_4$ are retained. The residual in (10.14) is zero in all 9 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(3, 5, 10)$. Source: https://oeis.org/A243513; column 4 of https://oeis.org/A243519. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 3x - 10x^2 + 6x^3.$$

10.5.6.31 A243514

Parameters: $q = 6$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010, 011\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 4$ and is proved for every $n \geq 5$, with coefficients

$$(c_1, \ldots, c_d) = (8, -21, 22, -8).$$

The complete matrix has $D = 11$ states, with $|W| = 6$ reachable states. All 66 coordinates of $v_0, \ldots, v_5$ are retained. The residual in (10.14) is zero in all 11 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(3, 5, 10)$. Source: https://oeis.org/A243514; column 5 of https://oeis.org/A243519. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 3x - 19x^2 + 33x^3 - 16x^4.$$

10.5.6.32 A243515

Parameters: $q = 7$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010, 011\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 5$ and is proved for every $n \geq 6$, with coefficients

$$(c_1, \ldots, c_d) = (12, -52, 102, -91, 30).$$

The complete matrix has $D = 13$ states, with $|W| = 7$ reachable states. All 91 coordinates of $v_0, \ldots, v_6$ are retained. The residual in (10.14) is zero in all 13 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(3, 5, 10)$. Source: https://oeis.org/A243515; column 6 of https://oeis.org/A243519. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 3x - 31x^2 + 106x^3 - 141x^4 + 60x^5.$$

10.5.6.33 A243516

Parameters: $q = 8$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010, 011\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 6$ and is proved for every $n \geq 7$, with coefficients

$$(c_1, \ldots, c_d) = (17, -111, 355, -584, 468, -144).$$

The complete matrix has $D = 15$ states, with $|W| = 8$ reachable states. All 120 coordinates of $v_0, \ldots, v_7$ are retained. The residual in (10.14) is zero in all 15 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(3, 5, 10)$. Source: https://oeis.org/A243516; column 7 of https://oeis.org/A243519. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 3x - 46x^2 + 258x^3 - 655x^4 + 739x^5 - 288x^6.$$

10.5.6.34 A243517

Parameters: $q = 9$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010, 011\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 7$ and is proved for every $n \geq 8$, with coefficients

```
23, -212, 1010, -2669, 3887, -2878, 840
```

The complete matrix has $D = 17$ states, with $|W| = 9$ reachable states. All 153 coordinates of $v_0, \ldots, v_8$ are retained. The residual in (10.14) is zero in all 17 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(3, 5, 10)$. Source: https://oeis.org/A243517; column 8 of https://oeis.org/A243519. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 3x - 64x^2 + 531x^3 - 2175x^4 + 4579x^5 - 4607x^6 + 1680x^7.$$

10.5.6.35 A243518

Parameters: $q = 10$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010, 011\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 8$ and is proved for every $n \geq 9$, with coefficients

```
30, -372, 2478, -9639, 22260, -29588, 20592, -5760
```

The complete matrix has $D = 19$ states, with $|W| = 10$ reachable states. All 190 coordinates of $v_0, \ldots, v_9$ are retained. The residual in (10.14) is zero in all 19 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(3, 5, 10)$. Source: https://oeis.org/A243518; column 9 of https://oeis.org/A243519. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 3x - 85x^2 + 976x^3 - 5849x^4 + 19574x^5 - 36095x^6 + 33305x^7 - 11520x^8.$$

10.5.6.36 A243600

Parameters: $q = 4$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 5$ and is proved for every $n \geq 6$, with coefficients

$$(c_1, \ldots, c_d) = (5, -5, -3, 3, 1).$$

The complete matrix has $D = 7$ states, with $|W| = 7$ reachable states. All 49 coordinates of $v_0, \ldots, v_6$ are retained. The residual in (10.14) is zero in all 7 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 10, 28)$. Source: https://oeis.org/A243600; column 3 of https://oeis.org/A243607. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 4x - 10x^2 - 2x^3 + 7x^4 + 2x^5.$$

10.5.6.37 A243601

Parameters: $q = 5$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 7$ and is proved for every $n \geq 8$, with coefficients

```
8, -18, 5, 17, -4, -6, -1
```

The complete matrix has $D = 9$ states, with $|W| = 9$ reachable states. All 81 coordinates of $v_0, \ldots, v_8$ are retained. The residual in (10.14) is zero in all 9 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 10, 29)$. Source: https://oeis.org/A243601; column 4 of https://oeis.org/A243607. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 4x - 22x^2 + 21x^3 + 24x^4 - 15x^5 - 13x^6 - 2x^7.$$

10.5.6.38 A243602

Parameters: $q = 6$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 9$ and is proved for every $n \geq 10$, with coefficients

```
12, -48, 65, 18, -74, -9, 26, 10, 1
```

The complete matrix has $D = 11$ states, with $|W| = 11$ reachable states. All 121 coordinates of $v_0, \ldots, v_{10}$ are retained. The residual in (10.14) is zero in all 11 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 10, 29)$. Source: https://oeis.org/A243602; column 5 of https://oeis.org/A243607. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 4x - 38x^2 + 101x^3 - 31x^4 - 129x^5 + 18x^6 + 63x^7 + 21x^8 + 2x^9.$$

10.5.6.39 A243603

Parameters: $q = 7$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 11$ and is proved for every $n \geq 12$, with coefficients

```
17, -106, 286, -245, -231, 329, 147, -106, -74, -15, -1
```

The complete matrix has $D = 13$ states, with $|W| = 13$ reachable states. All 169 coordinates of $v_0, \ldots, v_{12}$ are retained. The residual in (10.14) is zero in all 13 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 10, 29)$. Source: https://oeis.org/A243603; column 6 of https://oeis.org/A243607. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.40 A243604

Parameters: $q = 8$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 13$ and is proved for every $n \geq 14$, with coefficients

```
23, -206, 894, -1814, 902, 1940, -1536, -1306, 394, 526, 162, 21, 1
```

The complete matrix has $D = 15$ states, with $|W| = 15$ reachable states. All 225 coordinates of $v_0, \ldots, v_{14}$ are retained. The residual in (10.14) is zero in all 15 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 10, 29)$. Source: https://oeis.org/A243604; column 7 of https://oeis.org/A243607. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.41 A243605

Parameters: $q = 9$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 15$ and is proved for every $n \geq 16$, with coefficients

```
30, -365, 2297, -7771, 12461, -2366, -15807, 7237, 10936, -841, -3849, -1625, -307,
-28, -1
```

The complete matrix has $D = 17$ states, with $|W| = 17$ reachable states. All 289 coordinates of $v_0, \ldots, v_{16}$ are retained. The residual in (10.14) is zero in all 17 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 10, 29)$. Source: https://oeis.org/A243605; column 8 of https://oeis.org/A243607. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.42 A243606

Parameters: $q = 10$, $K = 2$, $s = 2$, and $\mathcal{B} = \{010\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 17$ and is proved for every $n \geq 18$, with coefficients

```
38, -603, 5165, -25592, 71533, -92714, -10048, 134330, -29886, -94396, -8366, 29547,
16394, 4033, 529, 36, 1
```

The complete matrix has $D = 19$ states, with $|W| = 19$ reachable states. All 361 coordinates of $v_0, \ldots, v_{18}$ are retained. The residual in (10.14) is zero in all 19 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 10, 29)$. Source: https://oeis.org/A243606; column 9 of https://oeis.org/A243607. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.43 A243635

Parameters: $q = 5$, $K = 2$, $s = 2$, and $\mathcal{B} = \{012\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 6$ and is proved for every $n \geq 7$, with coefficients

$$(c_1, \ldots, c_d) = (7, -14, 0, 21, -7, -6).$$

The complete matrix has $D = 9$ states, with $|W| = 9$ reachable states. All 72 coordinates of $v_0, \ldots, v_7$ are retained. The residual in (10.14) is zero in all 9 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 9, 21)$. Source: https://oeis.org/A243635; column 4 of https://oeis.org/A243641. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 4x - 19x^2 + 14x^3 + 30x^4 - 20x^5 - 12x^6.$$

10.5.6.44 A243636

Parameters: $q = 6$, $K = 2$, $s = 2$, and $\mathcal{B} = \{012\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 8$ and is proved for every $n \geq 9$, with coefficients

```
9, -25, 7, 64, -54, -48, 32, 16
```

The complete matrix has $D = 11$ states, with $|W| = 11$ reachable states. All 110 coordinates of $v_0, \ldots, v_9$ are retained. The residual in (10.14) is zero in all 11 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 9, 21)$. Source: https://oeis.org/A243636; column 5 of https://oeis.org/A243641. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 4x - 27x^2 + 40x^3 + 59x^4 - 126x^5 - 51x^6 + 80x^7 + 32x^8.$$

10.5.6.45 A243637

Parameters: $q = 7$, $K = 2$, $s = 2$, and $\mathcal{B} = \{012\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 10$ and is proved for every $n \geq 11$, with coefficients

```
11, -39, 21, 151, -217, -181, 339, 130, -154, -60
```

The complete matrix has $D = 13$ states, with $|W| = 13$ reachable states. All 156 coordinates of $v_0, \ldots, v_{11}$ are retained. The residual in (10.14) is zero in all 13 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 9, 21)$. Source: https://oeis.org/A243637; column 6 of https://oeis.org/A243641. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 4x - 35x^2 + 78x^3 + 87x^4 - 408x^5 - 16x^6 + 668x^7 + 57x^8 - 368x^9 - 120x^{10}.$$

10.5.6.46 A243638

Parameters: $q = 8$, $K = 2$, $s = 2$, and $\mathcal{B} = \{012\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 12$ and is proved for every $n \geq 13$, with coefficients

```
13, -56, 44, 305, -633, -476, 1772, 308, -2060, -368, 864, 288
```

The complete matrix has $D = 15$ states, with $|W| = 15$ reachable states. All 210 coordinates of $v_0, \ldots, v_{13}$ are retained. The residual in (10.14) is zero in all 15 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 9, 21)$. Source: https://oeis.org/A243638; column 7 of https://oeis.org/A243641. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 4x - 43x^2 + 128x^3 + 106x^4 - 976x^5 + 392x^6 + 2696x^7 - 1239x^8 - 3714x^9 + 363x^{10} + 2016x^{11} + 576x^{12}.$$

10.5.6.47 A243640

Parameters: $q = 10$, $K = 2$, $s = 2$, and $\mathcal{B} = \{012\}$. The source counts words of length $n+2$ for $n \geq 1$. Its empirical recurrence has order $d = 16$ and is proved for every $n \geq 17$, with coefficients

```
17, -99, 125, 931, -3185, -1785, 19063, -6754, -56936, 32132, 99764, -41096, -100896,
5152, 42048, 11520
```

The complete matrix has $D = 19$ states, with $|W| = 19$ reachable states. All 342 coordinates of $v_0, \ldots, v_{17}$ are retained. The residual in (10.14) is zero in all 19 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(4, 9, 21)$. Source: https://oeis.org/A243640; column 9 of https://oeis.org/A243641. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.48 A243724

Parameters: $q = 6$, $K = 2$, $s = 2$, and $\mathcal{B} = \{000, 012\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 7$ and is proved for every $n \geq 8$, with coefficients

```
5, 0, -29, 20, 51, -31, -30
```

The complete matrix has $D = 11$ states, with $|W| = 11$ reachable states. All 99 coordinates of $v_0, \ldots, v_8$ are retained. The residual in (10.14) is zero in all 11 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(3, 6, 12)$. Source: https://oeis.org/A243724; column 5 of https://oeis.org/A243729. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 3x - 9x^2 - 18x^3 + 52x^4 + 42x^5 - 74x^6 - 49x^7.$$

10.5.6.49 A243727

Parameters: $q = 9$, $K = 2$, $s = 2$, and $\mathcal{B} = \{000, 012\}$. The source counts words of length $n + 2$ for $n \geq 1$. Its empirical recurrence has order $d = 13$ and is proved for every $n \geq 14$, with coefficients

```
8, 0, -139, 196, 960, -1827, -3456, 7076, 7239, -12860, -9148, 9072, 5760
```

The complete matrix has $D = 17$ states, with $|W| = 17$ reachable states. All 255 coordinates of $v_0, \ldots, v_{14}$ are retained. The residual in (10.14) is zero in all 17 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(3, 6, 12)$. Source: https://oeis.org/A243727; column 8 of https://oeis.org/A243729. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$\begin{aligned} P(x) = 3x - 18x^2 - 36x^3 + 346x^4 + 99x^5 - 2696x^6 + 175x^7 + 10799x^8 \\ - 195x^9 - 22430x^{10} - 4582x^{11} + 19296x^{12} + 9401x^{13}. \end{aligned}$$

10.5.6.50 A244177

Parameters: $q = 3$, $K = 4$, $s = 4$, and $\mathcal{B} = \{00000, 01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 6$ and is proved for every $n \geq 7$, with coefficients

$$(c_1, \ldots, c_d) = (1, 3, 4, 5, 3, 2).$$

The complete matrix has $D = 23$ states, with $|W| = 23$ reachable states. All 184 coordinates of $v_0, \ldots, v_7$ are retained. The residual in (10.14) is zero in all 23 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(38, 108, 306)$. Source: https://oeis.org/A244177; column 2 of https://oeis.org/A244185. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 38x + 70x^2 + 84x^3 + 85x^4 + 50x^5 + 27x^6.$$

10.5.6.51 A244178

Parameters: $q = 4$, $K = 4$, $s = 4$, and $\mathcal{B} = \{00000, 01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 12$ and is proved for every $n \geq 13$, with coefficients

```
2, 6, 6, 3, -10, -23, -34, -36, -28, -19, -8, -3
```

The complete matrix has $D = 38$ states, with $|W| = 38$ reachable states. All 532 coordinates of $v_0, \ldots, v_{13}$ are retained. The residual in (10.14) is zero in all 38 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(48, 173, 650)$. Source: https://oeis.org/A244178; column 3 of https://oeis.org/A244185. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.52 A244179

Parameters: $q = 5$, $K = 4$, $s = 4$, and $\mathcal{B} = \{00000, 01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 18$ and is proved for every $n \geq 19$, with coefficients

```
4, 8, -5, -36, -86, -106, -52, 102, 337, 559, 701, 701, 582, 403, 227, 105, 34, 8
```

The complete matrix has $D = 53$ states, with $|W| = 53$ reachable states. All 1060 coordinates of $v_0, \ldots, v_{19}$ are retained. The residual in (10.14) is zero in all 53 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(49, 188, 790)$. Source: https://oeis.org/A244179; column 4 of https://oeis.org/A244185. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.53 A244180

Parameters: $q = 6$, $K = 4$, $s = 4$, and $\mathcal{B} = \{00000, 01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 24$ and is proved for every $n \geq 25$, with coefficients

```
7, 4, -44, -104, -72, 290, 1095, 2087, 2522, 1293, -2295, -8084, -14734, -20253,
-22955, -22128, -18438, -13301, -8283, -4402, -1946, -694, -179, -30
```

The complete matrix has $D = 68$ states, with $|W| = 68$ reachable states. All 1768 coordinates of $v_0, \ldots, v_{25}$ are retained. The residual in (10.14) is zero in all 68 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(49, 189, 811)$. Source: https://oeis.org/A244180; column 5 of https://oeis.org/A244185. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.54 A244181

Parameters: $q = 7$, $K = 4$, $s = 4$, and $\mathcal{B} = \{00000, 01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 30$ and is proved for every $n \geq 31$, with coefficients

```
11, -14, -109, -50, 561, 1905, 2485, -1765, -16073, -42234, -71345, -80685, -38274,
82534, 284947, 538989, 786421, 962575, 1022055, 956487, 794930, 587882, 386232,
224067, 113614, 49515, 18041, 5264, 1116, 144
```

The complete matrix has $D = 83$ states, with $|W| = 83$ reachable states. All 2656 coordinates of $v_0, \ldots, v_{31}$ are retained. The residual in (10.14) is zero in all 83 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(49, 189, 812)$. Source: https://oeis.org/A244181; column 6 of https://oeis.org/A244185. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.55 A244182

Parameters: $q = 8$, $K = 4$, $s = 4$, and $\mathcal{B} = \{00000, 01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 36$ and is proved for every $n \geq 37$, with coefficients

```
16, -57, -146, 471, 2030, 1606, -9926, -40696, -72180, -23242, 262788, 948320,
2028394, 3106669, 3270400, 1237472, -4174278, -13394281, -25599412, -38689072,
-49852288, -56526936, -57307380, -52390492, -43372994, -32559403, -22139372,
-13590504, -7488422, -3670971, -1580476, -586225, -181844, -44951, -8042, -840
```

The complete matrix has $D = 98$ states, with $|W| = 98$ reachable states. All 3724 coordinates of $v_0, \ldots, v_{37}$ are retained. The residual in (10.14) is zero in all 98 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(49, 189, 812)$. Source: https://oeis.org/A244182; column 7 of https://oeis.org/A244185. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.56 A244183

Parameters: $q = 9$, $K = 4$, $s = 4$, and $\mathcal{B} = \{00000, 01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 42$ and is proved for every $n \geq 43$, with coefficients

```
22, -139, 1, 1784, 1931, -12600, -50646, -48753, 215387, 1038438, 2236793, 1983939,
-4361986, -23908864, -62445686, -116557476, -164083468, -158681803, -35333656,
270032838, 791193237, 1504649448, 2319758240, 3094945131, 3677150092, 3949611368,
3868169748, 3471359234, 2861776683, 2168940385, 1510165759, 963934762, 562027256,
297750070, 142267724, 60680383, 22773619, 7366519, 1990304, 426740, 65808, 5760
```

The complete matrix has $D = 113$ states, with $|W| = 113$ reachable states. All 4972 coordinates of $v_0, \ldots, v_{43}$ are retained. The residual in (10.14) is zero in all 113 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(49, 189, 812)$. Source: https://oeis.org/A244183; column 8 of https://oeis.org/A244185. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.57 A244184

Parameters: $q = 10$, $K = 4$, $s = 4$, and $\mathcal{B} = \{00000, 01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 48$ and is proved for every $n \geq 49$, with coefficients

```
29, -277, 655, 3400, -7802, -49606, -30159, 403108, 1503590, 1635899, -5652490,
-31339466, -77219594, -98298345, 54151570, 669953408, 2120355686, 4618747926,
7785441378, 10142532705, 8820633444, -184894715, -21013605262, -56342602223,
-105733609417, -164683705023, -224995549990, -276580511002, -310166430329,
-319929413204, -305057789374, -269700758984, -221428750153, -168899208188,
-119626976395, -78560246496, -47719787532, -26718125970, -13723716584, -6426330933,
-2720546834, -1029660492, -343102672, -98490588, -23573519, -4460436, -603108, -45360
```

The complete matrix has $D = 128$ states, with $|W| = 128$ reachable states. All 6400 coordinates of $v_0, \ldots, v_{49}$ are retained. The residual in (10.14) is zero in all 128 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(49, 189, 812)$. Source: https://oeis.org/A244184; column 9 of https://oeis.org/A244185. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.58 A244694

Parameters: $q = 3$, $K = 4$, $s = 4$, and $\mathcal{B} = \{01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 7$ and is proved for every $n \geq 8$, with coefficients

```
3, 0, -1, 0, -1, 1, -1
```

The complete matrix has $D = 23$ states, with $|W| = 23$ reachable states. All 207 coordinates of $v_0, \ldots, v_8$ are retained. The residual in (10.14) is zero in all 23 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(39, 111, 317)$. Source: https://oeis.org/A244694; column 2 of https://oeis.org/A244702. The primary recurrence and the relevant parent formula block are explicitly empirical. The source rational generating function is also proved: $F = P/Q$, where $Q = 1 - \sum_{i=1}^{d} c_i x^i$ and

$$P(x) = 39x - 6x^2 - 16x^3 - 4x^4 - 11x^5 + 9x^6 - 14x^7.$$

10.5.6.59 A244695

Parameters: $q = 4$, $K = 4$, $s = 4$, and $\mathcal{B} = \{01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 13$ and is proved for every $n \geq 14$, with coefficients

```
5, -3, -5, 1, -3, 4, -4, 1, 3, 0, 2, -1, 1
```

The complete matrix has $D = 38$ states, with $|W| = 38$ reachable states. All 570 coordinates of $v_0, \ldots, v_{14}$ are retained. The residual in (10.14) is zero in all 38 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(49, 176, 661)$. Source: https://oeis.org/A244695; column 3 of https://oeis.org/A244702. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.60 A244696

Parameters: $q = 5$, $K = 4$, $s = 4$, and $\mathcal{B} = \{01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 19$ and is proved for every $n \geq 20$, with coefficients

```
8, -14, -9, 15, 7, 22, -9, 9, 13, -9, 0, -17, 0, -5, -6, -1, -3, 1, -1
```

The complete matrix has $D = 53$ states, with $|W| = 53$ reachable states. All 1113 coordinates of $v_0, \ldots, v_{20}$ are retained. The residual in (10.14) is zero in all 53 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(50, 191, 801)$. Source: https://oeis.org/A244696; column 4 of https://oeis.org/A244702. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.61 A244697

Parameters: $q = 6$, $K = 4$, $s = 4$, and $\mathcal{B} = \{01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 25$ and is proved for every $n \geq 26$, with coefficients

```
12, -41, 16, 81, 15, -19, -158, -58, -45, -106, -38, -53, 85, 57, 46, 67, 37, 56, 15,
15, 10, 3, 4, -1, 1
```

The complete matrix has $D = 68$ states, with $|W| = 68$ reachable states. All 1836 coordinates of $v_0, \ldots, v_{26}$ are retained. The residual in (10.14) is zero in all 68 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(50, 192, 822)$. Source: https://oeis.org/A244697; column 5 of https://oeis.org/A244702. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.62 A244698

Parameters: $q = 7$, $K = 4$, $s = 4$, and $\mathcal{B} = \{01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 31$ and is proved for every $n \geq 32$, with coefficients

```
17, -95, 160, 159, -307, -498, -480, 692, 1026, 917, 1051, 886, 1139, 217, -344, -582,
-767, -618, -822, -592, -433, -300, -169, -146, -50, -34, -15, -6, -5, 1, -1
```

The complete matrix has $D = 83$ states, with $|W| = 83$ reachable states. All 2739 coordinates of $v_0, \ldots, v_{32}$ are retained. The residual in (10.14) is zero in all 83 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(50, 192, 823)$. Source: https://oeis.org/A244698; column 6 of https://oeis.org/A244702. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.63 A244699

Parameters: $q = 8$, $K = 4$, $s = 4$, and $\mathcal{B} = \{01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 37$ and is proved for every $n \geq 38$, with coefficients

```
23, -190, 624, -299, -1735, -321, 3257, 7830, 2362, -6411, -12773, -15698, -15096,
-18309, -14657, -7203, 1319, 7754, 9212, 11865, 11915, 10867, 8583, 5912, 4520, 2834,
1796, 975, 505, 321, 117, 65, 21, 10, 6, -1, 1
```

The complete matrix has $D = 98$ states, with $|W| = 98$ reachable states. All 3822 coordinates of $v_0, \ldots, v_{38}$ are retained. The residual in (10.14) is zero in all 98 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(50, 192, 823)$. Source: https://oeis.org/A244699; column 7 of https://oeis.org/A244702. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.64 A244700

Parameters: $q = 9$, $K = 4$, $s = 4$, and $\mathcal{B} = \{01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 43$ and is proved for every $n \geq 44$, with coefficients

```
30, -343, 1785, -3418, -2904, 10172, 18515, -1513, -70620, -94811, -32588, 94041,
211477, 265558, 316407, 329408, 286665, 168596, 19643, -85855, -164056, -206321,
-227539, -216968, -179819, -141979, -104021, -74269, -48384, -29216, -17666, -9756,
-5441, -2576, -1217, -624, -231, -111, -28, -15, -7, 1, -1
```

The complete matrix has $D = 113$ states, with $|W| = 113$ reachable states. All 5085 coordinates of $v_0, \ldots, v_{44}$ are retained. The residual in (10.14) is zero in all 113 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(50, 192, 823)$. Source: https://oeis.org/A244700; column 8 of https://oeis.org/A244702. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.5.6.65 A244701

Parameters: $q = 10$, $K = 4$, $s = 4$, and $\mathcal{B} = \{01010, 01110\}$. The source counts words of length $n + 4$ for $n \geq 1$. Its empirical recurrence has order $d = 49$ and is proved for every $n \geq 50$, with coefficients

```
38, -574, 4276, -15029, 12263, 48204, -16847, -203774, -274320, 337901, 1264190,
1574718, 434569, -1814318, -3975030, -5698180, -6798034, -7216705, -6361941, -4141269,
-1415098, 1117645, 2963339, 4238015, 4880440, 4838799, 4315267, 3525637, 2742853,
2014392, 1390851, 910679, 565426, 343188, 195749, 106036, 55321, 27440, 13678, 5894,
2555, 1107, 410, 175, 36, 21, 8, -1, 1
```

The complete matrix has $D = 128$ states, with $|W| = 128$ reachable states. All 6528 coordinates of $v_0, \ldots, v_{50}$ are retained. The residual in (10.14) is zero in all 128 coordinates; the initial support lies in the verified closed set W. Thus (10.15) follows for every asserted index. The first three source values are $(50, 192, 823)$. Source: https://oeis.org/A244701; column 9 of https://oeis.org/A244702. The primary recurrence and the relevant parent formula block are explicitly empirical.

10.6 Verification, provenance and reuse

The archive contains this standalone manuscript, the numbered ledger, compact primary-source assertions, complete exact certificates, portable programs and internal review reports. After installing the dependency listed in `requirements.txt`, run

```
python3 r10_run_all.py
```

from the extracted directory. The checks require no network access and no previous-round program. Actual versions, counts and outcomes are recorded in the verification summary.

10.6.1 Why the finite certificates prove infinite assertions

For the word families, the state records the number of introduced letters and the equality pattern of the last few letters. Old letters absent from that suffix retain their exact multiplicity. This gives a finite transition matrix directly from the definition, before any empirical recurrence is consulted. The exact vector

$$z = T^b \mathbf{1} - \sum_{i=1}^{d} c_i T^{b-i} \mathbf{1}$$

vanishes on a transition-closed set containing the initial support. Multiplying by the initial row and every subsequent power of T proves every recurrence instance in the stated domain. It is not a fitted-prefix argument. Boundary shifts between word length and OEIS index are explicit.

The 65 selected compressed matrices have 2098 states. An independent implementation, which imports none of the compressed engine, keeps the actual letter names. It checks all 131159 states and 1052986 transitions in these full alphabets, including every row of the proposed compression. Direct enumeration of short words supplies 406 additional source checks. The main word verifier checks 1452 published terms and proves 31 source rational generating functions without increasing the count.

For the other generating functions, exact rational identities, radical operator identities, classical dimension formulas or defining polynomial recursions supply all-index arguments. Every inhomogeneous boundary polynomial is retained. The tatami specialization is derived at $z = -1$ from the defining polynomial quotient, rather than assumed from the conjectural generating function being proved.

The arithmetic and formal-series sections give written universal proofs. Numerical checks verify transcription, source coefficients and boundary cases. In particular, checking many periods is not the proof of A205572; the integral derivation argument proves all moduli and all indices at once. Reduction of non-locally-finite moment sums is performed only after finite Newton expansions have made the sums into rational functions.

10.6.2 Reproducibility and integrity

The delivered `.tex` embeds all proof sections and entry signatures; it has no external `\input` or `\include` dependencies. Compile with pdfLaTeX until references stabilize. `r10_build_dossier.py`

rebuilds the manuscript and catalogue. The README lists the retained inputs and reconstruction commands. An isolated regeneration check compares immutable mathematical outputs.

The archive manifest gives SHA-256 digests. Review reports bind the exact proof, source, code and certificate files inspected. The consistency check matches catalogue counts, proof labels, source offsets and retained mathematical inputs. The layout report identifies the PDF whose rendered pages were inspected. Timing fields and logs may change on a rerun; the mathematical assertions and exact certificates do not depend on timing.

10.6.3 Common references and limits

Individual OEIS links, original contributors and specialized references are given with the relevant statements. The general source is the OEIS Foundation's *On-Line Encyclopedia of Integer Sequences*, https://oeis.org/. The retained OEIS Open Appendix A comparison is https://arxiv.org/html/2608.11941v1; its scope is explicitly limited to the 97 unique identifiers in that appendix.

The component audits document named literature, known results removed, duplicate formulations and bounded alias screens. Such a screen cannot exclude every possible transformation between sequences. The resulting count is 100 precisely specified source-assertion packages with complete proofs or corrections. It is not a certificate of 100 worldwide-first discoveries, and it is not an official closure of 100 OEIS entries.

OEIS identifier locator

This locator includes all identifiers mentioned in the original manuscript bodies, including selected targets, supporting sequences, earlier work, uncounted supplements, and excluded candidates. Its size is not a selection count. The linked numbers identify the rounds containing the identifier; each round's catalogue gives the selected assertions and links to their proofs.

OEIS	Rounds	OEIS	Rounds	OEIS	Rounds	OEIS	Rounds
A000139	7	A121860	3	A210302	4	A243041	10
A000159	2	A123071	9	A210303	4	A243042	10
A000181	2	A126089	2	A210304	4	A243043	10
A000185	2	A126156	4	A210305	5	A243044	10
A000254	9	A127473	2	A210313	4	A243045	10
A000364	3	A127751	1	A210314	4	A243046	10
A000483	9	A127752	1	A210315	4	A243255	6
A000594	2	A127905	1	A210316	4	A243383	10
A000657	4	A128057	4	A210328	5	A243384	10
A000670	3	A129366	5	A210329	5	A243385	10
A000695	9	A129454	5	A210330	5	A243386	10
A000781	6	A129455	5	A210348	5	A243387	10
A000898	9	A130540	2	A210349	5	A243388	10
A000986	4	A131418	3	A210350	5	A243389	10
A001108	9	A132393	2	A210383	5	A243513	10
A001109	9	A133158	2	A210384	5	A243514	10
A001157	8	A133205	7	A210386	5	A243515	10
A001222	2	A133907	6	A210397	4	A243516	10
A001277	1	A133910	2	A210398	4	A243517	10
A001278	1	A134389	7	A210400	4	A243518	10
A001279	1	A134416	5	A210406	6	A243519	10
A001280	1	A134424	5	A210407	6	A243600	10
A001316	10	A135925	2	A210408	6	A243601	10
A001497	2	A136097	1	A210409	6	A243602	10
A001515	8	A136123	1	A210496	4	A243603	10
A001711	2	A140110	7	A210826	6	A243604	10
A001712	2	A141117	3	A212341	9	A243605	10
A001722	1	A141351	6	A212824	10	A243606	10
A001723	2	A141353	2	A212825	10	A243607	10
A001724	2	A141459	10	A212826	10	A243635	10
A001784	9	A142243	1	A212827	10	A243636	10
A001840	2	A143013	9	A212829	10	A243637	10
A001969	1	A143020	7	A214406	10	A243638	10
A002026	4	A143991	8	A216971	1	A243640	10
A002050	3	A144575	2	A217333	6	A243641	10
A002293	2	A146564	10	A218045	2	A243642	4
A002512	5	A146955	1	A218185	5	A243724	10
A002605	7	A152920	2	A220119	1	A243727	10
A002618	2	A153197	2	A220433	1	A243729	10
A002620	2	A153238	10	A220898	7, 8	A243763	5
A003586	8	A155051	7	A220899	7, 8	A243764	5
A003961	4	A155585	9	A224511	1	A243816	4
A004123	3	A155828	1	A224515	1	A244177	10
A004395	4	A156833	1	A225692	8	A244178	10
A004737	2	A156834	1	A226302	7, 10	A244179	10
A005046	3	A157003	6	A226602	5	A244180	10
A005329	6	A157021	2	A226787	8	A244181	10

OEIS	Rounds	OEIS	Rounds	OEIS	Rounds	OEIS	Rounds
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A005352	1	A157308	9	A227013	1	A244183	10
A005558	3	A157309	9	A227014	1	A244184	10
A005559	3	A157310	9	A227015	1	A244185	10
A005560	3	A157418	7	A227016	1	A244694	10
A005561	3	A158119	9	A227017	1	A244695	10
A005562	3	A158826	2	A227018	2	A244696	10
A005573	2	A159772	6	A227051	10	A244697	10
A005773	2	A160823	6	A228178	6	A244698	10
A006068	1	A161789	5	A228180	7	A244699	10
A006154	3	A162477	4	A228884	4	A244700	10
A006319	9	A162481	4	A228885	4	A244701	10
A006343	2	A162482	5	A229041	10	A244702	10
A006502	10	A162533	6	A229044	10	A244884	1
A006519	10	A162972	8	A230089	3	A244886	1
A006531	3	A163774	2	A230218	10	A246606	8
A006789	2	A163824	7	A233637	7	A247102	1
A007837	9	A164586	5	A233638	7	A247173	1
A008292	1	A164652	5, 9	A233639	7	A247471	2
A008364	1	A165201	7	A233640	7	A250917	2
A008365	1	A165641	10	A233646	8	A254314	6
A008486	1	A165886	10	A233647	8	A257104	1
A008578	1	A166228	4	A233653	8	A257388	6
A008930	6	A166306	2	A233675	8	A258409	4
A010694	9	A166756	2	A233676	8	A258899	3
A010786	5	A166757	2	A233677	8	A258903	3
A014533	8	A166758	2	A233682	8	A259708	10
A014865	3	A166759	2	A233684	7	A259709	10
A014866	3	A166760	2	A233685	7	A261363	9
A014868	3	A166761	2	A233686	7	A262735	6
A016789	1	A166762	2	A233687	8	A262768	5
A021009	2	A166776	2	A233691	8	A265649	2
A023887	8	A166777	2	A233710	7	A266489	7
A024172	2	A166778	2	A233711	7	A268149	8
A024180	2	A166779	2	A233718	8	A268293	7
A024219	1	A166780	2	A233724	8	A268402	8
A024220	1	A166781	2	A233726	8	A268600	1
A024385	1	A166782	2	A233727	8	A268601	1
A024388	1	A166783	2	A233728	8	A269027	1
A024389	1	A166786	2	A233733	8	A270490	5
A024390	1	A166796	2	A233749	7	A270530	5
A024399	1	A166797	2	A233750	7	A272261	2
A024401	1	A166798	2	A233787	9	A272462	10
A024403	1	A166799	2	A233792	9	A272468	10
A024719	8	A166800	2	A233877	9	A272651	10
A025060	4	A166801	2	A233878	9	A273228	6
A025256	9	A166802	2	A233879	9	A273351	5
A025258	2	A167137	3	A233880	9	A273905	10
A025259	5	A167138	9	A233883	9	A276646	5
A025260	9	A167139	9	A233885	7	A278023	6
A025268	6	A167141	9	A233886	7	A278083	5
A025269	6	A167532	9	A233910	7	A278085	9
A025270	6	A168268	9	A233911	7	A278472	7
A025272	5	A168452	2	A233921	6	A278618	8

OEIS	Rounds	OEIS	Rounds	OEIS	Rounds	OEIS	Rounds
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A025275	6	A168503	7	A233943	8	A279013	5
A026023	9	A168505	8	A233944	8	A279014	5
A026026	7	A171791	10	A233949	8	A280098	5
A026027	9	A171847	4	A233951	7	A284005	1
A026030	6	A171849	4	A233952	7	A285195	4
A026126	8	A171851	4	A233960	9	A286779	5
A026268	7	A173693	1	A233961	9	A288417	2
A026269	7	A173993	2	A233962	9	A290815	2
A026270	7	A174013	5	A233967	9	A291937	10
A026568	9	A174014	7, 10	A233982	6	A292177	10
A026569	9	A174015	10	A233983	6	A292394	7
A026570	9	A174016	7	A233984	6	A292395	7
A026571	6	A174107	6	A234029	9	A292461	2
A026572	9	A174169	8	A234030	9	A292877	10
A026582	9	A174193	1	A234031	9	A295012	5
A026641	2	A174490	8	A234032	9	A299406	6
A026725	9	A174491	8	A234036	9	A300616	8
A026732	9	A174808	6	A234107	9	A300617	8
A026741	9	A175505	7	A234108	9	A300732	7
A026743	6	A176604	5, 6, 7, 8	A234109	9	A300734	7
A026758	9	A176605	4, 5, 7, 8, 9	A234114	9	A300984	5
A026765	9	A176607	4, 5, 7, 8, 9	A234117	9	A301921	3
A027023	1, 2	A176610	4, 5, 7, 8, 9	A234118	9	A302180	8
A027027	2	A176611	4, 5, 7, 8, 9	A234119	9	A302181	8
A027028	2	A176612	4, 5, 7, 8, 9	A234122	9	A302184	8
A027050	1	A176615	4	A234124	9	A302186	8
A027051	2	A177258	1	A234125	9	A303290	10
A027052	1, 2	A177259	1	A234126	9	A305550	3
A027056	2	A177260	1	A234131	9	A306082	3
A027058	1	A177430	8	A234133	9	A308620	10
A027059	1	A177431	8	A234135	9	A308726	4
A027082	1, 2	A177510	6	A234140	9	A308815	8
A027087	2	A178113	2	A234146	7	A308833	10
A027088	1	A179504	8	A234147	7	A309263	7
A027089	1	A179648	4	A234154	9	A320192	4
A027113	2	A179873	7	A234155	9	A320352	3
A027116	2	A179929	5	A234156	9	A320949	10
A027117	2	A180195	1	A234157	9	A322327	2
A027170	1	A180654	9	A234158	9	A323599	9
A027175	1	A182879	5	A234161	9	A323689	1
A027182	1	A182892	8	A234162	6	A324300	1
A027907	6	A182894	5	A234163	6	A325286	10
A030204	5	A182899	4	A234170	9	A326478	10
A030212	7	A182904	5	A234171	9	A326691	8
A032115	2	A183586	6	A234175	9	A327096	2
A032123	2, 4	A183587	6	A234177	9	A328260	9
A032312	9	A183589	6	A234178	9	A329354	9
A033955	2	A183596	6	A234179	9	A329369	1
A034261	5	A184018	7	A234184	9	A329375	9
A034267	5	A184120	4	A234201	9	A329649	1
A034444	5	A184534	2	A234202	9	A329697	1
A035263	1	A185010	4	A234208	9	A330358	1
A037223	2	A185020	7	A234210	9	A334109	1

OEIS	Rounds	OEIS	Rounds	OEIS	Rounds	OEIS	Rounds
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A040003	9	A185072	10	A234212	9	A334592	1
A045406	4	A186185	1	A234217	9	A334595	2
A045742	1	A186335	2	A234220	6	A336838	4
A045743	4	A186338	4	A234221	6	A337500	1
A048651	6	A186776	5	A234222	6	A337758	10
A048702	3	A186940	4	A234241	8	A338328	10
A048868	3	A187011	5	A234245	8	A338632	5
A048896	9	A187250	1	A234259	8	A338636	5
A048993	2	A187252	1	A234260	8	A339100	10
A048994	2	A187256	4	A234261	8	A339384	5
A049803	1	A188570	3	A234266	8	A339387	5
A049804	1	A188571	3	A234269	1	A342912	2
A050168	4	A188572	3	A234307	2	A344502	8
A051176	9	A188573	3	A234326	8	A345651	1
A051283	2	A189176	5	A234327	8	A345993	8
A051699	5	A190160	2	A234328	8	A346074	1
A052174	3	A190166	5, 6	A234333	8	A346418	2
A052183	6	A190169	6	A234338	8	A347293	2
A052731	4	A190363	4	A234343	8	A348295	3, 9
A052736	9	A190725	7	A234444	8	A348299	3
A053532	2	A190736	7	A234445	8	A348410	2
A053644	1	A190788	7	A234450	8	A348713	6
A053645	1	A191107	5	A234452	8	A349802	2
A054120	1	A191319	7	A234453	8	A349890	6
A054523	7	A191398	7	A234458	8	A350512	1
A057660	2	A191579	2	A234483	8	A350557	1
A058087	2	A191792	4	A234484	8	A350900	2
A059259	2	A192069	9	A234490	8	A351933	1
A059279	7	A192248	9	A234491	8	A352117	8
A059297	2	A192249	9	A234492	8	A354242	3
A059837	1	A193679	8	A234497	8	A354253	3
A060043	7	A193758	9	A234557	6	A354766	9
A060791	9	A193844	2	A234558	6	A354933	2
A061002	6	A195194	3	A234559	6	A354995	2
A062803	7	A195986	10	A234651	7	A355282	1
A063081	10	A196523	8	A234652	7	A355372	9
A063087	10	A199716	3	A234653	7	A355401	1
A063098	10	A202943	9	A234659	9	A355407	9
A063105	10	A203150	1	A234660	9	A355414	9
A063110	10	A203157	1	A234661	9	A356459	7
A063116	10	A203159	1	A234665	9	A359643	7
A063118	10	A203160	1	A234667	9	A359914	4
A063126	10	A203163	1	A234668	9	A361552	4
A063128	10	A203164	1	A234672	9	A362560	1
A063140	10	A203165	1	A234674	8	A363409	9
A063147	10	A203166	1	A234675	8	A363411	9
A063148	10	A203167	1	A234681	8	A363412	9
A063168	10	A203231	1	A234683	9	A365095	8
A063233	10	A203233	1	A234684	9	A366561	7
A063247	10	A203234	1	A234690	9	A367865	8
A063647	10	A203235	1	A234697	9	A368628	3
A063946	1	A204243	6	A234698	9	A368630	4
A063956	6	A205355	3	A234699	9	A368631	4

OEIS	Rounds	OEIS	Rounds	OEIS	Rounds	OEIS	Rounds
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A065919	2	A205357	3	A234706	9	A368634	3
A065942	7	A205358	3	A234707	9	A369083	4
A068991	6	A205572	10	A234712	9	A371364	9
A069208	6	A206377	3	A234721	9	A372018	9
A071264	1	A206378	3	A234722	9	A372530	10
A071715	9	A206379	3	A234728	9	A372991	7
A072100	4	A206380	3	A234731	7	A373312	10
A075002	1	A208355	6	A234732	7	A373313	10
A075045	2	A209505	5	A234733	7	A373561	8
A077761	9	A209506	5	A234753	7	A375838	7
A078121	1	A209507	5	A234754	7	A376182	2
A078122	1	A209508	5	A234762	6	A376226	9
A078995	2	A209509	5	A234763	6	A376336	2
A078999	2	A209530	5	A234764	6	A376484	6
A079269	8	A209531	5	A234770	7	A376499	6
A079278	8	A209546	5	A234771	7	A377109	3
A079998	9	A209547	5	A234816	9	A377110	3
A080243	6	A209548	5	A234817	9	A377111	3
A080253	3	A209549	5	A234818	9	A377112	3
A080894	2	A209550	7	A234823	9	A377113	3
A081199	1	A209709	3	A234875	6	A377114	3
A081200	1	A209710	5	A234876	6	A377115	3
A081201	1	A209711	5	A234877	6	A377116	3
A081577	1	A209722	5	A234884	8	A377117	3
A082397	2	A209729	3	A234885	8	A377118	3
A082953	7	A209807	3	A234890	8	A377119	3
A083365	6	A209808	3	A234891	8	A377278	6
A084068	9	A209809	3	A234898	8	A378277	2
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A085361	5	A209811	3	A234911	8	A378578	4
A085765	1	A209822	3	A234984	8	A378580	3, 8
A086450	1	A209823	3	A234985	8	A378581	8
A087726	2	A209824	3	A234991	8	A379463	8
A089324	5	A209825	3	A234992	9	A379464	8
A089354	2	A209826	3	A234993	9	A379764	8
A091137	10	A209841	3	A234998	9	A379907	2
A091351	9	A209842	3	A235011	6	A380056	4
A091539	1	A209843	3	A235012	6	A380060	8
A092278	3	A209844	3	A235019	8	A380191	2
A092287	8	A209845	3	A235020	8	A380680	8
A092843	7	A209851	5	A235026	8	A380709	4
A094287	10	A209852	5	A235064	8	A380711	4
A094288	10	A209853	5	A235065	8	A381353	6
A095981	6	A209890	3	A235071	8	A381355	6
A096471	7	A209891	3	A235080	6	A381670	8
A096617	9	A209892	3	A235081	6	A381845	8
A096920	6	A209893	3	A235091	7	A382122	8
A097331	5	A209894	3	A235092	7	A382123	10
A097332	4	A209906	5	A235093	7	A382124	10
A097724	2	A209907	5	A235100	8	A382125	10
A098275	2	A209908	5	A235101	8	A383044	8
A098519	8	A209953	5	A235107	8	A383165	5
A098520	9	A209955	5	A235169	9	A383166	5

OEIS	Rounds	OEIS	Rounds	OEIS	Rounds	OEIS	Rounds
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A099553	10	A209962	3	A235175	9	A383700	7
A100095	4	A209963	3	A235180	8	A383701	7
A100311	1	A209964	3	A235181	8	A383702	7
A100631	2	A209965	3	A235186	8	A384270	9
A101601	8	A210005	4	A235191	8	A385001	7
A102086	2	A210006	4	A235192	8	A385252	6
A102318	4	A210007	4	A235193	8	A385590	6
A102319	5	A210008	4	A235198	8	A386830	7
A102880	6	A210045	3	A235205	9	A387084	5
A103138	8	A210046	3	A235206	9	A388390	7
A103517	10	A210047	3	A235212	9	A389425	3
A104624	5	A210048	3	A235216	7	A389536	3
A105696	1	A210054	4	A235217	7	A389537	6
A105794	2	A210055	4	A235218	7	A392211	1
A105864	6	A210056	4	A235233	9	A392976	1
A106391	7	A210057	4	A235234	9	A393170	6
A106407	10	A210063	5	A235239	9	A393171	6
A106608	9	A210070	5	A235251	8	A393172	3, 6
A107026	7	A210071	5	A235252	8	A393173	6
A107587	5	A210072	5	A235258	8	A393856	6
A107635	5	A210079	4	A235271	6	A393857	6
A108296	1	A210080	4	A235272	6	A393858	6
A108600	5	A210081	4	A235273	6	A393859	6
A108625	7	A210082	5	A235282	7	A394399	3
A108629	2	A210088	3	A235283	7	A394454	5
A108630	7	A210089	3	A235284	7	A394578	1
A110198	7	A210091	3	A235291	7	A394757	3
A110616	2	A210092	4	A235292	7	A394815	3
A110665	9	A210100	4	A235293	7	A395454	2
A110668	9	A210103	4	A235294	7	A395833	7
A111205	4	A210104	7	A235303	6	A396099	4
A111942	1	A210118	4	A235304	6	A396102	8
A111967	1	A210119	4	A235305	6	A396722	6
A112302	2	A210128	5	A235306	7	A396724	1
A112526	6	A210129	4	A235312	6	A396793	6
A112702	8	A210130	4	A235313	6	A396794	6
A113249	1	A210136	4	A235314	7	A396795	6
A113250	1	A210137	4	A239201	5	A396797	6
A113256	1	A210138	4	A239230	2	A396798	6
A113409	8	A210139	4	A240722	8	A396803	3
A114464	4	A210149	5	A242468	10	A396804	5
A114584	6	A210150	5	A242471	10	A396805	6
A114851	8	A210151	5	A242543	10	A396806	6
A116184	6	A210158	4	A242544	10	A396807	6
A116388	5	A210159	4	A242546	10	A396808	3
A116391	4	A210160	4	A242547	10	A396843	3
A117436	1	A210161	4	A242548	10	A396844	3
A118093	6	A210167	4	A242549	10	A396845	3
A119689	8	A210168	4	A242578	10	A396846	7
A119975	7	A210169	4	A242579	10	A397048	2
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A121553	1	A210176	6	A242581	10	A397242	5, 7
A121580	1	A210177	6	A242582	10	A397244	7

OEIS	Rounds	OEIS	Rounds	OEIS	Rounds	OEIS	Rounds
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A121584	1	A210179	7	A243017	10	A397345	7
A121639	1	A210260	4	A243022	6	A397346	7
A121696	1	A210261	4	A243027	10	A397347	7
A121697	1	A210262	4	A243028	10	A397348	7
A121745	1	A210263	5	A243029	10	A397349	7
A121747	1	A210269	5	A243033	10	A397356	3
A121751	1	A210270	5	A243038	10	A397588	9
A121752	1	A210271	5	A243039	10		
A121754	4	A210272	5	A243040	10		